

E-TENDER DOCUMENT

FOR

"Construction of Workman Housing Complex (BQ Area) on EPC mode -II of contract at OIL Duliajan, Assam"

PART – 1

E-Tender No: - RITES/NERPO/OIL/BQ-HOUSING/25

(TECHNICAL BID)

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Section 6B: Geotechnical Investigation Report (Indicative only)

Section 6C: Design Basis Report

Section 6D: Schedule of Finishes

AUGUST 2025



(A Government of India Enterprise)

Project Unit Head

4th floor, NEDFI House Ganeshguri

Dispur, Guwahati-781006

RITES LIMITED

Guwahati-781006

Notice Inviting e-Tenders

PU Head, Project Unit Guwahati, RITES Ltd. invites on behalf of **OIL India Limited**, online bid under EPC Mode-II of Contract on lumpsum component basis bids in two packet system for the following work(s):

NIT No.: RITES/NERPO/OIL/BQ-HOUSING/25

Name of Work: **Construction of Workman Housing Complex(BQ Area) on EPC mode -II of contract at OIL Duliajan, Assam.**

Estimated cost: **Rs. 160.02 Cr** Excl. GST

Earnest Money: Rs. 1,00,00,000.00

Period of Completion: 24 (Twenty-Four) Months,

Last time and date of submission of bid: 11:00 Hrs. (IST) on 18.09.2025

The bid forms and other details can be obtained from the website:
<https://etenders.gov.in/eprocure/app> and <https://rites.com/tenders>

Addendum/Corrigendum, if any, shall be hoisted online only.

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SECTION-1

NOTICE INVITING TENDER

AND

INSTRUCTIONS TO TENDERERS

SECTION-1

NOTICE INVITING TENDER AND INSTRUCTIONS TO TENDERERS

1.0 GENERAL

1.1 Tender Notice:

Tenders are invited through E-Tendering system by RITES Ltd., a Public Sector Enterprise under the Ministry of Railways, acting for and on behalf of **OIL India Limited** (Employer) as an Agent/Power of Attorney Holder, from eligible firms / agencies, satisfying the set eligibility and qualifying criteria, for the work of **“Construction of Workman Housing Complex (BQ Area) on EPC mode -II of contract at OIL Duliajan, Assam.”**

(Note: Throughout these bidding documents, the terms ‘bid’ and ‘tender’ and their derivatives are synonymous)

1.1(a) Category of this Tender: Works

1.2 Estimated Cost of Work:

The work is estimated to cost **Rs. 160.02 Cr (Rupees One Hundred Sixty Crore and Two lakhs Only)**. The Work estimate is generally based on CPWD PAR 2023 (excluding the element of GST on works contract service) and on market rates for non-schedule items. This Estimate, however, is given merely as a rough guide.

1.3 Time for Completion:

The time allowed for completion will be 24 (Twenty Four) months from the date of start which is defined in Schedule-F under Clause 5.1(a) of Clauses of Contract.

1.4 Brief Scope of Work:

- 1.4.1 The scope of work consisting “Design and Construction of various buildings Residential Blocks (Total-08 Towers each Tower (STILT+6) having total 192 Nos. dwelling unit), Guest House (G+3), Community Centre (G+1), Sub-station & Panel Room/Electrical Sub-station (Single Storey), External Development Works including other ancillary infrastructure (Guard rooms - 02 Nos. (Single Storey) including Control room for Main Gate, Security hut (Single Storey), Boundary Wall, Parking Sheds (Cantilever type), levelling work and raising original ground level work, internal roads & on-street parking areas (with & without shed) of the housing complex & external road, footpaths, Pump room & UGWT, signages, water supply system,

Storm water drainage, Sewerage system, STP, Horticulture, Landscaping etc.), having RCC framed structure for buildings and involves associated internal & external MEP services, installation, testing & commissioning of Elevators (lifts), HVAC, firefighting and other external development works etc.

- 1.4.2 The contractor is required to complete the project on EPC – Mode-II basis. The scope of work given is not exhaustive however, EPC Contractor is required to execute all the items as per Scope of work, approved drawings, Design basis reports (DBR), Technical specifications, Architectural Tender Drawings, Tender documents etc. covering the entire area in all respect to make it fit for its intended purpose.
- 1.4.3 The contractor shall prepare all coordinated drawings for all services and make all necessary co-ordination for satisfactory execution of all works/activities in time bound manner. The Contractor will appoint competent consultants having adequate experience in planning and design of Residential Buildings. The consultants will develop detail design and drawings based on the requirements specified in the tender document.
- 1.4.4 The Defects Liability Period (DLP) of 24 months will commence from the date of successful completion and handing over of the entire project. The defect liability period includes operation & maintenance of electromechanical equipments. The EPC Contractor shall deploy qualified personnel and provide necessary spares/tools during this period. A reporting mechanism and minimum service levels should be defined to ensure uninterrupted functioning of all systems.
- 1.4.5 The bidders may note that the work includes a number of specialized Civil/Electrical/Mechanical/Electronic Engineering services, Packaged type STP, Electrical Sub-Station, Gas Pipeline System as per PNGRB, Smart Metering, signages etc. which are to be executed as integral part of the project by engaging specialized agencies after approval by the Engineer-in-charge.
- 1.4.6 Maintenance will also be carried out by deploying the specialized agency with prior approval of Engineer-in-charge for all specialized works like PHE, firefighting, Electrical, HVAC, BMS and all other services etc.
- 1.4.7 A model, 3D view and walkthrough will be prepared & presented to RITES & Client within 15 days from award of work as per conceptual plan provided in tender. The latest software of design engineering including ETABS / STAAD PRO. (as desired by Engineer-in-charge), AutoCAD etc. will be used for design & Engg. Purpose. All softcopies shall be made available to RITES as required.
- 1.4.8 The Project is aimed for 3 (Three) Star GRIHA or above rating and EPC Contractor has to plan, execute and take final certification for the same.

- 1.4.9 The EPC contractor shall ensure that the design consultant/architect appointed by them is stationed at site during execution of work.
- 1.4.10 The bidder shall construct the building as per National Building Code 2016 to ensure that the building is accessible for people with disability.
- 1.4.11 All works must comply with the approved design, finishes, and relevant standards. Execution must be time-bound with full coordination among various stakeholders.
- 1.4.12 The details of building to be constructed under Workman Housing Complex (BQ Area) at OIL Duliajan are as follows:

S. No.	Buildings/Infrastructure	Tentative Plinth Area (Sqm)	Floor/Remarks
1	Residential Blocks (Total 08 Tower)	27355	Stilt+6
2	Guest House	1990	G+3
3	Community Centre	1035	G+1
4	Sub Station & Panel Room/Electrical Sub-station	368	Single Storey
5	External Development Works including other ancillary infrastructure (Guard rooms, Security hut, UGWT & Pump Room, STP etc.)	24,675 Sqm Ground area	As per approved drawings

The details of the buildings/Infrastructures, as tabulated above, are tentative. The building names, configurations, combinations, and phasing may be modified during the planning stage to comply with statutory norms, approvals from statutory authorities, and requirements arising from RITES/OIL India Limited, structural and MEP designs, or GRIHA compliance. However, such modifications shall not entail any change in the overall project cost, and no additional payment shall be admissible to the EPC Contractor on this account. The **rates quoted** by the EPC Contractor shall be deemed inclusive of all such changes and contingencies.

- 1.4.13 The detailed scope of work is provided at Section 2A (Scope of Work) of the Tender Document.

1.5 Availability of Site:

The site for the work is **available/** ~~shall be made available in parts, as specified below.~~

Duliajan is an industrial town of Dibrugarh district in the Indian State of Assam located in the upper north-east corner of India. The town is about 45 km east of Dibrugarh City. Nearest airport is Dibrugarh airport at a distance of 44 km.

1.6 Deadline for submission of bids is: 11:00 Hrs. on date: 18.09.2025

The Employer may extend the deadline for submission of Tenders by issuing an amendment in writing in accordance with Clause 6.3. The Employer may extend the deadline for submission of bids and/or the bid opening date and time, even otherwise, if it considers the same to be desirable / expedient. In case of such extension, all rights and obligations of the Employer and the Tenderer previously subject to the original deadline will be subject to new deadline. All Bidders are advised to see the website <https://etenders.gov.in/eprocure/app> for extension of deadline for submission of tenders and / or the bid opening date.

1.7 Eligibility Criteria for Participation in this Tender:

Subject to provisions in the Tender Document, participation in this Tender Process is open to all domestic bidders who fulfill the 'Eligibility' and 'Qualification criteria as prescribed in this tender document. Bidder should meet the following eligibility criteria as of the date of his bid submission and should continue to meet these till the award of the contract. Bidder shall be required to declare fulfillment of Eligibility Criteria in proforma-1. The Bidder, unless otherwise stipulated in the tender documents should meet following Eligibility Criteria:

- 1) The Bidder should be Private or Government Owned legal entity (individual, proprietary firm, firm in partnership, limited company (Private or Public Sector) or corporation).
- 2) The Bidder should not be (or proposes to be), a Joint Venture, **unless permitted explicitly as per clause 2.4 hereinafter and Critical Data Sheet.**
- 3) Bidder should have valid registration with Employees Provident Fund organization under 'EPF and Miscellaneous Provisions Act, 1952'.
- 4) Only Class –I Local Suppliers, as defined in Make-in-India policy, unless otherwise mentioned in these documents.
- 5) Only those contractors/ bidders, who commit themselves to Integrity Pact as per details in Clause 11.2 herein under, would be considered competent to participate

in the bidding process. **In other words, entering into this Pact would be a preliminary qualification. Any bid not accompanied by Integrity Pact duly signed by the bidder shall be considered to be a non-responsive bid and shall be rejected straight away.**

- 6) The Bidder should not have a conflict of interest, which substantially affects fair competition. The Bidder and any of its affiliates (associates, subsidiary, JV partner), should not have been engaged by Ministry/ Department/RITES/Employer to provide consultancy services for the preparation or implementation of this project. The prices quoted should be competitive and without adopting any unfair/ unethical/ anti-competitive means. No attempt should be made to induce any other bidder to submit or not to submit an offer for restricting competition.
- 7) The Bidder should not be insolvent, in receivership, bankrupt or being wound up, not have its affairs administered by a court or a judicial officer, not have its business activities suspended and must not be the subject of legal proceedings for any of these reasons.
- 8) The Bidder should not have been declared as Poor Performer by RITES and their name should not currently be in the 'Negative List' of RITES.
- 9) The Bidder (including their affiliates or subsidiaries or Contractors/ subcontractors for any part of the contract) should not be currently declared ineligible/suspended/ blacklisted/ banned/ debarred by RITES or by any Central/State Government Department/ Public Undertaking or Enterprise of Central/State Government and such ban should not be in force at the time of submission of the Bid or extended deadline for submission of bid.
- 10) The Bidder should not have an association (as a bidder/ partner/ director/ employee in any capacity) with
 - a. The near relations (defined as first blood relations, and their spouses, of the bidder or the bidder's spouse) of persons involved in decision making in the procurement as mentioned in Clause 17.1 hereinafter.
 - b. Of retired Gazetted or equivalent rank Officers of the Central or State Government or its Public Sector Undertakings as defined in clause 17.1 hereinafter
- 11) The Bidder must fulfill any other additional eligibility condition, if so prescribed, elsewhere in the Tender Document.

12) The Bidder should provide such evidence of their continued eligibility to the Employer/RITES Ltd if so requested.

1.8 Purchase Preference Policies of the Government

As detailed in the Tender Document, RITES reserves its right to grant preferences to eligible Bidders under various Government Policies/ directives (policies relating to Make in India; MSEs; Start-ups etc.).

1.9 CRITICAL DATA SHEET-

Sl. No.	Description	Details
1.	Published Date	17.30 Hrs. on 28.08.2025
	Bid Document Download / Sale Start Date	17.45 Hrs. on 28.08.2025
2.	Pre-bid Query Receipt Start Time & Date	10:00 Hrs & 29.08.2025
3.	Pre-bid meeting and Venue	11.00 Hrs. on 04.09.2025 Venue: RITES Ltd,NERPO, 4 th Floor, NEDFi House, Ganeshguri, Dispur, Guwahati-781006. Note: The meeting is intended to clarify the queries of bidders and hence only prospective bidders are informed to attend the meeting and not the manufacturers/ vendors.
4.	Pre-bid Query Receipt End Time & Date	17.00 Hrs & 04.09.2025
5.	Bid submission Start Date & Time	08.09.2025 & 11:00 Hrs
6.	Bid submission End Date & Time	18.09.2025 & 11:00 Hrs
7.	Bid Opening Date & Time	19.09.2025 & 11:30 Hrs
8.	Bid Validity (Refer Clause 8 below)	75 Days from Date of submission
9.	Tentative date of Opening of Financial Bid	10.10.2025 & 11:00 Hrs
10.	Tentative date of issue of Letter of Acceptance	27.10.2025 & 11:00 Hrs
11.	The work for which the Tender is being invited falls under the category of	Small*/Normal* /Large*

Sl. No.	Description	Details
	(Refer Clause 2.2 below)	(*Strike out whichever is not applicable)
12.	The work for which this Tender has been invited falls under (Refer Clause 2.3 below)	*Normal / *Difficult area. (*Strike out whichever is not applicable)
13.	In this tender Joint Venture is (Refer Clause 2.4 below)	*allowed / *not allowed . (*Strike out whichever is not applicable)
14.	Correction slips to GCC as applicable (Refer Clause 4.1 below)	Corrections Slip Nos.01 to 02
15.	Correction slips to SECTION-1 (Refer Clause 4.1 below)	Incorporated up to Correction Slip No.11
16.	Whether Clause 11.2 (Integrity Pact) shall be applicable	*YES / *NO (*Strike out whichever is not applicable)
17.	Cost of Tender Document: (Refer Clause 7 below)	Rs. 18,000/-Plus GST (GST @ 18 % i.e. Rs. 3240 /-) = Rs. 21240/- (Rupees Twenty-One Thousand Two Hundred Forty only)
18.	Earnest Money Deposit (EMD) – (Refer Clause 7 and 9 below)	Rs. 1,00,00,000.00 (Rupees One Crores only)
19.	i) Name, Address and email ID of IEM 1 ii) Name, Address and email ID of IEM 2 iii) Name, Address and email ID of IEM 3 (Refer Clause 11.2A below)	i) Sh. Amol Prabhakar Joshi, CES, (Retd.), 126A, Buty Layout C-604, Swapnil Trisha Apartments, Laxmi Nagar, Nagpur-440022. Email- amjosh_98@yahoo.com. ii) Sh. Jitendera Sharma, IFoS (Retd.) House No. 85, Sector 5-A, Ecocity 1, P.O.- Mullanpur Garibdas, New Chandigarh, Distt. SAS Nagar, Punjab-140901. Email- <u>Sharmaj7@gmail.com</u> iii) Mrs. Gauri Kumar, IAS(Retd.), Kalypso Court, Tower-1/502, JP

Sl. No.	Description	Details
		Wishtown, Sector – 128, Noida-201304. Email- ID: gkck1955@yahoo.co.in .
20.	Name, designation, address and email id of RITES' liaison officer for Integrity Pact (Refer Clause 11.2A below)	Sh. S. B. Samanta, Vertical Head (Finance), Shikhar, Plot No.1. Sector-29, Gurgaon-122001 (Haryana) Email: ed.fin@rites.com
21.	Eligibility of Local Suppliers and Non-local Suppliers as per Make in India Policy (Refer Clause- 12.15 below)	Class -I Local Suppliers
22.	Officer to be contacted regarding inspection of site. (Refer Clause-13 below)	Shri. Manjit Borkataki, Dy. General Manager (Civil), Project Unit-Guwahati-NERPO, Phone No. 6901582367,8876637833 Email: manjit.borkataki@rites.com
23.	Site Visit	14:00 Hrs to 17:00 Hrs From 30.08.2025 to 02.09.2025

**Bid opening date and time should not be less than 24 hours of the 'Bid submission End Date & Time'*

2.0 QUALIFICATION CRITERIA TO BE SATISFIED

- 2.1 The Qualification Criteria to be satisfied are given at Annexure I enclosed.
- 2.2 The Qualification Criteria to be satisfied will depend on the category of works, whether Small, Normal or Large. The work for which the Tender is being invited falls under the category of ~~Small~~/~~Normal~~/**Large** (****Strike out whichever is not applicable***).
- 2.3 The Qualification Criteria to be satisfied will also depend on whether the Work falls in Normal area or difficult area. Difficult area includes North East States, Jammu & Kashmir, Ladakh, Andaman & Nicobar Islands and the 60 districts requiring Integrated Action Plan of Government of India (List available at Annexure-IX). Normal area covers all areas other than difficult area.

The work for which this Tender has been invited falls under ~~*Normal~~/***Difficult** area.

*(*Strike out whichever is not applicable)*

2.4 In this tender Joint Venture is ~~*allowed/*not allowed~~.

*(*Strikeout whichever is not applicable; in case JV is not allowed, except the first line, the rest of Clause 2.4 shall be deleted).*

~~In case joint Venture is allowed the following will apply:~~

- ~~a) If JV is successful in the Bid, the Contract will be awarded in the name of JV. The JV Agreement should be executed within 15 days of receipt of Letter of Acceptance and the JV Agreement duly registered in accordance with law so as to be legally valid and binding on the members. The JV shall also open a Bank account in the name of JV and all payments due to the JV shall be credited by the Employer to that account only. To facilitate statutory deductions such as towards Income Tax, etc. made from the amounts due to the JV being credited to the concerned Government Departments, the JV shall arrange to obtain in the name of JV, PAN/TIN etc. as required.~~
- ~~b) Bid submitted by a Joint Venture of two or more firms as Partners/Members shall be accompanied by the following documents:~~
 - ~~I. A copy of Joint Venture MOU/Agreement duly notarized so as to be legally valid and binding on all the Partners/Members and incorporating the following provisions (**Suggested format at Annexure II**) should be uploaded:~~
 - ~~i. The Bid and, in case of a successful Bid, the Agreement shall be signed so as to be legally binding on all Partners/Members.~~
 - ~~ii. One of Partners/Members shall be nominated as being in charge and this authorization shall be evidenced by submitting Power of Attorney signed by legally authorized signatories of all the Partners/Members.~~
 - ~~iii. The Partner in charge/Lead Member shall be authorized to incur liabilities and receive instructions for and on behalf of any and all partners/members of the Joint Venture and entire execution of the Contract, shall be done exclusively with the Partner in charge.~~

~~iv. All the partners of the Joint Venture shall be liable jointly and severally for the execution of the contract in accordance with the Contract terms and a statement to this effect shall be included in the authorization through a Power of Attorney in favour of the Partner in-charge/Lead Member as well as in the Bid and in the Agreement (in case of a successful bid).~~

~~v. Indication of the precise responsibility of all the Partners/Members of the Joint Venture in respect of planning, design, construction equipment, key personnel and work execution of the Project duly indicating the percentage participation in JV by each Partner.~~

~~vi. In case of JV, the maximum number of Partners can be only three and the Partner in Charge/Lead Member shall have more than 50% participation in the JV and each of the other Members minimum 20% participation in the JV.~~

~~vii. All partners/members of the JV shall comply with the provisions in the Integrity Pact and any violation of the Pact by any partner/member shall be construed as a violation by the JV.~~

~~II. Power of Attorney in favour of the Partner in-charge/Lead Member on the lines mentioned in item “a” above. (Suggested format at Annexure IV)~~

2.5 The documents to be furnished by the Bidder to prove that he is satisfying the qualification Criteria laid down, should all be in the Bidder’s name, except in cases where through the name has changed, the owners continued to remain the same and in cases of amalgamation of entities.

3.0 FORMAT AND CHECK LIST FOR SUBMISSION OF INFORMATION ON QUALIFICATION CRITERIA

The information to be furnished and the documents to be enclosed shall be as per Clause 28.0 hereinafter. Documents/ information complete in all respects, in support of meeting the Qualification Criteria should be submitted in one go. Submission of additional documents shall not be permitted. Only clarifications and filling of gaps/missing information in the submitted documents, may be permitted.

4.0 CONTENTS OF TENDER DOCUMENT

- 4.1 Each set of Tender or Bidding Document will comprise the documents listed below and addendum issued in accordance with Clause 6 herein below:**

PART-1: - Technical Bid Packet

Section 1 Notice Inviting Tender and Instructions to Tenderers including Annexures
Section 2 Additional Conditions of contract
Section 2A Scope of Work
Section 3 Special Conditions of contract
Section 4 Proforma of Schedule (Schedules A to F)
Section 5 Technical Specifications
Section 6 Drawings
Section 6A Topographic Survey Report/Drawings (Indicative only)
Section 6B Geotechnical Investigation Report (Indicative only)
Section 6C Design Basis Report
Section 6D Schedule of Finishes

PART-2: - FINANCIAL BID PACKET

Schedule of Quantities (Bill of Quantities)

PART-3: - General Conditions of Contract EPC Projects

Section 7 Conditions of Contract
Section 8 Clauses of Contract
Section 9 RITES Safety Code
Section 10 RITES Model Rules for Protection of Health and Sanitary Arrangements for Workers
Section 11 RITES Contractor's Labour Regulations

- 4.2 Deleted**

- 4.3** The bidder is required to submit/upload only the documents mentioned in Clause 28.0 of Section I thereof. Rest of the Part-1 & Part-3 need not be uploaded. The bidder must, nevertheless, read the same. It shall be presumed that the bidder has read the contents of Part 1: Technical Bid Packet and Part 3: General Conditions of Contract for EPC Projects and the same will be binding upon him. The successful bidder will be required to sign the complete tender document i.e., Part 1, Part 2, Part 3 and Correction Slips, if any, thereto.

5.0 INSTRUCTIONS ON ACCESSING/PURCHASING OF BID DOCUMENTS AND SUBMISSION THEREOF

5.1 To participate in the E-Bid submission for RITES, it is mandatory for the bidders to get their firms registered with E-Procurement Portal <https://etenders.gov.in/eprocure/app>

5.2 The bidders are required to submit soft copies of their bids electronically on the CPP Portal, using valid Digital Signature Certificates. The instructions given below are meant to assist the bidders in registering on the CPP Portal, prepare their bids in accordance with the requirements and submitting their bids online on the CPP Portal.

5.3 REGISTRATION

- a) Bidders are required to enroll on the e-Procurement module of the Central Public Procurement Portal (URL:<https://etenders.gov.in/eprocure/app>) by clicking on the link “**Online Bidder Enrolment**” on the CPP Portal which is free of charge.
- b) As part of the enrolment process, the bidders will be required to choose a unique username and assign a password for their accounts.
- c) Bidders are advised to register their valid email address and mobile numbers as part of the registration process. These would be used for any communication from the CPP Portal.
- d) Upon enrolment, the bidders will be required to register their valid Digital Signature Certificate (Class III Certificates with signing key usage) issued by any Certifying Authority recognized by CCA India with their profile.
- e) Only one valid DSC should be registered by a bidder. Please note that the bidders are responsible to ensure that they do not lend their DSC’s to others which may lead to misuse.
- f) Bidder can log in to the site through the secured log-in by entering their user ID/Password and the password of the DSC/e-Token.

5.4 SEARCHING FOR TENDER DOCUMENTS

- a) There are various search options built in the CPP Portal, to facilitate bidders to search active tenders by several parameters. These parameters could include Tender ID, Organization Name, Location, Date, Value, etc. There is also an option of advanced search for tenders, wherein the bidders may combine a number of search parameters such as Organization Name, Form of Contract, Location, Date, Other keywords etc. to search for a tender published on the CPP Portal.
- b) Once the bidders have selected the tenders they are interested in, they may

download the required documents/tenders schedules. These tenders can be moved to the respective 'My Tenders' folder. This would enable the CPP Portal to intimate the bidders through SMS/E-mail in case there is any corrigendum issued to the tender document.

- c) The bidder should make a note of the unique Tender ID assigned to each tender; in case they want to obtain any clarification/help from the Helpdesk.

5.5 PREPARATION OF BIDS

- a) Bidder should take into account any corrigendum published on the tender document before submitting their bids.
- b) Bidder is advised to go through the tender advertisement/NIT and the tender document carefully to understand the documents required to be submitted as part of the bid. Bidder may please note the number of covers in which the bid documents have to be submitted, the number of documents – including the names and content of each of the document that need to be submitted. Any deviations from these may lead to rejection of the bid.
- c) Bidder, in advance, should get ready the bid document to be submitted as indicated in the tender document/schedule and generally, they can be in PDF/XLS/RAR/JPG formats. Bid documents may be scanned with 100 dpi with black and white option which helps in reducing size of the scanned document.
- d) To avoid the time and effort required in uploading the same set of standard documents which are required to be submitted as a part of every bid, a provision of uploading such standard documents (e.g. PAN Card copy, Annual Reports, Auditor Certificates etc.) has been provided to the bidders. Bidders can use "My Space" or "Other Important Documents" area available to them to upload such documents. These documents may be directly submitted from the "My Space" area while submitting a bid, and need not be uploaded again and again. This will lead to a reduction in the time required for bid submission process.

5.6 SUBMISSION OF BIDS

- a) Bid can be submitted only during validity of registration of bidder with CPPP E-Procurement Portal.
- b) Bidder should log into the site well in advance for bid submission so that they

can upload the bid in time i.e. on or before the bid submission time. Bidder will be responsible for any delay due to other issues.

- c) The Bidder has to digitally sign and upload the required bid documents one by one as indicated in the tender document.
- d) Bidder has to select the payment option as “offline” to pay the cost of tender document and EMD as applicable and enter details of the instruments.
- e) Bidder should prepare the financial instruments of the Cost of Tender Documents and EMD as per the instructions specified in Clause 7.0 (f) hereinafter. The originals should be posted/couriered/given in person to the concerned official, so as to reach him within a week from the date of opening. The details of the DD/any other accepted instrument, physically sent, should tally with the details available in the scanned copy and the data entered during bid submission time. If the date of issue of DD/any other accepted instrument, physically sent, is on or before the bid submission end date, the same shall also be accepted even if the details are different from the scanned copy uploaded along with the bid. Otherwise, the uploaded bid will be rejected.
- f) Bidders are requested to note that they should necessarily submit their financial bids in the format provided and no other format is acceptable. If the price bid has been given as a standard BOQ format with the tender documents, then the same is to be downloaded and to be filled by all the bidders. Bidders are required to download the BOQ file, open it and complete the Sky Blue coloured (unprotected) cells with their respective financial quotes and other details (such as name of the bidder). No other cells should be changed. Once the details have been completed, the bidder should save it and submit it online, without changing the filename. If the BOQ file is found to be modified by the bidder, the bid will be rejected.
- g) The server time (which is displayed on the bidders’ dashboard) will be considered as the standard time for referencing the deadlines for submission of the bids by the bidders, opening of bids etc. The bidders should follow this time during bid submission.
- h) All the documents being submitted by the bidders would be encrypted using PKI encryption techniques to ensure the secrecy of the data. The data entered cannot be viewed by unauthorized persons until the time of bid opening. The confidentiality of the bids is maintained using the secured Socket Layer 128 Bit encryption technology. Data storage encryption of sensitive fields is done. Any

bid document that is uploaded to the server is subjected to symmetric encryption using a system generated symmetric key. Further this key is subjected to asymmetric encryption using buyers/bid openers public keys.

- i) The uploaded tender documents become readable only after the tender opening by the authorized bid openers.
- j) Upon the successful and timely submission of bids (i.e. after clicking “Freeze Bid Submission” in the portal), the portal will give a successful bid submission message & abided to the bid. A bid summary will be displayed with the bid ID and the date & time of submission of the bid with all other relevant details.
- k) The bid summary has to be printed and kept as an acknowledgement of the submission of the bid. The acknowledgement may be used as an entry pass for any bid opening meetings.

5.7 ASSISTANCE TO BIDDERS

- a) Any queries relating to the process of online bid submission or queries relating to CPP Portal in general may be directed to the 24x7 CPP Portal Help Desk Number 0120-4200462, 0120-4001002, 0120-4001005, 0120-6277787, E-mail id: support-eproc@nic.in
- b) Bidders’ information useful for submitting online bids on the CPP Portal may be obtained at: <https://etenders.gov.in/eprocure/app?page=BiddersManualKit&service=page>
- c) It is mandatory for all bidders to have Class-III Digital Signature Certificate (DSC) in the name of the person along with name of Company who will digitally sign the bid from any of licensed Certifying Agency (CA). Bidders can see the list of licensed CAs from the link <https://www.cca.gov.in>
- d) Bidder shall ensure use of registered Digital Signature Certificate (DSC) only and safety of the same.
- e) In case the Digital Signature Certificate (DSC) holder who is digitally signing the bid and the person having Authority to Sign as per Clause 11 are different, even then all the terms and conditions of the tender document will be binding upon the bidder.

5.8 CLARIFICATIONS ON TENDER DOCUMENTS

A prospective Tenderer requiring any clarification on the Tender Document may

notify through queries, on line only within the specified period (refer clause 1.9-Critical Data Sheet herein before)

Request for clarifications including request for Extension of Time for submission of Bid, if any, must be received not later than 10 (ten) days prior to the deadline for submission of tenders. Details of such queries raised and clarifications furnished will be uploaded in CPP website <https://etenders.gov.in/eprocure/app> without identifying the names of the bidders who had raised the queries. Any modification of the Tender Document arising out of such clarifications will also be uploaded on CPP website.

6.0 AMENDMENT OF TENDER DOCUMENT

6.1 Before the deadline for submission of tenders, the Tender Document may be modified by RITES Ltd. by issue of addendum/corrigendum.

6.2 Addendum/ Corrigendum, if any, will be hosted on web site <https://etenders.gov.in/eprocure/app> and shall become a part of the tender document. All tenderers are advised to see the website for addendum/corrigendum to the tender document which may be uploaded up to 7 days prior to the deadline for submission of tender as finally stipulated.

6.3 To give prospective tenderers reasonable time in which to take the addenda/corrigenda into account in preparing their tenders, extension of the deadline for submission of tenders may be given before bid submission end date and time as considered necessary by RITES. Sometimes due to administrative reasons, the deadline for submission of tenders may be extended latest by the deadline for opening of tender as stipulated including extension given earlier. All tenderers are advised to see the website for extension of deadline for submission of tenders.

6.4 Tenderer who has downloaded the tender from Central Public Procurement Portal (CPPP) website <https://etenders.gov.in/eprocure/app> shall not tamper/modify the tender form including downloaded Price Bid Template in any manner. In case if the same is found to be tampered/ modified in any manner, tender will be completely rejected and EMD would be forfeited and tenderer is liable to be banned from doing business with RITES Ltd.

7.0 PREPARATION AND SUBMISSION OF BIDS

- a) Tender document may be downloaded from CPPP and the bids (Part-2 only) along with the information and documents specified in Clause 28 herein after, shall be submitted online following the instructions appearing on the screen. **Documents specified in Clause 28 of Section 1, Part-1 are required to be uploaded along**

with Part-2 (Financial Bid); the rest of Part-1 and the whole of Part-3 of the tender document need not be submitted online but it shall be deemed to have been submitted. Users are requested to map their system as per the System settings available on the link [https://etenders.gov.in/eprocure/app?page=BiddersManualKit &service=page](https://etenders.gov.in/eprocure/app?page=BiddersManualKit&service=page) on the CPP portal.

- b) After downloading/getting the tender document/ schedules from <https://etenders.gov.in/eprocure/app> the Bidder should go through them carefully and then submit the documents as asked, otherwise bid will be rejected. It is construed that the bidder has read all the terms and conditions before submitting their offer. Bidders are advised that prior to bid submission they should read the bid submission manual available on CPP Portal <https://etenders.gov.in/eprocure/app?page=BiddersManualKit&service=page> web site
- c) Bidders may ensure that all the pages of the documents mentioned in Clause 28 must be signed & stamped by authorized signatory and serially numbered. In case, it is found that bidder has not complied with the same, the documents shall be deemed to be signed and stamped as this is a digitally signed e-tender.
- d) The bids shall be submitted online following the instructions appearing on the screen. Bidders may insert their e-Token/Smart Card in their computer and Log onto CPP portal <https://etenders.gov.in/eprocure/app> using the User-Id and Password chosen during registration. Then they may enter the password of the e-Token/Smart Card to access the DSC.
- e) Prior to bid submission, bidder should get ready with the documents to be uploaded as part of the bid as indicated in the tender document/schedule. Generally, they can be in Excel/PDF/RAR/JPG formats. No other format is accepted. If there is more than one PDF document, then they can be clubbed together in a Zip file for uploading. There is no limit for uploading file. Bids shall be submitted online only at CPP website <https://etenders.gov.in/eprocure/app>

Tenderer/Contractor are advised to follow the instructions provided in the 'Instructions to the Contractors/Tenderer for the e-submission of the bids online through the Central Public Procurement Portal for e-procurement at <https://etenders.gov.in/eprocure/app>

Bid documents may be scanned with 100 dpi with black and white option which helps in reducing size of the scanned document.

Intending tenderers are advised to visit CPPP website <https://etenders.gov.in/eprocure/app> till the specified date and time of opening of tender to check if there is any extension of deadline of submission of tender.

f) **Cost of Tender Document & Earnest Money Deposit (EMD)**

During bid submission the bidder has to select the payment option as offline to pay the Cost of Tender Document and EMD and enter details of the instruments. In case of exemption from payment of cost of tender document and EMD as a matter of Govt. Policy, the scanned copy of document in support of exemption will have to be uploaded by the bidder during bid submission. The onus of proving that the bidder is exempted from payment of cost of tender document and/or EMD lies on the bidder. In this connection, it should be noted that mere opening of bid does not mean that the bid has to be considered by RITES as a valid bid. If later, it is discovered from the uploaded documents that bidder is not exempted from payment of cost of tender and/or EMD; his bid shall be treated as non-responsive. It may be noted that the benefits under Public Procurement Policy for Micro and Small Enterprises (MSEs) Order, 2012 are applicable to only Supply and Service Contracts only. **For this tender, the benefits under Public Procurement Policy for Micro and Small Enterprises (MSEs) Order, 2012 are not applicable**

- i. **Cost of Tender Document:** The Cost of Tender Document is **Rs. 18,000.00 Plus GST (GST@ 18% i.e. Rs. 3,240/-) =Rs. 21,240/- (Rupees Twenty-One Thousand Two Hundred and Forty Only)** which is non-refundable. It shall be in the form of an Account Payee Banker's Cheque/Pay Order/Demand Draft in favour of "RITES Ltd." issued by a scheduled Commercial Bank, payable at ~~Gurgaon/Delhi~~/ Guwahati. **No other mode of payment will be acceptable.**
- ii. **Earnest Money Deposit (EMD):** The total Earnest Money deposit (EMD) is **Rs. 1,00,00,000.00 (Rupees One Crore Only)**. The bids shall be accompanied by Earnest Money Deposit (EMD) of an amount and in the form specified in Clause 9 hereinafter.

Bidders are required to upload scanned copy of acceptable instruments for EMD and Cost of Tender document in different files (Either in PDF or Zip format) during on-line submission of Bid. These documents shall be deposited in "ORIGINAL" in a sealed envelope within a week from the date of opening to: **Project Unit Head, 4th floor, NEDFI House Ganeshguri, Dispur, Guwahati-781006**

Failing which the bid shall be rejected and the bidder shall be debarred from tendering

in RITES Ltd. for a period of **02 (two) years** unless the lapse is condoned by the Accepting Authority at the request of the bidder for valid reasons. The envelope should bear the tender details (Tender No., Tender Name etc.)

- (a) The bid should be submitted online in the prescribed format. No other mode of submission is accepted.
- (b) Bid shall be digitally signed by a representative of the bidder and submitted “on-line”. No hard copy of the documents (except those specifically asked for in the tender document) is required to be submitted.
- (c) The bidders will have to accept unconditionally the online user portal agreement which contains the Terms and Conditions of NIT including General and Special Terms & Conditions and other conditions, if any, along with on-line undertaking in support of the authenticity regarding the facts, figures, information and documents furnished by the bidder on-line in order to become an eligible bidder.
- (d) The bidder has to digitally sign and upload the required bid documents one by one as indicated. Bidders to note that the very act of using DSC for downloading the bids and uploading their offers shall be deemed to be a confirmation that they have read all sections and pages of the tender/bid document including terms and conditions without any exception and have understood the entire document and are clear about tender requirements which will be binding upon the bidder.
- (e) The bidders are requested to submit the bids through online e-tendering system before the deadline for submission of bids (as per Server System Clock displayed on the portal). RITES will not be held responsible for any sort of delay or the difficulties faced during online submission of bids by the bidders at the eleventh hour.
- (f) The bidder may seek clarification online only within the specified period. The identity of bidder will not be disclosed by the system. RITES Ltd. will clarify the relevant queries of bidders as far as possible. The clarifications given will be visible to all the bidders intending to participate in that tender. The clarifications may be asked from the day of “Pre Bid Query Receipt Start Date and Time” till “Pre Bid Query Receipt End Date and Time”.

8.0 TENDER VALIDITY

8.1 The Tender shall be valid for a period of **75 days** from the due date for submission of Tender or any extended date as indicated in sub para below.

8.2 In exceptional circumstances, during the process of evaluation of tenders and prior

to the expiry of the original time limit for Tender Validity, the Employer may request that the Tenderers may extend the period of validity unconditionally for a specified additional period. The request and the tenderer's response shall be made in writing/ e- mail. A Tenderer may refuse the request without forfeiting his Earnest Money. A Tenderer agreeing to the request will not be permitted to modify his Bid but will be required to extend the validity of the Earnest Money for the period of the extension.

8.3 In case, the bidder(s) refuse to extend the validity of bid on such request or withdraw its offer between the deadline for submission of bids and the expiration of the period of bid validity, following shall be applicable:

- (a) If the above event happens before completion of the Techno-commercial evaluation, then the Techno-commercial evaluation (including the withdrawn bids) shall be completed. If a withdrawn bid qualifies in techno-commercial evaluation, financial bid(s) of such bidders shall also be opened, and action shall be taken as per sub-para below:
- (b) If the above event happens after the techno-commercial evaluation but before the completion of the financial bid evaluation, then the financial bid evaluation (including withdrawn bids) shall be completed. If a withdrawn bid happens to be the L1 bidder (lowest acceptable bidder, who is techno-commercially qualified for delivery of services and would have been awarded a contract, but for his refusal to extend validity), the tender must be retendered. However, such L1 price of the withdrawn bids shall not be taken as precedence for determining price estimates or reasonableness.

9) EARNEST MONEY

9.1 The Tender should be accompanied by earnest money of **Rs. 1,00,00,000.00 (Rupees One Crore Only)** as per following:

Estimated Value of the Work in Rupees: 160.02 Cr	Earnest Money Deposit (EMD) in Rupees: Rs. 1,00,00,000.00
For works of estimated cost of any value	(2% of the estimated cost of the work rounded off to nearest Rs 1000/ subject to a maximum of ₹ 1 Crore)

The EMD shall be deposited in any of the following forms: -

Insurance Surety Bonds or Account Payee Banker's Cheque /Pay Order / Demand Draft or Fixed Deposit Receipts/ Bank Guarantee (including e-Bank Guarantee) issued by any Scheduled Commercial Bank drawn in favour of RITES Ltd. and payable at ~~Gurgaon / Delhi/~~ **Guwahati**.

Insurance Surety Bonds/ Fixed Deposit Receipts/ Bank Guarantee (including e-Bank Guarantee) should have validity for **180 days** or more from the last date of receipt of bids. The Bank Guarantee (including e-Bank Guarantee) should be submitted in the format given at Annexure-X.

Insurance Surety Bond- An Insurance Surety Bond (ISB) is a three-party agreement that provides financial assurance to one party (the beneficiary) by another party (the surety or bonding company) on behalf of a third party (the principal). ISB ensures that the principal fulfils their contractual obligations.

Unlike a Bank Guarantee, it is a type of premium-based insurance product and does not require a deposit of a collateral amount by the principal with the surety. Here are the key components:

1. **Principal:** The principal is the party that obtains the surety bond. The principal is typically the contractor or service provider who provides the Bid/ performance security to the Procuring Entity.
2. **Beneficiary:** The beneficiary is the party (Procuring Entity) that requires the Insurance surety bond. The beneficiary seeks financial protection in case the principal fails to meet their obligations.
3. **Surety Insurer:** The surety insurer is the bond issuing entity (Bank or Insurance company) that issues the bond. They act as a guarantor, assuring the beneficiary that the principal will perform as promised. If the principal defaults, the surety insurer assesses the extent of default and determines the amount payable under the bond. If the principal does not pay within 14 days, the surety insurer pays within 45 calendar days of receiving the necessary documentation.

9.2 Any Tender not accompanied by scanned copies of the instruments for payment of Earnest Money and Cost of tender document in an acceptable form (or, if applicable, the scanned copy of documents in support of exemption) shall be rejected by the Employer as non-responsive.

9.3 Refund of Earnest Money

The Earnest Money of the Tenderers whose Technical Bid is found not acceptable will be returned without interest within 30 days of declaration of result of technical evaluation after scrutiny of Technical Bid has been completed by the Employer subject to provisions of Clause 9.4(a) and Clause 9.4 (b). The Earnest Money of the Tenderers whose Technical Bid is found acceptable will be returned at the earliest after expiry of final bid validity period and latest by 30th day after the award of the contract subject to provisions of Clause 9.4(a) and Clause 9.4 (b). The bidder shall submit RTGS/NEFT Mandate Form as per Proforma given in Annexure VII, duly filled in.

The Earnest Money (EMD) of the successful bidder shall be returned when the bidder has furnished the required Performance Guarantee in terms of Clause 1 of the Clauses of Contract (Section-8), signed the Agreement in terms of Clause 9 (a) of Conditions of Contract (Section-7).

9.4 The Earnest Money is liable to be forfeited automatically without any notice to bidder/Contractor

- (a) If after bid opening, but before expiry of bid validity or issue of Letter of Acceptance, whichever is earlier, any Bidder
 - i. withdraws his tender; or
 - ii. makes any modification in the terms and conditions of the tender which are not acceptable to the Employer
 - iii. impairs or derogates from the tender in any respect within the period of validity of the tender
 - iv. If the bidder does not accept the correction of his price during evaluation
- (b) in case any information/document which may result in the Bidder's disqualification is concealed by the Bidder or any statement/information /document furnished by the Bidder or issued by a Bank/Agency/Third Party and submitted by the Bidder, is subsequently found to be false or fraudulent or repudiated by the said Bank/Agency/Third Party.
- (c) in the case of a successful Bidder, if the Bidder fails to furnish the Performance Guarantee within the period specified under Clause 1 of "Clauses of Contract".

In case of forfeiture of Earnest Money as prescribed here in above, the Bidder shall not be allowed to participate in the retendering process of the work.

10.0 MODIFICATION/ SUBSTITUTION/ WITHDRAWAL OF BIDS

- 10.1** The Tenderers shall submit offers which comply strictly with the requirements of the Tender Document as amended from time to time as indicated in Clause 6.0 above. Alternatives or any modifications by the tenderer shall render the Tender invalid.
- 10.2** The bidder can modify, substitute, re-submit or withdraw its E-bid after submission but prior to the deadline for submission of bids or the extended deadline, as the case may be. No Bid shall be modified, substituted or withdrawn by the bidder on or after the deadline for submission of bids or the extended deadline, as the case may be. Withdrawal of bid after such deadline would result in the forfeiture of EMD.

- 10.3** Any modification in the Bid or additional information supplied subsequently to the deadline for submission of bids or the extended deadline, as the case may be, unless the same has been explicitly sought for by RITES, shall be disregarded.
- 10.4** For modification of E-bid (Technical Bid), bidder has to detach its old bid from CPP portal and upload / re-submit digitally signed modified bid.
- 10.5** For withdrawal of bid, bidder has to click on withdrawal icon at CPP portal and can withdraw its E-bid.
- 10.6** After the bid submission on the portal, an acknowledgement number will be generated by the system which should be printed by the bidder and kept as a record of evidence for online submission of bid for the particular tender and will also act as an entry pass to participate in the bid opening.
- 10.7** The time settings fixed in the server side & displayed at the top of the tender site, will be valid for bid submission, in the e-tender system. The bidders should follow this time during bid submission.
- 10.8** All the data being entered by the bidders would be encrypted using PKI encryption techniques to ensure the secrecy of the data. The data entered will not be viewable by unauthorized persons during bid submission & will not be viewable by any one until the date & time specified for bid opening.
- 10.9** The bidder should logout of the tendering system using the normal logout option available in the portal and not by selecting the (X) exit option in the browser.
- 11.0 AUTHORITY TO SIGN**
- a) If the applicant is an individual, he should sign above his full type written name and current address.
 - b) If the applicant is a proprietary firm, the Proprietor should sign above his full type written name and the full name of his firm with its current address.
 - c) If the applicant is a firm in partnership, the Documents should be signed by all the partners of the firm above their full type written names and current addresses. Alternatively, the Documents should be signed by the person holding Power of Attorney for the firm in the Format at **Annexure-III**.
 - d) If the applicant is a limited Company, or a Corporation, the Documents shall be signed by a duly authorized person holding Power of Attorney for signing the Documents in the Format at **Annexure-III**.

- e) If the applicant is a Joint Venture, the Documents shall be signed by the Lead Member holding Power of Attorney for signing the Document in the Format at Annexure IV. The signatory on behalf of such Lead Partner shall be the one holding the Power of Attorney in the Format at **Annexure III**.
- f) If the tenderer is individual /sole proprietor/proprietary firm as per a) & b) above and if such tenderer wants to act through an agent; he/she should submit along with the tender, a copy of power of attorney duly stamped and authenticated by a Notary Public or by Magistrate in favour of the specific person in line with Clause 28(3) and as per Annexure III, specifically authorizing him to sign the tender, submit the tender and further to deal with the Tender/ Contract up to the stage of signing the agreement and thereafter till the complete discharge of all the contractual obligations, failing which tender shall be summarily rejected. If the tenderer is an individual, he/she may modify **Annexure III** accordingly.

11.1 Points to be kept in mind while preparing the bid

While filling in Qualification Information documents and the Financial Bid, following should be kept in mind:

- i. There shall be no additions or alterations except those to comply with the instructions issued by the Employer or as necessary to correct errors, if any, made by the Tenderers.
- ii. Conditional Offer/ Tender will be rejected. Unconditional rebate/discounts in the Financial offer will however be accepted.
- iii. The Employer reserves the right to accept or reject any conditional rebate/discounts. While evaluating the Bid Price, the conditional rebates/discounts which are in excess of the requirements of the bidding documents or otherwise result in accrual of unsolicited benefits to the Employer, shall not be taken into account.
- iv. The bidder has to quote value only in figures in the BOQ.
- v. In case of Item Rate Tenders, the bidders have to compulsorily quote rates of all the BOQ items as also all items of Item Rate Schedule/Sheet in a Mixed (Item Rate Schedule and Percentage Schedule) Tender.
- vi. In case of Item Rate Tenders, if the same item figures in more than one section/part of Schedule of Quantities, the Tenderer should quote the same rate

for that item in all sections/parts. If different rates are quoted for the same item, the least of the different rates quoted only shall be considered for evaluation of that item in all sections/parts of the Schedule of Quantities.

- vii. In case of item wise BOQ, the bidder is required to quote his rate for all items. For the items not quoted by the bidder, it will be presumed that the bidder has included the cost of that/those item(s) in the rates of other items and the rate for such item(s) shall be considered as Zero and the tender will be evaluated by the Employer accordingly and the work executed by the successful bidder accordingly.
- viii. In case of Percentage Rate BOQ, the bidder has to select Excess (+) or Less (-) and enter the valid percentage for that BOQ.
- ix. Deduction/recovery/credit items, if any, are placed in a separate sub-head and in a separate sheet of BOQ. In case of credit items/recovery items/deduction items for which the bidder has to pay the amount to Employer, the bidder is not allowed to make negative entry and the rate quoted by the bidder shall be taken as negatively default. The amount so calculated shall be considered as negative and deducted from the total of other sub-heads of BOQ to work out the total bid amount.

11.2 INTEGRITY PACT

- (i) The Bidder/ Contractor is required to enter into an Integrity Pact (Enclosed as Annexures-VI/1, VI/2 and VI/3) with the Employer. The Integrity Pact (Annexure-VI/1 and VI/3) duly signed by the Authorized representative of RITES Ltd. for and on behalf of Principal/Employer as its Agent/Power of Attorney Holder is enclosed herewith. Bidder shall take out the printout of the Integrity Pact (Annexures-VI/1, VI/2 and VI/3), sign on each page by authorized signatory of Bidder/ Each Member of JV/Each Partner of JV and duly witnessed, scan and submit/upload the same along with the Bid. **In case of failure to submit the Integrity Pact, as prescribed herein, duly signed and witnessed, along with the Bid, the Bid is likely to be rejected.**

(Note: - The integrity pact (Annexure-VI/1 and VI/3) signed by the authority competent to sign the contract Agreement on behalf of RITES will be uploaded as part of the tender document)

- (ii) The provision of this Clause 11.2 – Integrity Pact (IP), shall be applicable only when so provided in Clause 11.2A below which will also stipulate the names, addresses and e-mail ids of the Independent External Monitors as well as the

Name, designation, address and e-mail id of the official nominated by the Employer to act as the Liaison Officer between the Independent External Monitors and the Engineer-in-charge as well as the Contractor / Bidder.

11.2A Whether Clause 11.2 (Integrity Pact) shall be applicable ***YES/ *NO**

*(*Strike out whichever is not applicable)*

In case Integrity Pact is applicable, (When estimated cost put to tender is Rs.10 crores or more), the following Independent External Monitors who have been appointed by the Central Vigilance Commission, shall monitor implementation of IP -

Name, Address and e-mail ID of IEM 1

**Sh. Amol Prabhakar Joshi, CES, (Retd.),
126A, Buty Layout C-604, Swapnil Trisha Apartments,
Laxmi Nagar, Nagpur-440022.
Email- amjosh_98@yahoo.com**

Name, Address and e-mail ID of IEM 2

**Sh. Jitendera Sharma, IFoS (Retd.)
House No. 85, Sector 5-A, Ecocity 1,
P.O.- Mullanpur Garibdas, New Chandigarh,
Distt. SAS Nagar, Punjab-140901.
Email- Sharmaj7@gmail.com**

Name, Address and e-mail ID of IEM 3

**Mrs. Gauri Kumar, IAS (Retd.),
Kalypso Court, Tower-1/502, JP Wishtown,
Sector – 128, Noida-201304
Email ID: gkck1955@yahoo.co.in**

Note for Tender Preparing Authority *The names and addresses of all present IEMs are to be mentioned.

Complaints arising from tendering process shall be, as far as possible, referred to all the IEMs for their joint investigation and recommendation. In case the full panel is not available due to some unavoidable reasons, the available IEM(s) will conduct examination of the complaints. Consent of the IEM(s), who may not be available, shall be taken on record.

Name, Designation, Address and e-mail ID of RITES' Liaison Officer

Sh. S. B. Samanta,
Vertical Head (Finance),
Shikhar, Plot No.1. Sector-29,
Gurgaon-122001 (Haryana)
Email: ed.fin@rites.com

- 11.2B** The Guidelines on Banning of Business Dealings as per **Annexure-VIA** are applicable to all Tenders/contracts whether Integrity Pact is applicable as per Clause 11.2A above or not.

If business dealings with the Bidder/Contractor have been banned as per “The Guidelines on Banning of Business Dealings” as per Annexure-VIA, then such a Bidder/contractor individually and also any Joint Venture wherein such Bidder/contractor is a member, will not be eligible during the period till such ban is in force to participate in tenders of any work(s) which may be invited by RITES. In case the Bidder/ contractor is a Joint Venture, the JV as well as all members of the JV individually or as member(s) of any other Joint Venture (JV) will not be so eligible. In case the Bidder/contractor is a company then, in addition to the aforesaid provisions, the Associate Companies and Subsidiary Companies (as defined under Companies Act, 2013), of the company with whom Business Dealings have been banned, will also not be eligible”.

12.0 TENDER OPENING, EVALUATION AND CLARIFICATIONS

- 12.1** The Employer will open all the Tenders received, in the presence of the Tenderers or their representatives who choose to attend at 11:30 Hrs. on 19.09.2025 in the office of **Project Unit Head, 4th floor, NEDFI House Ganeshguri ,Dispur, Guwahati-781006, Assam**. In the event of the specified date of the opening being declared a holiday by the Employer, the Tenders will be opened at the appointed time and location on the next working day.
- 12.2** Opening of bids will be done through online process. RITES reserves the right to postpone or can cancel a scheduled bid opening at any time prior to its opening. Information of the same will be displayed at <https://etenders.gov.in/eprocure/appCPP> portal.
- 12.3** Bid opening committee will open the bids online in the presence of bidders or their authorized representatives who choose to attend on opening date and time. Also, the bidders can participate online during the bid opening process from their remote end through their dashboard. The bidder’s representatives, who are present, shall sign in an attendance register. At the time of technical bid opening, each bidder will be able

to view on-line through CPPP, the technical bids of the bidders who have participated in the tender and whose bids have been opened.

- 12.4** Bids will be opened as per date/time as mentioned in the Tender Critical Data Sheet unless the same is extended. On completion of Technical Bid Opening, each bidder will be able to view the technical bid documents of the bidders whose bids have been opened. Similarly, on the completion of Financial Bid Opening each bidder will be able to view the Financial as well as technical bid documents of the bidders whose bids have been opened.

(a) Mere Opening of Bid to be No Guarantee of its Validity

It may be noted that mere opening of a bid does not mean that the bid has to be considered by RITES as a valid bid. All bids will be evaluated to decide whether the bids are responsive or non- responsive.

- 12.5** RITES shall subsequently examine and evaluate the bids in accordance with the provisions set out in the tender document.
- 12.6** The results of technical and financial qualification of bidders will be available on the CPP Portal at <https://etenders.gov.in/eprocure/app> and intimated to the bidder through system generated email or SMS.
- 12.7** It will be the bidder's responsibility to check the status of their Bid on-line regularly after the opening of bid till award of work.
- 12.8** The bids will be checked for the eligibility as defined in clause 1.7 hereinbefore and for submission of documents as per clause 28.0 hereinafter and will be evaluated for qualification criteria as mentioned in Clause -2 hereinbefore and also in conjunction with provisions of Sub-Clauses 1.8, 12.15 and 12.16 and/or provisions given elsewhere in this Tender Document. RITES shall not be responsible for any postal delay in receipt of all original documents including the cost of tender document and EMD, as applicable. In case of non-receipt of these documents in original within the specified period, the bid will be treated as non-responsive.
- 12.9** Request for clarification/deficient documents from the bidder can be asked for either through the system or through E-mail. A system generated SMS alert will be sent to the bidder when clarifications/deficient documents are called through the system. In such a case, no separate communication will be sent in this regard. Non-receipt of email and/or SMS will not be accepted as a reason of non-submission of deficient documents or confirmatory documents within prescribed time. The date and time of submission of deficient documents cannot be extended.

12.10 After evaluation of Technical-Bid, the bidder will be able to view uploaded Tender Committee evaluation results as also the date and time of Financial Bid Opening.

12.11 The bidder will be able to view (through his Login Id) BOQ Sheets of other bidders, Comparative Chart and Financial Evaluation Summary uploaded by Tender Evaluation Committee. Without login, bidder will be able to view only Comparative Chart.

12.12 SINGLE PACKET SYSTEM- Not Applicable

~~Envelope 1 containing scanned copy of Earnest Money along with Mandate Form as per Annexure VII, Cost of tender document of all the Tenderers and Authority to Sign as per Clause 11.0 will be opened first and checked. If Earnest Money and Cost of Tender Document are not furnished as per tender stipulations, the Envelope 2 of Technical bid and Envelope 3 containing Financial bid will not be opened and the bid will be rejected as non-responsive unless the bidder has established that it is exempted from payment of Cost of Tender Document and Earnest Money Deposit. The Envelope 2 containing Technical Bid and Envelope 3 containing Financial Bid of other Tenderers who have furnished scanned copies of Earnest Money and cost of Tender document as per tender stipulations will then be opened simultaneously.~~

12.13 TWO PACKET SYSTEM - Applicable

- (a) Envelope 1 of Packet I containing scanned copy of Earnest Money along with Mandate Form as per Annexure VII, Cost of Tender Document of all the Tenderers and Authority to Sign as per Clause 11.0 and Integrity Pact document as per clause 11.2 will be opened first and checked. If Earnest Money and cost of Tender Document are not furnished as per tender stipulations, the Envelope 2 of PACKET-I (Technical Bid) and PACKET-II (Financial Bid) will not be opened and the bid will be considered as non- responsive and rejected, unless the bidder has established that it is exempted from payment of Cost of Tender Document and Earnest Money Deposit. The Envelope 2 of PACKET-I (Technical Bid) of other Tenderers who have furnished scanned copies of Earnest Money and cost of Tender document as per tender stipulations will then be opened.
- (b) PACKET-II (Financial Bid) of Tenderers whose Technical Bids are not found acceptable will not be opened. Such tenderers will be informed about non-acceptance of their Technical Bid through system generated SMS/E- mail. The tenderers whose Technical Bids are found acceptable will be advised accordingly and will also be intimated through e-mail the time and date and

place where and when PACKET-II (Financial Bid) will be opened.

- (c) At the appointed place, time and date, in the presence of the Tenderers or their representatives who choose to be present, the Employer will open the online PACKET-II (Financial Bid).

12.14 Evaluation of Bids

- (a) After opening of the Bids, as mentioned hereinbefore, The Employer will evaluate the Tenders, confidentially
 - (i) Information relating to the examination, clarification, evaluation and comparison of bids, and recommendations for the award of a contract, shall not be disclosed to bidders or any other person not officially concerned with such a process until the award to the successful bidder has been announced (unless mentioned otherwise in other clauses of this Tender document).
 - (ii) From the time of bid opening to the time of contract award, no bidder shall contact the Employer/RITES Ltd. on any matter related to the bid, except on request and prior written permission.
 - (iii) Any effort by the bidder to influence Employer/RITES Ltd. in bid evaluation, bid comparison or contract award decisions will vitiate the process and will result in the rejection of the bidder's bid.
- (b) The Employer will scrutinize the Technical Bids accepted for evaluation to determine whether each Tenderer
 - (i) Has submitted '**Authority to sign**' as per Clause 11.0 above and **Integrity Pact** (where applicable) duly signed and witnessed as per Clause 11.2 above.

A Bidder who has not submitted 'Authority to sign' and/or Integrity Pact (where applicable) duly signed by the bidder shall not be evaluated further and will be disqualified at this stage.

- (ii) Meets the Eligibility Criteria as defined in Clause 1.7 and has submitted documents as per clause 28.0 and meets the Qualification Criteria stipulated in Clause 2.0 and also in conjunction with Provisions of sub-clauses 1.8, 12.15, 12.16 and/or Provisions given elsewhere in Tender document.

- (c) If required, the Employer may ask any such Tenderer for clarifications/deficient documents on his Technical Bid through CPPP or through E-mail. The tenderer shall furnish the same online only in case clarifications are sought through CPPP and through E-mail if clarifications are sought through E-mail. If a Tenderer does not submit the clarification/document requested, by the specified time, the bid of such Tenderer is likely to be rejected.
- (d) The Employer, reserves its right to, but without any obligation to do so, to waive or seek any shortfall information/ documents, as mentioned above, only in case of historical documents which pre-existed at the time of the Bid Opening, and which have not undergone change since then and does not grant any undue advantage to any bidder.
- (e) Unless otherwise stipulated in these Tender Documents, the following are some of the crucial aspects for which a bid shall be liable to be rejected as nonresponsive:
 - (i) The tender is not in the prescribed format or is not submitted as per the stipulations in the Tender Document.
 - (ii) Required costs of tender document and EMD have not been submitted in the requisite form.
 - (iii) Bidder is not eligible to participate in the bid as per laid down eligibility criteria.
 - (iv) The bid validity is shorter than the required period.
 - (v) The bid departs from the essential requirements stipulated in the bidding document;
 - (vi) Non-submission or submission of illegible scanned copies of stipulated documents/ declarations
 - (vii) Any other conditions stipulated elsewhere in the tender documents
- (f) **The financial bid shall be opened as follows:**
 - (i) **Applicable to Single Packet System: – NOT APPLICABLE**
 - (ii) **Applicable to two packet system:** PACKET-II (Financial Bid) of Tenderers whose Technical Bids are not found acceptable will not be opened. Such tenderers will be informed about non-acceptance of their

Technical Bid through system generated SMS/E- mail. The tenderers whose Technical Bids are found acceptable will be advised accordingly and will also be intimated through e-mail the time and date and place where and when PACKET-II (Financial Bid) will be opened.

(g) **Financial Evaluation (Applicable to both single packet system and two packet system):** After opening of Financial Bids as mentioned in sub clause 12.14(f) (i) & (ii) as applicable, all responsive bids (Qualified in technical bid) are evaluated by the TC with a view to select the lowest (L1) bidder - the lowest evaluated, bid which meets the eligibility/ qualification criteria and techno-commercial aspects. While evaluating the financial bids the following procedure shall be followed:

- (i) In case the bidder does not quote his rate for any item(s) in Item Rate Tender or Mixed Tender containing one or more Item Wise Schedules, it will be presumed that the bidder has included the cost of that/those item(s) in the rates of other items and the rate for such item(s) shall be considered as Zero and the tender will be evaluated by the Employer accordingly and the work executed by the successful bidder accordingly.
 - (ii) In case of credit items/recovery items/deduction items for which the bidder has to pay the amount to RITES/Employer, the rate quoted by the bidder shall be taken as negative (bidder is allowed to make positive entry only) and the negative amount so calculated shall be considered to work out the total bid amount.
- (h) As part of evaluation of financial bid, written clarification from Lowest (L1) bidder including detailed price analysis of its Bid price in relation to scope, schedule, resource mobilization, allocation of risks and responsibilities and any other requirements of the bid document may be sought by Employer.

12.15 Preference to make in India

To encourage 'Make in India' and promote manufacturing and production of goods and services in India with a view to enhancing income and employment, Department of Promotion of Industry and Internal Trade (DPIIT), Ministry of Commerce and Industry, Government of India, issued Public Procurement (Preference to Make in India), Order 2017 (Latest order being Order No. No. P-45021/2/2017-PP (BE-II)-Part(4) Vol. II dated 19th July, 2024). The Order is applicable on the procurement of Goods, Works and Services. The Provisions of the Make in India Policy 2017 issued by Govt. of India, as amended from time to time shall be followed for consideration of the present tender.

The salient features of the policy and specific provisions as applicable to this tender are indicated below. In case of any contradiction of the provisions mentioned herein below and the latest policy as mentioned above, the provisions of the policy issued by Department for Promotion of Industry and Internal Trade (Public Procurement Section) shall take precedence:

1. Definitions:

'Local content' means the amount of value added in India which shall be the total value of the item procured (excluding net domestic indirect taxes) minus the value of imported content in the item (including all customs duties) as a proportion of the total value, in percent.

Explanatory notes for calculation of local content given above:

- a. Imported items sourced locally from resellers/distributors shall be excluded from calculation of local content.
- b. The license fees/royalties paid/ technical charges paid out of India shall be excluded from local content calculation.
- c. Procurement/supply of repackaged/refurbished/rebranded imported products as understood commonly shall be treated as reselling of imported products and shall be excluded from calculation of local content. The definition of repackaged/refurbished/rebranded imported products is as follows;

‘Refurbishing’ means repair of reconditioning of an imported product does not amount to manufacture because no new goods come into existence.

‘Repackaging’ means repacking of imported goods from bulk pack to smaller packs would not ordinarily amount to manufacture of a new item.

‘Rebranding’ means relabeling or renaming of change in symbol or logo/makes or corporate image of a company/organization/firm for an imported product would amount to rebranding.
- d. To ensure that imported items sourced locally from resellers/distributors are excluded from calculation of local content, RITES to obtain from bidders, the cost of such locally-sourced imported items (Inclusive of taxes) along with breakup on license/royalties paid/technical expertise cost etc. sourced from outside India. For items sold by bidder as reseller, OEM certificate for country of origin to be submitted.
- e. For contacts involving supply of multiple items, weighted average of all items to be taken while calculating the local content.

'Class-I local supplier' means a supplier or service provider, whose goods, services or works offered for procurement, has local content equal to or more than 50%, as defined under this clause.

'Class-II local supplier' means a supplier or service provider, whose goods, services or works offered for procurement, has local content more than 20% but less than 50%, as defined under this clause.

'Non - Local supplier' means a supplier or service provider, whose goods, services or works offered for procurement, has local content less than or equal to 20%, as defined under this clause.

'L1' means the lowest tender or lowest bid or the lowest quotation received in a tender, bidding process or other procurement solicitation as adjudged in the evaluation process as per the tender or other procurement solicitation.

'Nodal Ministry' means the Ministry or Department identified pursuant to public procurement (Preference to Make in India), Order 2017 (Latest order being Order No. P-45021/2/2017-PP (BE-II)-Part(4) Vol.II dated 19th July, 2024) in respect of a particular item of goods of services or works.

'Margin of Purchase Preference' means the maximum extent to which the price quoted by a "Class-I local supplier" may be above the L1 for the purpose of purchase preference.

'Procuring entity' means RITES Ltd

'Works ' means all works covered in the scope of work in this tender.

1.A Special treatment for items covered under PLI scheme

The manufacturers manufacturing an item under PLI scheme shall be treated as deemed Class II local supplier for that item unless they have minimum local content equal to or higher than that notified for Class-I local supplier for that item, provided the manufacturer has received incentive from the concerned PLI Ministry for that item. The above shall be applicable for the specific time period only, as notified by concerned PLI Ministry.

2. Eligible bidder in this tender: 'Class-I local supplier'

Notes for Guidance of Tender Preparing Authority and Tender document approving Authority:

Note 1- As per Para 3 (a) of the 'Make in India', Policy issued vide No. P-45021/2/2017-PP (BE-II)-Part (4) Vol.II dated 19th July, 2024 only 'Class-I Local Supplier' are eligible to Bid for "Works" tenders where sufficient local capacity and competence is available, irrespective of the Purchase value. Accordingly, only Class-I Local suppliers have been allowed to bid in this Tender.

Note 2- In case of special works which can allow participation of Class-II Local Supplier/ Non- Local Supplier besides Class-I Local suppliers, cases shall be dealt as per the provisions of 'Make in India Policy'. Accordingly, the Provisions of this clause 12.15 shall be prepared by Tender Preparing Authority and shall be checked by Tender document approving authority before according approval. For this purpose, Para 3A along with its Sub-paras of Public Procurement (Preference to Make in India), Order No. P-45021/2/2017-PP (BE-II)-Part (4) Vol.II dated 19th July, 2024 may be read carefully.

Note 3- ***Mandatory sourcing of items, with sufficient local capacity and competition, from Class-I local suppliers in SI/EPC/Turnkey Contracts/Service Tenders***

- a. *The items notified as having sufficient local capacity and competition, shall mandatory be sourced from Class-I local suppliers in SI/EPC/Turnkey Contracts/ Services tenders. This provision will be applicable only for those items which have been notified by the Nodal Ministry as Class I i.e. having sufficient local capacity and competition, with specific HSN codes.*
- b. *Notwithstanding above, if in any project, it is considered that it is not practically feasible to source such items from Class I local suppliers, it may take relaxation from such stipulation with the approval of Secretary of the administrative Ministry/ Department concerned or with the approval of the Competent Authority specified by the Administrative Ministry/Department, on case-specific basis.*

For RITES, Administrative ministry is 'Ministry of Railways'. Railway Board has notified the specific items for which sufficient local capacity and local competition exist through various letters/orders, which are available on Railway Board website (Railway Board Directorates → Stores → Circulars → Subject wise Circular → Make_In_India). Tender Preparing Authority and Tender document approving Authority are advised to follow the same with up-to-date amendments as notified by Railway Board from time to time. Railway Board has issued the latest letter No. 2020/RS(G)/779/2/Pt.1 (E3322671) dated 20.08.2024, which is available on its

website.

Note 4- Minimum local content: The 'local content' requirement to categorize a supplier as 'Class-I local supplier' is minimum 50%. For 'Class-II local supplier', the 'local content' requirement is minimum 20%. Nodal Ministry/ Department may prescribe only a higher percentage of minimum local content requirement to categorize a supplier as 'Class-I local supplier'/'Class- II local supplier'. For the items, for which Nodal Ministry/ Department has not prescribed higher minimum local content notification under the Order, it shall be 50% and 20% for 'Class-I local supplier'/'Class-II local supplier' respectively. Railway Board has notified minimum local content more than 50% for specific items through various letters/orders, which are available on Railway Board website (Railway Board Directorates → Stores → Circulars → Subject wise Circular → Make_In_India). Tender Preparing Authority and Tender document approving Authority are advised to follow the same with up-to-date amendments as notified by Railway Board from time to time. Railway Board has issued the latest letter No. 2020/RS(G)/779/2/Pt.1(E3322671) dated 20.08.2024, which is available on its website.

(a) Purchase Preference

In the present tender only 'Class-I local supplier', shall be eligible to bid irrespective of purchase value.

3. Type of this Tender/Work: Works

4. The minimum local content for this tender shall be equal to or more than 50%.

5. Verification of local content:

- (a) The 'Class-I local supplier' at the time of tender, bidding or solicitation shall be required to indicate percentage of local content and provide self-certification that the item offered meets the local content requirement for 'Class-I local supplier'. They shall also give details of the location(s) at which the local value addition is made.
- (b) In cases of procurement for a value in excess of Rs. 10 crores, the 'Class-I local supplier' shall be required to provide a certificate from the statutory auditor or cost auditor of the company (in the case of companies) or from a practicing cost accountant or practicing chartered accountant (in respect of

suppliers other than companies) giving the percentage of local content.

Note- Bidder's attention is invited to Para XV in Proforma I

- (c) The bidder shall give self-certification for local content in the quoted item (goods/works/services) at the time of tendering. However, at the time of execution of the project, for all contracts above INR 10 Crore, the contractor/ supplier shall be required to give local content certification duly certified by cost/ chartered accountant in practice. For cases where it is not possible to provide certification by Cost/Chartered Accountant at the time of execution of project, the supplier shall be permitted to provide the certificate for local content from Cost/ Chartered Accountant after completion of the contract, within time limit acceptable to the RITES. In case the contractor/ supplier does not meet the stipulated local content requirement and the category of the supplier changes from Class-I to Class-II/ Non-local or from Class-II to Non- local, a penalty upto 10% of the contract value may be imposed. However, contract once awarded shall not be terminated on this account.
- (d) Complaints about local content declarations/verification of local content shall be processed/addressed as per Public Procurement (Preference to Make in India), Order No. P-45021/2/2017-PP (BE-II)-Part(4) Vol.II dated 19th July, 2024.
- (e) In case of false declaration by Class-I local supplier or submission of false certificate, Banning of Business Dealings shall be done with defaulter as per the Guidelines given in Sub-clause 11.2B.

6. Exemption in sourcing of spares and consumables of closed systems:

Procurement of spare parts, consumables for closed systems and Maintenance/Service contracts with Original Equipment Manufacturer/Original Equipment Supplier/Original Part Manufacturer shall be exempted from Public Procurement (Preference to Make in India), Order 2017 (Latest order being Order No. P- 45021/2/2017-PP (BE-II)-Part(4) Vol.II dated 19th July, 2024).

8. Reciprocity Clause

Reciprocity Clause as mentioned in Public Procurement (Preference to Make in India), Order 2017 (Latest order being Order No. P-45021/2/2017-PP (BE-II)- Part (4) Vol.II dated 19th July, 2024) shall be applicable in this Tender.

Entities of countries which have been identified by the nodal Ministry/Department as not allowing Indian companies to participate in their Government procurement for any item related to that nodal Ministry shall not be, allowed to participate in Government procurement in India for all items related to that nodal Ministry/ Department, except for the list of items published by the Ministry/ Department permitting their participation.

Note to Tender Preparing Authority and Tender document approving Authority:

Tender Preparing Authority and Tender document approving Authority may kindly read provisions of Reciprocity Clause as given in Public Procurement (Preference to Make in India), Order 2017 (Latest order being Order No. P-45021/2/2017-PP (BE- II)-Part(4) Vol.II dated 19th July, 2024). Para 10 (d) (i) of Public Procurement (Preference to Make in India), Order No. P-45021/2/2017-PP (BE-II)-Part(4) Vol.II dated 19th July, 2024 is reproduced below for the attention of Tender Preparing Authority and Tender document approving Authority.

“When a Nodal Ministry/Department identifies that Indian suppliers of an item are not allowed to participate and/ or compete in procurement by any foreign government, due to restrictive tender conditions which have direct or indirect effect of barring Indian companies such as registration in the procuring country, execution of projects of specific value in the procuring country etc., it shall provide such details to all its procuring entities including CMDs/CEOs of PSES/PSUs, State Governments and other procurement agencies under their administrative control and GeM for appropriate reciprocal action.”

9. Manufacture under license/ technology collaboration agreements with phased indigenization: **Applicable/*~~Not Applicable~~ (*Strike out whichever is not applicable)*

Note to Tender Preparing Authority and Tender document approving Authority:

Tender Preparing Authority and Tender document approving Authority may kindly see if this provision is applicable to the subject tender and if this

provision is applicable, the same may be provided in the subject tender. For information of Tender Preparing Authority and Tender document approving Authority, the Para no. 13 and 13A of Public Procurement (Preference to Make in India), Order 2017 (Order No. P-45021/2/2017-PP (BE-II)-Part(4) Vol. II dated 19th July, 2024) are reproduced below:

"Para No. 13: While notifying the minimum local content, Nodal Ministries may make special provisions for exempting suppliers from meeting the stipulated local content if the product is being manufactured in India under a license from a foreign manufacturer who holds intellectual property rights and where there is a technology collaboration agreement/ transfer of technology agreement for indigenous manufacture of a product developed abroad with clear phasing of increase in local content.

Para No. 13A: In procurement of all goods, services or works in respect of which there is substantial quantity of public procurement and for which the nodal ministry has not notified that there is sufficient local capacity and local competition, the concerned nodal ministry shall notify an upper threshold value of procurement beyond which foreign companies shall enter into a joint venture with an Indian company to participate in the tender. Procuring entities, while procuring such items beyond the notified threshold value, shall prescribe in their respective tenders that foreign companies may enter into a joint venture with an Indian company to participate in the tender. The procuring Ministries/Departments shall also make special provisions for exempting such joint ventures from meeting the stipulated minimum local content requirement, which shall be increased in a phased manner. "

12.16 Restrictions on procurement from a bidder of a country which shares a land border with India

Orders (Order No. F-7/10/2021-PPD (1) Dated 23.02.2023 issued by Department of Expenditure, Ministry of Finance, Govt. of India (Order No. F-7/10/2021-PPD(1) Dated 23.02.2023) or any other latest order on the subject matter; also referred to as “the Order” in this clause) restricting procurement of Works from Bidders from certain countries that share a land border with India shall apply to this procurement.

Salient provisions of the policy are given below. However it will be the responsibility of the bidder to go through the complete policy and see if it is eligible in terms of the policy and meets the criteria required as per the policy. In case of any contradiction of the provisions mentioned herein below and the original policy as mentioned above, the provisions of the policy issued by

Department of Expenditure, Ministry of Finance, Government of India shall take precedence.

Any bidder from the countries sharing a land border with India will be eligible to bid in any procurement of works (including turnkey projects) only if the bidder is registered with the Competent Authority and if otherwise eligible to Bid as per eligibility conditions.

Any bidder (including an Indian bidder) who has a Specified Transfer of Technology (ToT) arrangement with an entity from a country which shares a land border with India will be eligible to bid only if the bidder is registered with the Competent Authority and if otherwise eligible-to Bid as per eligibility conditions.

The Competent Authority for registration will be the Registration Committee constituted by the Department for Promotion of Industry and Internal Trade (DPIIT), Government of India.

The provision of the Order on restrictions based on land borders will not be applicable for cases mentioned in Para 12 here in under:

Definitions

1. **"Bidder"** for the purpose of the Order (including the term 'tenderer', 'consultant' 'vendor' or 'service provider' in certain contexts) means any person or firm or company, including any member of a consortium or joint venture (that is an association of several persons, or firms or companies), every artificial juridical person not falling in any of the descriptions of bidders stated herein before, including any agency, branch or office controlled by such person, participating in a procurement process.
2. **"Tender"** for the purpose of the Order will include other forms of procurement, except where the context requires otherwise.
3. **"Transfer of Technology"** means dissemination and transfer of all forms of commercially usable knowledge such as transfer of know-how, skills, technical expertise, designs, processes and procedures, trade secrets, which enables the acquirer of such technology to perform activities using the transferred technology independently. (Matters of interpretation of this term shall be referred to the Registration Committee constituted by the DPIIT and the interpretation of the committee shall be final).
4. **"Specified Transfer of Technology"** means a transfer of technology in the sectors and/ or technologies, specified in the order.

5. **"Procuring entity"** means RITES Ltd.
6. **"Bidder (or entity) from a country which shares a land border with India"** for the purpose of the Order means
- a. An entity incorporated, established or registered in such a country; or
 - b. A subsidiary of an entity incorporated, established or registered in such a country; or
 - c. An entity substantially controlled through entities incorporated, established or registered in such a country; or
 - d. An entity whose beneficial owner is situated in such a country; or
 - e. An Indian (or other) agent of such an entity; or
 - f. A natural person who is a citizen of such a country; or
 - g. A consortium or joint venture where any member of the consortium or joint venture falls under any of the above.
7. **"Beneficial owner"** for the purpose of paragraph 6 (d) above will be as under:
- I. In case of a company or Limited Liability Partnership, the beneficial owner is the natural person(s), who, whether acting alone or together, or through one or more juridical person(s), has a controlling ownership interest or who exercises control through other means.

Explanation---

- a) "Controlling ownership interest" means ownership of, or entitlement to, more than twenty-five percent of shares or capital or profits of the company;
 - b) "Control" shall include the right to appoint the majority of the directors or to control the management or policy decisions, including by virtue of their shareholding or management rights or shareholders agreements or voting agreements;
- II. In case of a partnership firm, the beneficial owner is the natural person(s) who, whether acting alone or together, or through one or more juridical person, has ownership of entitlement to more than fifteen percent of capital or profits of the partnership;
- III. In case of an unincorporated association or body of individuals, the beneficial owner is the natural person(s), who, whether acting alone or

together, or through one or more juridical person, has ownership of or entitlement to more than fifteen percent of the property or capital or profits of such association or body of individuals;

- IV. Where no natural person is identified under (i) or (ii) or (iii) above, the beneficial owner is the relevant natural person who holds the position of senior managing official;
 - V. In case of a trust, the identification of beneficial owner(s) shall include identification of the author of the trust, the trustee, the beneficiaries with fifteen percent or more interest in the trust and any other natural person exercising ultimate effective control over the trust through a chain of control or ownership.
 - VI. To determine nationality while assessing the beneficial ownership of the bidder, the nationality mentioned in the beneficiary owner's passport should be considered. In case of the possibility of dual citizenship, nationality on all the passports should be considered through a suitable declaration. If nationality in any of the passports of the person whose beneficial ownership is being assessed is recorded to be from a country sharing a land border with India, the provision contained under this Order shall apply. Hong Kong and Macau are to be considered as part of China for the purpose of this Order.
8. **"Agent"** for the purpose of the Order is a person employed to do any act for another, or to represent another in dealings with third persons.

Note:

- 1. A person who procures and supplies finished goods from an entity from a country which shares a land border with India will, regardless of the nature of his legal or commercial relationship with the producer of the goods, be deemed to be an Agent for the purpose of this Order.
 - 2. However, a bidder who only procures raw material, components etc. from an entity from a country which shares a land border with India and then manufactures or converts them into other goods will not be treated as an Agent
9. **Sensitive Sectors/ Technologies (relevant only for the provisions on ToT arrangements):**

The Order may be referred for sectors/Technologies which have been identified as sensitive from the National Security Point of View. The bidder with ToT

arrangement in any sensitive Sector/Technology with a country which shares a land border with India shall require registration as indicated in Para 11 below. The Order may be referred for details of sensitive sectors/Technologies.

Requirement of registration

- 10.** Any bidder from a country which shares a land border with India will be eligible to bid in any procurement whether of goods, services (including consultancy services and non-consultancy services) or works (including turnkey projects) only if the bidder is registered with the Competent Authority i.e. Department for Promotion of Industry and Internal Trade (DPIIT).
- 11.** Further, any bidder (including bidder from India) having Specified Transfer of Technology (ToT) arrangement with an entity from a country which shares a land border with India will be eligible to bid in any procurement whether of goods, services (including consultancy services and non-consultancy services) or works (including turnkey projects) only if the bidder is registered with the Competent Authority i.e. Department for Promotion of Industry and Internal Trade (DPIIT). The requirement of registration for bidders covered in this paragraph shall apply to all tenders are issued/published after 01.04.2023.
- 12.** The provision of the Order on restrictions based on land borders will not be applicable for cases mentioned herein under:
 - a. In projects which receive international funding with the approval of the Department of Economic Affairs (DEA), Ministry of Finance, the procurement guidelines applicable to the project shall normally be followed, notwithstanding anything contained in this order and without reference to the competent Authority. Exceptions to this shall be decided in consultation with DEA.
 - b. This order shall not apply to procurement by Indian mission and by offices of government agencies/undertaking located outside India.
 - c. This order will not apply to bidders (or entities) from those countries (even if sharing a land border with India) to which the Government of India has extended lines of credit or in which the Government of India is engaged in development projects. Updated lists of countries to which lines of credit have been extended or in which development projects are undertaken are given in the website of the Ministry of External Affairs.

- d. Procurement of spare parts and other essential service support like Annual Maintenance Contract (AMC) / Comprehensive Maintenance Contract (CMC), including consumables for closed systems, from Original Equipment Manufacturers (OEMs) or their authorized agents, shall be exempted from the requirement of registration.

13. Sub-Contracting in Works Contract

In works contracts, including turnkey contracts, contractors shall not be allowed to sub-contract works to any **contractor** from a country which shares a land border with India unless such **contractor** is registered with the Competent Authority i.e. Department for Promotion of Industry and Internal Trade (DPIIT). The definition of “contractor from a country which shares a land border with India” shall be as in paragraph 6 above. This shall not apply to sub-contracts already awarded on or before 23.07.2020.

[Note: Procurement of raw material, components, etc. does not constitute sub-contracting.]

14. Validity of registration

In respect of tenders, registration should be valid at the time of submission of bids and at the time of acceptance of bids. In respect of supply otherwise than by tender, registration should be valid at the time of placement of order. If the bidder was validly registered at the time of acceptance/ placement of order, registration shall not be a relevant consideration during contract execution.

15. Model Certificates

Model Certificates, which may be inserted in tenders/obtained from Bidders are given as under:

a. Model Certificate for Tenders:

“I have read the clause regarding restrictions on procurement from a bidder of a country which shares a land border with India, I certify that this bidder is not from such a country or, if from such a country, has been registered with the competent Authority i.e. DPIIT. I hereby certify that this bidder fulfills all requirements in this regard and is eligible to be considered. (Where applicable, evidence of valid registration by the Competent Authority i.e. DPIIT shall be attached)”.

b. Model Certificate for Tenders for Works involving possibility of sub-

contracting:

“I have read the clause regarding restrictions on procurement from a bidder of a country which shares a land border with India and on sub-contracting to contractors from such countries; I Certify that this bidder is not from such a country or, if from such a country, has been registered with the Competent Authority i.e. DPIIT and will not sub-contract any work to a contractor from such countries unless such contractor is registered with the Competent Authority i.e. DPIIT. I hereby certify that this bidder fulfills all requirements in this regard and is eligible to be considered. (Where applicable, evidence of valid registration by the Competent Authority i.e. DPIIT shall be attached.)”

c. Model additional Certificate by Bidders in the cases of Specified ToT:

“I have read the clause regarding restrictions on procurement from a bidder having Transfer of Technology (ToT) arrangement. I certify that this bidder does not have any ToT arrangement requiring registration with the competent authority.”

OR

“I have read the clause regarding restrictions on procurement from a bidder having Transfer of Technology (ToT) arrangement. I certify that this bidder has valid registration to participate in this procurement”. (Evidence of valid registration by the Competent Authority i.e. DPIIT shall be attached)

Bidder’s attention is invited to Para xiv of Proforma-1 in this regard

13.0 INSPECTION OF SITE BY THE TENDERERS

Tenderers are advised to inspect and examine the site and its surroundings and satisfy themselves before submitting their Tenders, as to the nature of the ground and sub-soil (as far as is practicable), the form and nature of the site, the means of access to the site, local conditions, working hours, working conditions, availability of water, electricity, construction Labour, construction materials and other relevant requirements connected with work, the accommodation they may require and in general shall themselves obtain all necessary information as to risks, contingencies and other circumstances which may influence or affect their Tender. A Tenderer shall be deemed to have full knowledge of the site whether he inspects it or not and no extra charges consequent on any misunderstanding or otherwise shall be allowed. The Tenderer shall be responsible for arranging and maintaining at his own cost all materials, tools & plants, water, electricity, access, facilities for

workers and all other services required for executing the work unless otherwise specifically provided for in the contract documents. Submission of a tender by a Tenderer implies that he has read this notice and all other contract documents and has made himself aware of the scope and specifications of the work to be done and of conditions and rates at which stores, tools and plant etc. will be issued to him by the Employer and local conditions and other factors having a bearing on the execution of the work. The bidders may contact **Deputy General Manager (Civil), RITES Ltd., 4th floor, NEDFI House Ganeshguri, Dispur, Guwahati-781006 (Assam) & e-mail id: manjit.borkataki@rites.com and Mob. No (+)91-6901582367 between 9.00 Hrs. to 17:00 Hrs. regarding inspection of site. The bidder shall submit self-attested copy of Declaration about Site Inspection as per Annexure XII.**

14.0 EMPLOYER'S RIGHT ON ACCEPTANCE OF ANY TENDER

- (i) If required, the Employer may ask any Tenderer the breakdown of unit rates. If the Tenderer does not submit the clarification by the date and time set in the Employers request for clarification, such Tender is likely to be rejected.
- (ii) The competent authority on behalf of the Employer does not bind himself to accept the lowest or any other Tender and reserves to himself the authority to reject any or all the Tenders received without the assignment of any reason. All Tenders in which any of the prescribed conditions is not fulfilled or any condition is put forth by the Tenderer shall be summarily rejected.

15.0 CANVASSING PROHIBITED

Canvassing whether directly or indirectly, in connection with tenders is strictly prohibited and the tenders submitted by the Contractors who resort to canvassing will be liable to rejection.

16.0 EMPLOYER'S RIGHT TO ACCEPT WHOLE OR PART OF THE TENDER

The competent authority on behalf of the Employer reserves to himself the right of accepting the whole or any part of the tender and the Tenderer shall be bound to perform the same at the rates quoted.

17.0 MISCELLANEOUS RULES AND DIRECTIONS

- 17.1** The Tenderer shall not be permitted to tender for works if his near relations/relative is posted as Associated Finance Officer between the grades of AGM(F) and JM(F)

in the concerned SBU Unit of RITES or as an officer in any capacity between the grades of GGM/GM and Engineer (both inclusive) of the concerned SBU of the Employer/RITES. He shall also intimate the names of persons who are working with him in any capacity or are subsequently employed by him and who are near relatives to any Officer of Engineer rank and above in the organization of the Employer and RITES. Any breach of this condition by the Tenderer would render his Tender to be rejected.

No Officer of Engineer rank and above employed in Engineering, Finance or Administrative duties in an Engineering and Finance Department of the Organization of the Employer/ RITES is allowed to work as/or for a contractor for a period of Two years after his retirement from the Employer's/RITES service without the previous permission of the Employer/RITES in writing.

Further, The Bidder should not have an association (as a bidder/ partner/ director/ employee in any capacity), of retired Manager (of Gazetted Rank) or a retired Gazetted Officer of the Central or State Government or its Public Sector Undertakings if such a retired person has not completed the cooling-off period of two years after his retirement. However, this shall not apply if such managers/ officers have obtained a waiver of the cooling-off period from their erstwhile organization and bidder shall inform Employer/RITES in writing about such waiver.

The contract is liable to be cancelled if either the Contractor or any of his employees is found any time to be such a person who had not obtained the permission of the Employer/RITES/Erstwhile organization as aforesaid before submission of the tender or engagement in the Contractor's service.

Any breach of this condition by the contractor shall be treated as concealing of information and appropriate action will be taken for banning of business dealings with such contractors as per clause 11.2 B.

- 17.2** If required by the Employer, the Tenderers shall sign a declaration under the official Secret Act 1923, for maintaining secrecy of the tender documents drawings or other records connected with the work given to them. The unsuccessful Tenderers shall return all the drawings given to them.
- 17.3** In the case of any Item rate tender where unit rate of any item/ items appears unrealistic, such tender will be considered as unbalanced and in case the Tenderer is unable to provide satisfactory explanation, such a tender is liable to be disqualified and rejected.
- 17.4 Goods and Services Tax (GST)**

GST Registration Status:

- (a) All the prospective bidders should ensure that they are GST compliant and their quoted tax structure/ rates are as per GST Act/ Rules.
- (b) Bidder should be registered under GST and furnish GST Registration number and copy of GST Registration Certificate in their offer unless they are specifically exempted from GST registration under specific notification/ circular/ section/ rule issued under GST Law and to be clearly stated.
- (c) The prospective bidder has to furnish the GST Registration Number of the State / Verticals, in case of multiple registrations in a state, from which supply / services are to be made.
- (d) If the Bidder has opted for a composition levy under Section 10 of CGST ACT, he should declare the fact while bidding along with GST Registration number and copy of GST registration certificate.
- (e) If a bidder is not liable to take GST registration, i.e., having turnover below threshold at the time of bidding, he shall submit an undertaking along with a valid certificate from practicing Chartered Accountant (CA)/ Cost Accountant with Unique Document Identification Number (DIN) to the effect that Bidder fulfills all conditions prescribed in the notification exempting him from registration. Such bidder/ dealer shall not charge any GST (and/or cess wherever applicable) in the bill/ invoice. Further, Bidder should notify and submit to the employer within 15 days from the date of becoming liable to registration under GST Law.
- (f) State-wise registration of Employer is indicated in Annexure-XI
- (g) While quoting the basic rate, the bidder should offset the input tax credit available/ to be availed as per the GST Act, and shall inclusive of all taxes, duties, cess etc. other than GST.

Payment of GST under the contract:

- (a) The payment of GST and GST Cess to the contractor shall be made only on the latter submitting a GST compliant Tax invoice indicating the appropriate HSN code and applicable GST rate thereon duly supported with documentary evidence as per the provision of relevant GST Act and the Rules made there under. The delivery shall be shown being made in the name, location/ state, and

GSTIN of the place where supply is to be made, as per applicable law; irrespective of location/state of employer where LOA is issued.

- (b) While claiming reimbursement of duties, taxes etc. (like GST) from the Employer, as and if permitted under the contract, the contractor/supplier shall also certify that in case it gets any refund out of such taxes and duties from the concerned authorities at a later date, it (the contractor/supplier) shall refund to the Employer, Employer's share out of such refund received by him from the concerned authorities, immediately.
- (c) All necessary adjustment vouchers such as Credit Notes/ Debit Notes for any short/ excess supplies or revision in prices or any other reason under the contract shall be submitted to the Employer in compliance with GST provisions.
- (d) Liquidated damages or any other recoveries should be shown as deductions on the invoice, and GST shall be applicable only on the net balance payment due. In case of Gross Invoice, Credit Note shall be issued for the amount derived for Liquidated Damages before payment and payment shall be made on net.
- (e) GST shall be paid as per the rate at which it is liable to be assessed or has been assessed provided the transaction of the supply is legally liable to such taxes and is payable as per the terms of the contract subject to the following conditions:
 - (i) The Employer shall not be bound to pay a higher GST rate if leviable due to any misclassification of HSN number or incorrect GST rate incorporated in the contract due to supplier's / contractor's fault. Wherever the contractor invoices the Goods at GST rate or HSN number, which is different from that incorporated in the contract, payment shall be made as per GST rate, which is lower of the GST rates incorporated in the contract or billed.
 - (ii) The payment against Tax Invoice shall be subject to the TDS (Tax deducted at Source) U/s 51 of the CGST Act, 2017 in addition to applicable TDS under Income Tax Act, 1961. On deduction at source, Employer shall be depositing the same to the credit of the respective Government, file return and issue TDS Certificate in favour of the contractor to that respect.
 - (iii) Payment of GST on running account bill shall be made on regular basis taking the GST Compliance behavior of the contractor (i.e. regular/timely deposit of taxes to the credit of the Government and filing of regular and proper return under GST etc.). The GST so paid should get reflected in

employer's GST portal before the payment of GST in next RA bill **unless** both the invoices are raised in the same tax period, for which time limit for deposit of taxes have not elapsed and/or time limit for filing statutory return is not over. In such case, payment of GST may be made subject to GST Compliance Behaviour of the contractor and the claim is not against Final Bill.

- (iv) In case of final bill, GST amount shall only be paid/ reimbursed after through checking in GSTR-2B of the employer, showing eligible for ITC and correctness of GST amount reflected in all running bills so far paid.

Regarding Payment/reimbursement of GST against invoices/RA bill of the contractor, the decision of the Engineer-In-Charge shall be final and binding on the contractor.

- (v) Employer has the absolute right to check the GST Compliance behaviour of the contractor through GSTIN portal, its GSTR-2B, documentary evidence submitted by the contractor etc. while making payment/reimbursement.
- (vi) Employer shall have the right to recover any GST paid against invoices, from any amount payable to such agency unless the GST amount is reflected in the GSTR-2B (and shown as eligible) for the purpose of ITC. Amount to be recovered/withheld shall relate only to the extent of GST not deposited or adjusted within the due date of filing return.
- (vii) If Employer fails to take ITC, due to the non-deposit of GST/ wrong filing of return, the amount of GST paid shall be recovered from any amount payable along with interest, penalty, etc. and Contractor shall indemnify the employer all loss incurred, interest/penalty imposed by the authorities due to such bad tax behavior of the contractor.
- (viii) Employer shall not be held responsible for the contractor's tax payment or duty under a misapprehension of the law.
- (ix) Contractor is required to adjust his basic price to the extent required by a higher tax rate billed as per invoice to match the all-inclusive price mentioned in the contract.
- (x) In case of profiteering by the contractor/supplier relating to GST tax, the Employer shall treat it as a violation of the Code of Integrity in the contract

and avail any or all punitive actions there under, in addition to recovery and action by the GST authorities under the Act.

- (xi) The contractor shall issue Receipt vouchers, as per provision of GST, immediately on receipt of advances against provision of services as per terms of contract, wherever applicable.

Statutory Variation Clause:

Any increase or decrease in rate of taxes, duties levies, cess etc., under any law, including GST, under the period of contract on supply shall be governed by terms and conditions of contract. Any increase or decrease in rates of GST during the currency of the contract, on supply, shall be under Employer's account unless and otherwise stated in the contract.

- 17.5** Each Bidder shall submit only one Bid either as an individual or as a Proprietor in a Proprietary firm or as a Partner in a Partnership firm or as a Director of a Limited Company/Corporation or as a Partner in a Joint Venture. Any Bidder who has submitted a Bid for a work, shall not be a witness for any other Bidder for the same work. Failure to observe the above stipulations would render all such Tenders submitted as a Bidder and/or as a witness, liable to summary ejection.
- 17.6** The Contractor shall be fully responsible for all matters arising out of the Performance of the Contract and shall, at his own expense, comply with all laws/acts/enactments/orders/ regulations/ obligations whatsoever of the Government of India, State Government, Local Body and any Statutory Authority.
- 17.7** In case the bidder does not quote his rate for any particular/individual head(s) in EPC-II Mode Tender containing one or more sub head(s), it will be presumed that the bidder has included the cost of that/those sub head(s), in the rates of other sub head(s) and the rate for such sub head(s), shall be considered as Zero and the tender will be evaluated by the Employer accordingly and the work executed by the successful bidder accordingly.

18.0 SIGNING OF CONTRACT AGREEMENT

- 18.1** The Tenderer whose tender has been accepted will be notified of the award by the Employer by issue of a 'Letter of Acceptance' prior to expiration of the Bid Validity period.

The Letter of Acceptance will be sent to the Contractor in two copies one of which he should return promptly, duly signed and stamped. The Letter of Acceptance will

be a binding Contract between the Employer and the Contractor till the formal Contract Agreement is executed.

- 18.2** Within the period as specified in Clause 1 of ‘Clauses of Contract’, from the date of issue of Letter of Acceptance, the successful Tenderer shall deliver to the Employer, Performance Guarantee in the format prescribed.
- 18.3** The Tenderer whose Tender is accepted shall be required to submit at his cost stamp papers of appropriate value as per the provisions of Indian Stamp Act within 15 days of the date of issue of Letter of Acceptance.
- 18.4** At the same time the Employer notifies the successful Tenderer that his Tender has been accepted, the Employer will direct him to attend the Employer’s office within 28 days of issue of Letter of Acceptance for signing the Agreement in the proforma at Annexure V. The Agreement will however be signed only after the Contractor furnishes Performance Guarantee and hence, where justified, the period of 28 days stipulated above will be extended suitably.

18A Amendment to Contract

The conditions and clauses of this contract cannot be varied except through a written Supplementary Agreement with mutual consent of both the parties to the contract.

19.0 PRE-QUALIFICATION PROFORMA

The bidder shall fill the pre-qualification Proforma at **Annexure-VIII**. The bid will be evaluated considering only those details as mentioned in Annexure-VIII and corresponding documents uploaded along with the bid and no other details/certificate/document will be taken into consideration while evaluating the bid to decide whether the bidder is qualified or not.

20.0 BRIEF NOTICE INVITING E-TENDERS

PU Head, Project Unit Guwahati, RITES Ltd. invites on behalf of **OIL India Limited**, Online bid under EPC mode -II of Contract on lumpsum component basis bids in Two Packet System for the following work. No other mode of submission shall be accepted.

S. No.	NIT No.	Name of Work & Location	Cost of Tender Document	Estimated Cost put to Bid	Earnest Money	Period of Completion	Last Date & Time of submission of Bid, EMD, Cost of Tender Document and Other Documents as specified in the Press Notice	Time & Date of Opening of Bid
1	2	3	4	5	6	7	8	9
1.	BITES/NERPO/OIL/BQ-HOUSING/25 Dated 28.08.2025	Construction of Workman Housing Complex(BQ Area) on EPC mode -II of contract at OIL Duliajan, Assam	Rs. 18,000.00 Plus GST @ 18% i.e. Rs. 3240/- = Rs. 21,240/-	Rs 160.02 Cr	Rs. 1,00,00,000.00	24 (Twenty Four) Months	18.09.2025 & 11:00 Hrs	11:30 Hrs & 19.09.2025

21.0 The bid document consisting of tender drawings, specifications, the schedule of quantities of various types of items to be executed and the set of terms and conditions of the contract to be complied with and other necessary documents can be seen on CPP Portal <https://etenders.gov.in> free of cost.

22.0 After submission of the bid the bidder can re-submit revised bid any number of times but before last time and date of submission of bid as notified.

23.0 While submitting the revised/modified Financial bid, the bidder can revise/modify the rate of one or more item(s) any number of times (he need not re-enter rate of all the items) but before last time and date of submission of bid as notified.

24.0 The bid submitted shall become invalid if:

- (i) The bidder is found in eligible.
- (ii) The bidder does not upload all the documents as stipulated in the bid document.
- (iii) If any discrepancy is noticed between the documents as uploaded at the time of

submission of bid and hard copies as submitted physically by the lowest bidder in the office of bid opening authority.

- (iv) The bidder does not deposit physical instruments of EMD and Cost of Tender Document within a week of opening of technical bid.

25.0 Those contractors not registered on the website mentioned above, are required to get registered beforehand. If needed they can go to CPP Portal <https://eprocure.gov.in/eprocure/app?page=BiddersManualKit&service=page> bidder's manual kit.

26.0 The intending bidder must have valid Class-III digital signature to submit the bid.

27.0 On opening date, the bidder can login and see the bid opening process. After opening of bids, he will be able to view the competitors' bid documents.

28.0 List of Documents to be scanned and uploaded within the period of bid submission: -

1. Account Payee Banker's Cheque/ Pay Order/Demand Draft towards cost of Tender Document in accordance with Clause 7.0 (f) hereinbefore.

OR

~~Document in support of exemption from payment of cost of Tender Document, if applicable due to any Govt. Policy.~~

2. Insurance Surety Bonds or Account Payee Banker's Cheque/ Pay Order/ Demand Draft towards Earnest Money Deposit (EMD) in accordance with Clause 9hereinbefore and Bank Guarantee if applicable in accordance with Clause 9 hereinbefore

OR

~~Document in support of exemption from payment of EMD, if applicable as per any Govt. Policy.~~

3. Authority to Sign (if required as per Clause 11.0 hereinbefore) in the format given at **Annexure III** or (**Annexure III & Annexure IV**) as applicable.
4. Integrity Pact as per **Annexure VI/1, VI/2& VI/3** as mentioned in clause 11.2.
5. RTGS/NEFT details as per **Annexure-VII**. (Mandate form)

6. Self-attested copy of Documents in support of meeting the criterion of Annual Financial Turnover in accordance with Para 1 of **Annexure I**.
7. Self-attested copy of Certificates in support of meeting the criterion of Similar Work Experience in accordance with **Para 2(a) of Annexure I**.
8. Self-attested copy of Certificates in support of meeting the criterion of Construction Experience in key activities/specified components in accordance with **Para 2(b) of Annexure I**.
9. Self-attested copy of Documents (Audited standalone Balance Sheets, Statement of Profit and Loss and Note to accounts with UDIN) in support of meeting the Profitability criterion in accordance with **Para 4 of Annexure I**.
10. Self-attested copy of Documents in support of meeting the criterion of Net Worth in accordance with **Para 5 of Annexure I**.
11. Declaration by the Bidder in the format given in **Proforma 1 of Annexure I**, including certificates mentioned therein. In case of JV, each Partner of JV should submit **Proforma 1** duly signed, dated & with seal separately.
12. Self-attested copy of Guidelines on Banning of Business Dealings as per **Annexure-VIA**.
13. Pre-Qualification Proforma at **Annexure VIII** duly filled in.
14. Self-Attested Copy of Partnership Deed/Memorandum and Articles of Association of the firm.
15. Self-Attested copy of Corrigendum(s), if any/ minutes of Pre-bid meeting, if any.
- ~~16. Self-certified copy of the Joint Venture Agreement/Memorandum of Understanding as per **Annexure II** (where Joint Venture is allowed) to be executed by each member of JV individually.~~
- ~~17. Power of Attorney in favour of Lead Member as the Lead Member of JV (where Joint Venture is allowed) as per **Annexure IV**, to be executed by all the members of JV individually.~~
18. Self-attested copy of GST Registration Certificate (As applicable).
19. Self-attested copy of certificates as per requirement of sub-clause 6

(verification of local content) of clause 12.15 (Preference to Make in India)

20. Self-attested documents as per clause 12.16 (Restriction on Procurement from a bidder of a country which shares a land border with India) if applicable.
21. Self-attested copy of valid registration with EPFO under 'EPF and Miscellaneous Provisions Act, 1952'.
22. Self-attested copy of any other document if specified in tender document and the correction slips to the Tender Document.
- ~~23. Declaration by Each Partner of JV (in case of Large works/ Normal Works) as per Para 3 (Servicing of Loan/ Credit Limit) of Annexure-I.~~
24. Undertaking/ Declaration by Bidder/ Each Partner of JV in case of large works/ Lead Member of JV in case of Normal works that "Audited standalone Balance Sheets, Statements of Profit and Loss and other financial statements for the immediately preceding year are not available" as per Para 1 (e) (Annual Financial Turnover), Para 4 (Profitability) and Para 5 (Net Worth) of **Annexure-I**, if applicable.

Note: - Any clarification / deficient document(s) sought by RITES Ltd. as per provisions of these tender documents shall be submitted by the bidder online only.

29.0 List of Documents to be submitted physically by Lowest (L1) Bidder within a week of the opening of Financial Bid: -

1. Self-attested copy of PAN/TAN issued by income Tax Department.
2. Self-attested copy of registration under Labour Laws like PF, ESI etc.
3. Self-attested copy of the statement of the details of bidder's jurisdictional GST Assessing officer (designation, address and email address)
4. Self-attested copy of ISO 9000 Certificate. (if any)
5. Self-attested copies of all the documents specified in Clause 28.0 above.

30.0 RITES Ltd. may approach any Bank, Individual, Employer, Firm or Corporation, whether mentioned in the documents submitted by bidders or not, to verify the credentials and general reputation of the bidder and where JV is allowed the credentials and general reputation of lead member & each Member of Joint Venture.

31.0 With reference to Clause 3, Sub-clause 3(a), 3(a) (i), 3(b) and **Clause 48.7** of Clauses of Contract of GCC for EPC Projects (Section-8), the name of the contractor declared as “Poor Performer” as per clause 48.7 and the contractor, whose Contract is determined as per clause 3 and deemed to be declared as “Poor Performer” as per Sub-clause 3(a) (i) will be placed in RITES’ ‘Negative List of Contractors’ and then such a contractor individually and also any Joint Venture wherein such contractor is a member, will not be eligible for a period of up to two years, to participate in tenders of other works and in the tendering process for balance work as per sub-clause 3(b) which may be invited by RITES. In case the contractor is a Joint Venture, the JV as well as all members of the JV individually or as member(s) of any other Joint Venture (JV) will not be so eligible. In case the contractor is a company then, in addition to the aforesaid provisions, the Associate Companies and Subsidiary Companies (as defined under Companies Act, 2013), of the company included in the Negative list, will also not be eligible.

32.0 CONFIRMATION OF BG (Wherever Applicable)

In order to have a secured IT enabled verification/ confirmation of BGs from system of issuing Bank to RITES using SFMS facility through Indian Bank, following conditions shall apply: -

- a) Bank Guarantee issued by issuing Bank **must be routed through SFMS platform** and IFN 760 must be sent from BG issuing Bank to Indian Bank. In case of extension, BG must be **routed through SFMS platform** and IFN 767 must be sent from BG issuing Bank to Indian Bank.
- b) Bidder shall submit to RITES LIMITED the copy of SFMS message as sent by issuing Bank branch along with the original Bank Guarantee or extension of BG as the case may be.
- c) Particulars of RITES’ Bank are provided below so that the bidder can share these particulars with their bank guarantee issuing Bank.

Bank details are as follows:

- i) Bank address: **Indian Bank, GF-3 Block B, Vipul Square, Sushant Lok, Phase I, Gurgaon.**
- ii) Bank account No: **6597266376**
- iii) Branch IFSC: **IDIB000S149**
- iv) Branch MICR code: **110019038**

In case, the bidder shows its inability to submit the bank guarantee from issuing Bank, which is SFMS Compliant, RITES may accept paper Bank Guarantee sent to RITES directly by the issuing Bank under Registered Post (AD)/ Speed Post/ Courier/ By hand. However, a declaration that none of their bank branches is SFMS Compliant shall be obtained with such Bank Guarantees.

ANNEXURE – I

QUALIFYING CRITERIA FOR WORKS CONTRACTS

1. ANNUAL FINANCIAL TURNOVER

The bidder should have achieved a minimum average annual financial turnover of **Rs. 80.01 Crore** (Rupees Eighty Crore and One lakhs Only) during the last 3 Financial Years.

Notes:

- a) The Annual financial turnover shall be operating revenue (excluding Income from other sources) as given under the head "Revenue/Income" in audited standalone Statement of Profit & Loss. Share of bidder in Revenue from operation of Joint Arrangement entities which is not included in audited standalone Statement of Profit & Loss of bidder will not be considered.
- b) The Bidder should furnish Annual Financial Turnover for each of the last 3 Financial Years with average thereof in tabular form and give reference of the document (with page no.) relied upon in support of meeting the Qualification Criterion.
- c) The Bidder should submit self-attested copy of Audited standalone Balance Sheet and Profit and Loss Statement mentioning UDIN and Schedules, for last three financial years. Provisional Audited financial statement or certified statements will not be accepted.
- d) If the Audited standalone Balance Sheets, Statement of Profit and Loss and Notes to accounts for the immediately preceding year is not available in case of a work for which the last stipulated date for submission of the bids is before 30th September or in case, Government of India extends the date of conducting of AGM / Filing of Balance Sheet beyond 30th September, Audited standalone Balance Sheets, Statement of Profit and Loss and Notes to accounts of the three Financial Years immediately preceding the previous Financial Year shall be considered for evaluating the credentials of the Bidder. These provisions are also applicable in cases where JV is permitted.
- e) ~~In case JV is permitted the following provisions will apply:-~~
 - i) ~~Large Works~~

~~For each Partner, the average Annual Turnover during the last three Financial Years will be considered. This should not be less than the figure arrived at by multiplying the minimum Average Annual Turnover~~

~~stipulated for the Bidder in Qualification Criterion multiplied by the percentage participation by that partner in the JV. Each partner should satisfy this requirement and thus automatically JV will satisfy the criterion of minimum average Annual Turnover.~~

~~ii) Normal Works~~

~~The Partner in charge/ Lead. Member shall singly meet this criterion.~~

2. **WORK EXPERIENCE**

a) **Similar Works Experience**

~~(i) For works in Normal Areas (other than Difficult Areas)~~

~~The Bidder should have satisfactorily completed or substantially completed in his own name or proportionate share as a member of a Joint Venture, at least one similar work of minimum value of Rs. OR at least two similar works each of minimum value of Rs. OR at least three similar works each of minimum value of Rs. during the last 7 (Seven) years prior to the last stipulated date for submission of the Bid. Works completed prior to the cut-off date shall not be considered.~~

OR

(ii) **For Works in Difficult Areas (Refer Clause 2.3)**

The Bidder should have satisfactorily **completed or substantially completed** in his own name or proportionate share as a member of a Joint Venture, at least one similar work of minimum value of **Rs. 80.01 Crore** OR at least two similar works each of minimum value of **Rs.64.00 Crore** during the last **7 (Seven) years** prior to the last stipulated date for submission of the Bid. Works completed prior to the cut-off date shall not be considered.

(strike out (i) or (ii) as applicable)

Similar Works

Similar Works shall mean the work of Construction of Institutional/Educational/Residential/Commercial building (As defined in National Building Code) with RCC framed structure having minimum one 05 storey building comprising of all required Civil and E&M services including

internal & external water supply, plumbing, sanitary installations, Internal & external Electrical Installations, Elevators (Lifts) and Firefighting carried out in India under a single contract (including additional work carried out under the contract).

- i) **For the purpose of similar work, each basement, stilt constructed in the building shall be considered as a storey.**
- ii) **Go-downs/Warehouses/factory sheds/industrial building (As defined in National Building Code) shall not be considered as eligible similar works.**
- iii) **Mumty and machine room shall not to be considered in storey.**
- iv) **Own works/work under the same management /own certification of the bidder shall not be considered.**

In case the Bidder (Indian Company) wishes to rely on a work completed abroad, the value of such completed work in foreign convertible currency shall be converted into Indian Rupees. The conversion rate shall be decided by RITES based on the rates of currency on the date of completion of work (the bidder to also submit the currency conversion rate as on completion date of the Credential Certificate relied upon by the bidder for the purpose of work experience). Further, such a bidder (Indian Company) should have also completed at least one Similar work of value minimum 40% of estimated cost of work, in India in the last seven years prior to the last stipulated date for submission of the Bids.

~~aa) In case of Railway Works involving one or more components like Earthwork, Bridges, Track Linking, Track Laying, OHE, signaling etc., the Qualifying Criterion relating to Work Experience shall be as under:~~

~~(i) For works in Normal Areas (other than Difficult Areas)~~

~~The Bidder should have satisfactorily completed or substantially completed in his own name or proportionate share as a member of a Joint Venture, at least one similar work of minimum value of Rs. OR at least two similar works each of minimum value of Rs. during the last 7 (Seven) years prior to the last stipulated date for submission of the Bid. Works completed prior to the cut-off date shall not be considered.~~

OR

(ii) ~~For Works in Difficult Areas (Refer Clause 2.3)~~

~~The Bidder should have satisfactorily completed or substantially completed in his own name or proportionate share as a member of a Joint Venture, at least one similar work of minimum value of Rs. OR at least two similar works each of minimum value of Rs. during the last 7 (Seven) years prior to the last stipulated date for submission of the Bid. Works completed prior to the cut-off date shall not be considered.~~

~~(strike out (i) or (ii) as applicable)~~

Similar Works

~~Similar Works shall mean the work of carried out in India under a single contract (including additional work carried out under the contract).~~

~~In case the Bidder (Indian Company) wishes to rely on a work completed abroad, the value of such completed work in foreign convertible currency shall be converted into Indian Rupees. The conversion rate shall be decided by RITES based on the rates of currency on the date of completion of work (the bidder to also submit the currency conversion rate as on completion date of the Credential Certificate relied upon by the bidder for the purpose of work experience). Further, such a bidder (Indian Company) should have also completed at least one Similar work of value minimum 25% of estimated cost of work, in India in the last seven years prior to the last stipulated date for submission of the Bids.~~

Notes:

- I. A weightage of 7% (compounded annually from the date of completion of the work to the last stipulated date for submission of the Bids) shall be given for equating the value of works to the last stipulated date for submission of the Bids. For example, for a similar work of Rs 1,00,000/- completed on 31.12.2017 and the last stipulated date for submission of the bids was 31.08.2020, the value of work that shall be considered after giving weightage is $[Rs\ 1,00,000 \times 1.07 \times 1.07] \{1 + [(7/100) \times (244/366)]\} = Rs\ 1,19,832.87$, rounded off to Rs 1,19,833/-
- II. Works shall be considered as “Completed Works” where physical completion of entire work is over or commissioning of work has been done, whichever is earlier.

- III. Substantially Completed Work means an ongoing work in which payment equal to or more than 90% of the present contract value (excluding the payment made for adjustment of Price variation (PVC), if any) has been made to the contractor in that ongoing contract, Functional completion of the Project has been achieved and no proceedings of termination of contract on Contractor's default has been initiated.
- IV. In case of completed work, the value of final bill (gross amount) including the PVC amount (if paid) shall be considered as the completion cost of work. In case final bill is pending, only the total gross amount already paid including the PVC amount (if paid) shall be considered as the completion cost of work.
- V. In case of substantially completed work, the total gross amount already paid including the PVC amount (if paid), as mentioned in the certificate, shall be considered as the cost of substantially completed work.
- VI. Certificate for substantial completion of project/work/asset should contain three parts. Part-I shall contain financial value of work done, PVC amount and present contract value. Part-II shall contain 'certificate of functional completion of project/work/asset'. Part-III shall contain that no proceedings of termination of contract on contractor's default has been initiated. The credential certificate in this regard should have been issued not prior to 60 days of date of invitation of present tender.
- VII. In case a work is considered similar in nature for fulfillment of technical credentials, the overall cost including the PVC amount (if paid) of that completed work or substantially completed work, shall be considered and no separate evaluation for each component of that work shall be made to decide eligibility.
- VIII. Value of a completed work done by a Member in an earlier JV shall be reckoned only to the extent of the concerned member's share in that JV for the purpose of satisfying his/her compliance to the above-mentioned criteria in the tender under consideration.
- IX. In case of a Certificate from a Public Limited Company or its subsidiary, the Bidder should also submit documentary proof that the Public Ltd. Company was listed in Stock Exchange in India or Abroad when the work was executed for it.
- X. The Bidder should submit the details of such similar

completed/Substantially Completed works as per the format given in Annexure-VIII.

- XI. Similar Works carried out by another Contractor on behalf of the Bidder on a back-to-back basis will not be considered for satisfaction of the Qualification Criterion by the Bidder. Similar Works carried out by the bidder as a sub-contractor will be considered for satisfaction of Qualification criterion by the bidder subject to submission of experience certificate as per Sl. No. XII below.
- XII. Credential certificates issued by Government Organizations/Semi Government Organizations of Central or State Government; or by Public Sector Undertakings/ Autonomous Bodies of Central/State Government or their subsidiaries/by Public Ltd. Companies listed in Stock exchange in India or Abroad or subsidiaries of such companies shall only be accepted for assessing the eligibility of a Tenderer. Certificates issued by one of the aforesaid organizations as a lead member (having equity stake of 51% or more) of a Joint Venture (JV) for a work executed for the JV, will also be acceptable. In case of PPP projects, if the bidder has executed a work for a concessionaire who is one of the aforesaid organizations, then the certificate issued by such concessionaire or by the public authority concerned, will also be acceptable.
- XIII. The cutoff date shall be calculated backwards from the last stipulated date for submission of Tender i.e. for a Tender having the last stipulated date for submission on 06.08.2014, the cutoff date shall be 07.08.2007.

~~XIV. In case JV is permitted the following provisions will apply:-~~

~~Large Works~~

- ~~1. All the partners shall jointly meet this criterion.~~
- ~~2. Each partner should have satisfactorily completed in his own name or proportionate share as a member of a Joint Venture, **at least one similar work of minimum value of 10 % of the Estimated cost of work during the last 7 (Seven) years prior to the last stipulated date for submission of the Bid OR should have executed at least one of the Key Activities mentioned under Construction Experience in Key Activities.**~~

~~Normal Works~~

- ~~1. The Partner in charge/Lead Member shall singly meet this criterion.~~

~~2. Each partner should have satisfactorily completed in his own name or proportionate share as a member of a Joint Venture, at least one similar work of minimum value of 10 % of the Estimated cost of work during the last 7 (Seven) years prior to the last stipulated date for submission of the Bid OR should have executed at least one of the Key Activities mentioned under Construction Experience in Key Activities.~~

XV. The value of similar work shall be calculated excluding the element of GST on works Contract Service while evaluating on meeting the Qualifying Criteria of Work Experience.

b) Construction Experience in Key Activities

To qualify for award of the contract, each bidder in his own name or as a member of a Joint Venture should have, in the last 7 (Seven) years prior to the last stipulated date for submission of the bid, executed the following key activities carried out in India.

1. The bidder must have obtained at least one GRIHA 3-Star rating or higher certification for a building project.

This work can be part of eligible work at (a) above.

OR

GRIHA 3 star or above rating obtained by bidder under a separate contract shall also be considered for the purpose of assessing the technical competence only without adding its monetary value for determining the eligibility criteria.

Notes:

- I. Credential certificates issued by Government Organizations/Semi Government Organizations of Central or State Government; or by Public Sector Undertakings/ Autonomous Bodies of Central/State Government or their subsidiaries/by Public Ltd. Companies listed in Stock exchange in India or Abroad or subsidiaries of such companies shall only be accepted for assessing the eligibility of a Tenderer. Certificates issued by one of the aforesaid organizations as a lead member (having equity stake of 51% or more) of a Joint Venture (JV) for a work executed for the JV, will also be acceptable. In case of PPP projects, if the bidder has executed a work for a concessionaire who is one of the aforesaid organizations, then the certificate issued by such

concessionaire or by the public authority concerned, will also be acceptable.

- II. When key activities are expressed/ specified in monetary terms, a weightage of 7% (compounded annually from the date of completion of the Key activity to the last stipulated date for submission of the Bids) shall be given for equating the value of key activity to the last stipulated date for submission of the Bids. For example, for a key activity of Rs 1,00,000/- completed on 31.12.2017 and the last stipulated date for submission of the bids was 31.08.2020, the value of key activity that shall be considered after giving weightage is $[Rs\ 1,00,000 * 1.07 * 1.07] \{1 + [(7/100) * (244/366)]\} = Rs\ 1,19,832.87$, rounded off to Rs 1,19,833/-
- III. The work satisfying the criterion for a particular key activity may be different from a work satisfying the criterion for another key activity.
- IV. The Bidder should furnish with his Bid a tabular statement giving contract-wise quantities/amount of key activities (mentioning date of start and date of completion of key activities) executed in the last **7 (Seven)** prior to the last stipulated date for submission of the bids which meet the Qualification Criterion along with documentary proof in support thereof (indicating page nos.).
- V.
 - (a) Even if a work has not been completed but if the specified quantity/ specified value in monetary terms of the key activity has been completed in the **last 7 (Seven) years** prior to the last stipulated date for submission of the bids, the same shall be taken into consideration for the purpose of this criterion.
 - (b) Works carried out by another Contractor on behalf of the Bidder on a back-to-back basis will not be considered for satisfaction of the Qualification Criterion by the Bidder. Works carried out by the bidder as a sub-contractor will be considered for satisfaction of Qualification criterion by the bidder subject to submission of experience certificate as per Sl. No. I above.
- VI. Any key activity executed by the Bidder as a member of a Joint Venture will be accepted provided there is documentary proof in support of the same either in the MOU/Agreement of the JV or in a declaration by the Lead Member of that JV or the Client confirming that the Key activity

was actually executed by the Bidder.

~~In case JV is permitted the following provisions will apply:~~

Large Works

- ~~1. All the partners shall jointly meet this criterion.~~
- ~~2. Each partner should have satisfactorily completed in his own name or proportionate share as a member of a Joint Venture, at least one similar work of minimum value of 10% of the Estimated cost of work during the last 7 (Seven) years prior to the last stipulated date for submission of the Bid OR should have executed at least one of the Key Activities mentioned under Construction Experience in Key Activities.~~

Normal Works

- ~~1. All the partners shall jointly meet this criterion.~~
- ~~2. Each partner should have satisfactorily completed in his own name or proportionate share as a member of a Joint Venture, at least one similar work of minimum value of 10% of the Estimated cost of work during the last 7 (Seven) years prior to the last stipulated date for submission of the Bid OR should have executed at least one of the Key Activities mentioned under Construction Experience in Key Activities.~~

- VII. The value of Key Activity shall be calculated excluding the element of GST on works Contract Service.

3. **SERVICING OF LOAN / CREDIT LIMIT**

- (i) The bidder should furnish a declaration that he has not failed to service the principal amount or interest or both of a loan amount / credit limit from any Bank or Financial Institution during a period of one year prior to the dead line for submission of bids i.e. the last stipulated date for submission of bids.

Notes:

- a) In case a bidder has defaulted in servicing his loan/credit limit during a period of one year prior to the last stipulated date for submission of bids, he shall be disqualified.

b) The declaration may be included in the Declaration to be submitted in **Proforma 1 to this Annexure.**

~~e) In case JV is permitted the following provisions will apply:~~

~~Large Works~~

~~Each partner shall sign the declaration.~~

~~Normal Works~~

~~Each partner shall sign the declaration~~

4. PROFITABILITY

The Bidder should be a profit (net) making firm and should have made profit during any two of the past 3 Financial Years immediately preceding the last stipulated date for submission of bids. If the audited standalone Balance Sheets, Statement of Profit and Loss and Notes to accounts for the immediately preceding year is not available in case of a work for which the last stipulated date for submission of the bids is before 30th September or in case, Government of India extends the date of conducting of AGM / Filing of Balance Sheet beyond 30th September, audited standalone Balance Sheets, Statement of Profit and Loss and Note to accounts with UDIN of the three Financial Years immediately preceding the previous Financial Year shall be considered for evaluating the Profitability Criterion. The Profit Before Tax (PBT) shall be considered while evaluating the Profitability Criterion. These provisions are also applicable in cases where JV is permitted. The Bidder should furnish figures of Profit Before Tax (PBT) of last 3 years in a tabular form and submit attested copies of audited standalone Balance Sheets, Statement of Profit and Loss and Notes to accounts for the last three Financial Years. Specific reference with page no. of document which proves satisfaction of this Qualifying Criterion should be indicated in the tabular statement.

The Profitability shall be considered as per the audited standalone Statement of Profit & Loss of bidder. Share of bidder in Profit (PBT) of Joint Arrangement entities which is not included in audited standalone Statement of Profit & Loss of bidder will not be considered.

Notes:

~~— In case JV is permitted, the following provisions will apply:~~

~~Large Works~~

~~All the partners shall meet individually this criterion.~~

Normal Works

~~All the partners shall meet individually this criterion.~~

5. **NET WORTH**

The Bidder should have positive Net Worth of at least 15% of the estimated cost i.e. **Rs. 24.00 Cr (Rupees Twenty-Four Crore Only).**

Notes:

- a) Net Worth shall be computed from the bidder's audited standalone balance sheet of the financial year immediately preceding the last stipulated date for submission of the bids. If the Audited standalone Balance Sheet for financial year immediately preceding the last stipulated date for submission of the bids is not available in case of a work for which the last stipulated date for submission of the bids is before 30th September or in case, Government of India extends the date of conducting of AGM / Filing of Balance Sheet beyond 30th September, audited Balance Sheet of the Financial Year immediately preceding the previous Financial Year shall be adopted for computing the Net worth of the Bidder. These provisions are also applicable in cases where JV is permitted.

Share of bidder in Net Worth of Joint Arrangement entities which is not included in audited standalone Balance Sheet of bidder will not be considered.

- ~~b) In case JV is permitted, the following provisions will apply:~~

Large Works

~~All the partners shall jointly meet this criterion.~~

Normal Works

~~The Partner in charge/Lead Member shall singly meet this criterion~~

6. **POINTS TO NOTE ON SATISFACTION OF QUALIFYING CRITERIA IN CASE OF BOTH LARGE AND NORMAL WORKS**

- a) Sub-Contractor's Experiences and Resources

Sub-Contractors' Experiences and Resources will not be taken into account in determining the Bidder's compliance with the qualifying criteria.

- b) Experiences and Resources of the Parent Company and other subsidiary

companies

If the Bidder is a wholly owned subsidiary of a company, the experience and resources of the owner/parent company or its other subsidiaries will not be taken into account. However, if the Bidder is a Company, only the Experience (and not the Financial Resources) of its subsidiaries will be taken into consideration.

7. DISQUALIFICATION ON CERTAIN GROUNDS

Even though the Bidders may meet the above qualifying criteria, they are subject to be disqualified if they have

- a) Concealed any information/document which may result in the Bidder's disqualification or if any statement/information/document furnished by the Bidder or issued by a Bank/Agency/Third party and submitted by the Bidder, is subsequently found to be false or fraudulent or repudiated by the said Bank/Agency/Third Party. In such a case, besides Bidder's liability to action under Clause 9.4 of Instructions to Tenderers, the Bidder is liable to face the penalty of banning of business dealings with him by RITES.
- b) Records of any contract awarded to them, having been determined during the past three years prior to the dead line for submission of bids i.e. last stipulated date for submission of bids.
- c) Been declared as Poor Performer by RITES and their name is currently in the 'Negative List' of RITES.
- d) Their business banned or suspended by any Central/State Government Department/ Public Undertaking or Enterprise of Central/State Government and such ban is in force.
- e) Not submitted all the supporting documents or not furnished the relevant details as per the prescribed format.

A declaration to the above effect in the form of affidavit on stamp paper of Rs. 10/- duly attested by Notary/Magistrate should be submitted as per format given in Proforma-1 enclosed.

8. PAST CONTRACT PERFORMANCE AND LITIGATION

There shall be no instance of contract non-performance as a result of Applicant (Bidder)'s default during the last 5 (five) years prior to the last stipulated date for submission of the Bid.

Notes:

- a) Contract non-performance must be based on all information on fully settled disputes or litigation, i.e. dispute or litigation that has been resolved in accordance with the dispute resolution mechanism under the respective contract and where all appeal instances available to the applicant (bidder) have been exhausted.
- b) Bidder shall submit details of past contract performance and litigation(s) as per format placed at **Profoma-2** along with documents establishing details included in the **Proforma-2**.
- c) The cutoff date shall be calculated backwards from the last stipulated date for submission/opening of Tender i.e. for a Tender which is being opened on 06.08.2014, the cutoff date shall be 07.08.2009.

9. CONTRACT LITIGATION HISTORY

The aggregate amount of court/arbitration awards related to contractual litigation against the Applicant (Bidder), if any, during the last 5 (five) years prior to the last stipulated date for submission of the Bid, shall not exceed 25% of the Net Worth as per the latest financial statements.

Notes:

- a) The bidder shall provide accurate information in the Perfoma-3 about any litigation or arbitration resulting from contracts completed or ongoing under its execution during the last 5 (five) years prior to the last stipulated date for submission of the Bid along with documents establishing details included in the **Proforma-3**.
- b) The cutoff date shall be calculated backwards from the last stipulated date for submission/opening of Tender i.e. for a Tender which is being opened on 06.08.2014, the cutoff date shall be 07.08.2009.

Proforma-1

DECLARATION BY THE BIDDER

(Affidavit on Non-Judicial Stamp Paper of Rs.10/- duly attested by Notary/Magistrate)

This is to certify that We, M/s....., in submission of this offer confirm that: -

- i) We (including our affiliates or subsidiaries or constituents) are not insolvent, in receivership, bankrupt or being wound up, not have our affairs administered by a court or a judicial officer, not have our business activities suspended and are not the subject of legal proceedings for any of these reasons;
- ii) We are not convicted (within three years preceding the last date of bid submission) or stand declared ineligible/ suspended/ blacklisted/ banned/ debarred by appropriate agencies of Government of India from participation in Tender Processes, for offences mentioned in Tender Document in this regard. We have neither changed our name nor created a new “Allied Firm”, consequent to the above disqualifications.
- iii) Do not have any association (as bidder/ partner/ Director/ employee in any capacity) with such retired public official or near relations of such officials of Procuring Entity, as indicated, in the Tender Document.
- iv) We have no conflict of interest, which substantially affects fair competition. The prices quoted are competitive and without adopting any unfair/ unethical/ anti-competitive means. No attempt has been made or shall be made by us to induce any other bidder to submit or not to submit an offer to restrict competition.
- v) We have visited the site of work and seen the working conditions, approach road/path, availability of water, electricity, construction labour, construction materials and other relevant requirements connected with the work.
- vi) We have neither concealed any information/document which may result in our disqualification nor made any misleading or false representation in the forms, statements and attachments in proof of the qualification requirements;
- vii) During the past three years prior to the deadline for submission of bids, no contract awarded to us has been determined.
- viii) No Central/ State Government Department/ Public Sector Undertaking or Enterprise of Central/State Government has banned/ suspended business dealings with us as on date.
- ix) We have submitted all the supporting documents and furnished the relevant details as

per prescribed format and we agree to submit, without delay additional information/documents which may be demanded by RITES Ltd.

- x) List of Similar Works satisfying Qualification Criterion indicated in Annexure VIII does not include any work which has been carried out by us through a Subcontractor on a back to back basis.
- xi) The information and documents submitted with the Tender and those to be submitted subsequently by way of clarifications are correct and we are fully responsible for the correctness of the information and documents submitted by us.
- xii) We have not failed to service the principal amount or interest or both of a loan account/credit limit from any Bank or Financial Institution during a period of one year prior to the deadline for submission of bids.
- xiii) *The original instruments of EMD and Cost of Tender Document, in physical form shall be deposited by us with RITES Ltd. within a week from the date of opening of Technical Bid failing which RITES Ltd. may reject the bid and also take action to debar us from participating in Tenders invited by RITES Ltd. for a period of two years.

OR

~~*We are exempted from payment of cost of Tender Document and EMD and the certified copies of original documents in support of such exemptions shall be deposited by us with RITES Ltd. within a week from the date of opening of Technical Bids and original documents produced on demand, failing which RITES Ltd. may reject the bid and also take action to debar us from participating in Tenders invited by RITES Ltd. for a period of two years.~~

~~(* Delete whichever is not applicable)~~

- xiv) I/We have read the clause regarding restrictions on procurement from a bidder of a country which shares a land border with India and on sub-contracting to contractors from such countries and solemnly certify that I/we fulfill all requirements in this regard and are eligible to be considered. I/We Certify that: -
 - a) This bidder is not from such a country or, if from such a country, has been registered with the Competent Authority i.e. DPIIT (Where applicable, evidence of valid registration by the Competent Authority i.e. DPIIT shall be attached.)
 - b) I/We shall not sub-contract any work to a contractor from such countries unless such contractor is registered with the Competent Authority i.e. DPIIT.

(Where applicable, evidence of valid registration by Competent Authority i.e. DPIIT shall be attached)

- c) * "I have read the clause regarding restrictions on procurement from a bidder having Transfer of Technology (ToT) arrangement. I certify that this bidder does not have any ToT arrangement requiring registration with the competent authority."

OR

* "I have read the clause regarding restrictions on procurement from a bidder having Transfer of Technology (ToT) arrangement. I certify that this bidder has valid registration to participate in this procurement." *{Evidence of valid registration by the Competent Authority i.e. DPIIT shall be attached}*

* Delete whichever is not applicable

- xv) I have read the "Make in India Policy" and certify that I meet the eligibility criteria of Class-I Local Supplier and certify that percentage of Local content will be more than*(Fill in the percentage of Local content)* as specified in the tender document. **We are enclosing a certificate as per requirement of sub clause 6 (verification of local content) of clause 12.15 (Preference to Make in India)**
- xvi) I/we am/ are a startup firm registered by, Department of Industries Policy and Promotion(DIPP) and my registration number is, valid upto..... (Copy enclosed). We confirm that we are not holding more than 2 contracts under relaxed credentials at this point of time.
- xvii) We hereby undertake that observance and compliance related to Government/Regulatory/Statutory guidelines on Environmental, social and ethical aspects are ensured by us.
- xviii) We understand that in case any information/document which may result in our disqualification is concealed by us or any statement/information/document furnished by us or to be furnished by us in connection with this offer, or issued by Bank/Agency/Third Party is subsequently found to be false or fraudulent or repudiated by the said Bank/Agency/Third Party, business dealings with us may be banned.

(Note: In case of JV, each Partner/member of JV should submit Proforma 1 duly signed, dated & with seal separately.)

SEAL, SIGNATURE & NAME OF THE BIDDER

Signing this document

Proforma 2

Past Contract Non-Performance, in accordance with Annexure-I, Qualification Criteria for Works Contracts

☐ Contract non-performance did not occur since the date specified in Annexure-I , Qualification Criteria for Works Contracts, Sub para -8
☐ Contract(s) non-performance since the date specified in Annexure-I , Qualification Criteria for Works Contracts, Sub para – 8 as indicated below:

Year	Outcome as percentage of Net Worth	Contract Identification	Total Contract Amount (Rs.)
<i>[insert financial year]</i>	<i>[insert percentage]</i>	Contract Identification: <i>[indicate complete contract name, number, date, value and any other identification]</i> Name of Employer: <i>[insert full name]</i> Address of Employer: <i>[insert street/city/country]</i> Matter in dispute: <i>[indicate main issues in dispute]</i> Party who initiated the dispute: <i>[indicate “Employer” or “Contractor”]</i> Result of dispute: <i>[Indicate if resolution was treated by the Adjudicator, under Arbitration or dealt with by the Judiciary and whether resolved in favour or against the Applicant (Bidder)]</i>	<i>[insert amount]</i>

Proforma 3

Litigation History, in accordance with Annexure-I, Qualification Criteria for Works Contracts

☐ No litigation since the date specified in Annexure-I , Qualification Criteria for Works Contracts, Sub para - 9
☐ Litigation award(s) since the date specified in Annexure-I , Qualification Criteria for Works Contracts, Sub para – 9 as indicated below:

Year of award	Outcome as percentage of Net Worth	Contract Identification	Total Contract Amount (Rs.)
<i>[insert financial year]</i>	<i>[insert amount and percentage]</i>	Contract Identification: <i>[indicate complete contract name, number, date, value and any other identification]</i> Name of Employer: <i>[insert full name]</i> Address of Employer: <i>[insert street/city/country]</i> Reason(s) for non-performance: <i>[Indicate main reason(s)]</i>	<i>[insert amount]</i>

ANNEXURE-II

**DRAFT MEMORANDUM OF UNDERSTANDING EXECUTED BY MEMBERS OF THE
JOINT VENTURE**

(On each firm's Letter Head)

From

.....

.....

To

RITES Ltd.

.....

Dear Sir,

Ref: ~~RITES Tender Notice No..... dated for~~
~~..... (Name of Work)~~

~~We wish to confirm that our company/ firm (delete as appropriate) has formed a Joint Venture with (insert names of all other members of the group)~~

~~for purposes associated with your Tender No.....~~

~~(Members who are not the Lead Member of the Joint Venture should add the following paragraph) *~~

~~* The Joint Venture is led by (insert name of the Lead Member)~~

~~Whom we hereby authorize to act as leader on our behalf for the purposes of submission of Bid for (name of work) and to incur liabilities and receive instructions for~~

~~and on behalf of any and all the partners of the Joint Venture. For this purpose, we have executed a Power of Attorney in favour of (name of the Lead Member)~~

~~(Member who is the Lead Member of the Joint Venture should add the following paragraph)~~
~~**~~

~~** In this Joint Venture we act as Lead Member and for the purposes of bidding for the work, represent the Joint Venture.~~

~~Till the award of work, the Lead Partner shall furnish Bid bond and all other bonds/guarantees to the Employer on behalf of the Joint Venture, which shall be legally binding on all the partners of the Joint Venture.~~

~~In the event of our Joint Venture being awarded the contract we agree to be jointly with~~

~~..... (insert names of all other members of the Joint Venture) and severally liable to RITES, its successors and assigns for all obligations, liabilities, duties and responsibilities arising from or imposed by the contract subsequently entered into between~~

~~RITES and our Joint Venture.—~~

~~The precise responsibility of the Lead Member and other Members of the Joint Venture in respect of planning, design, construction equipment, key personnel and work execution including Percentage participation by each Member will be as indicated below and these shall not be varied/ modified subsequently without your prior approval.~~

S- No	Item	Responsibility of		
		JV Member 1 (Lead Member) *Mention name of JV lead- member	JV Member 2 *Mention name of JV member	JV Member 3 *Mention name of JV member
1.—	Planning			
2.—	Design			
3.—	Construction- Equipment 1.— 2.—			
4.—	Key Personnel 1.— 2.— 3.— 4.—			
5.—	Work Execution 1.— 2.— 3.—			
6.—	% Participation** (This shall be as per sub clause 2.4(b)(I)(vi))			

Note to bidders:

~~The above table has been given to give an idea to bidders. However, bidders may modify the details depending upon type of work and sharing of responsibilities as per JV agreement. The bidder shall mention Percentage Participation at Sl. No. 6 in the above table as per sub clause 2.4 (b) (I) (vi).~~

~~We further agree that entire execution of the contract shall be carried out exclusively through the Lead Member.~~

~~In case our Bid is successful, the Joint Venture Agreement incorporating the above provisions~~

~~will be executed within 15 days of receipt of Letter of Acceptance from you and shall be registered at the place where the Agreement will be signed, so as to be legally valid and binding on all Members of the Joint Venture.~~

~~We agree that the Joint Venture Agreement shall be valid during the entire currency of the Contract including the period of extension if any, and the maintenance period after the work is completed.~~

~~We further confirm that we shall open a Bank Account in the name of JV **and all payments due to the JV shall be made by you by crediting to that Account.** To facilitate statutory deductions such as towards Income Tax and VAT made from the amounts due to us against our bills, being credited to the concerned Government departments, we shall obtain PAN/TIN number etc. as required and advise you the details before claiming our first on account bill.~~

~~We affirm that in suppression of anything contrary contained herein before, the Integrity Pact with the Employer in the format at Annexure VI/2 & VI/3 (if applicable) **shall be duly signed by authorized signatory of each member/partner of the Joint Venture and will be duly witnessed.** All Members including the Lead Member shall comply with the provisions in the Integrity Pact and any violation of the Integrity Pact by any Member shall be construed as violation by the Joint Venture.~~

Yours faithfully,

Signature

(Name of Signatory).....

(Capacity of Signatory).....

Seal

Witness 1 _____ Witness 2 _____

Name _____ Name _____

Address _____ Address _____

Occupation _____ Occupation _____

Note:

- ~~1. To be executed by each Member of the Joint Venture individually.~~
- ~~2. In case of any inconsistency between MoU submitted by JV members, the MoU submitted by lead member of JV shall be considered correct.~~

FORMAT FOR POWER OF ATTORNEY TO AUTHORISED SIGNATORY

POWER OF ATTORNEY

(To be executed on non-judicial stamp paper of the appropriate value in accordance with relevant Stamp Act. The stamp paper to be in the name of the firm/company who is issuing the Power of Attorney).

We, M/s..... (name of the firm/company with address of the registered office) hereby constitute, appoint and authorize Mr./Ms..... (Name and residential address) who is presently employed with us and holding the position of and whose signature is given below as our Attorney to do in our name and our behalf all or any of the acts, deeds or things necessary or incidental to our bid for the work **Construction of Workman Housing Complex (BQ Area) on EPC mode -II of contract at OIL Duliajan, Assam**, including signing and submission of application/proposal, participating in the meetings, responding to queries, submission of information/documents and generally to represent us in all the dealings with RITES or any other Government Agency or any person, in connection with the works until culmination of the process of bidding, till the Contract Agreement is entered into with RITES and thereafter till the complete discharge of all the contractual obligations.

We hereby agree to ratify all acts, deeds and things lawfully done by our said Attorney pursuant to this Power of Attorney and that all acts, deeds and things done by our aforesaid Attorney shall always be deemed to have been done by us.

(Add in the case of a Joint Venture)

Our firm is a Member/Lead Member of the Joint Venture of,
..... and

Dated this day of 20.....

(Signature and name of authorized signatory being given Power of Attorney)

.....

(Signature and name in block letters of *All the partners of the firm, *Authorized Signatory for the Company)

*(*Strike out whichever is not applicable)* Seal of firm/ Company

Witness 1:

Witness 2:

Name:

Name:

Address:

Address:

Occupation:

Occupation:

Notes:

- In case the Firm/ Company is a Member of a Joint Venture, the authorized signatory has to be the one employed by the Lead Member.
- The mode of execution of the Power of Attorney should be in accordance with the procedure, if any, laid down by the applicable law and the charter documents of the executants (s) and when it is so required the same should be under common seal affixed in accordance with the required procedure.

**~~FORMAT FOR POWER OF ATTORNEY TO LEAD MEMBER OF JOINT-
VENTURE~~**

~~(To be executed on non-judicial stamp paper of the appropriate value in accordance with relevant Stamp Act. The stamp paper to be in the name of the company who is issuing the Power of Attorney)~~

~~Whereas RITES Ltd. has invited bids for (Name of work) for and on behalf of.....as an Agent /Power of Attorney Holder.~~

~~Whereas, the Members of the Joint Venture (JV) comprising of M/s....., M/s.....and M/s.....(the respective names and addresses of the Registered offices to be given) are interested in bidding for the work and implementing the same in accordance with the terms and conditions contained in the bid documents.~~

~~Whereas, it is necessary for the members of the JV to designate one of them as the Lead Member with all necessary power and authority to do, for and on behalf of the JV, all acts, deeds and things as may be necessary in connection with the JV's bid for the work.~~

~~NOW THIS POWER OF ATTORNEY WITNESSETH THAT~~

~~We, M/s.....,M/s.....and M/s.....hereby designate M/s..... being one of the members of the JV, as the Lead Member of the JV, to do on behalf of the JV, all or any of the acts, deeds or things necessary or incidental to the JV's bid for the work, including submission of application proposal, participating in meetings, responding to queries, submission of information/documents and generally to represent the JV in all its dealings with RITES or any other Government Agency or any person, in connection with the work until culmination of the process of bidding till the contract agreement is entered into with RITES and thereafter till the complete discharge of all the contractual obligations.~~

~~We hereby agree to ratify all acts, deeds and things lawfully done by our said Attorney pursuant to this Power of Attorney and that all acts, deeds and things done by our aforesaid Attorney shall and shall always be deemed to have been done by us/JV.~~

~~Dated this the day of.....20~~

~~.....~~

~~(Signature and Name in Block letters of *All the Partners of the firm / * Authorized Signatory for the Company)~~

~~.....~~

~~(* Strike out whichever is not applicable)~~

Seal of firm / Company

Witness 1 _____ Witness 2 _____

Name: _____ Name: _____

Address: _____ Address: _____

Occupation: _____ Occupation: _____

Notes:

- ~~_____ To be executed by all the members individually, in case of a JV.~~
- ~~_____ The mode of execution of the Power of Attorney should be in accordance with the procedure, if any laid down by the applicable law and the charter documents of the executant (s) and when it is so required the same should be under common seal affixed in accordance with the required procedure.~~

FORM OF AGREEMENT

(ON NON-JUDICIAL STAMP PAPER OF APPROPRIATE VALUE)

Agreement No. dated

THIS AGREEMENT is made on day of Two thousand between RITES Ltd. a Government of India Enterprise and a Company registered under Companies Act, 1956 having its registered office at SCOPE Minar, Laxmi Nagar, Delhi - 110092 and its Corporate Office at RITES BHAWAN, Plot No.1, Sector 29, Gurgaon (Haryana)-122001 representing through, RITES LIMITED acting for and on behalf of and as an Agent/ Power of Attorney Holder of.....hereinafter called the Employer (which expression shall, wherever the context so demands or requires, include their successors in office and assigns) on one part and M/s.....hereinafter called the Contractor (which expression shall wherever the context so demands or requires, include his/ their successors and assigns) of the other part.

WHEREAS the Employer is desirous that certain works should be executed viz. (brief description of work) and has by Letter of Acceptance dated accepted a tender submitted by the Contractor for the execution, completion, remedying of any defects there in and maintenance of such works at a total Contract Price of Rs..... (Rupees only)

NOW THIS AGREEMENT WITNESSETH as follows: -

1. In this Agreement words and expressions shall have the same meaning as are respectively assigned to them in the Conditions of Contract hereinafter referred to.
2. The following documents in conjunction with addenda/corrigenda to Tender Documents shall be deemed to form and be read and construed as part of this agreement viz.

The Letter of Acceptance dated

Priced Schedule (Bill) of Quantities

Notice Inviting Tender and Instructions to Tenderers.

Additional Conditions of contract

Scope of Work

Special Conditions of Contract.

Proforma of Schedule (Schedules A to F)

Technical Specifications

Drawings

Topographic Survey Report/Drawings (Indicative only)

Geotechnical Investigation Report (Indicative only)

Design Basis Report

Schedule of Finishes

Amendments to Tender Documents (List enclosed)

General Conditions of Contract for EPC Projects (~~read with Correction Slip Nos. 1 to~~
→) comprising of

- (i) Conditions of Contract
- (ii) Clauses of Contract
- (iii) RITES Safety Code
- (iv) RITES-Model Rules for the protection of Health and Sanitary arrangements for Workers
- (v) RITES – Contractor’s Labour Regulations

3. In consideration of the payment to be made by the Employer to the Contractor as hereinafter mentioned, the Contractor hereby covenants with the Employer to execute, complete, remedy defects therein and maintain the works in conformity in all respects with the provisions of the Contract.

4. The Employer hereby covenants to pay to the Contractor in consideration of the execution, completion, remedying of any defects therein and maintenance of the works, the contract price or such other sum as may become payable under the provisions of the contract at the time and in the manner prescribed by the Contract.

IN WITNESS whereof the parties hereto have caused their respective common seals to be hereinto affixed (or have herewith set their respective hands and seals) the day and year first above written.

SIGNED, SEALED AND DELIVERED BY

.....	
In the capacity of	Representing RITES LIMITED In the capacity of Agent/Power of Attorney Holder
On behalf of M/s	For and on behalf of
(The Contractor)	(The Employer)
In the presence of	In the presence of
Witness (Signature, Name & Designation)	Witness (Signature, Name & Designation)
1.	1.
2.	2.

ANNEXURE VI/1

To

All Bidders for the Subject Tender,

Subject: NIT No. **RITES/NERPO/OIL/BQ-HOUSING/25** for the work of **Construction of Workman Housing Complex(BQ Area) on EPC mode -II of contract at OIL Duliajan, Assam**

Dear Sir,

It is hereby declared that RITES LTD., acting for and on behalf of and as an Agent/Power of Attorney Holder of **OIL INDIA LIMITED** (The Principal/Employer) is committed to follow the principles of transparency, equity and competitiveness in public procurement.

The subject Notice Inviting Tender (NIT) is an invitation to offer made on the condition that the Bidder will sign the integrity Agreement, which is an integral part of tender/bid documents, failing which the tenderer / bidder will stand disqualified from the tendering process and the bid of the bidder would be summarily rejected.

This declaration shall form part and parcel of the Integrity Agreement and signing of the same shall be deemed as acceptance and signing of the Integrity Agreement on behalf of the RITES Ltd..

Yours faithfully

**RAJEN
SINGHA**

Digitally signed by
RAJEN SINGHA

Date: 2025.08.28

12:12:35 +05'30

(Rajen Singha)

AGM/PU-Head Guwahati/NERPO
RITES Ltd

for and on behalf of and as an

Agent/Power of Attorney Holder of **OIL India Limited**

Note- The RITES authority competent to sign the contract agreement, will sign this Annexure before uploading the tender.

INTEGRITY PACT

To,
Project Unit Head,
RITES Ltd., NERPO,
Project Unit- Guwahati,
4th floor, NEDFI House
Ganeshguri, Dispur
Guwahati-781006

**Name of work: Construction of Workman Housing Complex(BQ Area) on EPC mode -II
of contract at OIL Duliajan, Assam.**

Dear Sir,

I/We acknowledge that RITES Ltd. is committed to follow the principles thereof as enumerated in the Integrity Agreement enclosed with the tender/bid document.

I/We agree that the Notice Inviting Tender (**RITES/NERPO/OIL/BQ-HOUSING/25**) is an invitation to offer made on the condition that I/We will sign the enclosed integrity Agreement, which is an integral part of tender documents, failing which I/We will stand disqualified from the tendering process. I/We acknowledge that SUBMISSION OF THE BID SHALL BE REGARDED AS AN UNCONDITIONAL AND ABSOLUTE ACCEPTANCE of this condition of the NIT.

I/We confirm acceptance and compliance with the Integrity Agreement in letter and spirit and further agree that execution of the said Integrity Agreement shall be separate and distinct from the main contract, which will come into existence when tender/bid is finally accepted by RITES Ltd. and/or Client(*Strike out whichever is not applicable*).I/We acknowledge and accept the duration of the Integrity Agreement, which shall be in the line with Section 9 of the enclosed Integrity Agreement at Annexure-VI/3.

I/We acknowledge that in the event of my/our failure to sign and accept the Integrity Agreement, while submitting the tender/bid, RITES Ltd. and/or Client (*Strike out whichever is not applicable*), shall have unqualified, absolute and unfettered right to disqualify the tenderer / bidder and reject the tender/bid in accordance with terms and conditions of the tender/bid.

Yours faithfully

(Duly authorized signatory of the Bidder)

(Official Seal)

Place:.....

Date:

Witness 1:

(Name & Address).....

.....

Witness 2

(Name & Address).....

.....

Date:

*Note-1 The bidder will download this document and affix his signatures and upload such signed document, as explained in the tender document under para 11.2. In case of **Failure to submit the Integrity Pact duly signed and witnessed, along with the Bid, the Bid is likely to be rejected.***

*Note-2 In case of a Joint Venture, authorized signatory of each member/partner will submit this document duly signed and witnessed along with the Bid/Tender All authorized signatories of the members/partners of JY, can either sign on one document jointly or can sign on separate documents. In case of **Failure to submit the Integrity Pact duly signed and witnessed, along with the Bid, the Bid is likely to be rejected.***

INTEGRITY PACT

Between

RITES LTD. acting for and on behalf of and as an Agent/Power of Attorney Holder of **OIL INDIA LIMITED** (Name of Principal/Employer)

(Hereinafter called the "Principal/Employer" and which expression shall unless repugnant to the meaning or context hereof including its successors and permitted assigns)

AND.....

(Name of the Bidder)

(Hereinafter referred to as "The Bidder/Contractor" and which expression shall unless repugnant to the meaning or context hereof include its successors and permitted assigns)

Preamble

The Principal/Employer intends to award, under laid down organizational procedures, contract/s for **Construction of Workman Housing Complex(BQ Area) on EPC mode -II of contract at OIL Duliajan, Assam**. The Principal/Employer values full compliance with all relevant laws and regulations, and economic use of resources, and of fairness and transparency in his relations with the Bidder/s and/or contractor/s.

In order to achieve these goals, the Principal/Employer will appoint Independent External Monitor (s) (IEMs) who will monitor the Tender process and execution of the contract for compliance with the principles mentioned above.

Section 1- Commitments of the Principal/Employer

(1) The Principal/ Employer commits himself to take all measures necessary to prevent corruption and to observe the following principles:-

1. No employee of the Principal/Employer , personally or through family members, will in connection with the tender or for the execution of the contract, demand, take a promise for or accept, for self or third person, any material or immaterial benefit which the person is not legally entitled to.

2. The Principal/ Employer will, during the tender process, treat all Bidders with equity and reason. The Principal/Employer will in particular, before and during the tender process, provide to all Bidders the same information and will not provide to any Bidder confidential/additional information through which the Bidder could obtain an advantage in relation to the tender process or the contract execution.

3. The Principal/Employer will exclude from the process all known prejudiced persons.

(2) If the Principal/Employer obtains information on the conduct of any of his employees which is a criminal offence under the IPC (Indian Penal Code)/PC (Prevention of Corruption) Act, or if there be a substantive suspicion in this regard, the Principal/Employer will inform its Chief

**RAJEN
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Digitally signed by
RAJEN SINGHA

Date: 2025.08.28

12:13:44 +0530

राजन सिंह
अतिरिक्त महाप्रबंधक (सि) और
Addl. General Manager (C) & P.U. Head
राइट्स लि. / RITES LTD.
(भारत सरकार का प्रतिष्ठान) / (A Govt. of India Enterprise)
एनईआरपीओ, गुवाहाटी /NERPO, Guwahati

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Vigilance Officer and in addition can initiate disciplinary action.

Section 2 – Commitments of the Bidder/Contractor

(1) The Bidder/Contractor commits himself to take all measures necessary to prevent corruption. He commits himself to observe the following principles during his participation in the tender process and during the contract execution.

1. The Bidder/ Contractor will not directly or through any other person or firm, offer, promise or give to any of the Principal/Employer 's employees involved in the tender process or the execution of the contract or to any third person any material or other benefit which he is not legally entitled to, in order to obtain in exchange any advantage of any kind whatsoever during the tender process or during the execution of the contract.
2. The Bidder/Contractor will not enter with other Bidders into any undisclosed agreement or understanding, whether formal or informal. This applies in particular to prices, specifications, certifications, subsidiary contracts, submission or non-submission of bids or any other actions, to restrict competitiveness or to introduce cartelization in the bidding process.
3. The Bidder/Contractor will not commit any offence under the relevant IPC/PC Act; further the Bidder/Contractor will not use improperly, for purposes of competition or personal gain, or pass on to others, any information or document provided by the Principal/Employer as part of the business relationship, regarding plans, technical proposals and business details, including information contained or transmitted electronically.
4. The Bidder/Contractor will, when presenting his bid, disclose any and all payments he has made, is committed to or intends to make to agents, brokers or any other intermediaries in connection with the award of the contract.
5. Foreign bidders shall disclose the name and address of agents and representatives in India .
6. Indian Bidders shall disclose their foreign principals or associates.

(2) The Bidder/ Contractor will not instigate third persons to commit offences outlined above or be an accessory to such offences.

Section 3 – Disqualification from tender process and exclusion from future contracts

If the Bidder/Contractor, before award or during execution has committed a transgression through a violation of Section 2 above, or in any other form such as to put his reliability or credibility in question, the Principal/Employer is entitled to disqualify the Bidder/Contractor from the tender process or terminate/ determine the Contract, if already executed, as per clause 3 of clauses of contract or take action as per the procedure mentioned in the "Guidelines on Banning of Business Dealings" placed at Annexure- VIA".

Section 4 – Compensation for Damages

(1) If the Principal/Employer has disqualified in terms of the provisions in Section 3, the Bidder/Contractor from the tender process prior to the award of contract, the Principal/Employer is entitled to demand and recover the damages equivalent to Earnest Money Deposit/Bid Security.

**RAJEN
SINGHA**

Digitally signed by

RAJEN SINGHA

Date: 2025.08.28

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राजेन सिंघा/Rajen Singha
(नि) और प.ई. प्रमुख
आतिरिक्त महाप्रबंधक (नि) और प.ई. प्रमुख
Page 93 of 938

Addl. General Manager (C) & P.U. Head

राइट्स लि. / RITES LTD.

(भारत सरकार का प्रतिष्ठान) / (A Govt. of India Enterprise)
एनईआरपीओ, गुवाहाटी /NERPO, Guwahati

(2) If the Principal/Employer has terminated the contract during execution in terms of the provisions under Section 3, the Principal/Employer shall be entitled to demand and recover from the Contractor the damages equivalent to Earnest Money Deposit, Security Deposits already recovered and Performance Guarantee which shall be absolutely at the disposal of the Principal/Employer.

Section - 5 Previous transgression

(1) The Bidder/Contractor to disclose any transgressions with any other public/government organization that may impinge on the anti-corruption principle. The date of such transgression, for the purpose of disclosure by the bidders in this regard, would be the date on which cognizance of the said transgression was taken by the competent authority. The period for which such transgression(s) is/are to be reported by the bidders shall be the last three years to be reckoned from date of bid submission. The transgression(s), for which cognizance was taken even before the said period of three years, but are pending conclusion, shall also be reported by the bidders.

(2) If the Bidder/Contractor makes incorrect statement on this subject, he can be disqualified from the tender process or action can be taken as per the procedure mentioned in "Guidelines on Banning of Business Dealings".

Section - 6 Equal treatment of all Bidders/Contractors/Sub-Contractors

(1) The Bidder/Contractor undertakes to demand from all partners/sub-contractors (if permitted under the conditions/clauses of the contract) a commitment to act in conformity with this Integrity Pact and to submit it to the Principal/Employer before signing the contract.

(2) The Bidder/Contractor confirms that any violation by any of his partners/sub- contractors to act in conformity with the provisions of this Integrity Pact can be construed as a violation by the Bidder/Contractor himself, leading to possible Termination of Contract in terms of Section 4.

(3) The Principal/Employer will disqualify from the tender process all bidders who do not sign this Pact or violate its provisions.

Section 7- Criminal charges against violating Bidders/Contractors/Sub-Contractors

If the Principal/Employer obtains knowledge of conduct of a Bidder, Contractor or Partners/Sub- Contractor, or of an employee or a representative or an associate of a Bidder, Contractor or Sub- Contractor, which constitutes corruption, or if the Principal/ Employer has substantive suspicion in this regard, the Principal/Employer will inform the same to its Chief Vigilance Officer.

Section -8 Independent External Monitor/Monitors

(1) The Principal/Employer shall appoint competent and credible Independent External Monitor for this Pact. The task of the Monitor is to review independently and objectively, whether and to what extent the parties comply with the obligations under this agreement on receipt of any complaint by them from the bidder(s). The IEMs shall examine all the representations/grievances/complaints received by them from the bidders or their authorized

**RAJEN
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Digitally signed by
RAJEN SINGHA
Date: 2025.08.28
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राजन सिंघा / Rajen Singha
अतिरिक्त महाप्रबंधक (सि) और प्र. ई. प्रमुख
Addl. General Manager (C) & P.U. Head
राइट्स लि. / RITES LTD.
(भारत सरकार का प्रतिष्ठान) / (A Govt. of India Enterprise)
एनईआरपीओ, गुवाहाटी /NERPO, Guwahati

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representative related to any discrimination on account of lack of fair play in modes of procurement and bidding systems, tendering method, eligibility conditions, bid evaluation criteria, commercial terms & conditions, choice of technology/specifications etc.

For ensuring the desired transparency and objectivity in dealing with the complaints arising out of the tendering process, the matter should be examined as far as possible by the full panel of IEMs jointly or by the sole IEM as the case may be, who would look into the records, conduct an examination, and submit their joint recommendations to the Management. In case the full panel is not available due to some unavoidable reasons, the available IEM(s) will conduct examination of the complaints. Consent of the IEM(s), who may not be available, shall be taken on record.

(2) The Monitor is not subject to instructions by the representatives of the parties and will perform his functions neutrally and independently. He will report to the CMD/RITES Ltd.

(3) The Bidder/Contractor accepts that the Monitor has the right of access without restriction to all Project documentation of the Principal/Employer including that provided by the Contractor. The Contractor will also grant the Monitor, upon his request and demonstration of a valid interest, unrestricted and unconditional access to his project documentation. The same is applicable to Partners/Sub-Contractors. The Monitor is under contractual obligation to treat the information and documents of the Bidder/Contractor/Partners/Sub-Contractor with confidentiality.

(4) The Principal/Employer will provide to the Monitor sufficient information about all meetings among the parties related to the Project provided such meetings could have an impact on the contractual relations between the Principal/Employer and the Contractor. The parties offer to the Monitor the option to participate in such meetings.

(5) As soon as the Monitor notices or has reason to believe that violation of the agreement by the Principal/Employer or the Bidder/Contractor, has taken place, he will request the Party concerned to discontinue or take corrective action, or to take any other relevant action. The Monitor can in this regard submit non-binding recommendations. Beyond this, the Monitor has no right to demand from the parties that they act in a specific manner or refrain from action or tolerate action. The role of the Monitor will be advisory and his advice will be non binding on the Organaiztion. However, as IEMs are invariably persons with rich experience who have retired as senior functionaries of the government, their advice would help in proper implementation of the IP.

(6) The Monitor will submit a written report to the CMD/RITES Ltd. at the earliest from the date of reference or intimation to him by the Principal/Employer and should the occasion arise, submit proposal for correcting problematic situations.

(7) The word Monitor would include both singular and plural.

Section – 9 Pact Duration

This pact begins when both parties have legally signed it. It expires for the Contractor when his Security Deposit and Performance Guarantee are released and for all other Tenderers six months after the Contract has been awarded.

**RAJEN
SINGHA**

Digitally signed by
RAJEN SINGHA
Date: 2025.08.28
12:14:39 +05'30'

राजेन सिंघा / Rajen Singha
अतिरिक्त महाप्रबंधक (सि) और पी.ई. प्रमुख
Page 95 of 938
Addl. General Manager (C) & P.U. Head
राइट्स लि. / RITES LTD.
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If any claim is made/lodged during this time the same shall be binding and continue to be valid despite the lapse of this pact specified above, unless it is discharged/determined by CMD/RITES Ltd.

Section 10 Other Provisions

- (1) This agreement is subject to Indian Law. Place of performance and jurisdiction shall be as stated in the Contract Agreement.
 - (2) Changes and supplements as well as termination notices need to be made in writing.
 - (3) If the Bidder/Contractor is a Joint Venture, this agreement must be signed by legally authorized signatory of each Member/Partner of JV. The Memorandum of Understanding /Joint Venture Agreement will incorporate a provision to the effect that all Members of the Joint Venture will comply with the provisions in the Integrity Pact. Any violation of Section 2 above by any of the Partners/Members will be construed as a violation by the Joint Venture leading to possible Termination of Contract in terms of Section 3.
 - (4) Should one or several provisions of this agreement turn out to be invalid, the remainder of this agreement remains valid. In this case, the parties will strive to come to an agreement to their original intentions.
 - (5) A person signing the IP shall not approach courts while representing the matters to IEMs and he/she will await their decision in the matter.
 - (6) In case of sub-contracting, the principal contractor shall be solely responsible for the adherence to the provision of Integrity Pact (IP) by the sub-contractor.
 - (7) In the event of any dispute between Principal/ Employer and the contractor, in case, both the parties are agreeable, they may try to settle dispute through mediation before the panel of IEMs in a time bound manner. However, not more than five meetings shall be held for a particular dispute resolution. The fees/expenses on dispute resolution shall be equally shared by both the parties.
- In case, the dispute remains unresolved even after mediation by the panel of IEMs, the Principal/ Employer and contractor may take further action as per the terms & conditions of the contract.
- (8) The Bidder/Contractor undertakes (as part of Fall Clause, if applicable as per contract) that they have not and will not sell the same material/equipment at prices lower than the bid price.

**RAJEN
SINGHA**

Digitally signed by
RAJEN SINGHA
Date: 2025.08.28
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(Rajen Singha)

AGM/PU-Head Guwahati/NERPO

Rajen Singha /Rajen Singha

RITES Ltd.

अतिरिक्त महाप्रबंधक (सि) और प.ई. प्रमुख
Agent / Power of Attorney Holder

Addl. General Manager (C) & P.U. Head

(For & on behalf of the Principal/Employer)

(भारत (Official Seal) Govt. of India Enterprise)

एनईआरपीओ, गुवाहाटी /NERPO, Guwahati

(For the bidder/Contractor)

(Official Seal)

Place: Guwahati

Date:

Witness 1:

(Name & Address).....
.....

Witness 2

(Name & Address).....
.....

Note- 1 RITES authority competent to sign the contract agreement, will sign this Annexure before uploading the tender. The bidder will download this pre-signed document and affix his signatures and those of witnesses and upload such signed document, as explained in the tender document under para 11.2. **In case of Failure to submit the Integrity Pact duly signed and witnessed, along with the Bid, the Bid is likely to be rejected.**

2. In case of a Joint Venture, authorized signatory of each member/partner will sign the Integrity Pact and submit the same with the Bid/Tender. All authorized signatories of the members/partners of JV can either sign on one document jointly or can sign on separate documents. **In case of Failure to submit the Integrity Pact duly signed and witnessed, along with the Bid, the Bid is likely to be rejected.**

Guidelines on Banning of Business Dealings

1. Introduction

- 1.1 RITES, being a Public Sector Enterprise and ‘State’, within the meaning of Article 12 of Constitution of India, has to ensure preservation of rights enshrined in Chapter III of the Constitution. RITES has also to safeguard its commercial interests. It is not in the interest of RITES to deal with Agencies who commit deception, fraud or other misconduct in the execution of contracts awarded / orders issued to them. In order to ensure compliance with the constitutional mandate, it is incumbent on RITES to observe principles of natural justice before banning the business dealings with any Agency.
- 1.2 Since banning of business dealings involves civil consequences for an Agency concerned, it is incumbent that adequate opportunity of hearing is provided and the explanation, if tendered, is considered before passing any order in this regard keeping in view the facts and circumstances of the case.

2. Scope

- 2.1 The procedure of (i) Suspension and (ii) Banning of Business Dealing with Agencies, has been laid down in these guidelines.
- 2.2 It is clarified that these guidelines do not deal with the decision of the Management not to entertain any particular Agency due to its poor/inadequate performance or for any other reason.
- 2.3 The banning shall be with prospective effect, i.e., future business dealings.

3. Definitions

In these Guidelines, unless the context otherwise requires:

- i) ‘Bidder / Contractor / Supplier’ in the context of these guidelines is indicated as ‘Agency’.
- ii) ‘Competent Authority’ shall mean the Executive Director of the Division.
- iii) ‘First Appellate Authority’ shall mean Director in charge of the Division and
- iv) “Second Appellate Authority” shall mean the CMD, RITES.
- v) ‘Investigating Department’ shall mean any Department, Division or Unit investigating into the conduct of the Agency and shall include the Vigilance Department, Central Bureau of Investigation, the State Police or any other department setup by the Central or State Government having powers to investigate.

- vi) 'Banning Committee' shall mean a Committee constituted for the purpose of these guidelines **by the CMD**. The members of this Committee shall not, at any stage, be connected with the tendering process under reference.

4. Initiation of Banning/Suspension

Action for banning/suspension business dealings with any Agency should be initiated by the department/unit having business dealings with them after noticing the irregularities or misconduct on their part.

5. Suspension of Business Dealings

- 5.1 If the conduct of any Agency dealing with RITES is under investigation by any department, the Competent Authority may consider whether the allegations under investigation are of a serious nature and whether pending investigation, it would be advisable to continue business dealing with the Agency. If the Competent Authority, after consideration of the matter including the recommendation of the Investigating Department/Unit, if any, decides that it would not be in the interest to continue business dealings pending investigation, it may suspend business dealings with the Agency. The order to this effect may indicate a brief of the charges under investigation. The order of such suspension would operate for a period not more than six months and may be communicated to the Agency as also to the Investigating Department.

The Investigating Department/Unit may ensure that their investigation is completed and whole process of final order is over within such period.

- 5.2 As far as possible, the existing contract(s) with the Agency may be continued unless the Competent Authority, having regard to the circumstances of the case, decides otherwise.
- 5.3 If the Agency concerned asks for detailed reasons of suspension, the Agency may be informed that its conduct is under investigation. It is not necessary to enter into correspondence or argument with the Agency at this stage.
- 5.4 It is not necessary to give any show-cause notice or personal hearing to the Agency before issuing the order of suspension. However, if investigations are not complete in six months' time, the Competent Authority may extend the period of suspension by another three months, during which period the investigations must be completed.

6. Grounds on which Banning of Business Dealings can be initiated

- 6.1 If the security consideration, including questions of loyalty of the Agency to the State, so warrants;
- 6.2 If the Director/ Owner of the Agency, proprietor or partner of the firm, is convicted by a Court of Law for offences involving moral turpitude in relation to its business dealings with the Government or any other public sector enterprises or RITES, during the last five years;

- 6.3 If there is strong justification for believing that the Directors, Proprietors, Partners, owner of the Agency have been guilty of malpractices such as bribery, corruption, fraud, substitution of tenders, interpolations, etc.;
- 6.4 If the Agency employs a public servant dismissed/removed or employs a person convicted for an offence involving corruption or abetment of such offence;
- 6.5 If business dealings with the Agency have been banned by the Govt. or any other public sector enterprise;
- 6.6 In case any information/document which may result in the tenderer's disqualification is concealed by the Tenderer or any statement/ information/ document furnished by the Tenderer or issued by a Bank/ Agency/ third party and submitted by the tenderer, is subsequently found to be false or fraudulent or repudiated by the said Bank/Agency/Third Party.
- 6.7 If the Agency uses intimidation/ threatening or brings undue outside pressure on the Company (RITES) or its official in acceptance/performances of the job under the contract;
- 6.8 If the Agency indulges in repeated and/or deliberate use of delay tactics in complying with contractual stipulations;
- 6.9 Based on the findings of the investigation report of CBI/ Police against the Agency for malafide /unlawful acts or improper conduct on his part in matters relating to the Company (RITES) or even otherwise;
- 6.10 Established litigant nature of the Agency to derive undue benefit;
- 6.11 Continued poor performance of the Agency in several contracts;

(Note: The examples given above are only illustrative and not exhaustive. The Competent Authority *may decide to ban business dealing for any good and sufficient reason*).

7. Banning of Business Dealings

- 7.1 A decision to ban business dealings with any Agency shall apply throughout the Company.
- 7.2 If the Competent Authority is prima-facie of view that action for banning business dealings with the Agency is called for, a show-cause notice may be issued to the Agency as per paragraph 8.1 and an enquiry held accordingly.

8. Show-cause Notice

- 8.1 In case where the Competent Authority decides that action against an Agency is called for, a show-cause notice has to be issued to the Agency. Statement containing the

imputation of misconduct or mis-behavior may be appended to the show-cause notice and the Agency should be asked to submit within 30 days a written statement in its defense. If no reply is received, the decision may be taken ex-parte.

8.2 If the Agency requests for inspection of any relevant document in possession of RITES, necessary facility for inspection of documents may be provided.

8.3 On receipt of the reply of the Agency, or in case no reply is received within the prescribed time, the Competent Authority shall refer the case along with relevant details to the Banning Committee, which shall examine the reply of the Agency and other facts and circumstances of the case and submit its final recommendation to the Competent Authority for banning or otherwise. In case the action contemplated against the Agency includes forfeiture of EMD also besides Banning of Business Dealings, the Banning Committee will also examine whether Clause 9.4 of Tender and Contract Document is attracted and recommend forfeiture or otherwise of EMD considering all facts and circumstances of the case. A final decision on forfeiture or otherwise of the EMD and for Company-wide banning or otherwise shall be taken by the Competent Authority. The Competent Authority may consider and pass an appropriate speaking order:

- a) For completely exonerating the Agency; or
- b) For forfeiture of EMD but for not banning of business dealings with the Agency; or
- c) For forfeiture of EMD and banning the business dealing with the Agency.

8.4 The decision should be communicated to the Agency concerned along with a reasoned order. If it decided to ban business dealings, the period for which the ban would be operative may be mentioned.

9. Appeal against the Decision of the Competent Authority

9.1 The Agency may file an appeal against the order of the Competent Authority banning business dealing, etc. to the First Appellate Authority.

9.2 In the event the Agency is dissatisfied with the decision of the First Appellate Authority, it shall file a second appeal to the Second Appellate Authority.

9.3 The Appeals shall be preferred within one month from the receipt of the order of the Competent Authority or First Appellate Authority, as the case may be.

9.4 Appellate Authorities would consider the appeal and pass appropriate order which shall be communicated to the Agency as well to the Competent Authority.

10. Review of the Decision by the Competent Authority

Any petition/application filed by the Agency concerning the review of the banning order passed originally by Competent Authority under the existing guidelines either before or after filing of appeal before the Appellate Authority or after disposal of appeal by the

Appellate Authority, the review petition can be decided by the Competent Authority upon disclosure of new facts/circumstances or subsequent development necessitating such review.

11. Circulation of the names of Agencies with whom Business Dealings have been banned.

11.1 Depending upon the gravity of misconduct established, the Competent Authority of RITES may circulate the names of Agency with whom business dealings have been banned, to the Ministry of Railways and PSUs of Railways, for such action as they deem appropriate.

11.2 If Ministry of Railways or a Public Sector Undertaking of Railways request for more information about the Agency with whom business dealings have been banned a copy of the report of Inquiring Authority together with a copy of the order of the Competent Authority/Appellate Authority may be supplied.

12. Restoration

12.1 The validity of the banning order shall be for a specific time & on expiry of the same, the banning order shall be considered as "withdrawn".

12.2 In case any agency applies for restoration of business prior to the expiry of the ban order, depending upon merits of each case, the Competent Authority which had passed the original banning orders may consider revocation of order of suspension of business/lifting the ban on business dealings at an appropriate time. Copies of the restoration orders shall be sent to all those offices where copies of Ban Orders had been sent.

13. Power of Superintendence

CMD, RITES shall have overall power to take suo-moto action on any information available or received by him and pass such order(s) as he may think appropriate, including modifying the order(s) passed by any authority under these Guidelines.

MANDATE FORM

To

RITES Ltd.

.....

.....

Authorization for payments through Electronic Fund Transfer System (RTGS/NEFT)

Whereby authorize RITES Ltd. to make all our payments, including fund of Earnest Money, through Electronic Fund Transfer System (RTGS/NEFT). The details for facilitating the payments are given below:

(TO BE FILLED IN CAPITAL LETTERS)

1	NAME OF THE BENEFICIARY	
2	ADDRESS WITH PIN CODE	
3	(A) TELEPHONE NO. WITH STD CODE	
	(B) MOBILE NO.	
4	BANK PARTICULARS	
A	BANK NAME	
B	BANK TELEPHONE NO. WITH STD CODE	
C	BRANCH ADDRESS WITH PIN CODE	
D	BANK FAX NO. WITH STD CODE	
E	11 CHARACTER IFSC CODE OF THE BANK (EITHER ENCLOSE A CANCELLED CHEQUE OR OBTAIN BANK CERTIFICATE AS APPENDED)	
F	BANK ACCOUNT NUMBER AS APPEARING IN THE CHEQUE BOOK	
G	BANK ACCOUNT TYPE (TICK ONE)	SAVING/CURRENT/ LOAN CASH/ CREDIT/ OTHERS
H	IF OTHERS, SPECIFY	
I	PERMANENT ACCOUNT NUMBER (PAN)	
J	E-MAIL ADDRESS	

I / We hereby declare that the particulars given above are correct and complete. If the transaction is delayed or credit it not effected at all for reasons of incomplete or incorrect information, I / We would not hold RITES Ltd. responsible. Bank charges for such transfer will be borne by us.

Date:

Signature

Name:

(Authorized Signatory)

BANK CERTIFICATION

It is certified that the above-mentioned beneficiary holds bank account No with our branch and the Bank particulars above are correct.

Date:

SIGNATURE
(AUTHORISED SIGNATURE
Name.....
.....
OFFICIAL STAMP

PRE-QUALIFICATION PROFORMA

Name of work	Construction of Workman Housing Complex(BQ Area) on EPC mode -II of contract at OIL Duliajan, Assam				
Tender No.	RITES/NERPO/OIL/BQ-HOUSING/25 Dated:28.08.2025				
Name of the Bidder					
Details of Cost of Tender Document paid by Account Payee Banker's Cheque /Pay Order/ Demand Draft in favour of "RITES Ltd."	Name & Address of Issuing Bank	Amount	Date of Issue	Instrument placed at	Remarks
Details of EMD paid by Insurance Surety Bonds or Account Payee Banker's Cheque/ Pay Order/ Demand Draft/ Bank Guarantee (Indemnity e-Bank Guarantee) in favour of "RITES Ltd."					

Annual Financial Turnover

S. No.	Financial Years (The last three Financial Years Refer Annexure-I on Qualification Criteria)	Turn Over (Rs. In Lacs)	Documents placed at:	Remarks
1				
2				
3				
	Average Annual Turnover during last 3 financial years			

Profitability:

S. No.	Financial Years (The last three Financial Years Refer Annexure-I)	Profit with (+) sign or Loss with (-) sign (In Rs)	Document placed at:	Remarks
1				
2				
3				

Similar Work Experience:

(a) For Completed works:

S. No.	Name of work	Name of Client and address	Agreement/ Letter of Award No. and date	Actual Date of Start	Actual Date of Completion	Actual Completion Cost (Rs)	Completion Certificates placed at:	Remarks
1								
2								
3								
4								

Notes: In support of having completed above works, attach self-attested copies of the completion certificate from the owner/client or Executing Agency/Consultant appointed by Owner/Client indicating the name of work, the description of work done by the Bidder, date of start, date of completion (contractual & actual) and contract value as awarded and as executed by the Bidder.

(b) For substantially completed works

S. No.	Name of Work	Name & address of client	Agreement/ Letter of Award No. & date	Actual Date of start	Actual Cost of work completed(Rs)	Present Contract value(Rs)	% work completed	Status of functionality (whether functional or non-functional)	Whether any proceedings of termination of contract have been initiated. (Yes/No)	Substantial completion certificate placed at	Remarks
1											
2											
3											

Notes: Certificate for substantial completion of project/work/asset should contain three parts.
 Part-I shall contain financial value of work done, PVC amount and present contract value. Part-II shall contain certificate of functional completion of project/work/asset. Part-III shall contain that no proceedings of termination of contract on contractor's default has been initiated.

Construction Experience in Key Activities (if applicable)

S. No.	Key Activity/ Component	Quantity of Key Activity/ Component Executed	Amount of Key Activity/ Component Executed	Name of work	Name of Client	Actual date of Start of Key Activity	Actual date of Completion of Key Activity	Certificate placed at:	Remarks
1									
2									
3									

Net Worth:

S. No.	Financial Years	Net Worth (in Rs.)	Documents placed at:	Remarks
1	(Last Financial Year refer			

	Annexure I on Qualification criteria)			
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Other Documents to be submitted along with Tender Documents:

S. No.	Particulars	Documents placed at:	Remarks
1	Declaration by the Bidder in the format given in Proforma 1 of Annexure I, including certificates mentioned therein. In case of JV each Partner of JV should submit Proforma 1 duly signed, dated & with seal separately.		
2	Self-attested copy of Partnership Deed/Memorandum and Articles of Association of the Firm		
3	Self-attested copy of Written Power of Attorney of the signatory of the Tender on behalf of the tenderer. “(Annexure-III or (Annexure-III & Annexure IV) as applicable)”		
4	Self-attested copy of valid registration with EPFO under EPF and Miscellaneous Provisions Act, 1952.		
5	Integrity Pact as per Annexure-VI/1, VI/2 and VI/3 as mentioned in clause 11.2		
6	Self-attested copy of Guidelines on Banning of Business Dealings as per Annexure-VI A		
7	Self-attested copy of Corrigendum/Minutes of Pre-Bid Meeting, if any.		
8	RTGS/NEFT details as per Annexure-VII (Mandate form)		
9	Self-attested copy of GST Registration Certificate		
10	Self attested copy of the JV agreement/ MoU as per Annexure II (where JV is allowed) to be executed by each member of JV individually.		
11	Power of attorney in favour of Lead member as the Lead Member of JV (where JV is allowed) as per Annexure IV to be executed by all the members of JV individually.		

S. No.	Particulars	Documents placed at:	Remarks
12	Self-attested copy of certificates as per requirement of Sub – clause 6 (verification of Local content) of clause 12.15 (Preference to Make in India)		
13	Self-attested documents as per clause 12.16 (Restrictions on procurement from a bidder of a country which shares a land border with India), if applicable.		
14	Declaration by Each Partner of JV (in case of Large works/ Normal works) as per Para 3 (Servicing of Loan/ Credit Limit) of Annexure-I		
15	Undertaking/ Declaration by Bidder/ Each Partner of JV in case of Large works/ Lead Member of JV in case of Normal works that “Audited standalone Balance Sheets, Statement of Profit and Loss and other financial statements for the immediately preceding year are not available” as per Para 1 (e) (Annual Financial Turnover), Para 4 (Profitability) and Para 5 (Net Worth) of Annexure-I, if applicable.		
16	Self-attested copy of any other document if specified in tender document and correction slips to tender document		

List of 60 Districts covered under IAP

S. No.	State	District
1	Andhra Pradesh	Adilabad
2	Andhra Pradesh	Khammam
3	Bihar	Arwal
4	Bihar	Aurangabad
5	Bihar	Gaya
6	Bihar	Jamui
7	Bihar	Jehanabad
8	Bihar	Nawada
9	Bihar	Rohtas
10	Chhatisgarh	Bastar
11	Chhatisgarh	Bijapur
12	Chhatisgarh	Dantewada
13	Chhatisgarh	Jashpur
14	Chhatisgarh	Kanker
15	Chhatisgarh	Kawardha
16	Chhatisgarh	Koriya
17	Chhatisgarh	Narayanpur
18	Chhatisgarh	Rajnandgaon
19	Chhatisgarh	Surguja
20	Jharkhand	Bokaro
21	Jharkhand	Chatra
22	Jharkhand	Garhwa
23	Jharkhand	Gumla
24	Jharkhand	Hazaribagh
25	Jharkhand	Kodarma
26	Jharkhand	Latehar
27	Jharkhand	Lohardaga
28	Jharkhand	PachimSinghbhum
29	Jharkhand	Palamu
30	Jharkhand	PurbiSinghbhum

S. No.	State	District
31	Jharkhand	Ram Garh
32	Jharkhand	Saraikela
33	Jharkhand	Simdega
34	Madhya Pradesh	Anuppur
35	Madhya Pradesh	Balaghat
36	Madhya Pradesh	Dindori
37	Madhya Pradesh	Mandla
38	Madhya Pradesh	Seoni
39	Madhya Pradesh	Shahdol
40	Madhya Pradesh	Sidhi
41	Madhya Pradesh	Umaria
42	Maharashtra	Gadchiroli
43	Maharashtra	Gondiya
44	Orissa	Balangir
45	Orissa	Debagarh / Deogarh
46	Orissa	Gajapati
47	Orissa	Kalahandi
48	Orissa	Kandhamal / Phulbani
49	Orissa	Kendujhar / Keonjhar
50	Orissa	Koraput
51	Orissa	Malkangiri
52	Orissa	Mayurbhanj
53	Orissa	Nabarangapur
54	Orissa	Nuapada
55	Orissa	Rayagada
56	Orissa	Sambalpur
57	Orissa	Sonapur
58	Orissa	Sundargarh
59	Uttar Pradesh	Sonbhadra
60	West Bengal	Paschim Medinipur

PROFORMA FOR BANK GUARANTEE FOR EARNEST MONEY DEPOSIT (EMD)

(On Non-Judicial Stamp Paper of Appropriate Value)

To,

RITES LTD.

1. WHEREAS RITES Ltd. Acting for and on behalf as Agent/Power of Attorney Holder of OIL India Limited(hereinafter called 'The Employer') (which expression shall unless repugnant to the subject or context include its successors and assigns) has invited bids by way of E-tender Number- **RITES/NERPO/OIL/BQ-HOUSING/25 Dated:28.08.2025** (hereinafter called 'the Tender') for **Construction of Workman Housing Complex (BQ Area) on EPC mode -II of contract at OIL Duliajan, Assam** (*description of the tender work*);
2. AND WHEREAS (*name of the bidder*) (herein after called 'the Bidder') has submitted a bid for the Tender to the Employer.
3. AND WHEREAS a Bank Guarantee for Rs..... (Rupees only) (*Amount of EMD*) in favour of RITES Ltd (hereinafter called the "Beneficiary") valid till (*the date of validity of this Guarantee from the date of the submission of Tender's offer*) is required to be submitted by the Bidder as EMD along with the Tender.
4. We (*Name of the bank*) having our office at (*Address of the bank*) (hereinafter called 'the Bank') hereby give this Bank Guarantee and hereby agree unequivocally and unconditionally to pay to RITES Ltd. immediately on demand and without demur to the extent of the said sum of Rs. (Rupees only) for which payment well and truly to be made to the said Beneficiary, the Bank binds itself, its successors and assigns by these presents on default of any of the conditions of the Tender as follows:
 - i. if the Bidder after bid opening, but before expiry of bid validity or issue of Letter of Acceptance, whichever is earlier,
 - I withdraws his tender; or
 - II makes any modification in the terms and conditions of the tender which are not acceptable to the Employer
 - III impairs or derogates from the tender in any respect within the period of validity

of the tender

IV. If the bidder does not accept the correction of his price during evaluation

- ii. in case any information/ document which may result in the Bidder's disqualification is concealed by the Bidder or any statement/information/document furnished by the Bidder or issued by a Bank/Agency/Third Party and submitted by the Bidder, is subsequently found to be false or fraudulent or repudiated by the said Bank/Agency/Third Party.
 - iii. in the case of a successful Bidder, if the Bidder fails to furnish the Performance Guarantee within the period specified under Clause 1 of "Clauses of Contract".
5. We undertake to pay to the Beneficiary up to the above amount upon receipt of its first written demand, without the Beneficiary having to substantiate its demand, provided that in its demand the Beneficiary will note that the amount claimed by it is due to its owing to the occurrence of one or more of the conditions mentioned above, specifying the occurred condition or conditions.
6. NOT WITHSTANDING ANY THING CONTAINED HEREIN:
- I. Our liability under this Bank Guarantee shall not exceed Rs.(Rupees..... only)
 - II. This Bank Guarantee shall be valid upto.....
 - III. It is a condition of our liability for payment of the guaranteed amount or any part thereof arising under this Bank Guarantee that we receive a valid written claim or demand for payment under this Bank Guarantee on or before.....failing which our liability under the guarantee will automatically cease.

Seal:

(Authorized Signatory of the Bank)

Date:

Note:

- I. There should be adequate gap, preferably 06 months, in the Expiry/ Validity date and claim Period. In case, a bank is not agreeing for claim period, the validity of Bank Guarantee should be for the period as per the provision of Tender Document plus claim period of minimum three (03) months.
- II. In case of any cutting or overwriting on the Bank Guarantee, it should be properly authenticated under the signature and seal of the officer (s) who has/have signed the Bank Guarantee.

STATE / UNION TERRITORY IN WHICH GST REGISTRATION OF RITES

S. No	State	Regular GST Reg. No. (Final)
1	Jammu & Kashmir	01AAACR0830Q1ZK
2	Punjab	03AAACR0830Q1ZG
3	Haryana	06AAACR0830Q1ZA
4	Delhi	07AAACR0830Q1Z8
5	Rajasthan	08AAACR0830Q1Z6
6	Uttar Pradesh	09AAACR0830Q1Z4
7	Bihar	10AAACR0830Q1ZL
8	Tripura	16AAACR0830Q1Z9
9	Meghalaya	17AAACR0830Q1Z7
10	Assam	18AAACR0830Q1Z5
11	West Bengal	19AAACR0830Q1Z3
12	Jharkhand	20AAACR0830Q1ZK
13	Orissa	21AAACR0830Q2ZH
14	Chattisgarh	22AAACR0830Q1ZG
15	Madhya Pradesh	23AAACR0830Q1ZE
16	Gujarat	24AAACR0830Q1ZC
17	Maharashtra	27AAACR0830Q1Z6

STATE / UNION TERRITORY IN WHICH GST REGISTRATION OF RITES

18	Karnataka	29AAACR0830Q1Z2
19	Goa	30AAACR0830Q1ZJ
20	Kerala	32AAACR0830Q1ZF
21	Tamilnadu	33AAACR0830Q1ZD
22	Andaman & Nicobar Islands	35AAACR0830Q1Z9
23	Telengana	36AAACR0830Q2Z6
24	Andhra Pradesh	37AAACR0830Q1Z5

**DECLARATION ABOUT SITE INSPECTION
(On firms Letter Head)**

To
RITES Ltd.
for and on behalf of and as an
Agent/Power of Attorney Holder of
OIL INDIA LIMITED

Name of Work: “Construction of Workman Housing Complex(BQ Area) on EPC mode -II of contract at OIL Duliajan, Assam”

Tender No. RITES/NERPO/OIL/BQ-HOUSING/25 Dated:28.08.2025

Dear Sir,

It is hereby declared that as per Clause 13 of Section 1 (Notice Inviting Tender and Instructions to Tenderers) and conditions of this tender document, I/We, the bidder inspected and examined the subject site and its surrounding and satisfy myself / ourselves as to the nature of the ground and sub-soil (so far as is practicable), the forms and nature of the site before submitting the bid, the accommodation which may require and all necessary information as to risks, contingencies and other circumstances which may influence or affect our bid have been obtained. I/We, the bidder shall have full knowledge of the site and no extra charge consequent upon any misunderstanding or otherwise shall be claimed on later date. I/We, bidder shall be responsible for arranging and maintaining at own cost all materials, tools & plants, water, electricity access, facilities for workers and all other services required for executing the work unless otherwise specifically provided for in the contract documents. Submission of a bid by me/us implies that I /We, have read this notice and all other contract documents and has made myself /ourselves aware of the scope and specifications of the work to be done and local conditions and other factors having a bearing on the execution of the work.

.....

SEAL, SIGNATURE & NAME OF BIDDER

SECTION 2

ADDITIONAL CONDITIONS OF CONTRACT

SECTION 2A

SCOPE OF WORK

Scope of Work

1. The scope of work shall include (but not be limited to) the following works

- 1.1. The scope of work has been elaborated here broadly but this shall be read with details given elsewhere in this document and drawings attached herewith the tender documents.
- 1.2. The project involves the **Design and Construction** of the following components for the **OIL India Limited BQ Housing Complex at Duliajan**, to be executed on **EPC Mode-II** basis and the entire complex shall be constructed using RCC framed structures:

Buildings:

- a) 08 Residential Towers (Stilt + 6 Floors) - having total 192 Nos. dwelling unit
- b) Guest House (G+3)
- c) Community Centre (G+1)
- d) Sub-station & Panel Room/Electrical Sub-station (Single Storey)
- e) Guard Rooms (02 Nos., Single Storey, including Control Room/CCTV Surveillance Room at main gate)
- f) Security Hut (Single Storey)

Infrastructure and External Development Works:

- a) Levelling and raising of original ground levels
- b) Boundary Wall
- c) Cantilever-type Parking Sheds
- d) Internal roads and on-street parking (with and without sheds)
- e) External road works including connectivity from Entry/Exit-2 to the adjacent colony road near House No. TD+44 (TD+ Housing) and from Entry/Exit-1 to the existing public road.
- f) Footpaths, culverts, signage installation etc.

Utilities and Services including installation, testing & commissioning:

- a. Pump Room & Underground Water Tank (UGWT)
- b. Water supply system
- c. Storm water drainage
- d. Sewerage system
- e. STP with Treated underground water tank and pump room for horticulture and flushing.
- f. Gas supply system as per PNGRB
- g. HVAC/VRV/VRF systems
- h. Firefighting and Fire Alarm System & PA systems.
- i. Mechanical Ventilations & Fire Pressurization system.
- j. LV and ELV Services.
- k. Solar PV Plant.
- l. Elevators (Lifts)

- m. Internal and external MEP services etc.
- n. CCTV Surveillance system.

Green Development:

- a) Horticulture and Landscaping etc.

Note: The Electrical Sub-Station and Panel Room is to be constructed as a single-storey structure under the current scope of work, as detailed in the tender document. While executing the construction, due consideration must be given to ensure adequate space provision for the installation of electromechanical equipment—such as transformers, DG sets, electrical panels, etc.—as required for the DX Quarters in future. The layout and spatial planning should align with the tender drawings and accommodate future equipment placement as indicated therein.

- 1.3. The details of building/Infrastructure to be constructed at OIL Duliajan are as follows:

S. No.	Buildings /Infrastructures	Tentative Plinth Area (Sqm)	Floor/Remarks
1	Residential Blocks (Total 08 Tower)	27355	Stilt+6
2	Guest House	1990	G+3
3	Community Centre	1035	G+1
4	Sub Station & Panel Room/Electrical Sub-station	368	Single Storey
5	External Development Works including other ancillary infrastructure (Guard rooms, Security hut, UGWT & Pump Room, STP etc.)	24,675 Sqm Ground area	As per approved drawings

Note:

1. The details of the buildings/Infrastructures, as tabulated above, are tentative. The building names, configurations, combinations, and phasing may be modified during the planning stage to comply with statutory norms, approvals from statutory authorities, and requirements arising from RITES/OIL India Limited, structural and MEP designs, or GRIHA compliance. However, such modifications shall not entail any change in the overall project cost, and no additional payment shall be admissible to the EPC Contractor on this account. The rates quoted by the EPC Contractor shall be deemed inclusive of all such changes and contingencies.
- 1.4. A Defects Liability Period (DLP) of 24 months will commence from the date of successful completion and handing over of the entire project to OIL India Limited. The defect liability period includes operation & maintenance of electromechanical

equipments. The EPC Contractor shall deploy qualified personnel and provide necessary spares/tools during this period. A reporting mechanism and minimum service levels should be defined to ensure uninterrupted functioning of all systems.

- 1.5. The scope of work given is not exhaustive however, EPC Contractor is required to execute all the items as per Scope of work, Schedule of finishes, approved drawings, Design basis reports (DBR), Technical specifications, Architectural Tender Drawings, Special Conditions of Contract, other parts of Tender documents etc. covering the entire area in all respect to make it fit for its intended purpose. Work/Finishes shall be carried out as per the schedule of finishes, in coordination with the overall scope as per directions of Engineer-in-charge.
- 1.6. The EPC Contractor shall prepare Good for Construction (GFC) drawings for all structural and MEP services, which shall be duly vetted/proof-checked by IIT and approved by RITES/OIL India Limited prior to execution. In the event that, during the correlation of drawings and specifications, any minor or major components are found to be missing, the EPC Contractor shall be solely responsible for preparing the necessary additional designs/drawings, at no additional cost. Additionally, all required shop drawings shall also be prepared by the EPC Contractor and approved by RITES/OIL India Limited prior to execution.
- 1.7. The EPC Contractor shall prepare all coordinated drawings for all services and make all necessary co-ordination for satisfactory execution of all works/activities in time bound manner. The Contractor will appoint competent consultants having adequate experience in planning and design of Residential Buildings. The consultants will develop detail design and drawings based on the requirements specified in the tender document.
- 1.8. A model, 3D view and walkthrough will be prepared & presented by EPC Contractor to RITES & Client within 15 days from award of work as per conceptual plan provided in tender. The latest software of design engineering including ETABS/STAAD PRO. (as desired by Engineer-in-charge), AutoCAD etc. will be used for design & Engg. purpose. All softcopies shall be made available to RITES as required.
- 1.9. The EPC Contractor shall provide all relevant drawings in AutoCAD/Revit/PDF formats, as applicable, along with necessary inputs required for the development of Building Information Modelling (BIM) for various buildings and services. These drawings shall include detailed layout plans, sectional elevations, and other required service details. The EPC Contractor shall develop integrated/coordinated As-Built drawings using BIM tools and Revit software, after thoroughly identifying and resolving all clashes/conflicts among various services. This is to ensure smooth execution, installation, and future maintenance of the systems. No extra payment shall be admissible on this account. In addition, the EPC Contractor shall provide licenses (valid until completion of the project) each as mentioned for the following softwares:

One License:

- a) Revit
- b) ETABS / STAAD Pro (as desired by Engineer-in-charge)
- c) AutoCAD
- d) Navisworks

Two License:

- e) Primavera

The EPC Contractor shall ensure that valid licenses for the following software—**Revit, ETABS / STAAD Pro, AutoCAD, Navisworks, and Primavera**—are made available to **RITES/OIL India Limited** for use in connection with the project.

In the event of any delay in providing the required software licenses, a recovery of **Rs. 10,000/- (Rupees Ten thousand only) per month per software license** shall be imposed for each month of default, until such time the required licenses are made available and operational for project use.

- 1.10. The EPC Contractor shall ensure that the design consultant/architect appointed by them is stationed at site during execution of work.
- 1.11. The EPC Contractor shall construct the building as per National Building Code 2016 to ensure that the building is accessible for people with disability.
- 1.12. The work shall be carried out in accordance with true intent and meaning of the specifications, drawings etc. read together, regardless of whether the same may or may not be particularly shown on the drawings and/or described in the specifications, provided that the same can be reasonably inferred there from.
- 1.13. There may be several incidental works, which may not be mentioned explicitly in the contract document/specifications but are necessary to complete the item in all respect. All such incidental works shall be executed as per sound engineering practice and the direction of the RITES/OIL India Limited and shall be deemed to be included in the overall amount quoted by the EPC Contractor. Also no adjustment of rates shall be made on account of variation/change in actual working drawings which may result in change in incidental works or any other deviation in such element of work (which is incidental to the items of work and are necessary to complete such items in all respects) on account of the directions of RITES/OIL India Limited.
- 1.14. In case any item or its specifications are not explicitly mentioned in the bid documents, such items shall be executed in accordance with the latest CPWD Specifications, NBC-2016, IRC/Morth, relevant BIS Codes, Manufacturer's Specifications, or as per sound engineering practices, to ensure the complete functionality of the building and associated services. No claims whatsoever shall be entertained on this account. Notwithstanding any provisions in CPWD or any other specifications regarding items payable separately

or additionally, no extra payment shall be admissible. All such items/works shall be deemed included in the scope of work and considered subsumed within the quoted rates.

- 1.15. The EPC Contractor shall be fully responsible for obtaining all required statutory, municipal, and mandatory approvals/NOCs (including for tree cutting) necessary for the successful execution of the work. This includes approvals required before commencement, during execution, and after completion of the project, such as completion certificates, service connections, and occupation permissions. All such activities shall be carried out within the quoted price, and no additional payment shall be admissible on this account.

- 1.16. The EPC Contractor shall obtain all post construction statutory clearances, Occupancy Certificate, Consent to Operate (CTO) & other applicable approvals including service connections such as water supply, drainage, power supply, PNG Gas etc.

The EPC Contractor shall obtain all statutory approvals necessary for completion of all E&M works like Electrical Equipments installed in electrical substation, pollution, DG, Oil Storage, Elevators (Lifts), Fire safety certificate etc.

- 1.17. Statutory fee paid by the EPC Contractor to statutory authorities shall be reimbursed upon submission of all receipts/satisfactory evidence of having submitted the same to the statutory authorities. Necessary documents as and when required by Statutory Authority for obtaining statutory approvals shall be provided by RITES/OIL India Limited on written intimation by EPC contractor.

Deposits (including caution deposits, fee cost of pipelines/cables/panels, inspection charge and any other infrastructure charges) remitted by the EPC Contractor(s) towards getting new/additional water supply / power supply etc. connections shall be reimbursed upon submission of documentary evidence by the EPC Contractor(s). EPC Contractor is also required to liaise with external agency/ authority/OIL India Limited in connection with shifting of hidden services pipelines/cables/ power cable etc. wherever required.

- 1.18. The EPC Contractor is responsible to take periodic consent/ approvals from State pollution control department or Ministry of Environment for construction/ establishment etc. of any construction equipment, machinery or any other equipment during construction/ post construction/ two-year DLP period.
- 1.19. The cost of any dismantling & disposal (for all leads & lifts) of old plinth/ foundation/existing structures/fencing, etc. to such place where the same shall not cause nuisance and should be acceptable to the authorities concerned is deemed to be included in the overall amount quoted by the EPC Contractor. Similarly, removal/shifting of malba, Construction waste, existing dead/live utility services, cables, poles etc. as per the work requirement to the suitable locations. This is to be done in accordance with the instructions of and with prior approval from

RITES/OIL, and no extra payment shall be admissible on this account.

- 1.20. The EPC Contractor shall obtain all applicable permissions related to tree cutting and/or transplanting of trees, from the concerned authority of the State Government/OIL India Limited. The location for tree plantation/transplantation and number of trees to be planted /transplanted shall be as per instruction issued by RITES/OIL India Limited. The execution of cutting/removal and transplanting the trees as per issued permission or any other action including disposal of felled trees (to such place where the same shall not cause nuisance and should be acceptable to the authorities concerned) will be taken by the EPC Contractor without any extra cost.
- 1.21. The EPC Contractor is required to obtain all necessary statutory approvals and connect all the external services like Water Supply, Sewerage, storm water Drainage, Rain water harvesting, STP, Electric Supply, Substation, Street lighting, Elevators (Lifts), LAN, Telephone Lines, gas pipe lines etc. to the main lines of the authorities/service providers or any other agency, also connection of solar grid to main grid/any other services required for effective use of building and commissioning and this shall be considered as integral part of Scope of work and deemed to be included in the quoted price of the EPC Contractor.
- 1.22. The EPC Contractor shall ensure Project Management on BIM (Building Information Management) platform level 3.0 or higher/ M.S projects/ Primavera and will ensure training for smooth completion of the project. The EPC Contractor shall ensure availability of weekly reports on deployment of resources and progress.
- 1.23. CPWD specifications, relevant IS codes, National Building Code 2016, CPWD green Manual 2021, GHIRA Rating standards specification shall be followed in general except otherwise mentioned in bid document. Manufacture specifications shall be followed in case of item for which provision in above not available.
- 1.24. The EPC contractor shall Study the physical conditions of site for planning the work.
- 1.25. The Hardscape, pavement/ paths,landscape design drawings are also to be prepared and the EPC contractor shall do the required design & detailing for execution. All the detailed or shop drawings prepared by the EPC contractor shall be reviewed and approved by Engineer in charge or his authorized representative. However, it does not absolve EPC Contractor from design & code provisions/ statutory requirements. EPC Contractor shall prepare and get all the Shop drawings approved before execution. The EPC Contractor is required to submit all its submittals like Drawings, Documents, Reports, Schedules etc. (whether original or revised) in 4 (four) sets of Hard copies along with soft copy in (pen drive, Cloud storage, email etc) for review/approval/information of RITES/OIL India Limited.
- 1.26. All MEP services and Electro-Mechanical Equipments are to be executed by the agency according to technical & performance requirements elaborated in the document. The Technical Specifications, Drawings and detailed requirements for

each function viz., Architectural, Civil, Electrical & LV, Plumbing Sanitation & Fire Fighting, HVAC, External Development, External Facade, Elevators (Lifts), etc. are part of the work, elaborated in the document.

- 1.27. In case any modification in Architectural Design/ Civil/Structural design/ MEP drawing is needed as per requirement of statutory approvals/site conditions/Employer requirement, the agency shall do/redo the drawing without any extra cost as well as suggest solutions to the problems coming across during actual execution. The decision of the Engineer-in-charge or his authorized representative shall be final and binding. No compensation claim whatsoever will be entertained in this regard.
- 1.28. EPC contractor shall get the Design Mix of Concrete approved from any IITs/ NITs/ Govt. Engineering College/NCCBM. The same is to be got approved from engineer –in – charge or his representative before execution of work. In case of change of source/ quarry of material (as per approved design mix), the same shall be approved by Engineer In-charge or his authorized representative and revised Design Mix shall be approved from IIT's/ NIT's/ Govt. Engineering College/NCCBM.
- 1.29. EPC contractor shall also undertake supply, construction/ installation/ laying, testing, commissioning, trial run of all the required services such as internal & external Electrical, HVAC, Plumbing/ sanitary, PNG Gas, Networking, CCTV, firefighting. Scope shall also include rainwater harvesting system, External Development, roads with parking, pathways, culverts, gardens/lawn etc. The services mentioned herein are indicative and any other necessary facilities required for successful and functional occupation/operation of the campus is deemed to be inclusive of the scope.
- 1.30. Furniture Works (except for one sample unit, wardrobe & kitchen furniture of Residential Blocks as per tender document) are excluded from the scope of Work. Supply, Installation & Commissioning of Commercial Kitchen equipments for Guest House is included in the scope as per tender document.
- 1.31. The Agency is required to connect all the external services viz., Water Supply, Sewerage, Drainage, Electric Supply, LAN/WAN, Telephone Lines etc. to the main lines of the authorities/service providers/Municipality or any other agency and this shall be considered as integral part of Scope of work of EPC contractor.
- 1.32. The finalization of micro details of architectural items/ materials such as color scheme, Surface Finish (Matt, Glossy etc.) etc. shall be as per approval of Engineer-in-charge or his authorized representative at the time of execution.
- 1.33. EPC contractor shall make efforts and utilize practices/ processes that ensure minimum generation of waste during construction stage. The EPC contractor shall designate specific areas as per directions of Engineer-in-charge or his authorized representative at construction site for segregated collection of recycled materials as per statutory rules/ laws.

- 1.34. Minimum thickness of floors / finishes / pavements etc. required as per applicable codes, specifications & standards have to be maintained and the effect of any floor drops, terrace slopes, gradient in pavements is deemed to be included in the amount quoted by EPC contractor.
- 1.35. EPC contractor shall also undertake any minor modifications arising out of co-ordination/ interference of various services/ facilities during execution.
- 1.36. The EPC contractor shall undertake timely procurement of all the materials, equipment, Installations, accessories as required for timely completion and Installation & Commissioning of work for the project.
- 1.37. The EPC contractor shall get all Mock-up Sample & Finishes approved from the Engineer in charge or his authorized representative, OIL India Limited and Architect & set up a sample unit at the site for Samples & Mock-up.
- 1.38. The EPC Contractor shall ensure full compliance with all requirements necessary to achieve a minimum “Three Star” GRIHA rating, as per applicable GRIHA Guidelines. All related activities including necessary registration of work/project on behalf of RITES/OIL India Limited, analysis, documentation, coordination, submission, assistance etc. for obtaining this certification shall be carried out by the EPC Contractor, and no additional payment shall be admissible on this account. However, the necessary statutory fee for registration shall be reimbursed by RITES/OIL India Limited. It is the responsibility of contractor to get the final GRIHA certification for the project.

The EPC Contractor shall also provide necessary information in the form of BOQ, GCC, and Drawings to facilitate evaluation of whether the provisions contained in the tender document are adequate to meet the requirements for a “Three Star” GRIHA rating.

In case it is found that the existing provisions are inadequate, the EPC Contractor shall, at their own cost, undertake all necessary revisions, corrections, or modifications in methodology, materials, or any other aspect of the project to ensure compliance, without claiming any extra payment.

- 1.39. EPC contractor shall note that following requirements of GRIHA 3 star rating also needs to be implemented and the quoted rates of the EPC contractor are inclusive of the same.
 - Develop and implement spill prevention plan and erosion & sedimentation control plan.
 - Segregation & proper record keeping of “disposal of all construction waste materials like rocks, empty cement bags, plastic containers, scrap steel, sanitary waste” etc.
 - Proper housekeeping at site.
 - Ensuring efficient use of water during Construction.

- Controlling air pollution during construction and ensuring dust suppression through suitable water sprinkling, covering of construction materials.
 - Implementation of HSE (Health, Safety and Environment) procedures & policies effectively.
 - Ensuring proper and secure staging/scaffolding during construction.
 - Effective implementation of labour welfare measures/policies including well-ventilated and hygienic labour camps with adequate lighting, water supply & sanitary arrangements.
 - Submission of Monthly report detailing compliance to the GRIHA norms and attaching the data and photographs.
 - The EPC contractor shall deploy an experienced & qualified person exclusively for implementation of GRIHA LD 3 star rating requirements till the completion of work.
- 1.40. All the designs, drawings, documents etc. developed for the project will become the property of RITES/OIL India Limited. The drawing cannot be issued to any other person, firm or authority or used by the EPC contractor for any other project. No copies of any drawing or document shall be issued to anyone except RITES/OIL India Limited.
- 1.41. EPC contractor to follow the latest ECBC code, Harmonized guidelines & standards for universal accessibility in India 2021
- 1.42. Soil and ground water samples will be tested to ascertain that there is no threat to ground water quality by leaching of heavy metals and other toxic contaminants.
- 1.43. Top fertile soil to be preserved and to be later used in landscape.
- 1.44. The excavation/demolition debris must be disposed off in designated landfill areas or to be used within site for levelling purpose as per work requirement. Under no circumstance, the debris will be disposed in river bed/lakes etc.
- 1.45. Undertaking to be given that occupancy will be given only after drainage and water connections are in place.
- 1.46. Dust/smoke prevention measures such as wheel washing, water sprinkler, screening, barricading and debris chute must be installed.
- 1.47. The natural flow of existing storm water channel should not be altered or diverted.
- 1.48. Keeping in view the use of large quantities of water in curing, measures for reducing water demand during construction should be followed. Curing water should be sprayed on concrete structures; free flow of water should not be allowed for curing. After liberal curing on the first day, all concrete structures should be painted with curing chemical to save water Concrete structures should be covered with thick cloth/gunny bags and then water should be sprayed on them. This would avoid water rebound and will ensure sustained and complete curing. Ponds should be made using cement and sand mortar to avoid water flowing away from the flat surface while

curing.

- 1.49. The EPC contractor should ensure groundwater and municipal water meet the water quality norms as prescribed in the Indian Standards for various applications (Indian Standards for drinking [IS 10500-1991], irrigation applications [IS 11624-1986]).
- 1.50. The use of potable water during construction should be minimized.
- 1.51. Solid waste management to be taken care during construction phase should be:
 - Disposal of muck during construction phase should not create any adverse effect on the neighbouring communities and be disposed taking the necessary precautions for general safety and health aspects of people, only in approved sites with the approval of competent authority The Rules on the Solid Waste Management including Construction Waste issued by the MoEFCC as amended will be applicable.
 - Construction spoils, including bituminous material and other hazardous materials, must not be allowed to contaminate watercourses and the dump sites for such material must be secured so that they should not leach into the ground water.
 - Any hazardous waste generated during construction phase, should be disposed off as per applicable rules and norms with necessary approvals of the State Pollution Control Board.
 - Miscellaneous site debris such as broken tiles etc. shall be used on site for levelling /backfilling purpose.
 - Packaged STP /mobile toilets shall be provided for labour camp.
 - Polymer bags used for cement and gypsum shall be handed over to authorized recyclers.
 - Cardboard boxes and other packaging material will be handed over to authorized recyclers.
- 1.52. Air quality & noise levels to be taken care during construction phase should be:
 - The diesel required for operating DG sets shall be stored in underground tanks and clearance from Chief Controller of Explosives shall be taken, as applicable.
 - Ambient noise levels should conform to residential standards both during day and night as per Noise Pollution (Control and Regulation) Rules, 2000. Incremental pollution loads on the ambient air and noise quality should be closely monitored during construction phase. Adequate measures should be made to reduce ambient air and noise level during construction phase, so as to conform to the stipulated standards by CPCBI SPCB.

- Burning of waste to be banned.
 - The construction site DG to be maintained regularly so that the smoke emission and noise levels are as per permissible norms.
 - Regular P.U.C check for all construction machinery coming on site be done.
 - Noise cancellation and insulation devices such as mufflers, barricades etc to be used to avoid noise propagation to adjoining areas.
- 1.53. Air quality & noise levels to be taken care during construction & operation phase :
- The provisions of the Air (Prevention and Control of Pollution) Act, 1981 (14 of 1981), &, any latest Act/Code/ Direction shall be complied for control of noise pollution during construction and operation.
 - Setting up the barriers: National Building Code 2016 suggests that design solutions such as barrier blocks should be used to reduce external noise levels to at least 60-70 dB (A) at any point 10m from any inward-looking facade. Green belts and landscaping could act as an effective means to control noise pollution. In case of railway tracks, a minimum distance of 50m to 70m may be provided between the buildings and the tracks.
- 1.54. **Energy criteria to be taken care:**
- Appropriate processes and material be used to encourage reduction in carbon foot print.
 - Fly ash should be used as building material in the construction as per the provisions of Fly Ash Notification of September, 1999 and amended as on 27th August, 2003 and 3rd November, 2009.
 - Use of light coloured, reflective roofs having an SRI (solar reflectance index) of 50% or more should be promoted. The dark coloured, traditional roofing finishes have SRI varying from 5% to 20%.
- 1.55. Test commissioning of all electrical and mechanical systems by the respective supplier/OEM or EPC contractor to be done before handover.
- 1.56. **The scope of work also include (but not be limited to) the following works in relation to buildings/infrastructure mentioned in tender document:**
- a) **Detailed Survey:** EPC Contractor shall conduct detailed survey with respect to nearest available permanent benchmark, where proposed building and all other services like roads, paths, cycle track, drains, and water bodies are located /passes put. However, indicative topographic survey & trees details of the concerned areas where proposed building and roads and other development features/items are to be located is attached with bidding documents for pre knowledge of topography of the

site. EPC Contractor has also to verify authenticity of attached topographic survey himself along with missing/inaccurate details of existing infrastructure/features (over the ground/underground). No claim for extra /additional payments owing to/about the change in site condition shall be made entertained in future.

- i. The EPC Contractor shall make appropriate arrangements for storm water drainage during the execution of work to prevent water accumulation and ensure site safety.
 - ii. It shall be responsibility of the EPC Contractor to make proper approach road to the site for smooth operation.
- b) **Geotechnical Investigation:** Soil investigation report will be provided by RITES to EPC Contractor for reference purpose only. However, the EPC Contractor must carry out their own detailed geotechnical investigation, prepare the topographical contour map, and conduct soil testing at their own cost to satisfy themselves to meet design requirements. The uploaded report is only indicative, and no claims will be accepted based on it at any stage. EPC Contractors must also test the soil for harmful chemicals through qualified consultants and select suitable cement for foundations based on the findings to ensure protection against chemical reactions.
- c) Any required fee pertaining to any kind of Designing, Drawing, Soil investigation/ Geotechnical survey, Soil resistivity test, Proof checking, vetting, Design Mix, engaging any consultant or specialized agency, testing of material etc. will be borne by the EPC Contractor.
- d) Consultancy Services i.e., Architectural, MEP, HVAC/VRV/VRF, Interior, landscape planning and Engineering & Consultancy work etc. as per BIS/NBC/IPHS/NABH/GRIHA with latest correction slip etc.
- e) All Civil works such as execution and completion of Carriage of material, Earth work, Concrete work, Reinforcement, RCC, AAC Block work, Brick work, Stone work, Marble & granite work, Wood work, PVC work, Steel work including structural steel, railings, fire check doors, Flooring, Roofing, false ceiling, invisible grill, Structural glazing, Water proofing & Anti Termite Treatment, Wooden, aluminium, uPVC works, fittings, fixtures, Internal and external finishing works, etc.. Certain columns, fins, ribs, projections, ornamentations, elevation elements as required in RCC need to be finished smooth without any honeycombing.
- f) Fire Fighting and Fire detection and alarm works.
- g) Elevators (Lifts)
- h) Internal electrification & External electrification.
- i) Electrical Services, HT Panel, Transformers, HT/LT distribution system, DG sets, rising mains, UPS, Cable tray/raceways, aviation obstruction lights, Earthing, Lightning arrester, Bus ducts etc.

- j) VRV system for Community Centre & split AC system for Guest House.
- k) Mechanical ventilation system
- l) Fire Fighting system, Fire Alarm system & PA system.
- m) Low voltage systems (CCTV / LAN etc.) facilities, telephone for Guest House & community centre.
- n) Grid connected roof top Solar system
- o) Gas Piping system as per PNGRB.
- p) Decentralized power metering of power and air conditioning.
- q) Underground water tank with pump house.
- r) Sewage Treatment Plant including recycled water line and organic waste composter.
- s) External Lighting, facade lighting, Landscaping Lighting, Hybrid solar street light.
- t) Development of site: external services connections, Existing ground level and plinth level shall be adequately raised to prevent potential inundation during incessant rains (reference levels provided in tender drawings), Internal Roads, External roads for connectivity to Entry/Exits, pathways, footpaths, road/parking marking, culverts for road/footpath crossing over existing drains/nallahs etc., Boundary Walls, Parking sheds, Water supply (domestic, flushing, garden irrigation, fire-fighting), Plumbing, Sewer and storm water Drainage Works for both internal and external areas, STP, horticulture & landscape elements, internal & external signages, sign board.
- u) **Miscellaneous Works:**
 - i. Day light and Movement sensors-based lighting as per GRIHA.
 - ii. PA System
 - iii. Outdoor & Landscape Lighting inclusive of solar hybrid & normal street lights as per tender drawings/DBR and specifications.
 - iv. Façade Lightening
 - v. Signage Systems – all signages in all floors (Residential, Guest House & Community Centre) & External areas as per approved design and drawing by client/ RITES.
 - vi. Grid connected Solar Panels and Power generation system
 - vii. Security System Provision of CCTV Cameras in all staircases and corridors, common areas, entry/ exit, reception, entrance door, all other areas not

specifically mentioned in here, with all active and passive works complete.

- viii. Server Room Design with interior and MEP services.
- ix. Networking, EPABX, IT Data cabling, Optical Fibre Cables
- x. Entry and exit gate complex with manual swing doors, wicket gate and guard room
- xi. Boom Barriers at entry & exit gates
- xii. LED Glass/ Stainless Steel/ Acrylic Signages, the scope of work for the EPC Contractor would be to design non-digital and digital signage system to achieve an effective way finding scheme both for external and internal areas, as well as, for all common areas like Lobbies, Elevator (Lift) Lobbies, corridors, washrooms, service rooms etc
- xiii. Solar Hybrid Street lighting backed up by regular power to be provided along the road length as per directions of engineer in-charge. Landscaping & Horticulture works.
- v) Site Development & Landscape (including Irrigation lines of treated water, Horticulture, hardscape, softscape, landscaped terraces etc., landscaped roads, pathways, CC kerb stone.
 - i. All roads and parking area should be as per GRIHA norms complete as per specifications and directions of Engineer-In charge.
 - ii. The roads should be designed to carry light/car traffic and largest fire tender vehicle load as per IRC and applicable codes & manual.
 - iii. Reputed landscape consultant should be hired for all landscape works.
 - iv. Vehicular and Pedestrian Traffic circulation to be followed as per tender document.
- w) Interior fit out work in all floors as per tender specification and finishing schedule.
- x) Raceways wherever required in floors to accommodate all types of cables.

1.57. Other Works: Apart from above mentioned works following works are also in scope of work.

- a) EPC Contractor shall prepare and submit shop drawings for plumbing, drainage, sewerage, MS grills/railing/ladder/gates, Invisible Grill, SS railing, Wooden/Aluminium/UPVC windows, all types of doors frames and shutters, structural steel work, aluminum work, structure glazing work, etc. and detailed layout drawings including 3-D views for all civil and electrical services (duly checked and verified by suitable software), duly supported with design as required by engineer in charge.

- b) EPC Contractor shall submit item wise detailed measurements (RA & final bill) for record and other purpose as per the CPWD specification and mode of measurement as per direction of Engineer in charge.
- 1.58. Testing commissioning of all services, fittings and fixtures, equipment's and handing over.
- 1.59. **Completeness of work:**
- a) All Civil and E&M installations shall be completed in all respects and put in to operation including & commissioning.
 - b) All Civil and E&M works and services shall be declared as completed by RITES/OIL India Limited after completion of trail run of 1 month or date of completion and handing over of whole/ full work under the contract agreement, whichever is later.
- 1.60. EPC Contractor shall be responsible for successful execution of performance specifications like leakage /seepage free, structural defect free and full functioning of various fittings/ fixtures etc. commensurate to prestigious of National Importance. Penalties shall be recovered to the full cost of repair / replacement of the same in case of violation.
- 1.61. On completion of work, a project Completion Report (PCR) shall be submitted by EPC Contractor duly bringing out the Final Project Completion Cost, Total Time period taken to complete the work and also completed the Project Components as against awarded cost, Time and project Components. The Project Completion Report (PCR) shall be submitted along with Final Project Accounts within one month of settlement of final bill.
- 1.62. Before procurement of material, EPC Contractor shall submit data sheets and catalogues of reputed manufacturers of the equipment for approval of Engineer-in-charge.
- 1.63. The responsibility of Investigations Procurement, Construction, Safety, Quality, and Risk of Engineering lies with the EPC Contractor.
- 1.64. **EPC Contractor's Warranty of Design**
- a) The EPC Contractor shall be fully responsible, for the suitability, adequacy, integrity, durability, and practicality of the EPC Contractor's proposal in respect of design of building.
 - b) The EPC Contractor warrants that the Works have been or will be designed, manufactured, installed, and otherwise constructed and to the highest standards available using proven up-to-date good practice. By submitting the Drawings for review to the Engineer-in-charge, the EPC Contractor shall be deemed to have represented that it has determined and verified that the design and engineering,

including field construction criteria related thereto, are in conformity with the Scope of the Project, the CPWD Specifications and IS Standards and the Applicable Laws.

- c) The EPC Contractor warrants that the EPC Contractor's Proposals meet the requirements and is fit for the purpose thereof. Where there is any inadequacy, insufficiency, impracticality, or unsuitability in or of the Requirements or any part thereof, the EPC Contractor's Proposal shall consider, address or rectify such inadequacy, insufficiency, impracticality or unsuitability at EPC Contractor's own cost.
 - d) The EPC Contractor warrants that the Works will, when completed, comply with enactments and regulations relevant to the Works.
 - e) The EPC Contractor must get the vetted and approved structural design drawings, MEP drawings by IIT and shall also provide a structural stability certificate duly vetted from the Designer for the design for suitability, adequacy, and practicality of design for RITES/OIL India Limited requirements.
 - f) The EPC Contractor shall indemnify the RITES/OIL India Limited against any damage, expense, liability, loss, or claim, which the Employer might incur, sustain or be subject to arising from any breach of the EPC Contractor's design responsibility and/or warranty set out in this Clause and bid document.
 - g) Notwithstanding that such design may be or have been prepared, developed, or issued by the RITES, any of EPC Contractor's consultants, his sub-EPC Contractors and/or his qualified personnel/persons or cause to be prepared, developed or issued by others.
 - h) Notwithstanding any warranties, guaranties and/or indemnities that may be or may have been submitted by any other person.
 - i) Notwithstanding that the same have been accepted by the RITES/OIL India Limited.
 - j) The EPC Contractor shall be fully responsible for the Plants, Materials, goods, workmanship, preparing, developing, and coordinating all design Works to enable that part of the Works to be constructed and/or to be fully operational in accordance with the Contract's/bid document requirements.
 - k) No claim for additional payment or extension of time shall be entertained and/or no review and/or observation of the Engineer-in-charge and/or its failure to review and/or convey its observations on any Drawings shall relieve the EPC Contractor of its obligations and liabilities under this Agreement in any manner nor shall the Engineer-in-charge or the Employer be liable for the same in any manner; and if errors, omissions, ambiguities, inconsistencies, inadequacies or other Defects are found in the Drawings, they and the construction works shall be corrected at the EPC Contractor's cost, notwithstanding any review under this section.
- 1.65. The EPC Contractor shall get specifications of every item approved by OIL India Limited/RITES before commencement of work.
- 1.66. **Testing of materials:**

The EPC Contractor shall produce all the materials well in advance so that there is sufficient time for testing of the materials and clearance of the same before incorporation in the work.

Samples of all materials required for testing is included in the cost of work. Similarly, expenditure for testing in house or through outside laboratory approved by the Engineer-in- Charge shall be borne by the EPC Contractor. These laboratories may be in the Government sector, Semi Government or private sector NABL accredited laboratories.

All expenditures related to the factory inspection of materials including eletromechanical equipments at the manufacturer's premises— including Third Party agency inspection and testing charges, testing of all samples proposed to be used in construction, as well as associated activities such as packing, sealing, transportation, loading, and unloading—shall be borne entirely by the EPC Contractor at his own cost. No additional payment shall be made on this account under any circumstances.

All physical & chemical test shall be carried out as applicable for the materials as approved/required by Engineer-in-charge. Chemical analysis/tests for aggregates, water etc. shall also be carried out as per direction of Engineer-in-charge. Frequency of testing of material will be as per frequency laid down in CPWD Specification upto latest correction slip and as specified in agreement or as per the latest IS codes, IRC/Morth specifications. The cost of sample collection & submission of sample (materials) and the charges of testing of all materials shall be borne by the EPC Contractor and nothing extra shall be payable to EPC Contractor.

If any load testing or special testing is to be done for any sample whose strength is doubtful, the cost of the same shall also be borne by the EPC Contractor. All the hidden items such as water supply lines, drainage pipes, conduits, sewers, gas pipeline, fire fighting etc. are to be properly tested as per the design conditions/functionality before covering. Water tanks, taps, sanitary, water supply and drainage pipes, fittings and accessories should confirm to by laws and municipal body / corporation where CPWD Specifications are not available. The EPC Contractor should engage licensed plumbers for the work and get the materials (fixtures/fittings) tested by the Municipal Body/Corporation authorities wherever required at his own cost. The EPC Contractor shall give performance test of the entire installation(s) as per the standing technical specifications/approved drawings before the work is finally accepted and nothing extra whatsoever shall be payable to the EPC Contractor for the test.

The EPC Contractor shall arrange electricity at his own cost for testing of the various electrical installations as directed by RITES/OIL India Limited and for the consumption by the EPC Contractor for executing the work. Also, all the water required for testing various electrical installations, fire pumps, wet riser / firefighting equipment, fire sprinklers etc. and also testing water supply, sanitary and drainage lines, water proofing of underground sump, overhead tanks, water proofing treatment etc. shall be arranged by the EPC Contractor at his own cost. Nothing extra shall be payable on this account.

In case any test, as directed by RITES/OIL India Limited, cannot be conducted at the

site laboratory established by the EPC Contractor due to non-availability of required facilities, such tests shall be carried out at an approved Govt./NABL Accredited external laboratory. The entire cost of such tests, including materials, sampling, transportation, and testing charges, shall be borne by the EPC Contractor, and no additional payment shall be admissible on this account.

In case there is any discrepancy in frequency of testing as given in tender or BIS Codes and as shown in the list of mandatory tests and that in individual sub-heads of work as per CPWD Specifications, 2019 (Vol. I to Vol. II) with upto date correction slip the higher of the two frequencies of testing shall be followed, and nothing extra shall be payable to the EPC Contractor on this account.

EPC Contractor will prepare a quality assurance plan mentioning the list of mandatory testing and get it approved from the RITES/OIL India Limited.

- 1.67. It shall be deemed that the EPC Contractor has satisfied himself as to the nature and location of the work, general and local conditions and particularly those pertaining to transport including restriction of movement of traffic/ vehicles etc., handling, availability and storage of materials, availability of labour, weather conditions at site and general ground/sub soil conditions and the EPC Contractor has to quote his rates accordingly and nothing extra shall be payable on any reason whatsoever.
- 1.68. The EPC Contractor shall submit a fortnightly progress report of the work, including relevant photographs and videos, highlighting any delays in relation to the overall construction schedule, as well as identifying lapses or deviations in progress. The report shall also include support in devising remedial measures, preparation of revised schedules, and assessment of revised resource requirements, as applicable. This report is to be submitted to the Engineer-in-Charge.

In addition, a detailed Monthly Progress and Program Report shall be submitted to RITES/OIL India Limited by the 5th of every month, containing geo-tagged site photographs taken from vantage points to clearly reflect the progress of work. The format for this monthly report shall be as approved by the Engineer-in-Charge.

The EPC Contractor must provide enough CCTV cameras at site with LCD TVs in the site office/ control for monitoring of progress of work as per the direction of Engineer In-charge within the quoted cost.

- 1.69. All drawings/specifications/makes, shop drawings and construction methodology etc. shall be approved from RITES and their consultant prior to execution/ procurement. However, it does not imply that EPC Contractor absolves themselves from codal provisions/statutory requirements.
- 1.70. In the event of any discrepancies or inconsistencies among the Design Basis Report (DBR), Scope of Work, Technical Specifications, Drawings, CPWD Specifications, IS Codes, or other relevant standards, the most stringent specification shall prevail and be deemed applicable.

1.71. The facilities indicated on each floor are illustrative and may be subject to interchange, addition, deletion, or variation in size within the overall building envelope, as per the conceptual design on decision of Engineer-in-charge. Internal partitions and associated services may also be modified based on functional requirements as per the directions of the OIL India Limited/RITES.

1.72. The accommodation of interior elements, sanitary fixtures, and electrical fixtures shown in the tender drawings is indicative in nature. The make, design, and size of all such fixtures shall be subject to prior approval by RITES/OIL India Limited before procurement and installation.

1.73. Water Supply and Electricity

Arrangement of water for drinking purpose in addition to the water required for construction work is also to be arranged by the EPC Contractor at his own cost. The EPC Contractor shall also make his own arrangement for obtaining electric connection (s) during construction phase and make necessary payment directly to the department concerned. RITES will however make all reasonable recommendations to the authority concerned in this regard. The EPC Contractor shall make his own arrangement of water for completion of work and nothing will be paid on this account. The EPC Contractor shall get the water tested with regard to its suitability and conforming to the relevant IS Code. The EPC Contractor shall obtain written approval from the RITES/OIL India Limited before he proceeds by using the same for execution of work.

1.74. Bailing-Out Water, Shoring etc.

For execution of any items of work where bailing out water, shoring etc. are actually required but not specifically stated in the scope of item and/or tender document, it is to be understood that the contract amount quoted by the EPC Contractor shall cover such charges also and nothing extra on account of such charges, if any, shall be paid.

1.75. Mobilization of Men, Materials and Machinery

The Preparation of design/drawings & bill of quantities (of his stated scope of work) etc.; Bar Bending Schedule, Shop & Fabrication drawings for all work shall be supplied by the EPC Contractor as per agreed schedule/ as the work progresses.

However, it shall be the duty and responsibility of the EPC Contractor to bring to the notice of the RITES in writing as to any variation, discrepancy or any other changes required in Drawing, design etc. whether of RITES's Scope or EPC Contractor's Scope of work and to obtain/prepare (as the case may be) revised drawings and designs and / or approval of the RITES in writing for the same.

1.76. Quality Assurance Plan & Checklist for E&M Services

The decision on testing shall be as per direction of E-in-C or as per E&M quality checklist of CPWD vide OM no. 51(4)/CE(E) /CSQ/2016/293(H) dated 31/03/2016 as

applicable (available at CPWD website under publication section) shall be binding on EPC contractor.

1.77. Statutory Charges During DLP period

During the Defect Liability Period, all statutory charges incurred, such as water bills, electricity bills, NOC charges, etc., shall be borne and paid by the Client, i.e., OIL India Limited.

1.78. Running Diesel Generator Oil During DLP Period

The cost of consumable oil for the diesel generators shall be borne by the Client, i.e., OIL India Limited during DLP Period. However, the EPC contractor shall be responsible for informing the Client each time the generators are operated and must submit a monthly operational analysis report to the Client.

1.79. Repair during DLP

All consumables such as spare parts, defective equipment/component etc. during the DLP period shall be repaired or replaced by the EPC contractor at no additional cost.

If the EPC contractor fails to repair or replace the defect in the given time or satisfaction of Engineer In-charge, the same shall be rectified at the risk and cost of the EPC contractor, along with additional 20% administrative charges.

2. STRUCTURAL WORKS

2.1. The Scope of work of the EPC Contractor shall be as under: -

- a) Detail design/preparation of drawings based on DBR. Design of buildings shall be as per NBC 2016 for all loads and combinations and complying with IS 16700: 2023
- b) EPC Contractor shall be responsible for design coordination and technical support during the project within the finalized design.
- c) Preparation of Structural & Shop drawings and approval of Structural Design along with RCC design - Proof checking, Vetting of design/drawings from the IIT.
- d) Supply, fabrication, Construction, Installation and Commissioning of Structural Member.
- e) Obtaining all interim and completion Approval from applicable local body.

2.2. Further, details (in brief) are as under: -

- a) The detailed Civil/Structural Design/preparation of Drawings for above mentioned buildings /spaces are within the Scope of the EPC Contractor.
- b) The concept design (Architectural), Structural Design Philosophy of above buildings/spaces, structural framing is provided in tender documents based on which the EPC Contractor has to prepare detailed Civil/Structural/MEP drawings of each, and every aspect of the project required for construction of

buildings in all respect. The Structural Design & drawings shall be submitted to RITES for prior approval before execution. The EPC Contractor shall get the Structural Design & Drawings proof checked from IIT. The fee payable for proof checking shall be borne by EPC Contractor.

- c) The EPC Contractor shall not be absolved of their responsibility of structural stability and correctness of structural design as per latest applicable codes & practices and shall provide a structural safety certificate to RITES.
- d) Based on the DBR & the architectural drawings, the agency shall submit the structural design and drawings based on the Design Philosophy given in the DBR/Technical Specification, within the time as stipulated in Table of Milestone in Schedule-F of tender document. Every Civil/Structural drawing /detail shall be prepared conforming to detailed technical specification, CPWD specifications and relevant IS code. All dimensions of various structural members / slabs / columns / elements shall be gotten proof checked.
- e) The Civil/Structural Design & Drawings (whether RCC or Structural Steel or composite) expressly (but not limited to) includes the following-
 - i. Studying the Architectural Concept Design, Structural Design Philosophy, submitting proposed structural design framework of each building/floor/area and obtaining approval of RITES. Due care to be taken for integrating the structural drawing with the Architectural Drawings & with all MEP Services, Landscape features & Elevation features etc.
 - ii. Design on appropriate software like ETABS / STAAD Pro (as desired by Engineer-in-charge) and drawing on a software like Auto CAD including fabrication drawings, shop drawings, bar bending schedule etc. of each component of buildings/spaces within Scope of work on suitable scale (6 no. of sets of each drawing), including but not limited to: -
 - Construction of culverts.
 - Cutting and filling of land as per site conditions.
 - Sheet piling if required to safeguard existing trees.
 - Foundation Drawings/ Details & Schedule, whether raft or strip footing and/or combination of footings /Pile footing.
 - Retaining walls/ shear walls.
 - Ground Drawings with columns with capitals, slabs, flat slabs as per requirement.
 - Column Drawings/Schedule with capital and slabs, flat slabs as per requirement.
 - Suspended floors, roofs, landings, balconies, and access platform.
 - Lintels, beams, plinth beams, girders, bressummers and cantilevers.
 - Columns, Pillars, Piers, Abutments, Posts and Struts.
 - Stairs (including landings).
 - Curved floor plates in plan
 - Shafts.

- Vertical and horizontal fins individually or forming box louvers band, facias and eaves boards.
- All façade design elements in pre-cast/cast-in-situ RCC, GRC or FRP.
- All structures in landscape/building such as projection, pergolas, trellis, porch, fencing, boundary wall etc.
- Centring & shuttering using aluminium form work / 34 kg ply/new metal shuttering etc. with all due fitments as necessary and as per requirements.
- All steel structure including 2 hr. fire protection using fire paint wherever required.
- Exposed concrete work at locations as per design (to be replaced with texture paint finish, in case the EPC Contractor fails to achieve required finish in exposed concrete).
- Execution of complete work as per detail scope of work, technical specification, and drawings.
- The Structural system of the project shall be designed by the EPC Contractor in accordance with the Indian Code IS: 456 for concrete and IS 800 for steel structure with latest amendments, which is considered for concrete/steel structures respectively in the country.

3. PLUMBING & SANITARY WORKS

3.1 The Scope of work of the EPC Contractor shall be as under: -

- a) Detailed Designing/preparation of Shop drawings as per the DBR and its approval from the RITES/OIL India Limited.
- b) Supply, fabrication, Construction, Installation and Commissioning.

3.2 Further, details (in brief) are as under: -

- a) The Plumbing/Sanitary Design & Schematic/Indicative/Line drawings are included in Scope of work. Detailed fabrication & shop drawings, as required for execution at site are to be prepared by the EPC Contractor and got approved from RITES and their consultant before proceeding with the work.
- b) The preparation of Sanitary/Plumbing fabrication & shop drawings expressly (but not limited to) includes the following:
 - i) Integration of detailed shop drawings with Civil/Structural details prepared by EPC Contractor for each building and obtaining approval of RITES.
 - ii) Preparation of fabrication & shop drawings on a software like AutoCAD, etc. of each part of building having a sanitary/plumbing aspect within Scope of work on suitable scale (6 no. of sets of each drawing).
- c) The Execution of Plumbing/Sanitary work for buildings /spaces as per Scope of the work to be read with Technical Specifications and Drawings. The Sleeves, cut out Drainage arrangement, embedment, concealed

- pipings, hot & cold-water arrangement, overhead tanks, water recycling system etc. and connecting the service to the main line of OIL India Limited are expressly included (but not limited to) in the scope of work.
- d) There will be a city sewer connection, but all the sewerage generated is proposed to be fully treated within the project at the STP & wastewater is recycled.
 - e) The rainwater is proposed to be harvested & emphasis should be to create a zero-discharge project.
 - f) The toilets in all floors of all blocks are to be fitted & finished including all plumbing & sanitary fixtures & fitting as mentioned in tender document.
 - g) Toilets for drivers, security & maintenance staff are also be built & finished in the Ground as per requirements as mentioned in finishing schedule.
 - h) Design, installation, testing, commissioning, and handing over of STP, wastewater recycling plant etc. as per DBR, technical specification and drawings is included in EPC Contractor's scope of works.
 - i) Drainage system with automatic sump pumps to be designed /executed of Grounds and whole site.
 - j) 1 main pump and 1 standby pumps will be installed for drainage at open to sky road in slope area with power backup.
 - k) Calculate the requirements and develop the network of water supply along with planning & design of piped network system including storage connecting from Source(s) to take over point including metering.
 - l) From takeover point in campus for onward distribution network. Detailed design & layout as per approved concept plan of External & Internal Services within the Housing complex site.
 - m) Plan standby source of supply and underground water tank, etc.
 - n) Installation of Smart Water meters as per requirements of GRIHA.

4. FIRE FIGHTING

4.1 The Scope of work of the EPC Contractor shall be as under: -

- a) Detailed Designing/preparation of Shop drawings as per the DBR and its approval from RITES.
- b) Supply, fabrication, Construction, Installation and Commissioning.
- c) Obtaining approval of Fire department / OIL India Limited and any other relevant authority.

4.2 Further, details (in brief) are as under:-

- a) Detailed Design & engineering, supply, erection, testing, commissioning and handed over of complete fire fighting system for all the buildings and

premises in line with the stipulation of National Building Code 2016 and all other relevant codes.

- b) The work shall include providing and commissioning fire-fighting pumps i.e. all pumps, delivery line, firefighting electrical Panel, Fire Hydrants, sprinkler system (if required), piping, hose cabinet and landing valves, portable fire extinguisher and any other firefighting measure (including gas flood system in electrical panel rooms) complete as per requirement National Building Code 2016 and all other relevant codes.
- c) The system shall be supplied installed, tested and got approved from the local/statutory authority and handed over to employer as per approved Drawing and Technical Specifications.
- d) Listed below Fire Fighting and Fire Protection System Proposed as per NBC:
 - Fire Extinguisher system
 - Piping system confirming to IS: 1239 – MS Heavy Class
 - Wet Riser system / down comer system.
 - Terrace Water Pumps
 - External & Internal fire hydrants
 - Fire hose cabinet (FHC)
 - Overhead Fire water tank
 - Underground Fire water static storage has been provided in accordance with NBC requirement and as per IS: 3844

Fire department connection shall be provided on the external wall/ at designated location for Underground tank, Fire brigade draw-out connection shall be provided to draw-out water during the emergence of fire. The fire pumping system shall comprise of independent electrical hydrant pump, jockey pump, driven by the dedicated diesel engine pumps.

5. SEWERAGE TREATMENT PLANT:

5.1 The Scope of work of the EPC Contractor shall be as under:-

- a) Detailed Designing/preparation of Shop drawings as per DBR and its approval from the RITES.
- b) Supply, Fabrication, Construction, Installation and Commissioning.
- c) Obtaining statutory approval from any statutory body for Commissioning
- d) STP will be on O&M mode from day one.
- e) The STP shall be provided with tertiary treatment, ultra-filtration unit such that treated water is reused in flushing system.
- f) Treated water tank of STP should have sensor to monitor the treatment performance within the desired parameters.

5.2 Further, details (in brief) are as under:-

Providing one Sewerage Water Treatment Plants in prefab structure which will be used to collect, hold, recycle & redistribute water. The distribution will be done to

horticulture & to dual plumbing network in toilets for flushing & to any other identified locations by creating a suitable piping network.

Sewage disposal network including STP proper arrangements for Solid Waste Disposal, approved waste dump locations.

5.3 Solid Waste Management

- a) Packaging waste / Building materials
- b) Kitchen /Guest House/Community Centre dining area.
- c) Street sweepings.
- d) Provision of sprinkles for soft landscape for external development.

6. BULK ELECTRICAL SERVICES (ESS, DG, HSD, PANELS, DISTRIBUTION, INTERNAL ELECTRICAL WORKS, FIRE ALARM, CCTV, ETC)

The electrical services scope of works includes the following:

- a) HV Switchgear (HT Panel), Transformers and LV Switchgear.
- b) Main Power Distribution
- c) Standby Power transformer.
- d) UPS for LV systems, CCTV, EPABX & critical load etc
- e) General lighting for car parks and electrical, mechanical rooms.
- f) Lightning Protection System.
- g) Earthing system.
- h) Grid connected roof top solar generation systems.
- i) Internal and External Lighting including decorative poles with fitting and bollards light, façade light etc..
- j) Life safety and emergency evacuation system.
- k) Manual/Addressable Fire Alarm and Detection System.
- l) LAN Active and Passive System.
- m) IP EPABX System.
- n) PA System
- o) Rising Mains/ Bus duct System.
- p) Security & surveillance System
- q) Boom barriers at Entry and Exit points.
- r) VRV system & Split Acs.
- s) Mechanical Ventilation system

The total load demand of the BQ Complex shall be met by Electric substation through the 11/.433 KV, 2nos. of 2000KVA, Dry type transformers for the further sub distribution for a cluster of buildings as described in the layouts / SLD.

The earthing system will be in conformity with the IS: 3043. All non-current carrying metal parts forming part of the electrical system shall be connected to the grounding system. The requirement of Indian Electricity Rules and statutory requirement of local Electricity authority shall also be met fully.

- Automatic Power factor equipment (Capacitor Panel) if required shall be installed to enhance the power factor to the value of 0.95 or better in addition detuned filter shall be integrated with capacitor panel to suppress the

harmonics.

- Voltage Drop shall not be more than 3% from the output of the distribution transformer to the final distribution board. Cable sizes and quantity shall be arrived based on EPC Contractor working considering Voltage Drop as per IS codes.
- FRLSH PVC insulated copper wires will be used for all sub mains and final circuit wiring in conduit. All life safety equipment cabling shall be Fire Survival category.
- Heavy Duty PVC conduit will be used for wiring concealed in concrete. In all other areas like risers, plant rooms, car park and above false ceiling, surfaced mounted medium class MS conduit will be used.
- In all other areas surfaced mounted MS conduit will be used for extra low voltage wiring e.g. Telephone cable, data cable, and security co-axial cable, etc.
- Armoured cable will be used for external landscape lighting power.

Emergency Lighting / Escape lighting will be capable of

- Clearly indicating the escape routes.
- Providing adequate illumination along such routes to allow safe movement of persons towards and through the exits,
- Ensuring that fire alarm call points and firefighting equipment provided along the escape routes can be readily located.
- 20% External lighting will be designed on Hybrid solar power with batteries , mounted on 6m height/9m height GI pole and in case of bad weather it must be switched to main power automatically.
- The EPC Contractor should take all necessary approvals for connection to grid for excess electricity generation.

6.1 TRANSFORMER, DG & UPS

6.1.1 The Scope of work of the EPC Contractor shall be as under:-

- a) Detailed Designing/preparation of Shop drawings as per DBR and its approval from the RITES/OIL India Limited.
- b) Supply, fabrication, Construction, Installation and Commissioning.
- c) Obtaining approval from all statutory authorities.

6.1.2 Further, details (in brief) are as under:-

- a) Providing a complete end to end Electrical Supply System that will include Receiving HT 11 KV supply from OIL India Limited existing ESS, tapping point of HT Supply from OIL India Limited and planned ESS of BQ Housing shall be in the scope of EPC Contractor. Installation of Electrical Sub stations as per CEA, Indian Electricity norms in desired location as per architectural drawings of the project, HT panels, LT panels, Essential panel, floor distribution panels, bus duct, rising mains, DG set ensuring complete distribution as per internal electrical scheme.

- b) Preparation of Shop Drawings for Transformer, HT Panels, DG Set, Bus Trunking, rising mains, LT panels, floor distribution panels etc. and approval of the same taken from RITES/Client.
- c) Construction of Machine Foundations for all heavy Equipments.
- d) Supplying and fixing of all equipments like Transformer, HT Panels, LT Panel, APFC panel if required, DG Set, UPS, Bus Trunking, earthing, power cable, safety equipment's etc. as per technical specification & drawings and direction of RITES/Client.
- e) Testing, Commissioning and handing over to RITES/OIL India Limited.
- f) All transformers and DGs shall comply as ECSBC plus building standards (as per latest ECSBC (Commercial and Office building, 2024) or latest amendments).

6.1.3 Taking approval of installations from respective Electricity Supply Distribution Authorities .i.e. CEA

6.2 INTERNAL ELECTRICAL WORKS

6.2.1 The Scope of work of the EPC Contractor shall be as under:-

- a) Detailed Designing/ preparation of Shop drawings as per DBR and its approval from the RITES/OIL India Limited.
- b) Supply, fabrication, Construction, Installation, testing and Commissioning.

6.2.2 Further, details (in brief) are as under:-

- a) The internal electrification Design & Schematic/Line drawings as submitted by the. Any improvements recommended in drawings as per updated technology/product for benefit of project may be proposed by EPC Contractor at the time of execution to be approved by Engineer-In charge of RITES. However, detailed fabrication and shop drawings, as required for execution at site are to be prepared by the EPC Contractor in B.I.M. (Building Information Modelling) format and got approved from RITES before proceeding with the work.
- b) The Layout design (Scheme of Electrical arrangements of Electrical works for buildings/spaces and shop drawings in B.I.M. (Building Information Modelling) format of each aspect of the project, required for execution integrated with Civil/Structural /Architectural Drawings for each Building/Spaces and obtaining approval of RITES.
- c) Preparation of Drawing on a software like B.I.M. format etc. of each part of building having an Internal Electrical aspect within Scope of work on suitable scale (6 no. of sets of each drawing).
- d) The Execution of Internal Electrical work for buildings /spaces as per Scope of work read with Technical Specifications and Drawings. The following items are expressly included (but not limited to) in Internal Electrical execution.
 - i. Each floor/area slab/wall concealed electrical copper wiring/cablings/ conduiting (steel/PVC/GI) network including fan boxes, plastic conduits, wires/cables etc.
 - ii. Various type of LED Lighting fixture, ceiling fans, exhaust fans.
 - iii. Antimicrobial switches and switchboards, plugs, fans, exhaust fans, regulators, UPS, etc.

- iv. Distribution boards, Circuit breaker, MCB, RCCB, RCBO MCCB, Junction boxes, Smart Energy meters, Electrical Panels (Low Voltage/Medium Voltage/High Voltage), Earthing etc.
- v. L.T., M.V., Cables laying including GI cable tray for multiple services installation in Grounds, supporting accessories, fittings & fixtures etc.,
- vi. Diesel Generator sets with start-stop switches, alternator, 990 ltr. day oil fuel tank, exhaust system, acoustic & weatherproof enclosure etc including piping from day oil tank to DG set with valves & accessories.
- vii. Insulation of all equipment's/fittings etc. and commissioning the same
- viii. Installation & Commissioning of Dry Type Transformers, HT Panel, LT panel, Emergency Panel, DG set with AMF panel & Substation including HT & LT work in Ground levels.
- ix. All work to be carried out with prior approval of statutory authorities & part commissioning approvals also to be obtained from statutory authorities & distribution agencies.

6.3 INTERNAL LIGHTING WORKS

6.3.1 The Scope of work of the EPC Contractor shall be as under:-

- a) Detail design and preparation of drawings and approval from the RITES/OIL India Limited.
- b) Detailed Designing/preparation of Shop drawings and its approval from the RITES/OIL India Limited.
- c) Supply, fabrication, Construction, Installation and Commissioning.

6.3.2 Further, details (in brief) are as under:-

- a) Light Fixture Design
 - i. All the fixtures to be of very high finish (in terms of product design, appearance, powder coating, internal reflectors, diffusers, and overall finish) in order to give the interiors and Ground car parking a distinctive look and feel.
 - ii. All fixtures to have such fixing arrangements to the surfaces these have to connect, that orientation of fixtures remains architecturally aligned, fixtures are trim-less or have less than 3mm trim; connectors between lengths and to the mounting surfaces are quite robust and mostly click fixed.
 - iii. Fixture wattages have not been intentionally defined. Only minimum Lumen have been defined for each fixture. All fixtures (except for Aviation lights and Exit signs) have been rated for its Lumen package (coming out of fixture) and not system wattage. Lumen output from the fixtures have been defined to achieve the desired Lux levels and the manufacturer will offer lowest wattage possible as per the best product technology available at the time of procurement, to meet the Lumen requirement.

6.4 FIRE ALARM

6.4.1 The Scope of work of the EPC Contractor shall be as under:-

- a) Detailed Designing/preparation of Shop drawings as per DBR and its approval from the RITES/OIL India Limited.
- b) Supply, fabrication, Construction, Installation and Commissioning.
- c) Obtaining approval from applicable authorities.

6.4.2 Further, details (in brief) are as under

- a) Detailed fabrication & shop drawings, required in B.I.M (Building Information modelling format) as required for execution at site are to be prepared by the EPC Contractor and got approved from RITES before proceeding with the work.
- b) To provide fire alarm and detection system with associated communication and notification as per DBR/Drawing and Technical Specification.
- c) The system shall be supplied installed, tested, and got approved from the local/statutory authority and handed over to employer in an operational condition as per Drawing and Technical Specification.
- d) The emergency lighting will be provided to be put on within 1 sec of the failure of the normal lighting supply. Fire Alarm and Detection System will comprise of below mentioned components:
 - i. Manual/Intelligent addressable Fire Detection system with communication, notification & Interface capability.
 - ii. Manual call points, Hooters for activation of fire alarm system manually.
 - iii. Integrated Emergency Voice Evacuation System.
 - iv. Interlocking with other equipment's likes Elevators, Air-conditioning , Smoke Extraction system etc.
 - v. Talk back system.

6.5 CCTV

6.5.1 The Scope of work of the EPC Contractor shall be as under:-

- a) Detailed Designing/preparation of Shop drawings as per DBR and its approval from the RITES/OIL India Limited.
- b) Supply, fabrication, Construction, Installation and Commissioning.
- c) Obtaining approval from applicable authorities.

6.5.2 Further, details (in brief) are as under

- i. Providing, installation, testing & commissioning of IP based CCTV system in the Complex. The CCTV system shall monitor Entrance, lobbies, elevator (lift) lobbies, staircases, common areas, parking area in Grounds, entry and exit of the building / site, driveways, external areas, Service areas e.g., Pump room, STP Room, Sub-Station, DG Sets, etc. Intelligent video management software to be able to also actuate another services event based need to be installed and integrated. Power supply for CCTV on POE System.
- ii. The cameras should be on minimum configuration of 5MP and above and the

recorded images across all the cameras shall be stored locally for at least 30 days to facilitate investigations of a reported case. The Camera should have ONVIF Profile S&G Compliance.

- iii. All the outdoor proposed cameras shall be in IP-66 Housing with IP 66 Junction box.
- iv. Industrial Grade Switches should be used for CCTV network at the edge layer.
- v. Network connectivity for cameras having distance more than 75 meters, shall be on optical Fiber cable. Suitable provision shall be provided to connect with LAN infrastructure.
- vi. The video recording shall be non-embedded based recording server with video management software. The NVR solution shall be a centralized system deployed in the Server room. The viewing of the cameras shall be managed through IP network and can be made accessible at any location.
- vii. All the outdoor cameras shall be in IP-66 Housing with Junction box, media converter etc. is proposed in weather proof housing. The existing LAN network switches would be used for CCTV connectivity and will be connected to central CCTV server/ control room & All cameras shall have PoE connectivity ports.
- viii. Centralized Viewing at the command centre: The live camera feed should be visible in the command centre on the screens for all the cameras deployed across the housing.
- ix. LED Display Monitor: Adequate number of Full HD LED monitors with high resolution screen size of 55" (diagonal) are to be provided for viewing all live/recorded camera proceedings. The resolutions of 1920(H)x1080(V) pixels with comfortable viewing angle shall be provided for all monitors suitable for operation on 220-240V AC, 50 Hz power. The display LED monitors shall be equipped with all standard features such as Video-in, Audio-in, HDMI/DVI, RJ 45 Port etc.
- x. Individual/ dedicated LED Display Monitors shall be provided for each building/block and external areas of the campus as per requirement.
- xi. CCTV Control Room (with display monitor screens & video management server) shall be in the Security Room/ Control Room at entry gate of the BQ housing complex. The wiring inside the building shall be with CAT 6a cable in conduit and for Outdoor connectivity armoured Fiber Optic Cable shall be used. Optical Fiber cables shall be laid underground in HDPE/DWC pipes with suitable manholes for easy pulling and proper maintenance. PTZ cameras will be placed on roof top and on Street light Poles with necessary mounting arrangements in external areas. The video management server (VMS) shall have minimum 30 days storing capacity.
- xii. Suitable hardware/ software shall be provided for streaming live CCTV signal feeds of all indoor and outdoor cameras installed in the campus to remote locations outside the campus through Internet connectivity.

- xiii. EPC contractor shall prepare floor wise layouts of IP CCTV system mentioning location of different types of cameras, location of network rack including details of POE switches, Back bone connectivity from each floor to main rack, Details and locations of LED display, NVR & Server, Main rack, All interconnecting cable details etc.

6.6 VRV & MECHANICAL VENTILATION WORKS

6.6.1 The Scope of work of the EPC Contractor shall be as under: -

- a) Detailed Designing/preparation of Shop drawings as per DBR and its approval from the RITES/OIL India Limited.
- b) Supply, fabrication, Construction, Installation and Commissioning.

6.6.2 VRV/Split AC

- a) The VRV system for Community Centre, split AC for GuestHouse shall be provided as per scope of work. The Heat, Ventilation & Air-Conditioning (HVAC) Design & Schematic/Line drawings all as per DBR and technical specifications are included in Scope of work. Detailed fabrication & shop drawings, required in B.I.M (Building Information modelling format) as required for execution at site are to be prepared by the EPC Contractor and got approved from RITES before proceeding with the work.
- b) The Layout design (Scheme of HVAC arrangements) of HVAC works for above buildings/spaces is provided in tender documents based on which the EPC Contractor must design, prepare fabrication & shop drawings of each and every aspect of the project in required in B.I.M (Building Information modelling format) for execution suitably integrating with the structural & Architectural Drawings.
- c) Preparation of Drawing on a software like B.I.M (Building Information modelling format), etc. of each part of building having an HVAC aspect within Scope of work on suitable scale (6 no. of sets of each drawing).
- d) The Execution of HVAC work for buildings /spaces as per Scope of work read with Technical Specifications and drawings. The following items are expressly included (but not limited to) in HVAC works execution: -
 - Ducting including insulation, acoustic lining & fittings/fixtures of material as specified.
 - Piping, Insulation and valves etc.
 - Inline fans, drain piping & copper piping.
 - Dampers/grills, diffusers, louvers, remote controllers etc.
 - Installation of all equipment's/fittings etc. and commissioning the same
 - Indoor & outdoor units of VRV system & Split Air conditioner.
 - Air-conditioning with Central Control system.

6.6.3 MECHANICAL VENTILATION WORKS

- a) The Mechanical Ventilation Design & Schematic/Line drawings are excluded from Scope of work. Preparation of shop drawings in BIM (Building Information Modelling Format) integrating with the structural and Architectural Drawings for Ground ventilation and as per Drawing and Technical Specification, as required for execution at site are to be prepared by the EPC Contractor and got approved from RITES before proceeding with the work.
- b) Collection of data and preparation of pressurization calculation and data sheet for blowers and obtaining approval of RITES.
- c) Supplying and fixing of all equipment's as per technical specification, drawings, and design.
- d) Ducting, Insulation and Fixing of blowers etc.
- e) Commissioning of the entire system.
- f) Taking approval from respective authority on completion of work.
- g) All Mechanical ventilation equipment's to have provision for connectivity directly to IBMS for all their crucial parameters and operations.
- h) Work includes pressurization of all staircases, all elevators & all elevator lobbies.
- i) Work includes smoke exhaust ventilation from all floors of all blocks.
- j) Work includes mechanical ventilation as per fire norms – including fresh air supply & ventilation. Works includes provision of ventilation according to stack parking requirements & norms for supply & exhaust of carbon dioxide.
- k) Air Distribution
 - l) Air shall be distributed through GI Ducts and discharged through Powder coated grills / diffusers/ Thermal-fusers.
- m) Air-conditioning with Central Control system.
- n) Air will be extracted from the kitchens/pantry by scrubber unit via fireproof ductwork. No fire damper will be provided in kitchen/pantry extract duct. Kitchen/pantry duct shall be of 3mm thick black steel with welded construction. In addition to above spot cooling shall be provided in kitchen/pantry through treated fresh air unit.
- o) Exhaust from Toilets
 - p) Mechanical exhaust is provided for all toilets. Toilets are provided with in line exhaust. Make up will come from adjoining areas through door under cut or transfer grill.
- q) Exhaust from D.G Room

Mechanical Ventilation system is proposed in DG set room as per DG set heat dissipation to keep the DG room temperature within a limit to avoid any de-rating in DG output. DG room supply and exhaust air quantity shall be calculated based on DG set heat dissipation data. DG Room shall be provided supply and exhaust arrangement to maintained DG Room temperature as recommended by the manufacturer. In order to control the Noise level acoustic lining shall be provided in DG room.

- r) Exhaust from Plant Rooms

Transformer room, fire pump room, plumbing pump room, are provided with mechanical ventilation. Systems will be designed to ensure that heat emitted

from the equipment is effectively removed. All above area shall be provided ACPH Ventilation as per NBC.

s) Energy conservation features in Air conditioning:

Air-conditioning & ventilation system design shall comply of GRIHA requirements, proposed system is well equipped with energy conservation features to reduce energy consumption and operating costs where economically feasible, generally as follows:

- High COP (Low kW / TR) VRV system. (As per ECSBC plus building standards(as per latest ECSBC (Commercial and Office building, 2024) or latest amendments).
- High COP (Low kW / TR) VRV system.
- VRV/VRF distribution system.
- Energy efficient motors for indoor & outdoor units.
- Use of individual and multiple indoor unit which can be shut down during unoccupied or partially occupied periods through occupancy sensors.
- Selection of highly efficient fans for air handling unit and ventilation system.
- Variable speed drive on some selected indoor/outdoor/TFA units.

6.7 ELEVATORS

6.7.1 The Scope of work of the EPC Contractor shall be as under:-

- a) Detailed Designing/preparation of Shop drawings as per DBR and its approval from the RITES/OIL India Limited.
- b) Supply, fabrication, Construction, Installation and Commissioning.
- c) Obtaining approval of Lift Inspector.

6.7.2 Further, details (in brief) are as under

- a) Providing, installation, testing and commissioning and handing over of elevators of required capacity as per drawings and technical specification. Detailed fabrication & shop drawings, required in B.I.M (Building Information modelling format) as required for execution at site are to be prepared by the EPC Contractor and got approved from RITES before proceeding with the work.
- b) Obtaining Elevator license, approval from elevator inspectors before commissioning.
- c) All Elevators are provided from all Ground to upper floors. These shall be finished using highest quality maintenance friendly finishes & include all accessories. The touch panel shall be with LED display. Background Music shall be played. The internal finishes & all features of all passenger elevators shall be similar. The elevators/lift car floors shall be finished with Granite

and the walls shall be S.S mirror finish, elevator/lift car must be with c.c.t.v surveillance and final drawing as per approval from the RITES.

6.8 OUTDOOR & LANDSCAPE LIGHTING

6.8.1 The Scope of work of the EPC Contractor shall be as under:-

- a) Designing the system & obtain RITES approval.
- b) Detailed Designing/preparation of Shop drawings and its approval from RITES.
- c) Supply, Fabrication, Construction, Installation and Commissioning

6.8.2 Further, details (in brief) are as under

All external landscaped areas will be adequately lit with approved lighting fixtures to provide lighting to all pedestrian walkway, vehicular roads, entrances to complex, to buildings, to Grounds etc, to all fire exits. The landscape lighting will be done to create a pleasant world class environment in the evening for a safe & pedestrian friendly environment.

The lighting will accentuate trees by up lighting, direct traffic by motorable embedded lights, highlight ramps by wall embedded lights, highlight traffic directions by corner / curved street markings. There will be green planters that will have sufficient bollards / planters ground embedded lights to showcase the greenery in the evenings.

The design of lighting and fixture selection in these areas shall harmonize with the landscape.

Ornamental and pole-mounted light fixtures are to be appropriately scaled for each landscape situation. Alternates to pole lighting shall be considered to provide ambient and pathway lighting.

Accent lighting must be located to avoid viewing bright sources from buildings and public areas.

Provide adequate lighting and fixture selection for all directional, location and safety signage.

All the drawings will be as per the base drawing, the final drawing needs to be approved by the RITES.

6.9 FACADE & DIGITAL LIGHTING

6.9.1 The Scope of work of the EPC Contractor shall be as under:-

- a) Designing the system & obtain RITES approval.
- b) Detailed Designing/preparation of Shop drawings and its approval from RITES.
- c) Supply, Fabrication, Construction, Installation and Commissioning

6.9.2 Further, details (in brief) are as under

The facade lighting will be done on all levels of BQ housing complex. The facade lighting will be done in such a way that it highlights architecture features of the building.

The lighting will be on window and solid wall composition highlighting the pattern at night as per the base design attached below, final drawing to be approved by RITES/OIL India Limited.

Lighting of building facade is to be coordinated with the landscape, driveway, and pedestrian lighting.

Provide light sources with good to excellent colour rendering at the following locations:

- External signage
- Double Height Entrance space

6.10 BOOM BARRIERS

6.10.1 The Scope of work of the EPC Contractor shall be as under:-

- a) Detailed Designing/preparation of Shop drawings and its approval from RITES.
- b) Supply, Fabrication, Construction, Installation and Commissioning

6.10.2 Further, details (in brief) are as under

Providing & Fixing of Boom barriers at Entrance/Exit gate of the campus from main access road. These will be connected to card readers & sensors for automated access to smart tag users.

6.11 CUSTOMISED CABLE TRAY

6.11.1 The Scope of work of the EPC Contractor shall be as under:-

- a) Detailed Designing/preparation of Shop drawings and its approval from RITES.
- b) Supply, Fabrication, Construction, Installation and Commissioning

6.11.2 Further, details (in brief) are as under

6.11.3 These are suspended with cable tray systems used as a trunk service distribution & mounting integrated Tray. These will carry electrical supply, LV cables, CCTV, Digital lighting cable etc in separate compartments & will be suspended as boxed / closed cable trays from the ceiling / beam above. Typically, these will also have light, CCTV Camera, & Signage fixtures suspended / surface mounted on the bottom of these trays.

6.12 SIGNAGE SYSTEMS

6.12.1 The Scope of work of the EPC Contractor shall be as under:-

- a) Detailed Designing/preparation of Shop drawings and its approval from RITES.
- b) Supply, Fabrication, Construction, Installation and Commissioning

6.12.2 Further, details (in brief) are as under

The scope of work for the EPC Contractor would be to design non-digital and digital signage system to achieve an effective way finding scheme both for internal roads as well as, for all indoor common areas like Lobbies, Elevator/Lift Lobbies, corridors, washrooms, service rooms etc.

6.13 EXTERNAL DEVELOPMENT

6.13.1 EXTERNAL SERVICES, DISTRIBUTION & MISCELLANEOUS

- a) Installation, testing and commissioning of water supply system, sewerage system and drainage system as per Approved Drawings and Technical Specifications.
- b) Supply and Installation of pumps and construction of underground water tanks with pump room as per approved Drawing and Technical Specification.
- c) Providing and laying/fixing cabling from the main receiving station, rising main, meter, electrical panel etc. as per Drawing and Technical Specification.
- d) Streetlight with LED light, land scape light, all signage as per Drawing and Technical Specification.
- e) Multi-purpose poles in open area for integrated installation of Lighting, CCTV, PA & BGM system.
- f) Construction of Service ducts for electrical, water supply etc. as per Drawing and Technical Specification.
- g) Service Connection/tapping of the water supply, sewerage, drainage, electricity to the main/municipal system.

6.13.2 Site Development & Landscape (Including roads,pathways etc.)

Scope of work of EPC Contractor shall be as under: -

- a) Detailed Design for selected works such as boundary wall, Stainless-Steel, MS works etc.
- b) Detailed Designing/preparation of Shop drawings and its approval from RITES.
- c) Supply, fabrication, Construction, Installation and Commissioning.

Further details in brief are as under: -

- a) Levelling of land, cut & fill wherever required for construction of road, parking, footpaths and landscaping including construction of retaining walls if required,as per site condition and as/approval of engineer in-charge and specifications.
- b) Construction of Internal Roads/External roads (to be designed for fire tender as well), Berms, Pathways, Kerbs, open parking space etc. including connecting with the external road network as per Drawing and Technical Specification.
- c) All the roads shall have footpath along the length for pedestrian movement as/tender drawings and approval of engineer in-charge and specifications. All

footpaths shall have pre-cast kerb stone along the edge or as/site conditions/tender drawings or as per approval of engineer in-charge and specifications. Footpath and road network, designed as per IRC Codes, considering circulation and traffic movement including pedestrian, vehicular movement.

- d) A pre-cast/cast-in-situ service trench may be designed and constructed underneath the footpath, if required, for carrying sanitary, plumbing and electrical lines separately as/approval of engineer in-charge and specifications.
- e) Drain formation along the road length maintaining proper slope and drainage as/approval of engineer in-charge and specifications. All the stretches of storm water drain shall be RCC & covered. Roadsides may have suitable saucer drain and gully traps with gratings.
- f) Design & Construction of Boundary wall, M.S Manual Swing Gates, wicket gate, M.S Grills (enamel painted) if required, Boom Barrier etc. as per Drawing and Technical Specification.
- g) Work to include built in planters, bushes / ground covers in planters, street furniture, stone benches etc.

SECTION - 3

SPECIAL CONDITIONS
OF
CONTRACT

SECTION – 3

SPECIAL CONDITION OF CONTRACT

A. CONTRACT PRICES AND PAYMENT SCHEDULE

1. Mode of measurement of plinth area shall be as per CPWD PAR-2023.
2. For intermediate payments on “pro-rata basis”, amount shall be worked out on the basis of fraction of work done out of total scope of work under each activity / item. The assessment made by Engineer-in-Charge shall be final in this regard.
3. All efforts shall be made to release the payment within the period mentioned in clause 7.0 of GCC. RITES will not be responsible for any delay in releasing payment of running bill beyond the stipulated time as mentioned in clause 7.0 of GCC. The EPC contractor will not be entitled to any claim on OIL India Limited/RITES for delay in payment due to non- availability of fund. No interest will be paid on any delay on the payment of RA bills /Final bill.
4. Design including detailed topographical survey, geotechnical investigation, preparation of drawings for Structural, MEP (services & infrastructure required), shop drawings & any other design/drawing as indicated in tender document, proof checking/vetting of drawings from IIT is included in the scope of EPC contractor and nothing shall be payable extra on these account.
5. All the costs/expenses related to GRIHA is deemed to be included in accepted tendered value and nothing shall be payable extra. However, the necessary statutory fee for registration shall be reimbursed by RITES/OIL India Limited.
6. Any change in plinth areas, external development works and other ancillary infrastructure areas due to Structural/MEP/Design/Planning/GRIHA is deemed to be included in the accepted tendered value and nothing shall be payable extra.
7. **Variation in Scope of Work:**

In the event of any change in the original scope of work as defined in the Contract Agreement, which involves execution of **Extra Item**, the same shall be paid on ITEM RATE BASIS, and, the following provisions shall apply:

For Scheduled Items: Payment shall be made as per the rates specified in CPWD Delhi Schedule of Rates (DSR-2023) - with index as per prevailing.

For Non-Scheduled (Market Rate - MR) Items: Payment shall be made on the basis of Market Rate Analysis, supported by minimum three quotations/invoices, and duly

approved by the Engineer-in-Charge & CLIENT/OIL India Limited, in accordance with the applicable guidelines and procedures.

The EPC Contractor shall not carry out any such Extra Item without prior approval from the Engineer-in-Charge and AA & ES by the OIL India Limited/CLIENT.

No claim whatsoever on account of such variation shall be entertained, if any such work is executed without prior approval.

- 8. All running / intermediate & final payments shall be made to the EPC Contractor in accordance with the following schedule:**

8.1	Construction of Workman Housing Complex (BQ Area) on EPC mode -II of contract at OIL Duliajan, Assam. Residential Block (STILT+6, Total - 08 Towers) (Total Tentative Plinth Area of 08 towers = 27,355 Sqm)
Payment for Civil Component of Residential Blocks: The cost of the Civil Component (denoted as A) shall be calculated as: $A = 79\% \times \text{Tendered cost of Residential Blocks (as quoted under Item No. 1.1 of the Financial Bid (BoQ))}$	
Payment for Electrical & Mechanical Component of Residential Blocks: The cost of the Electrical & Mechanical Component (denoted as B) shall be calculated as: $B = 21\% \times \text{Tendered cost of Residential Blocks (as quoted under Item No. 1.1 of the Financial Bid (BoQ))}$	
NOTE: The payment for Civil Works pertaining to each of the 08 Residential Blocks shall be regulated based on the following formula: $\text{Payment per Residential Tower} = (A/8)$	

S. No.	Description of Item	Break up of total % age of A/8 calculated as above from item no. 1.1 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
8.1 A	Design & Construction of Residential Blocks Total 08 Towers (STILT+6) Floors buildings (Civil Works)			
1	Designing, Detailing and obtaining approvals. Investigation, Planning, Designing and obtaining the approvals for the scope of work, Technical specification & Drawings etc. complete			
1.1	On Submission of Inception report, Survey Report, Soil Test Report	0.40	1.00	1.00
1.2	On Preparation of structural design drawings and vetting of the same from IIT	0.10		
1.3	On Submission & approval of working drawings including Architectural & MEP services etc. including revision; if any	0.25		

S. No.	Description of Item	Break up of total % age of A/8 calculated as above from item no. 1.1 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
1.4	On Obtainng statutory clearance and approval of local bodies and concerned authorities	0.25		
2	On Completion of Foundation work up to Plinth level including site clearance Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in-Charge			
2.1	On completion of Jungle clearance, including uprooting of rank vegetation, grass, brushwood, Felling of trees of any girth and saplings. Removal of rubbish from outside the periphery of the area cleared, all cutting and filling in earth work as required including disposal of unsuitable/excess earth and supply of suitable/desired filling material for all leads and lifts, demolition of existing structures/foundations if any, anti-termite treatment, Earth work in excavation, Laying of PCC for foundation, RCC foundation & backfilling up to footings/raft/pile cap level etc. Complete in all respect as per Scope of work.	14.00	15.50	16.50
2.2	On Completion of the RCC frame structure work (from top of footings/raft/pile cap) up to Plinth level, plinth beam (including grade slab), filling in earth work as required and supply of suitable/desired filling material for all leads and lifts, anti-termite treatment. Complete in all respect as per Scope of work.	1.50		
3.	Civil & Structure Works: On Completion of the RCC frame work from Plinth level & up to top of the building. Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge.			

S. No.	Description of Item	Break up of total % age of A/8 calculated as above from item no. 1.1 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
3.1	On Completion of RCC Framework i.e. Column, beam, shear wall, staircase & GF/sitlt floor slab	5.00	35.50	52.00
3.2	On Completion of RCC Framework i.e. Column,beam,shear wall, staircase & 1st Floor slab	5.00		
3.3	On Completion of RCC Framework i.e. Column,beam,shear wall, staircase & 2nd floor slab	5.00		
3.4	On Completion of RCC Framework i.e. Column,beam,shear wall, staircase & 3rd floor slab	5.00		
3.5	On Completion of RCC Framework i.e. Column,beam,shear wall, staircase & 4th floor slab	5.00		
3.6	On Completion of RCC Framework i.e. Column,beam,shear wall, staircase & 5th Floor	5.00		
3.7	On Completion of RCC Framework i.e. Column,beam,shear wall, staircase & 6th Floor	5.00		
3.8	On Completion of RCC Framework i.e. Column,Beam,staircase, parapet wall & slab of mumty, elevator (lift) room & Overhead tank etc.	0.50		
4.	On Completion of Internal & External Brick work/AAC block & Partitioning walls, including RCC bends, lintels, sunshades works etc. Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge.			
4.1	On Completion of Stilt floor	0.30	9.00	61.00
4.2	On Completion of 1st Floor	1.35		
4.3	On Completion of 2nd floor	1.35		
4.4	On Completion of 3rd floor	1.35		
4.5	On Completion of 4th floor	1.35		
4.6	On Completion of 5th floor	1.35		

S. No.	Description of Item	Break up of total % age of A/8 calculated as above from item no. 1.1 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
4.7	On Completion of 6th floor	1.35		
4.8	On Completion of terrace level including parapet wall, mumty, elevator (lift) room & Overhead tank etc.	0.60		
5	Plaster (External & Internal) including drip course/grooving/chicken mesh jali on all junctions of RCC and Masonry work. Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge..			
5.1	On Completion of stilt floor	0.40		
5.2	On Completion of 1st Floor	0.40		
5.3	On Completion of 2nd floor	0.40		
5.4	On Completion of 3rd floor	0.40		
5.5	On Completion of 4th floor	0.40		
5.6	On Completion of 5th floor.	0.40		
5.7	On Completion of 6th floor	0.40		
5.8	On Completion of terrace level including parapet wall, mumty, elevator (lift) room & Overhead tank etc.	0.20		
6	Water proofing works. Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge.			
6.1	On Completion of 1st Floor	0.20		
6.2	On Completion of 2nd floor	0.20		
6.3	On Completion of 3rd floor	0.20		
6.4	On Completion of 4th floor	0.20		
6.5	On Completion of 5th floor.	0.20		
6.6	On Completion of 6th floor	0.20		
6.7	On Completion of terrace level including parapet wall, mumty, elevator (lift) room & Overhead tank etc.	2.80		
7	All types of Flooring such as Granite, all types of tiles, Kota Stone, CC flooring, skirting, dado, sill, platforms, counters, partitions, shelves, wall lining			

S. No.	Description of Item	Break up of total % age of A/8 calculated as above from item no. 1.1 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
	work, PCC as required etc. Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge.			
7.1	On Completion of stilt floor	0.90	7.00	75.00
7.2	On Completion of 1st Floor	0.90		
7.3	On Completion of 2nd floor	0.90		
7.4	On Completion of 3rd floor	0.90		
7.5	On Completion of 4th floor	0.90		
7.6	On Completion of 5th floor.	0.90		
7.7	On Completion of 6th floor	0.90		
7.8	On Completion of terrace level including parapet wall, mumty, elevator (lift) room & Overhead tank etc.	0.70		
8	Roofing works such as Gola, Proving & fixing all types of False ceilings, for roof covering etc. Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge.			
8.1	On Completion of 1st Floor	0.15	1.00	76.00
8.2	On Completion of 2nd floor	0.15		
8.3	On Completion of 3rd floor	0.15		
8.4	On Completion of 4th floor	0.15		
8.5	On Completion of 5th floor.	0.15		
8.6	On Completion of 6th floor	0.15		
8.7	On Completion of terrace level including parapet wall, mumty, elevator (lift) room & Overhead tank etc.	0.10		
9	Proving & Fixing of doors, windows, ventilators of all types complete with glazing, mesh (like wooden, Flush door, Aluminium, WPC), Fire rated doors, wardrobes, kitchen cabinet shutters etc. Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge.			
9.1	On Completion of stilt floor	0.90	7.00	83.00
9.2	On Completion of 1st Floor	0.90		

S. No.	Description of Item	Break up of total % age of A/8 calculated as above from item no. 1.1 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
9.3	On Completion of 2nd floor	0.90		
9.4	On Completion of 3rd floor	0.90		
9.5	On Completion of 4th floor	0.90		
9.6	On Completion of 5th floor.	0.90		
9.7	On Completion of 6th floor	0.90		
9.8	On Completion of terrace level including parapet wall, mumty, elevator (lift) room & Overhead tank etc.	0.70		
10	Finishing Work i.e Painting (Inside & outside) etc. Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge.			
10.1	On Completion of stilt floor	0.26	2.00	85.00
10.2	On Completion of 1st Floor	0.26		
10.3	On Completion of 2nd floor	0.26		
10.4	On Completion of 3rd floor	0.26		
10.5	On Completion of 4th floor	0.26		
10.6	On Completion of 5th floor.	0.26		
10.7	On Completion of 6th floor	0.26		
10.8	On Completion of terrace level including parapet wall, mumty, elevator (lift) room & Overhead tank etc.	0.18		
11	Other works, Viz. balcony railing, staircase railing, window grills, invisible grill and other steel works etc. (Internal & External). Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge.			
11.1	On Completion of stilt floor	0.26	2.00	87.00
11.2	On Completion of 1st Floor	0.26		
11.3	On Completion of 2nd floor	0.26		
11.4	On Completion of 3rd floor	0.26		
11.5	On Completion of 4th floor	0.26		
11.6	On Completion of 5th floor.	0.26		
11.7	On Completion of 6th floor	0.26		

S. No.	Description of Item	Break up of total % age of A/8 calculated as above from item no. 1.1 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
11.8	On Completion of terrace level including parapet wall, mumty, elevator (lift) room & Overhead tank etc.	0.18		
12	On Completion of Plumbing & sanitary piping works, installations, fittings, fixtures etc. Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge.			
12.1	On Completion of stilt floor	0.90	7.00	94.00
12.2	On Completion of 1st Floor	0.90		
12.3	On Completion of 2nd floor	0.90		
12.4	On Completion of 3rd floor	0.90		
12.5	On Completion of 4th floor	0.90		
12.6	On Completion of 5th floor.	0.90		
12.7	On Completion of 6th floor	0.90		
12.8	On Completion of terrace level including parapet wall, mumty, elevator (lift) room & Overhead tank etc.	0.70		
13	On compliance of GRIHA items specifications to obtain minimum GRIHA 3 Star Certification as per scope of work.	1.00	1.00	95.00
14	On Completion of Construction, Testing commissioning of all services, fittings and fixtures, equipment's, submission of all guarantee/warranty card, operation & maintenance manual, as-built drawings and handing over. Obtaining & submission of all necessary statutory approvals of completion plan from local authorities, Obtaining Fire clearance, Completion certificate, Occupancy Certificate from all concerned local bodies or any other authorities and consent to operate etc.			
14.1	On Completion of Construction, Testing commissioning of all services, fittings and fixtures, equipment's etc.	1.00	5.00	100.00
14.2	Obtaining & submission of all necessary	2.00		

S. No.	Description of Item	Break up of total % age of A/8 calculated as above from item no. 1.1 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
	statutory approvals of completion plan from local authorities, Obtaining Fire clearance, Completion certificate, Occupancy Certificate from all concerned local bodies or any other authorities and consent to operate etc. all complete.			
14.3	Submission of all guarantee/warranty cards, operation & maintenance manual, as-built drawings and handing over (After completion of 14.2)	2.00		

8.2	Construction of Workman Housing Complex (BQ Area) on EPC mode -II of contract at OIL Duliajan, Assam. Guest House (G+3) (Tentative Plinth Area = 1990 Sqm)
Payment for Civil Component of Guest House (G+3): The cost of the Civil Component (denoted as A) shall be calculated as: $A = 79\% \times \text{Tendered Cost of Guest House (as quoted under Item No. 1.2 of the Financial Bid (BoQ))}$	
Payment for Electrical & Mechanical Component of Guest House (G+3): The cost of the Electrical & Mechanical Component (denoted as B) shall be calculated as: $B = 21\% \times \text{Tendered Cost of Guest House (as quoted under Item No. 1.2 of the Financial Bid (BoQ))}$	

S. No.	Description of Item	Break up of total % age of A calculated as above from item no. 1.2 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
8.2 A	Design & Construction of Guest House (G+3) Floors building (Civil Works)			
1	Designing, Detailing and obtaining approvals. Investigation, Planning, Designing and obtaining the approvals for the scope of work, Technical specification & Drawings etc. complete			
1.1	On Submission of Inception report, Survey Report, Soil Test Report	0.40	1.00	1.00
1.2	On Preparation of structural design drawings and vetting of the same from IIT	0.10		
1.3	On Submission & approval of working drawings including Architectural & MEP services including revision; if any	0.25		
1.4	On Obtainng statutory clearance and approval of local bodies and concerned authorities	0.25		
2.	Civil & Structure Works: On Completion of Foundation, RCC frame work from Plinth level & up to top of the building including site clearance Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of			

S. No.	Description of Item	Break up of total % age of A calculated as above from item no. 1.2 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
	Engineer-in- Charge.			
2.1	On completion of Jungle clearance, including uprooting of rank vegetation, grass, brushwood, Felling of trees of any girth and saplings. Removal of rubbish from outside the periphery of the area cleared, all cutting and filling in earth work as required including disposal of unsuitable/excess earth and supply of suitable/desired filling material for all leads and lifts, demolition of existing structures/foundations if any, anti-termite treatment, Laying of PCC for foundation, RCC foundation, RCC frame structure work up to Plinth level, plinth beam, including grade slab etc. Complete in all respects.	9.50	47.70	48.70
2.2	On Completion of RCC Framework i.e. Column, beam, shear wall, staircase & GF slab	9.00		
2.3	On Completion of RCC Framework i.e. Column, beam, shear wall, staircase & 1st Floor slab	9.00		
2.4	On Completion of RCC Framework i.e. Column, beam, shear wall, staircase & 2nd floor slab	9.00		
2.5	On Completion of RCC Framework i.e. Column, beam, shear wall, staircase & 3rd floor slab	9.00		
2.6	On Completion of RCC Framework i.e. Column, Beam, staircase, parapet wall & slab of mumty, elevator (lift) room & Overhead tank etc.	2.20		
3.	On Completion of Internal & External Brick work/AAC block & Partitioning walls, including RCC bends, lintels, sunshades works etc. Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge.			
3.1	On Completion of GF floor	2.25	9.50	58.20
3.2	On Completion of 1st Floor	2.25		

S. No.	Description of Item	Break up of total % age of A calculated as above from item no. 1.2 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
3.3	On Completion of 2nd floor	2.25		
3.4	On Completion of 3rd Floor	2.25		
3.5	On Completion of terrace evel including parapet wall, mumty, elevator (lift) room & Overhead tank etc.	0.50		
4	Plaster (External & Internal) including grooving/chicken mesh jali on all junctions of RCC and masonry work. Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in-Charge.			
4.1	On Completion of GF floor	0.60	2.80	61.00
4.2	On Completion of 1st Floor	0.60		
4.3	On Completion of 2nd floor	0.60		
4.4	On Completion of 3rd Floor	0.60		
4.5	On Completion of terrace evel including parapet wall, mumty, elevator (lift) room & Overhead tank etc.	0.40		
5	Water proofing works. Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge.			
5.1	On Completion of GF floor	0.50	3.80	64.80
5.2	On Completion of 1st Floor	0.50		
5.3	On Completion of 2nd floor	0.50		
5.4	On Completion of 3rd Floor	0.50		
5.5	On Completion of terrace evel including parapet wall, mumty, elevator (lift) room & Overhead tank etc.	1.8		
6	All types of Flooring such as Granite, all types of tiles, Kota Stone, CC flooring, skirting, dado, sill, wall lining work, PCC as required etc. Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge.			
6.1	On Completion of GF floor	1.50	6.60	71.40
6.2	On Completion of 1st Floor	1.50		
6.3	On Completion of 2nd floor	1.50		

S. No.	Description of Item	Break up of total % age of A calculated as above from item no. 1.2 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
6.4	On Completion of 3rd Floor	1.50		
6.5	On Completion of terrace evel including parapet wall, mumty, elevator (lift) room & Overhead tank etc.	0.60		
7	Roofing works such as Gola, Providing & fixing all types of False ceilings for roof covering etc. Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge.			
7.1	On Completion of GF floor	0.22	0.95	72.35
7.2	On Completion of 1st Floor	0.22		
7.3	On Completion of 2nd floor	0.22		
7.4	On Completion of 3rd Floor	0.22		
7.5	On Completion of terrace evel including parapet wall, mumty, elevator (lift) room & Overhead tank etc.	0.07		
8	Proving & Fixing of doors, windows, ventilators of all types (like wooden, Flush door, Aluminium, UPVC, WPC), Fire rated doors, toughened glass door with partition etc. Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge.			
8.1	On Completion of GF floor	1.80	7.60	79.95
8.2	On Completion of 1st Floor	1.80		
8.3	On Completion of 2nd floor	1.80		
8.4	On Completion of 3rd Floor	1.80		
8.5	On Completion of terrace evel including parapet wall, mumty, elevator (lift) room & Overhead tank etc.	0.40		
9	Finishing Work i.e Painting (Inside & outside) etc. Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge.			
9.1	On Completion of GF floor	0.45	1.90	81.85
9.2	On Completion of 1st Floor	0.45		
9.3	On Completion of 2nd floor	0.45		
9.4	On Completion of 3rd Floor	0.45		

S. No.	Description of Item	Break up of total % age of A calculated as above from item no. 1.2 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
9.5	On Completion of terrace evel including parapet wall, mumty, elevator (lift) room & Overhead tank etc.	0.10		
10	Other works, Viz. balcony railing, staircase railing, window grills and other steel works etc. (Internal & External). Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge.			
10.1	On Completion of GF floor	0.20	0.95	82.80
10.2	On Completion of 1st Floor	0.20		
10.3	On Completion of 2nd floor	0.20		
10.4	On Completion of 3rd Floor	0.25		
10.5	On Completion of terrace evel including parapet wall, mumty, elevator (lift) room & Overhead tank etc.	0.10		
11	On Completion of Plumbing & sanitary piping works, installations, fittings, fixtures etc. Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge.			
11.1	On Completion of GF floor	2.00	9.50	92.30
11.2	On Completion of 1st Floor	2.00		
11.3	On Completion of 2nd floor	2.00		
11.4	On Completion of 3rd Floor	2.00		
11.5	On Completion of terrace evel including parapet wall, mumty, elevator (lift) room & Overhead tank etc.	1.50		
12	Kitchen equipments all Complete in all respect as per drawings, technical specifications, Schedule of finishes, Scope of work etc. and direction of Engineer-in- Charge.			
12.1	Supply, installation, testing commissioning and handing over to Client.	1.70	1.70	94.00
13	On compliance of GRIHA items specifications to obtain minimum GRIHA 3 Star rating Certification as per scope of work.	1.00	1.00	95.00

S. No.	Description of Item	Break up of total % age of A calculated as above from item no. 1.2 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
14	On Completion of Construction, Testing commissioning of all services, fittings and fixtures, equipment's, submission of all guarantee/warranty card, operation & maintenance manual, as-built drawings and handing over. Obtaining & submission of all necessary statutory approvals of completion plan from local authorities, Obtaining Fire clearance, Completion certificate, Occupancy Certificate from all concerned local bodies or any other authorities and consent to operate etc.			
14.1	On Completion of Construction, Testing commissioning of all services, fittings and fixtures, equipment's etc.	1.00	5.00	100.00
14.2	Obtaining & submission of all necessary statutory approvals of completion plan from local authorities, Obtaining Fire clearance, Completion certificate, Occupancy Certificate from all concerned local bodies or any other authorities and consent to operate etc. all complete.	2.00		
14.3	Submission of all guarantee/warranty cards, operation & maintenance manual, as-built drawings and handing over (After completion of 14.2)	2.00		

8.3	Construction of Workman Housing Complex(BQ Area) on EPC mode -II of contract at OIL Duliajan, Assam Community Centre (G+1) (Tentative Plinth Area = 1035 sqm)
Payment for Civil Component of Community Centre (G+1): The cost of the Civil Component (denoted as A) shall be calculated as: $A = 77\% \times \text{Tendered Cost of Community Centre (as quoted under Item No. 1.3 of the Financial Bid (BoQ))}$	
Payment for Electrical & Mechanical Component of Community Centre (G+1): The cost of the Electrical & Mechanical Component (denoted as B) shall be calculated as: $B = 23\% \times \text{Tendered Cost of Community Centre (as quoted under Item No. 1.3 of the Financial Bid (BoQ))}$	

S. No.	Description of Item	Break up of total % age of A calculated as above from item no. 1.3 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
8.3 A	Design & Construction of Community Centre (G+1) Floors buildings (Civil Works)			
1	Designing, Detailing and obtaining approvals. Investigation, Planning, Designing and obtaining the approvals for the scope of work, Technical specification & Drawings etc. complete			
1.1	On Submission of Inception report, Survey Report, Soil Test Report	0.40	1.00	1.00
1.2	On Preparation of structural design drawings and vetting of the same from IIT.	0.10		
1.3	On Submission & approval of working drawings including Architectural & MEP services including revision; if any	0.25		
1.4	On Obtainng statutory clearance and approval of local bodies and concerned authorities	0.25		

S. No.	Description of Item	Break up of total % age of A calculated as above from item no. 1.3 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
2.	Civil & Structure Works: On Completion of Foundation, RCC frame work from Plinth level & up to top of the building including site clearance Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge.			
2.1	On completion of Jungle clearance, including uprooting of rank vegetation, grass, brushwood, Felling of trees of any girth and saplings. Removal of rubbish from outside the periphery of the area cleared, all cutting and filling in earth work as required including disposal of unsuitable/excess earth and supply of suitable/desired filling material for all leads and lifts, demolition of existing structures/foundations if any, anti-termite treatment, Laying of PCC for foundation, RCC foundation, RCC frame structure work up to Plinth level, plinth beam, including grade slab etc. Complete in all respects.	9.00	39.00	40.00
2.2	On Completion of RCC Framework i.e. Column,beam,shear wall, staircase & GF/sitlt floor slab	13.00		
2.3	On Completion of RCC Framework i.e. Column,beam,shear wall, staircase & 1st Floor slab	13.00		
2.4	On Completion of RCC Framework i.e. Column,Beam,staircase, parapet wall & slab of mumty & Overhead tank etc.	4.00		
3.	On Completion of Internal & External Brick work/AAC block & Partitioning walls, including RCC bends, lintels, sunshades works etc. Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge.			
3.1	On Completion of GF floor	4.00	9.50	49.50
3.2	On Completion of 1st Floor	4.00		

S. No.	Description of Item	Break up of total % age of A calculated as above from item no. 1.3 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
3.3	On Completion of terrace evel including parapet wall, mumty & Overhead tank etc.	1.50		
4	Plaster (External & Internal) including grooving/chicken mesh jali on all junctions of RCC and AAC /brickwork. Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in-Charge.			
4.1	On Completion of GF floor	2.25	4.75	54.25
4.2	On Completion of 1st Floor	2.25		
4.3	On Completion of terrace evel including parapet wall, mumty & Overhead tank etc.	0.25		
5	Water proofing works. Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge.			
5.1	On Completion of GF floor	1.15	5.70	59.95
5.2	On Completion of 1st Floor	1.15		
5.3	On Completion of terrace evel including parapet wall, mumty & Overhead tank etc.	3.40		
6	All types of Flooring such as Granite, all types of tiles, Kota Stone, CC flooring, skirting, dado, wall lining work, PCC as required etc. Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge.			
6.1	On Completion of GF floor	4.00	9.50	69.45
6.2	On Completion of 1st Floor	4.00		
6.3	On Completion of terrace evel including parapet wall, mumty & Overhead tank etc.	1.50		
7	Roofing works such as Gola, Providing & fixing all types of False ceilings etc. Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in-Charge.			
7.1	On Completion of GF floor	1.25	2.85	72.30
7.2	On Completion of 1st Floor	1.25		
7.3	On Completion of terrace evel including parapet wall, mumty & Overhead tank etc.	0.35		

S. No.	Description of Item	Break up of total % age of A calculated as above from item no. 1.3 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
8	Proving & Fixing of doors, windows, ventilators of all types (like wooden, Flush door, Aluminium, UPVC, WPC), Fire rated doors, toughened glass door with partition etc. Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge.			
8.1	On Completion of GF floor	4.00	9.50	81.80
8.2	On Completion of 1st Floor	4.00		
8.3	On Completion of terrace evel including parapet wall, mumty & Overhead tank etc.	1.50		
9	Finishing Work i.e Painting (Inside & outside) etc. Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge.			
9.1	On Completion of GF floor	2.50	5.70	87.50
9.2	On Completion of 1st Floor	2.50		
9.3	On Completion of terrace evel including parapet wall, mumty & Overhead tank etc.	0.70		
10	Other works, Viz. balcony railing, staircase railing, window grills and other steel works etc. (Internal & External). Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge.			
10.1	On Completion of GF floor	0.40	0.90	88.40
10.2	On Completion of 1st Floor	0.40		
10.3	On Completion of terrace evel including parapet wall, mumty & Overhead tank etc.	0.10		
11	On Completion of Plumbing & sanitary piping works, installations, fittings, fixtures etc. Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge.			
11.1	On Completion of GF floor	1.90	5.60	94.00
11.2	On Completion of 1st Floor	1.90		
11.3	On Completion of terrace evel including parapet wall, mumty & Overhead tank etc.	1.80		

S. No.	Description of Item	Break up of total % age of A calculated as above from item no. 1.3 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
12	On compliance of GRIHA items specifications to obtain minimum GRIHA 3 Star rating Certification as per scope of work.	1.00	1.00	95.00
13	On Completion of Construction, Testing commissioning of all services, fittings and fixtures, equipment's, submission of all guarantee/warranty card, operation & maintenance manual, as-built drawings and handing over. Obtaining & submission of all necessary statutory approvals of completion plan from local authorities, Obtaining Fire clearance, Completion certificate, Occupancy Certificate from all concerned local bodies or any other authorities and consent to operate etc.			
13.1	On Completion of Construction, Testing commissioning of all services, fittings and fixtures, equipment's etc.	1.00	5.00	100.00
13.2	Obtaining & submission of all necessary statutory approvals of completion plan from local authorities, Obtaining Fire clearance, Completion certificate, Occupancy Certificate from all concerned local bodies or any other authorities and consent to operate etc. all complete.	2.00		
13.3	Submission of all guarantee/warranty cards, operation & maintenance manual, as-built drawings and handing over (After completion of 13.2)	2.00		

8.4	Construction of Workman Housing Complex (BQ Area) on EPC mode -II of contract at OIL Duliajan, Assam Sub Station & Panel Room/Electrical Sub-station (Single Storey) (Tentative Plinth Area: 368 Sqm)
Payment for Civil Component of Sub Station & Panel Room/Electrical Sub-station (Single Storey): The cost of the Civil Component (denoted as A) shall be calculated as: $A = 22\% \times \text{Tendered Cost of Sub Station \& Panel Room/Electrical Sub-station (as quoted under Item No. 1.4 of the Financial Bid (BoQ))}$	
Payment for Electrical & Mechanical Component of SUB STATION & PANEL ROOM/Electrical Sub-station (Single Storey): The cost of the Electrical & Mechanical Component (denoted as B) shall be calculated as: $B = 78\% \times \text{Tendered Cost of Sub Station \& Panel Room/Electrical Sub-station (as quoted under Item No. 1.4 of the Financial Bid (BoQ))}$	

S. No.	Description of Item	Break up of total % age of A calculated as above from item no. 1.4 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
8.4 A	Construction of Sub Station & Panel Room/Electrical Sub-station Floors buildings (Civil Works)			
1	Designing, Detailing and obtaining approvals. Investigation, Planning, Designing and obtaining the approvals for the scope of work, Technical specification & Drawings etc. complete			
1.1	On Submission of Inception report, Survey Report, Soil Test Report	0.40	1.00	1.00
1.2	On Preparation of structural design drawings and vetting of the same from IIT	0.10		
1.3	On Submission & approval of working drawings including Architectural & all MEP services including revision; if any	0.25		
1.4	On Obtainng statutory clearance and approval of local bodies and concerned authorities	0.25		

S. No.	Description of Item	Break up of total % age of A calculated as above from item no. 1.4 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
2	Sub Station & Panel Room (Civil work)			
2.1	On completion of Jungle clearance, including uprooting of rank vegetation, grass, brushwood, Felling of trees of any girth and saplings. Removal of rubbish from outside the periphery of the area cleared, all cutting and filling in earth work as required including disposal of unsuitable/excess earth and supply of suitable/desired filling material for all leads and lifts, demolition of existing structure if any, anti-termite treatment, Laying of PCC for foundation, RCC footings & backfilling up to plinth level, RCC work in superstructure, CC works, Masonry work, trenching works (External & Internal), foundation of electrical equipments, flooring, stone works, all finishing - Internal & External Plaster, painting, door-window-ventilators including their fittings and fixtures, Aluminium & glass work including fittings and fixtures, rolling shutter works, water proofing works, parapet walls on terrace, steel work, chequered plate for internal trenches, precast cover for external trenches, framework to support edges of covers & trenches & other structural steel work, plinth protection around the building, chajja & projection etc. including all incidental works to complete and making the facilities functional all complete as per scope of work, drawings, specifications etc. and direction of Engineer-in-Charge.	93.00	93.00	94.00
3	On compliance of GRIHA items specifications to obtain minimum GRIHA 3 Star rating Certification as per scope of work.	1.00	1.00	95.00

S. No.	Description of Item	Break up of total % age of A calculated as above from item no. 1.4 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
4	On Completion of Construction, Testing commissioning of all services, fittings and fixtures, equipment's, submission of all guarantee/warranty cards, operation & maintenance manual, as-built drawings and handing over. Obtaining & submission of all necessary statutory approvals of completion plan from local authorities, Obtaining Fire clearance, Completion certificate, Occupancy Certificate from all concerned local bodies or any other authorities and consent to operate etc.			
4.1	On Completion of Construction, Testing commissioning of all services, fittings and fixtures, equipment's etc.	1.00	5.00	100.00
4.2	Obtaining & submission of all necessary statutory approvals of completion plan from local authorities, Obtaining Fire clearance, Completion certificate, Occupancy Certificate from all concerned local bodies or any other authorities and consent to operate etc. all complete	2.00		
4.3	Submission of all guarantee/warranty cards, operation & maintenance manual, as-built drawings and handing over (After completion of 4.2)	2.00		

8.5	Construction of Workman Housing Complex (BQ Area) on EPC mode -II of contract at OIL Duliajan, Assam. External development works including other ancillary infrastructure (Total Tentative area = 24675 Sqm)
Cost of Civil, Electrical & Mechanical Component work for External Development including Other ancillary infrastructure = 100% x Tendered Cost of External development works including other ancillary infrastructure (as quoted under Item No. 1.5 of the Financial Bid (BoQ))	

S. No.	Description of Item	Break up of total % age of rate quoted under Item No. 1.5 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
	External Development Works (Civil works)			
A	Designing, Detailing and obtaining approvals. Investigation, Planning, Designing and obtaining the approvals for the scope of work, Technical specification & Drawings etc. complete			
1.1	On Submission of Inception report, Survey Report, Soil Test Report	0.40	1.00	1.00
1.2	On Preparation of structural design drawings and vetting of the same from IIT	0.10		
1.3	On Submission & approval of working drawings including Architectural & MEP services including revision; if any	0.25		
1.4	On Obtainng statutory clearance and approval of local bodies and concerned authorities	0.25		
B	On Completion of External Development Works & other ancillary infrastructure works including Jungle clearance, including uprooting of rank vegetation, grass, brushwood, Felling of trees of any girth and saplings, as required, Removal of rubbish from outside the periphery of the area cleared, all cutting and filling in earth work as required including disposal of unsuitable/excess earth and supply of suitable/desired filling material for all leads and lifts, levelling of site to desired levels as per site requirements and			

S. No.	Description of Item	Break up of total % age of rate quoted under Item No. 1.5 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
	filling to achieve the proposed formation level as per drawings, Internal & External Roads, culverts, pathways, Water supply, Drainage, Sewerage, STP, UG tanks & pump room, Rain Water Harvesting system, signages, landscaping & horticulture, Guard rooms, security hut, boundary wall, parking sheds, electrical & mechanical works, all types of Developments works etc. Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge.			
1	SITE DEVELOPMENT WORKS (Leveling & original ground raising works, Internal & External Roads, parking areas, footpaths, Sewer, Water Supply, Storm Water Drains, Horticulture Operations, Boundary Wall)			
1.1	On completion of demolition, earth in cutting, filling, levelling, raising of original ground and compaction & consolidation as per relevant applicable specifications of whole site area as per direction of Engineer-in- Charge			
1.1.1	On completion of 25% total area as per GFC.	1.75	7.50	8.50
1.1.2	On completion of 50% total area as per GFC.	1.75		
1.1.3	On completion of 75% total area as per GFC.	1.75		
1.1.4	On completion of 100% total area as per GFC.	1.75		
1.1.5	On Handing over to Client	0.50		
1.2	On completion of external sewerage consisting of excavation, PCC, laying HDPE (DWC) pipes, earth filling, constructing chambers / all depth manholes from Ist manhole of buildings to STP and its connections etc. complete			
1.2.1	On completion of 25% total length as per	0.80	3.50	12.00

S. No.	Description of Item	Break up of total % age of rate quoted under Item No. 1.5 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
	GFC.			
1.2.2	On completion of 50% total length as per GFC.	0.80		
1.2.3	On completion of 75% total length as per GFC.	0.80		
1.2.4	On completion of 100% total length as per GFC.	0.80		
1.2.5	On handing over to client	0.30		
1.3	On completion of the external water supply system — including laying, jointing, testing, and commissioning of CPVC pipes of all diameters for external works; connections from underground water tanks (domestic and STP-treated) to buildings and irrigation points; construction of chambers of all required sizes; installation of valves; sand filling, backfilling, levelling, and all related works — the system shall be deemed complete in all respects	0.70	0.70	12.70
1.4	Providing and constructing the external storm water drainage system, comprising excavation and backfilling; construction of RCC drains with PCC and RCC base and walls, concrete flooring at the bottom, and formwork; supply and fixing of precast RCC covers (perforated and non-perforated) supported with MS edge framework and continuous MS angle along trench edges; laying of NP-2 class RCC Hume pipes of required diameter for road crossings, including PCC bedding, jointing, and encasement as necessary; construction of manholes, chambers, and catch basins of all depths; and connection of the drainage network to the Rainwater Harvesting (RWH) system through culverts with all associated civil works, fittings, and accessories — all complete as per drawings, specifications, and directions of the Engineer-in-Charge.			
1.4.1	On completion of 25% total length as per GFC.	1.25	5.70	18.40

S. No.	Description of Item	Break up of total % age of rate quoted under Item No. 1.5 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
1.4.2	On completion of 50% total length as per GFC.	1.25		
1.4.3	On completion of 75% total length as per GFC.	1.25		
1.4.4	On completion of 100% total length as per GFC.	1.25		
1.4.5	On handing over to client	0.70		
1.5	On completion of Internal & External Roads, parking areas, footpaths as per GFC drawings and directions of the Engineer-in-Charge.			
	Internal Road works, Parking areas, Footpaths			
1.5.1	Filling with desired/suitable material, preparation of subgrade and berms, laying of Granular Sub-Base (GSB) and Water Bound Macadam (WBM), including all necessary earthwork, compaction, and leveling — all in accordance with relevant specifications, drawings, and as directed by the Engineer-in-Charge.	6.00	16.40	34.80
1.5.2	Completion of Bituminous road i.e. spreading of primer/tack coat, laying & compaction of Dense Bituminous Macadam, Bituminous Concrete etc. complete in all respect as per the relevant specifications and directions of the Engineer-in-Charge.	7.00		
1.5.3	Completion of road/parking markings, signages etc. for Road/parking etc. complete in all respect as per the relevant specifications and directions of the Engineer-in-Charge.	1.00		
1.5.4	Completion of Footpath/Ramp works (filling with desired/suitable material, Preparation of bed, PCC, Sand and Paver block etc.), kerb stones, etc. Complete in all respect and as per direction of Engineer-in-Charge.	2.40		
	External Road works			
1.5.4	Preparation of subgrade and berms, including earthwork, dressing, and	0.90	0.90	35.70

S. No.	Description of Item	Break up of total % age of rate quoted under Item No. 1.5 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
	leveling; laying, rolling, and compaction of Granular Sub-Base (GSB) and Water Bound Macadam (WBM); construction of bituminous road pavement layers; construction of culverts and all associated road works — all complete in every respect as per relevant specifications, drawings, and directions of the Engineer-in-Charge.			
1.6	On completion of construction of the boundary wall, including jungle clearance; all required cutting and filling earthwork; demolition of existing structures (if any); RCC and CC works; masonry work; provision of weep holes; external plastering and finishing; painting; construction of expansion joints; steel works; providing and fixing of MS gates as per approved drawings; development of entrance structures; and all other incidental works necessary to complete the facility and make it functional — the work shall be deemed complete in all respects.			
1.6.1	On completion of 25% total length as per GFC.	3.00	12.50	48.20
1.6.2	On completion of 50% total length as per GFC.	3.00		
1.6.3	On completion of 75% total length as per GFC.	3.00		
1.6.4	On completion of 100% total length as per GFC.	3.00		
1.6.5	Providing and fixing of MS gates as per approved drawings; development of entrance structures etc. complete in all respect as per GFC drawing, specification and directions of the Engineer-in-Charge.	0.50		
1.7	Horticulture & Landscaping Operations including 300 mm earth filling, grassing, tree plantation/shrubs and plotted plants etc. complete	0.90	0.90	49.10

S. No.	Description of Item	Break up of total % age of rate quoted under Item No. 1.5 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
2	Construction of STP with civil, MEP all complete as per scope of works & GFC drawings as per capacity mentioned in scope of work, DBR/technical specifications with ultra filtration unit etc. & as per direction of Engineer-in-charge.			
2.1	On designing, supply and construction of STP including civil structure, electrical, mechanical works.	4.00	5.25	54.35
2.2	On installation, Testing commissioning and handing over to Client complete including civil structure, electrical, mechanical all complete.	1.25		
3	External Signages Works all Complete in all respect as per drawings, technical specifications, Scope of work etc. and direction of Engineer-in- Charge.			
3.1	Supply, fabrication, installation External Signages Works, including foundations all complete and handing over to Client.	0.30	0.30	54.65
4	Construction of UG tank & pump room complete including fire tank all related items etc.			
4.1	On completion of Earth work in excavation, Laying of PCC for foundation, RCC footings/raft, walls & backfilling work up to 1.5 m Height tank wall etc. Complete in all respect.	2.00	7.50	62.15
4.2	On Completion of the RCC frame structure above 1.5 m to top of the UGT tank & pump room, Water proofing works, flooring works, finishing works, railing works, door-window works etc. Complete in all respect.	4.00		
4.3	On supply & installation of all pumps, pipes, DB, panels, wires, cables, all pipe fittings, valves and other electrical and plumbing accessories complete as per scope of work	1.30		

S. No.	Description of Item	Break up of total % age of rate quoted under Item No. 1.5 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
4.4	On testing and commissioning and handing over to client	0.20		
5	Construction of Guard room (2 Nos.) & security hut (1 No.) Complete including all related items etc.			
5.1	On completion of Jungle clearance, including uprooting of rank vegetation, grass, brushwood, Felling of trees of any girth and saplings. Removal of rubbish from outside the periphery of the area cleared, all cutting and filling in earth work as required including disposal of unsuitable/excess earth and supply of suitable/desired filling material for all leads and lifts, demolition of existing structure if any, anti-termite treatment, Laying of PCC for foundation, RCC footings & backfilling up to plinth level, RCC work in superstructure, CC works, Masonry work, flooring, sill, skirting, stone works, all finishing Internal & External Plaster, painting, door-window-ventilators including their fittings and fixtures, Aluminium & glass work including fittings and fixtures, water proofing works, parapet walls on terrace, steel work, plinth protection around the building, chajja & projection, water supply & sanitary fittings and their connection, water tank etc. including all incidental works to complete and making the facilities functional all complete as per scope of work, drawings, specifications and direction of Engineer-in-Charge.	1.20	1.50	63.65
5.2	On completion of supply, installation, testing & commissioning of Electrical Wiring for light Point, Power Points(16A/6A), AC points, UPS, Distribution Board fixing with Submain wiring, 4P MCB, RCBO, RCCB, SPMCB, Wiring connections etc. in all respect.	0.30		

S. No.	Description of Item	Break up of total % age of rate quoted under Item No. 1.5 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
6	Construction of Rain water harvesting system all complete as per scope of works & GFC drawings as per capacity mentioned in scope of work, DBR/ technical specifications.			
6.1	On supply/construction/installation of RWH and including civil structure all complete.	1.20	1.80	65.45
6.2	On Testing commissioning and handing over to Client complete including civil structure all complete.	0.60		
7	Construction of Parking shed all Complete			
7.1	On completion of 25% total area as per GFC.	3.25	13.00	78.45
7.2	On completion of 50% total area as per GFC.	3.25		
7.3	On completion of 75% total area as per GFC.	3.25		
7.4	On completion of 100% total area as per GFC.	3.25		
8	IP Based CCTV System (Area to be covered: All External area, Entry, Exit points).			
8.1	Planning, Detailed Design, drawings including proof checking/ vetting and approval from Engineer-in-charge.(For CCTV System)	0.10	4.70	83.15
8.2	On completion of supply and installation of the IP based CCTV equipments such as PTZ cameras in external area and bullet camera at Entry and Exit gate, Poles, NVR, cabling, PoE switch, LAN cables, rack, patch panel, patch cord, OFC cable etc. complete in all respectpet. as required as per approved drawings.	3.75		

S. No.	Description of Item	Break up of total % age of rate quoted under Item No. 1.5 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
8.3	On successful testing of CCTV system including integration with all buildings CCTV system after Installation.	0.17		
8.4	On completion of Final testing & Commissioning of the complete CCTV system complete in all respect.	0.38		
8.5	On completion of Handing over to End user/ Client.	0.30		
9	IP Based EPABX system, LAN cable,OFC cable, core switch, access switch, CAT6A cables, IP Telephone, face plate, patchcord, RJ45 jack, LIU, Rack etc. complete in all respect.		1.20	84.35
9.1	Planning, Detailed Design, drawings including proof checking/ vetting and approval from Engineer-in-charge.(For IP based EPABX & LAN cables)	0.10		
9.2	On completion of supply and installation of IP Based EPABX system, LAN cable,OFC cable, core switch, access switch, CAT6A cables, IP Telephone, face plate, patchcord, RJ45 jack, LIU, Rack etc. complete in all respect.	0.74		
9.3	On completion of Final testing & Commissioning of the complete CCTV system complete in all respect.	0.18		
9.4	On completion of Handing over to End user/ Client.	0.18		
10	Online 3 phase sine wave UPS system with 30 minutes back up including maintenance free batteries, interconnecting cables, battery racks etc. complete in all respect.			
10.1	Planning, Detailed Design, drawings including proof checking/ vetting and approval from Engineer-in-charge.(For UPS system)	0.01		

S. No.	Description of Item	Break up of total % age of rate quoted under Item No. 1.5 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
10.2	On completion of supply & installation of online 3 phase sine wave UPS system with 30 minutes back up including maintenance free batteries, interconnecting cables, battery racks etc. complete in all respect.	0.34	0.50	84.85
10.3	On completion of Final testing & Commissioning of the UPS complete in all respect.	0.1		
10.4	On completion of Handing over to End user/ Client.	0.05		
11	Electrical External Development works			
11.1	Street lights,Hybrid solar street lights,street light panels, street light poles,bollard light, gate light, façade light, post top light, projector light for park including poles for other lights etc. complete in all respect.		2.30	87.15
11.1.1	Planning, Detailed Design including proof checking/ vetting and approval from Engineer-in-charge.(ForExternal lighting,Panels, cables etc.)	0.10		
11.1.2	On completion of supply & installation of Street lights,Hybrid solar street lights,street light panels, street light poles,bollard light, gate light, façade light, post top light, projector light for park including poles for other lights etc. complete in all respect.	1.5		
11.1.3	On completion of final testing & commissioning of Street lights,Hybrid solar street lights,street light panels, street light poles,bollard light, gate light, façade light, post top light, projector light for park including poles for other lights etc. complete in all respect.	0.34		
11.1.4	On completion of Handing over to End user/ Client.	0.21		

S. No.	Description of Item	Break up of total % age of rate quoted under Item No. 1.5 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
11.2	Boom Barrier			
11.2.1	Planning, Detailed Design including proof checking/ vetting and approval from Engineer-in-charge.(For Exit and Entry gates. etc.)	0.015		
11.2.2	On completion of supply & installation of Boom Barrier for Exit & Entry gates.	0.105		
11.2.3	On completion of final testing and commissioning of boom barriers.	0.015		
11.2.4	On completion of Handing over to End user/ Client.	0.015		
12	PNG gas piping system for each unit of residential block and guest house kitchen including gas meters, laying in campus, connection with main/city supply all complete.			
12.1	Planning, Detailed Design, drawings including proof checking/ vetting and approval from Engineer-in-charge.	0.06		
12.2	On completion of supply & installation of PNG gas piping system for each unit of residential block and guest house kitchen including gas meters.	4.75		
12.3	On completion of final testing and commissioning of PNG gas piping system.	0.68		
12.4	On Completion of PNG gas piping system, submission of As-built drawings, Obtaining & submission of all necessary statutory approvals of completion system , Obtaining Fire clearance, Completion certificate, Occupancy Certificate from all concerned local bodies or any other authorities and consent to operate etc. complete in all	0.68		
			6.85	94.00

S. No.	Description of Item	Break up of total % age of rate quoted under Item No. 1.5 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
	respect.			
12.5	On completion of Handing over to End user/ Client.	0.68		
13	On compliance of GRIHA items specifications to obtain minimum GRIHA 3 Star rating Certification as per scope of work.	1.00	1.00	95.00
14	On Completion of Construction, Testing commissioning of all services, fittings and fixtures, equipment's, Submission of all guarantee/warranty cards, operation & maintenance manual, as-built drawings and handing over. Obtaining & submission of all necessary statutory approvals of completion plan from local authorities, Obtaining Fire clearance, Completion certificate, Occupancy Certificate from all concerned local bodies or any other authorities and consent to operate etc.			
14.1	On Completion of Construction, Testing commissioning of all services, fittings and fixtures, equipment's etc.	1.00		
14.2	Obtaining & submission of all necessary statutory approvals of completion plan from local authorities, Obtaining Fire clearance, Completion certificate, Occupancy Certificate from all concerned local bodies or any other authorities and consent to operate etc. all complete.	2.00	5.00	100.00
14.3	Submission of all guarantee/warranty cards, operation & maintenance manual, as-built drawings and handing over (After completion of 14.2)	2.00		

8.1	Construction of Workman Housing Complex (BQ Area) on EPC mode -II of contract at OIL Duliajan, Assam. Residential Block (STILT+6, Total - 08 Towers) (Total Tentative Plinth Area of 08 towers = 27,355 Sqm)
Payment for Electrical & Mechanical Component of Residential Blocks: The cost of the Electrical & Mechanical Component (denoted as B) shall be calculated as: $B = 21\% \times \text{Tendered Cost of Residential Blocks (as quoted under Item No. 1.1 of the Financial Bid (BoQ))}$	
NOTE: The payment for Civil Works pertaining to each of the 08 Residential Blocks shall be regulated based on the following formula: Payment per Residential Tower = (B/8)	

Sr. No.	Description of Item	Break up of total % age of B/8 calculated as above from item no. 1.1 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
8.1 B	Design & Construction of Residential Blocks Total 08 Towers (STILT+6) Floors buildings (Electrical & Mechanical Works)			
1	Planning, Detail design, Drawings, supply, installation, testing and commissioning of internal / External Electrical Installations of Building; roof top solar system, rising mains, tap off boxes, MS/PVC Conduits, Wiring, Switch & sockets, AC points, geysers points, call bell point with image display system ,cable TV point, Distribution boards, LED lighting fixtures , aviation lights, Bulk head lights, fans, Power Metering & Distribution Panels, emergency lighting, illuminated signages , Lighting conductors, Earthings, cables, Power & Point wiring , Smart Energy meter metering Panels, Essential Panel etc. in all respect.			
1.1	Planning, detailed design, drawings including proof checking / vetting and approval from Engineer-in-charge. (Internal/External Electrical works of Building)	0.45	0.45	0.45
1.2	On completion of installation of MS/PVC Conducting Work		4.40	4.85

Sr. No.	Description of Item		Break up of total % age of B/8 calculated as above from item no. 1.1 of the Financial Bid (BoQ)		Cumulative % age
			Item % age	Sub- Group % age	
	1.2.1	On completion of Ground floor	0.60		
	1.2.2	On completion of 1st floor	0.60		
	1.2.3	On completion of 2nd floor	0.60		
	1.2.4	On completion of 3rd floor	0.60		
	1.2.5	On completion of 4th floor	0.60		
	1.2.6	On completion of 5th floor	0.60		
	1.2.7	On completion of 6th floor	0.60		
	1.2.5	On completion of terrace	0.20		
2	On completion of installation Electrical Wiring for light Point, Power Points(16A/6A), AC points, Geyser Points & UPS, Distribution Board fixing with Submain wiring, 4P MCB,RCBO, RCCB, SPMCB, Wiring connections etc. in all respect.			7.15	12.00%
	2.1	On completion of Ground floor	0.95		
	2.2	On completion of 1st floor	0.95		
	2.3	On completion of 2nd floor	0.95		
	2.4	On completion of 3rd floor	0.95		
	2.5	On completion of 4th floor	0.95		
	2.6	On completion of 5th floor	0.95		
	2.7	On completion of 6th floor	0.95		
	2.8	On completion of terrace	0.50		
3	On completion of installation of LED lighting fixtures, ceiling fans, exhaust fan, Modular switch sockets (16A/6A), modular TV sockets, call bell with image viewer, bulk head lights, aviation lights with astronomical timer & other accessories etc. in all respect.				

Sr. No.	Description of Item		Break up of total % age of B/8 calculated as above from item no. 1.1 of the Financial Bid (BoQ)		Cumulative % age
			Item % age	Sub- Group % age	
	3.1	On completion of Ground floor	1.75	12.75	24.75%
	3.2	On completion of 1st floor	1.75		
	3.3	On completion of 2nd floor	1.75		
	3.4	On completion of 3rd floor	1.75		
	3.5	On completion of 4th floor	1.75		
	3.6	On completion of 5th floor	1.75		
	3.7	On completion of 6th floor	1.75		
	3.8	On completion of terrace	0.50		
4	On completion of testing of the above Sr. No. 1,2 &3 internal electrical works in all respect.		0.80	0.80	25.55
5	On completion of supply & installation of Floor distribution panel in each floor, essential Panel, cabling etc. complete in all respect.		6.46	6.46	32.01
6	On completion of supply & installation of emergency lighting, illuminated signages etc.		2.18	2.18	34.19
7	On completion of supply & installation of Earthing Electrodes, strips for all indoor & external equipment and lighting protection system of building etc.		2.25	2.25	36.44
8	On completion of supply & installation of EV chargers/ Sockets for car & scooter as per tender with all accessories, electrical wiring & control cable etc. complete in all respect. (Location: Parking Areas).		2.20	2.20	38.64
9	Firefighting system (Down comer, fire extinguisher etc. as per NBC-2016 & Fire Norms.)				
9.1	Planning, Detailed Design, drawings & Provisional Approval from Statutory Bodies for firefighting drawings/ scheme and including proof checking/ vetting done by OIL and		1.00	1.00	39.64

Sr. No.	Description of Item		Break up of total % age of B/8 calculated as above from item no. 1.1 of the Financial Bid (BoQ)		Cumulative % age
			Item % age	Sub- Group % age	
	approval from Engineer-in-charge. (for Firefighting system)				
9.2	On supply & laying of all hydrant pipes, valves, hose reels, fire extinguisher, FHC Cabinet including door etc. Complete in all respect				
	9.2.1	On completion of Ground floor & External Areas	0.50	3.05	42.69%
	9.2.2	On completion of 1st floor	0.40		
	9.2.3	On completion of 2nd floor	0.40		
	9.2.4	On completion of 3rd floor	0.40		
	9.2.5	On completion of 4th floor	0.40		
	9.2.6	On completion of 5th floor	0.40		
	9.2.7	On completion of 6th floor	0.40		
	9.2.8	On completion of terrace	0.15		
10	On supply & installation of terrace pumps, Automatic Electrical panels, cables, control cables etc. complete in all respect.		2.50	2.50	45.19
11	On successful testing of all fire fighting equipment after Installation		1.00	1.00	46.19
12	Fire Detection Alarm and Public address system				
12.1	Planning, Detailed Design, drawings including proof checking/ vetting and approval from Engineer-in- charge.		0.10	0.10	46.29
12.2	On completion of supply & installation of Fire Alarm Panels with power back, detectors, Manual call box, Hooters, cables, speakers, conduits, amplifier, earthing etc. complete in all respect.				
	12.2.1	On completion of Ground floor	0.35	2.80	49.09

Sr. No.	Description of Item		Break up of total % age of B/8 calculated as above from item no. 1.1 of the Financial Bid (BoQ)		Cumulative % age
			Item % age	Sub- Group % age	
	12.2.2	On completion of Ist floor	0.35		
	12.2.3	On completion on 2 nd floor	0.35		
	12.2.4	On completion on 3 rd floor	0.35		
	12.2.5	On completion on 4 th floor	0.35		
	12.2.6	On completion on 5 th floor	0.35		
	12.2.7	On completion on 6 th floor	0.35		
	12.2.8	On completion on terrace	0.35		
13	On successful testing of Fire alarm & PA System equipments after Installation		0.25	0.25	49.34
14	Supply, Installation, Testing and Commissioning of Passenger Elevators (Lifts) (02 Nos.)				
14.1	Planning, Detailed Design, drawings including proof checking/ vetting and approval from Engineer-in-charge.		0.10	0.10	49.44
14.2	On supply of 1 st 15 Passenger elevator (lift) (Speed 1.5 m/s. , 07 stops) equipments, cables, control panels complete etc.		8.00	11.00	60.44
14.3	On installation of 1 st , 15 Passenger elevator (lift) (Speed 1.5 m/s. , 07 stops) equipments, cables, control panels complete etc.		1.50		
14.4	On successful testing and commissioning of 1 st 15 Passenger elevators (lifts) (Speed 1.5 m/s. , 07 stops) after installation.		1.50		
14.5	On supply of 2 nd , 15 Passenger/Stretchor elevator (lift) (Speed 1.5 m/s. , 07 stops) equipments, cables, control panels complete etc.		8.00	11.00	71.44
14.6	On installation of 2 nd , 15 Passenger/Stretchor elevator (lift) (Speed 1.5 m/s. , 07 stops) equipments, cables, control panels complete etc.		1.50		

Sr. No.	Description of Item	Break up of total % age of B/8 calculated as above from item no. 1.1 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
14.7	On successful testing and commissioning of 2 nd 15 Passenger elevators (lifts) (Speed 1.5 m/s. , 07 stops) after installation.	1.50		
15	Pressurized mechanical ventilation system in the lift well of elevators (lifts) (as per NBC-2016 & fire norms).			
15.1	Planning, Detailed Design, drawings including proof checking/ vetting and approval from Engineer-in-charge.	0.25	2.36	73.80
15.2	On completion of supply and installation of the Axial flow fans, ducting, panels, integration with fire panel etc.	1.86		
15.3	On successful testing & commissioning of equipments after Installation	0.25		
16	On completion of supply & installation of Rising mains, tap off boxes in each floor, End feed unit, cabling from tap off box to floor distribution panel and other services on all floors, earthing of all equipments etc. complete in all respect.	5.65	5.65	79.45
17	On completion of supply & installation of Earthing Electrodes, strips, earth pits for all indoor & external equipments and lighting protection system of building etc. complete in all respect.	2.25	2.25	81.70
18	On completion of supply & installation of EV chargers for car & scooter as per tender drawings with all accessories, including electrical wiring, RCBO, RCCB, MCB box & control cable etc. complete in all respect. (Location: Parking Areas).	2.20	2.20	83.90
19	On completion of supply & installation of online 3 phase sine wave UPS system with 30 minutes back up including maintenance free batteries, interconnecting cables, battery racks etc. complete in all respect.	1.52	1.52	85.42

Sr. No.	Description of Item	Break up of total % age of B/8 calculated as above from item no. 1.1 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
20	IP Based CCTV System (Area to be covered: Staircases, Elevator (Lift) lobby, common area in each floor and Parking area, staircases, elevator (lift) lobby, common area, stilt floor area, Entry & Exit points etc complete in all respect)			
20.1	Planning, Detailed Design, drawings including proof checking/ vetting and approval from Engineer-in-charge.(For CCTV System)	0.10	3.79	89.21
20.2	On completion of supply and installation of the CCTV equipment such as cameras in each floor including ground floor, NVR, cabling, PoE switch, rack, patch panel, OFC cable complete in all respect etc. as required as per approved drawings.	3.19		
20.3	On successful testing of CCTV system after Installation.	0.50		
21	Obtaining of Griha-3 star rating			
21.1	On Completion of orientation workshop by Griha Team	0.10	1.45	90.66
21.2	After 1st site visit during construction by Griha Team	0.25		
21.3	After 2nd site visit during construction by Griha Team	0.25		
21.4	After 3rd site visit during construction by Griha Team	0.25		
21.5	On Obtaining Griha 3 star rating certificate	0.60		
22	Roof top solar system			
22.1	Planning, Detailed Design, drawings including proof checking/ vetting done and approval from Engineer-in-charge.(For roof top solar System)	0.10	5.79	96.45

Sr. No.	Description of Item	Break up of total % age of B/8 calculated as above from item no. 1.1 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
22.2	On completion of supply and installation of the solar system equipments such as Solar photovoltaic panels (Monocrystalline), inverter, DCDB, ACDB, GI structure, AC&DC cables, Lighting protection system, earthing etc complete in all respect as per approved drawings.	5.19		
22.3	On successful testing of solar system after Installation.	0.50		
23	On Completion of Construction, submission of As-built drawings of E&M services, Obtaining & submission of all necessary statutory approvals of completion plan , Obtaining Fire clearance, Completion certificate, Occupancy Certificate from all concerned local bodies or any other authorities and consent to operate etc. complete in all respect.	1.05	1.05	97.50
24	On completion of Final testing & Commissioning of the building including all Electrical & Mechanical services and Handing Over of the building in all respect.	2.50	2.50	100.00

8.2	Construction of Workman Housing Complex (BQ Area) on EPC mode -II of contract at OIL Duliajan, Assam.
	Guest House (G+3) (Tentative Plinth Area = 1990 Sqm)
Payment for Electrical & Mechanical Component of Guest House (G+3):	
The cost of the Electrical & Mechanical Component (denoted as B) shall be calculated as: $B = 21\% \times \text{Tendered Cost of Guest House (as quoted under Item No. 1.2 of the Financial Bid (BoQ))}$	

Sr. No.	Description of Item	Break up of total % age of B calculated as above from item no. 1.2 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
8.2 B	Design & Construction of Guest House (G+3) Floors buildings (Electrical & Mechanical Works)			
1	Planning, detail design, drawings, supply, installation, testing and commissioning of internal / External Electrical Installations of Building; providing & fixing of split air conditioners, solar system, CCTV, EPABX, rising mains, tap off boxes, MS/PVC Conduits, Wiring, Switch & sockets, AC points, geysers points, call bell point with image display system, cable TV point, Distribution boards, LED lighting fixtures, aviation lights, Bulk head lights, fans, Power Metering & Distribution Panels, emergency lighting, illuminated signages, Lighting conductors, Earthings, cables, Power & Point wiring, Smart Energy meter metering Panels, Essential Panel etc. in all respect.			
1.1	Planning, Detailed Design, drawings including proof checking/ vetting and approval from Engineer-in-charge. (Internal Electrical works of Building)	0.45	0.45	0.45
1.2	On completion of installation of MS/PVC Conducting Work			
	1.2.1 On completion of Ground floor	0.60	2.60	3.05
	1.2.2 On completion of 1st floor	0.60		

Sr. No.	Description of Item		Break up of total % age of B calculated as above from item no. 1.2 of the Financial Bid (BoQ)		Cumulative % age
			Item % age	Sub- Group % age	
	1.2.3	On completion of 2nd floor	0.60		
	1.2.4	On completion of 3rd floor	0.60		
	1.2.5	On completion of terrace	0.20		
2	On completion of installation of Electrical Wiring for light Point, Power Points(16A/6A), AC points, Geyser Points & UPS, Distribution Board fixing with Submain wiring, 4P MCB, RCBO, RCCB, SPMCB, Wiring connections etc. in all respect.				
	2.1	On completion of Ground floor	0.95	4.30	7.35
	2.2	On completion of 1st floor	0.95		
	2.3	On completion of 2nd floor	0.95		
	2.4	On completion of 3rd floor	0.95		
	2.5	On completion of terrace	0.50		
3	On completion of installation of LED lighting fixtures , ceiling fans, exhaust fan, Modular switch sockets (16A/6A), modular TV sockets, call bell with image viewer, bulk head lights, aviation lights with astronomical timer & other accessories etc. in all respect.				
	3.1	On completion of Ground floor	1.75	7.50	14.85
	3.2	On completion of 1st floor	1.75		
	3.3	On completion of 2nd floor	1.75		
	3.4	On completion of 3rd floor	1.75		
	3.5	On completion of terrace	0.50		
4	On completion of testing of the above Sr.No.1,2&3 internal electrical works in all respect.		0.80	0.80	15.65
5	On completion of supply & installation of Floor distribution panel in each floor, essential Panel, cabling etc. complete in all respect.		6.46	6.46	22.11

Sr. No.	Description of Item	Break up of total % age of B calculated as above from item no. 1.2 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
6	On completion of supply & installation of emergency lighting, illuminated signages etc.	2.18	2.18	24.29
7	On completion of supply & installation of Earthing Electrodes, strips for all indoor & external equipment and lighting protection system of building etc.	2.25	2.25	26.54
8	On completion of supply & installation of EV chargers/ Sockets for car & scooter as per tender with all accessories, electrical wiring & control cable etc. complete in all respect. (Location: Parking Areas).	2.20	2.20	28.74
9	Firefighting system (Down comer, fire extinguisher etc. as per NBC-2016 & Fire Norms.)			
9.1	Planning, Detailed Design, drawings & Provisional Approval from Statutory Bodies for firefighting drawings/ scheme and including proof checking/ vetting and approval from Engineer-in-charge. (for Firefighting system)	1.00	1.00	29.74
9.2	On supply & laying of all hydrant pipes, valves, hose reels, fire extinguisher, FHC Cabinet including door etc.complete in all respect			
	9.2.1 On completion of Ground floor & External Areas	0.50	1.85	31.59
	9.2.2 On completion of 1st floor	0.40		
	9.2.3 On completion of 2nd floor	0.40		
	9.2.4 On completion of 3rd floor	0.40		
	9.2.5 On completion of terrace	0.15		
10	On supply & installation of terrace pumps ,Automatic Electrical panels, cables, control cables etc. complete in all respect.	3.50	3.50	35.09
11	On successful testing of all fire fighting equipment after Installation	1.00	1.00	36.09

Sr. No.	Description of Item		Break up of total % age of B calculated as above from item no. 1.2 of the Financial Bid (BoQ)		Cumulative % age
			Item % age	Sub- Group % age	
12	Fire Detection Alarm and Public address system				
12.1	Planning, Detailed Design, drawings including proof checking/ vetting and approval from Engineer-in- charge.		0.10	0.10	36.19
12.2	On completion of supply & installation of Fire Alarm Panels with power back, detectors, Manual call box, Hooters, cables, speakers, conduits, amplifier, earthing etc. complete in all respect.				
	12.2.1	On completion of Ground floor	0.40	2.00	38.19
	12.2.2	On completion of 1st floor	0.40		
	12.2.3	On completion on 2nd floor	0.40		
	12.2.4	On completion on 3rd floor	0.40		
	12.2.5	On completion on terrace	0.40		
13	On successful testing of Fire alarm & PA System equipment after Installation		0.25	0.25	38.44
14	Supply Installation Testing Commissioning of Passenger Elevators (Lifts) (01 No.)				
14.1	Planning, Detailed Design, drawings including proof checking/ vetting and approval from Engineer-in-charge. (For Passenger elevators (lifts))		0.10	0.10	38.54
14.2	On supply of 1st, 15 Passenger/Stretchor elevator (lift) (Speed 1.5 m/s. , 07 stops) equipments, cables, control panels complete etc.		9.00	12.00	50.54
14.3	On installation of 1st, 15 Passenger/Stretchor elevator (lift) (Speed 1.5 m/s. , 07 stops) equipments, cables, control panels complete etc.		1.50		
14.4	On successful testing and commissioning of 1st 15 Passenger/stretchor elevator (lift) (Speed 1.5 m/s. , 07 stops) after		1.50		

Sr. No.	Description of Item	Break up of total % age of B calculated as above from item no. 1.2 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
	installation.			
15	Pressurized mechanical ventilation system in the lift well (as per NBC-2016 & fire norms).			
15.1	Planning, Detailed Design, drawings including proof checking/ vetting and approval from Engineer-in-charge.	0.25	2.36	52.90
15.2	On completion of supply and installation of the Axial flow fans, ducting, panels, integration with fire panel etc.	1.86		
15.3	On successful testing & commissioning of equipments after Installation	0.25		
16	On completion of supply & installation of Rising mains, tap off boxes in each floor, End feed unit, cabling from tap off box to floor distribution panel for building services on all floors, earthing of all equipments etc. complete in all respect.	5.65	5.65	58.55
17	On completion of supply & installation of Earthing Electrodes, strips, earth pits for all indoor & external equipments and lighting protection system of building etc. complete in all respect.	2.25	2.25	60.80
18	On completion of supply & installation of EV chargers for car & scooter as per tender drawings with all accessories, including electrical wiring & control cable etc. complete in all respect. (Location: Parking Areas).	2.20	2.20	63.00
19	On completion of supply & installation of online 3 phase sine wave UPS system with 30 minutes back up including maintenance free batteries, interconnecting cables, battery racks etc. complete in all respect.	3.52	3.52	66.52

Sr. No.	Description of Item	Break up of total % age of B calculated as above from item no. 1.2 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
20	IP Based CCTV System (Area to be covered: All Entry, Exit points, staircases, corridors, Elevator (Lift) lobby at each floor).			
20.1	Planning, Detailed Design, drawings including proof checking/ vetting and approval from Engineer-in-charge.(For CCTV System)	0.10	9.60	76.12
20.2	On completion of supply and installation of the CCTV equipment such as cameras in each floor including ground floor, NVR, cabling, PoE switch, rack, patch panel, OFC cable etc complete in all respectpet. as required as per approved drawings.	9.00		
20.3	On successful testing of CCTV system after Installation.	0.50		
21	IP Based EPABX system, LAN cable, OFC cable, core switch, access switch, CAT6A cables, IP Telephone, face plate, patchcord, RJ45 jack, LIU, Rack etc. complete in all respect.			
21.1	Planning, Detailed Design, drawings including proof checking/ vetting and approval from Engineer-in-charge.(For IP based EPABX & LAN cables)	0.10	10.09	86.21
21.2	On completion of supply and installation of IP Based EPABX system, LAN cable, OFC cable, core switch, access switch, CAT6A cables, IP Telephone, face plate, patchcord, RJ45 jack, LIU, Rack etc. complete in all respect.	9.49		
21.3	On successful testing of IP Based EPBAX & associated LAN cables system after Installation.	0.50		
22	Obtaining of Griha-3-star rating certificate			
22.1	On Completion of orientation workshop by Griha Team	0.10	1.45	87.66
22.2	After 1st site visit during construction by Griha Team	0.25		

Sr. No.	Description of Item	Break up of total % age of B calculated as above from item no. 1.2 of the Financial Bid (BoQ)		Cumulative % age
		Item % age	Sub- Group % age	
22.3	After 2nd site visit during construction by Griha Team	0.25		
22.4	After 3rd site visit during construction by Griha Team	0.25		
22.5	On Obtaining Griha 3 star rating certificate	0.60		
23	Roof top solar system			
23.1	Planning, Detailed Design, drawings including proof checking/ vetting and approval from Engineer-in-charge. (For roof top solar System)	0.10	8.79	96.45
23.2	On completion of supply and installation of the solar system equipments such as Solar photovoltaic panels (Monocrystalline), inverter, DCDB, ACDB, GI structure, AC&DC cables, Lighting protection system, earthing etc complete in all respect as per approved drawings.	8.19		
23.3	On successful testing of solar system after Installation.	0.50		
24	On Completion of Construction, submission of As-built drawings of E&M services, Obtaining & submission of all necessary statutory approvals of completion plan, Obtaining Fire clearance, Completion certificate, Occupancy Certificate from all concerned local bodies or any other authorities and consent to operate etc. complete in all respect.	1.05	1.05	97.50
25	On completion of Final testing & Commissioning of the building including all Electrical & Mechanical services and Handing Over of the building to end user in all respect.	2.50	2.50	100.00

8.3	Construction of Workman Housing Complex (BQ Area) on EPC mode -II of contract at OIL Duliajan, Assam Community Centre (G+1) (Tentative Plinth Area = 1035 sqm)
Payment for Electrical & Mechanical Component of Community Centre (G+1): The cost of the Electrical & Mechanical Component (denoted as B) shall be calculated as: $B = 23\% \times \text{Tendered Cost of Community Centre (as quoted under Item No. 1.3 of the Financial Bid (BoQ))}$	

Sr.No.	Description of Item	Break up of total % age of B calculated as above from item no. 1.3 of the Financial Bid (BoQ)		Cumulative age
		Item % age	Sub- Group % age	
8.3 B	Design & Construction of Community Centre (G+1) Floors buildings (Electrical & Mechanical Works)			
1	Planning, detail design, drawings, supply, installation, testing and commissioning of internal / External Electrical Installations of Building; MS/PVC Conduits, VRF HVAC system, roof top solar system, CCTV, LAN, IP based EPABX system, Wiring, Switch & sockets, AC points, geysers points, call bell point with image display system, cable TV point, Distribution boards, LED lighting fixtures, aviation lights, Bulk head lights, fans, Power Metering & Distribution Panels, emergency lighting, illuminated signages, Lighting conductors, Earthing's, cables, Power & Point wiring, Smart Energy meter metering Panels, Essential Panel etc. in all respect.			
1.1	Planning, Detailed Design, drawings including proof checking/ vetting and approval from Engineer-in-charge. (Internal / External Electrical works of Building)	0.45	0.45	0.45
1.2	On completion of MS/PVC Conducting Work		8.20	8.65
	1.2.1 On completion of Ground floor	4.00		
	1.2.2 On completion of Ist floor	4.00		

Sr.No.	Description of Item		Break up of total % age of B calculated as above from item no. 1.3 of the Financial Bid (BoQ)		Cumulative age
			Item % age	Sub- Group % age	
	1.2.3	On completion of terrace	0.20		
2	On completion of Electrical Wiring for light Point, Power Points(16A/6A), AC points,Geyser Points & UPS, Distribution Board fixing with Submain wiring, 4P MCB,RCBO, RCCB, SPMCB, Wiring connections etc. in all respect.				
	2.1	On completion of Ground floor	5.00	10.50	19.15
	2.2	On completion of 1st floor	5.00		
	2.3	On completion of terrace	0.50		
3	On completion of installation of LED lighting fixtures , ceiling fans, exhaust fan, Modular switch sockets (16A/6A), modular TV sockets, call bell with image viewer, bulk head lights, aviation lights with astronomical timer & other accessories etc. in all respect.				
	3.1	On completion of Ground floor	5.00	10.50	29.65
	3.2	On completion of 1st floor	5.00		
	3.3	On completion of terrace	0.50		
4	On completion of testing of the above internal electrical works in all respect.		0.80	0.80	30.45
5	On completion of supply & installation of Floor distribution panel in each floor, essential Panel, cabling etc. complete in all respect.		8.00	8.00	38.45
6	On completion of supply & installation of emergency lighting, illuminated signages at each floor etc. complete in all respect.		2.18	2.18	40.63
7	On completion of supply & installation of Earthing Electrodes, strips for all indoor & external equipment and lighting protection system of building etc.		5.00	5.00	45.63

Sr.No.	Description of Item	Break up of total % age of B calculated as above from item no. 1.3 of the Financial Bid (BoQ)		Cumulative age
		Item % age	Sub- Group % age	
8	On completion of supply & installation of EV chargers/ Sockets for car & scooter as per tender with all accessories, electrical wiring & control cable etc. complete in all respect. (Location: Parking Areas).	2.20	2.20	47.83
9	Firefighting system (Down comer, fire extinguisher etc. as per NBC-2016 & Fire Norms.)			
9.1	Planning, Detailed Design, drawings & Provisional Approval from Statutory Bodies for firefighting drawings/ scheme and including proof checking/ vetting and approval from Engineer-in-charge. (for Firefighting system)	1.00	1.00	48.83
9.2	On supply & laying of all hydrant pipes, valves, hose reels, fire extinguisher, FHC Cabinet including door etc. complete in all respect			
	9.2.1 On completion of Ground floor & External Areas	1.00	1.65	50.48
	9.2.2 On completion of 1st floor	0.50		
	9.2.3 On completion of terrace	0.15		
10	On supply & installation of terrace pumps ,Automatic Electrical panels, cables, control cables etc. complete in all respect.	3.50	3.50	53.98
11	On successful testing of all fire fighting equipment after Installation	1.00	1.00	54.98
12	Fire Detection Alarm and Public address system			
12.1	Planning, Detailed Design, drawings including proof checking/ vetting and approval from Engineer-in- charge.	0.10	0.10	55.08
12.2	On completion of supply & installation of Fire Alarm Panels with power back, detectors, Manual call box, Hooters, cables, speakers, conduits, amplifier, earthing etc. complete in all respect.			

Sr.No.	Description of Item		Break up of total % age of B calculated as above from item no. 1.3 of the Financial Bid (BoQ)		Cumulative age
			Item % age	Sub- Group % age	
	12.2.1	On completion of Ground floor	2.00	7.00	62.08
	12.2.2	On completion of Ist floor	2.00		
	12.2.3	On completion on terrace	2.00		
13	On successful testing of Fire alarm & PA System equipment after Installation		1.00		
14	Pressurized mechanical ventilation system in the elevator (lift) well (as per NBC-2016 & fire norms).				
14.1	Planning, Detailed Design, drawings including proof checking/ vetting and approval from Engineer-in-charge.		0.25	2.36	64.44
14.2	On completion of supply and installation of the Axial flow fans (for fresh & Exhaust air), inline fans for toilets, ducting, electrical panels, integration with fire panel etc.		1.86		
14.3	On successful testing & commissioning of equipments after Installation		0.25		
15	On completion of supply & installation of Earthing Electrodes, strips, earth pits for all indoor & external equipments and lighting protection system of building etc. complete in all respect.		2.25	2.25	66.69
16	VRF System				
16.1	Planning, Detailed Design, drawings including proof checking/ vetting and approval from Engineer-in-charge.		0.25	12.25	78.94
16.2	On completion of supply & installation of VRF system, indoor & outdoor units, ducting, Axial flow fans, Electrical panel, grill, diffusers etc. with all accessories, complete in all respect.		10.00		
16.3	On successful testing & commissioning of equipments after Installation		2.00		

Sr.No.	Description of Item	Break up of total % age of B calculated as above from item no. 1.3 of the Financial Bid (BoQ)		Cumulative age
		Item % age	Sub- Group % age	
17	IP Based CCTV System (Area to be covered: All Entry, Exit points, staircases, corridor, common hall etc. complete in all respect).			
17.1	Planning, Detailed Design, drawings including proof checking/ vetting and approval from Engineer-in-charge.(For CCTV System)	0.10	9.44	88.38
17.2	On completion of supply and installation of the CCTV equipment such as cameras in each floor including ground floor, NVR, cabling, PoE switch, rack, patch panel, OFC cable complete in all respectpet etc. as required as per approved drawings.	8.84		
17.3	On successful testing of CCTV system after Installation.	0.50		
18	IP Based EPABX system, LAN cable, OFC cable, core switch, access switch, CAT6A cables, IP Telephone, face plate, patchcord, RJ45 jack, LIU, Rack etc. complete in all respect.			
18.1	Planning, Detailed Design, drawings including proof checking/ vetting done by OIL and approval from Engineer-in-charge.(For IP based EPABX & LAN cables)	0.10	3.09	91.47
18.2	On completion of supply and installation of IP Based EPABX system, LAN cable, OFC cable, core switch, access switch, CAT6A cables, IP Telephone, face plate, patchcord, RJ45 jack, LIU, Rack etc. complete in all respect.	2.49		
18.3	On successful testing of IP Based EPBAX & associated LAN cables system after Installation.	0.50		
19	Obtaining of Griha-3 star rating certificate			
19.1	On Completion of orientation workshop by Griha Team	0.10	1.45	92.92

Sr.No.	Description of Item	Break up of total % age of B calculated as above from item no. 1.3 of the Financial Bid (BoQ)		Cumulative age
		Item % age	Sub- Group % age	
19.2	After 1st site visit during construction by Griha Team	0.25		
19.3	After 2nd site visit during construction by Griha Team	0.25		
19.4	After 3rd site visit during construction by Griha Team	0.25		
19.5	On Obtaining Griha 3 star rating certificate	0.60		
20	Roof top solar system			
20.1	Planning, Detailed Design, drawings including proof checking/ vetting and approval from Engineer-in-charge. (For roof top solar System)	0.10	3.53	96.45
20.2	On completion of supply and installation of the solar system equipments such as Solar photovoltaic panels (Monocrystalline), inverter, DCDB, ACDB, GI structure, AC&DC cables, Lighting protection system, earthing etc complete in all respect as per approved drawings.	2.93		
20.3	On successful testing of solar system after Installation.	0.50		
21	On Completion of Construction, submission of As-built drawings of E&M services, Obtaining & submission of all necessary statutory approvals of completion plan, Obtaining Fire clearance, Completion certificate, Occupancy Certificate from all concerned local bodies or any other authorities and consent to operate etc. complete in all respect.		1.05	97.50
22	On completion of Final testing & Commissioning of the building including all Electrical & Mechanical services and Handing Over of the building to end user in all respect.		2.50	100.00

8.4	Construction of Workman Housing Complex (BQ Area) on EPC mode -II of contract at OIL Duliajan, Assam Sub Station & Panel Room/Electrical Sub-station (Single Storey) (Tentative Plinth Area: 368 Sqm)
Payment for Electrical & Mechanical Component of SUB STATION & PANEL ROOM/Electrical Sub-station (Single Storey): The cost of the Electrical & Mechanical Component (denoted as B) shall be calculated as: $B = 78\% \times \text{Tendered Cost of Sub Station \& Panel Room/Electrical Sub-station (as quoted under Item No. 1.4 of the Financial Bid (BoQ))}$	

Sr. No.	Description of Item	Break up of total % age of B calculated as above from item no. 1.4 of the Financial Bid (BoQ)		Cumulative age
		Item % age	Sub- Group % age	
8.4 B	Construction of Sub Station & Panel Room/Electrical Sub-station Floors buildings (Electrical & Mechanical Works)			
	Electric Substation: Planning, Design, supply, installation, testing & commissioning of 11KV High Voltage Panels with 30 min battery power back up for control circuit, 11/.433KV Dry type step down Transformers, complete with associated auxiliaries as specified in technical specifications. High voltage cables for inter-connection between the H.V. panel and transformers including end terminations. M.V. Panels. Bus trunking (sandwich type)/cables for inter-connection between transformer's M.V. terminals and the M.V. Panel as specified. Automatic Power Factor Improvement Panel if required. active harmonic filters, TVSS (transient voltage suppression system), SPD (surge protection system), Earthing system for entire substation equipments. Safety Equipments, framed instruction chart & SLDs & fire suppression system as per technical specifications. Emergency M.V. Panel. DG set with AMF panel, MS structure			

Sr. No.	Description of Item	Break up of total % age of B calculated as above from item no. 1.4 of the Financial Bid (BoQ)		Cumulative age
		Item % age	Sub- Group % age	
	for DG exhaust, cables for inter connection between DG set to AMF panel, Main LT panel to AMF panel & AMF panel to Emergency panel, LT Power & Control cables including end terminations. External Electrical service connections, all electrical equipments foundations, Internal & External electrical works, fire detection and Alarm system etc. complete as required. Miscellaneous items required for successful testing and commissioning of substation equipments as per direction of Engineer-In-Charge.			
1	Planning,design & drawings of Substation equipments/materials,SLDs, all external lighting drawings & HT & LT Panel drawings, DG Exhaust structure drawing (E&M) etc. as required including proof checking/ vetting and approval from Engineer-in-charge.	0.50	0.50	0.50
2	On completion of installation & testing of Substation internal electrical works (MS conduiting, wiring, LED lighting, modular switch & sockets, fans, exhaust fans, DB,MCB etc. as required.	5.00	5.00	5.50
3	On supply of 11/433KV, Dry type Transformers with OLTC etc.	15.00	15.00	20.50
4	On supply of DG set with AMF Panel,including DG exhaust structure as per norms etc.	7.00	7.00	27.50
5	On supply of HT & LT panels,Main Emergency panel, APFC panel if required,sandwich type bus duct,HT/LT cables,end terminations,All building floor power distribution panels,building emergency panels,rubber mats, Exhaust fans, all equipment earthings with strips, safety equipments, Danger	30.00	30.00	57.50

Sr. No.	Description of Item	Break up of total % age of B calculated as above from item no. 1.4 of the Financial Bid (BoQ)		Cumulative age
		Item % age	Sub- Group % age	
	notice board, instruction chart, SLD with frame with all other associated equipment & materials etc. (including Civil foundations of all equipments etc.)			
6	On installation of Dry type transformers,DG set, AMF Panel, DG exhaust structure, HT & LT panels,Main Emergency panel, APFC panel if required,sandwich type bus duct,HT/LT cables,end terminations,All building floor power distribution panels,building emergency panels,rubber mats, Exhaust fans, all equipment earthings with strips, safety equipments, Danger notice board, instruction chart, SLD with frame with all other associated equipment & materials etc. (including Civil foundations of all equipments etc.)	12.50	12.50	70.00
7	On completion of supply and installation of fire suppression, fire detection & alarm system for substation as per fire norms	5.00	5.00	75.00
8	on completion of supply and laying of cable from main LT Panel in substation to residential block rising mains(08Nos.),guest house rising mains, community center building panels & for other services panels (HVAC, Fire Fighting, plumbing, STP panel & street light panels, interconnections of street,hybrid solar lights, facade lights other services panel etc.)and connection etc. complete in all respect.	12.50	12.50	87.50
9	Obtaining of Griha 3-star rating certificate	1.50	1.50	89.00

Sr. No.	Description of Item	Break up of total % age of B calculated as above from item no. 1.4 of the Financial Bid (BoQ)		Cumulative age
		Item % age	Sub- Group % age	
10	On Completion of Obtaining & submission of all necessary statutory approvals of completion plan from local authorities, Obtaining Fire clearance, site test reports, documents required to obtain CEA certificate, Completion certificate, Occupancy Certificate, CEA certificate, CPCB clearance from all concerned local bodies or any other authorities and consent to operate etc. in all respect.	1.00	1.00	90.00
11	On Completion of Construction, submission of As-built drawings of E&M services along with operation & maintenance manual, guarantee/Warrantee certificate etc. complete as required.	1.00	1.00	91.00
12	On completion of final testing and Commissioning of the substation equipments including all work services and Handing Over of the building to end user with all respect.	9.00	9.00	100.00

B. SPECIAL CONDITIONS OF CONTRACT - GENERAL

1. GENERAL REQUIREMENTS

- 1.1 The EPC Contractor shall abide to the restrictions, rules, regulations and timings imposed by OIL India Limited on the working and on movement of labour, materials etc.
- 1.2 No unauthorized person should be allowed to enter workspace without following the prescribed safety norms. All the dangerous moving parts of the portable / fixed machineries being used shall be adequately guarded. Inserting of bare wires for tapping the power from electrical sockets is completely prohibited. All unsafe conditions, unsafe acts identified by EPC Contractor, reported by site supervisors and / or safety personnel to be corrected immediately.

2. ENGAGEMENT OF AGENCY FOR SPECIALIZED NATURE OF ITEMS/ WORK:

- 2.1 For specialized items of works, the EPC contractor has to engage specialized agencies/firms who have work experience and submit work experience certificate of satisfactory completion of similar works as per following criteria during the last 07 (seven) years prior to the last stipulated date for submission of the bid. The decision of Employer or his representative in this regard shall be final and binding on the Agency.

- a) Three works each quantity not less than 40% of total amount in the work of similar work concerned.

OR

- b) Two works each quantity not less than 50% of total amount in the work of similar work concerned.

OR

- c) One work quantity not less than 80% of total amount in the work of similar work concerned.

2.2 Specialized items of work (Civil):

- i) Water Proofing Treatment (10 years guarantee period)
- ii) Post construction Anti-Termite chemical treatment (10 years guarantee period).
- iii) Sanitary, Plumbing and drainage work
- iv) Fire Doors
- v) False Ceiling & interior works.
- vi) Soil Investigation and Survey Work.
- vii) PEB BUILDING
- viii) Structural Design
- ix) Sewage Treatment Plant

- x) Rain Water Harvesting
- xi) Invisible Grill

(The EPC Contractor shall give specific guarantees that they shall be responsible for removal of any defect cropping up in these specialized works within the Guarantee period. The form in which the guarantee is to be executed by the EPC contractor on a stamp paper of the required value is at Annexure E-1 & E-2 of GCC (Part-3) respectively for Water Proofing Works and Anti Termite Treatment works).

2.3 Specialized items of work (Electrical):

- i) Fire Alarm System
- ii) Fire Fighting System
- iii) Elevator (Lift)
- iv) VRF and split
- v) CCTV & LAN system
- vi) Solar Photovoltaic Power Plant
- vii) DG set
- viii) Sub-station (Transformer, DG Set, HT/LT panels etc.)
- ix) Gas Pipe Line system

Note:

- i) The Contractor who possesses the expertise in any of the specialized work mentioned above shall be allowed for carry out such specialized work by Engineer-in-charge if documents establish the same.
- ii) The EPC Contractor shall submit the credential of specialized agency well in advance as per the direction of Engineer-in-charge. After verification of the same, written approval will be conveyed to the EPC Contractor in this regard. The EPC Contractor shall not change the specialized agency. However, if the change is warranted, he may do so, with permission of Engineer-in-charge. However, before making any such change, he has to enter into similar agreement as with previous agency & submit the same to Engineer-in-Charge for approval. This shall however be without any change in the accepted rates of the Contract Agreement and without any cost implications to RITES/ OIL India Limited. In case Engineer-in-Charge is not satisfied with the performance of any agency, he can direct the EPC Contractor to change the agency/contractor executing such items/components of works and this shall be binding on EPC Contractor. No claim of hindrance in the work shall be entertained from the EPC Contractor on this account.
- iii) It shall be the responsibility of EPC Contractor to sort out any dispute / litigation with the Specialized Agencies without any time & cost overrun to RITES/Oil India Limited. The EPC

Contractor shall be solely responsible for settling any dispute / litigation arising out of his agreement with the Specialized Agencies. The EPC Contractor shall ensure that the work shall not suffer on account of litigation/ dispute between him and the specialized agencies. No claim of hindrance in the work shall be entertained from the EPC Contractor on this account. No extension of time shall be granted and no claim whatsoever, of any kind, shall be entertained from the EPC Contractor on account of delay attributable to the selection/rejection of the Specialized Agencies or any dispute amongst them.

3. PROGRAM SCHEDULE

3.1 The EPC Contractor shall prepare an integrated program chart within fifteen days of issue of award letter including civil as well as E&M activities for the execution of work, showing clearly all activities from the start of work to completion, with details of manpower, equipment and machinery required for the fulfilment of the program within the stipulated period and submit the same for approval of the Engineer-In-Charge. The integrated program chart so submitted should not have any discrepancy with the physical milestones attached in the contract agreement. The program chart should include the following:

- (i) Descriptive note explaining sequence of various activities.
- (ii) Construction programme prepared on PRIMAVERA software, which will indicate resource in financial term, manpower and specialized equipment for every important stage. One planning engineer should be engaged in project who is familiar in PRIMAVERA software. Training for the latest PRIMAVERA software will be organized by EPC contractor for RITES/OIL engineers. No extra payment shall be made in this regard to the EPC contractor. Two (02) PRIMAVERA software licenses shall be provided by the EPC contractor to RITES along with program schedule for monitoring purpose. Nothing extra shall be paid in this regard.
- (iii) Program for procurement of materials by the EPC Contractor.
- (iv) Program for arranging and deployment of manpower both skilled and unskilled so as to achieve targeted progress.
- (v) Program of procurement of machinery/equipment having adequate capacity, commensurate with the quantum of work to be done within the stipulated period, by the EPC Contractor.
- (vi) Program for achieving fortnightly micro milestones and periodic milestones.
- (vii) In case of non-compliance/delay in compliance in this, a penalty @ Rs. 5,000/- per day will be imposed which will be recovered from the immediate next R/A Bill of the EPC Contractor.
- (viii) If at any time, it appears to the Engineer-In-Charge that the actual progress of work does not conform to the approved program referred above, the EPC Contractor shall produce a revised program showing the modifications to the approved program by additional inputs to ensure completion of the work within the stipulated time.

- (ix) The submission for approval by the Engineer-In-Charge of such program or the furnishing of such particulars shall not relieve the EPC Contractor of any of his duties or responsibilities under the contract. This is without prejudice to the right of Engineer-In-Charge to take action against the EPC Contractor as per terms and conditions of the agreement.

4. BUILDING INFORMATION MODELING (BIM)

All the 2D preliminary architectural drawings will be provided by RITES in soft copy of AutoCAD format. EPC contractor has to develop 3D models on any BIM application with minimum LOD 350 within one month after award of work and get it verified from the project consultant (appointed by EPC Contractor) & RITES. EPC contractor shall provide one (1) License exclusively without editing rights and one (1) license with editing rights for the RITES. During progress of the work, the EPC contractor will update actual progress in the BIM model and submit the required progress vis-à-vis actual progress through BIM modelling to the RITES fortnightly or as and when asked by the RITES.

If any clash deduction / discrepancy and missing information is found in already issued GFC or any other drawings and the drawings generated for 3D model from BIM application, the rectification in 3D model shall be done in consultation with the consultant Architect/ RITES and the final GFC drawings will be generated & will be used for construction.

The BIM professional to be appointed by the EPC contractor shall have the experience of development of 3D model LOD 350. The credential of BIM professional shall be submitted to Engineer in charge within 15 days from the award of work & got approved from Engineer -in- Charge.

5. MODE OF MEASUREMENT [FOR RECORD PURPOSE]

- 5.1 The EPC contractor shall prepare and submit a detailed estimate for the disciplines of Civil and Electrical within three months from date of start of work. The detailed estimate shall include all item of work necessary for completion of work as per relevant approved architectural, structural and services drawings and scope of work along with details of measurements of items. The detailed estimate shall be based on latest DSR (2023) published by CPWD and prevailing market rates.
- 5.2 The stage-wise measurement as per schedule of stage payment shall be recorded in E-Measurement Book/computerized Measurement Book by the EPC contractor.
- 5.3 In addition, the EPC contractor shall submit theoretical consumption statements for the items involving use of cement, steel reinforcement, chemical, paints, bitumen etc. as directed by the Engineer-in-Charge along with every running bill for record and reconciliation of materials issued, consumed and balance.
- 5.4 This measurement shall then be 100% checked by the authorized representative of the Engineer-in-charge. The EPC contractor shall incorporate all such changes or correction, as may be done during these checks, to his draft measurement book and submit the corrected measurement Book with its pages machine numbered to the Engineer-in-charge.

6. SUBMISSION OF AS BUILT DRAWINGS

- 6.1 EPC Contractor shall prepare & submit, on completion of construction work, 'as built' 3D BIM model to be given as soft copy & 'as-built' drawings extracted from 'as-built' 3D BIM model to Engineer-in-Charge for maintenance, repair & modification purposes.
- 6.2 The EPC Contractor shall make available five (05) sets (A0/A1 size as directed by Engineer in Charge) of completed Building Drawings, "As Built Drawings", Completion plans, Service plans etc. along with Inventory, Maintenance Manual / Standard Operating Procedure (SOP), for Equipment's and Plants, warranty certificates by manufacture etc. of various installed fittings, fixtures and equipment for the completed projects. This shall be the prerequisite for payment of final bill. The EPC contractor should also provide as-built drawings for E/M services besides civil & plumbing mentioned therein.
- 6.3 The EPC Contractor shall also make available five (05) sets of all drawings of internal and external services i.e., Water Supply, Sanitary line, Drainage lines, Firefighting etc. This shall be the prerequisite for payment of final bill. These drawings shall have the following information:
- a) Run off for all piping and their diameters including soil, waste pipes and vertical stacks.
 - b) Ground and invert level of all drainage pipes together with locations of all manholes and connections, up to outfall.
 - c) Run off for all water supply lines with diameters location of control valves, access panels etc.

7. THIRD PARTY QUALITY ASSURANCE (TPQA):

OIL India Limited may appoint third-party quality assurance agency (TPQA)/ committee to inspect the works and advice OIL India Limited on the quality, workmanship, processes, etc. The EPC Contractor shall assist the TPQA agency/Committee appointed by Oil India Limited in accessing information and conducting inspections. The EPC Contractor shall arrange to remove deficiencies brought out by TPQA in a reasonable time and shall submit compliance report to settle TPQA's paras / observations. Oil India Limited may also constitute a separate Review Committee consisting of a representative each from the Administrative Ministry, Finance (Internal Finance Wing) and the Executing Agency to review the progress of the work.

The EPC Contractor shall assist the RITES/OIL India Limited for reply to and comply with the observations pertaining to works made by CTE's branch, CAG, Vigilance and other competent authorities and remain liable for settlement of their observations/ objections/ actions, if any.

In case of observations on OIL India Limited by Internal Audit of the Ministry of Human Resource Development, CVC, etc., the EPC Contractor shall provide relevant information and assist RITES/OIL India Limited in replying to these statutory authorities.

All expenditure to be incurred for testing of all the samples e.g., materials for testing, packing, sealing, transportation, loading, unloading etc. including testing charges shall be borne by the EPC Contractor at his own cost and nothing extra shall be payable on this account.

8. AFTER PROJECT SERVICE AND WARRANTIES OF EQUIPMENT

The EPC contractor shall provide all back-to-back warranties and maintenance manual from the suppliers of all equipment & fixtures to RITES/ OIL India Limited on completion of work.

9. CONDITION OF NATIONAL GREEN TRIBUNAL

- 9.1 The EPC Contractor shall not store/ dump construction material or debris on the metalled roads within the campus as well as the peripheral roads around the campus.
- 9.2 The EPC Contractor shall get prior approval from Engineer-in-Charge for the area where the construction material or debris can be stored beyond the metalled roads. This area shall not cause any obstruction to the free flow of traffic /inconvenience to the pedestrians/public in general. It should be ensured by the agency that no accidents occur because of such permissible storage.
- 9.3 The EPC Contractor shall take appropriate protection measures like raising wind breakers of appropriate height as per NGT guidelines on all sides of the plot/area to ensure that no construction material dust fly outside the plot area, if required.
- 9.4 The EPC Contractor shall ensure that all the trucks or vehicles of any kind which are used for construction purposes or carrying construction material like cement, sand, earth and other allied material are fully covered. The EPC Contractor shall take every necessary precaution that the vehicles are properly cleaned including tyre washing at the exit gate and dust free to ensure that enroute their destination, the dust, sand or any other particles are not released in air or contaminate the air. The EPC Contractor has to make the tyre cleaning arrangements at the exit point of the site at his own cost.
- 9.5 The EPC Contractor shall provide mask to every worker on the construction site and involved in loading, unloading and carriage of construction material and construction debris to prevent inhalation of dust particles.
- 9.6 The EPC contractor shall provide all medical help, investigation and treatment to worker involved in the construction of building and carry off construction material and debris to dust emission.
- 9.7 The EPC contractor shall ensure that C& D waste is transported to the C&D waste site only and due records shall be maintained by the contractor.
- 9.8 The EPC Contractor shall compulsorily use wet jet in grinding and stone cutting.
- 9.9 The EPC Contractor shall comply with all the preventive and protective environmental steps as stated in the MoEF guidelines, 2019.

- 9.10 The EPC Contractor shall carry out on-Road-Inspection for black smoke generating machinery. The EPC Contractor shall use cleaner fuel.
- 9.11 The EPC Contractor shall ensure that all DG set comply emission norms notified by MoEF.
- 9.12 The EPC Contractor shall use vehicles having pollution under control certificate. The emissions can be reduced by a large extent by reducing the speed of a vehicle to 20 Kmph. Speed bumps shall be used to ensure speed reduction. In case where speed reductions cannot effectively reduce fugitive dust, the EPC Contractor shall divert traffic to nearby paved areas.
- 9.13 The EPC Contractor shall ensure that the construction material is covered by tarpaulin. The EPC Contractor shall take all other precaution to ensure that no dust particles are permitted to pollute air quality because of such storage.
- 9.14 The paving of the path for plying of vehicles carrying construction material is more permanent solution to dust control and suitable for longer duration projects.
- 9.15 Any violation of orders of MoEF including guidelines of State Government, SPCB or any officer of any department shall lead to stoppage of work for which EPC Contractor shall be responsible and no hindrance shall be accounted in this regard.

10. VARIOUS ANCILLARY PROVISIONS AT SITE

10.1 TEMPORARY BARRICADING

- 10.1.1 The EPC Contractor shall at his own expense, erect and maintain in good condition temporary barricading of minimum 3.0 height with coloured profile sheet and including MS frames /structural steel, nut, bolts, welding and all necessary T&P as per NGT norms and all around the working areas for segregating the construction site and also to control the dust pollution in the campus, as per directions of Engineer-in-Charge.
- 10.1.2 No extra payment shall be made for providing temporary barricading of required size/specification as the cost of barricading is deemed to be included in the quoted rates by the EPC Contractor. Entry to this entire site shall be control for proper security of man and materials and to avoid accidents. All management (including watch and ward) of barricades shall be the full responsibility of the contractor. It shall be dismantled and taken away by the EPC contractor after the completion of work at his cost with approval of the Engineer-in-charge. Nothing shall be payable on this account.

10.2 CLEANING OF ROADS

The EPC Contractor shall ensure the cleanliness of roads and footpaths by deploying proper manpower for the same. The EPC Contractor shall have to ensure proper brooming, cleaning washing of roads and footpaths, at all the time, throughout the entire stretch till the currency of the contract including disposal of seepage without any extra cost.

- 10.3 The EPC Contractor shall be deemed to have taken into account the quantum of excavation involved and that the surplus excavated earth remaining after use in

operations such as Horticulture/Landscaping, Gardening, backfilling etc. and to be disposed-off by him without any additional cost. The EPC Contractor shall store the excavated earth required for operations such as Horticulture/Landscaping, Gardening, backfilling etc., at his own place other than the project site under his safe custody at his own cost. Thereafter, the earth so stored shall be backfilled at site at the appropriate time. The cost of storage, transportation (to & from site), handling etc. shall be borne by the EPC Contractor.

10.4 It will be the responsibility of the EPC Contractor to get the permission for yard for dumping the surplus excavated earth from local authority if required. If any royalty/fee is payable to local authority, such royalty/fee shall also be borne by the EPC Contractor. Disposal shall be carried out strictly as per the regulations of local authority.

10.5 The EPC Contractor shall also make his own arrangement for the disposal of the spoils from the works to such place where the same shall not cause nuisance and should be acceptable to the authorities concerned without any cost to RITES/OIL India Limited.

10.6 ACCESS ROUTE

10.6.1 The EPC Contractor shall be deemed to have satisfied himself as to the suitability and availability of the access routes he chooses to use. The EPC Contractor shall be responsible for the maintenance of access routes. The Contractor shall provide at his cost signs or directions, which he may consider necessary or as instructed by Engineer-in-Charge for the guidance of his staff, labour and others. The Contractor shall obtain any permission concessions and related easement right that may be required from the relevant authorities for the use of such routes, signs and directions.

10.6.2 The RITES/OIL India Limited will not be responsible for any claims which may arise from the use or otherwise of any access route. The RITES/OIL India Limited does not guarantee the suitability or availability of any access route and will not entertain any claim for any non-suitability or non-availability for continuous use during construction of any such route.

10.6.3 The EPC Contractor shall use reasonable efforts to prevent any road or bridge from being damaged by the EPC Contractor's traffic or by the EPC Contractor's Personnel.

11. CO-OPERATION WITH OTHER EPC CONTRACTORS/SPECIALIZED AGENCIES/ SUB CONTRACTORS

- a. The EPC Contractor shall cooperate with and provide the facilities to other agencies working at site for smooth execution of the work. The EPC Contractor shall indemnify RITES/OIL India Limited against any claim(s) arising out of such disputes. The EPC contractor shall:
 - i) Allow use of scaffolding, toilets, sheds etc.
 - ii) Properly co-ordinate their work of their contractors.
 - iii) Provide control lines and benchmarks to his sub-contractors and the other contractors.

- iv) Provide electricity and water at mutually agreed rates.
 - v) Provide hoist and crane facilities for lifting material at mutually agreed rates.
 - vi) Co-ordinate with other contractors for leaving inserts, making chase, alignment of services etc. at site.
 - vii) Adjust work schedule and site activities in consultation with Engineer-in-charge and other Contractors to suit the overall scheduled completion.
 - viii) Resolve the disputes with other contractor/sub-contractors amicably and the
 - ix) Engineer-in-charge shall not be made an intermediary or an arbitrator.
- b. The work should be planned in a systematic manner so as to ensure proper co-ordination of various disciplines viz. water supply, drainage, rain water harvesting, electrical, firefighting, information technology, communication & electronics and any other services.
- c. Other Associated agencies may also simultaneously execute and install the works of sub-station / generating sets, air-conditioning, elevators (lifts) etc. for the work and the contractor shall afford necessary facilities for the same. The contractor shall leave such recesses, holes, opening trenches etc. as may be required for such related works at time of casting of concrete, stone work and brick work, if required, and nothing extra shall be payable on this account.
- d. The contractor shall conduct his work, so as not to interfere with or hinder the progress or completion of the work being performed by the contractor(s) or by the Engineer-in-Charge and shall as far as possible arrange his work and shall place and dispose off the materials being used or removed so as not to interfere with the operation of other contractor or he shall arrange his work that of the others in an acceptable and in a proper coordinated manner and shall perform it in proper sequence to the complete satisfaction of others.

12. INSPECTION OF WORK

- 12.1 In addition to the provisions of relevant clauses of the contract, the work shall also be open to inspection by Representatives/ Senior Officers of RITES/ OIL India Limited. The EPC Contractor shall at times during the usual working hours and at all times at which reasonable notices of the intention of the Engineer-in-charge or other officers as stated above to visit the works shall have been given to the EPC Contractor, either himself be present to receive the orders and instructions or have a responsible representative duly accredited in writing, to be present for that purpose. The Contractor shall provide necessary help for inspection and examinations and test of the materials and workmanship.
- 12.2 These representatives shall have full powers to inspect drawings of any portion of the work or examine the materials and workmanship of the plant at the Contractor's works or at any other place from where the material or equipment is to be obtained. Acceptance of

any material or equipment shall in no way, relieve the Contractor of his responsibility for meeting the requirement of the specifications.

12.3 The EPC Contractor shall, therefore, keep updated the following requirements and detailing.

- a) Display Board showing detail of work, weekly progress achieved with respect to targets, reason of shortfall, status of manpower, wages being paid for different categories of workers.
- b) Entrance and area surrounding to be kept cleaned.
- c) Display layout plans key plan, building drawings including plans, elevations and sections
- d) Up to date displays of Bar chart, CPM and PERT etc.
- e) Keep details of quantities executed, balance quantities, deviations, possible Extra Item, Substituted Item etc.
- f) Keep plastic / cloth mounted one set of building drawings.
- g) Set of Helmets and safety shoes for exclusive use for officers/ representatives visiting at site.

13. COMPLIANCE OF LAWS

13.1 The EPC Contractor shall keep himself fully informed of all acts and laws of the central and state government, all local bye laws, ordinances, rules and regulations and all orders and decree of bodies or, tribunals having any jurisdiction or authority which in any manner affect those engaged or employed on the work or which in any way affect the conduct of the works. EPC Contractor shall at all times, observe and comply with all such laws, ordinances, rules, regulations, orders and decrees, and shall give all notices and pay out of his own money any fees or charges to which he may be liable. He shall protect and indemnify RITES/ OIL India Limited and its representatives, officers and employees against any claim or liability arising out of violations of any such law, ordinances, legislations, order or decree, whether by himself or by his employees & authorized representatives.

13.2 Except as otherwise specifically stated elsewhere in the Contract:

- a) The EPC Contractor shall be deemed to have obtained all necessary information as to risks, contingencies and other circumstances which may influence or affect the works.
- b) By signing the Contract, the EPC Contractor accepts total responsibility for having foreseen all difficulties and costs of successfully completing the works.
- c) The Contract Price shall not be adjusted to take account of any unforeseen difficulties or costs.
- d) EPC Contractor's operations on Site: The EPC Contractor shall confine his operations to the site, and to any additional area which may be provided to the EPC

Contractor and agreed by the Engineer-in-Charge as working areas. The EPC Contractor shall take all necessary precautions to keep his personnel and equipment within the site and such additional areas, and to keep and prohibit them from encroaching on adjacent land.

14. EXISTING SERVICES & LAND FOR CONTRACTOR'S OFFICES, GODOWN, WORKSHOP WITHIN SITE

- 14.1 Existing drains, pipes, cables, overhead wires, sewer lines, water lines, domestic gas supply lines and similar services encountered in the course of the execution of the work shall be protected/ maintained against the damage by the EPC Contractor. Necessary cross-trenching shall be done at site by the EPC Contractor and shall identify all underground / overhead services and take necessary measures to protect the services before starting any excavation / activity. All temporary supports and other measures like removal/shifting of any live or dead service lines, cables, poles, gas supply lines etc. running through the plot area to protect and maintain the services during construction period as per the work requirement, to the suitable locations, is to be done in accordance with the instructions of and with prior approval from Engineer-in-charge/OIL India Limited, and shall be deemed to be included in the quoted rate / amount of the EPC Contractor and nothing extra shall be paid on this account. Engineer-in- Charge /OIL India Limited shall extend necessary support, as per statutory requirements, to the EPC Contractor for these approvals.

15. DESIGN MIX CONCRETE FROM BATCH MIX PLANT OR RMC

The EPC Contractor has to make their own arrangement for ready-mix cement concrete plant/batching plant at a place/ location separate from site, at their own cost. No extra payment shall be entertained by RITES/OIL India Limited in this regard as well as on account of transportation of Ready-Mix Concrete or any other material.

The EPC Contractor can arrange RMC (Ready Mix Concrete) to supplement the demand from RMC Plants after getting the same approved from Engineer-in-Charge for supplying the concrete.

Besides the strength to ensure durability of structure, grade of Design Mix Concrete shall be as specified, conforming to IS-456. Only Design Mix Concrete using admixtures with the prior approval of Engineer-in-Charge shall be used.

The Engineer-in-Charge reserve the right to inspect at any stage and reject the concrete if he is not satisfied about quality of product at the user's end.

The Engineer-in-Charge also reserves the right to exercise control over the followings:

- a) Ingredients, water and admixtures including corrosion inhibiting admixture purchased,

stored and to be used in the concrete including conducting of tests for checking quality of materials, recording of test results and declaring the materials fit or unfit for use in production of mix.

- b) Calibration check of the batch mix plant/RMC plant, and other equipment used for testing concrete and its ingredients.
- c) Weight and quantity check on the ingredients, water and admixtures added for batch mixing.
- d) Time of mixing of concrete.
- e) Testing of fresh concrete, recording of results and declaring the mix fit or unfit for use. This will include continuous control of the workability during production and taking corrective action, if required.

For exercising such control, the Engineer-in-Charge shall depute his authorized representative at the RMC/ BMC plant. It shall be responsibility of the EPC Contractor to ensure that all necessary equipment, manpower & facilities are made available to Engineer-in-Charge and/or his authorized representative at batch mix concrete plant/ RMC plant.

The concrete mix design will be carried out by the EPC Contractor, at his own cost, through one of the following laboratories/test houses to be approved by Engineer-in-Charge:

- a) IITs, NITs, NCCBM or any Govt. Engineering College.

16. DE-WATERING

De-watering required, if any shall be the Contractor's responsibility. Nothing extra shall be payable on this account.

17. FINAL TESTING OF THE INSTALLATION

The Contractor shall demonstrate trouble free functioning of all the Civil and E & M installations and services. The Engineer-in-Charge or his authorized representatives shall carry out final inspection of the various Civil and E & M services and installations. Any defect(s) noticed during demonstration shall be rectified by the Contractor at his own cost to the entire satisfaction of the Engineer-in-Charge. Nothing extra shall be payable on this account.

18. HANDING OVER THE WORKS ON COMPLETION

On satisfactory completion of all the works as per the provision of the Contract, the Contractor shall hand over buildings to OIL India Limited. The Contractor shall ensure that all development works, and bulk services are simultaneously carried out so as to make the buildings functional immediately on completion. The work, even if it is physically ready for occupations, shall be deemed to be completed from the date the services like

Water Supply, sewerage and electricity are made available to make the building functional. The completion certificate shall be issued only after the work is found functional in all respect. The safety and security of the completed buildings is the responsibility of the EPC Contractor till they are handed over to OIL India Limited. Nothing extra shall be payable on this account.

19. PARTICULAR SPECIFICATION / METHODOLOGY FOR GREEN BUILDING PRACTICE

The project is proposed for obtaining Minimum GRIHA 3 STAR or above and will be registered by EPC contractor & all the required data shall be provided by the bidder for obtaining the minimum GRIHA 3 Star or above. The GRIHA Team will be visiting the site as per requirements outlined in GRIHA manual and all associated cost such as transportation, fooding, lodging w.r.to GRIHA project officer's visit to be borne by EPC Contractor. Any test report if required by GRIHA rating system to be undertaken by EPC Contractor. Any cost associated with Third party commissioning & testing to be done by EPC Contractor.

The EPC Contractor needs to demonstrate the presence of a designated employee having GRIHA accreditation and experience that he is successfully completed at least one project under GRIHA 3 star or above, or have GRIHA empaneled firm as sub consultant to undertake documentation. All necessary documentation has to be done in the manner required to obtain GRIHA 3 STAR or Above Rating. All expenditure required for implementation of GRIHA provision for which contractor is responsible have to be considered by Contractor in quoted rates & nothing extra shall be paid on this account.

The contractor shall strictly adhere to the following conditions as part of his contractual obligations

19.1 SITE:

- (i) The contractor shall ensure that adequate measures are taken for the prevention of erosion of the topsoil during the construction phase. The contractor shall implement the Erosion and Sedimentation Control Plan (ESCP) provided to him by the Engineer-in-Charge as part of the larger Construction Management Plan (CMP). The contractor shall obtain the Erosion and Sedimentation Control Plan (ESCP) Guidelines from the Engineer in Charge and then prepare working plan for the following month activities as a CAD drawing showing the construction management, staging & ESCP. At no time soil should be allowed to erode away from the site and sediments should be trapped where necessary.
- (ii) The contractor shall ensure that all the topsoil excavated during construction works is neatly stacked and is not mixed with other excavated earth. The contractors shall take the clearance of the Engineer- in-Charge before any excavation. Topsoil should

be stripped to a depth of 20 cm from the areas to be disturbed, for example proposed area for buildings, roads, paved areas, external services and area required for construction activities etc. It shall be stacked only to a maximum height of 40 cm at designated areas, covered or stabilized with temporary seeding for erosion prevention and shall be reapplied to site during plantation, landscaping etc. of the proposed vegetation. Topsoil shall be separated from subsoil, debris and stones larger than 50 mm diameter. The stored topsoil may be used as finished grade for planting areas.

- (iii) The Contractor should follow the construction plan as proposed by the Engineer-in- Charge to minimize the site disturbance such as soil pollution due to spilling. Use staging and spill prevention and control plan to restrict the spilling of the contaminating material on site. Protect topsoil from erosion by collection storage and reapplication of topsoil, constructing sediment basin, contour trenching, mulching etc.
- (iv) No excavated earth shall be removed from the campus unless suggested otherwise by Engineer-in- Charge. All subsoil shall be reused in backfilling/landscape, etc. as per the instructions of the Engineer-in-Charge. The surplus excavated earth shall be disposed of by the contractor at his own cost for reuse after approval from Engineer-in-Charge. A certificate of reuse as required by the Engineer-in-Charge shall be submitted by the contractor.
- (v) The contractor shall not change the natural gradient of the ground unless specifically instructed by the Engineer-in-Charge. This shall cover all-natural features like water bodies, drainage gullies, slopes, mounds, depressions, etc. Existing drainage patterns through or into any preservation area shall not be modified unless specifically directed by the Engineer-in-Charge.
- (vi) The contractor shall not carry out any work which results in the blockage of natural drainage.
- (vii) The contractor shall ensure that existing grades of soil shall be maintained around existing vegetation and lowering or raising the levels around the vegetation is not allowed unless specifically directed by the Engineer-in-Charge.
- (viii) Contractor shall reduce pollution and land development impacts from automobiles use during construction.
- (ix) Overloading of trucks is unlawful and creates the erosion and sedimentation problems, especially when loose materials like stone dust, excavated earth, sand etc. are moved. Proper covering must take place. No overloading shall be permitted.
- (x) The dismantled material / building rubbish received from dismantling/demolishing shall be dumped to the dumping ground in properly covered truck with precaution. The EPC Contractor shall submit the hard copy of photograph showing the properly covered truck disposing the dismantled material/building rubbish. Failure of which shall be sternly dealt and a penalty @Rs.500/- per trip of truck shall be levied and the decision of Engineer-in-Charge shall be final and binding.

- (xi) The EPC Contractor shall not dump the construction material on the metalled road and shall keep the construction material on the physically demarcated space by the Engineer-in-Charge.
- (xii) All the building material responsible for pollution shall be brought at site from sources covered by tarpaulin and shall take all precautionary measure to ensure that no dust particles are permitted to pollute the air quality, failure of which the EPC Contractor shall be liable to pay damages as decided by Engineer-in-Charge. The decision of Engineer-in-Charge shall be final & binding.
- (xiii) There shall be no burning of leaves, plastic etc. at construction site.

19.2 CONSTRUCTION PHASE AND WORKER FACILITIES

The contractor shall specify and limit construction activity in pre-planned/ designated areas and shall start construction work after securing the approval for the same from the Engineer-in-Charge. This shall include areas of construction, storage of materials, and material and personnel movement.

19.3 PRESERVE AND PROTECT LANDSCAPE DURING CONSTRUCTION

The contractor shall ensure that no trees, existing or otherwise, shall be harmed and damage to roots should be prevented during trenching, placing backfill, driving or parking heavy equipment, dumping of trash, oil, paint and other materials detrimental to plant health. These activities should be restricted to the areas outside of the canopy of the tree or from a safe distance from the tree/plant by means of barricading. Trees will not be used for support, their trunks shall not be damaged by cutting and carving or by nailing posters, advertisements or other material. Lighting of fires or carrying out heat or gas emitting construction activity within the ground, covered by canopy of the tree is not to be permitted.

- 19.4 The contractor shall take steps to protect trees or saplings identified for preservation within the construction site.
- 19.5 Contractor should limit all construction activity within the specified area as per the Construction Management Plan (CMP) approved by Engineer-in-Charge.
- 19.6 The contractor shall avoid cut and fill in the root zones, through delineating and fencing the drip line (the spread limit of a canopy projected on the ground) of all the trees or group of trees. Separate the zones of movement of heavy equipment, parking, or excessive foot traffic from the fenced plant protection zones.
- 19.7 Contractor shall collect all construction waste generated on site. Segregate these wastes based on their utility and examine means of sending such waste to manufacturing units which use them as raw Material or other site which require it for specific purpose. Typical construction debris could be broken bricks, steel bars, broken tiles, spilled concrete and mortar etc.

- 19.8 The contractor shall provide potable water for all workers. Adequate measures for welfare of workers, staff and their children are to be taken. At camp /labour hut site, shelters/crutches shall be made with adequate facilities light, drinking water, teaching facilities for children. Adequate shelters for labour shall be also constructed at site of work as per direction of Engineer-in-Charge.
- 19.9 The contractor shall provide at least minimum level of sanitation and all safety facilities for the workers at their camp/labour site and at work sites. The contractor shall ensure cleanliness of workplace with regard to the disposal of waste and effluent, provide clean drinking water, water closets and urinals as per applicable standard. Adequate toilet facilities with signages shall be provided for the workmen within easy access of their place of work. The total numbers to be provided shall not be less than 1 per 25 employees in any one shift. Separate toilet facilities for ladies (staff and workers) are to be provided. Toilet connected with Septic tank facilities shall be provided from the start of building operations. Connection to a sewer line shall be made as soon as practicable. Every toilet shall be so constructed that the occupant is sheltered from view and protected from the weather and falling objects. Toilet facilities shall be maintained in a good sanitary condition. A sufficient quantity of disinfectant shall be provided. Natural or artificial illumination shall be provided.
- 19.10 The contractor shall ensure that air pollution due to dust/generators is kept to a minimum, preventing any adverse effects on the workers and other people in and around the site. The contractor shall ensure proper screening, covering stockpiles, covering brick and loads of dusty materials, wheel- washing facility, gravel pit, and water spraying. Contractor shall ensure the following activities to prevent air pollution during construction:
- i) Clear vegetation only from areas where work will start right away.
 - ii) Vegetate / mulch areas where vehicles do not ply.
 - iii) Apply gravel / landscaping rock to the areas where mulching / paving is impractical.
 - iv) Identify roads on-site that would be used for vehicular traffic. Upgrade vehicular roads (if these are unpaved) by increasing the surface strength by improving particle size, shape and mineral types that make up the surface and base. Add surface gravel to reduce source of dust emission. Limit amount of fine particles (smaller than 0.075mm) to 10 –20.
 - v) Water spray, through a simple hose for small projects, to keep dust under control. Fine mists should be used to control fine particulate. However, this should be done with care so as not to waste water. Heavy watering can also create mud, which when tracked onto paved public roadways, must be promptly removed. Also, there must be an adequate supply of clean water nearby to ensure that spray nozzles don't get plugged.
 - vi) Water spraying shall be done on:

- a) Any dusty materials before transferring, loading and unloading
 - b) Area where demolition work is being carried out
 - c) Any un-paved main haul road
 - d) Areas where excavation or earth moving activities are to be carried out
- vii) The contractor shall ensure that the speed of vehicles within the site is limited to 10 km/hr.
- viii) All material storages should be adequately covered and contained so that they are not exposed to situations where winds on site could lead to dust / particulate emissions.
- ix) Spills of dirt or dusty materials will be cleaned up promptly, so the spilled material does not become a source of fugitive dust and also to prevent of seepage of pollutant laden water into the ground aquifers. When cleaning up the spill, ensure that the clean-up process does not generate additional dust. Similarly, spilled concrete slurries or liquid wastes should be contained / leaned up immediately before they can infiltrate into the soil / ground or runoff in nearby areas.
- x) Provide barricading of appropriate height or as per direction of Engineer-in-Charge, along the site boundaries, next to a road, around storage facilities or other public area and using barricading of appropriate height and as per local body norms around building under construction.
- xi) Provide dust screens, sheeting or netting to scaffold along the perimeter of the building.
- xii) Cover stockpiles of dusty material with impervious sheeting
- xiii) Cover dusty load on vehicles by impervious sheeting before they leave the site
- xiv) Contractor should use equipment/techniques to avoid generation of C&D waste work. Contractor should use small equipment to make chases and avoid breakage by leveling or making arrangement for no cuttings for water supply lines in the walls/electrical conduits. Also, contractor shall use small equipment to drill holes and avoid levelling/hammering in walls to take out services.
- 19.11 Contractor shall be required to provide an easily accessible area that serves the entire building and is dedicated to the separation, collection and storage of materials for recycling including (at a minimum) paper, corrugated cardboard, glass, plastics, and metals. He shall coordinate the size and functionality of the recycling areas with the anticipated collections services for glass, plastic, office paper, newspaper, cardboard, and organic wastes to maximize the effectiveness of the dedicated areas. Consider employing cardboard balers, aluminum can crushers, recycling chutes, and collection bins at individual workstations to further enhance the recycling program.
- 19.12 The contractor shall ensure that no construction leachate (e.g. cement slurry etc.), is allowed to percolate into the ground. Adequate precautions are to be taken to safeguard

against this including, reduction of wasteful curing processes, collection, basic filtering and reuse. The contractor shall follow requisite measures for collecting drainage water run-off from construction areas and material storage sites and diverting water flow away from such polluted areas. Temporary drainage channels, perimeter dike/swale, etc. shall be constructed to carry the pollutant-laden water directly to the treatment device or facility (municipal sewer line).

- 19.13 Staging (dividing a construction area into two or more areas to minimize the area of soil that will be exposed at any given time) should be done to separate undisturbed land from land disturbed by construction activity and material storage.
- 19.14 The contractor shall comply with the safety procedures, norms and guidelines (as applicable) as outlined in the document Part 7, Constructional practices and safety, National Building code of India 2016 Vol-2, Bureau of Indian Standards. A copy of all pertinent regulations and notices concerning accidents, injury and first-aid shall be prominently exhibited at the work site. Depending upon the scope & nature of work, a person qualified in first aid shall be available at work site to render and direct first-aid to casualties. A telephone may be provided to first-aid assistant with telephone numbers of the hospitals displayed. Complete reports of all accidents and action taken thereon shall be forwarded to the competent authorities.
- 19.15 All workers of contractor and associate agencies, invariably and at all the times, must follow all safety norms, adopt safe construction practices and use all required safety gadgets in their working throughout the project duration.

The contractor shall issue Photo Identity Cards with unique numbers containing salient information of workers. Further the contractor shall establish a Time Office at the entry to demarcated area of site. The Time Office shall maintain a computerized record of all the workers allowed entry / working inside the demarcated area. This shall enable to exactly keep a record of persons at site at a given point of time.

Formation of Safety Monitoring Committee: The EPC Contractor within 15 days of start of work shall submit the names of Safety Engineer/Safety Manager and Safety Supervisors to the RITES/OIL India Limited, who shall notify a monitoring committee consisting of authorized representatives of the main working mandate of this monitoring committee will be to monitor and achieve the objectives of construction safety continuously, progressively and through affirmative action and to inculcate the safety culture among all stake holders. This committee will oversee implementation of safety program over the entire construction period.

Training and Awareness:

Training: The training shall be in two phases – first induction training and then periodic training / refresher workshop.

- i) **Induction Training:** All the workers shall have to undergo a training program of 16

hrs (8 hrs for 2 days) and to be declared satisfactorily trained by the Safety Manager before they are allowed to work on site.

- ii) Orientation Program: An orientation program shall be arranged for all people (other than workers) who normally work at or visit the site.
- iii) Workshops: Refresher workshops shall be arranged in every three months for all the workers on site.
- iv) Advance Training: For workers involved in high-risk activities, (to be identified by the Safety Monitoring Committee) an intensive training shall be kept once in a month.

The training modules shall be designed by the Safety Manager and approved by the Safety monitoring committee.

Training Methodology: The training methodology shall include both classroom and practical demonstration with audio visual techniques. For greater impact, demonstration with dummies will be done to highlight hazards of not following safe practices.

The training shall be imparted in vernacular language and may include means such as songs, theatre, puppetry etc. for better appreciation and assimilation by workers.

Implementation:

- i) The basic responsibility of implementation of safe practices shall be that of the safety manager and safety supervisors of the EPC Contractor at the first level and project team RITES on second level. The basic approach of implementation should be towards voluntary acceptability of safe practices by all stakeholders.
- ii) The safety arrangement made by the EPC Contractor shall be open to inspection by the safety officer or any other representative appointed by the RITES/OIL India Limited, and the observation made by him shall be complied with by the EPC Contractor.
- iii) All workmen are checked for their suitability before development by the respective Safety Manager and each Safety Supervisor. Workers physical fitness knowledge about the activity and his previous experience are checked before deployed. Workmen involved in physical activity (such as driver, operators, Height workers, Food handlers at Canteen and Pantries, welders) shall be subjected to pre-employment medical check-up, those who do not clear the medical examination shall not be employed.
- iv) Adequate number of safety equipment's and personals protective equipment (PPE) as per Indian Standards will be planned and procured.

Recommendations as per following table/Matrix should be followed:

Activity	Workmen Category	PPE's Recommended
General – Entry into work premises	All Employees	Safety Helmet, Safety Shoes & Reflective Jacket
Signalling	Security / marshal	Reflective Jacket
Working at Height – More than 1.8 metres	All	Full body harness Double lanyard
Involved with cement & Concrete Handling	All	Gum Boots & Rubber Hand Gloves
Breaking of ceramics & Agglomerate materials	Chippers	Eye protection – Clear Goggles
Welding & Gas Cutting	Welders & Cutters	Leather gloves, Safety shoe, Welding Shield with proper number
Working with slush	Un skilled & Excavation gang	Gum boots
Forming and Making shuttering materials	Carpenters and Wood workers	Face shield & Nose Mask
Rebar's handling & working	Bar benders	Cotton hand Gloves
Scaffolding	Scaffolders	Cotton hand gloves
Painting	Painters	Clear Goggles, Nose mask
DG Operators & Other Noise prone areas	Operators	Earmuff, Rubber Hand Gloves (Electrical Grade)
Electrical Maintenance & Repairs	Electricians	HV Rubber hand gloves
Concrete Batching Plant	Operators & Loaders	Nose Mask

Color Coding of Helmets:

Grey	All Staff of Contractor/other Respective Person
Green	Safety Inspectors
Red	Electricians & Signal men
Blue	Supervisors
Yellow	Workmen
Orange	New Workmen (for one month)
Purple	Visitor
White	RITES/OIL India Limited

Enforcement:

The safety team of the EPC Contractor and project team of RITES are entrusted with enforcement of safe practices. If safety program is not followed (as assessed by either RITES/OIL India Limited) then recovery as below shall be made:

- i) If the contractor does not employ and / or submit the names of Safety Manager and Safety Supervisors of specified numbers with appropriate qualification and or experience, then a recovery of Rs. 2000 per day and Rs. 1000 per day shall be affected for Safety Manager and Safety Supervisor respectively.
- ii) Contractor will ensure that no person shall be allowed to enter the demarcated area without adequate safety gadgets (as per occupation / purpose of visit).
- iii) Inside the work area it is the responsibility of the contractor to make sure all person will follow the safety instructions as per agreement.
- iv) In case of default from workers their employer (hiring agency) will have to pay a penalty of Rs 100 per offense.
- v) These fines will be collected into workers welfare fund. Money thus collected will be utilized for supporting welfare programmer for workers and their families.
- vi) In case the contractor / subcontractor do not impart training the same shall be provided by CPWD through an accredited agency. Thrice the amount of fee for such training shall be deducted from the contractor.
- vii) A display board shall be kept at site which would list the names of workers / teams and agencies following safety program in the best manner. This would be updated weekly.
- viii) During training and workshops, the names of persons / teams / agencies that are best following safety program shall be announced and they shall be felicitated.
- ix) On completion of the work, shields for best person / team / agencies following safety program in different categories shall be awarded.
- x) A biometric system/record of attendance will be put in place by the Contractor as per the direction of Engineer-in-Charge and the Safety Manager and all the Safety Supervisors will record their presence in the system. Recovery @ Rs. 500/- per instance will be made for failure to record such presence / non-availability of safety personals.

19.16 The contractor shall ensure the following activities for construction workers safety, among other measures:

- i) Guarding all parts of dangerous machinery.
- ii) Precautionary signs for working on machinery.
- iii) Maintaining hoists and lifts, lifting machines, chains, ropes, and other lifting tackles in good condition.

- iv) Durable and reusable formwork systems to replace timber formwork and ensure that formwork where used is properly maintained.
 - v) Ensuring that walking surfaces or boards at height are of sound construction and are provided with safety rails or belts.
 - vi) Provide protective equipment, helmets etc.
 - vii) Provide measures to prevent fires. Fire extinguishers and buckets of sand to be provided in the fire-prone area and elsewhere.
 - viii) Provide sufficient and suitable light for working during nighttime.
- 19.17 The storage of material shall be as per standard good practices as specified in Part 7, Section 2. Storage, stacking and Handling practices, NBC 2016 and shall be to the satisfaction of the Engineer-in-Charge to ensure minimum wastage and to prevent any misuse, damage, inconvenience or accident. Watch and ward of the Contractors materials shall be his own responsibility. There should be a proper planning of the layout for stacking and storage of different materials, components and equipment with proper access and proper maneuverability of the vehicles carrying the materials. While planning the layout, the requirements of various materials, components and equipment at different stages of construction shall be considered.
- 19.18 The contractor shall provide for adequate number of garbage bins around the construction site and the workers facilities and will be responsible for the proper utilization of these bins for any solid waste generated during the construction. The contractor shall ensure that the site and the workers facilities are kept litter free. Separate bins should be provided for plastic, glass, metal, biological and paper waste and labeled in both Hindi and English with suitable symbols.
- 19.19 The contractor shall prepare and submit spill prevention and control plans before the start of construction, clearly stating measures to stop the source of the spill, to contain the spill, to dispose the contaminated material and hazardous wastes, and stating designation of personnel trained to prevent and control spills. Hazardous wastes include pesticides, paints, cleaners, and petroleum products.
- 19.20 Contractor shall collect & submit the relevant material certificates for materials with high recycled (both post-industrial and post-consumer) content, including materials like RMC mix with fly-ash, glass with recycled content, calcium silicate boards etc.
- 19.21 Contractor shall collect the relevant material certificates for rapidly renewable materials such as bamboo, wool, cotton insulation, agri-fiber, linoleum, wheat board, strawboard and cork etc.
- 19.22 Where possible, the contractor shall select materials/vendors, harvested and manufactured regionally.
- 19.23 Contractor shall adopt an IAQ (Indoor Air Quality) management plan to protect the central air conditioning system during construction, control pollutant sources, and

interrupt pathways for contamination. He shall sequence installation of materials to avoid contamination of absorptive materials such as insulation, carpeting, ceiling tile, and gypsum wallboard. He shall also protect stored on-site or installed absorptive materials from moisture damage.

19.24 The contractor shall ensure that a flush out of all internal spaces is conducted prior to handover. This shall comprise an opening of all doors and windows for 14 days to vent out any toxic fumes due to paints, varnishes, polishes, etc.

19.25 Contractor shall make efforts to reduce the quantity of indoor air contaminants that are odorous or potentially irritating harmful to the comfort and well-being of installer and building occupants. Contractor shall ensure that the VOC (Volatile Organic Compounds) materials like paints, primer, coatings, adhesive, not exceeding the green building parameters/norms.

19.26 WATER USE DURING CONSTRUCTION

Contractor should spray curing water on concrete structure and shall not allow free flow of water. Concrete structures should be kept covered with thick cloth/gunny bags and water should be sprayed on them. Contractor shall do water ponding on all sunken slabs using cement and sand mortar bunds. The water used in construction and curing processes must meet the standards outlined in IS 456:2000, as updated by the latest amendment.

The Contractor shall remove from site all rubbish and debris generated by the Works and keep Works clean and tidy throughout the Contract Period. All the serviceable and non-serviceable (malba) material shall be segregated and stored separately. The malba obtained during construction shall be collected in well-formed heaps at properly selected places, keeping in a view safe condition for workmen in the area. Materials which are likely to cause dust nuisance or undue environmental pollution in any other way, shall be removed from the site at the earliest and till then they shall be suitable covered. Glass & steel should be dumped or buried separately to prevent injury. The work of removal of debris should be carried out during day. In case of poor visibility artificial light may be provided.

19.27 MATERIALS & FIXTURES FOR THE PROJECT

Contractor will produce wherever feasible certificate regarding distance of the source of the relevant material.

- i. The EPC Contractor shall procure PPC/OPC with fly ash conforming to IS: 1489 (Part-I) as required in the work. Cement required for the work shall be procured by the EPC Contractor.
- ii. The contractor has to comply as per MoEF issued notification 8.0.763I dated 14th Sept.1999 & latest notification of Jan. 2016 containing directive for greater fly ash utilization.
- iii. The contractor shall ensure that all paints, polishes, adhesives and sealants used both internally and externally, on any surface, shall be Low VOC products. The contractor shall get prior approval from the Engineer-in-Charge before the application of any

such material.

- iv. All plumbing and sanitary fixtures installed shall be as per the direction of the Engineer-in-Charge and shall adhere to the minimum LPM (liters per minute) and LPF (liters per flush) mentioned. The contractor shall employ 100 zero ODP (ozone depletion potential) insulation; HCFC (hydro-chlorofluorocarbon)/ and CFC (chlorofluorocarbon) free central air conditioning and refrigeration equipment and/halon-free fire suppression and fire extinguishing systems.

19.28 RESOURCES CONSUMED DURING CONSTRUCTION

- (a) The contractor shall ensure that the water and electricity is not wasted during construction. The Engineer in Charge can bring to the attention any such wastage and the contractor will have to ensure that such bad practices are corrected.
- (b) The contractor shall install necessary meters and measuring devices to record the consumption of water, electricity and diesel on a monthly basis for the entire tenure of the project.
- (c) The contractor shall ensure that all run-off water from the site, during construction is collected and reused to the maximum.
- (d) The contractor shall use treated recycled water of appropriate quality standards for construction, if available.
- (e) The contractor shall minimize the use of electricity.

19.29 CONSTRUCTION AND DEMOLITION WASTE

- (a) Contractor shall minimize the generation of construction waste as per the requirement for obtaining GRIHA 3 star or above.
- (b) All construction debris generated during construction shall be carefully segregated and stored in a demarcated waste yard. Clear, identifiable areas shall be provided for each waste type. Employ measures to segregate the waste on site into inert, chemical, or hazardous wastes.
- (c) All construction debris shall be used for road preparation, back filling, etc., as per the instructions of the Engineer in Charge, with necessary activities of sorting, crushing, etc.
- (d) No construction debris shall be taken away from the site, without the prior approval of the Engineer- in-Charge.
- (e) The contractor shall recycle the unused chemical/hazardous wastes such as oil, paint, batteries, and asbestos.
- (f) If and when construction debris is taken out of the site, after prior permissions from the Engineer in Charge, then the contractor shall ensure the safe disposal of all wastes and will only dispose of any such construction waste in approved dumping sites.

19.30 DOCUMENTATION

- (a) The contractor shall, during the entire tenure of the construction phase, submit the following records to the Engineer-in-Charge on a monthly basis:
 - i) Water consumption in liters
 - ii) Electricity consumption in kwh units
 - iii) Diesel consumption in liters
 - iv) Quantum of waste (volumetric/weight basis) generated at site and the segregated waste types divided into inert, chemical and hazardous wastes.
 - v) Digital photo documentation to demonstrate compliance of safety guidelines.
- b) The contractor shall, during the entire tenure of the construction phase, submit the following records to the Engineer-in-Charge on a fortnightly basis:
 - i) Quantities of material brought into the site, including the material issued to the contractor by the Engineer-in-Charge.
 - ii) Quantities of construction debris (if at all) taken out of the site
 - iii) Digital photographs of the works at site, the workers facilities, the waste and other material storage yards, pre-fabrication and block making works, etc. as guided by the Engineer-in- Charge.
- c) The contractor shall submit a document after construction of the buildings, a brief description along with photographic records to show that other areas have not been disturbed during construction. The document should also include brief explanation and photographic records to show erosion and sedimentation control measures adopted. (Document CAD drawing showing site plan details of existing vegetation, existing buildings, existing slopes and site drainage pattern, staging and spill prevention measures, erosion and sedimentation control measures and measures adopted for topsoil preservation during construction.)
- d) The contractor shall submit to the Engineer-in-Charge after construction of the buildings, a detailed as built quantification of the following:
 - i) Total materials used,
 - ii) Total topsoil stacked and total reused,
 - iii) Total earth excavated
 - iv) Total waste generated,
 - v) Total waste reused,
 - vi) Total water used,
 - vii) Total electricity, and
 - viii) Total diesel consumed.

- e) The contractor shall submit to the Engineer in Charge, before the start of construction, a site plan along with a narrative to demarcate areas on site from which topsoil has to be gathered, designate area where it will be stored, measures adopted for topsoil preservation and indicate areas where it will be reapplied after construction is complete.
- f) The contractor shall submit to the Engineer in Charge, a detailed narrative (not more than 250 words) on provision for safe drinking water and sanitation facility for construction workers and site personnel.
- g) Provide supporting document from the manufacturer of the cement specifying the fly ash content in PPC used in reinforced concrete.
- h) The contractor shall submit the following information to the Engineer-in-Charge at the end construction, for all material brought to site for construction purpose, including manufacturer's certification, verifying information and test data, where Specifications section require data relating to environment issue including but not limited to:
 - i) Source of products: Supplier and location of the supplier.
 - ii) Certifications from manufacturers of Low VOC paints, adhesives, sealant and polishes used at this particular project site.
 - iii) Certification from manufacturers of composite wood products/agro fiber products on the absence of added urea formaldehyde resin in the products supplied to them to this particular site.
- i) Submit environmental and pollution clearance certificates for all diesel generators installed as part of this project. Provide total support to Engineer in Charge and Green Building Consultants appointed for the work in completing all Green Building Rating related formalities, including signing of forms, providing signed letters in the contractor's letterhead whenever required.

19.31 EQUIPMENT

- a) To ensure energy efficiency during and post construction all pumps, motors and engines used during construction or installed, shall be subject to approval of the Engineer-in-Charge.
- b) All lighting installed by the contractor around the site and at the labour quarters during construction shall be LED bulbs of the appropriate illumination levels. This conditions a must, unless specifically prescribed. The contractor is to go through all other conditions of the GRIHA rating stipulations. Failure to adhere to any of the above-mentioned items, without approval of the Engineer in Charge, shall be deemed as a violation of contract and the contractor shall be held liable for penalty as per terms of the agreement. In case any penalty is imposed by any Hon'ble Court, NGT or any other authority due to non- compliance of any statutory order, or law or guidelines or pollution control or environmental norms, the same will be borne by the contractor.

20 SUBMISSION OF POLLUTION CONTROL PLAN

- a) The contractor shall submit the detailed action plan for control of pollution and for adherence to all the environmental guidelines/Laws/statutes/Court Orders/NGT

orders/orders of pollution control authorities through the entire period of construction at site. The detailed action plan shall be submitted to the Engineer-in-Charge within 15 days of the stipulated date of start of work and shall be got approved from the Engineer-in-Charge.

- b) The contractor shall arrange for control measures of all dust/noise/emission from the construction activities at site of work and shall install screens/curtains/ covers/dust trappers etc. as per guidelines/orders of the NGT/Court of law/ statutory authorities etc. No hindrance shall be allowed, arising out of any stay/stopping of work from any court/statutory authority/NGT/Govt. Authorities as a consequence of the contractor not adhering to any pollution control guideline/law/order of the state bodies during the construction period.

Nothing shall be paid to the contractor on account of expenses for any dust/pollution/emission control measures at the site of work or any delay in work due to any orders passed by any court/ statutory authority/Govt. Authorities during the period of construction.

A compensation of Rs. 1,000/- per day will be levied and recovered from the dues of the contractor for each day of delay beyond 15 days for non-submission of pollution control plan.

The contractor is strongly advised to study all dust/Noise/emission/pollution control norms / laws/Court Orders before bidding for the work and quote his rates accordingly for any liability which may arise on this account during the period of construction.

21 ACCESSIBILITY IN BUILT ENVIRONMENT

Contractors should follow norms of Barrier free built environment as per CPWD “Manual on Accessible Built Environment 2019” for persons with disability and elderly persons.

22 ADDITIONAL SPECIAL CONDITIONS FOR CEMENT, STEEL, CONCRETE, COVER BLOCKS & SHUTTERING OIL

The following additional conditions/specifications for Cement, Steel, Concrete, Cover block & Shuttering oil may be read with in addition to conditions/specifications mentioned elsewhere in the contract:

Cement OPC/PPC/PSC/Composite Cement/Portland Calcined Clay Limestone Cement shall meet the specifications of the latest edition of the relevant standard IS269, IS1489, IS455, IS16415 or IS18189. In addition to the above, the cement used shall have a minimum compressive strength of 43 MPa at 672 ± 4 hours measured as per IS4031 (Part 6).

All reinforcement shall meet all the requirements of the latest edition of IS1786. In addition to the above, the cross-section of the specimen, upon macroetching with nital (5 nitric acid in alcohol) as per ASTM standard E381, shall show a clear and continuous ring of martensite around a core of perlite or ferrite. The perlite or ferrite shall be present at no point on the periphery of the cross-section, i.e. on the surface of the reinforcement. Tests shall be carried out on at least 3 specimens drawn from each size of rebar to be used. All above tests shall be carried out before the acceptance of a certain make. Additionally, at least 3 specimens shall be tested for every 100 tonnes of reinforcement of each size.

The cover blocks shall be made of concrete and shall be at least 1 grade higher than the concrete in which they are to be embedded. Plastic cover blocks shall not be permitted.

Welding shall not be permitted in any reinforcement with a grade higher than 250 MPa.

All concrete used shall meet the relevant requirements of the latest edition of IS456. In addition to the above, the setting time of concrete, when measured as per IS8142 shall not be less than 2 hours, nor more than 8 hours. Testing of the setting time shall be carried out before the acceptance of the mix design. Additionally, the setting time shall be measured for every 1000 m³ of concrete delivered. The setting time shall be measured at a minimum frequency of once per month for each grade of concrete. The slump of concrete used shall not be less than 100 mm at the time of pouring of the concrete into the formwork. The strength of the concrete shall meet the relevant acceptance criteria in the latest edition of IS456. For these measurements, the cubes of concrete shall also be prepared at a location close to its final position, i.e. after all operations of transportation, pumping, etc. are complete. Compaction of the cubes shall be carried out using a table-vibrator. No compaction of cubes shall be carried out for any mixes that are self-compacting.

Any fly ash used in the production of concrete shall meet the relevant requirements of the most recent edition of IS1727. In addition to the above, the Blaine fineness of the fly ash measured as per IS1727 shall not be less than 320 m²/kg. The lime reactivity of the fly ash measured as per IS1727 shall not be less than 4.5 MPa. The loss on ignition of the fly ash measured as per IS1727 shall not be more than 4.

Any other pozzolanic or supplementary cementitious material used in the production of concrete shall meet the relevant requirements of the most recent edition of IS1727. In addition to the above, the Blaine fineness of the pozzolanic or supplementary cementitious material measured as per IS1727 shall not be less than 500 m²/kg. The lime reactivity of the material measured as per IS1727 shall not be less than 8.0 MPa. The loss on ignition of the material measured as per IS1727 shall not be more than 4.

Shuttering oil shall be of a non-foaming type and shall be tested for the same before

use.

All expenditure to be incurred for testing of all the samples e.g., packing, sealing, transportation, loading, unloading etc. including testing charges shall be borne by the EPC Contractor at his own cost and nothing extra shall be payable on this account.

23 PROFESSIONAL INDEMNITY INSURANCE

The Contractor shall effect professional Indemnity Insurance which shall cover the risk of professional negligence in the execution/construction of the Works including but not limited to safety, quality and soundness of the buildings/ sheds/ pavements/ other structures including structural elements beyond schedule completion period. This insurance shall be for a limit of not less than 5% of the Replacement Value per occurrence, with no limit on the number of occurrences. The Contractor shall maintain the professional indemnity insurance in full force and effect until ten (10) years after the Time for Completion for the whole of the Works. The Contractor undertakes to notify the Employer promptly of any difficulty in extending, renewing or reinstating this insurance.

The professional indemnity insurance procured by the Contractor:

- (a) must not be exhausted by any claim in respect of any other project or any other activities undertaken by the Contractor or any Subcontractor which are not related to this Project; and
- (b) shall extend to cover and include the professional activities of all other parties and persons who supply professional services to the Project such as project management, engineering supervision and procurement during the performance of the Works.

Additional obligations of the Contractor under this Clause

Nothing in this Clause limits the obligations, liabilities or responsibilities of the Contractor, under the other terms of the Contract or otherwise. Any amounts not insured or not recovered from the insurers shall be borne by the Contractor in accordance with these obligations, liabilities or responsibilities.

24 LIFE CYCLE COST:

The contractor shall be responsible for safety, quality and soundness of the structural elements beyond maintenance period. The contractor shall have obligation to rectify such defects minimum up to 35 (thirty-five) years from the date of completion of work. The defects have to be rectified within a reasonable time not exceeding forty-five days after issue of notice by Engineer- in- Charge. If contractor does not take corrective action within 45 days, then action

for debarring of the agency/Banning of Business shall be taken by the appropriate authority.

25 SPECIAL CONDITION FOR CIVIL & ELECTRICAL WORKS

25.1 Quality Assurance Policy shall be adopted as per QAP and Check List for E&M Services, including Inspection & Testing of Materials & Process at Manufacturer's Works, 3rd Party Lab. etc. No additional payment shall be made to the contractor for initial inspection /testing at the manufacturer's works by the representative of the Engineer-in-Charge. However, decision of Engineer- in-charge shall be final and binding on contractor/associate agencies for items/products/equipment not listed in this Checklist. The contractor/associate agencies shall have to produce certified copy of proof of procurement of material/service from Manufacturer/OEM/authorized dealer etc. before claiming payment.

i) Handing over of documents: All testing and commissioning shall be done by the Contractor to the entire satisfaction of the Engineer-in-Charge and all testing and commissioning documents shall be handed over to him. The Contractor shall also hand over all maintenance and operation manuals, technical specification, all certificates, and all other documentation as per the terms of the contract and requirement of Engineer-in-Charge.

ii) REJECTION OF INSTALLATION / PLANT

- Any item of plant or system or component which fails to comply with the requirements of this Specification in any respect whatsoever at any stage of manufacture, test, erection or on completion at site may be rejected by the Engineer In-charge either in whole or in part as he considers necessary/appropriate. Adjustment and/or modification work as required by OIL India Limited/RITES so as to comply with the Authority's requirements and the intent of the Specification shall be carried out by the Contractor at his own expense and to the satisfaction of the Authority Architect.
- After works have been accepted, the Contractor may be required to carry out assist in carrying out additional performance tests as reasonably required by the Architect/Employer.

iii) GUARANTEE / WARRANTY AND HANDOVER

- Contractor shall furnish a guaranty / warrantee for defect liability period against materials or workmanship. During guarantee / warranty period supplier is bound to replace the defective parts free of cost.
- The warrantee shall include for repair and / or replacement of any defects that appear in the supplied equipment / system at no additional expenses to the owner.

Furthermore, the warranty shall also cover the costs incurred in making the defective work good, including repair or replacement of building finishes, other materials, or damage to other equipment caused by such defects, or by subsequent replacement and repairs.

- All hardware including aluminium fittings, CP fittings & fixtures, sanitary fittings & fixtures, electrical fittings, and paints shall be guaranty / warrantee by the contractor for a minimum period of five (5) years, during which any defects shall be rectified or replaced by the contractor at no additional cost

25.2 Training and Operation: The contractor must arrange for training to the client staff and operate the system (if required) for a period of one month from the date of handing. No extra charges shall be paid for training.

25.3 Pre-delivery Inspection & Approval: The contractor shall offer the pre-delivery inspection of major equipment/ materials at manufacturers work to the Engineer-in-charge. The intimation for such inspections shall be given at least 15 days in advance from the date of proposed inspection. However, RITES along with OIL India Limited representative may inspect any/all the materials required in this project. All the testing facilities including consumables shall be borne by the contractor and nothing extra shall be paid on this account.

25.4 Final Testing and Inspection: The EPC Contractor shall demonstrate trouble free functioning of all the E & M installations and services. The Engineer-in-Charge or his authorized representatives shall carry out final inspection of the various E & M services and installations. Any defect(s) noticed during demonstration shall be rectified by the EPC Contractor at his own cost to the entire satisfaction of the Engineer-in-Charge. Nothing extra shall be payable on this account.

26 CONSTRUCTION OF SAMPLE FLAT

The EPC Contractor shall construct, within 12 (twelve) months from the date of commencement of work, a sample floor/unit of the Residential Block (STILT +6), complete in all respects as per Clause 51 of Section-8 (Clauses of Contract). This shall include all civil, electrical, and mechanical works, finishes, fixtures, equipment, furniture and services like cabling, Conduiting, Networking, raceways, switches, electrical system etc. as depicted in the tender drawings—whether explicitly mentioned or not—including all incidental works.

In addition, any furniture, equipment, or other items shown in the tender drawings (even if not specifically required as per the tender document) may also be provided by the Contractor in sample flat to facilitate its approval by the RITES/OIL India Limited. The Contractor shall submit high-quality photographs and video recordings of the completed sample flat for documentation purposes and shall ensure its maintenance including sample flat in proper condition until the final handing over of works.

All costs associated with the execution, presentation, and approval of the sample flat are deemed to be included in the Contractor's quoted price, and no additional payment

shall be made in this regard. Penalty as per Clause 51 of Section-8 (Clauses of Contract) shall be levied in case of non-compliance.

27 FINISH FLOOR LEVEL AND ROOF SLOPE PROVISION

The Finished Floor Level (FFL) shall be maintained minimum 50mm or as per applicable codes, specifications & standards above the structural slab level of all buildings & Infrastructure. In the Guest House (G+3) building, flooring thickness provided should accommodate the proposed raceway system. Screed Concrete shall be provided to maintain the flooring thickness required for buildings. These levels shall be finalized after incorporating all required floor drops and slopes for areas such as in balconies, toilets, etc.

Similarly, appropriate slope using cement concrete in accordance with design/specifications and site conditions shall be provided on roof terraces to ensure proper drainage of rainwater.

All costs associated with achieving the specified FFL, including floor drops, terrace slopes, screed concrete & cement concrete work, are deemed to be included in the EPC Contractor's quoted price, and no additional payment shall be made in this regard.

SECTION - 4

PROFORMA OF SCHEDULES

SECTION - 4

PROFORMA OF SCHEDULES

Schedule 'A'

Schedule of Quantity for quoting rates for Civil works & Electrical works of the Tender Document (As per Bill of Quantities attached)

(BOQ to be attached with Financial Bid)

Schedule 'B'

Schedule of materials to be issued to the contractor

(Refer Clause 10 of Clauses of Contract)

-Not Applicable-

Schedule 'C'

Tools and Plants to be hired to the Contractor

(Refer Clause 34 of Clauses of Contract)

-Not Applicable-

Schedule 'D'

Extra schedule for specific requirements / documents for the work, if any

-Not Applicable-

Schedule 'E'

Schedule of components of Cement, Steel, Other materials, POL, Labour etc. for Price Escalation.

(Refer Clause 10 CC of Clauses of Contract)

-Not Applicable-

Clause 10 CC – NOT APPLICABLE

Schedule 'F'

Reference to General Condition of Contract

Name of Work: **"Construction of Workman Housing Complex (BQ Area) on EPC mode-II of contract at OIL Duliajan, Assam "**

Estimated cost of work:	Rs. 160.02 Cr
Earnest money:	Rs. 1,00,00,000.00
Performance Guarantee (Ref. Clause 1)	5 % of Tendered Value
Security Deposit: (Refer clause 1A)	5% of Tendered Value

Notice Inviting Tender and Instruction to Tenders

Officer inviting tender:	PU Head, Project Unit Guwahati, RITES Ltd.
Applicable Mode of EPC Contract:	Mode-II
Type of Building:	Permanent
Construction Technology:	RCC Framed structure using steel formwork

CONDITIONS OF CONTRACT

Definitions

2 (iv) Principal/Employer	OIL India Limited
2 (v) Engineer-in-Charge	PU Head/GGM, RITES
2 (vii) Accepting Authority	Chairman & Managing Director (CMD), RITES

2 (ix) Percentage on Cost of materials and Labour to cover all overheads and profits	15%
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2 (x) Standard Schedule of Rates

Standard schedule of Rates as indicated below will be applicable to the Tender including latest amendments and corrections till the date of receipt of tender

S. No.	Item	Schedule Applicable
1	For execution of all civil & electrical engineering works of Buildings, external developments, Internal Roads etc.	1. CPWD Plinth Area Rate 2023 including latest amendments and corrections slip(s) till the date of receipt of tender. 2. CPWD DSR 2023 (Civil works) including latest amendments and corrections slip(s) till the date of receipt of tender.

		3. CPWD DSR 2022 (E&M) including latest amendments and corrections slip(s) till the date of receipt of tender 4. CPWD DSR 2019 (Wet Riser and Sprinkler System) including latest amendments and corrections slip(s) till the date of receipt of tender 5. CPWD DSR 2019 (E&M) Item for BEE 5 star rated, ceiling fan with Brush Less Direct Current (BLDC) Motor - including latest amendments and corrections slip(s) till the date of receipt of tender. 6. CPWD DSR 2019 for Solar Photovoltaic Power Plant-2019 including latest amendments and corrections slip(s) till the date of receipt of tender. 7. CPWD DSR- 2019 (E&M) for Façade Lighting-2019 including latest amendments and corrections slip(s) till the date of receipt of tender. 8. CPWD DSR- 2019 (Items for VRF/VRV Air-Conditioning System) – including latest amendments and corrections slip(s) till the date of receipt of tender. 9. Horticulture & Landscaping Schedule of rate 2020 with correction slip (s) till the date of receipt of tender.
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- 2 (xiii) Date of commencement of work: 15 days from the date of issue of LOA or the first day of handing over of site whichever is later.
- 9 (a) (ii) General Conditions of Contract: General Conditions of Contract as per Part 3 (General Conditions of Contract for EPC Project) of the Tender Document (attached).

CLAUSES OF CONTRACT

Clause-1

- (i) Time allowed for submission of P.G. from the date of issue of Letter of Acceptance (subject to maximum of 15 days)

15 days

Maximum allowable extension beyond the period provided in (i) above

07 Days

Maximum allowable extension with late fee at 0.1% per day of performance guarantee amount beyond the period provided in (i) above

Clause- 2

Authority for fixing compensation under Clause 2

Chairman & Managing Director, RITES

Clause-5

Time allowed for execution of work

24 Months from the date of start (Overall)

Date of start:

15 Days from the date of issue of Letter of Acceptance or the first date of handing over of site whichever is later

5.1 [b (i)] TABLE OF MILESTONE(S):

Sl. No.	Description of Milestone (Physical* / Financial*)	Time allowed in Months (from date of start)	Amount to be withheld in case of non-achievement of milestone
1.	Engagement of consultant by EPC contractor for Planning, Design & Engineering works , preparation of all architectural working drawings, landscape drawings, structural drawings and MEP / E&M services drawings and obtaining all clearances as may be applicable for the work including statutory approvals from the local body/authorities, vetting of structural design, tree cutting all complete to start the construction activities as per direction of engineer-in-charge. Dismantling of existing structures and removal of C&D waste to appropriate location outside the campus to be identified by the contractor at his risk and cost, clear the site for new construction. Contractor has to complete the following also: a) Complete site setup including testing laboratory and other preparation as per agreement condition b) Approval of Mix design, RMC plant c) Association with specialized agencies as per NIT and submission of relevant documents.	03 Months	0.10% of the accepted tendered value as per Financial Bid
2.	25% of Total work in Financial Terms	07 months	0.15% of the accepted tendered value as per Financial Bid.
3.	50% of Total work in Financial Terms	12 months	0.15% of the accepted tendered value as per Financial Bid.

Sl. No.	Description of Milestone (Physical*/ Financial*)	Time allowed in Months (from date of start)	Amount to be withheld in case of non-achievement of milestone
4.	75% of Total work in Financial Terms	18 months	0.15% of the accepted tendered value as per Financial Bid.
5.	90% of Total work in Financial Terms	22 months	0.25% of the accepted tendered value as per Financial Bid.
6.	All Civil, Electrical, Mechanical, IT and horticulture work completed in all respects for Residential Blocks (08 Towers), Guest House, Community Centre, Sub-station & Panel Room/Electrical Sub-station & External Development Works including Other Ancillary Infrastructure, all clearances from Local bodies, Completion Certificate, GRIHA certification, Occupancy certificate etc. required to declare buildings authorized for occupation and handing over of buildings/premises after completion of entire work with the satisfaction of Engineer-in-charge/OIL India Limited.	24 Months	0.20% of the accepted tendered value as per Financial Bid.

Note:

1. Withheld amount shall be released when subsequent milestone(s) is/are achieved within respective time specified. However, in case milestones are not achieved by the Bidder for the work, the amount shown against milestone(s) shall be withheld.
2. Monthly recovery for delay in submission of the monthly progress report within specified period – not exceeding **Rs. 1,00,000/- per month** for each default.

Schedule of handing over of the site		Site is available
Schedule of issue of Designs		All architectural drawings are the part of tender document.
5.4 Authority to decide:		
(i)	Extension of time	Vertical Head
(ii)	Rescheduling of milestones	Vertical Head

Clause-7

Gross work to be done together with net Payment/adjustment of advances for material collected, if any, since the last such payment for being eligible to interim payment: Rs. 5.00 Crore for all intermediate bills for Original Work.

Clause 8A

Authority to decide compensation on account if contractor fails to submit completion plans: PU Head/Regional Head/GGM/GM, RITES

Recovery rate for non-submission of completion plan by the contractor: 0.1 % (zero-point one percent)

Clause-10A

Whether Material Testing Laboratory is to be provided at site. Yes

If “YES” list of equipment to be provided As per Annexure B-1 of GCC, however, any other equipment or instrument required for the work, are to be brought and maintained at site as per directions of Engineer-in-Charge.

Clause-10B

Clause 10 B (ii) (Mobilization Advance) Yes

Whether Clause 10 B (ii) shall be applicable The Mobilization Advance shall be given as mentioned below:

(i) 1st Installment = 5%

(ii) 2nd Installment = 5%

Clause 10 B (iii) (Plant Machinery & Shuttering Material Advance)

Whether Clause 10 B (iii) shall be applicable **Not Applicable**

Clause-10 CC

Whether Clause 10 CC applicable **Not Applicable**

Clause-11

Specifications to be followed for execution of work

- 1) CPWD Specifications 2019, Volume-I and II and PAR-2023 with correction slips upto last date of submission of tender including extension, if any and manufacturer's specification.
- 2) CPWD General Specification for electrical Works with correction slips/ amendments up to last date of submission of tender.
 - i) Part I Internal 2023 with correction slips up to last date of submission of tender.
 - ii) Part II (External) 2023,
 - iii) Part III – (Lifts & Escalators) – 2003
 - iv) Part IV (Sub Station) 2013
 - v) Part V (Wet Riser & Sprinkler system) 2020
 - vi) Part VI Fire Detection and Alarm System 2018
 - vii) Part VII- DG Set-2013
 - viii) Part VIII- Gas based Fire extinguishing system-2013
 - ix) Heating, Ventilation & Air Conditioning (HVAC) – 2024
- 3) CPWD GRIHA Manual Green Rating Manual 2021.
- 4) CPWD Manual on Accessible Built Environment 2019.
- 5) Schedule of Rates, Analysis of Rates and Specifications (Horticulture & Landscaping) – 2020
- 6) ECBC 2017, National Building Code 2016 and relevant BIS codes with up-to-date corrections

All above specifications shall be applicable with correction slips up to last date of submission of bid.

Building information Model (BIM) is applicable: *YES, Applicable.*

Clause 12	Applicable
Deviation limit beyond which clauses 12.2 and 12.3 shall apply	3%

Clause-16

Competent Authority for deciding reduced rates.

CMD, RITES

CLAUSE 17

Damages and Defects Liability:

24 Months after the date of actual completion of work as recorded by Engineer-in-Charge

Clause 19 C

Rs. 5,000/- each default

Clause 19D

Rs. 5,000/- each default

Clause 19 G

Rs. 5,000/- each default (Subject to a maximum of 5% of the estimated cost put to tender)

Clause 19 K

Rs. 1000/- per tradesman per day

Clause-25

25 (i) Appellate Authority

Regional Head, RITES Limited, Guwahati
(The Authority immediately above Engineer-in-Charge to whom the Engineer-in- Charge report)

Appointing Authority

Chairman & Managing Director (CMD), RITES
(The Authority competent to appoint Arbitrator as per Item 9 of SOP)

Whether Clause 30 or 31 is applicable

Clause 30 applicable

Clause-32

Minimum Qualifications & Experience required and Discipline to which should belong

Designation	Minimum Qualification	Minimum Working Experience	Discipline to which should belong	Number
Principal Technical Representative	Graduate Engineer (Civil)	20 years (and having experience of one similar nature of work)	Civil	01
Deputy Technical Representative	Graduate Engineer (Civil)	12 years (and having experience of one similar nature of work)	Civil	02
Deputy Technical Representative	Graduate Engineer (Electrical)	12 years (and having experience of one similar nature of work)	Electrical	01
Engineers	Graduate/Diploma Engineer (Civil)	05 years for Graduate or 10 years for diploma	Civil	04
Engineers	Graduate/Diploma Engineer (Electrical)	05 years for Graduate or 10 years for diploma	Electrical	02
Engineers	Graduate Engineer (Civil)	08 years	Quality Engineer/Civil	01
Engineers	Graduate Engineer (Electrical)	08 years	Quality Engineer/Elect.	01
Diploma Engineer	Diploma Engineer (Civil)	08 years	Surveyor	01
Engineers	Graduate Engineer (Civil)	06 years	Project Planning/Billing	01
Engineers	Graduate Engineer (Electrical)	06 years	Project Planning/Billing.	01
BIM Professional	Graduate Engineer (Civil/Electrical)	05 years	BIM Professional Engineer	01

Clause-32(i) & (iv)

Recovery for non - deployment of Principal Technical Representative /Deputy Technical Representative / Project/ Site / Planning / Billing

Designation	Rate of Recovery (in Rs.) for non-deployment
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Principal Technical Representative	60,000.00 per month
Deputy Technical Representative	40,000.00 per month
Graduate/Diploma Engineer (5/10 years' Experience)	25,000.00 per month
Graduate Quality Engineer (8 years' Experience)	25,000.00 per month
Diploma engineer (Surveyor)	15,000.00 per month
Project Planning/billing Engineer	20,000.00 per month
BIM Professional	40,000.00 per month

Clause-38:

- i) Schedule/statement for determining theoretical quantity of cement & bitumen on the basis of Delhi Schedule of Rates printed by CPWD
- DSR 2023 – (For DSR Items)
- Wherever work is carried out as per specification other than CPWD specifications the cement/bitumen content will be calculated as per the nomenclature of the item/ as per the specifications applicable/ job mix formula approved.
- (ii) Variation permissible on theoretical quantities
- a) Cement 2% plus /minus
- b) Bitumen for all works 2.5% plus only & nil on minus side
- c) Steel Reinforcement and Structural Steel 2% plus/minus
Sections for each diameter, section and Category
- d) ~~All other materials~~ Nil

Sl. No.	Description of Item	Rates in figures and words at which recovery shall be made from the Contractor	
		Excess beyond permissible variation	Less use beyond the permissible variation
1	Cement	NIL	Not permitted

2	Steel Reinforcement	NIL	Not Permitted
3	Structural Sections	NIL	Not Permitted
4	Bitumen issued free	NIL	Not permitted
5	Bitumen issued at stipulated fixed price	NIL	Not permitted

Clause 42
Clause 42.10

The contractor shall, within **45 days** of award of work, establish and maintain the following facilities at site, **at no extra cost**:

1. Conference Room (12' x 20')

- 1 No. 20-seater conference table with 20 chairs
- 2.0 Ton AC (1 No.)
- Attached toilet with water supply and all fittings, fixtures

2. Site Office

- 3 rooms of size 10' x 16'
- 1 toilet connected to sewer/soak pit, ensuring hygienic disposal
- Office to be kept clean and functional during the construction period

3. IT Infrastructure

- 2 Nos. desktop computers (Core i9, 16GB RAM, 1TB HDD + 500GB SSD, 2GB dedicated graphics card)
- 1 A3 colour printer
- 2 Black & White laser printers
- All equipment to be new and fully operational with necessary software (MS Office, Primavera, BIM, etc.) & peripherals.

4. Furniture & ACs

- 3 Nos. 1.5 Ton ACs
- Furniture (all new) as below:

S. No.	Articles	Quantity
1	Executive table with side rack	2 No.
2.	Office table	2 nos.
3	Executive chair	2 Nos.
4	Office Chairs with visitor	10 Nos.

		chairs	
	5	Steel Almirah	2 Nos.
After completion of work contractor will take away these materials and site shall be cleaned free from encroachment. Nothing extra shall be paid on this account.			
Whether Clause 42.11.1A applicable	Yes		
Whether Clause 42.13.A applicable	Yes		

Clause-42.17

City of Jurisdiction of Court:

Guwahati, Assam

Clause-43.2.1

Sum for which Third Party Insurance to be obtained:

Rs. 10.00 Lakhs per occurrence with the number of occurrences limited to minimum four.

Clause-51:
Applicable.

Whether clause 51 shall be applicable.

Yes

Time allowed for completion of Sample Floor / Unit for each of building

Residential Block - 12 months from the date of start of the work.

SECTION-5

TECHNICAL SPECIFICATIONS

PART-A

TECHNICAL SPECIFICATIONS

1. General –

The work will be carried out as per the approved Architectural Drawings, structural drawings, MEP drawings & Schedule of Finishes, DBR etc. The items covered by the following General specifications shall be carried out as per these specifications. The standard specification shall be arranged by the contractor and will not be supplied by RITES along with the Tender Document.

2. STANDARD SPECIFICATIONS ISSUED BY CPWD (with correction slips upto last date of submission of tender including extension, if any)

a) Civil Engineering Works

Latest CPWD specifications 2019, Volume-I and II

b) Electrical Engineering Works

i) General Specifications for Heating, Ventilation & Air-Conditioning (HVAC) -2024

ii) CPWD General Specifications for Electrical Works

Part I Internal 2023

Part II External 2023

Part III Lifts & Escalators 2003

Part IV Substations 2013

Part V Wet Riser and Sprinkler Systems 2020

Part VI Fire Detection and Alarm System- 2018

Part VII D.G. Sets- 2013

Part VIII Gas based Fire Extinguisher System-2013

Part IX Heating, Ventilation & Air Conditioning Works 2017

c) Horticulture & Landscaping- Rates & Specification

Schedule of Rates, Analysis of Rates and Specifications (Horticulture & Landscaping)- **2020**

The above-mentioned documents are available as Priced Document issued by CPWD and in soft copy PDF Format in CPWD website.

3. Standard Specifications for Railway Works

a) Specifications for execution of all civil engineering works related to Building work, Road works, Horticulture works etc. (except Formation works, Bridge works and P. Way works): - Latest CPWD specifications as indicated in para 2 above.

b) Bridge Works, Formation and P. Way works

Indian Railway Unified Standard Specification-2021 (IRUSS-2021)

The IRUSS specifications are available on the Railway Board's website.

4. Standard Specifications Issued by Ministry of Road Transport and Highways (MORTH Specifications)

For Highways/ Hill Road/Major Road, Airport Pavements, Flyover, Bridge, Grade separator etc.

Specifications for Road and Bridge works (Fifth Revision) April 2013 have been published by Indian Road Congress as a priced document. These Specifications cover exhaustively various Road and Bridge works.

5. Special Specifications (Not Covered by any Standard Specifications)

(Include Special Specifications here, including Type of Tests, Lot Size, and Frequency of Tests to be carried out for the materials/Items).

6. The general conditions and special conditions of the contract including the drawings shall be read in conjunction with these technical specifications, and matters referred to, shown or described in one are not necessarily repeated in the other. These specifications are comprehensive but may exceed the requirements of this project.

7. Notwithstanding the sub-division of the specification into various headings, every part of it is to be deemed supplementary to every other part and is to be read with it, so far as it may be practicable so to do, or when the context so permits.

8. Items/materials to be used in Works:

- i. The agency/contractor should use "Make-in-India" products as per the Govt. of India "Make in India" policy updated up to the last date of submission of bids. The agency should also consider the availability of spare parts/components for maintenance purposes while proposing any brand/manufacturer.
- ii. In case of non-availability of 'Make in India' brands then only the proposal for use of international brands is allowed with the prior approval of Engineer-in-Charge complying with all Government of India norms.
- iii. The contractor shall supply ISI marked item/material. In case ISI marked item/material is not available, First Quality item/material shall be accepted. The samples of the item/material shall in either case have to be got approved from Engineer-in-Charge before procurement by Contractor.
- iv. The agency shall submit at least three ISI marked or 1st Quality make/brand (in case ISI marked item/material is not available) along with the current rates (the difference in rates shall not be more than 10% of the average of three ISI marked or 1st Quality make/brand) and specifications for the approval of Engineer-in-charge before procurement.
- v. The design of electrical equipments, wherever applicable, should be type tested by Central Power Research Institute (CPRI) / Electrical Research and Developments Associations (ERDA).
- vi. It will be contractor's responsibility to ensure timely submission of samples and relevant documents for approval of Engineer-in-Charge. Contractor will have to replace the defective and substandard materials at his own cost.

Part B

Technical Specification
for
Civil, Plumbing/Sanitary, Landscaping & Horticulture Works

PARTICULAR SPECIFICATIONS

a) General

- 1) The work shall be carried out in accordance with the architectural drawings, structural drawings, MEP drawings and approved shop drawings. Structural drawings, MEP drawings, shop drawings, and architectural drawings shall be properly correlated before executing the work. In case any discrepancy is noticed between architectural, MEP and structural drawings, the contractor shall obtain a final decision, in writing, from the RITES.
- 2) All work shall be carried out in accordance with Latest CPWD Specifications - 2019, Volume I and Volume II, along with all applicable correction slips up to the date specified, unless otherwise stated in the tender document.
- 3) For specifications of items not included and other related miscellaneous items, specifications shall be taken from latest relevant Latest CPWD Specifications 2019, Volume I and Volume II, IS/BIS Code along with all applicable correction slips up to the date specified or manufacturer specifications.
- 4) Notwithstanding anything stated in the CPWD or any other specifications regarding items to be paid extra or separately, no additional payment shall be admissible. All such items/works shall be deemed to be included in the scope of work and considered as subsumed within the quoted rate.
- 5) Wherever any reference to any Indian Standards occurs in the documents the same shall be inclusive of all amendments issued thereto or revisions thereof, if any, up to the date of receipt of tenders.
- 6) The work shall be executed and measured as per metric dimensions given in the drawings etc. (F.P.S. units wherever indicated are for guidance only).
- 7) 43 grade ordinary Portland cement conforming to IS 8112 or Portland Pozzolona Cement confirming to IS 1489 (Part I) shall be used for concrete, RMC work and other items. The Contractor shall procure cement only from reputed manufacturers.
- 8) The sources of course aggregate, fine aggregate & water to be used in the work shall be approved by the Engineer-in-Charge. The contractor shall ensure for himself their conformity to the relevant specifications and their availability before submitting such a proposal to the Engineer In charge.
- 9) Fine Aggregate: -The fine aggregates shall be coarse or fine as per required grading for intended use, as per Latest CPWD Specifications -2019 -Vol.I & Vol. II with correction slips up-to-date.
- 10) Coarse Aggregate: - It shall consist of crushed or broken stones. As per Latest CPWD Specifications -2019 -Vol.I & Vol.II with up-to-date correction slips.
- 11) Water: - Only potable water shall be used for curing and production of mortar and concrete. It shall confirm to requirements laid down in IS:456-2000 /Latest CPWD Specifications-2019-Vol.I & Vol.II with correction slips up to date.
- 12) Scaffolding
 - Wherever required, a double scaffolding system (cup lock type) for the working platform etc. shall be provided. The scaffolding system shall be stiffened with bracings, runners, connection with the building etc. wherever required for inspection of work at required locations with essential safety features for the workmen etc. Complete.

13) Embedding of pipes, conduits etc.

- All crossings, embedment etc. in RCC walls and floors for water supply, drainage and sanitary pipes, fittings etc. shall be provided with sleeves, recesses at exact locations as per the shop drawings for individual walls and floors so as to avoid cuttings of concrete, masonry work and floors. All such areas shall be made good, leakproof, during finishing and nothing extra shall be payable on these accounts.

14) Fittings & Fixtures for Doors, Windows, Ventilators, Sanitary & Plumbing etc.

- All the fitting & fixtures shall be procured from one of the approved/reputed manufacturers. The fittings shall be sourced from single brand and only one series or model to maintain the uniformity of the fittings and accessories. In case of any variance from the model / series is warranted, the decision of Engineer-in-charge shall be final and binding.
- All fittings and fixtures shall be got approved from the Engineer-in-charge before procurement well in advance and the approved samples shall be kept at site till completion of the work.
- All the fittings provided shall be inclusive of all accessories required to complete the item in working condition whether specifically mentioned or not in the specifications & elsewhere in this tender document.
- All the accessories including brackets, nuts, bolts, screws, bushes etc. shall be of the quality and make specified by the manufacture of the fitting.
- All the fittings including accessories shall be supplied with manufacturers test certificates with each lot of supply. The test certificate shall clearly indicate the lot numbers of the supplied fittings.
- The specification and method statement shall be submitted by the agency well in advance before the start of execution. Execution shall be allowed only after the due approval of engineer-in-charge.

Summary of Civil Specifications

SI No.	ITEMS OF WORK	SPECIFICATION
1.	Earthwork in Excavation for all works as per requirements	a. The work shall be executed as per CPWD /MORTH/relevant specification/IS/IRC Codes as per the work requirement including bailing and pumping out water, strutting etc. as may be applicable to the work for all leads & lifts. All relevant DSR-2023 items are applicable as per work requirements. b. Clearing jungle including site clearance, uprooting of rank vegetation, grass, brush wood, trees of all girths, and removal of roots outside the periphery of the area cleared from site complex at suitable location including all leads/lifts complete in all aspects as per direction of EIC. c. Earth work in excavation of all type of soil, boulders/rocks to require level/ Profile for Pavement/road work, footpath work, horticulture work, foundation and trenches, buildings & infrastructure etc. as per approved drawings and specification and as per direction of Engineer-in-Charge. d. Earth work in surface excavation not exceeding 30 cm in depth but exceeding 1.5 m in width as well as 10 sqm on plan including getting out and disposal of excavated earth for all leads & lifts, as directed by Engineer-in- Charge for all kinds of soil. e. Earth work in excavation by mechanical means (Hydraulic excavator)/ manual means over areas (exceeding 30 cm in depth, 1.5 m in width as well as 10 sqm on plan) including getting out and disposal of excavated earth for all leads & lifts, as directed by Engineer-incharge. f. Appropriate ground improvement or soil stabilization measures, if required, shall be undertaken based on the findings of the soil investigation report conducted by the EPC Contractor and the recommendations of the structural design, if any. The excavation work also includes dismantling & disposal of old plinth/ foundation/existing structures/fencing etc. and removal/shifting of existing dead/live utility services as per the work requirement to the suitable locations after taking due approval of RITES/OIL.

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		g. Excavation shall be done from top to bottom. Undermining or undercutting shall not be done. h. In case of excavation for foundation in trenches or over areas, the bed of excavation shall be to the correct level or slope and consolidated by watering and ramming. The excess depth shall be made good by the contractor with the concrete of the mix used for levelling/ bed concrete for foundations. Soft/defective spots at the bed of the foundations shall be dug out and filled with concrete. i. While carrying out the excavation for drain work care shall be taken to cut the side and bottom to the required shape, slope and gradient. The surface shall then be properly dressed. If the excavation is done to a depth greater than that shown on the drawing or as required by the Engineer-in-Charge, the excess depth shall be made good by the contractor at his own cost with stiff clay puddle at places where the drains are required to be pitched and with ordinary earth, properly watered and rammed, where the drains are not required to be pitched. In case the drain is required is to be pitched, the back filling with clay puddle, if required, shall be done simultaneously as the pitching work proceeds. The brick pitched storm water drains should be avoided as far as possible in filled-up areas and loose soils. j. Earth work by mechanical means involves careful planning keeping in view site conditions i.e. type of soil, nature of excavation, distances through which excavated soil is to be transported and working space available for employing these machines. The earth moving equipment should be accordingly selected. k. Surplus excavated / unsuitable earth/rock shall be taken out of the campus, at suitable locations as per the local norms/bye laws, for dumping/ disposing off for all leads/ lifts and transporting the earth/debris at his own cost after remittance of due royalty to the concerned authority as applicable and nothing extra will be paid to complete the work. l. Anti-termite treatment as per the necessity of work shall be carried

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		out as per relevant Indian standard codes/ CPWD specifications as per DSR 2023 item no. 2.34.1, 2.35.1.1, 2.35.3.1.
2.	Filling in EarthWork for all works as per requirements	a. The work in filling shall be executed as per CPWD /MORTH/relevant specification/IS/IRC Codes as per the work requirement including bailing and pumping out water, strutting etc. as may be applicable to the work for all leads & lifts. All relevant DSR-2023/MORTH items are applicable as per work requirements. b. Filling available earth/moorum or good earth/moorum brought from outside shall be done to level the ground as per requirement as per approved level /drawings. The material used shall be as per latest provisions of CPWD/MORTH/IRC specifications as applicable. c. Filling shall be done to require level/ Profile for Pavement/road work, footpath work, horticulture work, foundation and trenches, buildings & infrastructure etc. wherever required as per approved drawings and specification and as per direction of Engineer-in-Charge. d. For horticulture work good earth filling shall be done as per CPWD specifications & directions of engineer-in-charge. e. Anti-termite treatment as per the necessity of work shall be carried out as per relevant Indian standard codes/ CPWD specifications and as per DSR 2023 item no. 2.34.1, 2.35.1.1, 2.35.3.1. Filling in Trenches, Plinth, Under Floor, Sides Of Foundation Etc. Normally excavated earth from same area shall be used for filling. Earth used for filling shall be free from shrubs, rank, vegetation, grass, brushwood, stone shingle and boulders (larger than 75mm in any direction), organic or any other foreign matter. Earth containing deleterious materials, salt peter earth etc. shall not be used for filling. All clods and lumps of earth exceeding 8 cm in any direction shall be broken or removed before the earth is used for

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		filling. In case soil is required to be supplied from outside then same shall be arranged by EPC contractor for all leads & lifts. Filling The space around the foundations, trenches, plinth and drains in trenches shall be cleared of all debris, brick bats etc. The filling shall be done in layers not exceeding 20 cm in depth. Each layer shall be watered, rammed and consolidated. Ramming shall be done with iron rammers where possible and with blunt end of crow bars where rammers cannot be used. Special care shall be taken to ensure that no damage is caused to the pipes, drains, masonry or concrete in the trenches. Surplus excavated / unsuitable earth shall be taken out of the campus or shall be used in campus itself as per direction of the Engineer-in-charge. Additional good earth fit for filling, if required, for the work shall be procured from outside of the campus at his own cost, for all leads & lifts. In case of filling under floor, the finished level of filling shall be kept to the slope intended to be given to the floor. SAND FILLING IN PLINTH Sand Sand shall be clean and free from dust organic and foreign matter and its grading shall be within the limits of grading zone IV or V as per Latest CPWD specifications. Filling Sand filling shall be done in a manner similar to earth filling in plinth ,except that consolidation shall be done by flooding with water. The surface of the consolidated sand filling shall be dressed to the required level or slope and shall not be covered till the Engineer-in-Charge has inspected and approved the sand filling. Minimum 150mm Sand filling shall be done in plinth & flooring of all buildings/infrastructure (DSR –

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		2023 item 2.27). Earthwork in road, street parking & footpath area: Consolidation of surface/subgrade/existing ground/ excavated ground with power road roller of 8 to 12 tonne capacity after, dressing to camber and making good the undulations etc. for road work, parking areas and footpath areas. The work in embankment and subgrade shall be carried out as per CPWD/MORTH/IRC codes as per the requirement of the work. Available good earth/moorum or good earth/moorum brought from outside shall be filled to desired levels for levelling the ground as per requirement and as per approved level /drawings. The material used in embankment/subgrade etc shall be as per latest provisions of CPWD specifications /MORTH/IRC as applicable. Embankment Construction – As per latest provisions of CPWD specifications /MORTH/IRC as applicable and as per approved drawings. Subgrade Construction- As per latest provisions of CPWD specifications /MORTH/IRC as applicable and as per approved drawings.
3.	Sub-Structure - for all buildings, drain, culverts, retaining walls, UGWT & Pump room, Boundary Wall, Entrance structure at main gates, Parking sheds, trenches, service infrastructure & other allied RCC works etc.	a. Building/Infrastructure shall be earthquake resistant as per earthquake/seismic zone in which site is located and RCC structure with M30 or higher-grade concrete as per Structural design & approved Good for Construction drawings. b. RCC in Pile, raft, footings, foundations, wall, Plinth Beam, tie beam, Column, Pedestal, Boundary wall, drains, culverts, retaining walls, Grade Slab, tanks, pits, chambers, basements or any other structural elements of minimum M30 grade or higher as per the approved structural drawings with design mixed concrete (Ready mixed/site batched design mix with permissible admixtures), centering, shuttering, cover blocks & reinforcement. The work shall be executed as per latest provisions of CPWD specifications , IS: 456, IS: 1893, IS 13920 or any other applicable code and as per approved structural design. RCC leaner than M-30 shall not be used.

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		c. Reinforcement shall be Fe 500 D or more TMT (Thermo-Mechanically Treated) bars conforming to IS 1786:2008 including cutting, bending, placing, coupling, binding etc. Spacer bars, chairs, binding wire, supports bars etc. shall have to be provided to avoid sagging of reinforcement bars. d. Pile works shall include all equipments, machinery, material, accessories etc. all complete. However, for execution of piling works, polymer instead of bentonite slurry to be used. e. Initial test piles shall be provided by the contractor as per requirements including their testing. f. Testing of piles will be done as per the approved QAP for initial test, routine test, lateral load test & integrity testing etc. g. The wall below the Plinth Beam on outer periphery of the building shall suitably be provided, if required as per the design/site conditions. h. For RCC laid in or under water and/ or liquid mud, then first water shall be pump out or bail out and the surface shall be make clean by removing any slush etc., complete before placing reinforcement, centering, shuttering, concreting. i. Anti-termite treatment shall be carried for Buildings structure as per relevant latest provision of Indian standard codes/CPWD specifications. j. Filling in plinth with sand under floors, including watering, ramming of required depth (minimum 150mm) consolidating and dressing complete. k. Structural/Non-Structural grade slab shall be provided below flooring as per the necessity at the site, design requirements, and the functional requirements of the supported flooring. RCC slab of minimum 100 mm thickness shall be provided. The RCC floating slab shall be provided with nominal reinforcement in both directions (as per approved drawings). l. In RCC raft of electrical trenches necessary slope shall be provided with Cement concrete for draining out water through weep holes at every 1m c/c of 100mm dia uPVC pipes filled with aggregates of 20mm nominal size. Pipes fixed till PCC below the RCC Raft. m. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed. Note:-1. For Boundary Wall & RCC Drain work Minimum Grade of Concrete is M-25 or as per approved drawings/codal provisions.
4.	Super structure for all buildings, drain,	a. Building shall be earthquake resistant as per earthquake/seismic zone in which site is located and RCC structure with M30 or higher-grade

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	culverts, retaining walls, UGWT & Pump room, OHWT, Boundary Wall, Entrance structure at main gates, service infrastructure & other allied RCC works etc.	concrete as per Structural design & approved Good for Construction drawings for all levels. b. RCC in wall, Column, beams, shear walls, slabs, shafts, water tanks, Pedestal, Boundary wall, retaining walls, drains, pump room or any other structural elements of minimum M30 grade or higher as per the approved structural drawings with design mixed concrete (Ready mixed/site batched design mix with permissible admixtures), centering, shuttering, cover blocks & reinforcement. The work shall be executed as per latest provisions of CPWD specifications, IS: 456, IS: 1893, IS 13920 or any other applicable code and as per approved structural design. RCC leaner than M-30 shall not be used. c. Reinforcement shall be Fe 500 D or more TMT (Thermo-Mechanically Treated) bars conforming to IS 1786:2008 including cutting, bending, placing, coupling, binding etc. Spacer bars, chairs, binding wire, supports bars etc. shall have to be provided to avoid sagging of reinforcement bars. d. Expansion joints wherever required in buildings shall be as per DSR 2023 item no. 5.44, 5.45, 5.46. Bitumen impregnated fiber board to be filled in expansion joints as per applicable size in approved drawings and Directions of the Engineer in Charge. e. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed. Note: For Boundary Wall & RCC drain work Minimum Grade of Concrete is M-25 or as per approved drawings/ codal provisions.
5.	Plain Cement Concrete	The work of plain cement concrete shall be done as per the work requirement and as per the laid down procedure in Latest CPWD specifications in levelling coarse below foundation, pile cap, plinth beam, plinth protection, drains, trenches, paver blocks, kerb stones, below Grade slab etc. as per approved drawings and directions of Engineer-in-charge. Grade of PCC to be 1:3:6 (1 Cement : 3 coarse sand (zone-III) : 6 graded stone aggregate 20 mm nominal size) or as per approved drawing/specifications with minimum 100mm thickness to be provided as per requirements. For CC laid in or under water and/ or liquid mud, then first water shall be pump out or bail out and the surface shall be make clean by removing any slush etc., complete before placing centering, shuttering, concreting etc. Plinth protection along the perimeter of the buildings shall be provided

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		as per specific functional requirement for minimum 75cm wide or as directed by EIC as per DSR-2023 item 4.17. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
6.	Curing	Curing is the process of preventing loss of moisture from the concrete. The following methods shall be employed for effecting curing. Moist Curing Exposed surfaces of concrete shall be kept continuously in a damp or wet condition by ponding or by covering with a layer of sacking, canvas, Hessian or similar materials and kept constantly wet for at least 7 days from the date of placing concrete in case of ordinary Portland cement and at least 10 days where mineral admixtures or blended cements are used. The period of curing shall not be less than 10 days for concrete exposed to dry and hot weather conditions. In the case of concrete where mineral admixtures or blended cements are used, it is recommended that above minimum periods may be extended to 14 days. Membrane Curing: Approved curing compounds may be used in lieu of moist curing with the permission of the Engineer- in-Charge. Such compound shall be applied to all exposed surfaces of the concrete as soon as possible after the concrete has set. Impermeable membrane such as polythene sheet covering the concrete surface may also be used to provide effective barrier against the evaporation. Freshly laid concrete shall be protected from rain by suitable covering. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
7.	Form Work, Centering, Shuttering etc.	For all RCC & PCC works to be provided as per work requirements for all heights. Reference may be made to all items as stipulated in CPWD/DSR-2023 and direction of Engineer-in-charge. Water proofing ply Shuttering to be used as per the direction of engineer- in- charge.

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		Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
8.	Damp Proof Course	DSR-2023 item 4.10, 4.12 & 4.13 to be referred. Cement Concrete Layer This shall consist of cement concrete of specified proportions and thickness. The surface shall be levelled and prepared before laying the cement concrete. Edge of damp proof course shall be straight, even and vertical. Side shuttering shall consist of steel forms and shall be strong and properly fixed so that it does not get disturbed during compaction and the mortar does not leak through. The concrete mix shall be of workable consistency and shall be tamped thoroughly to make a dense mass. When the sides are removed, the surface should come out smooth without honey- coming. Continuity shall be maintained while laying the cement concrete layer and laying shall be terminated only at the predetermined location where damp proof course is to be discontinued. Curing Damp proof course shall be cured for at least seven days, after which it shall be allowed to dry. Application of Hot Bitumen Where so directed, hot bitumen in specified quantity shall be applied over the dried up surface of cement concrete, properly cleaned with brushes and finally with a piece of cloth soaked in kerosene oil. Bitumen of penetration A 90 or equivalent where used shall be heated to a temperature of $160^{\circ} \pm 5^{\circ}\text{C}$. The hot bitumen shall be applied uniformly all over, so that no blank spaces are left anywhere. It will be paid for separately. Latest CPWD specifications/relevant IS/BIS codes as applicable to be

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		followed.
9.	Autoclaved Aerated Concrete Block	a. External walls in superstructure of buildings are proposed with 225mm Autoclaved Aerated concrete (AAC) blocks masonry for all levels as per Latest CPWD specifications and other relevant latest applicable IS/BIS codes. b. Internal walls in superstructure of buildings as per approved drawings are proposed with 225/100mm Autoclaved Aerated concrete (AAC) blocks masonry for all levels as per Latest CPWD specifications and other relevant latest applicable IS/BIS codes. c. Parapet wall of 1200 mm high or decided by Engineer-in-charge, 225 mm thick Autoclaved Aerated concrete (AAC) blocks masonry with proper coping (1:2:4) with slope to drain of rainwater as per Indian Standards and Latest CPWD specifications. Note: 1. Autoclaved Aerated concrete (AAC) blocks masonry shall be provided with Grade - I AAC blocks of minimum density 551 kg/cum and conforming to IS 2185 (part 3) in super structure above plinth. DSR-2023 item 6.38, 6.47, 6.5 to be referred. 2. The R.C C bend shall be provided on 150mm to 300mm thick masonry to increase the strength and compatibility . The RCC bend shall be provided at sill level and at lintel level over throughout the wall. This thickness of the bend shall be approved by the Engineer in charge or as specified in approved drawings. Autoclave Aerated Concrete Block masonry shall be provided with polymer modified adhesive mortar. The polymer modified adhesive mortar shall be provided @ 30 kg per cum or with cement mortar 1:4 (1 cement : 4 coarse sand). 3. Autoclave Aerated Concrete Block with 100 mm thick masonry shall be laid in cement mortar 1:4 (1 cement : 4 coarse sand) and to be provided with two number 6mm dia reinforcement steel bar at every third course. 4. In case of any discrepancy, the decision of Engineer-in-charge will be final. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
10.	Brick Work	Brick work with common burnt clay F.P.S. (non modular) bricks of class designation 7.5 in Cement mortar 1:4 (1 cement : 4 coarse sand) are proposed in superstructure above plinth level (DSR-2023 item 6.4.1). Brick work with common burnt clay F.P.S (non modular) bricks of class designation 7.5 in Cement mortar 1:4 (1 cement : 4 coarse sand) have to

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		done in foundation and plinth (DSR item 6.1.1). Brick work to be provided for all floor levels as per approved drawings. Note: The brick work shall be done as per the work requirement i.e. in boundary wall, foundation & plinth, supporting walls for ramps, steps etc. as per approved drawings. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
11.	Plaster	a. Providing and applying 12 mm thick cement plaster (1 cement: 4 fine sand) on internal walls for all levels in smooth line and level etc. complete as per the latest CPWD specifications and relevant IS/BIS codes. b. 18mm thick plaster in two coats under layer 12 mm thick cement plaster 1:5 (1 cement : 5 coarse sand) and a top layer 6 mm thick cement plaster 1:6 (1 cement : 6 fine sand) on external walls for all levels and locations, finished in smooth line and level etc. complete as per the latest CPWD specifications and relevant IS/BIS codes. c. Providing and applying 6mm thick cement plaster, 1:3 (1 cement : 3 fine sand) applied on ceilings/slab for all levels and locations, finished in smooth line and level etc. complete as per the latest CPWD specifications and relevant IS/BIS codes. d. Chicken wire mesh or hexagonal I Wire Netting or Chicken Mesh is manufactured by Galvanized Iron, wires in sizes of opening from 12mm to 16 mm. These Chicken Mesh are made by 18 Gauge to 20 Gauge wire from reputed manufacturer and to be fixed with the help of nails along the length at junctions of R.C.C column, beam with Masonry work before plaster work. Note: 1. Necessary drip courses shall be provided in Chajja, Balcony, Projecting Roof, Beams etc. as per the latest CPWD specifications and relevant IS/BIS codes. 2. Groove of appropriate size have to be made at the junction of RCC Columns, Beams with Masonry work on external side. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
12.	Water Proofing	
a	Water Proofing (Terrace, Pipe cutouts, Toilets, Wet areas of	Providing and applying Self Healing Liquid Rubber Sealant For Water Proofing at all levels.

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	Kitchen)	Application – 1 to 1.2 Ltr/sqm - per two coats The properties of Self Healing Liquid Rubber Sealant: - Odourless, VOC free - Colour - Brown / Black - Specific gravity (liquid) g/cm³ - 1.0 Approx. - Chemical Resistance ASTM G 20 for resisting salt water and most inorganic solutions. - Volatile Organic Compound - Contains no solvents - Viscosity, Brookfield (cps) - 12000 to 15000 - pH – 10 to 12 - Accelerated Weathering Xenon arc) ASTM G 155s, 250 hrs - No deterioration of the film, >90% retention of original tensile strength - Complies with Biological Resistance as per ASTM E154, ASTM D412 - Complies with Impact Resistance in reference to CGSB 37 – GP – 56 @ 23°C - Should Pass Water Tightness After Impact - Tensile Strength ASTM D 412, Kpa (psi) > 830 - Elongation percentage > 950% - Recovery ASTM D 412, percentage > 95% - Adhesion to concrete DeFelsco, kPa (psi) > 900 - Should pass Puncture Resistance in reference to CGSB 37-GP-56 - Blistering ASTM D714, 1000Hrs exposure - None - Abrasion/Slip resistance ASTM F 1677 - Very Good - Passes Low temperature mandrel -10°C - Hardness, Durometer 00 - 83 to 85 - Hardness, Shore A - 18 to 20 - Hardness, Shore D - 3 to 4 - Thermal Resistance ASTM C518-04 (2.66/ply)- 0.028 - UL Fire rating (Ballasted) - Class A A. On terrace - Haunching/Fillet/Gola of minimum 75x75mm in Cement concrete 1:2:4 including finishing with cement mortar 1:3 (1 cement: 3 fine sand) to be made at junction of Slab & wall towards slab side.

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		B. The waterproofing system on terrace, toilets, wet areas of kitchen, cutouts to be executed by the manufacturer or his approved applicator which will consist of following operations: - i) Apply 10-15mm levelling cement mortar 1:3 on green concrete as per requirement. ii) Brushing and Vacuum cleaning the surface to make free from any loose material, oil, grease, burr, dirt, dust etc. iii) Repairing the surface if required, as per recommendation of manufacturer or his approved applicator to make it water tight. It is to be ensured that surface should be dry & free from moisture before application. iv) Providing and applying by spraying/brushing or as recommended by manufacturer water base green liquid rubber as under: - First coat of Liquid rubber - Immediately after first coat apply reinforcing fabric on all surface with special focus on gaps, cracks, joints, bolt holes, parapet wall corner, drainage area. - Apply second coat on reinforcement fabric after first coat is touch dry (4-6 hrs). - Repeat application for 6 coats consuming liquid rubber @ 2.80 to 3.2 Litre/sqm (to achieve dry film thickness of more than 300 microns). - The final coat (after 6th coat) should be puncher resistant. v) Every two coats consumes 1 to 1.2 litre/sqm of liquid rubber. vi) A layer of non-woven geotextile of 50 gsm with overlap of minimum 100mm shall be laid over final coating surface for further grading/tiling etc. vii) Number of coats and procedure of application including curing time will be as per manufacturer's to achieve minimum dry film thickness of 300 microns of self healing liquid rubber such that waterproofing system is provided with 10 years of system guarantee against leakages. If dry film thickness more than 300 microns is required for achieving 10 years guarantee then same should be provided with applicable number of coats. C. On terrace - After application of water proofing system screed of cement concrete 1:2:4 (1 cement : 2 coarse sand : 4 graded stone aggregate 20mm nominal size) of minimum thickness 50 mm at Khurra with required slope towards Khurra to ridge of roof will be provided. D. On Vertical Surface of terrace, toilets, wet areas of kitchen, waterproofing as above up to 300mm as above shall be provided. Necessary water ponding test to be carried out. E. After that the following flooring work shall be carried out on terrace

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		as under: Providing and fixing Heat Resistant Terrace Tiles (300 mm x 300 mm x 20 mm) with SRI (solar refractive index) > 78, solar reflection > 0.70 and initial emittance > 0.75 on waterproof and sloped surface of terrace, laid on 20 mm thick cement sand mortar in the ratio of 1:4 (1 cement : 4 coarse sand) and grouting the joints with mix of white cement & marble powder in ratio of 1:1, including rubbing and polishing of the surface upto 3 cuts complete, including providing skirting upto 150 mm height along the parapet walls in the same manner. F. Injection Grouting (at damp / leak points If Required) - Fix GI/PVC/MS Nozzles to the required depth of slab @ leak/ damp points in areas marked for treatment after water ponding test. - Grout cement slurry admixed with non-shrink grouting admixture of approved make at a dosage as per manufacturer's specifications. - Grout till material oozes out from neighbor nozzle/ the nozzle refuse to accept further material. - Once the grouting is complete, cut the nozzles and seal the holes with polymer modified mortar and bring it to smooth finish. Sealing/Packing joints in Outlets/Inlet Pipes, Bore Packing: Providing Coving at pipe periphery of of size 50mm x 50mm minimum at junctions of pipe and Slab using 1:3 cement and sand along with bonding agent. Providing and laying Geo fabric 45gsm of 300mm over the Fillet/ Gola. Plumbing system shall be treated with the epoxy primer and silica sand. Packing the Outlet sleeves using free flow non-shrink grout along with bonding agent. In overall the work should be completed to provide waterproofing at cutouts/outlet sleeves in slab/walls for plumbing system. Waterproofing to be done as per manufacturer or its approved applicator after approval of Engineer-in-charge so as to provide with 10 years of system guarantee/warranty against leakages.
b	Waterproofing Treatment of UGT/ OHWT /Elevator (Lift) Pit/Pump room	Providing and applying integral crystalline slurry of hydrophilic in nature for waterproofing treatment to the RCC structures, prepared by mixing in the ratio of 5 : 2 (5 parts integral crystalline slurry : 2 parts water) for vertical surfaces and 3 : 1 (3 parts integral crystalline slurry : 1 part water) for horizontal surfaces and applying the same from negative (internal) side with the help of synthetic fiber brush. The material shall meet the requirements as specified in ACI-212-3R-2010 i.e by reducing permeability of concrete by more than 90% compared with control concrete as per DIN 1048 and resistant to 16 bar hydrostatic pressure on negative side. The crystalline slurry shall be capable of self-healing of cracks up to a width of 0.50 mm. The work shall be carried out all complete as per specification and the direction of the engineer-in-charge. The product performance shall carry guarantee for 10 years

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		against any leakage. For vertical surface two coats @ 0.70 kg per sqm per coat sqm For horizontal surface one coat @1.10 kg per sqm. Pipe peripheral treatment: Filling the annular gap between pipe periphery and concrete / masonry with non-shrink ready to use grouting compound including cleaning the same properly and sealing the inside joint mouths with a moisture insensitive epoxy putty in a continuous, 10mm thick gasket around the pipes to a width of approximately 10mm. Protective Plastering work: Applying 10-12mm avg. thick cement Plaster (1:4) as per direction of EIC. Screed Concrete work (Floor area): laying of avg. 50mm screed concrete 1:1.5:3 admixed with integral waterproofing compound over the floor area, as per manufacturer's specification & direction of EIC. Providing and applying of swellable type water stop tape, 19mm x 25mm thick in linear meter (expansive nature) for construction joints treatment of RCC structure such as raft slab, retaining walls, water storage tank and at the junctions of raft slab with the retaining walls etc.. After cleaning the surface, one coat of required primer for swellable water stop tape shall be applied throughout the length of the joint @3.78 litre per 240 running meter. Over the primed surface swellable type water stop tape shall be placed. The work shall be carried out all complete as per specification and the direction of the Engineer-In-Charge. The product performance shall carry guarantee for 10 years against any leakage. Food Grade Epoxy Coating All inside walls and floors of UGT and OH tanks Providing and applying 2 coats of Food grade solvent free high build thixotropic with a very high chemical resistance with solid content 100% substrate moisture content shall be less than 4% before application of epoxy coating , coating shall be applied by mixing 2 components using slow speed heavy duty drilling machine fixed with mixing paddle, applying first coat using brush or roller and allowing it to dry for 16-18 hrs, applying second coat and allowing it to dry All inside walls and floors of UGT and OH tanks Providing and applying 2 coats of Food grade solvent free high build thixotropic with a very high chemical resistance with solid content 100% substrate moisture content shall be less than 4% before application of epoxy coating , coating shall be applied by mixing 2 components using slow speed heavy duty drilling machine fixed with mixing paddle, applying first coat using brush or roller and allowing it to dry for 16-18 hrs, applying second coat and allowing it to dry completely, etc and complete as per manufacturer's specification. [The properties of membrane are:- Pot life :>30 min;

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		Mixed density : 1.2kg/Ltrs 27° C; The coating shall be approved/certified to be used in potable water. All waterproofing system to be provided with 10 years of system guarantee/warranty against leakages.
13.	Flooring & skirting/dado /facias/cladding/counter/ shelves etc.	
a	RCC Slab with broom finish	Providing and laying RCC Grade slab from RMC batching plant. The ready mixed concrete shall be laid and finished with vibrator complete as per Latest CPWD specifications and directions of Engineer-in- charge (Refer DSR-2023 item 5.33 & 5.33.1). The nominal reinforcement has to be provided in both directions (as per approved drawings) as per DSR-2023 item 5.22.6. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
b	Cement Plaster skirting	The skirting around the walls & columns have to executed with cement plaster skirting 30 cm high & 18 mm thick in Cement plaster, with cement mortar 1:3(1 cement : 3 coarse sand), finished with a floating coat of neat cement) Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
c	Kota stone flooring and skirting	a) Kota stone of 25 mm thickness slab flooring over 20 mm (average) thick base laid over and jointed with grey cement slurry mixed with pigment to match the shade of the slab, including rubbing and polishing complete with base of cement mortar 1 : 4 (1 cement : 4 coarse sand). b) Kota stone slabs 20 mm thick in risers of steps, skirting, dado and pillars laid on 12 mm (average) thick cement mortar 1:3 (1 cement: 3 coarse sand) and jointed with grey cement slurry mixed with pigment to match the shade of the slabs, including rubbing and polishing complete. Size of Kota stone slab for flooring as per decision of Engineer-in-charge. In steps 03 nos. anti-skid grooves to be provided near nosing as directed by Engineer-in-charge. Single slab of kota stone to be used in treads and risers of steps. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.

SI No.	ITEMS OF WORK	SPECIFICATION
d	Rectified Ceramic anti-skid glazed floor tiles	Providing and laying rectified Ceramic anti skid glazed floor tiles of 300mmX300mm (thickness to be specified by the manufacturer), of 1st quality conforming to IS : 15622, of approved make, in all colours, shades, except White, Ivory, Grey, Fume Red Brown, laid on 20 mm thick bed of cement mortar 1:4 (1 Cement : 4 Coarse sand), jointing with grey cement slurry @ 3.3 kg/sqm including pointing the joints with white cement and matching pigments etc., complete. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
e	Ceramic Glazed Wall Tiles	Providing and fixing 1st quality ceramic glazed wall tiles conforming to IS: 15622 (thickness to be specified by the manufacturer), of approved make, in all colours, shades except burgundy, bottle green, black of any size as approved by Engineer-in-Charge, in skirting, risers of steps and dados, over 12 mm thick bed of cement mortar 1:3 (1 cement : 3 coarse sand) and jointing with grey cement slurry @ 3.3kg per sqm, including pointing in white cement mixed with pigment of matching shade complete upto false ceiling height. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
f	Polished Glazed Vitrified Floor Tiles	Providing and laying Vitrified tiles in floor in required sizes (thickness to be specified by the manufacturer) with water absorption less than 0.08% and conforming to IS:15622, of approved brand & manufacturer, in all colours and shade, laid on 20 mm thick cement mortar 1:4 (1 cement:4 coarse sand) jointing with grey cement slurry @3.3 kg/sqm including grouting the joints with white cement and matching pigments etc. The tiles must be cut with the zero chipping diamond cutter only . Laying of tiles will be done with the notch trowel, plier, wedge, clips of required thickness, leveling system and rubber mallet for placing the tiles gently and easily. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
g	Matt Finished/Anti Skid Vitrified Floor Tiles	Providing and laying Vitrified tiles in floor in required sizes (thickness to be specified by the manufacturer) with water absorption less than 0.08% and conforming to IS:15622, of approved brand & manufacturer, in all colours and shade, laid on 20 mm thick cement mortar 1:4 (1 cement:4 coarse sand) jointing with grey cement slurry @3.3 kg/sqm including grouting the joints with white cement and matching pigments etc. The tiles must be cut with the zero chipping diamond cutter only . Laying of tiles will be done with the notch trowel, plier, wedge, clips of required thickness, leveling system and rubber mallet for placing the tiles gently and easily. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
h	Vetrified tiles in skirting/dado/cladding	Providing and laying Vitrified tiles in different sizes as decided by Engineer-in-charge (thickness to be specified by manufacturer), with

SI No.	ITEMS OF WORK	SPECIFICATION
		water absorption less than 0.08 % and conforming to I.S. 15622, of approved make, in all colours & shade, in skirting, riser of steps, over 12 mm thick bed of cement mortar 1:3 (1 cement: 3 coarse sand), jointing with grey cement slurry @ 3.3 kg/ sqm including grouting the joint with white cement & matching pigments etc. complete. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
i	Polished Granite stone slab	a. Providing and laying Polished Granite stone flooring in required design and patterns, in linear as well as curvilinear portions of the building all complete as per the approved architectural drawings with 18 mm thick stone slab over 20 mm (average) thick base of cement mortar 1:4 (1 cement : 4 coarse sand) laid and jointed with cement slurry and pointing with white cement slurry admixed with pigment of matching shade including rubbing, curing and polishing etc. all complete as specified and as directed by the Engineer-in-Charge. Granite stone slab colour black, Cherry/Ruby red or equivalent as decided by Engineer-in-charge. b. <u>In staircase/steps</u> - 18 mm thick gang saw cut, premoulded and prepolished, machine cut of required size, approved shade, colour and texture laid over 20 mm thick base cement mortar 1:4 (1 cement : 4 coarse sand), joints treated with white cement, mixed with matching pigment, epoxy touch ups, including rubbing, curing, moulding and polishing to edges to give high gloss finish etc. complete at all levels. Granite stone slab colour black, Cherry/Ruby red or equivalent as decided by Engineer-in-charge. 15 mm pre- moulded double nosing in staircase/steps projecting beyond finished surface of riser to be provided. In Stair case/steps 03 nos. anti-skid grooves to be provided near nosing as directed by Engineer-in-charge. Single piece granite stone slab shall be provided in treads/risers of stair case. c. <u>Skirting, risers of steps/staircase</u> – providing & fixing gang saw cut/ machine cut, polished 18 mm thick Granite stone backfilled with grout of 12 mm (average) thick cement mortar 1:3 (1 cement : 3 coarse sand) laid and jointed with cement slurry @ 3.3 kg/sqm including joints treated with white cement, admixed with matching pigment, epoxy touch ups including rubbing, curing, polishing etc. all complete as per direction of Engineer-in-charge. Granite stone slab colour black, Cherry/Ruby red or equivalent as decided by Engineer-in-charge.

SI No.	ITEMS OF WORK	SPECIFICATION
		Single piece granite stone slab shall be provided in treads/risers of stair case. Note: The same colour/pattern slab shall be used in Flooring/Landing and skirting/riser/treads. In ramps 18 mm Thk. Polished Granite 100mm anti-skid strips to be provided. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
j	Mirror Granite cladding, Sill, Railing counter, shelves, facias etc.	Polished dado, Window sill, shelves, facias etc. a. <u>Sill, Vanity counters, facias</u>- Providing and fixing 18 mm thick gang saw cut, mirror polished, premoulded and prepolished, machine cut of required size, approved shade, colour and texture laid over 20 mm thick base cement mortar 1:4 (1 cement : 4 coarse sand), joints treated with white cement, mixed with matching pigment, epoxy touch ups, including rubbing, curing, moulding and polishing to edges to give high gloss finish etc. complete at all levels. Granite stone slab colour black, Cherry/Ruby red or equivalent as decided by Engineer-in-charge. Single piece granite stone slab shall be provided in window sill. b. Contractor has to provide opening of required size & shape for wash basin/kitchen sink in platform, vanity counter and similar location in Granite/stone work, including necessary holes for pillar taps etc. including moulding, rubbing and polishing of cut edges etc. complete. c. Fixing granite stone in facia and drops of width upto 150 mm with epoxy resin based adhesive, including cleaning etc. complete. d. <u>On elevator (lift) façade/dado/cladding</u> – providing & fixing gang saw cut/ machine cut, mirror polished 18 mm thick Granite stone backfilled with grout of 12 mm (average) thick cement mortar 1:3 (1 cement : 3 coarse sand) laid and jointed with cement slurry @ 3.3 kg/sqm including joints treated with white cement, admixed with matching pigment, epoxy touch ups including rubbing, curing, polishing etc. all complete. For fixing granite for cladding/facade, fastener may be used of suitable size, if required and as per direction of Engineer-in-charge. Granite stone slab colour black, Cherry/Ruby red or equivalent as decided by Engineer-in-charge. e. <u>For Kitchen Counter/platforms, shelves, partitions in flats</u>

SI No.	ITEMS OF WORK	SPECIFICATION
		• <u>Counter/platforms</u> – 2 nos. 18 mm thick granite slabs gang saw cut, mirror polished to be attached using epoxy resin adhesive and supported on granite partitions/vertical slabs (30mm thick gang saw cut, mirror polished on both sides, edges moulded). The counter granite will be both sides polished, edge moulding, finishing, rubbing, curing etc. all complete. Additionally, brackets of appropriate size for stability to be provided including anchorage fasteners & drilling of holes for fixing granite with wall all complete as per site/structural requirement. Counter/platform shall be projecting 50mm beyond under-counter/vertical support. • <u>For Kitchen under-counter & over-head shelves/partitions</u> – 18mm thick granite shelves/partitions, gang saw cut, mirror polished (both sides), edges moulded, rubbing, curing etc. to be fixed using brackets and angles of appropriate size for stability including anchorage fasteners, drilling of holes for fixing granite with wall all complete as per site/structural requirement. For partitions (horizontal & vertical) in shelves similar kind of arrangement to be made. Single piece granite to be used in shelves & partitions. • Heavy-duty MS powder coated L-shaped/or any other shape as suitable for brackets for fixing granite counter/shelf/partitions with wall. Brackets of appropriate length covering at least 2/3rd of the countertop/shelve depth, minimum 6 inches wide and a minimum thickness of 10 mm shall be provided, as per requirement, to support the top counter and overhead/undercounter shelves. These brackets shall ensure adequate structural support and prevent sagging or cracking of the granite countertop & overhead shelves. Number of brackets to be fixed will be such that stability and strength of granite is maintained. • Granite stone slab colour black, Cherry/Ruby red or equivalent as decided by Engineer-in-charge. • The epoxy grout will be applied between wall & granite as well as granite counter/shelf with vertical partition. f. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
k	Granite Moulding & Polishing	Providing edge moulding to 18 mm & 30mm thick stone slabs, including machine polishing to edge to give high gloss finish etc. complete as per design approved by Engineer-in-Charge. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
l	Hardonite flooring	62 mm thick cement concrete flooring with concrete hardener topping, under layer 50 mm thick cement concrete 1:2:4 (1 cement : 2 coarse sand

SI No.	ITEMS OF WORK	SPECIFICATION
		: 4 graded stone aggregate 20mm nominal size) and top layer 12mm thick cement hardener consisting of mix 1:2 (1 cement hardener mix : 2 graded stone aggregate, 6mm nominal size) by volume, hardening compound mixed @ 2 litre per 50 kg of cement or as per manufacture's specifications including cost of cement slurry, nosing of steps etc. complete.
m	Wooden Flooring & skirting	25 mm wooden planking, tongued and grooved in flooring , including fixing with iron screws complete with Secong Class teak wood. Providing & fixing 12 mm thick 2nd class teak wood plain lining tongued and grooved, including wooden plugs complete with necessary screws and priming coat on unexposed surface.
14.	False Ceiling	
a	Calcium silicate tiles	Providing and Fixing 15 mm thick densified tegular edged eco friendly light weight calcium silicate false ceiling tiles of approved texture of size 595 x 595 mm in true horizontal level, suspended on inter locking metal grid of hot dipped galvanised steel sections (galvanizing @ 120 grams per sqm including both side) consisting of main 'T'runner suitably spaced at joints to get required length and of size 24x38 mm made from 0.33 mm thick (minimum) sheet, spaced 1200 mm centre to centre, and cross "T" of size 24x28 mm made out of 0.33 mm (Minimum) sheet, 1200 mm long spaced between main'T' at 600 mm centre to centre to form a grid of 1200x600 mm and secondary cross 'T' of length 600 mm and size 24 x28 mm made of 0.33 mm thick (Minimum) sheet to be inter locked at middle of the 1200x 600 mm panel to from grid of size 600x600 mm, resting on periphery walls /partitions on a Perimeter wall angle pre-coated steel of size (24x24X3000 mm made of 0.40 mm thick (minimum) sheet with the help of rawl plugs at 450 mm centre to centre with 25mm long dry wall screws @ 230 mm interval and laying 15 mm thick densified edges calicum silicate ceiling tiles of approved texture in the grid, including, cutting/ making opening for services like diffusers, grills, light fittings, fixtures, smoke detectors etc., wherever required. Main 'T' runners to be suspended from ceiling using G.I. slotted cleats of size 25x35x1.6 mm fixed to ceiling with 12.5 mm dia and 50 mm long dash fasteners, 4 mm G.I. adjustable rods with galvanised steel level clips of size 85 x 30 x 0.8 mm, spaced at 1200 mm centre to centre along main 'T', bottom exposed with 24 mm of all Tsections shall be pre-painted with polyster baked paint, for all heights, as per specifications, drawings and as directed by Engineer-in-Charge. The calcium silicate ceiling tile shall have NRC value of 0.50 (Minimum), light reflection > 85%, non- combustible as per B.S. 476 part IV, 100% humidity resistance and also having thermal conductivity <0.043 w/mK. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.

SI No.	ITEMS OF WORK	SPECIFICATION
b	Gypsum board false ceiling	Providing and fixing false ceiling at all height including providing and fixing of frame work made of special sections, power pressed from M.S. sheets and galvanized with zinc coating of 120 gms/sqm (both side inclusive) as per IS : 277 and consisting of angle cleats of size 25 mm wide x 1.6 mm thick with flanges of 27 mm and 37mm, at 1200 mm centre to centre, one flange fixed to the ceiling with dash fastener 12.5 mm dia x 50mm long with 6mm dia bolts, other flange of cleat fixed to the angle hangers of 25x10x0.50 mm of required length with nuts & bolts of required size and other end of angle hanger fixed with intermediate G.I. channels 45x15x0.9 mm running at the spacing of 1200 mm centre to centre, to which the ceiling section 0.5 mm thick bottom wedge of 80 mm with tapered flanges of 26 mm each having lips of 10.5 mm, at 450 mm centre to centre, shall be fixed in a direction perpendicular to G.I. intermediate channel with connecting clips made out of 2.64 mm dia x 230 mm long G.I. wire at every junction, including fixing perimeter channels 0.5 mm thick 27 mm high having flanges of 20 mm and 30 mm long, the perimeter of ceiling fixed to wall/partition with the help of rawl plugs at 450 mm centre, with 25mm long dry wall screws @ 230 mm interval, including fixing of gypsum board to ceiling section and perimeter channel with the help of dry wall screws of size 3.5 x 25 mm at 230 mm c/c, including jointing and finishing to a flush finish of tapered and square edges of the board with recommended jointing compound, jointing tapes, finishing with jointing compound in 3 layers covering upto 150 mm on both sides of joint and two coats of primer suitable for board, all as per manufacturer's specification and also including the cost of making openings for light fittings, grills, diffusers, cutouts made with frame of perimeter channels suitably fixed, all complete as per drawings, specification and direction of the Engineer in Charge. 12.5 mm thick tapered edge gypsum moisture resistant board conforming to IS 2095- (Part-1): 2011 (Board with BIS certification marks).
15.	Paint & Finishing works	
a	Putty	Providing and applying white cement based putty of average thickness 1 mm, of approved brand and manufacturer, over the plastered wall surface to prepare the surface even and smooth complete. On external wall water proof putty to be provided. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
b	Plaster of paris	Providing and applying plaster of paris putty of 2 mm thickness over plastered surface to prepare the surface even and smooth complete. Latest CPWD specifications/relevant IS/BIS codes as applicable to be

SI No.	ITEMS OF WORK	SPECIFICATION
		followed.
c	Wall & Ceiling finish in Balcony/Covered Entry porch/Plot Boundary wall	Finishing walls with ready mixed Premium acrylic emulsion paint having VOC less than 50 gm/litre and UV resistance as per IS 15489:2004, Alkali & fungal resistance, dirt resistance exterior paint of required shade (Company Depot Tinted) with silicon additives. Two or more coats applied @ 1.43 litre/ 10 sqm. Over and including priming coat of exterior primer applied @ 0.90 litre/10 sqm. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
d	External Wall Finish	Finishing walls with textured exterior paint of required shade with two or more coats applied @ 3.28 ltr/10 sqm)over and including priming coat of exterior primer applied @ 2.20kg/10 sqm Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
e	Internal finish ceilings	Two or more coats of painting with acrylic emulsion paint of interior grade, having VOC (Volatile Organic Compound) content less than 50 grams/ litre of approved brand and manufacture, including applying additional coats wherever required to achieve even shade and colour with water thinnable cement primer of approved brand and manufacture on surface having VOC content less than 50 grams/litre. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
f	Internal finish Walls	Two or more coats of painting with acrylic emulsion paint of interior grade, having VOC (Volatile Organic Compound) content less than 50 grams/ litre of approved brand and manufacture, including applying additional coats wherever required to achieve even shade and colour with water thinnable cement primer of approved brand and manufacture on surface having VOC content less than 50 grams/litre. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
g	Internal finishing of Shafts	White washing with lime with three or more coats to give even shade. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
h	Structural Steel/MS Painting	Applying two priming coats of With ready mixed red oxide zinc chromate primer and two or more coats of synthetic enamel paint of approved brand & colour on steel galvanised iron/ steel works. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.

SI No.	ITEMS OF WORK	SPECIFICATION
i	Wood work Painting	Painting with synthetic enamel paint of approved brand and manufacture of required colour to give an even shade with Two or more coats on new work over an under coat of suitable shade with ordinary paint of approved brand and manufacture With ready mixed pink or Grey primer on wood work (hard and soft wood) having VOC content less than 50 grams/ litre. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
16.	DOOR & WINDOWS	
a	DOOR/WINDOW FRAME	
a1	2 nd Class Teak Wood Door/Window/Ventilator Frame	2nd class teak wood door/window/ventilator frame of required size, wrought framed and fixed in position with hold fast lugs or with dash fasteners of required dia & length. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
a2	2 nd Class Teak Wood Beading on door/window/ventilator wooden frames	Fixing 2 nd class teak wooden moulded beading of required size to door and window frames/Ventilator frames with iron screws, plugs and priming coat on unexposed surface etc. complete. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
a3	WPC (Wood polymer composite) solid Door Frame	Factory made single extruded WPC (Wood polymer composite) solid frame with 50 x 100 mm size comprising of virgin PVC polymer of K value 58-60 (Suspension Grade), calcium carbonate and natural fibers (wood powder/ rice husk/ wheat husk) and nontoxic additives (maximum toxicity index of 12 for 100 gms) fabricated with miter joints after applying PVC solvent cement and screwed with full body threaded star headed SS screws having minimum frame density of 750 kg/cum, screw withdrawal strength of 2200 N (Face) & 1100 N (Edge), minimum compressive strength of 58 N/mm ² , modulus of elasticity 900 N/mm ² and resistance to spread of flame of Class A category with property of being termite/borer proof, water/moisture proof and fire retardant and fixed in position with M.S hold fast/lugs/SS dash fasteners of required dia and length complete as per direction of Engineer-in- Charge. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
a4	Fire doors Frame & shutter for shafts, Staircase/Exit pathway (complete assembly).	Electrical Shafts - The work has to be executed as per CPWD DSR-2023 Item No. 9.162A or 9.162B as decided by Engineer-in-charge and latest CPWD specification/relevant NBC/IS codes. FHC shaft – as per NBC/CPWD guidelines. Staircase/Exit pathway – as per CPWD DSR -2023 item no. 9.162F.

SI No.	ITEMS OF WORK	SPECIFICATION
		SS 304 fire rated mortice latch and lock with levers and a pair of lever handles of approved quality with necessary screws etc. complete - <u>one no. for each door</u> to be provided as per requirements and direction of Engineer-in-charge. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
a5	Aluminium Fixed Frame	Providing and fixing aluminium work for door, windows, ventilators, partitions with extruded built up standard tubular sections/ appropriate Z sections and other sections of approved make conforming to IS: 733 and IS: 1285, fixing with dash fasteners of required dia and size, including necessary filling up the gaps at junctions, i.e. at top, bottom and sides with required EPDM rubber/neoprene gasket etc. Aluminium sections shall be smooth, rust free, straight, mitred and jointed mechanically wherever required including cleat angle, Aluminium snap beading for glazing / paneling, C.P. brass / stainless steel screws, all complete as per architectural drawings/shop drawings and the directions of Engineer-in-charge. Powder coated aluminium (minimum thickness of powder coating 50 micron). Filling Gap between aluminium frame & adjacent RSS/Masonry/stone by providing weather silicon sealant over backer rod of approved quality. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
a6	MS frame, grills	1. Providing and fixing M.S. Tubular frames for doors, windows, ventilators and cupboard with rectangular/ L-Type sections, made of 1.60 mm thick M.S. Sheet, joints mitred, welded and grinded finish, with profiles of required size, including fixing of necessary butt hinges and screws and applying a priming coat of approved steel primer & fixing with carbon steel galvanised dash fastener of required dia and size. 2. Providing and fixing to openings with rawl plugs, screws etc. M.S. grills of required pattern in frames of windows etc. with M.S. flats, square or round bars etc. including priming coat with approved steel primer all complete. Minimum 12mm bars, 50x6mm flat to be used. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
b	DOOR/WINDOW/Ventilator SHUTTER	
b1	Flush Door shutter	Providing and fixing ISI marked flush door shutters 35mm thick conforming to IS : 2202 (Part I) decorative type, core of block board construction with frame of 1st class hard wood and well matched teak 3 ply veneering with vertical grains or cross bands and face veneers on

SI No.	ITEMS OF WORK	SPECIFICATION
		both faces of shutters including ISI marked stainless steel butt hinges. Providing lipping with 2nd class teak wood battens 25mm minimum depth on all edges of flush door shutter & finished with ready mixed wood primer (pink or grey primer having VOC content less than 50gms/litre) & synthetic enamel paint on all edges of approved brand & manufacture of required colour to give an even shade. 1 mm thick decorative high pressure Laminated sheet of plain / wood grain in gloss/ matt/ suede finish with high density protective surface layer and reverse side of adhesive bonding quality conforming to IS : 2046 Type S, on both sides/faces of flush door shutter. Following fittings per door single leaf: All doors to have: - Aluminium fitting, ISI marked, anodized (anodic coating not less than grade AC 10 as per IS: 1868), transparent or dyed to required colour or shade, with necessary screws etc. complete: - Tower bolt 250 x 10 mm - <u>one no. for each door</u> - Hanging floor door twin rubber stopper - <u>one no. for each door leaf</u> - Chromium plated brass 100 mm mortice latch and lock with 6 levers and a pair of lever handles of approved quality with necessary screws etc. complete - <u>one no. for each door</u>. - PVC door buffer (3 inch) – <u>one no. for each door leaf</u> - Minimum 04 Nos. SS 304 butt hinges of size 125*64*2.50 mm for each door leaf. - SS 304 Bright/Matt Finished Stainless Steel Handles 125 mm - <u>two nos. for each door leaf (to be provided only in Community Centre door D4)</u>. - Aluminium extruded section body tubular type universal hydraulic door closer (having brand logo with ISI, IS : 3564, embossed on the body, door weight upto 36 kg to 80 kg and door width from 701 mm to 1000 mm), with double speed adjustment with necessary accessories and screws etc. complete – <u>one no. for each door except that for doors of flats in residential blocks</u>. Note: Stainless Steel (SS) Magic Eye to be provided & fixed on flush door shutter of Main Entrance to units/flats of Residential blocks. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
b2	2 nd class teak wood shutters with Wire gauge for doors/windows & Glazing for windows	30mm thick Shutter of 2nd class teak wood with panels of Stainless Steel grade 304 fly proof wire gauge Mesh with wire of dia. 0.5mm and average width of aperture 1.4mm in both directions fixed with 2nd class teak wood beading as required with necessary screws and ISI marked stainless steel butt hinges of required size.

SI No.	ITEMS OF WORK	SPECIFICATION
		30mm thick shutter of 2nd class teak wood with glazing of 5 mm thick float glass panes (weight not less than 12.50 kg/sqm) fixed with 2nd class teak wood beading as required with necessary screws and ISI marked stainless steel butt hinges of required size. The wood work to be painted with ready mixed wood primer (pink or grey primer having VOC content less than 50gms/litre) and synthetic enamel paint all around of approved brand & manufacture of required colour to give an even shade. Following fittings per door/window single leaf: All door to have: - Aluminium fitting, ISI marked, anodized (anodic coating not less than grade AC 10 as per IS: 1868), transparent or dyed to required colour or shade, with necessary screws etc. complete: - Tower bolt 250 x 10 mm - <u>one no. for each door</u> - Tower bolt 150x x10 mm - <u>one no. for each door</u> - Hanging floor door twin rubber stopper - <u>one no. for each door</u> - Sliding door bolts 300 x16 mm – <u>02 nos. for main entrance door and one no. for balcony doors.</u> - SS 304 Bright/Matt Finished Stainless Steel Handles 125 mm - <u>two nos. for each door.</u> - Minimum 04 Nos. SS 304 butt hinges of size 125*64*2.50 mm for each door. - Aluminium extruded section body tubular type universal hydraulic door closer (having brand logo with ISI, IS : 3564, embossed on the body, door weight upto 36 kg to 80 kg and door width from 701 mm to 1000 mm), with double speed adjustment with necessary accessories and screws etc. complete – <u>one no. for each door (for doors of flats in residential blocks door closer not to be provided).</u> All windows to have: - Aluminium fitting, ISI marked, anodized (anodic coating not less than grade AC 10 as per IS: 1868), transparent or dyed to required colour or shade, with necessary screws etc. complete: - Tower bolt 100 x 10 mm - <u>two nos. for each shutter</u> - Casement stay 250mm - <u>one no. for each shutter</u> - SS 304 Bright/Matt Finished Stainless Steel Handles 75 mm - <u>one nos. for each shutter.</u> - Minimum 02 Nos. SS 304 butt hinges of size required size for each shutter. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.

SI No.	ITEMS OF WORK	SPECIFICATION
b3	WPC (Wood polymer composite) solid flush door shutter	Factory made single extruded WPC (Wood polymer composite) solid decorative type 30 mm thk. flush Door shutter comprising of virgin polymer of K value 58-60 (Suspension Grade), calcium carbonate and natural fibers (wood powder/ rice husk/wheat husk) and non toxic additives (maximum toxicity index of 12 for 100 gms) having minimum density of 650 kg/cum and screw withdrawal strength of 1800 N (Face) & 900 N (Edge), minimum compressive strength 50 N/mm², modulus of elasticity 850 N/mm² and resistance to spread of flame of Class A category with property of being termite/borer proof, water/moisture proof and fire retardant. WPC to be laminated with PVC foil of minimum 14 microns thick of approved design pasted with hot melt adhesive on both faces of shutter and fixing with stainless steel butt hinges of required size with necessary full body threaded star headed counter sunk S.S screws, all as per direction of Engineer-In- Charge. All toilet doors to have: a. Aluminium fitting, ISI marked, anodized (anodic coating not less than grade AC 10 as per IS: 1868), transparent or dyed to required colour or shade, with necessary screws etc. complete: • Tower bolt 150x x10 mm - <u>one no. for each door</u> • Hanging floor door twin rubber stopper - <u>one no. for each door</u> • Sliding door bolts 250 x16 mm – <u>02 nos. for each doors.</u> • 50 cm long aluminium kicking plate of size 100x3.15 mm – <u>inside of Divyang toilet door shutter.</u> b. SS 304 Bright/Matt Finished Stainless Steel Handles 100 mm - <u>two nos. for each door.</u> c. Minimum 04 Nos. SS 304 butt hinges of required size for each door. d. PVC Door Buffer (3 inch) – <u>one no. for each door</u> Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
b4	Aluminium shutters & Louvers	Providing and fixing aluminium work for door, windows, ventilators, partitions with extruded built up standard tubular sections/ appropriate Z sections and other sections of approved make conforming to IS: 733 and IS: 1285, fixing with dash fasteners of required dia and size, including necessary filling up the gaps at junctions, i.e. at top, bottom and sides with required EPDM rubber/neoprene gasket etc. Aluminium sections shall be smooth, rust free, straight, mitred and jointed mechanically wherever required including cleat angle, Aluminium snap beading for glazing / paneling, C.P. brass / stainless steel screws, all complete as per architectural drawings/shop drawings and the directions of Engineer-in-charge. Powder coated aluminium (minimum thickness of powder coating 50 micron) including providing and fixing hinges/pivots and making provision for fixing of fittings wherever required including EPDM rubber/neoprene gasket as required.

SI No.	ITEMS OF WORK	SPECIFICATION
		Necessary handles for shutters & powder coated concealed/touch locking arrangement to be provided for interlocking all types of shutters with frames. Aluminium fittings for aluminium louvers of plumbing shaft - ISI marked, anodized (anodic coating not less than grade AC 10 as per IS: 1868), transparent or dyed to required colour or shade, with necessary screws etc. complete: a. 01 no. Tower bolt 250 x 10 mm to be provided. b. aluminium handle of 125mm. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
b5	Glazing/Wire Mesh for Aluminium Shutters/frames/Partitions	Float Glass panes of 5mm thickness (Weight not less than 12.50 Kg/m) Frosted glass panes 4mm thick (weight not less than 10kg/sqm) Stainless Steel grade 304 fly proof wire gauge Mesh with wire of dia. 0.5mm and average width of aperture 1.4mm in both directions fixed with aluminium shutter with necessary accessories. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
b6	MS shutter	1mm thick M.S. sheet door shutter with frame of 40x40x6mm angle iron and 3 mm M.S. gusset plates at the junctions and corners, all necessary fittings complete (like locking arrangement & handles, rivets etc.), including applying a priming coat of approved steel primer with M.S. angles 40x40x6 mm for diagonal braces and fixing shutters with outermost frame (fixed to wall with dash fastener) etc. all complete. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
b7	Frameless Toughened glass door & Partition	Providing and fixing mill finish powder coated System Profile 12 mm thick toughened glass partition with 12 mm thick toughened glass doors, aluminium system profile size (25X45mm) and every glass joint use I-Profile frame for glass partition, with all necessary accessories like door handle, SS pivot with pin (SS 304), hydraulic floor spring, door lock etc. All fittings & fixtures suitable to door & partition size. Providing & fixing upto 70% of the toughened glass door & partition with self-adhesive vinyl or polyester privacy films in frosted or patterned finish such that privacy is maintained. Pattern/frost look to be approved by Engineer-in-charge. H-Type pull handle: Material: Stainless Steel 304 Grade Finish: Satin Stainless Steel (SSS)

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		Diameter: 25 mm Length: 900 mm Fixing Type: Back-to-back mounting (handles on both sides of the door) Compatibility: Suitable for Glass, Wood, and Metal Doors Features: Corrosion-resistant, durable Comfortable grip with a 25 mm diameter Sleek, modern design Ideal for commercial use Installation: Requires drilling for back-to-back fixing, suitable for glass doors. Double action hydraulic floor spring of approved brand and manufacture conforming to IS : 6315, having brand logo embossed on the body / plate with double spring mechanism and door weight upto 125 kg or as required, for doors, including cost of cutting floors, embedding in floors as required and making good the same matching to the existing floor finishing and cover plates with brass pivot and single piece M.S. sheet outer box with slide plate etc. complete with stainless steel cover plate minimum 1.25mm thickness. Offset Plastic seal: Type: Offset plastic seal (lip or Y-type profile) Material: Soft PVC or TPR (flexible and durable) Glass Fit: Designed for 10mm–12mm glass Installation: Clip-on / press-fit (tool-free, no adhesive) Length: Usually 2.1m to 2.5m Color: Clear / transparent (frosted options available) Purpose: Seals gaps between overlapping glass panels or glass-to-wall Features: Water & dust resistant Sound insulation Glass Clamp Rail: Material: Zinc (corrosion-resistant) Length: 1500 mm (1.5 meters) Finish: Zinc-plated for added durability Purpose: Secures glass panels in sliding or fixed glass door systems Mounting Type: Wall or ceiling-mounted
b8	uPVC Sliding doors	Providing and fixing factory made uPVC glazed/wire mesh windows/doors comprising of lead free uPVC multi-chambered frame, sash and mullion/coupler (where ever required) extruded profiles having minimum wall thickness of 1.70 mm for Series R1 and R2 profiles and 2.10 mm for Series R3 and R4 profiles conforming to EN: 12608 in any shape, colour and design duly reinforced with galvanized mild steel section made of required shape & size as per CPWD Specification, uPVC extruded glazing beads, interlocks and Inline sash adaptor (where

SI No.	ITEMS OF WORK	SPECIFICATION
		ever required) of appropriate dimension, EPDM gasket, hardware, SS 304 grade fasteners of nimum 8 mm dia with countersunk head, comprising of matching polyamide PA6 grade sleeve for fixing frame to finished wall as per IS 1367 : Part 1 to 14, plastic packers, plastic caps and necessary stainless steel screws etc. Profile of frame, sash & mullion (if required) shall be mitred cut and fusion welded/mechanically jointed duly sealed at all corners, including drilling of holes for fixing hardware and drainage of water etc. After fixing frame the gap between frame and adjacent finished wall shall be filled with weather proof silicon sealant over backer rod of approved size and quality, all complete as per approved drawing conforming backer rod of approved size and quality, all complete as per approved drawing conforming to CPWD specification & direction of Engineer- in-Charge. Section of steel reinforcement and cross sections of uPVC profiles to be as per design approved by Engineer-in-Charge. SS Wire mesh / Glazing of plain/ toughened/ laminated/ double glass unit with / without high performance coatings as per design requirements and conforming to IS: 3548 & IS: 16231. Note: Structural design proof checked from a Government Engineering Institute, to be provided by the manufacturer for : (i) Sites with basic wind speed > 45 m/sec as per IS 875 – Part 3 (ii) Sites with structure height more than 20m for all wind speeds a. With Three track three panels sliding door with two glazed & one wire mesh panels with Aluminium channel for roller track, wool pile, zinc alloy (zamak) powder coated handle on two panels along with multi- point locking system, adjustable nylon rollers with SS 304 body. Using R3 series with frame (98mm & above) x (40mm & above) & both glazed and fly screen sash (30mm & above) x (74mm & above) for Height upto 2.5 metre. b. Glazing: with 6 mm thk. Toughened glass panel. c. Wire Mesh: SS 304 using wire gauge with average width of aperture 1.4mm in both directions with wire of 0.50mm all complete with 12mm mild steel U beading. If height comes out to be more than 2.5m as per requirements then DSR – 2023 item no. 9.147B3.2 shall apply with R4 series. Necessary fitting, fixtures, handles, locks (zinc alloyed (powder coated) touch lock with necessary screws etc. complete) shall be provided. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.

SI No.	ITEMS OF WORK	SPECIFICATION
17.	Wardrobes in Flats of Residential Blocks	a. <u>Shutter</u> – Providing & fixing 19mm thick one side decorative and other side balancing lamination factory pressed BWP grade marine ply as per IS 710 of approved brand including 2mm thick PVC edge banding tape with hot glue by edge bending machine. (DSR-2023 item 9.168) b. Providing and fixing stainless steel soft closing spring hinges at 0 degree hinges (hydraulic type) of approved make/brand to cupboard shutters with full threaded steel screws including making necessary recess in board and finished etc. complete as per direction of Engineer-in-charge (DSR-2023 item no. 9.171) – <u>minimum 04 nos. for each shutter leaf.</u> c. Providing and fixing 19mm thick both side balancing lamination factory pressed BWP grade marine ply as per IS 710 of approved brand boxes,shelves,racks,almirah,cupboard and drawer etc. including necessary nails,screws etc. complete as per direction of Engineer-in-charge (DSR-2023 item 9.169). d. Providing and fixing chromium plated brass 50 mm cupboard or wardrobe knobs with nuts complete (DSR-2023 item 9.91). e. Providing and fixing double strip (horizontal type) magnetic catcher of approved quality in cupboard /ward robe shutters, including fixing with necessary screws etc. complete (DSR-2023 item 9.114.2). f. Providing and fixing special quality chromium plated brass cupboard locks with six levers of approved quality including necessary screws etc. complete (DSR-2023 item 9.90.3). g. 20mm dia. Aluminium hanger rod with slotted (non-screw type) brackets. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed and work as per approved drawings & direction of Engineer-in-charge.
18.	Kitchen cabinets shutter in Flats of Residential Blocks	a. <u>Shutter</u> – Providing & fixing 19mm thick one side decorative and other side balancing lamination factory pressed BWP grade marine ply as per IS 710 of approved brand including 2mm thick PVC edge banding tape with hot glue by edge bending machine. (DSR-2023 item 9.168). b. Providing and fixing stainless steel soft closing spring hinges at 0 degree hinges (hydraulic type) of approved make/brand to cupboard shutters with full threaded steel screws including making necessary recess in board and finished etc. complete as per direction of Engineer-in-charge (DSR-2023 item no. 9.171) – <u>minimum 02 nos. for each shutter leaf.</u> c. Providing and fixing chromium plated brass 50 mm cupboard or wardrobe knobs with nuts complete (DSR-2023 item 9.91). d. Providing and fixing double strip (horizontal type) magnetic catcher of approved quality in cupboard /ward robe shutters, including fixing with necessary screws etc. complete (DSR-2023 item 9.114.2). all fittings/fixtures complete.

Sl No.	ITEMS OF WORK	SPECIFICATION
		Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed and work as per approved drawings & direction of Engineer-in-charge.
19.	Curtain Rod	Curtain rods with required accessories of SS grade 304 in all windows & doors in all rooms except kitchen, Toilets.
20.	Invisible grill	Supply & Installation of Invisible Grill in Balconies of flats of residential blocks (8 nos. Stilt+6) as under: 3mm thick 316 Grade stainless steel Invisible grills with all required / involved activities like fixing aluminium track/channels or brackets etc. as required all complete, assembling, grinding, lifting & shifting of grills/material at designated location manually, transportation, equipment's charges for work, tools & tackles, cleaning (wherever applicable), etc. all complete. Specifications Required (these are minimum criteria): • Gap between two vertical wire ropes – 50 mm (2 inches approx.) • Desired no. of Horizontal Ropes equally distributed @ 2.5 feet height and fixed using (cross) nickel plated stiffener SS 316 grade with vertical wire at each crossing of wire ropes. • 7 x 7 Core Wire SS 316 L/SS 316 Marine Grade Nylon Coated (Nylon coating 0.5 mm thick). • 1.25 Inch Aluminium Track (Duly White PU Coated). • SS 304 Fittings, SS 316 Stiffener • Aluminium Rivet for Fixing cover with section • Wire rope diameter - 3mm • Nickel Plating Stiffener SS316 • Screw Fastener HRD C M8*80mm or relevant SS anchor fasteners of approved make. • SS 304 locking screws/bolts with nylon washers fitted on Aluminium Profile section. • Each end of SS 316/316L grade wire shall be crimped with 2-3mm thick aluminium ferrule. The work is to be done to ensure safety of residents. The work is of specialized nature and to be executed by engaging specialized agencies after approval by the Engineer-in charge.
21.	STEEL WORKS AND RAILINGS	
a	Staircase/balcony /Buildings Entry Ramps	Providing and fixing stainless steel railing (Grade 304 of as per IS 6911:2017) made of Hollow tubes, channels, plates etc., including welding, grinding, buffing, polishing and making curvature (wherever required) and fitting the same with necessary stainless steel nuts and bolts complete, i/c fixing the railing with necessary accessories & stainless steel dash fasteners , stainless steel bolts etc., of required size, on the top of the floor or the side of waist slab or at knee/stub wall of Balcony/Lobby/corridors, entry ramps to buildings and wherever

SI No.	ITEMS OF WORK	SPECIFICATION
		proposed in the architectural drawings etc. with suitable arrangement as per approval of Engineer-in-charge complete. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
b	Structural Steel	1. Structural steel work riveted, bolted or welded in built up sections, trusses and framed work, including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer all complete as per approved drawings. 2. Steel work welded in built up sections/ framed work, including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer using structural steel etc. as required in stringers, treads, landings etc. of stair cases, including use of chequered plate wherever required, all complete and in gratings, frames, guard bar, ladder, railings, brackets, gates and similar works as per approved drawings. 3. Steel work in built up tubular (round, square or rectangular hollow tubes etc.) trusses, posts/column etc., including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer, including welding and bolted with special shaped washers etc. complete. The structural steel tubes shall conform to IS 1161/1239/4923. 4. MS Hand rail of approved size by welding etc. to steel ladder railing, staircase railing and similar works, including applying priming coat of approved steel primer as per approved drawings. 50mm dia pipe to be provided in handrail or as per approved drawings. Necessary fasteners/fixtures/plates/other structural steel members etc. as required to complete the work shall be provided. All steel work to be finished with two priming coats of with ready mixed red oxide zinc chromate primer and two or more coats of synthetic enamel paint of approved brand & colour. Reference for requirement in some structures: a. In internal electrical trench, cover to be provided for the whole stretch of trench as under: • frame work for covers not more than 0.9m long along the length of trench to be provided with minimum 8mm thick chequered plate for the required width of trench. • All sides of cover frame to be provided with angle sections along with angle section in middle as well along the length of trench for fixing chequered plate on top such that chequered plate is flushed with adjacent flooring. • 02 nos. suitable MS collapsible hooks/handles to be fixed to

SI No.	ITEMS OF WORK	SPECIFICATION
		chequered plate of every cover as lifting arrangement. • For placement of covers suitable angle sections along the length of trench should also be provided at time of casting of RCC trench to protect both the edges of trench/floor along with angle section support through the width of trench (fitted/welded as per size of cover) such that covers with angle sections are placed/fitted on these properly. b. For chainlink fencing around DG sets, STP etc.: • Top, bottom and cross bracing of suitable angle sections to be provided and fixed/welded for 1.5 metre high chain link fencing along with angle post at every 3m C/C with angle strut at every 4th angle post. • Similarly, gate of 1.5 m high with frame made of angle section all around with cross bracings to be provided and fixed/welded with chain link. Necessary sliding bolt for closing/locking arrangement of gate also to be provided. c. MS ladder on terrace for mumty and OHWT • Width of ladder to be for comfortable climbing. • Rod/Square Bar 20mm spaced at maximum 300mm c/c to be provided in rungs/steps and welded with side stringer/rails of 75x75x6mm angle. • Suitable brackets with angle section 75x75x6mm to be provided for welding with plates anchored to wall/beam. • Safety cage with MS flat 50x4mm (hoop rings) to be provided @ 1m c/c and fixed with vertical flats 50x4mm 3 nos. • Suitable Anchor bolts/fasteners to be provided for secure fixing to wall. • Side rails/hand rail to be extended minimum 1100mm above Mumty/OHWT roof level for safe transition. Structural steel sections and their grade for all locations/structures including mentioned above will be as per approved drawings to be developed as per structural requirements fitting the intended purpose/site conditions. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
c	Rolling shutter	Supplying and fixing rolling shutters of approved make, made of required size M.S. laths, interlocked together through their entire length and jointed together at the end by end locks, mounted on specially designed pipe shaft with brackets, side guides and arrangements for inside and outside locking with push and pull operation complete, including the cost of providing and fixing necessary 27.5 cm long wire springs manufactured from high tensile steel wire of adequate strength conforming to IS: 4454 - part 1 and M.S. top cover of required thickness

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		for rolling shutters. (80x1.25 mm M.S. laths with 1.25 mm thick top cover). Providing and fixing ball bearing for rolling shutters as per the requirements of shutters size and with mechanical device for operating rolling shutters. Providing mechanical device for operating rolling shutters of all sizes. As per requirements grills shall also be provided in rolling shutters manufactured out of 8 mm dia M.S. bar instead of laths as per drawing. Suitable ventilation grills for 500mm height to be provided in rolling shutters of Electrical Sub-station and Panel Room of UGWT & Pump room. Handles, locks, fasteners and all fittings to be provided. All steel work to be finished with two priming coats of With ready mixed red oxide zinc chromate primer and two or more coats of synthetic enamel paint of approved brand & colour. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
d	Parking shed	The sub-structure shall be of RCC foundations with anchor /foundations bolts provided in pedestal for fixing base plate for structural steel super structure as per the design. The structural steel may be built up tubular or sections/framed work as under: 1. Structural steel work riveted, bolted or welded in built up sections, trusses and framed work, including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer all complete as per approved drawings. 2. Steel work in built up tubular (round, square or rectangular hollow tubes etc.) trusses, posts/column etc., including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer, including welding and bolted with special shaped washers etc. complete. The structural steel tubes shall conform to IS 1161/1239/4923. Necessary members like purlins, rafters etc. will be provided as per approved drawings. The entire structure is to be designed as per IS 800:2007 or any other relevant IS/BIS codes with latest correction slips. Roof sheeting shall be corrugated/profiled GS sheet (conform to IS 277) including vertical / curved surface fixed with polymer coated J or L hooks, bolts and nuts 8 mm diameter with bitumen and G.I. limpet

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		washers or with G.I. limpet washers filled with white lead, including a coat of approved steel primer and two coats of approved paint on overlapping of sheets complete (up to any pitch in horizontal/ vertical or curved surfaces), including cutting to size and shape wherever required. The minimum 0.63 mm thick roof sheet with zinc coating not less than 275 gm/sqm shall be used as per approved structural Drawings, Latest CPWD specifications 2019 & directions of the Engineer in Charge. Primary structural connections are to be with hot-dip galvanized high tensile bolts of class 8.8/4.6 or as per approved structural drawings. Fabrication and tolerances shall be as per IS 7215 & IS 12843 respectively or as per relevant applicable IS/BIS codes/ latest CPWD specifications. Anchor Bolts/Foundation bolts Anchor bolts of class 8.8/4.6 shall conform to IS 5624/IS 1367/IS 3757 or as per relevant applicable IS/BIS codes and as per approved structural drawings. Nuts & lock nuts conform to IS 1363 (part 1 to 3) and washer shall conform to IS 2016. All anchor bolts shall have lock nuts. The portion of anchor bolts projecting above concrete (inclusive of the part within grout) shall be mechanically cleaned and painted as per the specification after erection. The total no of anchor bolts shall not be less than 04 nos in any base plate. The gap between plate and concrete to be suitably filled with cementitious grout as per approved structural drawings. All steel work to be finished with two priming coats of With ready mixed red oxide zinc chromate primer and two or more coats of synthetic enamel paint of approved brand & colour. Structural steel sections and their grade will be as per approved drawings to be developed as per structural requirements fitting the intended purpose/site conditions. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
e	MS Main Entry/Exit/Wicket Gate	Steel work in built up tubular (round, square or rectangular hollow tubes etc.) trusses, posts/column etc., including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer, including welding and bolted with special shaped washers etc. complete. The structural steel tubes shall conform to IS 1161/1239/4923. Structural steel work in riveted, bolted or welded in built up sections,

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		trusses and framed work, including cutting, hoisting, fixing in position and applying a priming coat of approved steel primer all complete as per approved drawings will also be provided to complete the work. Necessary MS hinges supporting the swing of gates to be fixed to columns as per approved drawings. All steel work to be finished with two priming coats of With ready mixed red oxide zinc chromate primer and two or more coats of synthetic enamel paint of approved brand & colour. The structure to be designed aligning with the entrance and other structures. Structural steel sections and their grade will be as per approved drawings to be developed as per structural requirements fitting the intended purpose/site conditions. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
f	Concertina coil fencing in Boundary Wall	Providing and fixing concertina coil fencing with punched tape concertina coil 600 mm dia 10 metre openable length (total length 90 m), having 50 nos rounds per 6 metre length, upto 3 m height of wall with existing angle iron 'Y' shaped placed 2.4 m or 3.00 m apart and with 9 horizontal R.B.T. reinforced barbed wire, stud tied with G.I. staples and G.I. clips to retain horizontal, including necessary bolts or G.I. barbed wire tied to angle iron, all complete as per direction of Engineer-in-charge, with reinforced barbed tape(R.B.T.) / Spring core (2.5mm thick) wire of high tensile strength of 165 kg/ sq.mm with tape (0.52 mm thick) and weight 43.478 gm/ metre. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
g	Chain link fencing	Providing and fixing with G.I. wire of dia. 4mm, chain link fabric fencing of required width in mesh size 50x50 mm including strengthening with 2 mm dia wire or nuts, bolts and washers as required complete as per the direction of Engineer-in-charge. The chain link fencing shall be fixed for DG set area of Electrical Substation, around STP, UGWT etc. as per approved drawings. Latest CPWD specifications/relevant IS/BIS codes as applicable to be followed.
22.	Roads, Footpaths	
a	Kerb stone	Along the road, footpath to retain paver blocks and as per drawings. Providing and laying at or near ground level factory made kerb stone of M-25 grade cement concrete in position to the required line, level and curvature, jointed with cement mortar 1:3 (1 cement: 3 coarse sand), including making joints with or without grooves (thickness of joints except at sharp curve shall not to more than 5mm), including making

SI No.	ITEMS OF WORK	SPECIFICATION
		drainage opening wherever required complete etc. as per direction of Engineer-in-charge (Precast C.C. kerb stone shall be approved by Engineer-in-charge). Latest CPWD specifications/IRC/Morth/relevant IS/BIS codes as applicable to be followed.
b	Paver blocks (footpath, building entry ramps)	Providing and laying factory made M-30 Paver Block having 60 mm thickness chamfered edge Cement Concrete paver blocks in footpath, parks, lawns, etc, of required strength, thickness & size/ shape, laid in required colour & pattern over 50mm thick compacted bed of sand, compacting and proper embedding/laying of inter locking paver blocks into the sand bedding layer through vibratory compaction by using plate vibrator, filling the joints with sand and cutting of paver blocks as per required size and pattern, finishing and sweeping extra sand. complete all as per direction of Engineer-in-Charge. Paver Block with 50mm sand bed has to be provided as over 100 mm thick 1:3:6 PCC & as per CPWD specifications. The bedding sand & joint filling sand should comply with the CPWD/IRC norms as applicable. (Design, colour of paver block shall be approved by Engineer-in-charge). Latest CPWD specifications/IRC/Morth/relevant IS/BIS codes as applicable to be followed.
c	Tactile flooring	Providing and laying tactile tile (for vision impaired persons as per standards) of size 300x300x9.8mm having with water absorption less than 0.5% and conforming to IS:15622 of approved make in all colours and shades in for outdoor floors such as footpath, court yard, multi modals location etc., laid on 20mm thick base of cement mortar 1:4 (1 cement : 4 coarse sand) in all shapes & patterns including grouting the joints with white cement mixed with matching pigments etc. complete as per direction of Engineer-in-Charge. (DSR-2023 item 16.90). Latest CPWD specifications/IRC/Morth/relevant IS/BIS codes as applicable to be followed.
d	Internal/External Road & On-street Parking area	a. Providing and laying design mix cement concrete of M-30 grade vacuum dewatered concrete, using cement content as per design mix, using coarse sand and graded stone aggregate of 40 mm nominal size in appropriate proportions as per approved & specified design criteria, providing dowel bars with sleeve/ tie bars wherever required, laying at site, spreading and compacting mechanically by using needle and surface vibrators, levelling to required slope/ camber, floating, trowelling, finishing with required texture/broom finish, including steel form work with sturdy M.S. channel sections, curing, making provision for contraction/ expansion, construction & longitudinal joints (10mm wide x 50 mm deep) by groove cutting machine, providing and filling joints with approved joint filler,

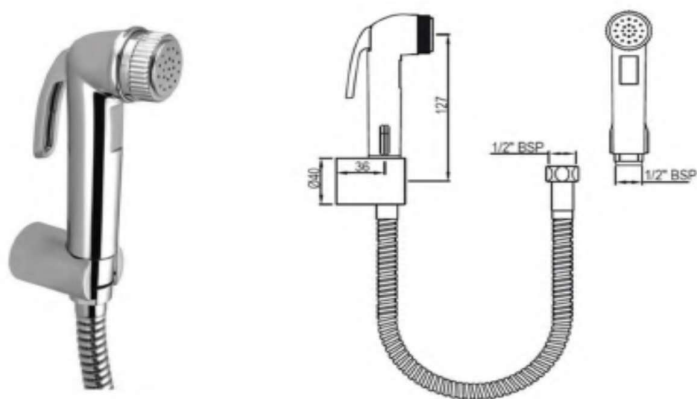
SI No.	ITEMS OF WORK	SPECIFICATION
		sealants, complete all as per direction of Engineer-in-charge. b. Providing and fixing pre-moulded joint filler in expansion joints of RCC roads / CC pavements after making the joints dust free with high pressure air jet cleaners, all complete as per direction of the Engineer-in-Charge. (Pre-moulded joint fillers shall be made of bitumen hot sealing compound impregnated fibre board having impregnation more than 35%, conforming to IS:1838 for fibre board and IS: 1834 for hot sealing bitumen compound grade A.). c. Providing and filling in position rubberized bitumen hot sealing compound Grade A confirming to IS 1834 for sealing of expansion/contraction/longitudinal/traverse joints in roads / pavements all complete as per direction of the Engineer-in-Charge. d. Providing and applying 2.5 mm thick road marking strips (retroreflective) of specified shade/ colour using hot thermoplastic material by fully/ semi automatic thermoplastic paint applicator machine fitted with profile shoe, glass beads dispenser, propane tank heater and profile shoe heater, driven by experienced operator on road surface including cost of material, labour, T&P, cleaning the road surface of all dirt, seals, oil, grease and foreign material etc. complete as per direction of Engineer-in-charge and accordance with applicable specifications. The Thermoplastic paint shall be used for marking parking strips, Parking symbol for PWD person, edges of road/parking, etc. as per direction of Engineer-in-charge. The minimum CBR of 8% is recommended for the 500mm subgrade. Latest CPWD specifications/IRC/Morth/relevant IS/BIS codes as applicable to be followed.
e	WMM	Providing, laying, spreading and compacting graded stone aggregate (size range 53 mm to 0.075 mm) to wet mix macadam (WMM) specification including premixing the material with water at OMC in for all leads & lifts, laying in uniform layers with mechanical paverfinisher in sub- base / base course on well prepared surface and compacting with vibratory roller of 8 to 10 tonne capacity to achieve the desired density, complete as per specifications and directions of Engineer-in-Charge. Latest CPWD specifications/IRC/Morth/relevant IS/BIS codes as applicable to be followed.
f	GSB	Construction of granular sub-base by providing close graded Material conforming to specifications, mixing in a mechanical mix plant at OMC, carriage of mixed material by tippers to work site, for all leads & lifts, spreading in uniform layers of specified thickness with motor grader on prepared surface and compacting with vibratory power roller to achieve

SI No.	ITEMS OF WORK	SPECIFICATION
		the desired density, complete as per specifications and directions of Engineer-in-Charge. With material conforming to Grade-I (size range 75 mm to 0.075 mm) having CBR Value-30. Latest CPWD specifications/IRC/Morth/relevant IS/BIS codes as applicable to be followed.
g	WBM	Supplying, stacking, laying, spreading and compacting stone aggregate of specified sizes to WBM specifications in uniform thickness, hand picking, rolling with 3 wheeled road/vibratory roller 8-10 tonne capacity in stages to proper grade and camber, applying and brooming requisite type of screening / binding material to fill up interstices of coarse aggregate, watering and compacting to the required density. WBM III to be provided as per requirements. Latest CPWD specifications/IRC/Morth/relevant IS/BIS codes as applicable to be followed.
23.	Over Head Water Tank	• OH Tank for Fire (As per NBC 2016) • OH Tank for Domestic water (As per Water requirement) • OH Tank for Flushing water (As per Water requirement) • Water proofing with fixing ceramic tiles on internal walls & floors for RCC water tanks as per specifications.
24.	Underground Water Tank	RCC UG Tank for Fire & Domestic water (As per Water requirement).
25.	Pre-cast RCC covers	Providing and placing in position 100 mm thick factory made machine batched & machine mixed Precast RCC Rectangular Covers of various sizes, of minimum M-25 grade cement concrete for RCC work, including cost of centering, shuttering, reinforcement of 8 mm dia TMT bars of Fe 500 grade @ maximum 100mm c/c on both ways, neat cement punning on finished surface, properly encased on all edges with 1.6 mm thick, 100 mm wide MS sheet duly painted over priming coat, reinforcement to be welded at edges with MS sheet and providing 2 Nos. 12 mm dia bar for hooks etc i/c cost of cartage, all leads & lift, handling at site etc. all complete as per approved drawings & direction of Engineer-in-Charge. The precast covers to be provided on external Electrical trench. Latest CPWD specifications/IRC/Morth/relevant IS/BIS codes as applicable to be followed.

SUMMARY OF PLUMBING/SANITARY SPECIFICATIONS

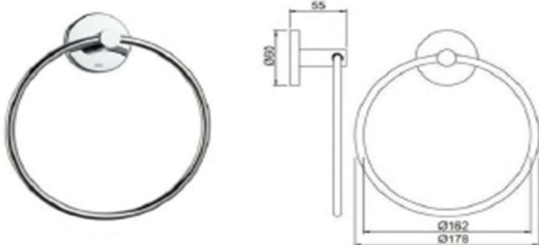
SI NO.	ITEMS OF WORK	SPECIFICATION
1.	Water Closet- Wall Mounted	Providing & Fixing white vitreous china extended wall mounting water closet of size 780x370x690 mm of approved shape including providing & fixing white vitreous china cistern with dual flush fitting, of flushing capacity 3 litre/ 6 litre (adjustable to 4 litre/ 8 litres), including seat cover, and cistern fittings, nuts, bolts and gasket etc complete as per DSR Item No. 17.78 and latest CPWD Specifications. (Reference Image is given below) 
2.	Water Closet- Floor Mounted	Providing & Fixing floor mounted, white vitreous china single piece, double traps syphonic water closet of approved brand/make, shape, size and pattern including integrated white vitreous china cistern of capacity 10 litre with dual flushing system, including all fittings and fixtures with seat cover, cistern fittings, nuts, bolts and gasket etc including making connection with the existing P/S trap, complete in all respect as per directions of Engineer-in-Charge as per DSR Item No. 17.81 and latest CPWD Specifications. (Reference Image is given below)

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3.	Urinal	Providing & Fixing white vitreous china battery based infrared sensor operated urinal of approx. size 610 x 390 x 370 mm having pre & post flushing with water (250 ml & 500 ml consumption), having water inlet from back side, including fixing to wall with suitable brackets all as per manufacturers specification and direction of Engineer-in-charge as per DSR Item No. 17.80 and latest CPWD Specifications.
4.	Urinals Division plate	Providing & fixing Urinal Division Plate of granite with table rubbed, edges rounded and polished, of size 900 x 450 mm, fixed in urinal partitions by cutting a chase of appropriate width with chase cutter and embedding the stone in the chase with epoxy grout or with cement concrete 1:2:4 (1 cement: 2 coarse sands: 4 graded stone aggregate 6 mm nominal size).
5.	Flat Back Washbasin	Providing and fixing White Vitreous China Wash basin size 550x400 mm approx. with C.I. brackets, 32 mm C.P. brass waste of standard pattern, approved quality and make, including painting of fittings and brackets, cutting and making good the walls wherever required with a 15 mm CP Brass single hole basin mixer. Ref. as per DSR Item No. 17.7A and latest CPWD Specifications. Net weight of wash basin should be equal to or more than 12 Kgs. (for basin mixer refer CP brass mixture for Wash basin)
6.	Under Counter Wash Basin	Providing & Fixing Oval Shaped under counter White Vitreous china wash basin (product dimension 605x430x200 mm & basin inner dimension 545x365x160 mm) with vertical outlet, ID of waste outlet hole dia. $\varnothing 45 \pm 3$ mm, Horizontal distance between the Bowl edge and the center line of the waste outlet hole ≤ 114 mm including Over flow hole cap, UC Screw(clamp), CP brass waste, Template. Wash basin fixed with Granite Top, premoulding at edges all compete with all fittings and brackets, cutting and making good the walls wherever required as per latest CPWD Specifications & other relevant specifications. (for basin

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		mixer refer CP brass mixture for Wash basin).
7.	Bottle Trap	Providing & fixing CP Brass 32 mm size Bottle Trap of approved quality & make and as per the direction of Engineer-in-charge as per DSR Item No. 17.22A and latest CPWD Specifications. The bottle trap with 300mm & 190mm long wall connection pipes & wall flange to be provided.
8.	Angle Valve	Providing & fixing 15mm nominal bore C.P. brass angle valve with wall flange for basin mixer, geyser points etc. of approved make & quality conforming to IS:8931. Ref. as per DSR item no. 18.53 and latest CPWD Specifications. (Reference Image is given below)
		
9.	P.V.C. connection pipe	Providing and fixing uplasticised PVC connection pipe with brass unions 45 cm length 15mm nominal bore as per DSR Item No. 18.21 and latest CPWD Specifications.
10.	Health Faucet	Providing & fixing health faucet (including hose, handset & bracket) of chrome-plated brass with anti-corrosive properties. The flexible hose made of SS 304 and of length 1.0 to 1.2 m with internal EPDM rubber lining or equivalent as per relevant IS/BIS codes or other specifications. • Wall-mounted hook or holder with screws and plastic plugs. • Complete set shall include spray gun, flexible hose, wall hook and to be installed at a height convenient for use, generally 450 mm to 600 mm above finished floor level. • (Reference Image is given below)
		

SI NO.	ITEMS OF WORK	SPECIFICATION
11.	Long Body Bib Cock	Providing & fixing 15mm nominal bore C.P. brass long body bib cock with wall flange, foam flow technology and of approved make & quality conforming to IS standards and weighing not less than 690 gms. Reference as per DSR Item No. 18.51 and latest CPWD Specifications. (Reference Image is given below) 
12.	2-way Bib Cock	Providing and fixing 2-way bib cock, 15 mm nominal bore, C.P. brass body with wall flange, foam flow technology at main spout and of approved make and quality, conforming to IS standards, and weighing not less than 750 gms, all complete as per latest CPWD & other relevant Specifications and as directed by the Engineer-in-Charge. (Reference Image is given below) 
13.	Long Nose Bib Cock	Providing & fixing 15mm nominal bore C. P. Brass long nose bib cock with wall flange, foam flow technology and of approved make & quality conforming to IS standards and weighing not less than 810 grams. Reference as per DSR Item No. 18.50 and latest CPWD Specifications.
14.	CP Brass mixture for Washbasin	Providing & fixing CP Brass mixture, 15mm nominal bore for hot & cold water with single lever quarter turns faucet with foam flow technology, of approved make & quality fixed with washbasin conforming to IS standards and weighing not less than 1340 grams and as per latest CPWD & other relevant specifications.
15.	CP Brass Bath & Shower Mixture	Providing & fixing CP Brass hot and cold bath & shower Mixer, 15mm nominal bore, 3-in-1 System with Provision for Hand shower & Overhead Shower Complete with 115mm Long Bend Pipe, connecting

SI NO.	ITEMS OF WORK	SPECIFICATION
		legs, wall flanges, foam flow technology for spout, Wall Mounted with all fittings & fixtures of approved make & quality conforming to IS standards and weighing not less than 3200 grams. (Reference Image is given below) 
16.	Shower rose	Providing & fixing C.P. brass shower rose with 15 or 20 mm inlet and 150 mm dia with shower tail length of 450 mm as per DSR Item No. 18.22 and latest CPWD Specifications.
17.	Kitchen Sink with drain board and swan tap for hot & cold mixture	Providing & Fixing Stainless Steel A ISI 304 (18/8) kitchen sink as per IS:13983 with C.I. brackets and stainless steel plug 40 mm, including painting of fittings and brackets, cutting and making good the walls wherever require with drain board of size 510x1040mm bowl depth 200mm as per DSR Item No. 17.10 and latest CPWD Specifications. (Reference Image is given below). Kitchen sink to be 1mm thick premium stainless steel. CP brass 15mm nominal bore size Swan type bib cock for hot and cold mixer, spout with foam flow technology and of approved make & quality conforming to IS standards and weighing not less than 750 grams. 
18.	P.V.C. waste pipe	Providing & Fixing flexible 32mm dia. P.V.C. waste pipe for sink or wash basin including P.V.C. waste fittings complete as per DSR Item No. 17.28 and latest CPWD Specifications.
19.	Toilet paper holder	Providing & fixing SS 316 Toilet Paper Holder with wall flange and concealed fittings as per relevant IS/BIS Code of size & design as approved by Engineer-in-charge.

SI NO.	ITEMS OF WORK	SPECIFICATION
20.	Towel Rail & Towel Ring	Providing & fixing 600mm long SS 316 Towel Rail with wall flange as per relevant IS/BIS Code of complete with brackets, CP brass screws with concealed fittings as approved by Engineer-in-charge.  Providing & Fixing SS 316 grade Towel ring with round flange all complete with screws/fasteners (Reference image is given below) 
21.	Grab Bars	Providing & fixing 38mm dia. grab bar with minimum 40mm clearance from wall for length of 600mm horizontally and 900mm length vertically in common toilets of 16 flats of first floor of Residential Towers as directed by Engineer-in-charge. In PWD toilets grab bars (horizontal & vertical) to be provided as per NBC requirements. All grab bars to be made of Stainless Steel grade 304 with wall flange and fittings, screws/fasteners all complete.
22.	Mirror & Glass shelf	Providing & fixing Mirror of superior glass with plastic moulded frame and 6mm thick hard wood backing supported on CP brass screws. (Size 450X600mm) as per latest CPWD Specifications. Providing and fixing 600x120x5 mm glass shelf with edges round off, supported on anodised aluminium angle frame with C.P. brass brackets and guard rail complete fixed with 40 mm long screws, rawl plugs etc., complete as per DSR Item No. 17.33 and latest CPWD Specifications.
23.	Liquid Soap Container & Soap Dish Holder	Providing & Fixing liquid soap container 109 mm wide, 125 mm high and 112 mm distance from wall of standard shape with bracket of the same materials with snap fittings of approved make & quality and colour, weighing not less than 105 gms. Providing & fixing CP Brass Soap Dish Holder having length of 138 mm, breadth 102 mm, height of 75 mm with concealed fitting arrangements, weighing not less than 106 grams of approved make & quality and colour

SI NO.	ITEMS OF WORK	SPECIFICATION
24.	CPVC Pipe- Internal work-Exposed on wall	Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply and all CPVC plain & brass threaded fittings, including fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and testing of joints complete as per direction of Engineer in Charge.as per DSR Item No. 18.7 and latest CPWD specifications. Nominal dia. shall be as per drawings & requirements as approved by Engineer-in-charge.
25.	CPVC Pipe- Concealed work	Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply, including all CPVC plain & brass threaded fittings and fixing the pipe with clamps at 1.00 m spacing. This includes jointing of pipes & fittings with one step CPVC solvent cement and the cutting chases and making good the walls etc. including testing of joints complete as per direction of Engineer in Charge.as per DSR Item No. 18.8 and latest CPWD specifications. Nominal dia. shall be as per drawings & requirements as approved by Engineer-in-charge.
26.	CPVC -External Work	Providing and fixing Chlorinated Polyvinyl Chloride (CPVC) pipes, having thermal stability for hot & cold water supply including all CPVC plain & brass threaded fittings This includes jointing of pipes & fittings with one step CPVC solvent cement, trenching, refilling & testing of joints complete as per direction of Engineer in Chargeas per DSR Item No. 18.9 and latest CPWD Specifications. Nominal dia. shall be as per drawings & requirements as approved by Engineer-in-charge.
27.	GI-Pipes- Internal work - Exposed on wall	Providing and fixing G.I. pipes complete with G.I. fittings and clamps, including cutting and making good the walls etc. Internal work - Exposed on wall as per DSR Item No. 18.10 and latest CPWD Specifications. Nominal bore dia. shall be as per drawings & requirements as approved by Engineer-in-charge.
28.	GI-Pipes- Concealed work	Providing and fixing G.I. Pipes complete with G.I. fittings and clamps, including making good the walls etc. concealed pipe, including painting with anti corrosive bitumastic paint, cutting chases and making good the wall as per DSR Item No. 18.11 and latest CPWD Specifications. Nominal bore dia. shall be as per drawings & requirements as approved by Engineer-in-charge.
29.	GI-Pipes- External work	Providing and fixing G.I. pipes complete with G.I. fittings including trenching and refilling etc as per DSR Item No. 18.12 and latest CPWD Specifications. Nominal bore dia. shall be as per drawings & requirements as approved by Engineer-in-charge.
30.	Smart Water Meter	Providing and fixing smart water meter as per GRIHA norms at Muncipal Supply, Borewell, Treated Water outlet from STP, Captured rainwater for reuse in project, UGT, OHT (Flushing & Domestic Water) with respect to bore size of pipes.
31.	P trap	Providing and fixing uPVC P Trap with 50mm water seal with rubber seal on inlet to push fit hopper for waste. A P-trap for SWR (Soil, Waste & Rainwater) adhere to IS 14735 for fittings and IS 13592 for pipes. These standards specify dimensions, material properties and performance requirements for P-traps and other SWR components. Size shall be as per

SI NO.	ITEMS OF WORK	SPECIFICATION
		drawings & requirements as approved by Engineer-in-charge.
32.	SS grating	Providing and fixing 100% stainless steel grating of approved make and quality grade 304 on top of floor trap. The rate includes the cost of all ancillary works and material as required to complete the works as per relevant IS codes. The shape, size shall be as approved by Engineer-in-charge.
33.	UPVC Soil , waste & vent pipes & fittings	Supply, installation, testing, and commissioning of uPVC soil, waste, and vent (SWV) pipes (8Kg/sq.cm) and fittings conforming to IS 13592:1992 (amended) and IS 14735:1999 for fittings. Pipes shall be self-extinguishing, corrosion-resistant, non-toxic, lead-free and manufactured using high-grade unplasticized PVC with plain or socketed ends, designed for solvent cement or rubber ring jointing. Pipes and fittings shall be ISI-marked and shall have smooth inner surfaces to ensure minimum flow resistance and no sediment deposition. The work shall include making proper connections to horizontal and vertical stacks, water closets (WC), washbasins (WB), urinals, and other sanitary fixtures using all required accessories such as bends, tees, reducers, couplers, sockets, plugs, clamps, expansion joints, Y branches, rodding eyes, thrust blocks, and hangers. Pipes shall be fixed to walls/ceilings using suitable supports or clamps at specified intervals and laid to proper gradient as per manufacturer's guidelines and approved drawings. All exposed pipes shall be painted with two coats of synthetic enamel paint over a primer of approved make and shade. Including Cutting and making good walls, floors, and RCC elements, Jointing with approved solvent cement or rubber ring (as applicable) All materials and workmanship shall comply with relevant IS codes and shall be executed as per latest CPWD specifications and instructions of the Engineer-in-Charge.
34.	UPVC for Rain Water pipes & fittings	Providing and fixing on wall face unplasticised Rigid PVC rain water pipes conforming to IS : 13592 Type A, including jointing with seal ring conforming to IS : 5382, leaving 10 mm gap for thermal expansion, (i) Single socketed pipes of 150 mm dia. (working pressure: 6 kg/sqcm). All materials and workmanship shall comply with relevant IS codes and shall be executed as per latest CPWD specifications and instructions of the Engineer-in-Charge.
35.	Coupler/ Single pushfit Coupler/ Single tee with door/ Single tee without door/ Bend 87.5°/ Shoe (Plain)	Providing and fixing on wall face unplasticised - PVC moulded fittings/ accessories for unplasticised Rigid PVC rain water pipes conforming to IS : 13592 Type A, including jointing with seal ring conforming to IS : 5382, leaving 10 mm gap for thermal expansion as per latest relevant IS/BIS Codes.
36.	UPVC Pipes clips	Providing and fixing unplasticised -PVC pipe clips of approved design to unplasticised – PVC rain water pipes by means of 50x50x50 mm hard wood plugs, screwed with M.S. screws of required length, including cutting brick/block work and fixing in cement mortar 1:4 (1 cement : 4 coarse sand) and making good the wall etc. complete as per latest relevant IS/BIS Codes.

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37.	UPVC Clean Out Plug	Providing and fixing clean out pipe (door piece) of UPVC for cleaning- 110 mm outer dia
38.	UPVC Floor Drain	Providing and fixing UPVC floor drain (FD) with 50 mm inlet & 50mm outlet.
39.	Cast iron Grating in the mouth of Rain Water	Providing and fixing to the inlet mouth of rain water pipe cast iron grating 15 cm diameter and weighing not less than 440 grams. as per DSR Item No. 12.44 and latest CPWD Specifications.
40.	Khurra	Making khurras 45x45 cm with average minimum thickness of 5 cm cement concrete 1:2:4 (1 cement : 2 coarse sand : 4 graded stone aggregate of 20 mm nominal size) over P.V.C. sheet 1 m x1 m x 400 micron, finished with 12 mm cement plaster 1:3 (1 cement : 3 coarse sand) and a coat of neat cement, rounding the edges and making and finishing the outlet complete. as per DSR Item No. 12.22 and latest CPWD Specifications.
41.	Gate Valve & Ball valve	Providing and fixing gun metal gate valve with C.I. wheel of approved quality (screwed end) as per DSR Item No. 18.17 and latest CPWD Specifications. Size shall be as per drawings & requirements as approved by Engineer-in-charge. Providing & fixing ball valve (brass), of approved make & quality, with socketed or threaded connection & PTFE (Teflon) seats for leak proof sealing with quarter turn lever handle having corrosion resistant coating all complete as per latest CPWD & other relevant specifications. Size shall be as per drawings & requirements as approved by Engineer-in-charge.
42.	Float valve	Providing and fixing ball valve (brass) of approved quality, High or low pressure, with plastic floats complete as per DSR Item No. 18.18 and latest CPWD Specifications. Size shall be as per drawings & requirements as approved by Engineer-in-charge.
43.	Brass ferrule	Providing and fixing brass ferrule with C.I. mouth cover including boring and tapping the main as per DSR Item No. 18.20 and latest CPWD Specifications. Size shall be as per drawings & requirements as approved by Engineer-in-charge.
44.	Butterfly valve	Supplying, fixing, testing and commissioning of butterfly valve of PN 1.6 rating with bronze/gunmetal seat duly ISI marked complete with nuts, bolts, washers, gaskets conforming to IS 13095 of sizes as required as per drawings, DSR and latest CPWD Specifications.
45.	Gully Trap	Providing and fixing square-mouth S.W. gully trap class SP-1 complete with C.I. grating brick masonry chamber with water tight C.I. cover with frame of 300 x300 mm size (inside) the weight of cover to be not less than 4.50 kg and frame to be not less than 2.70 kg as per standard design With common burnt clay F.P.S.(non modular) bricks of class designation 7.5 as per DSR Item No. 19.4 and latest CPWD Specifications. Size shall be as per drawings & requirements as approved by Engineer-in-charge.
46.	DWC HDPE	Providing, trenching and laying Double Wall corrugated (DWC) SN 8 pipes as per IS 16098 (Part-2) with plain and in-line socketed end including all fittings like fabricated tees, cross, bends, Y etc. The pipes

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		shall be fixed and jointed with EPDM rubber ring (as per IS 5382). Bed concrete shall be 1:4:8 (1 cement : 4 coarse sand : 8 graded stone aggregate 40 mm nominal size) for grading up to haunches of DWC as per standard design and relevant IS Codes. Size shall be as per drawings & requirements as approved by Engineer-in-charge.
47.	Foor Rest	Providing orange colour safety foot rest of minimum 6 mm thick plastic encapsulated as per IS : 10910 on 12mm dia steel bar conforming to IS : 1786 having minimum cross section as 23 mmx25mm and over all minimum length 263 mm and width as 165mm with minimum 112 mm space between protruded legs having 2 mm tread on top surface by ribbing or chequering besides necessary and adequate anchoring projections on tail length on 138 mm as per standard drawing and suitable to with stand the bend test and chemical resistance test as per specifications and having manufacture's permanent identification mark to be visible even after fixing, including fixing in manholes with 30x20x15 cm cement concrete block 1:3:6 (1 cement : 3 coarse sand : 6 graded stone aggregate 20 mm nominal size) complete as per design. as per DSR Item No. 19.16 and latest CPWD Specifications.
48.	Polyethylene water storage tank	Providing and placing on terrace (at all floor levels) polyethylene water storage tank, IS : 12701 marked, with cover and suitable locking arrangement and making necessary holes for inlet, outlet and overflow pipes but without fittings and the base support for tank as per DSR Item No. 18.48 and latest CPWD Specifications.
49.	Brick Masonry Manhole	Constructing brick masonry manhole of inside size 80 x 90 cm with common burnt clay F.P.S. (non modular) bricks of class designation 7.5 of required depth in cement mortar 1:4 (1 cement : 4 coarse sand) with R.C.C. top slab with 1:1.5:3 mix (1 cement : 1.5 coarse sand (zone-III) : 3 graded stone aggregate 20 mm nominal size), foundation concrete 1:4:8 mix (1 cement : 4 coarse sand (zone-III) : 8 graded stone aggregate 40 mm nominal size), inside plastering 12 mm thick with cement mortar 1:3 (1 cement : 3 coarse sand) finished with floating coat of neat cement and making channels in cement concrete 1:2:4 (1 cement : 2 coarse sand : 4 graded stone aggregate 20 mm nominal size) finished with a floating coat of neat cement complete as per standard design as per DSR Item No. 19.7 & 19.8 and latest CPWD Specifications. The depth of manholes shall be as per drawings & requirements as approved by Engineer-in-charge.
50.	Brick Masonry 0.91m Dia Circular type Manhole	Constructing brick masonry circular type manhole 0.91 m internal dia at bottom and 0.56m dia at top in cement mortar 1:4 (1 cement : 4 coarse sand), inside cement plaster 12 mm thick with cement mortar 1:3 (1 cement : 3 coarse sand) finished with a floating coat of neat cement, foundation concrete 1:3:6 mix (1 cement : 3 coarse sand : 6 graded stone aggregate 40 mm nominal size), and making necessary channel in cement concrete 1:2:4 (1 cement : 2 coarse sand : 4 graded stone aggregate 20 mm nominal size) finished with a floating coat of neat cement, all complete as per standard design : Required depth with SFRC cover and frame (heavy duty, HD- 20 grade designation) 560 mm internal diameter conforming to I.S. 12592, total weight of cover and frame to be not less than 182 kg., fixed in cement concrete 1:2:4 (1 cement : 2 coarse sand : 4 graded stone aggregate 20

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		mm nominal size) including centering, shuttering all complete With common burnt clay F.P.S. (non modular) bricks of class designation 7.5 as per DSR Item No. 19.9 & 19.10 and latest CPWD Specifications. The depth of manholes shall be as per drawings & requirements as approved by Engineer-in-charge.
51.	Brick Masonry 1.22m Dia Circular type Manhole	Constructing brick masonry circular manhole 1.22 m internal dia at bottom and 0.56 m dia at top in cement mortar 1:4 (1 cement :4 coarse sand) inside cement plaster 12 mm thick with cement mortar 1:3 (1 cement : 3 coarse sand) finished with a floating coat of neat cement foundation concrete 1:3:6 (1 cement : 3 coarse sand : 6 graded stone aggregate 40 mm nominal size) and making necessary channel in cement concrete 1:2:4 (1 cement : 2 coarse sand : 4 graded stone aggregate 20 mm nominal size) finished with a floating coat of neat cement, all complete as per standard design. Required depth with SFRC Cover and frame (heavy duty HD 20 grade designation) 560 mm internal diameter conform ing to I.S. 12592, total weight of cover and frame to be not less than 182 kg. fixed in cement concrete 1:2:4 (1 cement : 2 coarse sand : 4 graded stone aggregate 20 mm nominal size) including centering, shuttering all complete With common burnt clay F.P.S. (non modular) bricks of class designation 7.5 as per DSR Item No. 19.11 & 19.12 and latest CPWD Specifications. The depth of manholes shall be as per drawings & requirements as approved by Engineer-in-charge.
52.	NP2 RCC PIPE	Providing and laying non-pressure NP2 class (light duty) R.C.C. pipes with collars jointed with stiff mixture of cement mortar in the proportion of 1:2 (1 cement : 2 fine sand) including testing of joints etc. complete as per DSR Item No. 19.6 and latest CPWD Specifications. The dia. of pipe shall be minimum 200mm as per drawings & requirements as approved by Engineer-in-charge.
53.	Brick Masonry 30x30x50 cm	Constructing masonry Chamber 30x30x50 cm inside, in brick work in cement mortar 1:4 (1 cement :4 coarse sand) for stop cock, with C. I. surface box 100x100 x75 mm (inside) with hinged cover fixed in cement concrete slab 1:1.5:3 mix (1 cement : 1.5 coarse sand : 3 graded stone aggregate 20 mm nominal size), including necessary excavation, foundation concrete 1:5:10 (1 cement : 5 fine sand : 10 graded stone aggregate 40mm nominal size) and inside plastering with cement mortar 1:3 (1 cement : 3 coarse sand) 12mm thick, finished with a floating coat of neat cement complete as per standard design : With common burnt clay F.P.S.(non modular) bricks of class designation 7.5 as per DSR item no. 18.32 and latest CPWD specifications.
54.	Masonry Chamber 60x60x75 cm	Constructing masonry Chamber 60x60x75 cm inside, in brick work in cement mortar 1:4 (1 cement : 4 coarse sand) for sluice valve, with C.I. surface box 100mm top diameter, 160 mm bottom diameter and 180 mm

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		deep (inside) with chained lid and RCC top slab 1:1.5:3 mix (1 cement : 1.5 coarse sand : 5 graded stone aggregate 20mm nominal size), including necessary excavation, foundation concrete 1:5:10 (1 cement : 5 fine sand : 10 graded stone aggregate 40 mm nominal size) and inside plastering with cement mortar 1:3 (1 cement : 3 coarse sand) 12 mm thick, finished with a floating coat of neat cement complete as per standard design : With common burnt clay F.P.S.(non modular) bricks of class designation 7.5 as per DSR item no. 18.33 and latest CPWD specifications.
55.	Masonry Chamber 90x90x100 cm	Constructing masonry Chamber 90x90x100 cm inside, in brick work in cement mortar 1:4 (1 cement : 4 coarse sand) for sluice valve, with C.I. surface box 100 mm top diameter, 160 mm bottom diameter and 180 mm deep (inside) with chained lid and RCC top slab 1:1.5:3 mix (1 cement : 2 coarse sand : 4 graded stone aggregate 20 mm nominal size), including necessary excavation, foundation concrete 1:5:10 (1 cement : 5 fine sand : 10 graded stone aggregate 40 mm nominal size) and inside plastering with cement mortar 1:3 (1 cement : 3 coarse sand) 12 mm thick, finished with a floating coat of neat cement complete as per standard design : With common burnt clay F.P.S.(non modular) bricks of class designation 7.5 as per DSR item no. 18.34 and latest CPWD specifications.
56.	Masonry Chamber 60x45x50 cm	Constructing masonry Chamber 60x45x50 cm inside, in brick work in cement mortar 1:4 (1 cement : 4 coarse sand) for water meter complete with C.I. double flap surface box 400x200x200 mm (inside) with locking arrangement and RCC top slab 1:1.5:3 mix (1 cement : 1.5 coarse sand : 3 graded stone aggregate 20 mm nominal size), including necessary excavation, foundation concrete 1:5:10 (1 cement : 5 fine sand:10 graded stone aggregate 40 mm nominal size) and inside plastering with cement mortar 1:3 (1 cement : 3 coarse sand) 12 mm thick, finished with a floating coat of neat cement complete as per standard design : With common burnt clay F.P.S.(non modular) bricks of class designation 7.5 as per DSR item no. 18.37 and latest CPWD specifications.
57.	GULLY CHAMBER	Constructing brick masonry road gully chamber 50x45x60 cm with bricks in cement mortar 1:4 (1 cement : 4 coarse sand) including 500x450 mm pre-cast R.C.C. horizontal grating with frame complete as per standard design: With common burnt clay F.P.S.(non modular) bricks of class designation 7.5 as per DSR Item No. 19.27 and latest CPWD Specifications.
58.	Fire Hydrant Chamber	Constructing masonry Chamber 60x60x75 cm inside, in brick work in cement mortar 1:4 (1 cement : 4 coarse sand) for fire hydrant, with C.I. surface box 350x350mm top and 165 mm deep (inside) with chained lid and RCC top slab 1:1.5:3 mix (1 cement : 1.5 coarse sand : 3 graded stone aggregate 20mm nominal size), including necessary excavation, foundation concrete 1:5:10 (1 cement : 5 fine sand : 10 graded stone aggregate 40 mm nominal size) and inside plastering with cement mortar 1:3 (1 cement : 3 coarse sand) 12 mm thick, finished with a floating coat of neat cement complete as per standard design: With common burnt clay F.P.S.(non modular) bricks of class designation 7.5 as per DSR Item No.

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		18.36 and latest CPWD Specifications.
59.	Air Release Valve	Providing and fixing C.I. double acting air valve of approved quality with bolts, nuts, rubber insertions etc. complete including tail pieces, tapers etc. if required as per DSR Item No. 18.59 and latest CPWD Specifications. Diameter of valve shall be as per requirements as approved by Engineer-in-charge.
60.	Boring/Drilling	Boring/drilling bore well of required dia for casing/ strainer pipe, by suitable method prescribed in IS: 2800 (part I), including collecting samples from different strata, preparing and submitting strata chart/ bore log, including hire & running charges of all equipments, tools, plants & machineries required for the job, all complete as per direction of Engineer-in-charge, upto 90 metre depth below ground level. In all types of soil 300mm dia. As per DSR item no. 23.1 and latest CPWD specifications.
61.	UPVC CM PIPE	Supplying, assembling, lowering and fixing in vertical position in bore well, unplasticized PVC medium well casing (CM) pipe of required dia, conforming to IS: 12818, including required hire and labour charges, fittings & accessories etc. all complete, for all depths, as per direction of Engineer –in-charge 150 mm nominal size dia. as per DSR Item No. 23.3 and latest CPWD Specifications.
62.	UPVC RMS PIPE	Supplying, assembling, lowering and fixing in vertical position in bore well unplasticized PVC medium well screen (RMS) pipes with ribs, conforming to IS: 12818, including hire & labour charges, fittings & accessories etc. all complete, for all depths, as per direction of Engineer-in-charge 150mm nominal size dia. as per DSR Item No. 23.4 and latest CPWD Specifications.
63.	BOULDERS	Supplying, filling, spreading & leveling stone boulders of size range 5 cm to 20 cm, in recharge pit, in the required thickness, for all leads & lifts, all complete as per direction of Engineer-in-charge as per DSR Item No. 23.5 and latest CPWD Specifications.
64.	GRAVELS	Supplying, filling, spreading & leveling gravels of size range 5 mm to 10 mm, in the recharge pit, over the existing layer of boulders, in required thickness, for all leads & lifts, all complete as per direction of Engineer-in-charge as per DSR Item No. 23.6 and latest CPWD Specifications.
65.	COARSE SAND	Supplying, filling, spreading & leveling coarse sand of size range 1.5 mm to 2 mm in recharge pit, in required thickness over gravel layer, for all leads & lifts, all complete as per direction of Engineer –in-charge.as per DSR Item No. 23.7 and latest CPWD Specifications.
66.	GRAVEL PACKING	Gravel packing in tubewell construction in accordance with IS: 4097, including providing gravel fine/ medium/ coarse, in required grading & sizes as per actual requirement, all complete as per direction of Engineer-in-charge as per DSR Item No. 23.8 and latest CPWD Specifications.
67.	BAIL PLUG	Providing and fixing 150mm dia. Bail plug/ Bottom plug of required dia to the bottom of pipe assembly of tubewell as per IS:2800 (part I) as per DSR Item No. 23.15 and latest CPWD Specifications.
68.	MODULAR RAIN WATER HARVESTING PIT	Supply of Recycled Polypropylene Geo cellular modular unit Aquacell L1200 x W600 x H400 mm in size according to EN17152-1 with min 95% void ration with clip-in together with all required accessories like

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		bottom plate, side plate, adapter, Inspection shaft, Geotextile, Filter to create a below-ground geo cellular structure for attenuation or rainwater which offers the optimal solution for fastest installation and total access for inspection and cleaning. Number of units should be selected based on storage/infiltration requirement. The system selection should be based on traffic load, Cover depth and soil condition. Maximum installation depths – to base of units/cover depth should be considered based on soil condition/load. The installation and selection should be as per manufacturer specification and guidelines.
69.	Activated Carbon Filter	Providing and fixing vertical activated carbon filter comprising of activated carbon filter media fabricated from high performance M.S plate of minimum 8mm thick for shell & 10mm thick for dished ends with top and bottom flanged openings including top and bottom hub and lateral distribution and collection system complete with initial charge of filter media, face piping, valves, accessories, painting, testing and commissioning complete. (Tested to 6 Kg/m²). Discharge : 9000 lph Filtration Rate : 15000 lit./m²/h Suggested dia app. : 1000 mm
70.	WATER SOFTNER	Providing and fixing 'Cation' ION exchange water softener fabricated from minimum 8mm thick M.S. plate on shell and 10mm thick M.S. plate on dished ends complete with initial charge of filter media and ION exchange resins, face piping accessories, piping testing and commissioning complete with resins of approved quality and make and brine tank of required capacity. (Bidder must indicate type and quantity of resin used and quantity of salt required per regeneration). Minimum expected rate of flow: 9000 lph Total soft water to be produced between two regeneration: app. 115 Cu.m Expected raw water hardness : app. 500 ppm Expected treated water hardness: less than 100 ppm Max. working pressure : 4 Kg/cm² Total soft water to be produced between two regeneration: app. 115 Cu.m
71.	SALT SATURATOR	Providing and fixing M.S salt saturator, internally rubber lined as per requirement and externally painted with suitable anti-corrosive paint, with all required accessories for holding and supplying salt for softener complete in all respects, with electrically operated stirrer and of capacity 200 litres. For domestic water hydro pumps, raw water pump, auto chemical doser and sump pumps etc. in plant room. Two incoming main isolation 100A TPN MCCB (35ka). 25A TPN MCCB outgoing for domestic water transfer pump-2 Nos 25A TPN MCCB outgoing for Softener/filter feed pumps - 2 Nos. 16A TP MCB of for each sump pumps - 1 Nos. Panel type voltmeter 150mm with rotary selector switch for reading voltage between phases. Panel type Ammeter 150mm with rotary selector switch for reading currents between phases.

SI NO.	ITEMS OF WORK	SPECIFICATION
		Neon phases indicating lights on the incoming mains. All internal wiring colour coded from incoming mains to various switchgear, starter, meter, indicating lamp and bus bar. All the cable entry to the panel shall be from the top. (All for above from item (a) to (i) complete set). (The shop drawing to be prepared and approved before fabrication)
72.	MOTOR PUMP CONTROLLER	Providing and fixing motor pump controller cum water level indicating panel in powder coated sheet metal box in 16 gauges M.S including the following: Electronic water level controller working on magnetic type sensing probes with necessary length of armored cable up to control panel. Electronic minimum 4-level indicator with magnetic type sensing probes and necessary length of armored cable upto control panel. Arrangement of alternate switching mechanism for operational and standby pumps. For Water level indicators with following details: Underground tanks Minimum 4 level Multi level indicators with magnetic type sensor probes complete. (Each UGT)
73.	STP	135 KLD STP plant with treated water tank for the use of Irrigation and flushing system to be constructed along with pump room, plant room as per DBR & Technical specifications with tertiary treatment integrated with ultra-filtration facility. Treated water tank of STP should have sensor to monitor the treatment performance within the desired parameters.
74.	Storm water Drainage	Covered RCC drain to be constructed with reinforcement as per Item No. 5.33.1, 5.22.6 with 1:3:6 100mm thick PCC levelling course as per DSR Item. 4.1.5. Roadsides may also have suitable precast saucer drain (M-30 Grade) with smooth finish and 100mm thick PCC base as per DSR Item. 4.1.5. The saucer drain precast elements shall be supplied & placed in proper alignment & slope, jointed with cement mortar 1:3 (1 cement: 3 coarse sand) and connected to covered drain channel as per site requirements. Necessary gully traps with gratings to be provided.gully traps with gratings At location of road/footpath crossing suitable RCC (M-30) box drain to be provided as per requirements.

SI NO.	ITEMS OF WORK	SPECIFICATION
75.	RCC cover on Drains	Providing and placing in position 100 mm thick factory made machine batched & machine mixed Precast RCC Rectangular Covers on drains of footpath of various sizes, of minimum M-25 grade cement concrete for RCC work, including cost of centering, shuttering, reinforcement of 8 mm dia TMT bars of Fe 500 grade @ maximum 100 mm c/c on both ways, neat cement punning on finished surface, properly encased on all edges with 1.6 mm thick, 100 mm wide MS sheet duly painted over priming coat, reinforcement to be welded at edges with MS sheet and providing 2 Nos. 12 mm dia bar for hooks etc including cost of cartage, all leads & lift, handling at site etc. all complete as per direction of Engineer-in-Charge.
76.	Perforated RCC drain Covers	Providing and placing in position 100 mm thick factory made precast RCC perforated drain covers, having concrete of strength not less than M-25, of various sizes, including cost of centering, shuttering, reinforcement of 8 mm dia TMT bars of Fe 500 grade @ maximum 100 mm c/c on both ways, neat cement punning on finished surface, including providing 50 mm dia perforations @ 100 to 125 mm c/c, including properly encased on all edges with 1.6 mm thick, 100 mm wide MS sheet duly painted over priming coat, reinforcement to be welded at edges with MS sheet etc including cost of cartage, all leads & lift, handling at site etc. all complete as per direction of Engineer-in-Charge.

PLUMBING AND SANITARY WORK - GENERAL

○ TECHNICAL SPECIFICATION FOR PLUMBING WORKS GENERAL

The Plumbing Installation work shall be carried out in accordance with Indian Standard Code of Practice. It shall also be in conformity with the current Indian Plumbing rules and regulations and requirements of the Local Authority and Fire assurance regulations, so far as these become applicable to the installation. Plumbing work in general shall be carried out as per following CPWD General Specifications amended up to date. **All sanitary and plumbing work must be carried out through licensed plumber .**

IS CODE FOR PLUMBING WORKS

1. IS:1729 Specification for sand cast iron spigot and socket soil, waste and ventilating pipes, fittings and accessories.
2. IS:1536 Specification for centrifugally cast (spun) iron pressure pipes for water, gas and sewage.
3. IS:1538 (Part-I to XXIII) Specification for cast iron fittings for pressure for water, gas and sewage.
4. IS:3714 Code of practices for laying C.I pipes
5. IS:782 Specification for caulking lead
6. IS:1879 Specification for malleable cast iron pipe fittings.
7. IS:4984 High-density polythene pipe for potable water supplies, sewage and Industrial effluents.
8. IS:783 Width and depth of trench for R.C.C pipes.
9. IS:4127 Width and depth of trench for S.W pipes.
10. IS:780 Specification for sluice valve for water works purposes.
11. IS:7558 Code of practice for domestic hot water installation.
12. IS:2064 Code of practice for selection, installation and maintenance of Sanitary appliances
13. IS:2065 Code of practices for water supply in building
14. IS:2183 (Part-I) Code of practice for Plumbing in multi-storied buildings.
15. IS:1239 Specifications for mild steel tubes tubular and other wrought steel fittings. (5th

Rev.)

16. IS:5312 (Part-I) Specifications for swing check type reflux (Non return) valve
17. IS:3114 Code of Practice for laying of C.I. pipes (2nd Rev.)
18. IS:456 Code of practice for plain and reinforced concrete (3rd Rev.) (Amendment 2)
19. IS:1172 Code of basic requirements for water supply, drainage & sanitation (4th Rev.)
20. IS:1200 (Part-16) Code of practice for methods or measurements of building and Civil Engineering works: Part 16 Laying of water and sewer lines including appurtenant items (3rd Rev.)
21. IS:1200 (Part-19) Code of practice for methods or measurements of building and Civil Engineering works: Part 19 Water supply, plumbing and drains (3rd Rev.)
22. IS:3989 Centrifugally cast (spun) iron spigot and socket soil, waste and ventilating pipes, fittings and accessories (2nd Rev.) (Amendment)
23. IS:13095 Butterfly valves for general purposes
24. IS:458 Precast Concrete pipes (with or without reinforcement) (3rd Rev.) (Amendment 2)
25. IS:1726 C.I. Manhole covers & frames (3rd Rev.)
26. IS: 1916 Steel cylinder pipe with concrete lining and coating (1st Rev.).
27. IS:12592(part1) Pre-cast concrete manhole covers and frames: Part 1 Cove
28. IS:12592(part2) Pre-cast concrete manhole covers and frames: Part 2 Frames
29. IS:6392 Steel pipe flanges (Amendment 1)
30. IS: 6418 C.I and malleable C.I. flanges for general engineering purposes.
31. IS:4985 Unplasticized PVC pipes for potable water supplies (2nd Rev.) (Amendment 2)
32. IS:7181Horizontally cast double flanged pipes for water, gas and sewage (1st Rev.) (Amendment 1)
33. IS: 210 Grey iron casting. (4th Rev.)
34. BS EN 1057 Copper pipes
35. BS EN 1254 Copper Fittings
36. IS:4985 UPVC pipes

37. IS:15778 CPVC pipes
38. I.S.13592/4985 Specification for Upvc soil, waste and ventilating pipes, fittings and accessories.
39. IS: 1239 (Part-II) Specification for mild steel tubes, tubular and other wrought steel filling.
40. IS: 4984 High-density polythene pipe for potable water supplies, sewage and Industrial effluents.
41. IS: 783 Width and depth of trench for R.C.C pipes.
42. IS: 4127 Width and depth of trench for S.W pipes.
43. IS: 1742 Code of practice for building drainage.
44. IS: 2064 Code of practice for selection, installation and maintenance of sanitary appliances.
45. IS: 2065 Code of practices for water supply in building.
46. IS: 2183 (Part-I) Code of practice for Plumbing in multistoried buildings.
47. IS: 1239 Specifications for mild steel tubes 140ubular and other wrought steel fittings. (Fifth Revision).
48. IS: 778 Specifications for copper alloy gate, globe and check valves for water works purposes.
49. IS:5312 (Part-I) Specifications for swing check type reflux (Non return) valve.
50. IS:456 Code of practice for plain and reinforced concrete (3rd Rev.) (Amendment 2)
51. IS:12820 Code of practice for dimensional requirements of rubber gaskets for mechanical joints and push on joints for use with cast iron pipes and fittings for carrying water, gas & sewage.

1.1 PREAMBLE

1. Work under this contract shall consist of furnishing all labour, materials, equipment and appliances necessary and required to completely furnish all the Plumbing and other specialized services as described hereinafter and as Indicated.
 - a) Without restricting to the generality of the foregoing Sanitary installations shall include the following:
 - a. Sanitary Fixtures & CP Fittings
 - b. Soil, Waste, Rain Water and Vent Pipes.

- c. Internal and External Water Supply System
- d. Domestic water distribution system from plant room to various units.
- e. Flushing water distribution system from STP to various units.
- f. External Garden hydrant system including Garden hydrant point in connection from treated effluent from STP.
- g. Internal and External Sewerage and Storm water drainage system with excavation of pipe.
- h. Modular Rain water harvesting system

1.2 TECHNICAL CONDITIONS

A.- GENERAL INSTRUCTIONS

1. All works specified in the specification have to be executed in accordance with:
 - a) The latest DSR & guidelines of Latest CPWD Specifications as maximum wherever possible.
 - b) The rules and regulations of Local Authority Having Jurisdiction, and as per the statutory regulations applicable.
 - c) Applicable norms laid down by the relevant sections of latest editions of National Building Code (NBC) and all relevant codes of Bureau of Indian Standards shall be followed as applicable.
 - d) The codes of the Uniform Plumbing Code of India shall be used as a general guide for good engineering practice, design and workmanship norms.
2. All materials used in the works shall have Bureau of Indian Standards valid certification stamped, marked or cast on the material in an acceptable and approved manner, as specified hereinafter.
3. It is the contractor's responsibility to ensure the competence of design to meet the above requirements.
4. Drawings issued are schematic and indicate the concept. Contractor shall make his shop drawings on basis of Architectural and Interior design drawings issued by the Engineer-In- Charge. Work will be executed only as per approved shop drawings.
5. Contractors shall furnish detailed Shop drawings, hydraulic and other design calculations for approval.
6. Work under this contract shall be carried out strictly in accordance with Specifications.

B. LICENSE AND PERMITS

1. Contractor must keep constant liaison with all relevant authorities and shall be responsible for obtaining all approvals relating to water supply, sewerage, storm-water drainage system including rainwater harvesting complete.
2. Contractor shall obtain No Objection Certificate before commencement of work, from the local authorities all related to his work as required for the building.
3. Contractor shall obtain, from the local authorities all related completion certificates with respect to his work as required for occupation of the building

C-METRIC CONVERSION

1. All dimensions and sizes of materials and equipment given in this document are commercial metric sizes.
2. Any weights, or sizes given in the specification having changed due to metric conversion, the nearest equivalent sizes accepted by Indian Standards shall be acceptable.

D. REFERENCE POINTS

1. Contractor shall provide permanent Bench Marks, Flag Tops and other reference points for the proper execution of work and these shall be preserved till the end of the work.
2. All such reference points shall be in relation to the levels and locations given in the architectural drawings.

E. DRAWINGS

1. Service drawings are diagrammatic but shall be followed as closely as actual construction permits.
2. Architectural drawings shall take precedence over Plumbing or other services drawings as to all dimensions.
3. Contractor shall verify all dimensions at site and bring to the notice of the Engineer-In- Charge, all discrepancies. Engineer-In-Charge decision shall be final.
4. Large size details and manufacturers dimensions for materials to be incorporated shall take precedence over small-scale drawings.
5. All drawings/sketches issued by the RITES for the works are the property of the Engineer- In- Charge and shall not be lent, reproduced or used on any works other than

intended without the written permission of the Engineer-In-Charge.

6. All corrections and changes made on the site shall be shown on these reference drawings for final incorporation in the completion drawings. All changes to be made shall be got installed from the Engineer-In-Charge.

F. SHOP DRAWINGS

1. The Contractor shall submit Shop Drawings to RITES in required numbers for Plumbing works as an Advance Copy to the Engineer-In-Charge for approval before start of work. Subsequent to the approval of the shop drawings, the Contractor shall submit Shop Drawings for execution to the Engineer-In-Charge. The Contractor shall also submit the Technical Specifications for all items, including pump curves, single line diagrams etc. as relevant.
2. All Sanitary Engineering drawings will be prepared by the Contractor. Contractor shall also obtain the necessary architectural, structural and other services drawings, based on which they shall prepare their shop drawings as per site conditions.
3. Shop drawings shall incorporate the following:
 - a) All Structural supports/hanging/laying and jointing details for all types of pipes as required.
 - b) Typical details for Toilets & Fixtures required.
 - c) Plumbing layout plans as required and for any changes in the layout of Plumbing/ Architectural Drawings.
 - d) Equipment & piping layout for Mechanical and Electrical equipment's as required, SLDs, mounting details of circuit breakers, location of panels, installation of terminals and faucets etc.w.r.t. finishes, surrounding levels & locations.
 - e) Contractor's fabrication drawings.
4. The Contractor can only commence the work after the approval of above documents by Engineer-In-Charge.
5. Contractors shall ensure that the Shop drawings are approved by the Engineer-In-Charge prior to any execution.

G. COMPLETION DRAWINGS & DOCUMENTS

1. On completion of work contractor shall submit complete set of as Built drawings to the Engineer-In-Charge. These drawings shall have the following information.
 - a) Run of all pipes with diameters and length on all floors and vertical stacks.

- b) Ground and Invert levels of all plumbing services pipes.
 - c) Location of all valves.
 - d) Location of all Mechanical equipment with layout and piping connection.
2. Contractor shall provide, in required numbers, Test certificate, Routine Type Test certificates for Motors, Dynamic balancing certificate for Impellers, Calibration certificate for instrument, Operation and Maintenance Manuals, performance data and list of spare parts supplied together with the name and address of the Manufacturers for all Mechanical and Electrical equipment's provided by him in the form of a Book or Manual.
3. All Warranty / Guarantee" cards / certificates in original issued by the manufacturers shall be handed over to the Engineer-In-Charge also in the form of a comprehensive record book/ documents.

H. MATERIALS (SUPPLIED BY THE CONTRACTOR)

- 1. All materials used in the works shall conform to the specifications.
- 2. As far as possible all materials shall be bearing I.S. certification marks as per approval of the Engineer-In-Charge.
- 3. All materials shall bear the necessary certification marks, conforming to the specifications / Indicative Items / Drawings requirements.
- 4. Unless otherwise specified and expressly approved in writing by the Engineer-In-Charge.

I. INSPECTION AND TESTING OF MATERIALS

- 1. All materials before being allowed to be brought into the store will be preliminary / visually inspected at the entry gate of the project site before the security personnel. All materials shall be got inspected from the Engineer-In-Charge before receiving. This inspection will be conducted with the help of the quality approval format as per RITES.
- 2. Contractor shall be required, if requested, to produce manufacturers Test Certificate for the particular batch of materials supplied to him. The tests carried out shall be as per the relevant Indian Standards.
- 3. For examination and testing of materials and works at the site Contractor shall

provide all Testing and Gauging Equipment necessary but not limited to the followings:

- a) Theodolite
 - b) Dumpy level
 - c) Steel tapes
 - d) Weighing machine
 - e) Plumb bobs, Spirit levels, Hammers
 - f) Micrometers
 - g) Thermometers, Stoves
 - h) Hydraulic test machine
 - i) Smoke test machine
4. All such equipment shall be tested for calibration at any approved laboratory, and the test and calibration certificate shall be submitted to the Engineer-In-Charge.
5. All Testing Equipment shall be preferably located in special room meant for the purpose.

J. MOCK UP AND TRIAL ASSEMBLY

1. The installation of Sanitary Fixtures and fittings shall be as per the shop drawings approved by Engineer-In-Charge.
2. The Contractor shall have to assemble at least each type of Sanitary Fixtures and Fittings in order to determine precisely the required supply and disposal connections. Relevant instructions from manufacturers shall be followed as applicable. This trial assembly shall be developed to determine the location of puncture holes, holding devices etc. which will be required for final installation of all Sanitary fixtures and fittings. The above assembly shall be subject to final approval by Engineer-In- Charge.
3. The Fixtures in the trial assembly can be reused for final installation.

K. EXECUTION OF WORK

1. The work shall be carried out in conformity with the Plumbing drawings and within the requirements of Architectural, HVAC, Electrical, Structural / Green Building and other specialized services drawings.
2. The Contractor shall make provision for hangers, sleeves, structural openings and

other requirements well in advance to prevent hold up of progress of the construction program.

3. Contractor shall be responsible for co-ordination with other agencies working on the project relating to their scope of work and shall take approval from the Engineer-In-Charge wherever required.

L. CUTTING & MAKING GOOD

No structural member shall be chased or cut without the written permission of the Engineer- In- Charge.

M. TESTING

1. Piping and drainage works shall be tested as specified under the relevant clauses of the specifications.
2. Tests shall be performed in the presence of the Engineer-In-Charge.
3. All materials and equipment found defective shall be replaced and whole work tested to meet the requirements of the specifications.
4. Contractor shall perform all such tests as may be necessary and required by the local authorities to meet Municipal or other by-laws in force.
5. Contractor shall provide all labour, equipment and materials for the performance of the tests.
6. All appliances, fixtures and fittings shall be tested before and after installation.
7. Water seals of all appliances shall be tested. The Contractor shall block the ends of waste and ventilation pipes and shall conduct air test.

N. SITE CLEARANCE AND CLEANUP

1. The Contractor shall, from time to time clear away all debris and excess materials accumulated at the site.
2. After the Fixtures, equipment and appliances have been installed and commissioned, Contractor shall clean-up the same and remove all plaster, paints stains, stickers and other foreign matter of discoloration leaving the same in a ready to use condition.
3. On completion of all works, Contractor shall demolish all stores, remove all surplus

materials and leave the site in a broom clean condition.

O. FINAL INSTALLATION

The Contractor shall install all sanitary fixtures and fittings in their final position in accordance with the approved trial assemblies and as shown on the Drawings. The installation shall be complete with all supply and waste connections. The connection between building and piping system and the sanitary fixtures shall be through proper unions and flanges to facilitate removal / replacement of Sanitary Fixtures without disturbing the built-in piping system. All unions and flanges shall match in appearance with other exposed fittings.

Fixtures shall be mounted rigid, plumb and true to alignment. The outlet of water closet pans and similar appliances shall be examined to ensure that outlet ends are butting and the receiving pipes before making the joint. It shall be ensured that the receiving pipes are clear of obstruction. When Fixtures are being mounted, attention shall be paid to the possibility of movement and settlement by other causes. Overflows shall be made to ensure that necessary anchoring devices have been provided for supporting water closets, wash basins, sinks and other appliances.

P. PROTECTION AGAINST DAMAGE

The Contractor shall take every precaution to protect all Sanitary fixtures against damage, misuse, cracking, staining, breakage and pilferage by providing proper wrapping and locking arrangement till the completion of the installation and handling over. At the time of handling over, the Contractor shall clean, disinfect and polish all the fixtures and fittings. Any Fixtures found damped, cracked, clipped, strained or scratched shall be removed and new fixtures and fittings free from defects shall be installed.

Q. GUARANTEE / WARRANTY

1. The contractor shall submit a warranty for all equipment, materials and accessories supplied by him against manufacturing defects, malfunctioning or under capacity functioning.
2. The form of warranty shall be as approved by the Engineer-In-Charge.
3. The warranty shall be valid for a period as specified in SCC.
4. The warranty shall expressly include replacement of all defective or under capacity equipment. Engineer-In-Charge may allow repair of certain equipment if the same is

found to meet the requirement for efficient functioning of the system.

5. The warranty shall include replacement of any equipment found to have capacity lesser than the rated capacity as accepted in the contract. The replacement equipment shall be approved by the Engineer-In-Charge.
6. The contractor shall operate all mechanical equipment for a period as mentioned in the SCC.

1.3 SANITARY ENGINEERING SERVICES

1.3.1 INTRODUCTION

Water supply and wastewater disposal constitute a very important part of the services in a building. Maintenance of high levels of hygiene and cleanliness are indispensable to the wellbeing as a whole building services.

This report intends to highlight the details of the following proposed services, which are to be provided from the point view of Sanitary Engineering, Fire Fighting and other allied services.

1.3.2 SANITARY AND OTHER APPLIANCES

Without restricting to the generality of the foregoing, sanitary and other appliances shall include the following:

- 1.3.2.1 All Sanitary appliances and fittings shall be as per Green Rating for Integrated Habitat Assessment (GRIHA) Rating.
- 1.3.2.2 Chromium plated brass fittings.
- 1.3.2.3 Accessories e.g. towel rods, towel rails, coat hooks, etc.
- 1.3.2.4 Whether specifically indicated or not the Contractor shall provide for all appliances and fixtures, all fixing devices, nuts, washers, Teflon tape, sealant, cement, brackets, supports, paints, connectors, Chromium Plated (CP) riser pipes, adopters, bolts, screws, hangers, etc., as required.
- 1.3.2.5 All exposed pipes within toilets and near appliances/ fixtures shall be of CP brass or copper unless otherwise specified
- 1.3.2.6 All appliances, fixtures and fittings shall be provided with all such accessories as are required to complete the item in working condition whether specifically mentioned or not in the Schedule of Quantities, specifications, and drawings.

Accessories shall include proper fixing arrangement, brackets, nuts, bolts, washers, screws and required connection pieces.

- 1.3.2.7 Fixing screws shall be half round head CP brass screws, with CP brass washers unless otherwise specified.
- 1.3.2.8 CP fittings shall be cast brass CP of the best quality approved by the Engineer.
- 1.3.2.9 All materials shall be rust proofed; materials in direct or indirect contact shall be compatible to prevent electrolytic or chemical (bimetallic) corrosion.
- 1.3.2.10 The appliance shall be placed in the correct position or marked out in order that pipe work can be fixed or partially fixed first.
- 1.3.2.11 The appliance shall be fixed in a manner such that it will facilitate subsequent removal, if necessary.
- 1.3.2.12 Pipe connections shall be made with demountable unions. Pipe work shall not be fixed in a manner that it supports or partially supports an appliance.
- 1.3.2.13 Appliances shall be fixed true to level firmly fixed to anchor or supports provided by the manufacturer and additional anchors or supports where necessary.
- 1.3.2.14 When directed, Contractor shall install Fixtures and accessories in a mock-up room for the approval of the Engineer-In-Charge. Sample room Fixtures may be reused on the works

1.3.3 WATER SUPPLY SYSTEM

The total water requirement is proposed to be catered through Bore well and Municipal Connection. The total quantity of water per day is estimated based on the proposed occupancy and activities catered.

As a general rule the following rates per capita per day may be considered for domestic and non- domestic needs as per NBC 2016-part 9 plumbing services point for water supply requirements.

- 1. The water from the sources will be brought into the firefighting underground Tank then will carry raw water Tank, which will serve exclusively as static underground water storage tanks for raw water and domestic.
- 2. Domestic water supply from the underground tank to fill the overhead tank through the pumping system. From the overhead tank domestic water supply for the entire building through the gravity system.

1.3.3.1 Scope of work

1. Work under this section consists of furnishing all labour, materials equipment and appliances necessary and required to completely install the water supply system as required by the drawings, specified here.
2. Without restricting to the generality of the foregoing, the water supply system shall include the following:
 - All water lines to different parts of building and making connection from source etc.
 - Pipe protection and painting.
 - Control valves, masonry chambers and other appurtenances.
 - Connections to all toilets, kitchen equipment's, storage tanks and appliances.
 - Excavation and refilling of pipe trenches, wherever required.
 - Trenches for taking pipe lines for these services.

1.3.3.2 GENERAL REQUIREMENTS

1. All materials shall be new of the best quality conforming to specifications. All works executed shall be to the satisfaction of the Engineer-In-Charge.
2. Pipes and Fittings shall be fixed truly vertical, horizontal or in slopes as required in a neat workman like manner.
3. Short or Long bends shall be used on all main pipe lines as far as possible. Use of Elbows shall be restricted for short connections. As far as possible all Bends shall be formed by means of a hydraulic pipe bending machine for pipes up to 65mm dia.
4. Pipes shall be fixed in a manner so as to provide easy accessibility for repair and maintenance and shall not cause obstruction in shafts, passages etc.
5. Pipes shall be securely fixed to walls and ceilings by suitable clamps at intervals
6. Valves and other appurtenances shall be so located as to provide easy accessibility for operations, maintenance and repairs.

1.3.3.3 CPVC PIPES & FITTINGS & VALVES FOR COLD WATER/HOT WATER SUPPLY DISTRIBUTION SYSTEM (Internal/External Plumbing)

All the internal pipes including the pipes dropping down through shafts for various fixtures to be used for water supply shall be CPVC pipe to IS: 15178

- All branch water supply pipes: - 20mm, 25mm, 32mm, 40mm, 50mm, 65mm, 80mm
- Valves on branches, main line and pumps shall have ball valve/gate valve/butterfly valve
- Mention the pressure of each pipes.

Water Distribution System shall be CPVC (Chlorinated Polyvinyl Chloride) Water

Supply Piping System with pipe as per CTS SDR-11 at maximum working pressure of 400 psi (28.1 Kg/cm²) at 23oC and 100 psi (7.03 Kg/cm²) at 82oC (from ½” - 2”), using solvent welded CPVC fitting i.e. Tees, elbows, couplers, unions, reducers, bushing etc. including transition fitting (connection between CPVC and metal pipe / G.I.) i.e. Brass adapter (both male and female threaded) all conforming to ASTM D-2846 with only CPVC solvent cement 5 conforming to ASTM F-493 - with only Clamps/ structural metal supports as required/directed at site including cutting chases and filling the same with cement concrete/cement mortar as required. All termination points for installation of faucets shall have brass termination fittings. Installation shall be to the satisfaction of Engineer in charge/manufacturers of pipes and fittings

- Outside Diameters and Wall Thicknesses for CPVC 4120, SDR 11 Plastic Pipe

TABLE 18.16

Sl.No	Nominal size	Nominal outer diameter	Mean Outside Diameter		Outer diameter at any point		Wall Thickness					
							Class 1, SDR 11			Class 3, SDR 17		
			Max	Min	Max	Min	Avg. Max	Min	Max	Avg. Max	Min	Max
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
(i)	15	15.9	15.8	16.0	15.8	16.0	2.2	1.7	2.2	-	-	-
(ii)	20	22.2	22.1	22.3	22.0	22.4	2.5	2.0	2.5	-	-	-
(iii)	25	28.6	28.5	28.7	28.4	28.8	3.1	2.6	3.1	-	-	-
(iv)	32	34.9	34.8	35.0	34.7	35.1	3.7	3.2	3.7	-	-	-
(v)	40	41.3	41.2	41.4	41.1	41.5	4.3	3.8	4.3	-	-	-
(vi)	50	54.0	53.9	54.1	53.7	54.3	5.5	4.9	5.5	-	-	-
(vii)	65	73.0	72.8	73.2	72.2	73.8	-	-	-	4.8	4.3	4.8
(viii)	80	88.9	88.7	89.1	88.1	89.7	-	-	-	5.9	5.2	5.9
(ix)	100	114.3	114.1	114.5	113.5	115.1	-	-	-	7.5	6.7	7.5
(x)	150	168.3	168.0	168.6	166.5	170.1	-	-	-	11.1	9.9	11.1

Notes

1. For CPVC pipes SDR is calculated by dividing the average outer diameter of the pipe in mm by the minimum wall thickness in mm. If the wall thickness calculated by this formula is less than 1.52 mm, it shall be increased to 1.52 mm. The SDR values shall be rounded to the nearest 0.5.

- Pressure Ratings for CPVC, SDR 11 Plastic Pipe

Nominal Size		Pressure Rating, PSI (Kg/cm ²)			
(in.)	(mm)	73.4°F (23°C)	180°F	(82°C)	
½	15	400 (28.1)	80	(7.0)	
¾	20	400 (28.1)	100	(7.0)	
1	25	400 (28.1)	100	(7.0)	
1-1/4	32	400 (28.1)	100	(7.0)	
1-1/2	40	400 (28.1)	100	(7.0)	
2	50	400 (28.1)	100	(7.0)	

1.3.3.4 JOINING CPVC PIPES AND FITTINGS

- Cutting

Pipe shall be cut with either with a wheel type plastic pipe cutter hacksaw blade, and care shall be taken to make a square cut, which provides optimal bonding area within a joint.

- Deburring/Beveling

Burrs and fillings should be removed from the outside and inside of pipe with a pocket knife or file otherwise burrs and fillings may prevent proper contact between pipe and fittings during assembly.

- Fitting/Preparation

A clear dry rag/cloth should be used to wipe dirt and moisture from the fitting sockets and tubing end. The tubing should make contact with the socket wall 1/3 or 2/3 of the way into the fitting socket.

- Assembly

After applying the solvent cement on both pipe and fitting socket, pipe should be inserted into the fitting socket within 30 seconds and rotating the pipe 1/4 to 1/2 turn while inserting so as to ensure even distribution of solvent cement with the joint. The assembled system should be holed for 10 seconds (approximately) in order to allow the joint to set up.

Set and Cure Times: remake the joint to avoid potential solvent cement set and cure times shall be strictly adhered to as per the below mentioned table.

✓ Minimum cure prior to pressure testing at 150 psi :

Ambient Temperature During Cure Period	Pipe Sizes	
	1/2" - 1"	1 1/4" - 2"
Above 15°C	1 Hr	2 Hrs
4 - 15 °C	2 Hrs	4 Hrs
Below 4 °C	4 Hrs	8 Hrs

Special care shall be exercised when assembling flow guard systems in extremely low temperature (below 4 C) or extremely high temperature (above 45 C) in extremely hot temperatures, make sure that both surfaces to be joined are still wet with cement solvent when putting them together.

1. Testing:

Once an installation is completed and cured or per above mentioned recommendations, the system should be hydrostatically pressure tested at 150 psi (10 bar) for one hour. During pressure testing, the system should be filled with water and if a leak is found, the joint should be cut out and replacing the same with new one by using couplers.

2. Transition of CPVC to Metals:

When making a transition connection to metal threads, special Brass (Male and female adapters) should be used.

3. Threaded Sealants:

Teflon tap shall be used to make threaded connections leak proof.

4. Solvent Cement:

CPVC solvent cement conforming to ASTM F 493 should be used for joining pipe with fittings and valves. CPVC cement solvent have a minimum shelf life of 1 year. Solvent have a minimum shelf life of 1 year. Aged cement solvent will often change colour or began to thicken and become gelatinous or jelly to like and when this happens, the cement should not be used. The cement solvent should be used within 30 days after opening the company's seal and tightly close the seal after using in order to avoid its freezing. The freezed cement solvent should be discarded immediately and fresh one should be used.

✓ Hangers and Supports:

Most hangers designed for metal pipe are suitable for CpvC pipes. Hangers should not have rough or sharp edges which come in contact with the tubing.

Supports should be as per the below mentioned table:

Horizontal and Vertical Support									
Spacing									
Size of pipe		21 deg. C (70°F)		49 deg. C (120°F)		71 deg. C (160°F)		82 deg. C (180°F)	
Inch	(mm)	Ft.	cm	Ft.	cm	Ft.	cm	Ft.	Cm
½"	(15)	5.5	(167.7)	4.5	(137.16)	3.0	(91.44)	2.5	(76.2)
¾"	(20)	5.5	(167.7)	5.0	(152.4)	3.0	(91.44)	2.5	(76.2)
1"	(25)	6.0	(182.88)	5.5	(167.7)	3.5	(106.68)	3.0	(91.44)
1 ¼"	(32)	6.5	(198.12)	6.0	(182.88)	3.5	(106.68)	3.5	(106.68)

1 ½"	(40)	7.0	(213.36)	6.0	(182.88)	3.5	(106.68)	3.5	(106.68)
2"	(50)	7.0	(213.36)	6.5	(198.12)	4.0	(121.92)	3.5	(106.68)

- **CLAMPS**

G.I. pipes in shafts and other locations shall be supported by M.S. clamps of design approved by Engineer-In-Charge. Pipe in wall chases shall be anchored by iron hooks. Pipes at ceiling level shall be supported on structural clamps fabricated from M.S. structural as described in the sub section. Pipes in typical shafts shall be supported on Slotted Angles/Channels as specified elsewhere.

- **UNIONS**

Contractor shall provide adequate number of unions on all pipes to enable dismantling later. Unions shall be provided near each Gunmetal Valve, Stop Cocks, or Check Valves and on straight runs as necessary at appropriate locations as required and/or directed by Engineer- in- Charge.

- **FLANGES**

- a) Flanged connections shall be provided on pipes where shown on the drawings, all equipment connections as necessary and required or as directed by Engineer-In- Charge.
- b) Connections shall be made by the correct number and size of the bolts and made with 3 mm thick insertion rubber washer. Bolt hole dia for flanges shall conform to match the valves.

- **CUTTING CHASES IN MASONARY WALLS**

Cold water distribution pipes to fixtures and equipment exposed to view in the bathrooms, kitchens, and sanitary compartments shall be chased into walls or floors or placed in wall cavities. The Contractor shall be responsible for cutting all notches, chases, and recesses in walls and floors and only a diamond cutter shall be used. The maximum size of conduit or pipe permitted to be concealed in floor slabs shall be 32 mm diameter unless otherwise approved by the Engineer-In- Charge.

The chases upto 7.5 x 7.5 cm shall be made in the walls for housing GI pipes etc. These shall be provided in correct positions as shown in the drawings or directed by the Engineer- In-Charge. Chases shall be made by chiseling out the masonry to proper line and depth. After the pipes etc. are fixed in chases, the chases shall be

filled with cement mortar 1:2:4 or as may be specified, and made flush with the masonry surface. The concrete surface shall be roughened with wire brush to provide a key for plastering.

Where pipes pass through beams or structural walls, subject to the approval of the Engineer-In- Charge, the Contractor shall ensure that sizes and locations of openings required are formed in when the relevant beams or walls are cast.

- GUNMETAL VALVES

- a) Valves 65mm dia and below shall be heavy Gunmetal. Ball valves shall be tested at manufacturer's works and the same stamped on it. All Valves shall be approved by the Engineer-In-Charge before they are allowed to be used on work.

- BALL VALVES

- a) Ball Valves have body material as Forged Brass Chrome plated with Spindle Brass Nickel Plating & Lever handle Steel Chrome plated with green plastic cover. The valve is suitable for water maximum working pressure up to 25 bar (PN 25). The valve is operated by turning. The rotation from open to close is a quarter turn (90°) which closes in a clock-wise direction.

- BUTTERFLY VALVES

- a) Butterfly valves of approved quality for pressure rating of 230 P.S.I. with locking arrangement and gearbox with handle operated (for above 150mm dia) shall be provided or as specified.
- b) Butterfly valves shall be of specified quality conforming to IS:13095 or BS:5155.
- c) Joints for butterfly valves shall be made with suitable tail /socket pieces on the pipe line and flanged joints made with 3mm thick insertion rubber gasket with appropriate number of bolts, nuts and washers.
- d) Butterfly valves shall be provided on all branches as required.

- PRESSURE REDUCING VALVES

- a) This valve protects House hold installation against excessive pressure from the Overhead supply. By installing a PRV, Pressurization damage is avoided and water consumption is reduced. The set pressure is also maintained constant even when there is wide inlet pressure fluctuation.

- b) Reduction of the operating pressure and maintaining it at a constant level minimizes flow noise in the installation.

- **NON-RETURN VALVES**

- a) Non-return valves shall be of Cast Iron body and Bronze/Gunmetal seat. They shall conform to class of IS: 5312 with pressure rating PN 1.6 and have flanged ends. They shall be swing check type in horizontal runs and lift check type in vertical runs of piping. They shall not be spring-loaded type. An arrow mark in the direction of flow shall be marked on the body of the valve.

- **AIR RELEASE VALVES**

- a) Air release valves shall be single acting type air valves with cast iron body and bronze/gunmetal internal parts and plastic float.
- b) Each air release valve shall be provided with isolating ball valve of specification given above.

- **CAST IRON PIPES**

- a) Pipes for water supply mains where specified shall be cast iron pipes, centrifugally cast spun iron pipes, class LA conforming to I.S. 1536. Quality certificates shall be furnished.

1.4 Fittings and Inspection Chambers Fittings and chambers shall be provided as required.

1.4.1 Anchor Block

Suitable anchor blocks shall be provided at all bends and tees to encounter the excessive thrust developed due to water hammer.

1.4.2 Rubber Joints

Joints between two pipes shall be made by pre molded rubber joints with suitable tackles in a manner recommended & approved by the manufacturer. No joints shall be covered until the lines are hydraulically tested.

1.4.3 VALVE CHAMBERS

Contractor shall construct chambers for all full way valves, butterfly valves and other type of valves as specified. These shall be made, in brick masonry in cement mortar 1:4 (1 cement: 4 coarse sand) on cement concrete foundations 150mm thick 1:5:10 mix (1 cement: 5 coarse sand: 10 graded stone aggregate 40mm nominal size) 12 mm thick cement plaster 1:3(1 cement : 3 coarse sand) inside finished with a floating coat of neat cement with 8mm thick M.S surface box with hinged cover and locking arrangement, 150 mm thick reinforcement

cement concrete top slab of 1:2:4 (1 cement : 2 coarse sand: 4 graded stone aggregate 20mm nominal size), as specified and as required, including excavation, back filling rammed complete or as specified.

Valve chambers shall be constructed as specified in Indicative Items but generally shall be of the following sizes:

Length (mm), Width (mm), Depth (mm) for pipes dia. Up to 80 mm, 600,600 and 1000 For pipes dia. 80 mm and above 750,750 and 1000

1.4.4 TESTING

- a) All pipes, fittings and valves shall be tested by hydrostatic pressure of min. 1.5 times, the working pressure and subject to minimum of 7 kg/cm² in any case and with the consent of Engineer-In- Charge.
- b) Pressure shall be maintained for a period of at least two hours without an appreciable drop in the pressure after fixing at site. (+10 %). A test register shall be maintained, and all entries shall be signed and dated by Contractor(s) and Engineer-In-Charge.
- c) In addition to the sectional testing carried out during the construction, Contractor shall test the entire installation after connections to the overhead tanks or pumping system or mains. He shall rectify all leakages and shall replace all defective materials in the system. Any damage done due to carelessness, open or burst pipes or failure of fittings, to the building, furniture and Fixtures shall be made good during the defect's liability period.
- d) After completion of the water supply system, Contractor shall test each valve by closing and opening it a number of times to observe if it is working efficiently. Valves which do not effectively operate shall be replaced by new ones and the same shall be tested as above.

1.4.5 DISINFECTION

After completion of the work Contractor shall flush clean the entire system with the city's water filtered after the connection has been made.

After the first flushing, commercial bleaching powder is to be added to achieve a dosage of 2 to 3 mg/l of water in the system added and flushed. This operation should be performed twice to ensure that the system is fully disinfected and usable.

1.4.6 Y- strainer

Stainless steel Y- strainer fabricated out of 1.6 mm thick stainless steel, Grade

304, sheet with 3 mm dia holes with stainless steel flange

1.4.7 Dirt box strainer

The purpose of a strainer is to remove larger, unwanted suspended particles from a liquid, primarily to protect downstream equipment, like pumps, from damage and must be positioned in the pipeline upstream of the equipment requiring protection. C.I. dirt box strainer for bulk type water meter with nuts bolts, rubber insertions etc.

1.4.8 Pressure gauge

A pressure gauge is used to check for faults in the tanks or systems by measuring the force of pressure in the water or air and shall be used with gun metal/ brass valves completely

1.4.9 Pressure Reducing Valve

This valve protects Household installation against excessive pressure from the Overhead supply. By installing a PRV, Pressurization damage is avoided, and water consumption is reduced. The set pressure is also maintained constant even when there is wide inlet pressure fluctuation. Reduction of the operating pressure and maintaining it at a constant level minimizes flow noise in the installation.

1.4.10 Expansion Bellow

Expansion bellows, also known as expansion joints, will be used in water tank applications to absorb thermal expansion and contraction of pipes, accommodate misalignment, and reduce vibration and noise transmission with Material Rubber and GI Flange, Thread Size as per site condition

1.4.11 Water Tank Alarm

A water tank alarm, also known as a water level indicator or Automatic water Level Indicator with Musical Alarm on Tank Full and Buzzer on Tank Empty, will be used to prevent water damage from flooding on Tank Full and Buzzer on Tank Empty with 20 metres of Sensor Wire

1.4.12 Water storage Tank

RCC OHT of required capacity with boosting arrangement with necessary installation for supply of water including standby Pump as per drawings, design and scheme as approved by the Engineer- in charge. This shall consist of provisions for drinking water chambers, firefighting chambers and for unfiltered water supply for horticulture, flushing and other uses.

1.4.13 Puddle flange

Puddle flanges are crucial for creating a watertight seal where pipes penetrate the tank's walls, preventing leaks and ensuring structural integrity.

G.I fabricated hot dipped galvanized coating puddle flange comprising of 600 mm dia long GI class 'C' pipe (screwed end up to 50 mm & flange for more than 50 mm dia) with 6 mm thick GI plates of required size including cutting holes in Centre of plate, inserting pipe & making welded (Arc) joint complete. Same shall be installed with TEC welded joint in position with reinforcement bars of RCC tank wall/slab including making cutout in shuttering during concrete casting of tank of following diameter. (For underground tank) will be following size 50mm dia, 80mm dia, 150mm dia and 200mm dia.

1.4.14 Hydro pneumatic Pumps

Compact Self-Contained Skid Mounted Hydro pneumatic pump System will be used at for regulating the pressure with all accessories will be used for water supply and detailed technical specifications is available under Electrical sub-head in the tender document.

1.5 PRE COMMISSIONING

1. Ensure that all pipes are free from debris and obstruction.
2. Check all valves and fire hydrant for effective opening and closing action. Defects should be rectified or valves replaced.
3. Ensure that all Connections to Branches has been made.
4. Ensure that mains have been connected to the respective pumps, underground and overhead tanks.
5. Water supply should be available at main Underground tank.
6. All main line Valves should be closed.

1.6 COMMISSIONING

1. Fill Underground tank with water. Add fresh bleaching powder after making a solution to be added near inlet.
2. Start the Water Supply Pump and allow water to fill the main Underground tank. Water will first fill the fire tank and then overflow to the Raw Water tanks.
3. After filling the Overhead Reservoir drain the same to its one forth capacity through tank scour valve. (This is to ensure removal of all mud, debris etc. from the tank).
4. Fill Overhead tank to full.
5. Release water in the main lines by opening Valves in each circuit. Drain out water in the system through scour valves or fire hydrant in lower regions. Ensure clean water is now coming out of the system.
6. Open valves for individual clusters. Observe for leakages or malfunctions, check pressure & flow at end of line by opening Hydrants etc. Remove and rectify defects noticed.

7. Check all outlet points for proper operation by opening each valve and allowing water to flow for a few minutes. Also check for effective closure of valve.
8. The entire water supply system should be disinfected with bleaching powder and system flush cleaned.
9. Send four samples of water drawn from four extreme locations for testing for bacteriological test in sterilized bottles obtained from the concerned laboratory. (Laboratory personnel may collect the samples themselves).

1.7 SOIL, WASTE & VENT PIPES - General

Soil waste from water closets and urinals etc. will be collected by horizontal and vertical soil pipes and discharged directly to the manholes. Waste water from wash basins, sinks, and from other waste fixtures shall be collected separately by waste pipes and be discharged through gully traps into the manhole of the external sewerage system.

The external sewerage system shall be running around the building periphery having manholes in front of each shaft. The main sewer line will carry the whole sewage by gravity up to the Sewage Treatment Plant or Municipal Sewer Line

1.8 SOIL, WASTE & VENT PIPES – (Internal Plumbing)

1. Work under this section shall consist of furnishing all labour, materials, equipment and appliances necessary and required to completely install all soil, waste, vent and rainwater pipes as required, specified hereinafter and given in the Indicative Items.
2. Without restricting to the generality of the foregoing, the soil, waste, vent and rainwater pipes system shall include the following:
 - I. Vertical and horizontal Soil, Waste and Vent Pipes, Rainwater Pipes and Fittings, Joints Clamps and connections to Fixtures.
 - II. Connection of pipes to Gully Traps & Manholes etc.

1.8.1 GENERAL REQUIREMENTS

- 1.8.1.1 All materials shall be new of the best quality conforming to specifications and subject to the approval of Engineer-In-Charge.
- 1.8.1.2 Pipes and fittings shall be fixed truly vertical, horizontal or in slopes as required in a neat workmanlike manner.
- 1.8.1.3 Pipes shall be fixed in a manner to provide easy accessibility for repair and maintenance and shall not cause obstruction in shafts, passages etc.
- 1.8.1.4 Pipes shall be securely fixed to walls and ceilings by suitable clamps at intervals specified.
- 1.8.1.5 Access doors for fittings and cleanouts should be so located that they are easily accessible for repair and maintenance.

All works shall be executed as directed by the Engineer-In Charge.

1.8.2 PIPES & FITTINGS FOR SOIL/WASTE DRAINAGE

- 1.8.2.1 Pipes & Fittings
- 1.8.2.2 Pipes for soil / waste drainage shall be conforming to uPVC I.S.13592 as specified in schedule of quantities.
- 1.8.2.3 Pipes shall be clamped to the wall with approved type U clamps/ Angle Supports.
- 1.8.2.4 Fittings for the pipes shall be injection moulded with approved type of sockets and 'O' rings joints/solvent welded joints as per recommendations of the manufacturers.
- 1.8.2.5 Access door shall be made up with 3 mm thick insertion rubber washer and white lead. The bolts shall be lubricated with grease or white lead for easy removal later. The fixing shall be air and watertight.

1.8.3 UPVC PIPES & FITTINGS FOR RAINWATER

1.8.3.1 Pipes

- a. UPVC pipes for Rainwater pipe system shall be un-plasticized (rigid) PVC pipes conforming to I.S:4985 as specified in schedule of quantities.
- b. Fittings for the pipes shall be injection molded with approved type of sockets and 'O' rings joints/solvent welded joints as per recommendations of the manufacturers.
- c. Jointing shall be done as per the manufacturer's recommendation. The pipes and fittings must have matching dimensions for a perfect joint. Loose or excessively tight joints in the system shall not be accepted. Fittings must have sufficient gap (approx. 10 mm) for permissible thermal expansion of pipes. uPVC pipes shall be clamped to the wall with approved type uPVC saddle clamps.

1.8.3.2 Fittings

- a) Fittings shall conform to the same Indian Standard as for pipes. Contractors shall use pipes and fittings to match specifications.
- b) Fittings shall be of the required degree of curvature with or without access to the door.
- c) Access door shall be made up with 3 mm thick insertion rubber washer and white lead. The bolts shall be lubricated with grease or white lead for easy removal later. The fixing shall be air and watertight.

1.8.3.3 Fixing

- a) All vertical pipes shall be fixed by M.S. clamps, truly vertical. Branch pipes shall be connected to the stack at the same angle as that of the fittings. No collars shall be used on vertical stacks. Each stack shall be terminated at top with a cowl (terminal guard).
- b) Horizontal pipes running along the ceiling shall be fixed on structural adjustable clamps of special design shown on the drawings or as directed. Horizontal pipes shall be laid to uniform slope and the clamps adjusted to the proper levels so that the pipes fully rest on them.
- c) Contractor shall provide all sleeves, openings, hangers, and inserts during the construction. He shall provide all necessary information to the building Contractor for making such provision in the structure as necessary. All damages shall be made good to restore the surfaces

1.8.3.4 CLAMPS

- a) Holder bat clamps shall be of standard design and fabricated from M.S. flats 40x3 mm thick and 12 mm dia M.S. Rod and 6 mm nuts and bolts. They shall be painted with two coats of black bitumen paint before fixing. Holder bat clamps shall be fixed in cement concrete 1:2:4 mix blocks 10x10x10 cms deep.
- b) Where holder bat clamps are to be fixed in RCC column or slotted angles, walls or beam they shall be fixed with 40x3 mm flat iron "U" type clamps with anchor fasteners of approved design or 6 mm nuts and bolts.
- c) Structural clamps shall be fabricated by M.S. structural members e.g. rods, angles, and channels flats as per detailed drawing or as directed. Contractors shall provide all nuts, bolts, welding material and paint the clamps with one coat of red oxide and two or more coats of black enamel paint.
- d) Slotted angle/channel supports on walls shall be provided wherever shown on drawings. Angles/channels shall be of sizes shown on drawings or specified in schedule of quantities. Angles/channels shall be fixed to brick walls with bolts embedded in cement concrete blocks and to RCC walls with suitable anchor fasteners. The spacing of support bolts horizontally shall not exceed 1 m.

- e) Wherever M.S. clamps are required to be anchored directly to brick walls, concrete slabs, beams or columns, nothing extra shall be payable for clamping arrangement and making good with cement concrete 1:2:4 mix (1 cement :2 coarse sand :4 mm stone aggregate 20 mm nominal size) as directed by the Engineer-in-Charge.

1.8.3.5 TESTING

- a) Before using it at site all uPVC soil and waste pipes shall be tested by filling up with water for at least 10 minutes. After filling, pipes shall be struck with a hammer and inspected for blow holes and cracks. All defective pipes shall be rejected and removed from the site within 48 hours. Pipes with minor sweat may be accepted at the discretion of the Engineer-in- Charge.
- b) Pipes shall be tested after installation, by filling up the stack with water. All opening and connections shall be suitably plugged. The total head in the stack shall be, however, not 3 m exceed.
- c) Alternatively, Contractor may test all soil and waste stacks by a smoke testing machine. Smoke shall be pumped into the stack after plugging all inlets and connections. The top end shall, however, be left open. The stack shall then be observed for leakages and all defective pipes and fittings removed or repaired as directed by the Engineer-in-Charge.
- d) A test register shall be maintained, and all entries shall be signed and dated by Contractors and Engineer-in-Charge.

1.8.4 TRAPS

- a) Nahni trap or floor traps: Nahni traps or floor traps shall be uPVC with an effective seal of 50 mm. The trap and waste pipes shall be set in cement concrete blocks firmly supported on the structural floor. The blocks shall be in 1:2:4 mix (1 cement :2 coarse sand :4 stone aggregate 20 mm nominal size) and extended to 40 mm below finished floor level. Contractor shall provide all necessary shuttering and cantering for the blocks. Size of the block shall be 30x30 cms of the required depth.
- b) Floor trap inlet: Bath room traps and connections shall ensure free and silent flow of discharging water. Where specified, Contractor shall provide a special type inlet fitting hopper fabricated from 100 mm pipe without or with one, two or three inlet sockets to receive the waste pipe (s). Joint between waste and hopper inlet socket shall be Properly joint. Hopper shall be connected to P or S trap with at least 50 mm seal (hopper and traps shall be paid for separately.) Floor trap inlet hoppers and the traps shall be set in cement concrete blocks as specified in Para above without extra charge.

- c) Floor Trap Grating: Floor traps shall be provided with 75-150mm square or round C.P./Stainless steel grating, with rim of approved design and shape. Minimum thickness shall be 4 mm (for C.P. brass) or 1.2 mm (for SS), or as specified in the Schedule of Quantities.
- d) P trap: P-traps shall be of uPVC construction with an effective water seal of 50 mm. The trap and connected waste pipes shall be embedded in cement concrete blocks, firmly supported on the structural floor or wall as applicable. The concrete blocks shall be of mix proportion 1:2:4 (1 part cement : 2 parts coarse sand : 4 parts 20 mm nominal size stone aggregate) and shall extend up to 40 mm below the finished floor level or flush with the finished wall surface, as required. The contractor shall provide all necessary shuttering and centering for the casting of these blocks. The size of each block shall be 300 mm × 300 mm, with the required depth to suit the trap and pipe alignment.

1.8.5 WASTE PIPE FROM APPLIANCES

- a) Waste pipe from appliances e.g. washbasins, sinks shall be of uPVC.
- b) All pipes shall be fixed in gradient towards the outfalls of drains. Pipes inside a toilet room shall be in chase unless otherwise shown on drawings. Where required pipes may be run at ceiling level in suitable gradient and supported on structural clamps.

1.8.6 CUTTING AND MAKING GOOD

Pipes shall be fixed and tested as buildings proceeds. Contractor shall provide all necessary holes cut outs and chases in structural members as building work proceeds. Wherever holes are cut or left originally, they shall be made good with cement concrete 1:2:4 (1 cement: 2 coarse sand: 4 stone aggregate 20 mm nominal size) or cement mortar 1:2 (1 cement: 2 coarse sand) and the surface restored as in original condition.

1.8.7 INSPECTION & TESTING

A) INSPECTION

Work should be inspected during installation and tests applied on completion, care being taken that, all work which is to be encased for concealed is tested before it is finally enclosed. Inspection should be carried out to ensure the following:

- a) Work accords with the drawing and specifications.

- b) All pipe brackets, clips etc. are securely fixed.
- c) Fixtures are correctly spaced.
- d) Pipe is protected where necessary.
- e) Embedded pipe work is properly protected before sealing-in
- f) All access covers, caps or plugs: -
 - Are accessible
 - Are so made that the internal faces truly complete in internal bore.
 - Cause no obstruction in the pipe bore
 - Are well joined.

B) TESTING

The soil, waste piping system and rain water should be tested after installation as follows:

C) WATER TEST

The pipes shall be tested after installation & before the appliances are connected, preferably in sections so as to limit the static head of 4.5m. The pipe shall be filled with water for at least 10 minutes. After filling, pipes shall be struck with a hammer and inspected for blow holes and cracks. Then it will be necessary to seal all openings and leaks at joints immediately as observed during the test and all defective pipes shall be rejected and removed from the site.

D) SMOKE TEST

The smoke test shall be conducted as follows:

Smoke shall be introduced into the stack after sealing all inlets and connections at the lowest points. This shall be done using a smoke testing machine, which comprises a bellows and a burner. The material typically burned is greasy cotton waste, which produces dense, pungent smoke that is easily detectable both visually and by smell. Any leaks in the pipe system will allow the smoke to escape, indicating the location of the defect. The top end of the stack shall be left open to allow the smoke to pass freely through the system, ensuring the entire length of the stack is tested for potential leakage.

EXTERNAL PLUMBING WORK

2. SEWERAGE / DRAINAGE SYSTEM GENERAL

The Plumbing Installation work shall be carried out in accordance with Indian Standard Code of Practice. It shall also be in conformity with the current Indian Plumbing rules and regulations and requirements of the Local Authority and Fire assurance regulations, so far as these become applicable to the installation. Plumbing work in general shall be carried out as per following CPWD General Specifications amended upto date.

2.1 IS CODE FOR PLUMBING WORKS

S. No. I.S. Code Description / Title

1. IS: 1729 Specification for sand cast iron spigot and socket soil, waste and ventilating pipes, fittings and accessories.
2. IS: 1536 Specification for centrifugally cast (spun) iron pressure pipes for water, gas and sewage.
3. IS: 1538 (Part-I to XXIII) Specification for cast iron fittings for pressure for water, gas and sewage.
4. IS: 3714 Code of practices for laying C.I pipes
5. IS: 782 Specification for caulking lead
6. IS: 1239 (Part-II) Specification for mild steel tubes, tubular and other wrought steel filling.
7. IS: 1879 Specification for malleable cast iron pipe fittings.
8. IS: 4984 High-density polythene pipe for potable water supplies, sewage and Industrial effluents.
9. IS: 783 Width and depth of trench for R.C.C pipes.
10. IS: 4127 Width and depth of trench for S.W pipes.
11. IS: 780 Specification for sluice valve for water works purposes.
12. IS: 7558 Code of practice for domestic hot water installation.
13. IS: 1742 Code of practice for building drainage
14. IS: 2064 Code of practice for selection, installation and maintenance of

Sanitary appliances

15. IS: 2065 Code of practices for water supply in building
16. IS: 2183 (Part-I) Code of practice for Plumbing in multi-storied buildings.
17. IS: 1239 Specifications for mild steel tubes 140mm and other wrought steel fittings. (5th Rev.)
18. IS: 778 Specifications for copper alloy gate, globe and check valves for waterworks purposes.
19. IS: 5312 (Part-I) Specifications for swing check type reflux (Non return) valve
20. IS: 3114 Code of Practice for laying of C.I. pipes (2nd Rev.)
21. IS: 456 Code of practice for plain and reinforced concrete (3rd Rev.) (Amendment 2)
22. IS: 12820 Code of practice for dimensional requirements of rubber gaskets for mechanical joints and push on joints for use with cast iron pipes and fittings for carrying water, gas & sewage.
23. IS: 1172 Code of basic requirements for water supply, drainage & sanitation (4th Rev.)
24. IS: 1200 (Part-16) Code of practice for methods or measurements of building and Civil Engineering works: Part 16 Laying of water and sewer lines including appurtenant items (3rd Rev.)
25. IS: 1200 (Part-19) Code of practice for methods or measurements of building and Civil Engineering works: Part 19 Water supply, plumbing and drains (3rd Rev.)
26. IS: 3989 centrifugally cast (spun) iron spigot and socket soil, waste and ventilating pipes, fittings and accessories (2nd Rev.) (Amendment)
27. IS: 13095 Butterfly valves for general purposes
28. IS: 458 Precast Concrete pipes (with or without reinforcement) (3rd Rev.) (Amendment 2)
29. IS: 1726 C.I. Manhole covers & frames (3rd Rev.)
30. IS: 1916 Steel cylinder pipe with concrete lining and coating (1st Rev.).

31. IS: 12592(part1) Pre-cast concrete manhole covers and frames:Part 1 Cove
32. IS: 12592(part2) Pre-cast concrete manhole covers and frames: Part 2 Frames
33. IS: 6392 Steel pipe flanges (Amendment 1)
34. IS: 6418 C.I and malleable C.I. flanges for general engineering purposes.
35. IS: 4985 Unplasticized PVC pipes for potablewater supplies (2nd Rev.) (Amendment 2)
36. IS: 7181 Horizontally cast double flanged pipes for water, gas and sewage (1st Rev.) (Amendment 1)
37. IS:210 Grey iron casting. (4th Rev.)
38. BS EN 1057 Copper pipes
39. BS EN 1254 Copper Fittings
40. IS:4985 UPVC pipes
41. IS:15778 CPVC pipes
42. IS:8329 Ductile iron

2.2 SCOPE OF WORK

- 2.2.1 Work under this section shall consist of furnishing all labour, Materials, Equipment's and Appliances necessary and required to completely finish Sewerage/Drainage system as required by the drawings and specified hereinafter or given in the Indicative Items.
- 2.2.2 Without restricting to the generality of the foregoing, the sewerage system shall include: -
 - a) Installation of all sewer lines / effluent lines
 - b) Installation of all storm water drainage lines
 - c) Construction of all catch basins, chambers, manholes & other related civil works
 - d) Rain water harvesting system

2.3 GENERAL REQUIREMENTS

- a) All materials shall be new of the best quality conforming to specifications and subject to the approval of the Engineer-In-Charge.
- b) Drainage lines shall be laid to the required gradients and profiles.
- c) All drainage work shall be done in accordance with the local Municipal bye laws.
- d) Location of all manholes, catch basins etc., shall be got confirmed by the Contractor from the Engineer-In-Charge before the actual execution of work at site.
- e) All works shall be executed as directed by Engineer-In-Charge.

2.4 ALIGNMENT AND GRADE

- a) The sewer pipes shall be laid to alignment and gradient but subject to such modifications as shall be ordered by the Engineer-In-Charge from time to time to meet the requirements of the works. No change from the lines, depths of cutting or gradients of sewers shown on the plans and sections shall be permitted except by the express direction in writing of the Engineer-In-Charge.

2.5 DWC PIPES – for external Sewer system

All underground Sewer pipes where specified (other than those specified) shall be DWC pipes conforming to IS 16098(PART-2). Pipes shall be true and straight with uniform bore. Throughout cracked, warped pipes shall not be used on the work. All pipes shall be tested by the manufacturer and the Contractor shall produce, when directed a certificate to that effect from the manufacturer. For road crossing DWC grade shall be used at 1m depth.

2.5.1 LAYING

DWC (Double Wall Corrugated) pipes shall be laid on a well-prepared bedding of compacted soil, sand, or granular material, or on a cement concrete bed as specified. Where directed, concrete cradles may be provided. These cradles, if used, may be precast and shall be sufficiently cured to avoid cracks or breakage during handling. The invert of the cradle shall be kept approximately 12 mm below the invert level of the pipe to ensure uniform support and to prevent disturbance to the pipe alignment.

The DWC pipe shall then be carefully placed on the prepared bed or cradles and aligned to the specified line and gradient using sight rails, boning rods, or other suitable equipment. The use of concrete bedding or cradles may be omitted if instructed by the Engineer-in-Charge, based on site conditions and design requirement

2.5.2 JOINTING

After setting out the DWC pipes to the correct line and level, the pipe ends shall be cleaned thoroughly to remove any dirt, dust, or debris. The rubber ring (EPDM/SBR) shall be placed correctly in the groove of the socket end. The spigot end of the next pipe shall then be aligned and inserted into the socket with a firm, steady push to ensure a proper push-fit joint.

Lubricant, as recommended by the manufacturer, may be applied to the spigot end to facilitate jointing and prevent damage to the rubber ring. The joint shall be checked to ensure that the insertion depth is correct, and that the rubber ring is not displaced.

All joints shall be properly aligned, and no angular deflection beyond the manufacturer's allowable limits shall be permitted. No mortar, gaskin, or collars are required for DWC pipe joints. The completed joint shall be inspected to ensure watertightness and structural integrity.

2.6 Storm Water Drainage System

The RCC storm water drain shall be **designed and constructed** to safely collect and convey **surface runoff**, including **rainwater** and other forms of storm water, from paved areas, roads, open grounds, and adjoining premises to the Rain water harvesting without causing flooding, erosion, or structural damage.

- All Storm water drain channel shall be RCC covered with precast RCC covers (perforated/non-perforated as per approved drawings & site requirements). Where storm water drains are required to cross under roads, driveways, or other traffic-bearing surfaces, suitable **RCC box culverts** shall be provided to carry the designed peak discharge. The culverts shall be designed to withstand the specified traffic loading (IRC Class AA or as directed) and soil cover as per **IRC:SP:13** and relevant IS codes. The foundation shall be prepared to the

required bearing capacity. Proper inlet and outlet transitions, wing walls, and aprons shall be constructed to prevent erosion and ensure smooth flow. All joints shall be watertight, and adequate freeboard shall be maintained

- All rainwater drainage lines carrying water from the terrace and passing under buildings, floors after the nearest catch basin chamber of the building as per drawings and passing under the ground, roads with heavy traffic, or in exposed positions above ground shall be of RCC (NP-2) pipes and connected with main RCC drain channel. The routing and positioning of such pipes shall be in accordance with site requirements and approved drawings.
- Precast saucer drains of approved dimensions and shape shall also be provided as per site requirements and laid to the required slope for efficient surface runoff collection. The drains shall be manufactured using reinforced cement concrete of minimum M25 grade, with smooth internal finish and adequate thickness to ensure durability and load-bearing capacity. Joints between precast units shall be properly aligned, sealed, and made watertight. The saucer drains shall be securely connected to the RCC storm water drainage system through properly constructed inlets, gully traps, or junction chambers, ensuring free flow without backflow or leakage. All works shall be carried out in accordance with the approved drawings and instructions of the Engineer-in-Charge.
- Providing and placing in position 100 mm thick factory made machine batched & machine mixed Precast RCC Rectangular Covers on drains of various sizes, of M-25 grade cement concrete for RCC work, including cost of centering, shuttering, reinforcement of 8 mm dia TMT bars of Fe 500 grade @ maximum 100 mm c/c on both ways, neat cement punning on finished surface, properly encased on all edges with 1.6 mm thick, 100 mm wide MS sheet duly painted over priming coat, reinforcement to be welded at edges with MS sheet and providing 2 Nos. 12 mm dia bar for hooks etc including cost of cartage, all leads & lift, handling at site etc. all complete as per direction of Engineer-in-Charge
- Providing and placing in position 100 mm thick factory made precast RCC perforated drain covers, having concrete of strength not less than M-25, of various sizes, including cost of centering, shuttering, reinforcement of 8 mm

dia TMT bars of Fe 500 grade @ maximum 100 mm c/c on both ways, neat cement punning on finished surface,, including providing 50 mm dia perforations @ 100 to 125 mm c/c, including properly encased on all edges with 1.6 mm thick, 100 mm wide MS sheet duly painted over priming coat, reinforcement to be welded at edges with MS sheet etc including cost of cartage, all leads & lift, handling at site etc. all complete as per direction of Engineer-in-Charge.

2.7 FITTINGS AND INSPECTION CHAMBERS

Cleanout plugs shall be provided on head of each drain and at location indicated on plans or directed by Engineer-In-Charge. Cleanout plugs shall be of size matching the full bore of the pipe.

2.7.1 TESTING

- (a) All lengths of the drain shall be fully tested for water tightness by means of water pressure maintained for not less than 30 minutes. Testing shall be carried out from manhole. All pipes shall be subjected to a test pressure of at least 1.5-meter head of water. The test pressure shall, however, not exceed 1.5-meter head at any point. The pipes shall be plugged preferably with standard design rubber plugs on both ends. The upper end shall, however, be connected to a pipe for filling with water and getting the required head. The tolerance figure of two liters per millimeter of dia per kilometer may be allowed during a period of ten minutes

- (b) Trapped Air

Allowance shall be made for (i) by adding water until absorption has ceased after which the test proper should commence. Any leakage will be visible and the defective part of the work should be cut out and made good. The excessive sweating from a particular pipe or joint shall be watched for and taken as indicating a defect to be made good.

- (c) Sewer and Drain Pipelines shall be tested for straightness by:

Inserting a smooth ball 12mm less than the internal diameter of the pipe. In the

absence of obstructions such as yarn or mortar projecting at the joints the ball should roll down the invert of the pipe and emerge at the lower end.

Means of a mirror at one end and a lamp at the other end. If the pipe line is straight the full circle of light will be seen otherwise obstruction or deviation will be apparent.

- (d) The Contractor shall give a smoke test to the drains and sewer, if directed by the Engineer- In-Charge.
- (e) A test register shall be maintained which shall be signed and dated by Contractor, Engineer- In- Charge.

2.8 CEMENT CONCRETE AND MASONRY WORKS FOR MANHOLES AND CHAMBERS ETC.

All materials shall be as per indicative items and as per specification for civil works (CPWD).

2.8.1 CEMENT CONCRETE FOR PIPE SUPPORT

1. Wherever specified, all pipes shall be supported in concrete bed all round or in haunches.
2. Unless otherwise directed by the Engineer-In-Charge cement concrete for bed, all round or in haunches shall be laid as follows: -
 - Details for Cement Concrete
 - Description Upto 3 M depth
 - Pipes in open ground (No sub soil water) All round (1:5:10)
 - Pipes (all) in sub soil water condition All round (1:3:6)
 - Pipes under the building or at road crossing or under public places All round (1:2:4)
 - (1=1=1 cement, 2-3-5 coarse sand, 4-6-10) stone aggregate 20 mm nominal size)
3. R.C.C. pipes may be supported on brick masonry or precast R.C.C.
4. Pipes in loose soil or above ground shall be supported on brick or RCC anchor blocks.

2.8.2 MANHOLES AND CHAMBERS

All manholes, chambers and other such works as per Latest CPWD Specification.

2.9 EXTERNAL WATER SUPPLY

2.9.1 SCOPE

The scope of this section comprises the supply, installation, testing and commissioning of piping network for water supply for external services as follows:

- a) Domestic Water Supply.
- b) Connection to various mechanical equipment's to be supplied and installed.

The Contractor shall make all necessary application and arrangements for his work to be inspected by the Local Authorities.

The Contractor shall be solely responsible for obtaining the Authorities approval of his works prior to the handing over of the complete water supply / distribution installation to the Engineer- In- Charge.

2.9.2 PIPING MATERIALS

All the external pipes to be used for water supply and flushing supply shall be CPVC pipes.

No unsightly exposed runs shall be permitted. Outside the building the piping shall be installed as per CPWD specifications.

- **TRENCHES**

- a) The galvanized iron/CPVC pipes and fittings shall be laid in trenches. The width and depth the trenches for the different diameters of the pipes shall be as follows:

Details for Trenches

Dia of Pipe	Width of Trench	Depth of Trench
15mm to 50mm	30 cms	60 cms
65mm to 100mm	45 cms	75 cms

- b) At joints the trench width shall be widened where necessary. The work of excavation and refilling shall be done true to line and gradient in accordance with general specifications for earthwork in trenches.
- c) When excavation is done in rock, it shall be cut deep enough to permit the pipes to be laid on a cushion of sand minimum 7.5 cm deep.

2.9.3 PIPING INSTALLATION SUPPORT

Galvanised Iron Pipes & Fittings-As per Latest CPWD Specification. Laying and

Jointing of GI Pipes-As per Latest CPWD Specification

CPVC pipes laying and jointing As per Latest CPWD Specifications

2.9.4 WATER METERS (Buildings & Infrastructure)

Providing and fixing smart water meter as per GRIHA norms at Municipal Supply, Borewell, Treated Water outlet from STP, Captured rainwater for reuse in project and UGT(Install meters at inlet & outlet to track storage & distribution, OHT (Flushing & Domestic Water) for:

- Each Tower (G+6)- Meter at main riser or pump outlet supplying water to the tower
- Guest House (G+3)- Meter at inlet to monitor the total consumption.
- Community Centre- Meter at inlet to monitor the total consumption.

Water meters design shall be supplied for installation at required locations as shown in the tender drawings and as per DBR.

All water/British thermal unit meters should have an RS-485/RS-232 port with publicly available/industry standard protocol (Modbus, etc.) and register maps.

Installation of Smart Water Meter and Stop Cock

The G.I. or CPVC pipeline shall be cut to the required length at the designated location for installation of the smart water meter and stop cock. Suitable fittings, such as sockets, unions, and jam nuts, shall be used to securely connect the components.

The stop cock shall be installed on the **inlet side** of the water meter for isolation and control. The protective paper disc inserted in the ripples of the water meter (if provided) shall be removed prior to installation.

The meter shall be installed in a **horizontal or vertical** position, as permitted by the manufacturer, ensuring the flow direction aligns with the arrow cast on the meter body.

Care shall be taken to ensure that the **factory seal of the meter remains intact** and is not tampered with.

In cases where the meter is to be installed on a newly laid pipeline, the entire line shall be thoroughly flushed and cleaned before installing the meter to prevent

damage or obstruction of Smart Water Meter and Stop Cock.

2.9.5 TESTING

- a) The Contractor shall notify the Engineer-In-Charge in advance of any test so that the Engineer-In-Charge can witness the tests if he so wishes.
- b) All water supply system shall be tested to hydrostatic pressure test of at least one and a half (1.5) times the maximum pressure but not less than 10Kg/Sq.cm for a period of not less than 8 hours. All leaks and defects in joints revealed during the testing shall be rectified and got approved at site by retest. Piping required subsequent to the above pressure test shall be retested in the same manner.
- c) System may be tested in sections and such sections shall be entirely retested on completion.
- d) The Contractor shall make sure that proper noiseless circulation of fluid is achieved through the entire piping network of the system concerned. In case of improper circulation, the contractor shall rectify the defective connections.
- e) In addition to the sectional testing carried out during the construction, contractor shall test the entire installation after connections to the overhead tanks or pumping system or mains. He shall rectify all leakages and shall replace all defective materials in the system. Any damage done due to carelessness, open or burst pipes or failure of fittings, to the building, furniture and fixtures shall be made good by the contractor during the defects liability period.
- f) After commissioning of the water supply system, contractor shall test each valve by closing and opening it a number of times to observe if it is working efficiently. Valves which do not effectively operate shall be replaced by new ones and the same shall be tested as above.
- g) A test register shall be maintained and all entries shall be signed and dated by Contractor(s) representative and Engineer-In-Charge.

2.9.6 DISINFECTION OF PIPING SYSTEM AND STORAGE TANKS

Before commissioning the water supply system, the contractor shall arrange to disinfect the entire system as described in the succeeding paragraph.

The water storage tanks and pipes shall first be filled with water and thoroughly flushed out. The storage tanks shall then be filled with water again and disinfecting

chemical containing chlorine added gradually while tanks are being filled to ensure thorough mixing. Sufficient chemical shall be used to give water a dose of 50 parts of chlorine to one million parts of water.

If ordinary bleaching powder is used, the proportions will be 150 gm of power to 1000 liters of water. The power shall be mixed with water in the storage tank. If a proprietary brand of chemical is used, the proportions shall be specified by the manufacturer. When the storage tank is full, the supply shall be stopped and all the taps on the distributing pipes are opened successively working progressively away from the storage tank. Each tap shall be closed when the water discharged begins to smell of chlorine. The storage tank shall then be filled up with water from supply pipe and added with more disinfecting chemical in the recommended proportions. The storage tank and pipe shall then remain charged at least for three hours. Finally, the tank and pipes shall be thoroughly flushed out before any water is used for domestic purpose.

The pipe work shall be thoroughly flushed before supply is restored.

2.9.7 STERILIZATION OF MAIN

After the pipe work has been tested and approved, but before it is coupled, it shall be sterilized with a solution of chloride of lime.

2.9.8 VALVES

All valves (gate, globe, check, safety) shall be of gun metal suitable for the particular service as specified. All valves shall be of the particular duty and design as specified. Valves shall either be of screwed type or flanged type, as specified, with suitable flanges and non-corrosive bolts and gaskets. Tail pieces as required shall be supplied along with valves. Gate, globe and check valves shall conform to Indian Standard IS:776 and non- return valves and swing check type reflux to IS:5312.

Sluice valves, where specified shall be flanged sluice valves of cast iron body. The spindle, valve seat and wedge nuts shall be gunmetal. They shall generally have non-rising spindle and shall be of the particular duty and design as specified. The valves shall be supplied with suitable flanges, non- corrosive bolts and asbestos fiber gaskets. Sluice valves shall conform to Indian standard IS: 780 and IS: 2906.

- (a) Brass Ball Valves:

Valves 50mm dia and below shall be screwed type ball valves with stainless steel balls spindle Teflon seating and gland packing tested to a hydraulic pressure of 20 Kg/Sq.cm. and accompanying couplings and steel handles to BIS – 5351.

- (b) Valves above 50mm dia. shall be Butterfly Valves. Cast iron (IS 210), grade FG 260 butterfly valves conforming to PN 1.6, heavy duty cast iron disc with anti-corrosive nickel plating, nitrile seat and stainless steel 410 stem with lever/gear operation and powder coated finish.

Float Valves to be fixed in storage tanks shall consist of cast brass lever arm having copper balls (26 SWG) screwed to the arm integrally. The copper ball shall have bronze welded seams. The closing/opening mechanism incorporating the piston and cylinder shall be non-corrosive metal and include washers. The size and construction of ball valves and float shall be suitable for desired working pressure operating the supply system. Where called for brass valves shall be supplied with brass hexagonal back nuts to secure them to the tanks and a socket to connect to supply pipe. All valves shall be suitable for the working pressure involved.

- (c) Pressure Gauge: The pressure gauge shall be constructed of die cast aluminum and stove enameled. It shall be weather proof with an IP 55 enclosure. It shall be a stainless-steel Burden tube type pressure gauge with a scale range from 0 to 16 Kg / cm square and shall be constructed as per IS:3524. Each pressure gauge shall have a siphon tube connection. The shut off arrangement shall be by Ball Valve.

2.9.9 WATER FITTINGS

Unless otherwise specified all Gunmetal fittings such as gate, globe, check & safety valves shall be fitted in pipe line. Necessary unions shall be provided on both ends of the valves for easy replacement. The joints between fittings and pipes shall be leak-proof when tested to desired pressure rating. The defective fittings and joints shall be replaced.

2.9.10 CONNECTIONS TO RCC WATER TANKS

The contractor shall provide all inlets, outlets, washouts, vents, ball cocks, overflows control valves and all such other piping connections including level indicator to water storage tanks as called for. All pipes crossing through RCC work shall have puddle flanges fabricated from GI pipes of required size and length and

welded to 6/8 mm thick MS plate. All puddle flanges must be fixed in true alignment and level to ensure further connection in proper order.

Full way gate valves shall be provided as near the tank as practicable on every outlet pipe from the storage tank except the overflow pipe. Overflow and vent pipes shall terminate with mosquito proof grating.

The overflow pipe shall be so placed to allow the discharge of water being readily seen. The overflow pipe shall be of size as indicated. A stop valve shall also be provided in the inlet water connection to the tank. The outlet pipes shall be fixed approximately 75mm above the bottom of the tank towards which the floor of the tank is sloping to enable the tank to be emptied for cleaning.

2.9.11 THRUST BLOCKS

In case of bigger pipes (80 mm dia and above), thrust blocks of cement concrete 1:2:4 (1 cement: 2 coarse sand: 4 graded stone aggregate of 20 mm nominal size) shall be constructed on all bends or as directed by Engineer-In-Charge.

2.9.12 MASONRY CHAMBER

- a. All masonry chambers for stop cocks, sluice valves and meter etc. shall be built as per shop drawings.
- b. The excavation for chambers shall be done true to dimension and level indicated on plans or as directed by the Engineer-In-Charge.
- c. Concrete shall be of cement concrete 1:3:6 (1 cement: 3 coarse sands: 6 graded stone aggregate 40 mm nominal size).
- d. Brick shall be of class designation 75 in cement mortar 1:6 (1 cement: 6 fine sand).
- e. Inside Plastering not less than 12 mm thick shall be done in cement mortar 1:3 (1 cement: 3 fine sand) finished with a floating coat of neat cement.

2.9.13 SHIFTING OF EXCAVATED SURPLUS MATERIAL

Contractor shall make his own arrangement to shift the surplus excavated material within the site limits as directed by Engineer-In-Charge.

1) Fittings and Inspection Chambers

Fittings and chambers shall be provided as required.

2) Anchor Block

Suitable anchor blocks shall be provided at all bends and tees to encounter the excessive thrust developed due to water hammer.

3) Rubber Joints

Joints between two pipes shall be made by premoulded rubber joints with suitable tackles in a manner recommended & approved by the manufacturer. No joints shall be covered until the lines are hydraulically tested.

2.9.14 VALVE CHAMBERS

Contractor shall construct chambers for all full way valves, butterfly valves and other type of valves as specified in Indicative Items. These shall be made, in brick masonry in cement mortar 1:4 (1 cement: 4 coarse sand) on cement concrete foundations 150mm thick 1:5:10 mix (1 cement: 5 coarse sand: 10 graded stone aggregate 40mm nominal size) 12 mm thick cement plaster 1:3(1 cement : 3 coarse sand) inside finished with a floating coat of neat cement with 8mm thick GI surface box with hinged cover and locking arrangement, 150 mm thick reinforcement cement concrete top slab of 1:2:4 (1 cement : 2 coarse sand:4 graded stone aggregate 20mm nominal size), as specified and shown in drawings, including excavation, back filling rammed complete or as specified in Indicative Items.

Valve chambers shall be constructed as specified in Indicative Items

2.9.15 TESTING

- a) All pipes, fittings and valves shall be tested by hydrostatic pressure of min. 1.5 times, the working pressure and subject to minimum of 7 kg/cm² in any case and with the consent of Engineer-In-Charge.
- b) Pressure shall be maintained for a period of at least two hours without appreciable drop in the pressure after fixing at site. (+10 %). A test register shall be maintained and all entries shall be signed and dated by Contractor and Engineer-In-Charge.

- c) In addition to the sectional testing carried out during the construction, Contractor shall test the entire installation after connections to the overhead tanks or pumping system or mains. He shall rectify all leakages, and shall replace all defective materials in the system. Any damage done due to carelessness, open or burst pipes or failure of fittings, to the building, furniture and Fixtures shall be made good during the defect liability period.
- d) After completion of the water supply system, Contractor shall test each valve by closing and opening it a number of times to observe if it is working efficiently. Valves which do not effectively operate shall be replaced by new ones and the same shall be tested as above.

2.9.16 PRE COMMISSIONING

- a) Ensure that all pipes are free from debris and obstructions.
- b) Check all valves and fire hydrant for effective opening and closing action. Defects should be rectified or valves replaced.
- c) Ensure that all Connections to Branches have been made.
- d) Ensure that mains have been connected to the respective pumps, underground and overhead tanks.
- e) Water supply should be available at main Underground tank.
- f) All main line Valves should be closed.

2.9.17 COMMISSIONING

- a) Fill Underground tank with water. Add fresh bleaching powder after making a solution to be added near inlet.
- b) Start Water Supply Pump and allow water to fill main Underground tank. Water will first fill the fire tank and then overflow to the Raw Water tanks.
- c) After filling Overhead Reservoir drain the same to its one forth capacity through tank scour valve. (This is to ensure removal of all mud, debris etc. from the tank).
Fill Overhead tank to full.
- d) Release water in the main lines by opening Valves in each circuit. Drain out water in the system through scour valves or fire hydrant in lower regions. Ensure clean water is now coming out of the system.
- e) Open valves for individual clusters. Observe for leakages or malfunctions, check pressure & flow at end of line by opening Hydrants etc. Remove and rectify defects

noticed.

- f) Check all outlet points for proper operation by opening each valve and allowing water to flow for a few minutes. Also check for effective closure of valve.
- g) The entire water supply system should be disinfected with bleaching powder and system flush cleaned.
- h) Send four samples of water drawn from four extreme locations for testing for bacteriological test in sterilized bottles obtained from the concerned laboratory. (Laboratory personal shall collect the samples themselves).

2.10 SEWAGE TREATMENT PLANT (FRP PACKAGED UNIT with ULTRA FILTRATION)

2.10.1 SCOPE OF WORK GENERAL

The scope of work under this contract shall include the Design, Engineering, Fabrication, Supply, Installation, Testing, Commissioning, and Operation & Maintenance (O&M) of a Packaged Sewage Treatment Plant (STP) of 135 KLD capacity based on Moving Bed Biofilm Reactor (MBBR) technology followed by Ultra Filtration (UF) system.

2.10.1.1 The scope of work includes

- a) Design, Supply, Fabrication, Installation, Testing, Commissioning, and Operation & Maintenance of a 135 KLD FRP Packaged STP based on MBBR Technology with integrated Ultra Filtration (UF).
- b) The STP unit shall be compact, factory-fabricated (prefabricated), modular, skid-mounted and delivered in ready-to-install condition.
- c) The plant should ensure 100% recycling of treated wastewater for irrigation, flushing and horticulture, meeting MoHUA discharge norms.
- d) The Contractor shall be responsible for:
 - Design & drawing preparation (P&ID, GA, layout)
 - Manufacturing, supply and installation
 - Testing, commissioning, trial run
 - Pollution Control Board approvals (if applicable)
 - Operation and Maintenance as per SCC
 - Training of site staff
 - Submission of O&M Manual

- e) All mechanical equipment shall be housed within the FRP unit with required piping, valves, and instrumentation.
- f) The STP must function efficiently even at **20% loading**, with automatic flow regulation.

2.10.2 General Information

The maximum safe bearing capacity of soil strata shall be taken considering the soil available and ground water table at site.

2.10.3 STP Technical detail

Project objective is to treat domestic sewage generated from bathroom, flushing, kitchen etc. The sewage shall be treated to meet the CPWD specifications, relevant IS codes and Guidelines of Central/State Pollution Boards and standards as per directions of Hon'ble NGT vide orders dated 30.04.2019. The capacity of sewage treatment plant shall be 135 KLD.

In this project, domestic sewage generated from bathrooms, flushing systems, kitchens, etc., shall be collected and conveyed to the equalization section of the packaged sewage treatment plant (135 KLD capacity). The packaged FRP STP shall carry out all stages of treatment—biological and tertiary including ultra-filtration—within a compact, skid-mounted unit. The raw sewage shall be treated to achieve the desired outlet parameters as per statutory norms. The treated water shall be reused for gardening, flushing, and other non-potable applications. Waste sludge generated during treatment shall be collected in the sludge holding section of the packaged unit and shall be dewatered periodically using sludge bags or external sludge drying beds, and disposed of as per applicable regulations.

2.10.4 Scope of the work

The scope of work includes the design, engineering, supply, installation, testing, commissioning, and performance demonstration of a Sewage Treatment Plant (STP) of 135 Kilo Litres per Day (KLD) capacity, based on Moving Bed Biofilm Reactor (MBBR) technology, including all associated civil, mechanical, electrical, and instrumentation works. The STP shall be designed to treat domestic wastewater generated from bathrooms, kitchens, and flushing systems within the premises, and produce treated effluent that meets the prescribed norms of the National Green Tribunal (NGT) and the Central Pollution Control Board (CPCB).

2.10.5 Technical Proposal

Domestic sewage generated from bathroom, flushing, kitchen etc. shall be collected in the Collection/Equalization tank. Collected raw sewage shall be treated to meet desired treated sewage parameters and after complete treatment treated sewage shall be reused for gardening, flushing and other miscellaneous activities.

2.10.6 Design Basis

- a) Average Flow: 135 KLD

- b) Operation time: 24 hours
- c) Source of wastewater: bathroom, flushing, kitchen etc.
- d) End use of treated sewage: Treated sewage shall be collected in the treated sewage tank which can be reused for gardening, flushing or for other miscellaneous activities.

2.10.7 Key treatment stages:

- a) **Screening & Grit Removal:** Prevents blockages by removing solids and grit.
- b) **Equalization Tank:** Balances flow and pollutant loads.
- c) **Anaerobic Digester:** Effective for soap and detergent breakdown from greywater, reducing BOD/COD and preventing foaming issues.
- d) **MBBR Reactor:** Biological treatment using bio-media for efficient degradation of organic pollutants.
- e) **Secondary Clarification:** Settles biological sludge for safe disposal.
- f) **Tertiary Treatment:**
 - Pressure Sand Filter (PSF) removes suspended solids.
 - Activated Carbon Filter (ACF) removes color, odor, and dissolved organics.
 - Sodium hypochlorite dosing ensures pathogen kill.

The treated water will be reused for gardening, toilet flushing, and other non-potable purposes, promoting water conservation and sustainability.

2.10.8 Inlet and Outlet parameters

- a) The treated effluent quality parameters & their limit for utilizing it for gardening and miscellaneous inhouse activities, for which STP has designed are as follow.

Parameter	Inlet	Outlet
Bio-Chemical Oxygen Demand	225-250 mg/L	< 5 mg/L
Chemical Oxygen Demand	350-450 mg/L	< 50 mg /L
Total Suspended Solids	250-350 mg/L	< 10 mg/L
Total Phosphorus-P	8 mg/L	<1 mg/L
Fecal Coliforms	10 ⁷ MPN/100mL	Desireable-100 MPN/100mL Permissible-230 MPN/100mL
Nitrogen- Total	50 mg/L	<10 mg/L
PH	6.5-8.5	6.5-8.5
Oil & Grease	<15	<3 mg/L

- As per order by Ministry of Housing and Urban Affair date 18/12/2023, above mentioned STP outlet parameters are applicable to all mode of disposals.

- The plant is to be designed based on the above-mentioned parameters.

2.10.9 Proposed Treatment Process

2.10.9.1 Bar & Fine Screen Chamber

Raw sewage enters the STP at a rate of 135 KLD and first passes through a Bar Screen followed by a Fine Screen Chamber, where large debris, floating solids, and coarse materials such as plastics, rags, and sticks are removed. This prevents clogging and damage to downstream equipment and pumps.

2.10.9.2 Oil Skimmer Unit

The screened sewage then flows into an Oil Skimmer Unit, which removes free-floating oil and grease from the wastewater. This step is essential for reducing the organic load and improving the efficiency of biological treatment stages.

2.10.9.3 Collection Tank

Post-skimming, the sewage is collected in a Collection Tank, where initial flow equalization and homogenization take place. A Raw Sewage Transfer Pump is used to lift the sewage to the next stage.

2.10.9.4 Segregation cum Equalization Tank

The sewage is pumped into a Segregation cum Equalization Tank, which stabilizes fluctuations in flow and quality. This helps in achieving a uniform load for biological treatment. Continuous mixing may be provided to prevent settling of solids.

2.10.9.5 Anoxic Reactor Tank

The equalized sewage flows into the Anoxic Reactor, where denitrification takes place. In the absence of oxygen, denitrifying bacteria convert nitrate (NO_3^-) to nitrogen gas (N_2), thereby reducing nitrogen levels in the effluent. A Sludge

Recirculation Pump ensures mixed liquor from downstream aerobic zones is returned to maintain denitrification efficiency.

2.10.9.6 MBBR (Moving Bed Biofilm Reactor) Tank

From the anoxic tank, sewage enters the MBBR Reactor, an aerobic zone equipped with floating biofilm media and supplied with oxygen via air blowers and diffusers. Here, aerobic microorganisms degrade organic matter (BOD/COD), and nitrification (conversion of ammonia to nitrate) occurs.

2.10.9.7 Secondary Clarifier

The treated sewage from the MBBR tank flows into a clarifier, where biomass (biological sludge) settles at the bottom. The clear supernatant is drawn from the top and passed to the tertiary treatment stage. A portion of the settled sludge is recirculated to the anoxic tank, while excess sludge is sent to sludge management.

2.10.9.8 Filter Feed Tank

The clarified water is collected in the Filter Feed Tank, which serves as a buffer tank for tertiary treatment. A Filter Feed Pump delivers the water to the filtration system.

2.10.9.9 Tertiary Treatment – Pressure Sand Filter (PSF) & Activated Carbon Filter (ACF)

The tertiary treatment unit is an skid-mounted outdoor system that includes:

- Pressure Sand Filter (PSF): Removes remaining suspended particles, turbidity, and colloidal matter.
- Activated Carbon Filter (ACF): Adsorbs residual organic compounds, color, odor, and traces of heavy metals.
- Disinfection Unit (Hypochlorite Dosing): Doses sodium hypochlorite for microbial **disinfection** before final reuse/discharge.

2.10.9.10 Ultrafiltration Unit

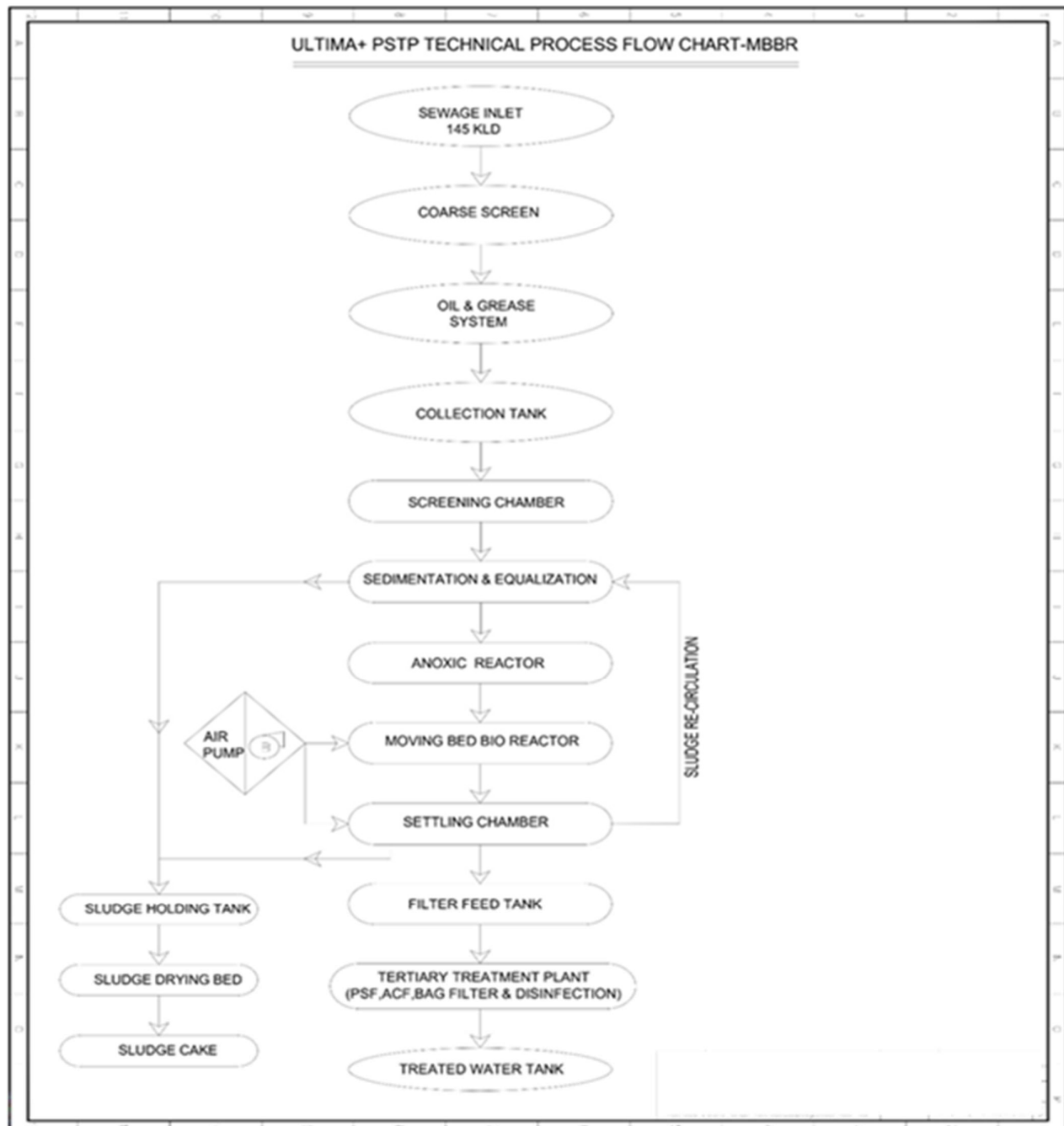
The supernatant is further polished using Ultra Filtration (UF) membrane units to achieve high-quality treated water suitable for reuse. The final treated water is stored in the treated water tank with inline disinfection for applications such as flushing, gardening, or other non-potable uses.

2.10.9.11 Treated Water Tank

The final **treated** effluent, meeting reuse standards (BOD < 5 mg/L, COD < 10 mg/L, TSS < 10 mg/L, pH 6.5–8.5), is stored in a Treated Water Tank. It can be reused for gardening, toilet flushing, or other non-potable purposes, or safely disposed of as per regulatory norms.

2.10.9.12 Sludge Management System

The settled sludge at the bottom of the first chamber of PSTP is collected in the sludge bag system. Packaged STPs are designed for minimal sludge generation. In general, desludging is required **in** the frequency of 4 to 6 months once. Sludge is pumped into a Sludge Bag system. The outflow goes back into the collection tank and the sludge accumulated in the bags are dried to form dried cakes.



Typical process flow diagram of Sewage Treatment Plant

2.10.10 Moving Bed Biological Reactor (MBBR) with ultra filtration unit

The Moving Bed Biofilm Reactor (MBBR) technology, originally developed in Norway by the Norwegian University of Science and Technology in collaboration with Kaldnes (now AnoxKaldnes AS), is utilized in the proposed FRP Packaged STP. In this advanced biological treatment process, microorganisms exist in both suspended form and as biofilm attached to specially designed carrier media within the aeration chamber.

This dual-mode microbial presence enables efficient degradation of organic matter through both suspended growth and attached growth mechanisms. After preliminary treatment via integrated bar screen and grit chamber, the wastewater enters the aerobic MBBR zone within the FRP unit, where biological treatment takes place over the designed hydraulic retention time (HRT).

Post aerobic treatment, the effluent flows into a compact clarifier or tube settler, where the biomass (biological sludge) is separated from the treated water. The settled sludge is collected from the bottom section and routed to the sludge holding tank for further processing, while the clarified supernatant is directed to the tertiary treatment section.

The supernatant is further polished using Pressure Sand Filter (PSF), Activated Carbon Filter (ACF), and Ultra Filtration (UF) membrane units to achieve high-quality treated water suitable for reuse. The final treated water is stored in the treated water tank with inline disinfection for applications such as flushing, gardening, or other non-potable uses.

2.10.11 Scope of Work

The scope of work shall include Design, Engineering, Construction, Fabrication, Procurement, Supply, Manufacture, Fabrication, Supply, Transportation, Storage, Handling, Erection, Testing and Commissioning of the Sewage Treatment Plant of capacity 135 KLD.

135 KLD MBBRwith UF Based Packaged Sewage Treatment Plant			
SN	Item Description	Unit	Quantity
1	Bar Screen		
	Coarse Screen	Nos	1
	Type - Standard		
	MOC - MS Epoxy Coated		
	Spacing - 10 mm		
2	Bar Screen		

	Fine Screen	Nos	1
	Type - Standard		
	MOC - MS Epoxy Coated		
	Spacing - 5 mm		
3	Oil skimmer Unit	Nos	1
	Belt Width : 100 mm x 2000 mm		
	Make : Standard		
	Collection Tray : 50 liters		
4	Raw sewage transfer Pump	Nos	2
	1W+1S		(1w+1s)
	Flow rate: 8 m3/hr		
	Head: 12 m		
	Type: Surface pump /Submersible Pump		
	Moc: Ci		
6	135 KLD MBBR PSTP Tank	Lot	1
	Capacity - 135 KLD		
	Thickness - 12mm or more		
	MOC – Filament Wound FRP		
	FRP Tanks		
	a. Screening tank		
	b.Anerobic tank		
	b. Aeration tank		
	c.Settling tank		
	d.Filterfeed tank		
7	Air Blower - Aeration Tank	Nos	12
	Type - Magnatic diaphragm		
	Air flow Rate - 180 m3/hrs		
	Moc: Ci Type : 1 Phase		

8	Hypohlorite Dosing System		
	Dosing Tank	Nos	1
	Capacity - 100 L		
	MOC - LDPE		
	Dosing Pump	Nos	1
	Capacity - 0 to 6 lph		
	MOC - PP		
9	Pressure sand filter		
	Type : Cylindrical Vertical	Nos	1
	Operating Flow : 8 m3/hr		
	MOC : FRP		
	Accessories : Multiport Valves		
10	Activated Carbon filter		
	Type : Cylindrical Vertical	Nos	1
	Operating Flow : 8 m3/hr		
	MOC : FRP		
	Accessories : Multiport Valves		
11	Bag Filter		
	Flow Rate - Upto 8 m3 /hr		
	MOC- PP	Nos	1
12	Rotameter		
	Range : F 10000 L	Nos	2
13	Electrical Pannel & Accessories with IOT device	lot	1
14	Filter Feed Pump	Nos	2
	1W+1S		(1w+1s)
	Flow rate: 8 m3/hr		

	Head: 25 m		
	Type: Centrifugal Monoblock		
	Moc : SS		
15	Sludge Transfer Pump	Nos	2
	Flow rate: 6 m ³ / hr		1W+1S
	Head: 10 m		
	Type: Surface pump		
16	Flow Meter		
	Type - Electromagnetic Digital Flow meter		
	Position : Raw inlet & Outlet	Nos	2
17	UF System with complete set	lot	1
	Flow rate: 7m ³ / hr		

Note: Treated water tank of STP should have sensor to monitor the treatment performance within the desired parameters.

2.10.12 Technical Specification

a) Area requirement: as per design

Note: The treated water tank shall be constructed as per the DBR and requirements , and with pump rooms.

2.11 FOR HOTRICULTURE

Pipes

CPVC Pipe confirming to IS 15778 of required dia for garden irrigation system cut to required length pipe to be laid below level in trenches up to required depth, including excavation in all kind of soil (hard rock) dewatering, refilling, watering, ramming and removing the surplus excavated material and putting 100 m sand all-around and making good the same.

Riser pipe

Brass Sprinkler including riser pipe & accessories for garden/green area complete as

required will be used.

Lawn hydrant

25 mm dia Lawn hydrant consisting of 25 mm dia Ball Valve, G.I nipple and threaded hose receiver complete as required will be used.

Rain Water Harvesting Pit

Rain water harvesting pit with the following accessories will be constructed

- Boring/drilling bore well of required dia for casing/ strainer pipe, by suitable method prescribed in IS: 2800.
- Fixing of 200 mm dia in vertical position in bore well, unplasticized PVC medium well casing (CM) pipe of required dia, conforming to IS: 12818, including required hire and labour charges, fittings & accessories etc. all complete.
- Fixing of 200 mm dia in vertical position in bore well unplasticized PVC medium well screen (RMS) pipes with ribs, conforming to IS: 12818, including hire & labour charges, fittings & accessories etc.
- Filling, spreading & leveling stone boulders of size range 5 cm to 20 cm, in recharge pit.
- Filling, spreading & leveling gravels of size range 5 mm to 10 mm, in the recharge pit, over the existing layer of boulders.
- Filling, spreading & leveling coarse sand of size range 1.5 mm to 2 mm in recharge pit, in required thickness over gravel layer,
- Gravel packing in tubewell construction in accordance with IS: 4097, including providing gravel fine/ medium/ coarse, in required grading & sizes as per actual requirement.
- Fixing Bail plug/ Bottom plug of required dia to the bottom of pipe assembly of tubewell as per IS:2800 (part I).
- Recharge well including earth work in excavation in over area and supply & installation of Recycled Polypropylene Geo cellular modular unit Aquacell L1200 x W600 x H400 mm in size according to EN17152-1 with min 95% void ration with clip-in together with all required accessories like bottom plate, side plate, adapter, Inspection shaft, Geotextile, Filter to create a below-ground geo cellular structure for attenuation

or rainwater which offers the optimal solution for fastest installation and total access for inspection and cleaning. Number of units should be selected based on storage/infiltration requirement. The system selection should be based on traffic load, Cover depth and soil condition. Maximum installation depths – to base of units/cover depth should be considered based on soil condition/load. The installation and selection should be as per manufacturer specification and guidelines.

- De-Silting Chamber including earth work in excavation in over area, foundations in all kinds of soil/ordinary rock including all lifts and lead, backfilling, including PCC (1:4:8) work, masonry work, RCC M15 grade work in top slab including reinforcement, centering, shuttering, 12mm cement plastering (1:3) finished with a float coat of neat cement, as per technical specifications, drawings, all complete.
- Providing and placing in position 100 mm thick factory made precast RCC perforated & Non-perforated drain covers, having concrete of strength not less than M-25, of various sizes, including cost of centering, shuttering, reinforcement of 8 mm dia TMT bars of Fe 500D grade @ maximum 100 mm c/c on both ways, neat cement punning on finished surface,, including properly encased on all edges with 1.6 mm thick, 100 mm wide MS sheet duly painted over priming coat, reinforcement to be welded at edges with MS sheet etc including cost of cartage, all leads & lift, handling at site etc. all complete as per direction of Engineer-in-Charge.

LIST OF BUREAU OF INDIAN STANDARDS (BIS) CODES FOR SANITARY INSTALLATIONS

S. No.	BIS. No.	Subject
1.	IS 771 (Pt.1)	Specification for glazed fire clay sanitary appliances: Part 1: General requirements.
2.	IS 771 (Pt.-2)	Specification for glazed fire clay sanitary appliances: Part 2: Specific requirements of kitchen and laboratory sink.
3.	IS 772	Specific action for general requirements for enameled cast iron sanitary appliances.
4.	IS 774	Flushing cisterns for water closets and urinals (Other than plastic cistern)-Specifications.
5.	IS 1300	Phenolic moulding materials.- Specifications
6.	IS 13983	Providing and fixing stainless steel sink
7.	IS 1570 (Part-5)	Schedule for Wrought steel (Stainless Steel and heat resisting steels)
8.	IS 1703	Water fittings- copper alloy float valves (horizontal plunger type) - Specification.
9.	IS 1729	Cast iron /Ductile Iron Drainage Pipes and pipe fittings for Over ground non-pressure pipe line Socket and Spigot Series.
10.	IS 1795	Specification for pillar taps for water supply purposes.
11.	IS 1865	Iron casting with spheroidal or nodular graphite
12.	IS 2267	Polystyrene moulding and extrusion materials - Specifications
13.	IS 2326	Specification for Automatic Flushing Cisterns for Urinals (Other than plastic cisterns)
14.	IS 2548 (Part-1)	Plastic seats and covers for water closets Part 1: Thermo set seats and covers - Specifications
15.	IS 2548 (Part-2)	Plastic seats and covers for water closets Part 2: Thermoplastic seats and covers.- Specifications
16.	IS 2556	Vitreous sanitary appliances (vitreous china) -Specifications
17.	IS 2556 (Part-1)	Part-1: General requirements.
18.	IS 2556 (Part-2)	Part-2: Specific requirements of wash-down water closets.
19.	IS 2556 (Part-3)	Part-3: Specific squatting pans.
20.	IS 2556 (Part-4)	Part-4: Specific requirements of wash basins.
21.	IS 2556 (Part-5)	Part-5: Specific requirements of laboratory sinks.
22.	IS 2556 (Part-6)	Part-6: Specific requirements of Urinals & Partition plates
23.	IS 2556 (Part-7)	Part-7: Specific requirements of accessories for sanitary appliances
24.	IS 2556 (Part -14)	Part-14: Specific requirements of integrated squatting pans.
25.	IS 2556 (Part -15)	Part-15: Specific requirements of universal water closets.
26.	IS 2556 (Part -16)	Part-16: Specific requirements of wash down wall mounted

		water closets
27.	IS 2963	Specification for Copper alloy waste fittings for wash basins and sinks.
28.	IS 3076	Specification for low density polyethylene pipes for potable water supplies.
29.	IS 3389	Urea formaldehyde moulding materials. - Specifications
30.	IS 3989	Specification for centrifugally cast (spun) iron spigot and socket soil, waste and ventilating pipes fittings and accessories.
31.	IS 4827	Specification for electroplated coating of nickel and chromium on copper and copper alloys.
32.	IS 4984	Specification for high density polyethylene pipes for potable water supplies.
33.	IS 4985	Unplasticised P.V.C. pipes for potable water supply - Specifications.
34.	IS 7231	Plastic flushing cisterns for water closets and urinals - Specifications.
35.	IS 13983	Stainless steel sinks for domestic purposes -Specifications.
36.	IS 15905	Centrifugally Cast Iron Hubless pipes & fittings

LIST OF BUREAU OF INDIAN STANDARDS (BIS) CODES FOR WATER SUPPLY

S. No.	BIS. No.	Subject
1	IS 554	Pipe threads where pressure tight joints are required on the threads- Dimensions, tolerances and designation.
2	IS 778	Specification for copper alloy gate, and check valves for water works purposes
3	IS 779	Water meters (domestic type) -Specification
4	IS 14846	Specification for sluice valves for water works purposes (50 to 1200 mm size)
5	IS 781	Specification for cast copper alloy screw down bib taps and stop valves for water services
6	IS 782	Specification for caulking lead
7	IS 909	Underground fire hydrant, sluice valve type-Specification
8	IS 1239 (Part 1)	Steel tubes tubular and other wrought steel fittings, Part 1- Steel tubes- Specification
9	IS 1239 (Part 2)	Specification for mild steel tubes tubular and other wrought steel fittings, Part 2-Mild street tubular and other wrought steel pipe fittings
10	IS 1536	Centrifugally cast (spun) iron pressure pipes for water gas and sewage- Specification
11	IS 1537	Specification for vertically cast iron pressure pipes for water, gas and sewage
12	IS 1538	Cast iron fittings for pressure pipes for water, gas and sewage - Specification
13	IS 1703	Water fittings - copper alloy float valves (horizontal plunger type) - Specification
14	IS 2692	Ferrules for water services- Specification
15	IS 3950	Specification for surface boxes for sluice valves
16	IS 4736	Specification for Hot-dip Zinc Coatings on mild steel tubes
17	IS 5312 (Part 1)	Swing type reflex (non return) valves for water works purposes. Part 1- Single door pattern
18	IS 5312 (Part 2)	Swing type reflex (non return) valves for water works purposes. Part 2- Multi door pattern
19	IS 5382	Rubber sealing rings for gas mains, water mains and sewers
20	IS 9762	Specification for polyethylene floats (spherical) for float valves

21	IS 9763	Plastic Bib taps and stop valves (rising spindle) for cold water services- specifications
22	IS 15450	PE-AL-PE Pipes for hot and cold water supplies-Specifications
23	IS 15778	Chlorinated Polyvinyl Chloride (CPVC) pipes for potable hot and cold water distribution supplies-specifications.
24	IS 15801	Polypropylene- Random Copolymer Pipes for hot and cold water supplies-Specifications
25	IS 2685	Code for selection/Installation and Maintenance of Sluice Valve

Summary of Technical Specification for Landscaping & Horticulture Works

HORTICULTURE & LANDSCAPING WORKS AS PER DRAWINGS/DSR'23, CPWD SPECIFICATIONS 2019, HORTICULTURE & LANDSCAPING - 2020 OF DSR (HSOR 2020) & DIRECTIONS OF THE ENGINEER IN CHARGE			
Sl. no.	Description	Areas/ Locations/Horticulture & Landscaping Schedule of Rates 2020 & DSR 2023 Item No.	
1	EARTHWORK, SLUDGE, MANURE, GRASSING, DOOB GRASS TURF, ANTITERMITE TREATMENT.	Area/Location as per drawing	Work includes all items of Horticulture & Landscaping DSR of CPWD as contemplated in HSOR 2020 as may be applicable to the work. Reference may be made to HSOR 2020 item no. – 2.1, 2.2, 2.3, 2.4.1, 2.4.2, 2.4.3, 2.5, 2.7, 2.8, 2.9, 2.10, 2.10.2, 2.14, 2.14.1, 2.14.2, 2.14.3, 2.14.4, 2.16, 2.34, 2.56 and DSR 2023 item no. – 2.6.1 It is to be noted that supply of Good Earth or disposal of excavated earth is for all leads and lifts.
2	Trees, Palm, Specimen Plants, Shurbs, Ground Covers (Supply, stacking & Installation)	Area/Location as per drawing	Work includes all items of HSOR as contemplated in HSOR 2020 as may be applicable to the work like Supply of Trees, stacking, installation etc. Reference may be made to HSOR item no. – 2.57.1, 2.57.2, 2.57.3, 7.75, 7.5, 7.58, 7.88, 7.53, 7.50, 7.55, 8.36, 6.4, 6.12, 8.4.

PART-C

TECHNICAL SPECIFICATIONS

(E&M Works)

1. General

Design, Planning, Supply, Installation, Testing and Commissioning of Electrical & Mechanical Works/Services shall be carried out in accordance with CPWD Specifications, CEA, Govt. Building Bylaws, NBC 2016, ECBC 2017, Central Pollution Control Board, relevant IS Codes, International standards as applicable, Indian Electricity Rules and Acts as amended up to date.

The requirement of the energy efficiency measures shall be complied as per GRIHA 3-star rating and ECBC-2017 as amended up to date. Clearances from local authorities as per norms shall be with the scope of work of EPC contractor.

Electrical & Mechanical work shall be carried out as per following CPWD General Specifications as amended up to date.

1.	Part-I- Internal- 2023
2.	Part-II- External -2023
3.	Part-III-Lift & Escalators -2003
4.	Part-IV- Sub Station –2013
5.	Part V- Wet Riser & Sprinkler Systems –2020
6.	Part VI- Fire Detection and Alarm System –2018
7.	Part VII- D.G. Sets - 2013
8.	Part VIII- Gas Based Fire Extinguishing System 2013
9.	General Specifications for Heating, Ventilation & Air-Conditioning (HVAC)-2024

List of Applicable IS/IEC CODE		
Sr. No.	IS No.	Title
List of IS Codes of Electrical Works		
1)	IEC-61439, Part-1 & 2	Low-voltage switchgear and control gear assemblies
2)	IEC-62271 (Latest)	High-voltage switchgear and control gear assemblies
	IS:2147, 1966	Degree of protection
3)	IS: 13947 (Part 4, Sec. I), 1993 BS: 60947-4-1, 1992: IEC:158	Contactor for voltage not exceeding 1000V AC
4)	IS:375, 1993	Marking and arrangement of bus bars
5)	IS:694, 1990 & IS:8130,1984	PVC Insulated cables and aluminum conductor

6)	IS:1248,1991	Directing electrical indicating instruments
7)	IS:13118 (All parts), 1991	Alternating current circuit breakers
8)	IS:2705 (Part 1 to 4), 1992	Current transformers
9)	IS:3156 (Part 1 to 3), 1992	Voltage transformers
10)	IS-732:	Code of practice for electrical wiring installation system voltage not exceeding 650V.
11)	IS-3043:	Earthing.
12)	IS-2309:	Code of practice for the protection of buildings and allied structure against Lightning
13)	IEC61439 (Part 1 to 6)	Low -voltage switchgear and control gear assemblies
14)	IS-3716:	Insulation co-ordination application guide.
15)	IS-8130:	Conductors for insulated electrical cables and flexible cords.
16)	IS-5831:	PVC insulation and sheath of electric cables.
17)	IS-3975:	Mild steel wire, strips & tapes for armouring cable.
18)	IS-3961:	Current rating of cables
19)	IS-694:	PVC insulated (heavy duty) electric cables for working. Voltage up to and including 1100 volts.
20)	IS-424- 1475 (F-3):	Power cable flexibility test.
21)	IEC-439/IS- 7098: Part 1	Cross-linked polyethylene insulated pvc sheathed cables for working voltages up to and including 1100 volts
22)	IEC-439/IS- 7098: Part 2	Cross-linked polyethylene insulated thermoplastic sheathed cables for working voltages from 3.3 kv up to and including 33 kv
23)	IEC-439/IS- 7098: Part 3	Cross-linked polyethylene insulated thermoplastic sheathed cables specification part 3 for working voltages from 66 kv up to and including 220 kv
24)	IS-1554:	PVC insulated cables up to 1100 volts.
25)	IS-10810:	Test procedures for cables.
26)	IS-6121:	Cable glands.
27)	IS-10418:	Cable drums.

28)	IEC-754(1):	FRLS PVC insulated cable.
29)	ASTM-D-2863:	Standard method for measuring minimum oxygen concentration to support candle-like combustion of plastic (oxygen index).
30)	ASTM-D-2843:	Standard test method for measuring the density of smoke from burning or decomposition.
31)	ASTM E- 662/IEC 754(A)	Standard test method for specific optical density of smoke generated by solid materials.
32)	IEEE-383:	Standard for type test class-IE, electric cables, field splicers and connections for power generation station.
33)	IS 13947/IEC 947:	Air circuit breaker/moulded case circuit breaker.
34)	IS-8623:	Specification for factory-built assemblies of switch gear and control gear for voltage up to and including 1000vac/1200vdc
35)	IS 1018:	Switchgear and control gear selection/installation and maintenance
36)	IS-1248:	Direct acting indicating analogue electrical measuring instruments and testing accessories.
37)	IS-13779:	Digital measuring instruments and testing accessories.
38)	IS-3156:	Voltage transformer
39)	IS-2705:	Current transformer for metering and protection with classification burden and insulation.
40)	IS -2147:	Degree of protection provided by enclosures for low voltage.
41)	PART I, II, III	Switchgear and control gear
42)	IS-3427:	Metal enclosed switchgear and control gear
43)	BS-162:	Safety clearance
44)	IS-3202:	Code of practice for climate proofing of electrical equipment.
45)	IS-375:	Marking and arrangement for switchgear, bus bars, main connections, and auxiliary wiring.
46)	IS-722:	Ac electric meters
47)	IS-3231 / IEC- 255:	Electrical relays for power system protection.
48)	IS-5082:	Electrolytic copper/aluminium bus bars

49)	IS-2834:	Capacitors
50)	IS-2713:	Steel tubular pole
51)	IS-335:	Specification for insulating oil
52)	IS-3837: (PART I, II, III)	Specifications for accessories for rigid steel conduit for electrical wiring. GI/STEEL /PVC conduit pipe for electrical wiring.
53)	IS-2274:	Code of practice for electrical wiring installation system voltages exceeding 650 volts.
54)	IS-6665:	Code of practice for industrial lighting
55)	IS-3646:	Interior insulation part 1&2
56)	IS-1944:	Code of practice for lighting of public through fares.
57)	IS-7752:	Guide for improvement of power factor consumer's installation.
58)	IS-13346:	General requirement for electrical for explosive gas atmosphere.
59)	IS-13408:	Code of practice for the selection, installation and maintenance of electrical apparatus for use in potentially explosive atmospheres
60)	IS-12360:	Voltage and frequency for ac transmission & distribution system.
61)	IS-5572:	Classification of hazardous area for electrical installations.
62)	IS-5571:	Guide for selection of electrical equipment for hazardous area.
63)	IS-4201:	Application guide for Current Transformer
64)	IS-4146:	Application guide for Voltage Transformer
65)	IS-10028:	Code of practice for installation and maintenance of transformer
66)	IS-8478:	Application guide for on load tap changer
67)	IS-10561:	Application guide for power transformer
68)	IS-1646:	Code of practice for fire safety of buildings electrical installation

69)	IS-3034:	Code of practice for fire safety of industrial building- electrical generating and distribution station
70)	IP-30:	National electrical code (NEC) BIS publication.
71)	IS-4722:	Rotating electrical machines.
72)	IS-4889:	Method of determination of efficiency of rotating electrical machines.
73)	IS-325:	Three phase induction motors.
74)	IS-4729:	Measurement and evaluation of vibration of rotating electrical machines.
75)	IS-900:	Installation and maintenance of induction motors.
76)	IS-4029:	Air break switches.
77)	IS-2208-9224:	HRC cartridge fuses.
78)	IS-2959:	Contactors.
79)	IS-9537:	Rigid steel conduit.
80)	IS-1030-1982:	Specification for carbon steel castings for general engineering purpose.
81)	IS 2026 - Part I to V -	power transformers
82)	IS 335	Transformer oil.
83)	IS 10028 (Part II & III)	Installation and Maintenance of Transformers
84)	IS 2099	Bushings.
85)	IS 6600	Guide for loading of oil immersed transformers
List of IS Codes of Fire Fighting and Fire Alarm System		
86)	IS-8757	Glossary of terms associated with Fire safety
87)	IS-884	Specification for first-aid hose reel for firefighting
88)	IS-901	Specification for couplings, double male and double female instantaneous pattern for firefighting,

89)	IS-902	Specification (for suction hose couplings for firefighting Purposes (third revision)
90)	IS-903	Specification for fire hose delivery couplings, branch pipe, nozzles and nozzle spanner. (fourth revision)
91)	IS-904	Specification for two and three- way suction collecting heads for firefighting purposes.
92)	IS-907	Specification for suction strainers, cylindrical type For firefighting purpose. (second revision)
93)	IS-908	Specification for fire hydrant, stand post type (Second revision)
94)	IS-909	Specification for underground fire hydrant. Sluice valve type
95)	IS-636	Specification for non-percolating flexible firefighting delivery hose.
96)	IS-7637	Glossary of terms for firefighting equipment.
97)	IS-937	Specification for washers for water fittings for firefighting purposes.
98)	IS-1641	Code of practice for fire safety of buildings (general): General Principles for fire grading and classification.
99)	IS-1642	Code of practice for fire safety of buildings (General): Details of construction. (second revision)
100)	IS-1643	Code of practice for fire safety of buildings (general) ' Exposure hazard. (first revision)
101)	IS-1644	Code of practice for fire safety of buildings (general): Exit requirements and personal hazard. (second revision)
102)	IS-1646	Code of practice for fire safety of buildings(general): Electrical installations (Third revision)
103)	IS-2871	Specification for branch pipe, universal for Firefighting purposes.
104)	IS-2930	Functional requirements for hose Laying tender for fire brigade use.
105)	IS-5290	Specification for landing valves.
106)	IS-8090	Specification for couplings, branch pipe, nozzle, used in hose reel tubing for firefighting.
107)	IS-8442	Specification for stand post type water and foam monitor for firefighting (first revision)
108)	IS-9972	Specification for automatic sprinkler heads. (first revision)
109)	IS-11101	Specification for extended branch pipe for fire brigade use
110)	IS -12349	Fire protection Safety sign.
111)	IS -12407	Graphic symbols for fire protection plan.
112)	IS -9668	Code of practice for provision and maintenance of water supplies and firefighting.

113)	IS -3844	Code of practice for installation and maintenance of internal fire hydrants and hose reel on premises.
114)	IS -12585	Specification for thermoplastic house (Textile Reinforced) for Water General purpose.
115)	IS -10221	Code of practice for coating and wrapping of underground mild steel pipe lines. (first revision)
116)	IS-15105	Design and installation of fixed automatic sprinkler Fire extinguisher system-Code of Practice.
117)	IS -325	Three phase induction motors.
118)	IS-1822	AC Motor starter for voltage not exceeding 1000 volts.
119)	IS -3624	Pressure and vacuum gauges.
120)	IS-1520	Horizontal centrifugal pumps for clear, cold, fresh water.
121)	IS-1239	Mild steel tubes, tubulars and other wrought steel fittings.
122)	IS -3589	Specification for Steel pipes for water, and sewage
123)	IS -6392	Steel pipe flanges.
124)	IS -778	Specification for copper alloy gate, globe and check valves and water works purpose (fourth revision)
125)	IS -2592	Recommendation for methods of measurement or fluid flow by means of orifice plates and nozzles.
126)	IS -732	Code practice for electrical wiring installations.
127)	IS 900	Code of practice for installation and maintenance of induction motors.
128)	IS -1248	Direct acting electrical indicating analogue electrical Measures and their accessories general requirement instruments.
130)	IS-2516	A. C. Circuit breakers for voltages not exceeding 1000 volts.
131)	IS -4047	Heavy duty air break switches and composite units of air break switches and fuses (for voltage not exceeding 1000 volts.
132)	IS -2208	HRC cartridge fuse links up to 650 volts.
133)	IS -1554 (Part I)	PVC insulated (heavy duty) electric cables for working voltage up to and including 1100 volts.
134)	IS:1536	Specification for Centrifugally Cast (Spun) Iron pressure pipes for water, gas & sewage (first revision)
135)	IS 1537	Specification for Vertically Cast Iron pressure pipes for water, gas & sewage (first revision)
136)	IS:1538	Specification for Cast Iron Fittings for pressure pipes for water, gas & sewage (third revision)

137)	IS:780	Sluice valve for water works purposes (50 to 300 mm size)
138)	IS:13095	Butterfly valves for general purpose.
139)	IS:13039	Code of practice for provision and maintenance of External Hydrant System (first revision)
Pumps		
140)	IS:1520	Horizontal centrifugal pumps for clear, cold and fresh water.
141)	BS:599	Methods of testing pumps.
142)	PTC:8	ASME Power Test Codes - Centrifugal Pumps
Motor		
143)	IS:325	Induction motors, three-phase
144)	IS:900	Induction motors, installation and maintenance
145)	IS:7816	Guide for testing insulation resistance of rotating machines.
146)	IS:4029	Guide for testing three phase induction motors.
147)	IS:3043	Code of practice for earthing.

SPECIFICATIONS FOR E& M WORKS

- 1) All Electrical & Mechanical works shall be executed as per technical specifications provided in the tender documents, CPWD Specifications, National Building Code 2016, and ECBC code 2017 & 2024. In case of difference, in case of any discrepancy between two specifications, the more stringent or higher standard shall prevail and be followed. "In the absence of applicable CPWD Specifications, relevant IS Codes, local by-laws, local authority rules (as amended up to date), the provisions of the National Electrical Code (2011), and the National Building Code (2016), sound engineering practices shall be followed. The order of precedence for these references shall be as defined in the Notice Inviting Tender (NIT)."
- 2) The requirement of the energy efficiency measures shall be complied as per the provisions of local body and the provisions as per minimum three-star GRIHA rating and ECBC 2017, 2024 and any other latest amendments.
- 3) The scale of amenities and the provisions for different services shall be as per the client's requirement/ approved yardstick if any / local body /statutory requirements as applicable.
- 4) Clearances from local authorities for CEA, Lift inspector, Fire, pollution control body etc. shall be in the scope of EPC contractor.
- 5) Variations in the scope or quantity of incidental works, resulting from revisions to working drawings or as necessitated for the proper execution and completion of the primary works, shall not entitle the Contractor to any rate adjustments or additional payments. Such incidental works shall be deemed to be included in the contract scope and executed as per the instructions of the Engineer-in-Charge.
- 6) Unless expressly excluded in the contract documents, the quoted price shall be deemed to include all costs necessary for the complete execution, testing, commissioning, and full functionality of electrical and mechanical services for the BQ Housing Complex.
- 7) All costs associated with providing and completing electrical and mechanical services to the BQ Housing Complex, ensuring full functionality in all aspects, unless otherwise expressly stated in the contract documents, are included in the tender cost for this work.
9. The scope of works shall comprise all Civil, Electrical, Mechanical, and Safety-related components, along with any other works necessary for the construction, completion, and commissioning of the project. The EPC Contractor shall ensure that the Works are executed in their entirety to the satisfaction of the Engineer-in-Charge.
10. The minimum quantities and activities required for completion of the project and associated Electrical & Mechanical (E&M) works are outlined in the following pages. However, if additional quantities or activities are required to meet applicable standards, codes, or ensure the full functionality of the system, the contractor shall execute the same without any extra cost.

11. All major electrical equipment such as HT panel, transformers, LT Panels, Emergency panel, AMF Panel HT/LT breakers, DG sets, UPS systems, Elevators (Lifts) and other critical components shall be subjected to routine and type testing at the manufacturer's works, in accordance with relevant IS/IEC standards and approved QAP (Quality Assurance Plan). The Contractor shall coordinate with the manufacturer and provide 15 days prior notice to the Engineer-in-Charge to witness the testing. The cost of such testing, including arrangements for travel, boarding, lodging, and local transportation of representatives of the Employer/Engineer-in-Charge for witnessing the tests, shall be borne by the EPC Contractor and deemed included in the quoted price. No extra payment shall be admissible on this account. Only equipment that has successfully passed all required tests and obtained the Engineer's approval shall be dispatched to site. The test reports, certificates, and inspection documents shall be submitted to the Engineer-in-charge/ Employer prior to shipment.

12. The general guidelines and specifications to be followed for the execution of all Electrical and Mechanical (E&M) works are outlined below:

Sr.No.	Description	Brief Specifications
1.	Internal Electrical Installations	As per CPWD General Specification for Electrical Works Part I (Internal) -2023 amended up to date.
1.1	Conduits & Cable Trays	The work will be carried out in recessed/surface PVC/ MS conduit wiring system with modular socket in accordance with CPWD General Specifications for Electrical Works Part-I(Internal)-2023 amended up to date; Hot Dip Galvanized GI Cable trays (perforated or ladder type) for cable laying. Data and communication cables shall be laid in the Recessed/surface MS/PVC conduit or laid in the metallic raceways/ channels with SS 304 cover and matching junction boxes. Any conduit planned for installation over surface shall be MS or as per fire norms.
1.2	Wires & Cables	FRLSH PVC insulated Copper conductor cables will be used for points, circuit & sub-main wiring; Power & light point wiring shall be with copper conductor of size as per CPWD specifications & relevant BIS standards; RG-6 Cable TV wiring; Cat-6A Cabling/ telephone wiring as Required for EPABX/Data networking / CCTV requirements etc.
1.3	Switches & Sockets	All switches, sockets, stepped type electronic fan regulators, bell push, data & communication outlets etc. and accessories along with matching mounting boxes shall be of modular type.

1.4	MCB DBs & LT Panels	Meter Boards & Main Distribution Boards as per CPWD specification/ requirement; Size of distribution board shall be as per number of light / power circuits. All distribution boards shall be MS powder coated MCB based, Single phase/ 3 phase and suitable incomer as per individual DB requirement, double door type with RCCB/RCBO of required capacity and MCCB/MCB of suitable rating as required as per the given tender specification. LT panels shall be as per specifications mentioned in the tender document, CPWD specifications & IS/IEC61439 with MCB/MCCB/ACB as required.																														
1.5	Light fixtures & Fittings	LED indoor light fixtures in all spaces, common areas on all floors, staircases, corridors, lobbies, service rooms, basement etc. shall be provided. For basement and wet areas, the degree of protection for the fittings shall be IP-65. The minimum illumination requirement shall be as per the NBC 2016 amended up to date. All LED indoor lighting fixtures should be confirmed to the following parameters <table border="1"> <thead> <tr> <th>Sr.No.</th><th>Description</th><th>Parameters</th></tr> </thead> <tbody> <tr> <td>1</td><td>CRI</td><td>> 80</td></tr> <tr> <td>2</td><td>THD</td><td>≤ 10%</td></tr> <tr> <td>3</td><td>System efficacy</td><td>Minimum 110 lm/W at temp 0 to +45°C and humidity 10% to 90%.</td></tr> <tr> <td>4</td><td>System life time (Burning hour with driver)</td><td>50000 hours @L70 as per LM 70 and testing as per LM 80.</td></tr> <tr> <td>5</td><td>Driver voltage</td><td>220-240V</td></tr> <tr> <td>6</td><td>Power factor</td><td>> 0.9</td></tr> <tr> <td>7</td><td>Efficiency</td><td>≥ 85%</td></tr> <tr> <td>8</td><td>Surge protection</td><td>(L-N) 1KV, (E-L/N) 2KV.</td></tr> <tr> <td>9</td><td>Colour Temperature</td><td>As per site requirement.</td></tr> </tbody> </table>	Sr.No.	Description	Parameters	1	CRI	> 80	2	THD	≤ 10%	3	System efficacy	Minimum 110 lm/W at temp 0 to +45°C and humidity 10% to 90%.	4	System life time (Burning hour with driver)	50000 hours @L70 as per LM 70 and testing as per LM 80.	5	Driver voltage	220-240V	6	Power factor	> 0.9	7	Efficiency	≥ 85%	8	Surge protection	(L-N) 1KV, (E-L/N) 2KV.	9	Colour Temperature	As per site requirement.
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1.6	Fans/Exhaust fans	1200/1400 mm 5-star Ceiling fans [BLDC] BEE 5star as per the given specification & exhaust fans/ fresh air fans in residential units, guest house, community center, services rooms, bathrooms/toilets and in common areas or as per NIT drawings shall be provided as per CPWD specifications and related BIS.																														
1.7	Earthing	Copper/GI plate earth pits and complete earthing system as required as per CPWD specifications.																														

1.8	Lightning Arrestor	Lightning Protection System as per IS/IEC 62305 standard & National Building Code (NBC 2016).																		
2	External Lighting & Power distribution	As per CPWD General Specification for Electrical Works Part-II (External) -2023 and relevant BIS, amended up to date. All external lighting shall be controlled with astronomical timer/photo sensor-based switching.																		
2.1	Cables	XLPE armoured cables copper/aluminium between Main LT panel & Essential panel to floor distribution panel, rising mains, outdoor feeder pillars, other building distribution panels, distribution boards, STP panel, firefighting panel, plumbing panel, termination box of pole etc. and wiring of suitable size between termination box to fitting i/c their termination. The power cabling should be sized to ensure that the distribution losses do not exceed 3% of total power usage.																		
2.2	Feeder Pillars	Outdoor type Feeder Pillars shall be suitable for 3 phase, 50Hz, 415volts, A.C. system. Rating and size of Feeder pillar shall be as per designed load and requirement. The feeder pillars shall be as per CPWD specifications with MCB/MCCB/ACB as required. To be provided with astronomical timer/ photo sensor-based switching for outdoor and street lighting, to save Energy.																		
2.3	Fittings	LED outdoor light fixtures (IP-66/67) of suitable lumen output/wattage for compound lighting, street lighting, landscape lighting, facade lighting, security light fittings at perimeter on boundary wall etc. as required. The minimum illumination requirement shall be as per the NBC 2016 amended up to date. All LED Outdoor lighting fixtures should be confirmed to the following parameters																		
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		8	Surge protection	(L-N) 1KV, (E-L/N) 2KV.	
		9	Colour Temperature	As per site requirement.	
		10.	Degree of protection of the fittings	Minimum IP 66/IP67.	
2.4	Street Light Poles	Octagonal GI poles with sliding door/maintenance window, Single/double bracket arm along the road as per IS standards. Ornamental/Decorative poles and bollards with LED fittings may also be provided as per site requirements.			
2.5	Earthing	Copper/GI plate earth pits and complete earthing system as required as Per CPWD specifications.			
3	Electrical Sub-station	As per CPWD General specifications for Electrical works Part IV Substation-2013 and relevant BIS amended up to date covering 11KV/0.433KV substation equipment comprising HT Panel (2 I/C(1250 Amp) +6O/G(800 Amp)+1B/C) with 30min power backup for control circuit, 2 Nos. 2000 KVA Dry type transformers(1W+1Stand by) with OLTC etc. as required, HT cables, sandwich bus trunking/ suitable cabling from transformer to LT panel etc as required, Main LT panel & Essential panel (TTA as per IEC 61439), other Electrical distribution panels(IS/IEC 61439), automatic power factor correction panel if required, active harmonic filters,TVSS (transient voltage suppression system), Essential panels, earthing, required interconnections, substation safety equipments, Wooden framed SLD & instruction chart as per required size, HT/LT Rubber mat, first aid box, danger plates, sign boards, hand gloves etc as required as per CEA & Indian electricity rules as amended up to date.LT cabling from substation to the buildings panels & rising mains etc as required.			
3.1	HT Panel	HT Panel board with 30 min power pack: (2I/C (1250 Amp) +6 O/G (800 Amp) +1B/C (1250Amp)) and other accessories and safeties. HT incoming and out goings with VCBs, metering (in all panels and suitable for net metering), copper bus bar and protection equipments/accessories and compatible for SCADA/BMS as required.			
3.2	Transformer	2*2000KVA, Dry type Transformer, complete with ON load tap changer and other accessories and safeties, cable end box etc. as per relevant IS and having Maximum losses as per table of ECBC 2017 code (as per ECBC building/required GRIHA rating or			

		better) with latest amendments if any. The entire work shall comply with the latest CPWD specifications for substation and electrical works.
3.3	LT Panel	Main LT panel (TTA, IEC 61439) (normal supply) for receiving supply from Transformers and having ACB's/MCCBs as outgoing, Main LT panel (Essential supply) for receiving supply from Main LT panel and DG sets and ACB's/MCCBs as outgoing; Main Sub LT Panel for supplying power to smaller loads having MCCB's as outgoing. The incomers rating of LT breakers of main panel shall match with transformer rating or next available standard ratings of ACB with requisite breaking capacity of for all the incoming and outgoing switch gears as per applicable fault level.
3.4	APFC Panels	APFC panel if required shall be provided on separate sections of main LT panel and having switching off arrangement in case of DG supply in the main panel; Designed capacity to maintain overall P.F as applicable for ECBC building category or better (presently 0.97 or better).
3.5	Bus trunking	All Bus trucking and rising mains shall be compact Sandwich type, of required breaking capacity, complete with joints, fire barriers, tap off boxes, mounting arrangement etc. and of required degree of protection. Bus trunking shall be used for the connections between Transformers to Main LT Panel and rising mains shall be used in residential blocks & guest house building including all accessories etc. or as per specific site requirements.
3.6	Cables	HT Panel and transformers will be connected through three core 11KV HT XLPE(E) of requisite capacity based on the current carrying and breaking (fault level) capacity, HT cable end terminations shall be Heat shrinkable type; These cables shall have individually screened cores and be manufactured and tested according to IS:7098(Part-II)-1973 amended up to date, The Power cabling should be sized to ensure that the distribution losses do not exceed 3% of total power usage.
3.7	Earthing	Earthing (Copper/GI plate) sets, connections etc. shall be as CPWD specifications and relevant BIS, NBC 2016 etc.

4	Diesel Generating Set	380 KVA salient type DG Set as per CPWD General Specification for Electrical Works Part-VII, DG Sets-2013 as amended up to date, relevant code, CPCB IV+ norms and local body norms covering DG sets, AMF panel, DG set Acoustic enclosure ventilation/ smoke exhaust as required, earthing of DG set system, control cabling, fuel tank/ piping, DG set exhaust piping with MS structure, stack/exhaust chimney as per CPCB norms, civil works connected with DG set including civil foundation as required. DG sets shall be silent type, minimum rating/capacity specified is nominal capacity at standard test conditions. The DG set capacity shall be suitable for the load covering common area lighting, elevators (lifts), water supply pumps, sewerage and drainage pumps, firefighting essentials, and other essential power supply requirements as per the site requirements.
4.1	Engine	The engine shall be capable of driving the alternator continuously capable of taking 10% over loading for one hour after 12 hours of continuous operation with full load under the prevailing operating conditions.
4.2	Alternator	Generator Sets shall be 415V, 3Phase, 4wire, 50Hz, 0.8PF with acoustic enclosure.
4.3	AMF Panel	Generator sets shall start automatically in the event of a power failure and shall transfer power to emergency/critical loads automatically and shall switch off after some time delay in case of restoration of power supply.
4.4	Exhaust Pipe	As per CPWD General specifications for Electrical works Part VII DG sets-2013, relevant codes and CPCB/local body norms.
4.5	Cables	Complete Electrical and control wiring for various accessories etc.
4.6	Earthing	Earthing (copper plate) sets as per CPWD Specification and relevant BIS.
4.7	Day oil tank	990 Ltr. Capacity.
5	Fire Fighting and wet riser System	Firefighting system including wet risers, down comer, internal hydrants, first aid hose reels, valves & accessories, door & frame of FHC shafts including accessories, electrical and diesel engine driven pumps, terrace pumps, hose pipe and branch pipe with nozzle, air vessels, fire service connection, fire service inlet, control system with control components (like pressure switches, pressure transducers, air vessel, air release valve flow switches,

		level indicator, alarm etc. designed and provided as per provision given in NBC-2016, relevant BIS codes, local fire by laws & CPWD specifications part V-2020, all with up-to-date amendments. Pipe materials and other fixtures for firefighting, plumbing work shall be as per CPWD specifications including necessary approval, OC as required from respective local bodies / CFO etc. Note: The detail designing of the firefighting system shall be in the scope of the EPC contractor.
6	Fire Alarm System	As per CPWD general specification for Electrical works, Part-VI fire detection and alarm system-2018, NBC 2016, local fire Bye laws, IE rules, BIS/IEC, Indian standards amended up to date; covering Manual/Automatic fire alarm system complete with items like detectors, manual call point/boxes, cabling/wiring, fire panels, talkback system, RI, Repeater panels etc. as required including integration of FAS panel with smoke ventilation & pressurization fans, AHUs, Elevators (Lifts), sprinkler monitoring panel and as required for any other services.
6.1	Cables	Either copper conductor Armoured Fire survival cables of suitable size or Copper wire in heavy duty MS conduit of suitable size shall be used compliant to the relevant IS.
6.2	PA System	Public Address System with Audio Amplifiers, speakers & required wiring shall be provided to cover all the areas as per the requirement of CPWD specifications, NBC 2016, and local bye laws as amended up to date.
7	IP Based EPABX System	IP based EPABX system comprising of core switches & L2 switches with 10G, 10 giga SFP modules, industry standard appliance server, cloud based, enterprise-grade UC solution, MID/Entry level IP/SIP phone with dual 1 giga ports, racks, CAT-6A cable, patch panels, cable managers, stacking cables, patch cords, and OFC cables etc. complete as required.
8	CCTV Surveillance System	As per good Engineering practice, NIT specifications or as required under the codes or by local/statutory System authorities compliant to relevant Indian standard 13252. IP Based CCTV system for Residential blocks including guest house, community center building, building security comprising of PTZ, Dome & bullet camera, cabling, digital recording display system with minimum display of 5" X8" per camera and hardware & software support both for indoor and outdoor/ external surveillance IP based Closed circuit surveillance system

		is proposed to monitor the activities at critical/desired areas. The video shall be continuously transported to existing server, using IP cameras; The items supplied by the EPC contractor should have specifications compatible with each other required for satisfactory operation of the system. CCTV cameras and storage server/NVR shall have ONVIF Profile-S compliance for interoperability.
9	Mechanical Ventilation System	The system shall be provided as per NBC code of india-2016 and requirement of the local body, CPWD General specifications for Heating, Ventilation, and air conditioning work-2024 and relevant IS codes as amended up to date. The system shall be designed and provided to achieve rate of air change in various ventilated areas as prescribed in NBC-2016 and local building bye laws/site requirements.
9.1	Fans	All the fan motors which will operate in normal mode shall have IE-3 efficiency. The fan motors required to be operated during fire mode shall be minimum IE-2 efficiency.
9.2	Ducting	The staircase, elevator (lift) lobby & lift well pressurization shall be designed and provided as per provision given in NBC2016, CPWD specification, and local by law as amended up to date. Scope of works include GSS ducting, grills, louvers, flexible connection/canvas connection, MS supports and all associated works and shall be executed as per CPWD specifications. The pressurization system shall be integrated with fire alarm system for automatic operation on command from fire detection system.
10.	Elevators (Lifts)	The elevators (lifts) shall be with power operated center opening doors and AC variable voltage & variable frequency controls, Car safety & Governor, 3 side hand rails in car, Guides, guide rails, ropes, car platform, cabin finishes (stainless steel), car & landing doors (stainless steel), hoist motors, braking system, Automatic Rescue Device (ARD) etc. Complete as per the CPWD Specifications, NBC 2016 and related IS standards amended up to date. (The elevators (lifts) should be of machine room less type. Provision of elevators (lifts) will be made as per National Building Code-2016 / relevant BIS amended up to date) considering no. of floors, height of the building and speed of elevators; At least one elevator (lift) in each block be made compliant to barrier free and accessibility requirements as per CPWD specifications & capacity to be minimum 15 passengers for barrier free elevator (lift); One elevator (lift) in each tower

		should be 15 Passenger cum Bed/ stretcher lift, with size and capacity of lift car to be designed accordingly. Provision of fire elevators (lifts) shall also be ensured as per local fire authorities requirements/ bye laws. IP-based CCTV camera (1 No.) shall be provided in all Lift Cars of all buildings for surveillance/monitoring of elevator (lift) control panel. The CCTV cameras, installed in the elevators (lifts), shall be connected/ integrated with the IP based CCTV system with suitable cabling, hardware items etc. as required.
11	Hydro pneumatic water supply system. (Domestic & Flushing system)	Hydro pneumatic Water Supply System shall be provided in pump room complete with suitable capacity of two pump set, one working and one standby, pneumatic Water Supply tank, micro-processor-based control panel, VFD, interconnecting pipes, valves, System cabling and switchgears. Water supply pumping stations with auto control panels shall be provided for filling terrace water tanks of residential blocks, guest house & community center building etc. as per site requirement. It shall be designed and provided to meet water demand as per provisions given in NBC-2016. Each pump sets shall be with variable frequency drive and each pump set should have at least 3 nos. of pumps (2W+1S). All pump motors shall be of minimum IE3 efficiency rated (except dewatering pumps, where IE2 is acceptable). Each pump shall have an efficiency of more than 70% at selection point; Type, Pressure, Head and discharge of the pumps, sizes and type of valve shall be designed as per site requirement for proper service. Note: The detail designing of the water pumping system shall be in the scope of the EPC contractor.
12	Roof top Solar PV Power Plant	Roof top grid interactive solar photo voltaic plant system shall be inclusive of the PV Power Plant Solar panels of prescribed efficiency, mounting structure (GI), all cabling, inverter, panel, ACDB, DCDB, metering devices, earthing, lightning protection complete as per state-of-the-art technology as per good engineering practice compliant to relevant Indian standards/local/statutory authorities/MNRE guidelines.; The capacity of plant shall be derived as per available space on terrace of the buildings & minimum capacity of plant shall be 375 KWp. High efficiency Monocrystalline PV cells with high efficiency invertors shall be used. Minimum power output

		guarantee offered for the SPV Module shall not be less than rated for 25 years; Invertors / Power conditioning unit (PCU suitable for mains as well as DG supply) as required. However, capacity of each inverter should not be more than 30 KWp; The inverter output shall always follow the grid in terms of voltage and frequency; AC collection Panel with metering to monitor generation at the terrace and AC cabling from Panel to receiving switch; Measuring devices i/c net metering, indication devices, protection devices, switching devices, all necessary wiring, DC and AC cabling including feeding power to essential supply by providing necessary tap off boxes; Mounting structure of GI on terrace of the building, protection of system for external / internal causes and from all kind of surges etc.; Earthing and Lightning protection of Solar system etc. as required.
13	Boom Barrier	Electromechanical boom barrier with all accessories up to 6 meter length; consisting of a fixed housing and a movable arm; in two pieces without central support, electrically operated from security/ guard- room through push buttons; Barrier Housing Unit: Powder Coated; Boom: Powder Coated White RAL 9010m or approved color with Red reflective strips having IP 44 (Barrier Body) & IP 67 (Control Card Box); All Aluminium Housing with Base frame in SS-304 for high protection against corrosion; Operator console, necessary hardware, software, Power and control cabling etc. as required It should have anti-crush safety mechanism to suspend the motion of barrier when met with any obstruction; Quick opening and closing of barrier with soft landing; Speed of barrier should be programmable while installing as per frequency of traffic; Provision to integrate with other peripheral devices like Access Control System, Beam sensors; It should be activated by a single push button and with an option to operate wirelessly through remote. In addition, it must have provision to operate on electronic signals from Access Control System; Safety device like beam sensor should be provided with the barrier; Barrier should have clamped mechanism in open and closed positions; Control system (gear unit, crank drive, and control unit etc.) should be housed inside a stainless-steel pole stand casing; It should have distinct visible marking/ paint work. The barrier body should be treated with 100 micro epoxy zinc plating anti

		corrosion paint.
14	VRV/VRF AC system	VRF/VRV system shall be provided for community Centre building. Variable Refrigerant Flow (VRF) shall meet or exceed the efficiency requirements System specified as per minimum "ECBC Building" as per ECBC 2017 and the work shall be executed as per CPWD General Specifications for Air Conditioning Works-2024, relevant IE rules, relevant IS codes as amended up to date; It should consist of suitable nos. of outdoor and indoor units (capacity mentioned on the NIT drawings), refrigeration pipes and drain pipes, all necessary civil work for installing including 1st charging of refrigerant gas for proper and specified functioning of the VRV system, AHUs, Fan coil unit, TFA, cassette, ductable split etc., plumbing works, ducting, volume dampers, Thermal insulation for pipes and ducting, Controls, power cables, electrical panels complete as required. VRV/VRF system shall be communicable type seamlessly integrated. All indoor and outdoor units shall be BMS compatible. Ensure that all the insulation used in the building envelope and for HVAC systems; refrigerant used in the HVAC systems and refrigeration equipment are HFC free.
15	UPS System	The UPS system shall be online 3 Phase, 50Hz with 30 min backup or more including batteries, inter connecting cable, battery racks. Each UPS module must have the following built-in parts/features: IGBT PWM Rectifier based input charger (float cum equalizing). IGBT /technology-based Inverter. Automatic Bi directional Static switch; Inbuilt Manual Bypass Switch; K 13 Copper Wound Galvanic Isolation transformer at UPS Output after the static Galvanic inverter output isolation transformer (inbuilt). switch as per IEEE 1100-2005 in a separate cabinet (inbuilt); Fully Microprocessor Controlled Circuitry; Inverter with stipulated Switching Frequency with PWM controlled using DSP logic. Analog control shall not be acceptable; A UPS shall be tested against EMC (Electromagnetic compatibility) category C2; Inbuilt/External Back Feed protection shall be provided. Input phase reversal protection & Correction (The system should

		run in mains operation in spite of phase sequence reversal) & same should be shown is SLD. Provision for separate Input for rectifier and for Bypass. Event Monitoring & Diagnostics Last 100 events with exact date & time should be monitored from the front LCD panel of the UPS & up to 900 events from UPS system memory using Laptop. The input voltage window must be from 340 V to 460 Vat full load. The UPS system shall be continuous duty, highly reliable and solid state. The system shall be modular in design so that any individual unit can be easily isolated/taken out for repair or any additional added in future for increasing the capacity; The UPS shall be fully microprocessor-controlled till the level of rectifier/inverter. UPS system output must be independently brought out to the UPS and synchronized (voltage, phase angle and frequency must be equal). The provision of parallel operation and automatic bypass shall be provided as per the site specific requirements. The design manufacture, inspection, testing and installation of the UPS System covered under this specification shall conform to the latest international standards such as: EN 62040-1 - General Safety Regulations. EN 62040-2 - EMC Regulations.
15.1	Battery	Battery should be designed to provide 30 minutes back up or more as required. Battery should be 12V lead acid sealed maintenance free type. The UPS module shall have the Battery Circuit breaker mounted in the battery Rack. When this breaker is opened no battery voltage should be present in the UPS enclosure. The UPS module should be disconnected automatically when the battery reaches to the minimum discharge voltage level or when signalled by other control functions.
16	LAN System	High speed Data Networking system is to be provided. Work shall include all passive and active components excluding server; LAN System shall comprise of core switches & L2 switches with 10G, 10 giga SFP modules, network management software, racks, CAT-6A cable, patch panels, cable managers, stacking cables, patch cords, OFC etc. complete as required.

17.	Emergency light Illuminated signage	& Exit signs shall not exceed 5W per phase as per ECBC-2017; Self-contained rechargeable emergency exit light with minimum 6 hours battery backup; IP 20 rated; Confirms to IS 10322 (part-5/ sec1): 2012, Single side /Double side facia as per requirement of location installed. Emergency light & Illuminated signage shall be installed as per site requirements.
18.	Smart Energy meter	Each individual flat shall be equipped with a Smart Energy Meter incorporating a load limiting mechanism. Smart energy meter should have the following features: Digital energy metering with real-time monitoring capability. Remote reading and communication interface (e.g., RF, PLC, or GSM/GPRS) compatible with utility/distribution company requirements. Tamper detection and alert functionality. Prepaid or postpaid billing support, as applicable. The meter shall be capable of automatically disconnecting the power supply when the connected load exceeds the predefined limit. Load threshold values shall be programmable and remotely configurable through the utility backend or local interface. Auto-reconnection feature once the load falls within the permissible range or upon user reset. Installation & Commissioning The EPC Contractor shall supply, install, test, and commission the smart energy meters in all individual flats. All associated accessories, wiring, communication modules, and integration with the building's metering network shall be included in the scope of EPC contractor. The smart meters must comply with relevant BIS/IS/IEC standards.

1. Internal & External Electrical works:

The work shall be carried out as per CPWD General Specifications of Electrical Works Part-I (Internal) 2023, Part-II (External)-2023, relevant IE rules, NBC 2016, IS standard as amended up to date. It shall also be in conformity with the current Indian Electricity rules and regulations and requirements of the Local Electricity Supply Authority and Fire safety regulations, so far as these become applicable to the installation. In case for any part of the work specification is not available in the aforesaid mentioned documents then part of the work will be carried out in accordance with sound engineering practice and as per directions of Engineer-in-charge.

(A) SCOPE OF WORK:

The scope of work includes Planning, Design & preparation of Drawings, obtaining approvals from the Engineer-In-Charge before execution. , supply, Installation, testing and commissioning of Internal and External Electrical work such as Power wiring, Telephone/ TV conduiting, LED light Fittings, call bell, Exhaust fans, Ceiling fans, Switch boards, Modular switch & sockets, MCB's, MCCBs, MCB DBs, Rising Mains, lightning & protection requirement, earthing, aviation lights, hybrid solar street lighting, facade lights, bollards lights, Feeder pillars and associated electrical panels etc. for the buildings as per CPWD specifications as amended up to date as described in the details of specifications of this sub head. Complete wiring in all the buildings for light, power etc. shall be done in appropriate size MS (mild steel) conduits of minimum medium class conforming to relevant IS Code and for LV system (Data, telephone, TV, CCTV etc.) the PVC conduits of minimum medium class conforming to relevant IS Code shall be used.

(B) GENERAL REQUIREMENTS:

1. All the material to be used on this work by the EPC contractor shall be got approved from the Engineer-in-Charge in advance before procurement or installation at the site.
2. All the equipment shall be delivered with; Manufacturer's test certificate, Manufacturer's technical catalogues and Installation / Instruction (O&M) manuals.
3. The work shall be carried out according to approved drawing in stages as per instruction of the Engineer-in-Charge who will have the right to change the layout as per requirement at site and the contractor shall not have any claim due to change in layout.
4. All damages caused to the building during execution of electrical work shall be the responsibility of the EPC Contractor and shall be repaired immediately at his own cost to the

satisfaction of the Engineer-in-Charge.

5. The imperfect workmanship will not be accepted, and defects shall be rectified by the EPC contractor without any additional cost and up to the satisfaction of the Engineer-in-Charge. The program of electrical works is to be coordinated in accordance with the building work.
6. All debris from electrical works should be removed and the site should be cleaned immediately by the EPC Contractor after each day's accumulation of debris. Similarly, if there is any rejected material it must be immediately removed from the site by the contractor.
7. The size of MS/PVC conduit and wiring shall be got approved from the Engineer-in-Charge before the execution of work.
8. The EPC contractor shall make his own arrangement at his own cost for Electrical/ General tools and plants required for the work.
9. The contractor has to make his own arrangements for stores and watch and ward and no extra claim for this will be entertained.
10. The contractor shall make his own arrangements for electrical power supply for the construction activities. No extra payments for the same will be made.
11. The wiring and conduit routes shall be marked out in the first drawing by the contractor, and got the same approved by the Engineer-in-Charge.
12. The entire installation shall be at the risk and responsibility of the EPC contractor until these are testing & commissioning and handed over to the Oil India Ltd., Duliajan, Assam. The watch & ward shall be the responsibility of the contractor till handing over.
13. All items of interrelated works considered necessary to make the scope complete and operative are deemed to be included, shall be provided by the contractor at no extra cost.
14. The connection inter connection, earthing and inter earthing shall be done by the EPC contractor wherever required and nothing extra shall be paid on this account.
15. Nothing extra shall be paid for inter connections with thimbles/wires/tapes strips etc. Used on the work.
16. The EPC Contractor shall, on demand of the Engineer-in-Charge, produce evidence to the satisfaction of the Engineer-in-Charge regarding purchase of wiring, modular switches, and accessories, MCBs, MCCBs, DBs, fans and fixtures and accessories and other electrical items from the authorized outlets of the manufacturer.
17. All MS/ PVC conduits accessories shall be of the same make as conduits and shall be ISI

marked. The conduits shall be terminated as switch boxes/metallic junction boxes with suitable glands/check nuts.

18. All the sub main and circuits wiring including loose wire in loose wire box for connections inside switch boxes and MCB DBs. No payment for these loose wires & loose wire box shall be made.
19. To facilitate drawing of wires, 18 SWG GI fish wire shall be provided along with laying of recessed conduit for which no extra payment shall be made.
20. The connection between incoming switch/isolator and bus bar shall be made with suitable size of thimble and cable at no extra cost.
21. Copper conductor of insulated cables of size 1.5 Sq.mm and above shall be stranded and terminals provided with crimped lugs. All hardware items such as screws, thimbles, GI wire etc. which are essentially required for completing an item as per specification will be deemed to be included in the item even when the same have not been specifically mentioned.
22. All hardware items such as nuts/bolts/screws/washers etc. to be used in work shall be zinc/cadmium plated iron.
23. While laying conduit, suitable size junction boxes shall be provided for pulling the wire.
24. Materials to be used in work are to be ISI marked. The materials to be used in the work shall be got approved by the Engineer in Charge. The Engineer-in-Charge shall reserve the right to instruct the contractor to remove the material which, in his opinion, is not acceptable.
25. All the individual power points 5 Amps and 16 Amps shall have separate circuit wiring with minimum 2.5 sq mm & 4 sq mm wire respectively.
26. Switch for geyser point shall be provided outside the toilet on switch board & socket inside the toilet near geyser location both shall be of minimum 25 Amps rating with 6 sq. mm wiring.
27. The design, supply, installation, testing and commissioning of lightning protection of each building is in the scope of EPC contractor. The design shall be done as per NBC-2016, CPWD specifications as amended up to date. All components in a wiring installation shall be of appropriate ratings of voltage, current and frequency, as required at the respective sections of CPWD specification of the electrical installations in which they are used.
28. Minimum 1.5 sq mm PVC insulated FRLS copper conductor wire will be used for point wiring.
29. Minimum 2.5 sq mm PVC insulated FRLS copper conductor wire will be used for 6A switch & socket. and 4 sq. mm PVC insulated FRLS copper conductor wire will be used for 16-amp

switch & socket wiring and 6 sq mm PVC insulated FRLS copper conductor cable will be used for 25 A, AC socket wiring.

30. Submain wiring shall not be used below 10 sq mm PVC insulated FRLS copper conductor cable.
31. 300 mm sweep 900 RPM metallic exhaust fan with metallic louvers (and not wire mesh) shall be used in Kitchens.
32. 1200/ 1400 mm Sweep BLDC Ceiling fan 5-star rating with regulator/remote shall be used as per approved make.
33. One number 16 Amp Socket outlet with switch and One LED bulk head in elevator (lift) shaft at each floor/as per the requirement of elevator (lift) vendor shall be provided for maintenance purpose.
34. Size of distribution board shall be as per number of light / power circuits. All distribution boards shall be double door type. RCBO of suitable rating shall be provided as main incomer in all DBs. Loose wire box shall be provided as per requirement. Distribution board shall have minimum 20% space extra available for fixing MCBs / MCCBs for future expansion. All Distribution Boards (DB) shall be prewired as per direction of Engineer-in-charge.
35. Essential DBs shall be provided on each floor for common services and have separate sub – main wiring. The size of DBs at one location shall be same irrespective of number of circuits connected from the DBs. The wiring and conduit route plan/drawings shall be submitted by the EPC Contractor and shall be got approved from the Engineer-in-charge.
36. Stair case lighting shall be controlled in loop as well as through switches on each floor.
37. Socket outlets modular type shall be 6A 3 pin, 16 Amp 3 pin or 16/6 Amp 6 pin. 5 pin socket outlets will not be permitted.
38. Modular type switches & sockets and stepped type electronic fan regulators 2 module / sockets / telephone outlets / TV sockets are to be provided wherever indicated in the drawings. Modular cover plates of switches, sockets, telephone, and TV sockets etc. shall be in two parts and the switch boxes shall be of the same manufacturer.
39. Lighting & Fan Design should be carried out as per NBC 2016 /CPWD specification /ECBC guidelines. Wherever range of illumination for space is mentioned, higher side of Lux level must be taken for design purpose.
40. Ceiling fans & Exhaust fans shall be provided as per the locations mentioned in NIT drawings

and as per the drawings approved by Engineer-in-charge.

41. Cubicle type compartmentalized floor distribution panel for each floor of all blocks to be provided made of minimum 2.00 mm thick CRCA sheet duly powder coated with incoming switch gears, bus bar chamber, metering chambers, 6mm thick phenolic laminated sheet base for meters, cut outs, MCBs etc., outgoing DP/TPN MCBs suitable ratings for each quarter in separate compartment, earthing etc. as required will be provided by EPC Contractor. Marking shall be provided on each electrical circuit/feeder of the panel. The Panel shall have provision for two incoming supplies. The EPC contractor will also take approval of drawing of Floor distribution Panel from Engineer-in-charge before procurement.
42. Essential Service Panel for common services (for Common area lighting, elevators (lifts) Control Room, CCTV System, stilt floor lighting Fire Panels, External lighting, and spare feeders etc.) will be provided in each block of residential building, guest house and community center building by EPC Contractor. The Panel shall have provision for two incomers will be in the scope of the EPC Contractor. EPC contractor shall take approval of this panel from Engineer-in-charge before procurement.
43. Power supply for indoor VRV units in community center building shall be from separate double door TPN Distribution board at each floor and separate circuit shall be provided for the same (one suitable MCB for each unit).
44. All switches, sockets, Telephone socket, Data sockets, stepped type electronic fan regulators, bell push switch, hot dip galvanized switch box and accessories shall be of modular type. Make and model no. shall be approved by Engineer-in-charge.
45. There shall be separate shafts for Electrical works, LV System (CCTV, LAN, TV, Telephone, Fire alarm, PA etc.), and Firefighting System (piping & hydrant) etc. as required.
46. For all E&M works mentioned in the contract, the agency has to also comply with the provisions of NBC 2016, ECBC-2017 and more stringent provisions will be followed for design consideration.
47. The minimum scale of electrical inventory to be provided in each of the buildings as per NIT drawings. These are only indicative, minimum, and not exhaustive. It is the responsibility of the EPC Contractor to provide the required number of fixtures, fittings, power sockets and equipments to cater to the intended requirement of a particular building and as per direction of Engineer-in-charge.

48. Heavy duty Metallic exhaust fan of 900 RPM, double ball wearing with metallic louvers as per the requirement of ventilation shall be provided in Electrical Panel room, Electrical- Substation rooms, utility rooms etc. The qty of exhaust fan/ventilation fan shall be calculated on the basis of air change required as per NBC 2016 as amended up to date. However minimum size of exhaust fan shall not be less than 300 mm sweep.
49. Call bell system with image viewer shall be provided in each unit of residential blocks as per direction of Engineer-in-Charge.
50. Bulk-head fittings pressure die-cast housing with IP 65/IP67 protection surface mounted: Elevators (Lifts) shaft, entrance of pump room, and in terrace area outside Mumty etc. as required as per site shall be provided as per direction of Engineer-in-charge.
51. 16 Amp, 6 pin Power socket outlet with switch shall be provided at all floors with 4 sq mm FRLS copper wiring inside elevator (lift) shaft or as per requirement of elevators (lifts) manufacturer.
52. GI Metal trunking raceways 1.6mm thick (including raceways junction box and 3mm thick SS 304 cover plate for each junction box, couplers, jointing sleeve, floor fixing supports etc. complete as required.) for drawing LAN cables /power cables/telephone cables in guest house and community center building shall be provided by EPC contractor. Separate suitable size raceway shall be provided for power & LAN cables/telephone cables.
53. Rising mains shall be of sandwich type rising mains with Aluminium busbar for use on 3 phase 4 wire 415 V, 50Hz A.C. supply with GI/CRCA enclosure having IP-54 rating after fixing the tap off boxes and all accessories, made of 1.6mm thick steel sheet duly powder coated in convenient sections complete with 4 Nos Aluminium bus bars having current density of 130 A/ sq. cm at nominal current rating, necessary joints & expansion joints, fire barrier at each floor, provision of tapping at every meter, adopter box and copper flexible for joints, continuous earthing with 2 Nos. Aluminium strip of not less than 20 x 3 mm up to 400 Amp & 20 x 5 mm for higher capacity (one on each side) including, GI clamping brackets, angle iron bracket, steel fasteners, connecting to earthing system etc. as required including adopter box, End feed unit, tap- off box with suitable rating MCCBs. Rising mains shall be provided by EPC contractor for Each residential blocks and guest house building.
54. Rating of rising mains shall be decided as per maximum load of the buildings and 20% future expansion and approved by Engineer -In-Charge. Rising mains shall be conforming to IEC

- 61439 as amendment up to. Rating of rising mains in Residential block & Guest house shall be minimum 400 amps ($I_{sc}=25kA$ for 1 second) connected load and future expansion.
55. For feeding solar power to the rising mains, suitable rating tap-off box shall be provided at the top of each rising main by the EPC Contractor.
 56. The rupturing capacity of the MCB's shall not be less than 10kA, higher capacity shall be provided if required in the detailed design. The MCB's shall have ISI mark and rupturing capacity of the MCCB's shall be as per design of fault level, but shall not be less than 25 kA.
 57. Make of MCB/MCCB, DBs shall be approved by Engineer-In-charge.
 58. Earthing: All earthing for equipment (except Lighting Arrestor) shall be done with copper plate earthing unless otherwise specially mentioned. Copper plate earthing system comprising of earth electrode, earth conductor, earth bus, protective conductor etc. shall be as per provision laid down in CPWD specifications for electrical works part – 1, 2023 as amended up to date. Earthing system should be designed such as to maintain earth resistance as specified in CPWD specifications. Earth resistance shall be checked / tested in harsh climatic conditions.
 59. Lightning Arrestor: Lightning conductor shall be provided for buildings & Substation. The Lightning protection system shall be designed as per rolling sphere method as per NBC 2016 as amended up to date.
 60. All LT and HT cables to be laid in open area outside the building shall be laid either in RCC Hume pipes or in ground with sand cushioning and brick protection or in RCC trench or in DWC pipes as per CPWD specifications/ instructions of Engineer-in-Charge.
 61. Each building shall be provided with adequate number of aviation obstruction lights (LED type) having total lumen output not less than 3000 lumens and step-up transformers with suitable mounts, brackets, extension rods, hardware etc. The lights will be installed at the highest point of the building as per the requirements of NBC 2016 and electrical wiring for the same will also be taken care of.
 62. Required illumination level for general lighting shall be achieved as per CPWD General Specification for Electrical (Part-I)-2023, NBC-2016, ECBC-2017 and relevant IS standard as amended up to date. Wherever range of illumination for space is mentioned, higher side of Lux level shall be taken for design purpose and decision of Engineer-in-Charge shall be final in that regard.
 63. Cable trays above false ceiling on all other places shall be Perforated Hot Dipped

Galvanized Iron (galvanization thickness i.e. average mass of Zinc coating shall not be less than 65 microns for 2 mm thick & 50 microns for 1.6 mm thick as per IS standard) with perforation not more than 17.5%, in convenient sections, joined with connectors, suspended from the ceiling with G.I. suspenders including G.I. bolts & nuts, etc. as required. Suitable size as per site conditions shall be used with thickness 1.6 mm for cable trays with width ≤ 300 mm & thickness 2 mm for cable trays with width > 300 mm.

64. Wiring for cable TV with coaxial cable RG-6 grade, 0.7 mm solid copper conductor PE insulated with fine tinned copper braided & protected with PVC sheath recessed as required.
65. LT panels shall be as per CPWD specifications & IS/IEC 61439 with MCB/MCCB/ACB as required. The drawing of panel boards must be got approved from Engineer-in-charge before fabrication work. The panel board shall consist of MCCB/ACB as incomer and outgoing, copper bus bar, digital type ammeter, voltmeter or multifunction meter, selector switches, LED type indication lamps etc. as per standard sound engineering practice. Every multifunction meter should have RS485 port.
66. The building should be provided with DG Set power backup for common area, Corridor, stair case & lift lobby, Elevators (Lifts), STP, Water supply pumps, LV system, essential services, and security system etc. as required as per site and direction of Engineer-in-charge.
67. Each house shall be provided with independent 8-way TPN DB, double door type and MCB as per circuit requirement. The supply shall be connected from meter panel to DBs at different locations suitable size copper conductor wires/ cables.
68. After completing the work, necessary test results as envisaged in CPWD General Specifications Part-I (Internal)-2023 & Indian Electricity Rules 1956 as amended up to date shall be recorded and submitted to the department. And testing required to obtain CEA certificate shall be submitted by EPC contractor. Test report shall be duly signed by authorized person of EPC contractor.
69. All items of interrelated works considered necessary to make the scope complete and operative are deemed to be included, shall be provided by the contractor at no extra cost.
70. Facade lighting shall be provided at the Entry and Exit gates of the campus, as well as the Community Center and Guest House buildings. The lighting design shall enhance architectural aesthetics while ensuring adequate illumination and safety. Fixtures shall

include energy-efficient LED luminaires suitable for outdoor and facade applications, with appropriate ingress protection (minimum IP65), corrosion-resistant finish, and architectural-grade mounting hardware. Lighting shall be uniform, glare-free, and coordinated with the overall landscape and building elevations. All wiring, control gear, supports, junction boxes, and protective devices shall be included. The system shall also integrate with the central lighting control panel, wherever applicable.

71. Hydro-pneumatic pumping systems shall be provided for supplying water to the domestic and flushing water distribution networks. The system shall be designed to maintain constant pressure and ensure uninterrupted water supply across all connected outlets. The system shall include pressure vessels, booster pumps (with duty-standby arrangement), control panels with automatic pressure management, non-return valves, pressure switches, and all associated fittings and accessories as per approved design. The entire setup shall be complete in all respects and fully integrated with the building's plumbing services, including interfacing with the control panel, wherever applicable. The design and installation shall conform to relevant IS codes and best engineering practices.
72. All major electrical and mechanical equipment—including Transformers, Motors, Pumps, HVAC Indoor and Outdoor Units, Elevators (Lifts), and associated systems—shall comply with energy efficiency standards as prescribed in the Energy Conservation Building Code (ECBC) 2017, as amended up to date.
73. An air scrubber and air washer system shall be provided for the guest house kitchen to ensure effective removal of fumes, smoke, grease particles, and odor, thereby maintaining acceptable indoor air quality. The air scrubber shall be designed to filter airborne contaminants using a multi-stage filtration system including pre-filters, activated carbon filters, and wet scrubbers, as required. The air washer shall provide additional air cleaning and humidity control, enhancing ventilation performance. The system shall be adequately sized based on the kitchen exhaust load and conform to relevant IS/ASHRAE standards. It shall include all associated ducting, fans/blowers, pump sets, nozzles, controls, and drainage arrangements. The entire system shall be integrated with the kitchen exhaust and fresh air system, ensuring continuous and efficient operation. All equipment shall be corrosion-resistant, energy-efficient, easy to maintain, and compliant with environmental and health safety norms.

74. Treated Fresh Air (TFA) units shall be provided for the Community Centre building to ensure adequate ventilation and maintain indoor air quality as per ASHRAE/IS standards. The TFA units shall be designed to supply pre-cooled and filtered fresh air to air-conditioned spaces, maintaining positive pressure and compensating for exhaust air losses. Each unit shall comprise a multi-stage filtration system (including pre-filters and fine filters), cooling coil, centrifugal blower, insulated casing, drain tray, and necessary control systems. The units shall be double-skin, energy-efficient, and designed for low noise operation. The capacity of each unit shall be determined based on occupancy load, fresh air requirements, and system zoning. All associated ducting, supports, vibration isolators, drain piping, power/control cabling, and integration with the Building Management System (BMS), wherever applicable, shall be included in the scope of work.
75. A Central Control Unit (CCU) shall be provided for the VRF (Variable Refrigerant Flow) air-conditioning system in the Community Center building to enable centralized monitoring, scheduling, and control of all indoor units.
76. **Provisioning of EPABX Room and CCTV Surveillance Control Room.**
- a. An air-conditioned room with adequate space shall be provided in guest house for the installation and operation of the EPABX (Electronic Private Automatic Branch Exchange) system. The room must meet the following requirements:
 - b. Location & Accessibility: Strategically located for ease of access and maintenance, with restricted entry for authorized personnel only.
 - c. Environmental Conditions: Equipped with proper air conditioning (24x7) to maintain suitable temperature and humidity levels for sensitive electronic equipment.
 - d. Space & Layout: Sufficient floor area and clearances shall be provided to accommodate EPABX racks, patch panels, cabling trays, and associated accessories, with room for future expansion.
 - e. Power Supply: Provision of dedicated power supply with UPS backup for uninterrupted operation. Separate earthing for equipment and safety shall be ensured.
 - f. Lighting & Ventilation: Adequate lighting and ventilation in accordance with industry standards and local building codes.

- g. **CCTV Surveillance Control Room:** A separate CCTV surveillance control room shall also be provided at Entry gate for housing the CCTV surveillance system, including monitors, recording units (NVR/DVR), network switches, and storage devices. It shall also be air-conditioned and equipped with ergonomic consoles and operator seating. Provision of dedicated power supply with UPS backup for uninterrupted operation. Separate earthing for equipment and safety shall be ensured. Adequate lighting and ventilation in accordance with industry standards and local building codes.

All civil, electrical, and IT infrastructure works associated with these rooms shall be within the scope of the EPC Contractor, including false flooring/ceiling, fire protection and structured cabling etc.

77. Outdoor CCTV surveillance shall provide comprehensive coverage of the entire campus, with special attention to key areas such as the Central Park, Underground (UG) Tank and Pump Room, Guest House, Community Center, inside elevator (lift) cars, car parking zones, and all corners and vulnerable points along the perimeter.

78. **Automatic Lighting Control for Common Areas**

To ensure energy efficiency and automation, the following provisions shall be made under the scope of the EPC Contractor:

a) **Lux Sensor Switches:**

Automatic lux sensor-based switches shall be provided for controlling common illumination points such as:

- Street lighting.
- Park lighting.
- Stilt floor lighting.
- Staircase lighting.
- These sensors shall automatically switch the lights **ON at dusk** and **OFF at dawn**, based on ambient light levels.
- Sensors must be weatherproof and compliant with relevant IP (Ingress Protection) standards for outdoor installations.

b) **Programmable Timer Switches:**

- In addition to lux sensors, **programmable timer switches** shall be installed for park and similar areas where lighting is required only during specific evening hours.

- The timer switches should allow flexible programming to turn **lights OFF automatically at a pre-set time**, reducing unnecessary energy consumption.
 - The timer devices shall have battery backup to retain settings in case of power outages.
 - All sensors, timers, and associated wiring and controls shall be included in the contractor's scope. The system shall be designed to ensure **seamless and reliable automatic operation** with provisions for **manual override** wherever necessary.
79. Floor evacuation plans shall be provided on each floor of the residential towers, community centre, and guest house to ensure clear guidance during emergencies.
80. An IG-541 gas-based fire suppression system shall be provided for the substations, along with the associated fire detection system, to ensure effective fire protection and compliance with applicable safety standards.
81. A First Aid Box shall be installed on the ground floor of each building (Residential, guest house & community center building) at a clearly visible and easily accessible location, ensuring availability to all occupants in case of emergencies.
82. Provisions shall be made in the Security Control Room for centralized monitoring of all fire safety and emergency communication systems, including fire alarms, smoke detectors, heat detectors, manual call points, hooters, and the Talk Back system.

h. **Technical specifications of light fixture placed as under:**

1.		Supply of 385mm Top Dia & 182mm Height Dark Sky Certified unique spherical cap shape design post top & completely made in India luminaire. The luminaires shall be made of Aluminum top spun cap with die cast heatsink, duly powder coated with gray color pure polyester. The complete luminaire has IP66 & IK08 protection and it comprises of polycarbonate Fresnel lens/street optics lens to achieve Nil up-light and reduced light pollution on driveway as well as footpath. The LEDs shall have minimum CRI ≥ 70, CCT-3000K & shall have SDCM ≤ 5. The luminaire shall have wattage 35W (for parking side) and 45W (for driveway side) as per lux requirement and shall deliver 100 lm/W (efficacy). Beam angle of the Luminaire shall be asymmetric type to cover driveways. Fixture shall operating temperature range of 0°C to 45°C. Input Voltage Range of 150-300V AC. The luminaire shall have declared life of 50,000 burning hours at L70. Luminaire and Driver must be made in India with same brand. Luminaire manufacturer must have In-house NABL accredited lab to conduct LM79. The successful bidder must submit the NABL accreditation certificate for Luminaire manufacturer along with technical bid. LM80, RoHS compliance issued by LED luminaire manufacturer, BIS certificate for LED Luminaire. Brand name shall be embossed / engraved on each and every light fixtures. Suitable for external pole dia. 65mm to 75mm. Luminaire and Driver should be Made in India.
2.		Supply of 11W LED Decorative Bollard with System efficacy of 100 lm/W (1100 lumens). Luminaire shall be made up of pure polyester power coated RAL7021 casted Aluminium housing with seamless polycarbonate diffuser, having dimension of dia (Ø) 150mm with height of 800mm. All external hardware must be of SS304 grade. LM 80-08 compliant LEDs of EMC/ceramic package from reputed makes should be provided. LEDs used in the product shall comply with EN 62471 for Photo-biological safety and certificate for the same from manufacturer shall be provided. Luminaire shall be with IP65 and IK08 protection. The LED shall be compliant with LM80-08 standard with L70 life of 50000 Hrs tested at maximum current (Complete LM 80 test report for LED should be submitted for 10000 hrs of testing). The

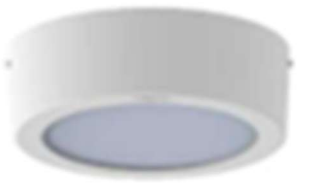
		LEDs used should be with CCT of 3000 K, CRI of Minimum 80 and SDCM <5. LEDs must be driven at max 70% of the rated current. MCPCB with Aluminium core. LED Driver shall be silicone potted, Integrated type, Isolated, Constant Current with range 140-280V AC. Driver Efficiency > 80% & 3 KV internal surge protection, Power factor greater than 0.95 and total harmonic distortion (THD) of less than 10% should be integral to the luminaire. Manufacturer shall have inhouse lab approved by NABL or ministry of science of govt of India. LM79 and LM80 reports need to be submitted from a NABL/UL accredited lab to verify above parameters. Compliance to Indian Standards of IS10322 Part 5 / Sect 5:1967; 16105: 2012; IS 16106: 2012; IS 15885 (Part2/Sec13): 2012; IS 16108: 2012 should be there.
3.		SITC of recess mounted 2X2 LED panel, made of CRCA housing with high efficiency PS Diffuser. LED Used shall be SMD type and fixture should have minimum efficacy at System level ≥ 120 lumens/watt with Minimum system Lumens 4320 & Max System wattage of 36W, Life of fixture : 50000 burning Hrs. @ L70 Lumen maintenance, CCT-4000K/5700K/6500K, (SDCM ≤ 3), CRI Ra ≥ 80 , THD $< 5\%$, PF > 0.95 , an operating Voltage Range of 150 - 270 VAC. Minimum Internal Surge Protection 4.0KV & IP20/IK02 protection, Fixture should comply with to IEC61000-3-2 ed.3.2, 2009 for Harmonics, IEC61347 -2 -13, 2006 in Conjunction with IEC61347-1 ed.2.0, 2007 for Electrical Safety, IEC62384 ed.1.1, 2011 for performance and IEC61547 ed.2.0, 2009, IEC 60598-2-2, CISPER 15 for EMC / EMI compliance with flicker free operations ripple $< 5\%$. Manufacturer shall have inhouse lab with mirror type photogoniometer approved by NABL or ministry of science of govt of India. LM 79 report needs to be submitted from a NABL/UL accredited lab to verify above parameters.
4.		Supply of 30W LED Decorative Post Top type luminaire of hexagonal top with System efficacy of 125 lm/W (3750 lumens), suitable for external poles of dia. 65mm to 75mm. Luminaire shall be made up of pure polyster power coated RAL7021 pressure die cast Aluminium housing with Polycarbonate Fresnel lens for symmetrical light distribution. All external hardware must be of SS304 grade. LM 80-08 compliant LEDs of EMC/ceramic package from reputed makes should be provided. LEDs used in the product shall comply with EN 62471 for Photo-biological safety and

		certificate for the same from manufacturer shall be provided. Luminaire shall be with IP66 and IK10 protection. The LED shall be compliant with LM80-08 standard with L70 life of 50000 Hrs tested at maximum current (Complete LM 80 test report for LED should be submitted for 10000 hrs of testing). The LEDs used should be with CCT of 4000/5700K, CRI of Minimum 70 and SDCM <5. LEDs must be driven at max 70% of the rated current. MCPCB with Aluminium core and copper thickness of 35 micron to be used for LED mounting. LED Driver shall be silicone potted, Integrated type, Isolated, Constant Current with range 140-290V AC. Driver Efficiency > 85% & 5KV internal surge protection, Power factor greater than 0.95 and total harmonic distortion (THD) of less than 10% should be integral to the luminaire. Manufacturer shall have inhouse lab approved by NABL or ministry of science of govt of India. LM79 and LM80 reports need to be submitted from a NABL/UL accredited lab to verify above parameters. Compliance to Indian Standards of IS10322 Part 5 / Sect 5:1967; 16105: 2012; IS 16106: 2012; IS 15885 (Part2/Sec13): 2012; IS 16108: 2012 should be there.
5.		Supply of 70W SMD LED Streetlight fixture with System efficacy of 130 lm/W (9100 lumens). Luminaire shall be made up of pure polyster powder coated die cast Aluminium housing with high purity IP66 PC lens without any glass cover, of single module of same make. The luminaire should be in rectangular shape with dimension 235mm x 160mm x 100mm (± 2mm). It should not weight more than 1.6Kg ± 0.2 Kg. The housing should be made of LM6/ADC12 Alloy, non-corrosive Single Piece pressure die-cast aluminum, to withstand extreme environments. All external hardware must be of SS304 grade. The name of the brand should be embossed/engraved & should not be mentioned using sticker/screen printing. Luminaire shall be designed with street optic LENS for uniform light distribution. Maximum LENS dome top temperature shall be less than 95°C at 45°C ambient temperature. LM 80-08 compliant LEDs of EMC/ceramic package of wattage of 4W-5W from reputed makes should be provided. LEDs used in the product shall comply with EN 62471 for Photo-biological safety and certificate for the same from manufacturer shall be provided. Luminaire shall be with IP66 and IK10 protection. The LED shall be compliant with LM80-08 standard

		with L70 life of 100000 Hrs tested at maximum current (Complete LM 80 test report for LED should be submitted for 10000 hrs of testing). The LEDs used should be with CCT of 5700 K, CRI of Minimum 70 and SDCM of < 5. LEDs must be driven at max 70% of the rated current. MCPCB with Aluminium core and copper thickness of 35 micron to be used for LED mounting. LED Driver shall be silicone potted, Integrated type, Isolated, Constant Current with range 140-270V AC. Driver Efficiency > 85% & 5 KV internal surge protection along with external SPD of 10 KV mounted inside the streetlight and firmly fixed to the housing, Power factor greater than 0.95 and total harmonic distortion (THD) of less than 10% should be integral to the luminaire. LED Driver must have short circuit, open circuit, Auto cut-off protection for low and high input voltage. Driver must have 440V input voltage protection for 8 hours. Driver must be of die cast / extruded Aluminium casing. Manufacturer shall have inhouse lab approved by NABL or ministry of science of govt of India. LM 79 and LM80 reports need to be submitted from a NABL/UL accredited lab to verify above parameters. Compliance to Indian Standards of IS10322 Part 5 / Sect 3:2013; 16105: 2012; IS 16106: 2012; IS 15885 (Part2/Sec13): 2012; IS 16108: 2012.
6.		Supply of 90W SMD LED Streetlight fixture with System efficacy of 135 lm/W (12150 lumens). Luminaire shall be made up of pure polyester powder coated die cast Aluminium housing with high purity IP66 PC lens without any glass cover, of single module of same make. The luminaire should be in rectangular shape with dimension 325mm x 184mm x 110mm (± 2mm). It should not weight more than 1.9Kg ± 0.2 Kg. The housing should be made of LM6/ADC12 Alloy, non-corrosive Single Piece pressure die-cast aluminum, to withstand extreme environments. All external hardware must be of SS304 grade. The name of the brand should be embossed/engraved & should not be mentioned using sticker/screen printing. Luminaire shall be designed with street optic LENS for uniform light distribution. Maximum LENS dome top temperature shall be less than 95°C at 45°C ambient temperature. LM 80-08 compliant LEDs of EMC/ceramic package of wattage of 4W-5W from reputed makes should be provided. LEDs used in the product shall comply with EN 62471 for Photo-biological safety and certificate for the same from manufacturer shall be provided. Luminaire shall be with IP66 and IK10

		protection. The LED shall be compliant with LM80-08 standard with L70 life of 100000 Hrs. tested at maximum current (Complete LM 80 test report for LED should be submitted for 10000 hrs. of testing). The LEDs used should be with CCT of 5700 K, CRI of Minimum 70 and SDCM of < 5. LEDs must be driven at max 70% of the rated current. MCPCB with Aluminium core and copper thickness of 35 micron to be used for LED mounting. LED Driver shall be silicone potted, Integrated type, Isolated, Constant Current with range 140-270V AC. Driver Efficiency > 85% & 5 KV internal surge protection along with external SPD of 10 KV mounted inside the streetlight and firmly fixed to the housing, Power factor greater than 0.95 and total harmonic distortion (THD) of less than 10% should be integral to the luminaire. LED Driver must have short circuit, open circuit, Auto cut-off protection for low and high input voltage. Driver must have 440V input voltage protection for 8 hours. Driver must be of die cast / extruded Aluminium casing. Manufacturer shall have inhouse lab approved by NABL or ministry of science of govt of India. LM 79 and LM80 reports need to be submitted from a NABL/UL accredited lab to verify above parameters. Compliance to Indian Standards of IS10322 Part 5 / Sect 3:2013; 16105: 2012; IS 16106: 2012; IS 15885 (Part2/Sec13): 2012; IS 16108:2012.
7.		Supply of All-in-One Hybrid LED solar street light consists of solar photovoltaic module, battery, battery management system, control electronics, LED driver, interconnecting cables and pole mounting piece integrated into the luminaire as a single unit. Solar streetlight with IP65 ingress protection having a system wattage $70W \pm 10\%$ Watt with system lumens output $11900 \pm 10\%$, 95Wp high efficiency monocrystalline silicon panel, highly efficient MPPT based charge controller and 25.6V, 24Ah Lithium Iron Phosphate (LiFePO₄) battery with proper battery management system which provides higher life of >2000 cycles. Light fixture should operate at Full Power up to 6Hrs. (At fully charged battery) and Saving mode up to 27Hrs.(At fully charged battery). LED system efficacy:>170lm/W and MPPT charge controller with 90% efficiency. Fixture should have no load current less than 10mA. Output power shall be $63W \pm 10\%$ with output current approx 4.02A at full power. Typical correlated colour temperature should be 5700K with CRI>70. It should have Integral PIR sensor for motion detection to ensure battery optimisation with help of a

		suitable dimming profile. Luminaire shall have Automatic Dusk to Dawn feature. Luminaire shall have inbuilt hybrid power supply option that can charge the battery during non-sunny days automatically from the grid. The luminaire should be made from single piece extruded aluminium & IK08 impact protection (without motion sensor) certified housing, with anti-corrosive coating and must have a UV stabilized polycarbonate cover. The luminaire shall have a specially designed pole mounting piece that enables installation at varying tilts from 0 to 15 degrees for optimal light distribution. Luminaire shall be suitable for external pipe dia. of 62 – 72mm. System should have Smart self-diagnostic feature for indicating battery charging, cut off & fault if any and Fixture should have over voltage protection. Dimension: 1260X410X60 mm with Tolerance: ± 5mm, Weight: 27.5Kg (approx.)
8.		Supply of IP65 rated 200W Image projector with construction material high quality aluminium alloy with black color coating. Aluminium with fan cooling for Better heat dissipation method, long lasting light source life of 50,000 hours. Luminous flux of min. 12000 lumen and color temperature shall be 8000K, Projection distance is 15m for dark ambience environment. Beam angle shall be 15deg, 25 deg, 30 deg options. It can project sharp customizable logos in color, monochrome or two tones as per requirement. Gobo size is 37/25mm in diameter, focus shall be manual and control mode through DMX 512. Net weight shall be 5KG ± 0.5KG and dimensions 250mmx310mmx185mm. It shall be supplied with DMX Master controller and software. including Programming & Commissioning charges for projector and accessories.
9.		Supply of decorative wall bracket having canopy arm chrome lamp cup with E27 lamp holder. Dimension: L180XW120XH300mm ± 2mm.
		Supply of suitable for general lighting surface downlighter made of Pressure die cast aluminium suitable for general lighting.

10		Luminaire should have die cast heat sink with polycarbonate cover. The luminaire should have CRI > 80 and CCT of 4000K, SDCM<3 as per ANSI Bin C78.377A. The luminaire shall be compliant with IP40 classification. The fixture should have a minimum system efficacy of 100 lumen/Watt and a minimum system lumen output of 1800 lumens and maximum system wattage of 18 Watts. The luminaire shall be designed so as to ensure lumen depreciation of up to 30% over 50k burning hours @ design ambient temp 45 deg C. The isolated constant current electronic driver used shall have a power factor >0.95@full load, THD <5%, Surge protection 2.5KV, Operating voltage range: 100-265V AC, Fixture should have class I insulation. The fixture should comply with the parameters as per IS10322. Manufacturer shall have inhouse lab approved by NABL or ministry of science of govt of India. LM 79 and LM80 reports need to be submitted from a NABL/UL accredited lab to verify above parameters. Including 3 core 2.5 sq.mm FRLS PVC insulated and PVC sheathed copper conductor cable and earthing etc. up to suitable point complete as required. Dimensions: Ø152x71mm (±2mm), Weight should not be more than 0.45Kg.
11.		Supply of suitable for general lighting surface downlighter made of Pressure die cast aluminium suitable for general lighting. Luminaire should have die cast heat sink with polycarbonate cover. The luminaire should have CRI > 80 and CCT of 4000K, SDCM<3 as per ANSI Bin C78.377A. The luminaire shall be compliant with IP40 classification. The fixture should have a minimum system efficacy of 100 lumen/Watt and a minimum system lumen output of 1200 lumens and maximum system wattage of 12Watts. The luminaire shall be designed so as to ensure lumen depreciation of up to 30% over 50k burning hours @ design ambient temp 45 deg C. The isolated constant current electronic driver used shall have a power factor >0.95@full load , THD <5%, Surge protection 2.5KV, Operating voltage range: 100-265V AC, Fixture should have class I insulation. The fixture should comply with the parameters as per IS10322. Manufacturer shall have inhouse lab approved by NABL or ministry of science of govt of India. LM 79 and LM80 reports need to be submitted from a NABL/UL accredited lab to verify above parameters. Including 3 core 2.5 sq.mm FRLS PVC insulated and PVC

		sheathed copper conductor cable and earthing etc. upto suitable point complete as required. Dimensions: Ø118x64mm (±2mm), Weight should not be more than 0.40Kg.
12		Supply of 2ft length, 10W LED Mirror luminaire wall mounted delivering minimum system lumen of 1150 lumens & the luminaire shall be with system efficacy 115 lm/W. The luminaire Housing shall be made of extruded aluminium & the diffuser shall be of PMMA/PC. The operating CCT shall be 4000K with CRI of ≥ 80 , SDCM ≤ 5 . The luminaire shall be with BIS Registered electronic driver with input voltage range of 150 to 270V, THD $\leq 10\%$ & PF ≥ 0.95 . LED Life with L70 Criteria should be 50,000 hours. IP20 protection. The LED Driver shall be easily available in India for repair and service. The LED shall be SMD type. Luminaire shall have minimum surge protection of 2.5KV. The luminaire shall be with following certifications: LM79 & LM80 issued by LED manufacture along with Photo Biological Safety Standard.
13		Supply of 4 feet, ceiling or wall mounted, SMD LED batten of 20W LED luminaire with System efficacy of 150 lm/W and 3000 system lumens, operating in ambient temperature of 0°C to 45°C. Luminaire shall be made up of extruded anodised aluminium housing with polycarbonate high transmissivity diffuser, ensuring uniform illumination glare free lighting, excellent heat dissipation, high durability, scratch resistance and non-rusting features. LEDs used in the product complies with IEC 62471 for Photo-biological safety. With screwless design, luminaire is with IP40 protection with stainless steel mounting clamps. LM 80-08 compliant LEDs from reputed makes should be provided. The LED shall comply with L70 life of 50000 burning hours tested at maximum current. The LEDs used shall be with CCT of 4000/5700 K and CRI of minimum 80, with Mac Adam CCT binning < 3 . Luminaire shall have FR4 PCB and its LED Driver shall be of Integrated type prewired in product housing, Isolated, Constant Current with range 150-270V AC with internal surge protection of 3kV. Driver of Efficiency $> 85\%$, Power factor of minimum 0.95 and total harmonic distortion (THD) of less than or equal to 10% shall be integral to the luminaire. Manufacturer shall have inhouse lab approved by NABL or ministry of science of govt of India. LM 79

		and LM80 reports need to be submitted from a NABL/UL accredited lab to verify above parameters. Compliance to Indian Standards of IS10322 Part 5 / Sect 5:1967; 16105: 2012; IS 16106: 2012; IS 15885 (Part2/Sec13): 2012; IS 16108: 2012. Dimensions: 1190 X 40 X 45mm, Weight: 0.55Kg
14		Supply of 4 feet, ceiling or wall mounted, SMD LED batten of 32W LED luminaire with System efficacy of 150 lm/W and 4800 system lumens, operating in ambient temperature of 0°C to 45°C . Luminaire shall be made up of extruded anodised aluminium housing with polycarbonate high transmissivity diffuser, ensuring uniform illumination glare free lighting, excellent heat dissipation, high durability, scratch resistance and non-rusting features. LEDs used in the product complies with IEC 62471 for Photo-biological safety. With screwless design, luminaire is with IP40 protection with stainless steel mounting clamps. LM 80-08 compliant LEDs from reputed makes should be provided. The LED shall comply with L70 life of 50000 burning hours tested at maximum current. The LEDs used shall be with CCT of 4000/5700 K and CRI of minimum 80, with MacAdam CCT binning < 3. Luminaire shall have FR4 PCB and its LED Driver shall be of Integrated type prewired in product housing, Isolated, Constant Current with range 150-270V AC with internal surge protection of 3kV. Driver of Efficiency > 85%, Power factor of minimum 0.95 and total harmonic distortion (THD) of less than or equal to 10% shall be integral to the luminaire. Manufacturer shall have inhouse lab approved by NABL or ministry of science of govt of India. LM 79 and LM80 reports need to be submitted from a NABL/UL accredited lab to verify above parameters. Compliance to Indian Standards of IS10322 Part 5 / Sect 5:1967; 16105: 2012; IS 16106: 2012; IS 15885 (Part2/Sec13): 2012; IS 16108: 2012. Dimensions: 1190 X 40 X 45mm, Weight: 0.55Kg
15.		Supply of 4 feet surface mounted SMD LED batten of 40W LED fixture with System efficacy of 100 lm/W and 4000 system lumens, operating in ambient temperature of 0°C to 45°C. Luminaire shall be made up of co-extruded polycarbonate white housing with polycarbonate UV stabilized high transmissivity co-extruded diffuser, ensuring uniform illumination glare free lighting, excellent heat dissipation, high durability, scratch resistance and very light weight with non-rusting features. LEDs used in the product complies

		with IEC 62471 for Photo-biological safety. Luminaire should have inbuilt microwave sensor of mounting height 4mtr (max) along with pre-programmed driver with maximum hold time of 30min. Luminaire is with IP65 and IK10 protection with stainless steel mounting clamps. LM 80-08 compliant LEDs from reputed makes should be provided. The LED shall comply with L70 life of 50000 burning hours tested at maximum current. The LEDs used shall be with CCT of 5700 K and CRI of minimum 80, with MacAdam CCT binning < 5. LED Driver shall be of Integrated type prewired in product housing, Isolated, Constant Current with range 150-260V AC with internal surge protection of 5kV. Driver of Efficiency > 85%, Power factor of minimum 0.95 and total harmonic distortion (THD) of less than or equal to 10% shall be integral to the luminaire. Manufacturer shall have inhouse lab approved by NABL or ministry of science of govt of India. LM 79 and LM80 reports need to be submitted from a NABL/UL accredited lab to verify above parameters. Compliance to Indian Standards of IS10322 Part 5 / Sect 5:1967; 16105: 2012; IS 16106: 2012; IS 15885 (Part2/Sec13): 2012; IS 16108:2012.
16		Supply of Integral, SMD 2x2 Backlit LED Panel, IP 20, 24W LED recessed mounted fixture with system efficacy of minimum 150lm/W and system lumen output of minimum 3600lm, operating in ambient temperature of 0°C to 45°C at 50Hz frequency. Luminaire shall have CRCA housing coated in white (RAL9016) with translucent HET diffuser for uniform illumination and wider light distribution. LEDs used in the product shall comply with IEC 62471. LM 80-08 compliant LEDs from reputed makes should be provided. The LED shall comply with L70 life of 50000 burning hours tested at maximum current. The LEDs used should be with CCT of 4000 K and CRI of Minimum 80. LED Driver shall be Isolated type, Constant Current topology driver with proper heat sink for heat dissipation suitable to operate in input voltage range of 150V to 270V (nominal rated voltage – 240V) with 4KV internal surge protection, Power factor of minimum 0.95 and total harmonic distortion (THD) of less than or equal to 10% should be integral to the luminaire. Manufacturer shall have inhouse lab approved by NABL or ministry of science of govt of India. LM 79 and LM80 reports need to be submitted from a NABL/UL accredited lab to verify above parameters.

		Compliance to Indian Standards of IS10322 Part 5 / Sect 5:1967; 16105: 2012; IS 16106: 2012; IS 15885 (Part2/Sec13): 2012; IS 16108: 2012
17.		Supply of SMD LED, IP 40, 18W Recessed downlighter with System efficacy of 100lm/W and system lumen output 1800 lumen, operating in an ambient temperature of 0°C to 45°C at 50Hz frequency. Luminaire shall have aluminium pressure die cast housing in white with high transmittivity diffuser for uniform illumination and reduced glare. EMC/PCT package of wattage of 0.5-1 W LEDs used in the product shall comply with IEC 62471 for Photo-biological safety. LM 80-08 compliant LEDs from reputed makes should be provided. The LED shall comply with L70 life of 50000 burning hours tested at maximum current. The LEDs used should be with CCT of 4000 K and CRI of Minimum 80, with McAdam CCT binning <3 mounted on 1.6mm MCPCB/FR4 with 35-micron cu layer. LED Driver shall be external, Isolated type, Constant Current topology driver with proper heat sink for heat dissipation suitable to operate at nominal rated voltage of 240V with 3kV internal surge protection, Power factor of minimum 0.95, driver efficacy of minimum 85%, and total harmonic distortion (THD) of less than or equal to 7% should be integral to the luminaire. Manufacturer shall have inhouse lab approved by NABL or ministry of science of govt of India. LM 79 and LM80 reports need to be submitted from a NABL/UL accredited lab to verify above parameters. Compliance to Indian Standards of IS10322 Part 5 / Sect 5:1967; 16105: 2012; IS 16106: 2012; IS 15885 (Part2/Sec13): 2012; IS 16108: 2012.
18		PC/PMMA Diffuser with Input voltage 240V, 50 Hz Operating Voltage 140V-270V with Colour Temperature of 5700K more than 1000 Lumens of IP65, Power Factor >0.95, Luminaries and Driver make should be same CRI>80 making connection and testing, complete all as specified and as directed by engineer in-charge.
19		Supply of 9 metre GI Octagonal pole of 3mm thickness with 70mm top diameter and 155mm. The Base plate size should be of 260 mm and thickness of the base plate should be 16mm. The foundation bolt diameter should be of M24 and total length of 750 mm. 4 numbers

		of foundation bolt. Material of pole sections are hot rolled MS Plate of BS EN 10025: 2019, Gr. S355JO / JR IS 2062: 2011, Gr.- E350 BR and for Base plate and foundation bolt as per IS 2062: 2011, Gr E250 A. Metal protection with hot dip galvanized as per IS 4759: 1996 / IS 2629: 1985. It should have 1000 mm of Single arm/ double arm bracket to mount the streetlight.
20		SITC of 4.0 mtr Aluminium step tubular pole, Pole should have Anti rust coating & should be PU Painted, 4 nos of foundation nut & bolt(16mm dia & 600 mm long with templates) with a base plate of 250mmx250mm, Base Plate thickness- 15 mm. Pole Thickness- 3.0mm. Pole Base OD up to 1 mtr -1.2 mtr of 102mm dia, then 6 mm OD Pole should have a din rail for MCB mounting provision with Bakelite sheet and 32 mm Electrical Connector with Bubble sheet Packing suitable for BOQ complete as required.

Note:

1. For all LED fittings CCT (5700K - 6500K or 4000K) shall be chosen based on final design, aesthetics, functionality and as per direction of Engineer-in-charge.
2. The building is a 3 Star GRIHA rated Building; therefore, the energy consumption of the building should be minimum but without affecting the functional requirements of the building. So, the fittings, fixtures and fans must be energy efficient. All the LED light fixtures shall be energy efficient as per ECBC-2017 requirement as amended up to date. Fittings shall be designed for continuous trouble-free operation under any atmospheric conditions without reduction in life or without deterioration of materials and internal wiring. Outdoor fittings shall be weather and rain proof.
3. Lighting shall be based on average lighting level considerations, which are as per NBC2016, ECBC-2017 as amended up to date.
4. Facade and Landscape lighting shall be state-of-the art through a professional lighting designer. The intent shall be to showcase the architectural elements of the building during night time from the arterial roads leading into the building.
5. All luminaries and fans should have min. 5 years manufacture warranty.

Additional Specifications for External Electric Installation

1. The quantities worked out in schedule of quantities are based on equipment considered at design stage and drawing. Roads are to be constructed under Civil work and accordingly street light must be provided along the road work.
2. Security Lights, Road Lighting, Parking and Area lighting shall be provided as per drawing attached in NIT and functional requirement.
3. Pathways, Garden, Landscape, Facade lighting where ever required shall be installed

keeping in mind the Aesthetic considerations and Efficient working of street lights and staircase lighting is required for security of the colony and safety and convenience of the residents. Therefore, adequate street lighting, staircase lighting is to be provided. Back lanes of residential blocks are also to be covered by basic street lighting for security.

4. Lighting luminaries for street / compound and gate lighting shall be of LED type and illumination level shall be as per latest CPWD specification and NBC-2016 as amended up to date.
5. The luminance level for road lighting is governed by IS-1944 (Parts 1 & 2); the level of luminance should be adequate to provide visibility which guarantees for the user a maximum of safety and sufficient visual comfort. Reference may be made to Clause 6.10.2 of CPWD specification PART-II EXTERNAL-2023 regarding illumination values to be provided on the road surfaces.
6. The Minimum Technical Data for LED Luminaires (Outdoor Application) of All LED Indoor lighting fixtures shall be as indicated in guidelines.
7. All lights such as street, compound, Landscape, Facade, Gate, and any other external lighting shall be controlled by an independent Indoor/ Outdoor control and distribution panel equipped with suitable rating control gears like MCCB, MCB, Timer switch, Contactor, relay, astronomical time switch, Meter units or any other items as necessary for functional requirement. There shall be arrangement of bypass switch so that in case of failure of time switches, the lights can be operated after bypassing the same. The control panel should have 20% spare outgoing switches (MCCBs) with compartment.
8. Separate cables of suitable sizes shall be laid from ESS to street light feeder panels, Essential panel of each building, compound lighting, jogging track etc.
9. The height and distance between each pole shall be maintained as per provision laid down in CPWD specifications as amended up to date.
10. The scope of the EPC Contractor shall include the complete work related to tapping of electrical power from the existing source up to the proposed substation. This includes, but is not limited to:
 - i. Survey, design, and engineering of the power tapping system.
 - ii. Procurement and laying of High Tension (HT) XLPE cables of appropriate voltage grade.
 - iii. Supply and installation of all associated cable accessories such as terminations, straight-through joints, and clamps.
 - iv. Execution of the complete feeder work including trenching, cable laying, cable protection, route markers, and backfilling.
 - v. Restoration of surfaces after cable laying, as per applicable norms and local authority requirements.
 - vi. Securing **Right of Way (RoW)**, including obtaining all necessary permissions and clearances from landowners, municipal authorities, and other statutory bodies.
 - vii. Handling of all **statutory approvals**, scrutiny and sanction procedures from relevant utility or government departments.
 - viii. Payment of all associated **fees**, including application, inspection, scrutiny, and sanction charges.

- ix. Coordination with the local power distribution/transmission company or authority to ensure seamless integration and energization.
- x. The entire scope shall be the sole responsibility of the EPC contractor and shall be executed on a turnkey basis, ensuring compliance with applicable standards, regulations, and safety codes.

2. ELECTRICAL SUBSTATION (11KV/0.433 KV)

The scope of work includes Planning, Design & preparation of drawings, obtaining statutory approvals from the concerned department, Supplying, installation, testing and commissioning of 11 kV/0.433 kV substation equipment comprising HT panel, Dry type transformers with OLTC , HT cable, bus trunking/armoured cable from transformer to LT panel, LT panels, automatic power factor correction panel if required, active harmonic filters, TVSS (transient voltage suppression system), SPD (surge protection system), essential panel, earthing, required inter-connections, substation safety equipment including LT cabling from substation to the various buildings panels (Essential and Non Essential).

2.1 Items of Work

A new Sub-Station shall be required for BQ housing complex. The sub-station installation work shall comprise of supply, installation, testing and commissioning of the following:

- a. High Voltage Panels (HT Panel) (2 I/C +1BC+6 O/G) with 30 min battery power back up for control circuit.
- b. 11/0.433KV,2000KVA, 50Hz Dry type Transformers (1Working+1Stand by)) complete with associated auxiliaries as specified.
- c. High voltage cables for inter-connection between the H.V. panel and transformers including terminations.
- d. M.V. Panels (Essential and Non-Essential).
- e. Bus trunking (sandwich type) for inter-connection between transformer's M.V. terminals and the M.V. Panel as specified.
- f. Automatic Power Factor Improvement Panel if required.
- g. Earthing system for entire substation equipment.
- h. Safety Equipments & fire suppression system.
- i. LT Power & Control cables.

j. Miscellaneous items.

Any other items/equipments required for successful testing and commissioning of the all-electrical works will be provided by the EPC Contractor without any extra cost.

All electrical in connection with installation of electric sub-stations shall be carried out in accordance with the provisions of Indian Electricity Act, 2003 and the Indian Electricity Rules, 1956 amended up to date.

The work shall be executed as per CPWD specifications of electrical works (Part -I), CPWD specifications of electrical works (Part -IV) and up to date correction slip, NBC-2016 (Latest Amendment), ECBC-2017 (Latest Amendment) and as per directions of Engineer-in-charge. The sub-station equipment and their installation shall conform to relevant Indian standards. **The substation equipment shall be tested for routine/acceptance tests as per relevant standard at manufacture's work before dispatch at site.**

The substations shall be planned to locate outside the buildings. Sub-Station shall be designed complete with 11 KV HT Panel Board, Dry type Transformers with OLTC, Main LT Panel, Essential Panel, Capacitor Panel if required, fire suppression system, safety equipment, UPS with batteries for backup power for HT Panel control system and all other items / accessories as required as per CPWD specifications for substations, to comply fire norms & GRIHA-3 or better.

The minimum and indicative capacity of the Sub-Station equipment selected based on estimated load of the buildings is as follows however, detail designing must be done by the EPC contractor. (Ref. SLD drawing No. RITES/BLD/EL/TEN/206 to 208)

- a. HT VCB Panel (09 Panel VCB)- As per SLD (01 Nos.)
- b. Transformer – 11KV/0.433KV, 2000KVA, Dry Type with On Load tap Changer – (02 Nos.)
- c. Bus Duct (Sandwich type)/cable From Transformer to Main LT Panel (IP-66 rating, Class F insulation).
- d. Main LT Panel (TTA) – As per SLD – (01 No.)
- e. APFC Panel if required.
- f. Building Rising Mains – FP 630A,50kA Rating, (08 Nos.) + Guest house-FP,400A, 25kA rating (01 No.) as per SLD drawing No. RITES/BLD/EL/TEN/208.
- g. 3 Phase online UPS – 60KVA for LV system.
- h. Essential panel (TTA) (Ref. SLD drawing No. RITES/BLD/EL/TEN/206 to 208).
- i. Firefighting panel, plumbing panel and street light panels.

The EPC contractor shall cover all cost like earth filling work, civil foundation work required for installation of the transformer in ESS, MS structure required for installing bus ducting, cables etc.

outdoor cable trenches, fencing, earthing, Lighting Arrestors, and other associated items required in outdoor locations for substation.

2.2 Monitoring of Power Consumption:

As part of GRIHA 3-star rating requirement for calculating energy consumption of every service, all the outgoings in all LT and HT Panels shall be provided with Energy Meter with a communication port.

The energy meters for monitoring as per GRIHA norms are required to be with IBMS compatibility and shall be having provision of meter reading by RS485 communication port which shall be communicating required parameters to a required DCU or gateway feeding the IBMS software. The energy meters of each panel (Essential/Non-Essential) shall be looped with RS 485 ports and the communication shall be done by laying CAT 6A cables which will be terminating to the GATEWAY required for wireless/wired communication. Near every panel where communication via RS 485 is required, a LAN port shall be provided for ethernet connectivity to gateways.

Also, microprocessor based over current, earth fault & short circuit protection relays shall be provided in the Main LT Panel.

2.3 HIGH VOLTAGE PANEL (HT PANEL), 9 PANEL VCB:

a. Scope of work

This Specification covers the design, supply, installation, testing & commissioning, packaging & forwarding to site of HT Panel Set with 30 min power pack having Incomer, bus coupler Outgoing, Indoor Type, Metal Clad Floor Mounted of 11 KV, incorporating Vacuum Circuit Breaker, Panels along with remote control & equipped with all specified accessories i.e. CTs, PTs (Indoor Type), metering and protection system including termination etc. The panel shall be designed as per IEC-62271-200 up to latest amendment and as per CPWD specifications for Electrical Works (Part – I) and (Part- IV) up to latest amendment. However, minimum/indicative incomer, outgoing & Bus coupler VCB panel required in HT panel shall be as **per Ref. SLD drawing No. RITES/BLD/EL/TEN/206 to 208.**

b. CODES AND STANDARDS

- i. The equipment shall comply with the requirements of latest revision of the following standards issued by BIS, unless otherwise specified.

IS: 1248 Direct acting indicating analogue electrical measuring instruments and their accessories.

IS: 2071 High Voltage Test Technique.

IS: 2544 Porcelain post-insulators for systems with nominal voltage greater than 1000V.

IS: 2705 Current transformers.

IS: 3618 Phosphate treatment of iron and steel for protection against corrosion.

IS: 5082 Wrought aluminium and aluminium alloy bars, rods, tubes, and sections for electrical purposes.

- IS: 5578 Guide for marking of insulated conductors
 - IS: 6005 Code of practice for phosphating of iron and steel
 - IS: 10601 Dimensions of terminals of high voltage switchgear and control gear.
 - IS: 11353 Guide for uniform system of marking and identification of conductors & apparatus terminals.
 - IS: 13703 Low voltage fuses for voltages not exceeding 1000V AC or 1500V DC.
 - IEC 60282-1 High voltage fuses — Current Limiting fuses
 - IEC 60358 Surge Capacitor
 - IS/IEC 60470 High Voltage Switchgear Alternating Current Contactors and Contactor based motor starters
 - IS/IEC 60529 Degree of Protection provided by Enclosure.
 - IS/IEC 62271 High Voltage Switchgear and Control gear (Part 1, 100, 102, 103, 105, 200,201,301)
 - IEC 62271 High Voltage Switchgear and Control gear.
 - IEC/IEEE 62271- High Voltage Switchgear and Control gear Part 37-013: Alternating 37-013 current generator circuit breakers
 - IEEE C37.013 AC Generator Circuit Breaker Rated on a Symmetrical Current Basis.
- ii. In case of imported equipment, standards of the country of origin shall be applicable, if these standards are equivalent or stringent than the applicable Indian standards.
 - iii. The equipment shall also conform to the provisions of CEA Regulations with latest amendments and other statutory regulations currently in force in the country.
 - iv. In case Indian standards are not available for any equipment, standards issued by IEC/ BS/VDE/ IEEE/ NEMA or equivalent agency shall be applicable.
 - v. In case of any contradiction between various referred standards/ specifications/ data sheet and statutory regulations, most stringent requirement shall govern and decision of CLIENT in this regard shall be final and binding.

c. VCB Panel Technical specifications:

12 KV, 1250 Amps, 31.5/40kA/3sec, MV Indoor Air insulated draw out type folded metal clad switchgear panels. Power frequency withstand voltage: 28 KV rms for 60 sec, Rated lightning impulse withstand voltage (peak):1.2/50 micro sec: 75 KV, Rated short circuit breaking current (rms): 31.5kA/40 kA for 3sec., Rated short circuit making current (peak): 80 kA.

Internal Arc fault withstand capacity:31.5/40 kA for 1 sec. Panel enclosure shall be self-supporting floor mounted, Indoor, dust and vermin proof, totally enclosed as per IEC 62271-200. Special attention must be given so that all the apparatus shall be designed to obviate the risk of accidental short circuit due to insects, mites, rodents, or microorganisms.

VCB shall have inbuilt anti-pumping feature, separate anti-pumping relay is not acceptable.

The inspection window on the VCB door shall be made of toughened glass. As per IEC 62271-200, it should bear the same mechanical strength as that of the enclosure. Circuit breaker shall be mounted on a withdrawable truck which shall be horizontally draw out from service position to isolated position i.e. **“Complete Roll-on floor type of design.”** Cassette type VCB design is not acceptable. Switchgear shall be fully type tested for extended mechanical endurance of M2 class as per IEC 62271. Switchgear shall be E2 and C2 electrical parameter compliance.

Each unit of VCB Panel Board should be divided into functional individual compartment such as Bus Bar, Instrument/relay & LT control, Circuit breaker, Cable termination and Instrument transformers (CT, PT). There shall be metallic partitions between individual compartments. All these compartments should have proper access for maintenance & should be segregated from each other by means of earthed metallic sheet steel/safety shutters. Safety shutters should be fully automatic and mechanically interlocked with the movement of the truck assembly having danger plates and pad locking arrangements. It shall have both electrical and mechanical flag indication for Isolate/Test/Service position. The switchgear shall be of high-quality galvanized sheet steel construction with sheet not less than 2.5 mm thickness (front door, load bearing/structural members/gland plate) and not less than 2 mm thickness for non-load bearing section & shall totally enclosed dust and vermin proof, free standing, and floor mounting extendable type. Front door hinges shall be of heavy-duty type. **Half Moulded SMC 40 % Glass housing between phases should be provided. Bakelite, Lamtuf sheet or any other phase separation sheet is not acceptable between phases.**

d. Interlocks:

The following interlocks shall be available between panel and breaker:

- A breaker in closed condition cannot be inserted into a panel in the Service position or cannot be taken out from panel when in SERVICE position.
- A breaker cannot be operated in between TEST & SERVICE position.
- The breaker shall not be moved either test to service or vice versa in ON position.
- The breaker shall not be switched ON in any position other than test or service.
- A breaker cannot be inserted into a panel in the SERVICE position or cannot be taken out from a panel when in SERVICE position without connecting the control circuit.
- Mechanical interlocking of the breaker with panel doors are required so that panel door shall not be opened with breaker in ON position. It shall be FOOLPROOF.
- VCB's circuits should have provisions for remote operations for closing & tripping of VCB's.
- The bus shutters shall be of self-closing type which shall close automatically when the truck is withdrawn.
- The cradle shall have proper design and construction so as to permit smooth withdrawal and insertion of the breaker. The movements shall be free of jerks, easy to operate and shall be on steel balls/rollers. The cradle shall have the markings for the two distinct and separate positions viz., Service and Test of the circuit breaker.

e. secondary disconnecting switch (sds) interlocks

- VCB cannot be racked in unless the SDS plug is inserted.
- SDS plug cannot be removed when VCB is in service or intermediate position.
- SDS plug can be removed only when the VCB is in test or isolated position.

A spacious cable chamber shall be located at the rear of the panel capable to accommodate 2 nos. 11 KV XLPE cable. Cable chamber should be accessible through removable rear cover backed by a framed wire mesh cover for additional safety. There should be 600 mm minimum vertical clearance in cable compartment between bottom gland plate and cable connection terminal. Bottom plate in cable compartment should be detachable type and cut into two halves to facilitate easy entry of cable for future use. Arrangement must be done for proper sealing of gland plate.

The panel front access shall be single stroke opening type with pad lockable arrangement to ensure operator safety and rear access shall be with folded frame bolted covers with handles. **It should be possible to open/close the door of the VCB compartment in a single latch action. A special provision should be provided so that rear door cover should be interlocked with same panel**

front door. Switchgear panel shall be with horizontal isolation and horizontal draw out type Vacuum Circuit Breaker. The panel shall be possible to extend the switchgear in either direction in the future date. There shall be detachable side covers.

As per site condition, there is moisture condensation problem during rainy season and high humidity condition. Hence Special provisions should be made for anti-condensation/moisture proof to eliminate high voltage tracking phenomenon, corrosion effect and improve insulation life. Design each feeder with additional space heater arrangement. Parts such as hangers, supports shall be hot dip galvanized as per IS: 2629 (latest edition) & IS: 4826 with zinc plating and olive-green passivation. The material for springs of shutter mechanism etc. shall be rust proof. All metal bolts and nuts to be used in busbars, panels and earthing work shall be made of stainless steel (SS) with heavy duty double flat washer (SS) and locking spring washer. Each individual feeder panel shall be provided with metal enclosure on all sides and shall be sent as individual panel but shall be suitable for assembly at site to form three/four set of switchboards as per existing site constraints. Loss of service continuity shall be of LSC2B category with compartments segregated from each other by metallic partitions (PM).

Degree of protection: High voltage Compartment: IP55, all components inside the cubicle shall be provided with traffolite material labels for identification.

Internal arc classification: IAC AFLR Each power compartments should be fitted with individual pressure relief flaps on its top surface. Switchgears shall be equipped with all safety interlocks for upmost safety of personnel and equipment. Special Deflector/Gas duct type of arrangement should be provided at top of every breaker confirming to IAC test for 1 sec. i.e. as per IEC 62271-200 ensuring full safety. Type test certificate from NABL accredited lab should be submitted by Manufacturer at the time of Inspection.

Accessibility to compartments: Busbar compartment (tool based), Switching device compartment (Interlock controlled), Connection compartment (tool based) Lateral metallic wiring ducts for laying Control cables shall be used wherever it is passing through HV compartment. Whereas in LV compartment control wire to be carried through PVC wire duct, having comb arrangement and in such a way that while putting the cover, all combs should align automatically.

f. **CURRENT TRANSFORMER**

The current transformer shall comply with IS: 2705 part I, II and III with all latest amendments.

- All Current transformers supplied shall be of dry type epoxy resin insulated.
- The accuracy class of the current transformer used for metering shall be class 1.0 and class 5P10 for protection as per latest relevant Indian standards.

- The current transformer shall have the required VA capacity to supply connected meters and relays without exceeding the accuracy limits specified above. They shall have a rated secondary current of 1A as per the metering / relay requirements.
 - All meters supplied shall be suitable for operation from current transformers having nominal secondary rating of 1A.
 - All CTs shall have undergone routine tests as specified in relevant Indian standards.
- g. **POTENTIAL TRANSFORMER** shall be Draw out type (Ratio: 11KV/ $\sqrt{3}$ /110V/ $\sqrt{3}$ Volts), epoxy resin cast moulded. The High voltage winding of PT transformer shall be protected by current limiting HT fuses duly encapsulate by cast resin barrel i.e. fully insulated with spring loaded tip.

Installation of Line Potential Transformer of Cast resin insulated draw out type with HT primary fuses at below of VCB panel only with bushings and shutters to the connection compartment. Line PT shall be mounted on VCB truck, however rack in rack out operation of PT, inside the panel, shall be independent of racking operation of VCB. Racking operation of the PT shall be possible only in closed door condition like VCB. The complete switchgear panel surfaces inside as well as outside should be powder coated with epoxy resin, standard colour RAL 7035 with 7 tanks pre-treatment process, after application of anti-rusting paint.

Auxiliary wiring & Terminals: All wiring for control, signaling, protection and instrument circuit should be of 1100 volts grade FRLS insulated multistranded copper conductors. The wiring in the panel should be laid through proper cable trays with proper identification ferules. The CT wiring shall be of 4.0sq. mm and all other controls as well as PT wiring shall be of 1.5sq. mm multistrand copper. All inter panel control wiring within each shipping section shall be through pre terminated plugin type terminals so as to reduce the time/mistake during commissioning.

Each panel shall be provided with transparent acrylic type labels of 40mm Height, nut bolted on front and rear of a non-removable part, clearly indicating the panel number and name. Details of name plate shall be given after installation of panels. Suitable danger/warning stickers should be provided wherever access to live parts from outside is possible. Labels indicating the function of internal devices and terminal block numbers and wire groups shall be provided. The same numbers or letters as used in the schematic and wiring diagrams shall identify all internal devices such as relays, fuses, MCBs, switches. Instructions shall be screen printed on the VCB door, instead of Stickers/ plates.

Indication Lamps: Each circuit Breaker panel should be provided with indication lamps for Breaker ON/OFF, Auto trip, Spring charged, trip circuit healthy, DC fail, VCB position i.e. Service, Test etc. All indication lamps should be properly labelled and LED type only. All LED shall be LVGP (low voltage glow protection) & industrial type. All indicating lamp labels should be mounted such that they are tilt free. Each feeder of the panel should be provided with panel illumination lamp LED type

only 230V AC with door switch & fuse, 2 Nos. space heaters 100 watts (230 Volt AC) continuously rated tubular/strip type with manual ON/OFF switch & thermostat cut off, spring charging motor ON/OFF switch, local/remote switch, and TNC switch. Each feeder should be provided with modular five pin 5/15 Amp, 230 V AC socket and switch backed by MCB.

Terminal blocks for Current transformer secondary lead wires shall be heavy duty, stud type disconnecting terminals and provided with external plug type shorting link and earthing facility. Low voltage compartment front cover shall house a terminal block with screwed cover arrangement duly CT and PT connection with plug shorting facility for safe external testing/calibration purpose. Each terminal block should not have more than one incoming and outgoing wire termination. Terminals for AC circuit and DC circuit shall be separately mounted in different DIN rails. The number of terminals making up a terminal block shall include at least six spare terminals. Totally 20% spare terminals shall be provided in each panel. All wiring from external connections shall be brought out to terminal on a readily accessible terminal block.

- h. **Bus Bar and Connectors:** Air insulated, Electrolytic grade tinned copper flat rectangular bus bar of 2200 Amp. Continuous rating. The main Busbars and Jumpers are of copper material and grade of ETP. Busbar shall have smooth surfaces free from burrs, sharp edges to minimize dielectric stress concentration.
- i. **Cable Termination:** There shall be a clear space allowed between the terminal block and cable entry for the spreading and terminations of external conductors. Positioning of cable termination shall avoid obstruction of other cable terminations, removable covers etc., and provide easy access for terminating cables. System for verification of safe isolation from supply for feeder and Busbar by means of capacitive voltage detecting system with panel front closed. All Hinged doors/ frames and movable parts shall be earthed through adequately sized (10 Sq.mm minimum) flexible tinned copper braids to the fixed frame to the switch board.

Separate holes with rubber grommet shall be provided on LV compartments for inter panel wiring & communication wiring. Mechanical door stopper arrangement to be provided for LV door such that door does not open beyond 95 Degrees. Single actuating mechanism shall be provided for electrical and mechanical indication of VCB position status. Firm shall supply 2 sets of all special tools i.e. (racking handle, spring charging handle etc.) required for day-to-day operation in a separate package. Manufacturer should provide separate instruction label for torque tightening of Busbar coupling. VCB breaker lifting trolley of appropriate load capacity shall be provided by the manufacturer.

- j. **EARTHING:** There shall be a continuous copper earth bus as per relevant standards running at the bottom of the panel. Earth bus shall be robust and shall be capable of carrying full short circuit current for 1 second. Doors, covers and all non-current carrying metallic parts shall be earthed through flexible copper wires. This also includes instrument casting and cable armour which are also connected to the earth bus. Earth bus must be type tested for short circuit

withstand capacity at 40KA for 1 sec. Earth bus bar of size as per statutory requirement shall be provided throughout the full length of the panel at the bottom and projected outside on both ends of the panel so as to facilitate termination of earth conductors from earth pits.

- k. **ERECTION & COMMISSIONING:** Complete erection & commissioning work shall be executed by Licensed Electrical Contractor. Firm must submit the license certificate and details of above contractor through which they intent to execute the work. The erection & commissioning work must be carried out under the supervision of OEM. Commissioning will only be carried out by the firm after erection and successfully carried out pre commissioning checks. After complete installation of switchgears and insulation of high voltage system, firm shall arrange high voltage test kit to check the power frequency insulation level (28KV), secondary injection test kit for checking CT, protection & metering system. All functional tests like Auto/ Local/ Remote operations, relay testing, finishing of wirings/ terminations and painting, etc. Practical trial will start after ensuring that all statutory/ mandatory requirements are met. A commissioning manual will be prepared by the firm and produced with detailed commissioning activities. Once the equipment has been installed and commissioned, site acceptance tests will be performed by the firm to confirm the systems operation and compliance with the requirements specifications. The recorded document confirms that the testing has been carried out successfully, has been duly checked and signed by both the firm and the representative and will be available for review/inspection prior to commencement of system handed over to end user.

l. **SWITCHBOARD WISE DESCRIPTION**

The Switchboard as complete consist of 2 Nos. of Incomer + 6 Nos. of Outgoing +1 No. Bus Coupler Panel with 30 min power pack.

A) **11kV 1250A Incomer Panel (2 Nos.)**

- **BREAKER:** 11kV, 1250A, 31.5/40kA, 24V DC TC CC, 230V AC motor spring charged mechanism type Vacuum Circuit breaker.
Make as approved by Engineer-in-charge
- **CURRENT TRANSFORMER:** Cast resin dual core 3 CTs, 400-200/5A of 15VA burden and accuracy Class – 1.0 for metering and class 5P10 for protection.
Make as approved by Engineer-in-charge
- **POTENTIAL TRANSFORMER:** Ratio: (11KV/ $\sqrt{3}$ /110V/ $\sqrt{3}$ Volts), Volts, Accuracy class is 1.0. Rated Burden 50VA. Make as approved by Engineer-in-charge
- **Meters:** MFM of Accuracy class 1.0 monitoring all electrical parameters like per phase, voltage, current, demand, W, VA, VAR, all instantaneous, integrated and demand parameters, Neutral

current, Frequency, PF, % load, Phase angle, THD working. Make as approved by Engineer-in-charge

- **Digital Ammeter** with Selector switch – 1 no. Make as approved by Engineer-in-charge
- **Digital Voltmeter** with Selector switch – 1 no. Make as approved by Engineer-in-charge
- **RELAYS:** The offered relays shall be multifunction numerical type. All Numerical Relay shall have minimum following features:
 - Relay shall be suitable for 1A CT secondary.
 - Relay shall be draw out type
 - Relay setting change.
 - Relay Fault, Event & Disturbance recorder extraction.
 - Reading the measurement values.
- Relay shall be provided with an alpha numeric LCD display and keypad for navigation and setting.
 - Make as approved by Engineer-in-charge
- **High speed master trip relay** – 1 No., Make as approved by Engineer-in-charge
- **Trip circuit supervision relay** – 1 No., Make as approved by Engineer-in-charge
- Accessories like local remote selector switch, TNC switch, Space heater with thermostat – 2 nos., DC & AC control circuit through MCB, Indicating LED lamps, Spring Charged indication, DC supply healthy, Breaker ON/OFF, in service, test, disconnected position etc. should be provided.

B) 11kV 800A Outgoing Panel (6 nos.)

- **BREAKER:** 11kV, 800A, 40kA, 24V DC, 230V AC motor spring charged mechanism type Vacuum Circuit breaker.
Make as approved by Engineer-in-charge
- **CURRENT TRANSFORMER:** Cast resin of dual core 3 CTs 150- 200/5 A of 15VA burden and accuracy Class-1.0 for metering and class 5P10 for protection.
Make as approved by Engineer-in-charge
- **Meters:** Microprocessor based flush type digital multifunction energy meter with accuracy class 1.0 and with RS485 port with MODBUS protocol for data logging/downloading shall be provided for all VCB panels. The meter shall be of size 96mmx96mm and shall measure the following electrical parameters: frequency, voltage, current, power factor, KVA, KVA_r, KWh, phase angle and harmonic components. Multi-function meter (MFM) of Accuracy class 1.0 should be as per IEC 62052-11/IEC 62053-22 monitoring all electrical parameters.
Make as approved by Engineer-in-charge
- **Digital Ammeter** with Selector switch – 1 no.
Make as approved by Engineer-in-charge
- **RELAYS:** The offered relays shall be multifunction numerical type. All Numerical relays Relay shall have minimum following features:

- a. Relay shall be suitable for 1A CT secondary.
- b. Relay shall be draw out type
- c. Relay setting change.
- d. Relay Fault, Event & Disturbance recorder extraction.
- e. Reading the measurement values.
- f. Relay shall be provided with an alpha numeric LCD display and keypad for navigation and setting.
- g. Make as approved by Engineer-in-charge
- **High speed master trip relay** – 1 No., Make as approved by Engineer-in-charge
- **Trip circuit supervision relay** – 1 No., Make as approved by Engineer-in-charge
- **Transformer fault Alarm/Trip Aux. Relay** (For Transformer feeder) – 2 nos.
Make as approved by Engineer-in-charge
- Accessories like local remote selector switch, TNC switch, Space heater with thermostat – 2 nos., Hooter, DC & AC control circuit through MCB, Indicating LED lamps, Spring Charged indication, DC supply healthy, Breaker ON/OFF, in service, test, disconnected position etc. should be provided.

C) Bus Coupler Panel(1No.)

- **BREAKER:** 11kV, 1250A, 31.5/40kA, 24V DC, 230V AC motor spring charged mechanism type Vacuum Circuit breaker.
Make as approved by Engineer-in-charge.
- Accessories like local remote selector switch, TNC switch, Space heater with thermostat – 2 nos., Hooter, DC & AC control circuit through MCB, Indicating LED lamps, Spring Charged indication, DC supply healthy, Breaker ON/OFF, in service, test, disconnected position etc. should be provided.
- 3 Nos. LED showing R, Y, & B voltage, Breaker 'ON, OFF, Trip' indicating light, breaker in-service / test position and 16 Amp TNC switch for ON / OFF/ Close VCB required. Suitable nos. and appropriate rating of HRC fuses/ MCB required for protection of LED, multifunction meter and auxiliary power supply, control circuit for closing, tripping and indication circuit and numerical relay etc., on bus coupler feeder.
- Cast resin of dual core 3 CTs, 400-200/5 A of 15VA, burden and accuracy Class - 0.5 for metering and class 5P10 for protection.
- 1 No. Microprocessor based non directional, numerical relay with O/L, E/F and S/C
- protection for bus-coupler feeder with synchronizing check facility.
Make as approved by Engineer-in-charge

The following minimum safety devices shall be provided to ensure the safety of operating personnel:

- a) Individual explosion vents for Bus bars/Breaker/Cable and CT chambers on the top/Side of the panel to let out the gases under pressure generated during unlikely event of a fault inside the panel.
- b) Cubicle with front door/panel pressure tested for arc faults.
- c) CB and metal enclosure earthed in accordance with relevant IS / IEC.
- d) Self-operating shutters, shielding live fixed contacts, shall be provided which closes automatically when truck is withdrawn to test position. Locking arrangement should be provided for the shutters.
- e) Breaker shall not be moved in ON condition from service to test position & vice versa.
- f) The CB cannot be switched ON' when the truck is in any position between test and service.
- g) All nut & bolts used inside the panel should be of high tensile, bright zinc plated, hexagonal headed, metric size, manufacture to DIN-931 of steel, tensile strength as per standard, coarse threaded with two nos. bright zinc plated flat and spring washers.
- h) Lifting hooks shall be provided for the panels.

D) Panel Markings, Rating plates, Nameplates and Labels

The switchgear panel shall have the following identification markings in a permanent manner:

- a) Panel name both in front and rear side.
- b) Caution boards conforming to IS-2551 (as amended up to date) both in front and rear sides.
- c) CT specification name plate on CT and at panel cover at rear.
- d) Incoming & outgoing cable box.
- e) The markings and identifications of conductors, apparatus terminals shall be as per IS-5578 (as amended up to date) & IS-11353 (as amended up to date).
- f) Main rating plate / nameplate shall be stainless steel. Other tag plates shall be either stainless steel or Traffolyte.
- g) Nameplate language shall be English. Dual language nameplate shall be provided if specified in the requisition / Data Sheet.
- h) Rating plate shall contain all information in line with IS/ IEC 62271-200.

E) Cable Terminal Box

HT cable boxes with termination links for termination of incoming and outgoing HT cables should be provided in the rear side of the unit. Rear incoming cable box should be of suitable size for safe entry of two nos. of incoming cables and should have suitable terminal links for safe termination of both the incoming cables for loop in/ loop out connection as is done in case of ring main unit. The termination arrangement should be such that it should be possible to disconnect one cable in the event

of fault in that cable and power-up the unit with the other incoming cable. Link rating shall be 600 amp (min).

Panel shall have incomer connections suitable for 2 x 3C x 300 sq. mm. XLPE 11KV cable (cable entry from bottom side) with indoor kit, end termination with heat shrinkable jointing kit etc. as required.

m. INSPECTION, TESTING & ACCEPTANCE

- All inspection and Testing shall be as per approved Inspection and Testing Plan (ITP).
- The tests carried out at the switchgear manufacturer's works shall be in accordance with the requirements specified in IS/IEC 62271-200.
- Factory Acceptance test (FAT) witness testing will take place only after all the latest relevant drawings, technical specifications, certificates and completed data sheets have been submitted and approved by the Engineer-in-charge.
- The manufacturers shall complete all internal inspection and routine tests prior to inspection by the RITES/CLIENT, to ensure that the witnessed testing is not prolonged by fault finding and rectification. Faults or unsatisfactory operations that result in abortive witness tests shall be rectified at the expense of the manufacturer. Any expense incurred by the RITES/ CLIENT resulting from abortive tests will be chargeable to SUPPLIER.
- During fabrication, equipment shall be subject to inspection by RITES/CLIENT or by an agency authorized by the CLIENT. SUPPLIER shall furnish all necessary information concerning the supply to RITES/ CLIENT. RITES/CLIENT shall have free access to the SUPPLIER's works for the purpose of inspecting the process of manufacture in all its stages and he will have the power to reject any material, which appears to him to be of unsuitable description or of unsatisfactory quality.
- All test instruments shall be calibrated and copy of calibration certificates should be provided on request.
- All parts and components shall be tested and certified, fully in accordance with the referenced codes and standards and the requisition.
- For equipment bought from other sub-suppliers, certified test reports of tests carried out at the sub-supplier's works shall be submitted. Normally all routine tests as specified in the relevant standards shall be conducted by the sub-supplier at his works.

n. REJECTION CRITERIA

The CLIENT/Contractor may reject any panel, if the following is found during FAT:

- a. Temperature of the bus bars during Heat run test exceed the guaranteed values (including positive tolerance).
- b. Contact resistance of breaker differs from the guaranteed value (including positive tolerance).

- c. Panel fails any test.
- d. Any deviation from specification is found not acceptable.

2.4 CAST RESIN DRY TYPE TRANSFORMER, 11/.433KV,2000KVA

A. Scope of work

- This document specifies the minimum acceptable requirements by EPC contractor for design, engineering, procurement, fabrication, assembly, inspection, testing, commissioning, and delivery to site of 11/.433KV, 2000 KVA, Resin cast Epoxy, Dry type Transformer for its installation including earthing by connecting suitable earth strips to the existing earth pits/grid.
- Energizing the Transformer and testing with connected load in the building.
- Obtaining approval related to Transformer & HT Panel from CEA etc. as required.
- 11 kV HT Cable, laying, termination and connection to 11 kV HT panel & Transformer Primary shall also be in the scope of EPC contractor along with the required length of Cable.
- LT Bus duct connection and termination to Transformer secondary to main LT Panel shall also be in the scope of EPC contractor.
- Control Cable, laying, termination, testing and connection to 11 kV HT panel & 2000 kVA Transformer Marshalling panel with the required length of Cable.
- Civil/excavation & back filing, sand and bricks laying work required for 11 kV HT cable shall be in the scope of EPC contract.
- This technical specification shall be read in conjunction with the standard conditions of the contract with correction slips, as are relevant for commercial aspects, as well as schedules and drawing and requirements under these specifications.
- In the event of any discrepancy between these specifications and inter connected documents, the technical requirements as per the contract specifications shall be followed and deemed to be having overriding value.
- All electrical works in connection with installation of electric sub-stations shall be carried out in accordance with the provisions of Indian Electricity Act, 2003 and the Indian Electricity Rules, 1956 amended up to date. Wherever I.E. rule numbers have been indicated, they are based on I.E. Rules, 1956 amended up to date.

B. Codes and Standards

The design, manufacture and performance of transformer shall comply with all currently applicable statutory regulations and safety codes in the locality, where the equipment will be installed. Nothing in this specification shall be construed to relieve the vendor of his responsibility.

Unless otherwise specified, equipment shall conform to the following latest applicable Standards and equivalent IEC Standards.

IS Codes			Description
IS	11171	1985	Specification of Dry Type Power Transformers
IS	2026 Part I	2011	Specification of Power Transformers
IS	2026 Part 2	2010	Temperature Rise
IS	2026 Part 3	2009	Insulation levels and Dielectric Tests
IS	2026 Part 4	1977	Terminal Markings, Tapping and connections
IS	2026 Part 5	2011	Transformer / Reactor Bushings - Minimum External clearance in Air Specification
IS	10028	1985	Code of practice for Installation & Maintenance of Transformers.
IEC	60076-1	2011	Power Transformer Part 1 – General.
IEC	60076-2	2011	Power Transformer Part 2 – Temperature Rise.
IEC	60076-3	2013	Power Transformer Part 3 – Insulation level and Dielectric test.
IEC	60076-3-1	1987	Power Transformer Part 3-1 – External clearances in air.
IEC	60076-4	2002	Power Transformer Part 4 – Tapping and connections.
IEC	60076-5	2011	Power Transformer Part 5 – Ability to withstand short circuit.
IS	2099	1986	Bushings for alternating voltages above 1000 Volts.
IS	3347	1979	Dimensions for Porcelain and transformer bushing.

IS	3639	1966	Fitting & Accessories for Power Transformers
IS	1271	2012	Thermal evaluation and classification of electrical insulation.
NEMA	TR1	2013	Transformers, Step Voltage Regulators and Reactors

C. **Technical Data**

a. **Rating**

The rating, voltage ratio, vector group and type of cooling of transformer, tap changing gear are specified as Appendix A data sheet. Transformer shall also meet the requirements of Indian Electricity Rules, Fire Insurance Association and Electrical Inspectorate.

b. **Temperature Rise**

Temperature rise by resistance of winding shall not exceed as per specified value in data sheet, over an ambient temperature of specified value in data sheet.

c. **Temperature indication devices and alarm**

- Winding temperature indicator (4 inches dial type thermometer) shall be provided with two electrical contacts (one for alarm and one for trip). CT of suitable ratio shall be provided for WTI.
- The tripping contacts of winding temperature indicators shall be adjustable to close between 60°C and 120°C and alarm contacts to close between 50°C and 100°C accordance to class of insulation class and both shall reopen when the temperature falls about by 10°C.
- All connection of control / power & communication cable Connections shall be brought from the device to marshalling box by capillary tube, cables etc. as applicable.

d. **Condition of Installation**

Transformer shall be suitable for indoor installation as specified in data sheet.

D. **Construction Features**

i **General**

Epoxy-resin dry-type transformers shall be provided for indoor applications. These transformers shall be supplied by the original transformer manufacturer. License products etc. are not acceptable.

The mechanical and electrical design of the transformers shall be based on the following conditions and requirements:

- The design shall be based on site and service conditions as specified.
- The maximum permissible temperature rise of the transformer windings is to be selected by the transformer manufacturer in accordance with the thermal classes of the insulation as specified in Data Sheet.
- The transformers shall be capable of operating continuously at full load at any tap position within their temperature rise limit.
- The transformers shall be capable of withstanding external short circuits for at least 1 seconds.
- Neutral points shall be brought out and grounded.
- Noise level shall be limited to 68 dB as per NEMA standard TR1.
- Doors shall be suitable to be opened from outside.
- There should be minimum three hinges per door from top to bottom.

ii. Core

- The magnetic core shall be made of laminations of non-ageing, cold-rolled, grain-oriented, silicon steel of high permeability without burrs. Each lamination shall be insulated with high quality insulation coating.
- The joints of limbs and yokes shall be designed and constructed to keep the no-load losses and the hot spot temperature in the magnetic core as well as the noise level as low as possible.
- The core and its clamping plates shall form a rigid unit structure, which shall maintain its form and position under the severe stresses encountered during shipment, installation, and short circuits.
- Care shall be taken to secure uniformly distributed mechanical pressure over all the laminations to prevent setting of the core and to limit noise and vibrations to a minimum under service conditions.
- The maximum magnetic flux density in the legs and yokes of the core shall not exceed 1.72 Tesla at rated voltage and frequency.

iii. Earthing

- Internal earthing of all metal parts of transformer except for the individual core laminations, core bolts associated individual clamping plates shall be earthed.
- Flexible earthing braid shall be provided between all metal parts, joined with gaskets.
- 2 Nos. separate Earthing pad/terminal shall be provided on the cable box for armour earthing from inside and Earthing grid connection from outside.
- Earthing strip from Earthing Bushing up to earthing terminal shall be provided.
- Doors shall be earthed by suitable PVC insulated multistranded copper wire of suitable size.

iv. Winding

- The transformers shall be Delta-Star connected, and the vector group shall be Dyn11 in accordance with IS 2026 (2011).
- All HV windings shall be insulated & values shall be furnished in Technical Data Sheets.
- The windings shall be made of copper of high conductivity to increase the capability of the transformer of withstanding the stresses under short-circuit conditions. Modern technology in design and construction shall be applied, and the current density shall not exceed 2.48 A/mm² in any part of the windings.
- The insulation material shall be based on an epoxy-resin/ powdered quartz mixture, which makes the windings maintenance-free, humidity-resistant and tropicalized as well as fire-resistant and self-extinguishing. The insulation material of windings and connections shall be free from insulation compounds subject to shrinking or collapse during service.
- The thermal class shall be "F" for all windings, and the temperature rise limits shall be limited to 90°C.
- The coils must be capable of withstanding movement and distortion caused by all operating conditions. Adequate barriers shall be provided between windings and core as well as between high voltage and low voltage windings.
- All leads or bars from the windings to the termination points shall be rigidly supported. Stresses on coils and connections must be avoided.
- PT 100 RTD should be embedded inside the coil of each phase and it should be wired up to a junction box. 1.5 Sq.mm copper Teflon insulated lead shall be used for connection between junction box and scanner. The scanner shall be mounted inside the Marshalling box. The scanner shall be provided with RS-485 port. Auxiliary supply shall be suitable for 230 V AC.
- All HV & LV winding lead connections shall be provided with check nuts.

v. Housing

- The unit shall be housed in protective steel housing with sufficient ventilation offering a degree of protection class minimum IP23. The unit shall be fitted with plain bi-directional rollers.
- Protective housing shall be made of zinc coated sheet steel. The unit with housing shall form one moveable and liftable unit (with transformer). Adequate ventilation shall be provided by vermin proof louvers.
- The performed bottom plate shall have sufficient clearance from the floor level to guarantee sufficient flow of natural air ventilation. Hand hole shall be provided with transparent cover to operate the tap changer. The hand hole cover shall be operatable with special Key/Tool only.
- A danger sign with cautionary notice both in regional language and in 'ENGLISH' shall be fixed on the
- cover. Temperature indicator and alarm shall be provided in easy-accessible location. Facility shall be available to install the alarm remote to the unit.
- Housing shall be provided with door limit switch, such that in event of opening of any door of the Transformer housing, the limit switch shall give trip command to switchgear. Wiring from limit switch to marshalling box shall be carried out by the EPC contractor.

vi. Structure

- The structure shall consist of suitable C-profiles rigidly fixed onto the upper and lower yoke of the core. Four lifting lugs shall be provided on the upper frame as well as traverses under the bottom as a mounting base. Two earthing terminals of adequate size shall be provided and installed diagonally at the lower frame.
- Lifting lugs shall be required for lifting core coil assembly.
- Bi-directional flat wheels with blocking facilities shall be provided. Appropriate support metal/rubber elements for structure-borne noise insulation of the transformer shall be installed underneath the rollers.

vii. Temperature Monitoring

- To protect the transformer against overheating two temperature monitoring systems (alarm and trip) shall be provided each to be connected to the LV windings by sensors to be selected according to the thermal class of insulation involved.
- Separate MCBs shall be provided for control supply for alarm and trip circuits. The control supply shall not be taken from secondary side of the transformer.

- RTD shall be provided with PT100 temperature sensor having nominal resistance of 100 ohms at zero degree centigrade. The PT100 temperature sensor shall have three wire ungrounded system. The calibration shall be as per IS 2848 or equivalent.

viii. Name Plates and other Designation Plates

- The following plates made of engraved polished stainless steel shall be supplied (to be subject for approval):
- A rating plate of weather proof material bearing the data specified in clause-8 of IS 11171: 1985 & IS 2026(2011) as amended up to date.
- A diagram plate shows in an approved manner the connections and, in addition, a plan views of the transformer giving the correct physical relationship of the terminals. This terminal marking plate for phases and neutral on HV/LV sides shall be of stainless steel and engraved as above and shall be permanently fixed.

ix. Terminals

- The transformer housing shall be fitted with cable boxes. The cable bushings on both HV and LV sides shall allow cable termination from the bottom.
- The HV cable terminals shall have clamp and holes suitable for terminating relevant type & size cable as indicated in data sheet
- The LV cable terminals shall be suitable for terminating relevant type & size cable / Bus duct as indicated in data sheet.
- The transformer shall be supplied complete with dry type heat shrinkable termination of approved make along with all the necessary cable lugs, grip type glands, etc. Cable support shall be provided in such a way as to relieve the strain on cable termination.
- The HT cable box shall not be detachable and shall have bottom entry only.
- The LT box shall have removable gland plates so that entry can be made from the bottom as well as top.
- Phase colour coding required on insulating sleeves on both HV & LV side terminals.
- The Link connection should be properly marked. The link connection diagram in separate metal sheet rather than on the cast resin HV winding so as to avoid any burning of the links and resin arising out of loose connection due to frequent changing of the links or due to human error and same should also be mentioned in Rating & Diagram plate.

x. Method of Cooling

The transformer itself shall be provided with a self-cooled type of cooling system (AN).

xi. OLTC tap changing Equipment

- The OLTC tap changing shall be maintained by means of suitable terminal links to enable the transformer to be matched to the prevailing system conditions. OLTC tapping range shall be provided as specified in Data Sheet.
- OLTC Tap changing links shall be shrouded with cover made from insulating material to prevent deposit of dust.

xii. Marshalling Box

- A sheet steel vermin proof well-ventilated and weather proof cabinet marshalling box shall be provided for transformer ancillary apparatus.
- The marshalling box shall accommodate temperature indicators, control and protection equipment in marshalling box, along with terminals and gland plates for incoming & outgoing cables.
- Marshalling box shall be provided with toughened glass visor to facilitate the reading of temperature indicators.
- Marshalling box shall be provided with cubicle illuminating lamp along with control switch & door limit switch.
- Marshalling box shall be provided with locking facility.

xiii. SURGE ARRESTOR

- 3** Nos. Surge Arrestor to be provided at the HV side of the transformer to limit surge in the system.

xiv. COOLING

The transformer shall be suitable for working in a well-ventilated protective enclosure with natural air cooling inside/outside the enclosure and shall conform to IS-2026 (Part – II) ANAN Type.

xv. TEMPERATURE RISE

- i) The limits of permissible temperature rise for windings, core and other parts shall be in accordance with the clause 3 of IS – 2026 (Part – II) for class H temperature class of insulation.
- ii) The maximum permissible temperature rise limits over maximum ambient temperature of 50 Deg. C shall be as below:

Particulars	Cast Resin Dry Type Transformer
a) Winding (Temp. rise measured by resistance method)	90° C (Class H of insulation)
b) Core, metallic parts & adjacent material.	Temperature shall in no case reach a value that will damage the core itself, other parts, or adjacent material.

xvi. Painting

Steel surfaces exposed to the weather shall be thoroughly cleaned and applied first coat of zinc chromate, second coat of oil and weather resistant paint and final coat of glossy oil and weather resistant non-fading paint of shade as specified in data sheet. The overall paint thickness shall be 100 microns (minimum) for Transformer & all accessories.

xvii. Accessories

- Transformer shall be provided with all necessary accessories like Winding temperature indicator, Rollers, two earthing pads, lifting lugs, jacking pads, Off-circuit Tapping switch, Terminal marking plate, additional neutral bushing, Marshalling Box, Rating & Diagram Plate etc.
- Rating and diagram plate shall be provided near Marshalling box.
- Details for above items shall be as specified in this specification.

xviii. Impedance, Losses, and Efficiency

- Impedance value and allowable limit of tolerance shall be provided as per IS 2026 (Part 1) (1977).
- No-Load & Copper Losses value shall be limited to value as specified in data sheet. Allowable limit value of tolerance shall be provided as per IS 2026 (Part 1) (1977).
- Transformer shall be designed to achieve Maximum efficiency at 50% loading.
- Power transformers of the proper ratings and design must be selected to satisfy the minimum acceptable efficiency at 50% and full load rating.

xix. Tests, Inspection and Guarantee

a) Routine Tests

- Measurement of voltage ratio at all tap positions.

- Check of vector group by voltmeter method
- Measurement of winding resistance at all tap positions.
- Measurement of impedance voltage on all taps Measurement of no-load losses.
- Measurement of load losses at principal tap and maximum current tap as specified in the IEC standards.
- Measurement of insulation resistance.
- Induced over voltage withstand test with measurement of partial discharges.
- Separate source AC voltage withstand test.
- Magnetic Balance Test.
- Measurement of capacitance & Tan Delta for HV & LV bushings.
- Polarity test

b) TYPE TEST:

The manufacturer should produce copy of Test Certificates on the following TYPE TESTS, which were already carried out at CPRI or at any NABL Accredited testing Lab for similar / higher rating of Transformer.

- Short Circuit Test.
- Lightning Impulse Test
- Temperature Rise Test.
- Each rating of the transformer should be type tested, including noise level measurement short-circuit testing, in any recognized international testing laboratory. All types of test certificates should be submitted by the manufacturer of the transformer.

c) Inspection

- Engineer-In-Charge/OIL or his authorized representative reserves the right to witness all the above tests and the manufacturer shall inform Engineer-In-Charge/OIL or his authorized representative of the same 15 days prior to commencement of the inspection.
- All equipment, instruments etc. required for the test shall be supplied by the manufacturer and shall be checked and tested for accuracy during the 6 months preceding the test and shall bear the tag of the competent authority.

xx. Technical Data sheet

The enclosed “Annexure- A” Technical data sheets indicate detailed technical and quantity requirements.

Annexure – A		
(Technical Particulars for 2000KVA,11/.433KV Dry Type Transformer)		
Sr.No.	Description	Specified value
1	Application	Distribution
2	Quantity No.	02 (two) Nos.
3	IS specification to be complied with	IS 11171 and IS 2026 (Part 1,2,3,4&5) amended up to date.
4	Installation	Indoor
5	Transformer Type	Cast Epoxy Resin Dry Type
6	Rating	2000 KVA
7	Voltage Ratio	11KV/433V
8	Frequency	50 Hz $\pm$ 3%
9	Vector Group	Dyn 11
10	Cooling type	AN
11	Material of winding	Electrolytic Copper of high conductivity
	Core material	non-ageing CRGO (Cold Rolled grain silicon Steel)
12	Winding Connection Primary / Secondary	DELTA / STAR
13	Insulation Class (Minimum)	H
14	Impedance at 75 °C	6.0% $\pm$ tolerance as per relevant IS
15	No of Phases	3
16	Fault Level at HV Side	26.3 kA for 1 Sec.
17	Insulation Level	
i)	Impulse withstands	
	a) H. V	75KVp
	b) L. V	6KVp
ii)	Power Frequency with stand Voltage	
	a) H. V	28 KV for 1 min
	b) L. V	3 KV for 1min
18	System Ground	Solidly earthed.
19	Tapping	On Load Tap Changer + RTCC + AVR at H.V side.
		Taps: - 5% to + 15% at 2.5% steps 9 positions.

20	Aux. Suppliers	For Space heater - 230V, 1Ph, 50 Hz AC. Controller – 230V AC Supply.
21	Tap Changer	
a)	Manually and Auto	
	i) Link Type	On Load Type
	ii) Gear Driven	Yes
b)	Remote Motorized	Yes
22	i) H.V side termination arrangement	Suitable for Cable connection
	ii) LV side termination arrangement	Suitable for Cable connection
23	H.V & L.V side disposition	
24	Transformer Loss	As per ECBC 2017 & relevant IS
25	Neutral Current Transformer	Required.
26	Neutral Grounding	Solidly grounded.
26	Miscellaneous	
a	WTI, required with trip & alarm contact.	Required
b	Enclosure	Min. IP 23
c	Wheels	Plain (Bi-directional)
f	Magnetic flux density at rated voltage & frequency Tesla	1.72
g	Constant power at all Taps	Yes
h	Configuration of Primary /Secondary	90° Apart
i	Extra Neutral Bushing	Required
j	Thermal insulation class of Winding	Class H
k	Noise level at measuring distance of 1 Mt.	Within IS Tolerance
l	Sheet steel thickness of enclosure	2mm

xxi. Drawings

The EPC Contractor is required to submit the following drawings of the transformer for approval from the Engineer-in-Charge before manufacturing of the same at manufacturer place.

- General arrangement drawing, showing plan, elevation, and side view, indicating all dimensions, transformer fittings & accessories etc.
- Civil foundation drawing for transformer.
- Rating and Diagram plate.
- Tap Changing Diagram Plate.
- G.A. of Marshalling box.
- G.A. of Primary & Secondary cable box as applicable.
- Bus throat connection diagram if applicable.
- Primary and Secondary Bushing drawing.
- Schematic wiring diagram of marshalling box.
- Index Sheet of the above drawing.

All the above drawings shall contain following data for reference:

- Owner P.O. / Contract reference No. & Date.
- Name of Client:
- Name of consultant:
- Name of EPC contractor:
- Name of Project:

The EPC contractor shall cover all cost like earth filling work, transformer civil foundation work required for installation of the transformer, RCC trench in ESS, all RCC trench in ESS shall be covered by 5 mm thick MS chequered plate including MS structure and other associated items required. Nothing extra shall be paid on this account.

2.5 MAIN LT PANEL:

A. SCOPE

The scope of this section covers design, manufacture, supply/installation, testing and commissioning of 415V Power Distribution Panels including ACBs/ MCCBs/ MCBs. The panel should be Totally type tested Assembly (TTA) / Design verified as per IEC 61439, Part 1 and 2 and other relevant code. The rating and configuration of incoming and outgoing feeders will be as per the SLD of the electrical panels, **drawing ref. no. RITES/BLD/EL/TEN/216 to 217** complete with all accessories such as Multifunction meter, Energy meter, protection relays, control wiring, auxiliary contacts, indicating lamps etc.

B. STANDARDS

In general, the equipment shall conform to all relevant IS/IEC standards. In case of any contradiction between the IS/IEC and this specification, the more stringent of the two shall apply.

STANDARD	DESCRIPTION
IEC 61439	Low-voltage switchgear and control gear assemblies
IEC 60228	Conductors of insulated cables
IEC 60255	Measuring relays and protection equipment
IEC 60529	Degrees of protection provided by enclosures (IP Code)
IEC 60831	Shunt power capacitors of the self-healing type for AC systems having a rated voltage up to and including 1000 V
IEC 60871	Shunt capacitors for AC power systems having a rated voltage above 1000 V
IEC 60898	Electrical accessories – Circuit-breakers for over current protection for household and similar installations
IEC 60947-6-1/EN 60947-6-1	low-voltage and control gear Multiple function equipment. Automatic transfer switching equipment.
IEC 60947-2/EN 60947-2	Specification for low-voltage switchgear and control gear circuit breakers
IEC 60947-1	Specification for low-voltage switchgear and control gear. Contactors and motor- starters. Electromechanical contactors and motor-starters.
IEC 61008	Residual current operated circuit-breakers without integral over current protection for household and similar uses (RCCBs)

IEC 62262	Degrees of protection provided by enclosures for electrical equipment against mechanical impacts (IK code)
IEC 61641	Enclosed low-voltage switchgear and control gear assemblies - Guide for testing under conditions of arcing due to internal fault.
IEC 61869/BSEN 61869	Instrument transformers
IS 13779	ac Static Watt-hour Meters, Class 1 and 2
IS 13947-5-2	Low-Voltage Switchgear and Control gear, Part 5: Control Circuit Devices and Switching Elements, Section 2: Proximity Switches
IS 13947-5-1	Low-Voltage Switchgear and Control gear, Part 5: Control Circuit Devices and Switching Elements, Section 1: Electromechanical Control Circuit
IS 13947-4-1	Low-Voltage Switchgear and Control gear: Part 4 - Contractors and Motor-Starters
IS 13947-3	Low voltage switchgear and control gear, part 3: switches, disconnectors, switch-disconnectors, and fuse combination units
IS 13947-2	Low-Voltage Switchgear and Control gear, Part 2: Circuit Breakers
IS 13947-1	Low-voltage switchgear and control gear, Part 1: General rules
IS 5553	Reactors – Specification

C. Design And Performance Requirement

All the 415V AC, devices/equipment like bus support insulators, circuit breakers, VTs, etc., mounted inside the switchgear shall be suitable for continuous operation and satisfactory performance under the following supply conditions:

- a) Variation in supply voltage: $\pm 5\%$.
- b) Variation in supply frequency: $\pm 3\%$.

D. Switchboard Constructional Features

- The switchboards shall be manufactured by authorized franchises of OEM based on OEM's design.
- The enclosures shall be designed to take care of normal stress as well as abnormal electro-mechanical stress due to short circuit conditions.

- The Switchboards shall be metal clad totally enclosed, floor mounted freestanding, fully compartmentalized bolted type of modular extensible design suitable for indoor mounting.
- Switchboards shall be made up of requisite vertical sections, which when coupled together, shall for continuous dead front switchboards.
- All covers and doors provided shall offer adequate safety to operating persons and provide ingress protection as per BOQ.
- Ventilating openings and vent outlets, if provided, shall be arranged such that same ingress protection is retained.
- Switchboard panels and cubicles shall be fabricated with CRCA Sheet Steel of thickness not less than 1.6 mm in general and load bearing members with 2 mm and shall be folded and braced as necessary to provide a rigid support for all components. The doors and covers shall be fabricated from CRCA sheet steel of thickness not less than 1.6 mm.
- Switchboard shall be readily extendible on both sides by addition of vertical sections after removal of the end covers.
- Sheet steel barriers shall be provided between two adjacent vertical panels running to the full height of the switchboard, except for the horizontal busbar compartment. Each shipping section shall have full metal sheets at both ends for transport and storage.
- All equipments associated with a single circuit except MCB circuits shall be housed in a separate compartment of the vertical section. The Compartment shall be sheet steel enclosed on all sides. The front of the compartment shall be provided with the hinged door, with locking facilities.
- All doors and covers shall also be fully gasketed with D-type gaskets to prevent any ingress of dust and vermin.
- The enclosure protection shall not be less than IP43 unless other specified in BOQ.
- There should be generous availability of space for ease of installation and maintenance with adequate safety for working in one vertical section without coming into contact with any live parts.
- The temperature rise limits of horizontal & vertical bus bars when carrying rated current along its full run shall be as per type tested design of OEM.
- All identical circuit breakers and module chassis of same test size shall be fully interchangeable without having to carry out modifications.
- The MCCBs shall be arranged in multi-tier formation whereas the Air Circuit Breakers shall be arranged in Single or Double tier formation only to facilitate operation and maintenance.
- Wherever two breaker compartments are provided in the same vertical section, insulating barriers and shrouds shall be provided in the rear cable compartment to avoid accidental touch with the live parts of one circuit when working on the other circuit.
- All boards shall be single front only.
- The Switchboard shall be configured with Air Circuit Breakers, MCCBs, and other equipment as called for in the schedule of quantities.
- The LV switchboards shall be minimum of Form 3B separation, except MCB feeders.

- The LT Panel shall have sufficient space to house all electrical equipment allowing spare Space capacity of 10% for the future expansion.
- The switchgear assembly/sub-assemblies or panels shall be termite and rodent proof.
- All non-current carrying metallic components shall be permanently connected to earth.
- The maximum temperature of the bus bars and the bus connections shall be as per IEC 61439 standard or latest amended up to date.
- Panel shall be provided with “Danger Notice Plate” conforming to relevant Indian Standards.
- Clearances between phase-phase, phase –Earth/ neutral should be in line with IEC.

E. Switchboard Dimensional Limitations

- A base channel 100 mm x 5 mm thick shall be provided at the bottom.
- The overall height of the switchboard shall be limited to 2200 mm.
- The height of the operating handle, push buttons etc. shall be restricted between 250 mm to 1800 mm from finished floor level.

F. Switchboard Compartmentalization

- For compartmentalized switchboards, separate totally enclosed compartments shall be provided for horizontal bus bars, vertical bus bars, ACBs, MCCBs and cable alleys.
- Hinged lockable doors for each separate compartment shall be provided and duly interlocked with the breaker in "ON" and "OFF" position.
- For all Circuit Breakers separate and adequate compartments shall be provided for accommodating instruments, indicating lamps, control contactors and control MCB etc. These shall be accessible for testing and maintenance without any danger of accidental contact with live parts of the circuit breaker, bus bars and connections.
- Each switchgear cubicles shall be fitted with label in front and back identifying the circuit, switchgear type, rating, and duty. All operating device shall be located in front of switchgear only. Minimum height from floor level for any device mounted on panel cover shall be 300 mm.
- Cable compartments shall be of adequate size for easy termination of all incoming and outgoing cables entering from bottom or top.

G. Switchboard Bus Bars

- For all ratings up to 4000A, Bus bars shall be made of high conductivity, high strength, high quality aluminium alloy, complying with requirements of grade E 91E of IS 5082 – 1981. Design of bus bar system shall comply to IS 5578 and IS 11353. Bus bars shall be of rectangular cross sections suitable for full load current for phase bus bars and half rated current for neutral

bus bar or as stipulated in schedule of quantities. Busbar shall be suitable to withstand the stresses of fault level as specified in schedule of quantities.

- The ratings and configuration of bus bars shall be as per type tested design of OEM.
- The bus bars shall be extensible on either side of the switchboard.
- The bus bars shall be supported on non-breakable, non-hygroscopic SMC insulated supports at regular intervals, to withstand the forces arising from a fault level as stipulated in single line diagram (SLDs) of panels. They should have minimum Comparative Tracking Index (CTI) of 600V (as per IS 2824).
- All bus bars shall be color-coded.
- The bus bar supports should be supplied by the OEM.
- Minimum clearances between phases / live parts shall be 25mm and phases/ live parts/ neutral to ground shall be 19mm except on the equipment terminals.
- The bus bars will be of Aluminium with a current density of 1.0 Amp per Sq. mm. The bus bars will be insulated with heat shrinkable sleeves and properly supported. All live parts shall be shrouded by means of acrylic / steel /FRP sheets to ensure no accidental contact with live parts during maintenance and provide simultaneous inspection. The busbar supports will be designed to withstand short circuit current mentioned in the drawings. The bus bars will be adequately insulated and protected to prevent accidental contact during operation and maintenance.

H. Earthing

- One earthing terminal shall be provided on each side of switchboard.
- An earth bar size must be at least 1: 50X6 mm Aluminium (same can be provided with Cu also, if required). Provision for arrangement for 2: 50X6 mm shall be possible, if required.
- The earth bar shall be electrically continuous and shall run the full extent of each board. This earth shall be on the same side as the cable entry.
- Each unit shall be constructed to ensure satisfactory electrical continuity between all metal parts which are not intended to be alive.
- Suitable holes with bolts and nuts shall be provided at each end of earth bar of switchgear for connection to a main earthing grid.
- The earth bar shall be accessible in each cable entering compartment either directly or through a branch extension to ground the cable armour and shields.
- Door earthing shall be provided for all doors.
- Material used for the earthing busbar shall be same as the main busbar.

I. Instrument Accommodation

- Instruments and indicating lamps shall not be mounted on the Air Circuit Breaker Compartment door. A separate and adequate compartment shall be provided and the instrumentation shall be

accessible for testing and maintenance without danger of accidental contact with live parts of the Switchboard.

- For MCCBs, instruments and indicating lamps can be provided on the compartment doors.
- The current transformers for metering and for protection shall be mounted on the solid aluminium bus bars with proper supports.
- All MCCB shall be provided with rotary operating mechanism & padlocking arrangement.

J. Wiring

- All wiring for relays and meters shall be with PVC insulated FRLS copper conductor wires.
- The wiring shall be coded and labelled with approved ferrules for identification.
- The minimum size of copper conductor control wires shall be 2.5 sq. mm.
- Runs of wires shall be neatly bunched and suitably supported and clamped.
- Means shall be provided for easy identification of wires.
- Identification ferrules shall use at both end of wires.
- All control wires meant for external connections are to be brought out on a terminal board.

K. Cable Termination

- Knockout holes of appropriate size and number shall be provided in the Switchboard in conformity with the location of incoming and outgoing conduits/cables.
- The cable terminations of the Circuit Breakers shall be brought out to terminal cable sockets suitably located.
- The cable terminations for the MCCB's shall be brought out to the rear in the case of rear access switchboards.
- Removable gland plates shall be provided for power and control cables. The gland plates shall be 3 mm thick and for single core cables shall be of non-magnetic material.

L. Painting And Finishing

- Sheet steel used in the fabrication of switchboards shall undergo a rigorous cleaning and surface treatment seven tank/Nine tank process comprising of alkaline degreasing, descaling in dilute sulphuric acid and a recognized phosphating process after which a coat of primer paint with the final paint shall be applied over the treated surface. Final paint coat of oven baked powder coating, of minimum 80–120-micron thickness shall then be provided.

M. Name Plates and Labels

- Suitable engraved white on black nameplates and identification labels of metal for all Switchboards and Circuits shall be provided according to the latest edition of relevant Indian Standard specification. These shall indicate the feeder number and feeder designation.
- Panel shall be provided with “Danger Notice Plate” conforming to relevant Indian Standards.

N. Testing And Inspection

- The switchgear shall be completely assembled, wired, adjusted and all routine tests as specified by the applicable standard code shall be conducted.
- Visual Inspection and Dimensional Check.
- Verification of Bill of Material.
- Check of conformity with wiring diagrams and plans.
- Mechanical operation tests, and checking of interlocks.
- Testing of the interchange ability of moving parts.
- HV test on power and control wiring / bus bars.
- Functional test for control circuits.
- Tightness of screwed/bolted connections.
- Electrical & mechanical operational checks.
- For equipment bought from other sub suppliers, certified test reports of tests carried out at the manufacturer's works shall be submitted. Normally, all routine tests as specified in the relevant standards shall be conducted by the sub supplier at his works.

M. Technical Data Sheet for panel:

Sr. No	Description	Recommended Specification
(1)	Type of Enclosure	Main Enclosure minimum of form 3b separation
(2)	Fault kA	65 kA -1 Sec.
(3)	Thickness of CRCA sheets	
a.	Structural uprights	2.0 mm or as per IEC 61439
b.	Covers and doors	2.0 mm or as per IEC 61439
c.	Base channel with anti-termite treatment	MS Channel 100mm x 50mm x 5 mm (Minimum)
d.	Gland plate	3.0 mm or as per IEC 61439
e.	Load bearing plate	2.0 mm or as per IEC 61439
f.	Painting/ Process	Epoxy Powder coating with seven-tank/Nine tank pre-treatment process.

g.	Paint shade; Inside & outside	Preferred RAL7035
(4)	Busbars	4000A, 65kA for 1 sec, Aluminium
(5)	Cable Entry	As per site requirement.
(6)	Enclosure Protection/ Ventilation	IP -43
(7)	Control Wiring/ Power Wiring	Insulated 660/1100Volts FRLS Cu wire. / Insulated 1100Volts FRLS Cu wire.
a.	Voltage Circuit (P.T.)	Minimum 2.5 sq mm
b.	Current Circuit (C.T.)	Minimum 2.5 sq mm
c.	Minimum size of Power wiring	4.0 sq mm
(8)	Cable Lugs / Cable Glands & termination	Tinned copper lugs and brass glands.
(9)	Operating Height	As per IEC 61439.
(10)	Type of CTs	Cast Resin type
(11)	Design Ambient Temp	45 Deg. Centigrade
(12)	Makes of the components to be used	
a.	Make of Panel / Switchboard	As per Approved by Engineer-In-Charge
b.	ACBs	As per Approved by Engineer-In-Charge
c.	MCCBs	As per Approved by Engineer-In-Charge
d.	Multi-function Meters	As per Approved by Engineer-In-Charge
e.	Ammeter/ Voltmeter selector switches	As per Approved by Engineer-In-Charge
f.	Wires	As per Approved by Engineer-In-Charge
g.	CT/PT	As per Approved by Engineer-In-Charge
h.	Control Fuses	As per Approved by Engineer-In-Charge
i.	Indication lamps/selector switch and other Equipments as required.	As per Approved by Engineer-In-Charge
(13)	Components rating	As per SLD approved by Engineer-in-charge.

2.6 DG SET

GENERAL:

1. The location of the DG sets shall be kept in such a way so that during the operation it should not disturb the campus day to day functioning. The DG sets will be kept in the open area near the substation building on suitable foundation and will be connected with suitable sizes of bus trunking/ cables as per CPWD specification to the AMF panel. AMF panel will be connected to Essential panel with suitable size of bus trunking/cables as per CPWD specifications. The incomers of the panels shall be properly interlocked with each other and with couplers to ensure any short circuit should not occur.
2. The DG set shall be silent type with acoustic enclosure and as per CPCB norms.
3. The silencer of the DG set shall be hospital Grade type.
4. The height of the exhaust pipe shall be as per CPWD specification and CPCB norms whichever is stringent. If required, supporting MS structure of suitable size shall also be provided which is included in the scope of work of EPC contractor.
5. Approval/clearance of the complete installation shall be obtained by the contractor from CPCB/state pollution Control Boards/ Local Bodies/ Central Electricity Authority (CEA)/other licensing authorities wherever required. Any statutory fee, as applicable, shall be paid by Contractor.
6. The tenderer shall submit detailed schematic diagram for the approval of the Engineer-In-Charge. He will make sure that the equipment offered shall fulfil the design conditions. All the equipments and their installation shall be suitable for the environmental conditions encountered at the location as indicated in specifications.
7. The entire work shall be carried out as per following CPWD General Specifications for Electrical works wherever applicable and as amended up-to-date. CPWD General Specification for Electrical works (Part- VII) DG Set 2013, CPWD General Specification for Electrical works (Part-I) Internal – 2023 CPWD General Specification for Electrical works, (Part-II) External - 2023 CPWD General Specification for Electrical works, (Part-IV) Sub-Station- 2013 as amended up to date.

8. Indian Electricity Act and Rules:

All electrical works in connection with installation shall be carried out in accordance with the provision of Indian Electricity Act 2003 and Indian Electricity Rules 1956, both amended up-to date.

9. Indian Standards:

All the components shall confirm to relevant Indian Standard specifications, wherever existing, amended to date

10. Fire Regulation:

The installation shall be carried out in conformity with the local Fire Regulations and Rules there under wherever they are in force and the provisions in local bye-laws, if any.

11. Safety Code and Labour Regulation:

In respect of all labour employed directly or indirectly on the work for the performance of contractor 's part of work, the contractor at his own expense, will arrange for the safety provisions as per the statutory provisions, B.I.S. recommendations, factory act, workman 's compensation act, CPWD Code and instructions issued from time to time. Failure to provide such safety requirements would make the EPC contractor liable for penalty for each violation. In addition, the Engineer-in-Charge, shall be at liberty to make arrangements and provide facilities as aforesaid and recover the cost from the contractor.

The contractor shall provide necessary barriers, warning signals and other safety measures while executing the work of DG set installation, transformer installations, HT& LT Panel installations, HT< cables etc. or wherever necessary so as to avoid accident. He shall also indemnify RITES against claims for compensation arising out of negligence in this respect. Contractor shall be liable, in accordance with the Indian Law and Regulations for any accident occurring due to any cause. The department shall not be responsible for any accident occurred or damaged incurred or claims arising there from during the execution of work. The contractor shall also provide all insurance including third party insurance as may be necessary to cover the risk. No extra payment would be made to the contractor due to the above provisions thereof.

12. Central Pollution Control Board (CPCB) NORMS:

The DG Sets shall be meet the requirements of environmental of latest CPCB norms regarding emission and noise norms amended up to date.

The firm need to furnish the certificate of Type approval and certificate of conformity of production issued by the Central Pollution Control Board (CPCB) or any authorized agency as prescribed in the above law.

DG set Scope of work

The scope of work shall cover complete supply, transportation, foundation, installation, testing and commissioning of 380 KVA dg set. All minor civil works including Foundation, electrical and other works associated with the testing, installation and commissioning of the sets shall be carried out by the contractor as per specification and drawings approved by Engineer-in-charge.

The tenderer should quote for complete job to be executed under the works contract. The contractor is advised to inspect the site to obtain first-hand information of all site conditions before tendering. Item included (but not limited to) in the contract are abstracted below:

- a) Diesel engines directly coupled with alternators mounted on a rigid fabricated base frame with resilient anti vibration mountings.
- b) AMF Panel for DG Set.
- c) Auto Synchronizing cum load sharing panels.
- d) DG cooling system as per relevant norms.
- e) Exhaust piping as per norms.
- f) Acoustic Enclosure as per CPCB Norms.
- g) Erection, testing and final checking up of the installation at site, commissioning and putting the DG set into successful operation in co-ordination with Main Panel cum AMF Panel.
- h) The DG Sets shall be capable of delivering desired KVA at 0.8 PF for the external load after meeting its own requirement for auxiliaries etc. and after accounting for de-rating due to various factors, site conditions.
- i) Necessary earthing, comprising of system neutral earthing, Body earthing and loop earthing etc.

Diesel Engine: -

1. Rating: -

- a. The engines shall be direct injection 6 cylinder, In line, 4 stroke cycles, radiator cooled, turbo charged diesel engine developing suitable BHP for giving desired prime power output at 0.8 PF at the load terminals of the respective alternators (exclusive of the power requirements of auxiliaries deriving power from the engine) (as per ISO 8528 part-I).
- b. The engines shall be capable for delivering specified prime power rating at variable loads for PF of 0.8 lag with 10% overload available more than specified output for one hour in every 12 hours. The average load factor of the engine over period of 24 hours shall be 0.85 (85%) for prime power output. The engines shall conform to BS 5514/ BS 649/ IS10000/ ISO: 3046 amended up to date.

- c. Necessary certificates indicating the compliance of the above capacity requirement for the engine model so selected along with compliance of Noise and Emission norms as per latest CPCB guidelines for DG set capacity should be furnished from the manufacturers.
- d. The engine shall be fitted with following accessories subject to the design of the manufacturer:
 - i. Dynamically balanced Fly wheel
 - ii. Necessary flexible coupling and guard for alternator and engine (applicable only for double bearing alternator)
 - iii. Air cleaner (dry/oil bath type) as per manufacturer standard.
 - iv. A mechanical/electronic governor to maintain engine speed at all condition of load. (see Clause 2.2.1.4 of CPWD General Specification for Electrical Works Part-VII, DG Set-2013, Page-16).
 - v. 990 ltr. daily fuel service tank, fabricated from minimum 3mm M.S sheet with inlet, outlet connections air vent tap, drain plug and level indicator(gauge) M.S. fuel piping from tank to engine with valves, unions, reducers, flexible hose connection and floor mounting pedestals, twin fuel filters and fuel injectors. The location of the tank shall depend on standard manufacturers design.
 - vi. Dry exhaust manifold with suitable exhaust hospital grade silencer to reduce the noise level.
 - vii. Suitable self-starter for 12 V/24 V DC.
 - viii. Battery charging alternator unit and voltage regulator, suitable for starting batteries, battery racks with interconnecting leads and terminals.
 - ix. Necessary gear driven oil pump for lubricating oil, priming of engine bearing as well as fuel systems as per manufacturer recommendation.
 - x. Naturally aspirated/turbo charger (as per manufacturer standard).
 - xi. Lubrication oil cooler
 - xii. Lubrication oil filters with replaceable elements.
 - xiii. Crank case heater as per manufacturer recommendations.
 - xiv. Fuel injection: Engine should have suitable fuel injection system in order to achieve low fuel consumption.
 - xv. Fuel control solenoid.
 - xvi. Fuel pump with engine speed adjustment.
- e. **Engine Control Panel:** Microprocessor based generator set monitoring, metering and control system with LCD display for following: -
 - Start/stop key switch.
 - Remote start-stop
 - Lube oil pressure Indication
 - Water temp indication

- RPM indication
- Engine Hours Indications
- Battery charging indication
- Low lub. Oil trip indication
- High water temp. indication
- Over speed indication
- Engine Hours indication
- Engine Metering: Oil pressure, Engine temperature, starting battery voltage, Engine running hours.
- AC Alternator Metering: L-L Voltage and L-N Voltage, Current (phase and total), kVA (phase and total) and Frequency. kwh, Total & per phase (kw & kVA), PF, Utility Voltage and Frequency.
- Engine Protection: Low lube oil pressure, High/Low coolant temperature, Battery High/Low/Weak Volts, Fail to Crank/Start, Sensor failure, Cranking lockout, Low fuel level.
- AC Alternator Protection: Over/Under Voltage, Over/Under Frequency, Loss of AC sensing. Overspeed, Over Current, kW Overload.
- Data Logging: Engine hours, Control hours and up to 5 recent fault codes.
- 12/24 Volt DC operation.
- Sleep mode
- Modbus interface (RS485 RTU)
- In Power compatible (PC based service tool)
- Certifications - meets the requirement of relevant ISO, EN,
- Mil Std. and CE standards
- Maintenance due alarm based on Engine Run Time and due date.
- AMF Functionality
- Electronic Governing
- f. All moving parts of engine shall be mechanically guarded in such a manner that a human finger cannot touch any moving part.
- g. Radiator
- h. Bus bars with suitable capacity with incoming/outgoing terminals.
- i. Indicating lamps for 'Load ON' and 'Set Running' Instrument fuses duly wired and ferruled.
- j. MCCB of suitable rating with short circuit protections.
- k. AC/DC separation inside control panel for safety.
- l. Any other item not included/ specified but is a standard design of the manufacturer

2. Fuel System

It shall be fed through engine driven fuel pump. A replaceable element of fuel filter shall be suitably located to permit easy servicing. The daily service tank shall be complete with necessary supports, gauges, connecting pipe work etc. Pipe sealant should be used for sealing all connections. No Teflon tape is to be used. If pipeline is greater than 10mtr the detailed Engineering is required in consultation with OEM.

3. Lubricating oil system

It shall be so designed that when the engine starts after a long shut down lubrication failure does not occur. Necessary priming pump for the lub. oil circuit as per recommendation of manufacturer shall be installed, to keep bearings primed.

4. Starting System: -

This shall comprise of necessary set of heavy-duty batteries (four nos. 12 V D.C.), and suitable starter motors, axial gear to match with the toothed ring on the fly wheel. Battery capacity shall be suitable for meeting the needs of the starting system as well as the requirements of control panel, indications, and auxiliaries such as priming pump etc. the scope shall cover all cabling, terminals including initial charging etc. The system shall be capable of starting the DG Set within 20-30 sec, even in winter condition with an ambient temperature down to 00 C.

5. Piping work: -

All pipe lines and fittings and accessories required inside the enclosure and outside for exhaust piping shall be provided by the contractor. This shall include necessary flexible pieces in the exhaust, fuel, lube oil and water as are necessary in view of the vibration isolation requirement in the installation. Piping of adequate size shall be used for lub. Oil of the materials as per manufacturer standard. However, only MS pipes for the exhaust and fuel oil lines shall be used. For fuel lines within the acoustic enclosure, PVC braided pipe as per manufacturer recommendations can be used. However, for fuel lines outside the acoustics enclosure only. The MS pipe be used shall be as per pipe work clause no. 2.2.1.10 of CPWD Specs.

The pipe work shall be inclusive of all fittings and accessories required such as bends, reducers, elbows, flanges, flexible connections, necessary hardware 's etc. The installation shall cover clamps, supports, hangers etc. as are necessary for completing the work. Welding or brazing will be permitted in the installation. However, the work shall be sectionalized with flanged connections as are necessary for easy isolation for purposes of maintenance of unit as approved by Engineer-in-charge.

All M.S. pipe work shall be medium class. The work includes necessary lagging of exhaust pipe work using Rock wool & AL sheet Cladding.

6. Common bed plate

Engine and alternator shall be coupled by means of flexible coupling and both units shall be mounted on a common bed plate together with all ancillaries to ensure perfect alignment of engine and alternator with all minimum vibrations. The bed plate shall be such that it will be feasible to remove the engine sump without disturbing the engine assembly. Fabricated bed plate will be acceptable. The bed plate shall be suitable for installation on suitable antivibration mounting system.

7. Battery charger

The battery charger shall be suitable to charge required numbers of batteries at 12V/ 24 volts complete with, transformer, rectifier, charge rate selector switch, indicating ammeter & voltmeter etc. Connections between the battery charger & batteries shall be provided with suitable copper leads with lugs etc.

8. Exhaust System

- i. **Exhaust piping:** All MS pipes for exhaust lines shall be confirming to relevant IS. The runs forming part of factory assembly on the engine flexible connections up to exhaust silencer shall be exclusive of exhaust piping item. The work includes necessary cladding of exhaust pipe and insulation work. The exhaust pipe support structure shall be got approved by engineer-in-charge before execution.
- ii. Exhaust system should create minimum back pressure.
- iii. Number of bends should be kept minimum and smooth bends should be used to minimize back pressure.
- iv. Pipe sleeve of large dia should be used while passing the pipe through concrete wall & gap should be filled with felt lining.
- v. Exhaust piping inside the Acoustic Enclosure should be lagged with asbestos rope along with aluminium sheet cladding to avoid heat in put to the enclosure.
- vi. Exhaust flexible shall have it 's free length when it is installed.
- vii. Class C 'MS pipes and long bend / elbows should be used.
- viii. The exhaust outlet should be in the direction of prevailing winds and should not allow exhaust gases to entire air inlet / windows etc.
- ix. When tail end is horizontal, 45 Degree downward cut should be given at the end of the pipe to avoid rain water entry into exhaust piping.

- x. When tail end is vertical, there should be rain trap to avoid rain water entry. If rain cap is used, the distance between exhaust pipe and rain cap should be higher than diameter of pipe. Horizontal run of exhaust piping should slope downwards away from engine to the condensate trap. Silencer should be installed with drain plug at bottom.
- xi. Optimum Silencer Location: Location of the silencer in exhaust system has very definite influence on both reduction of noise and back pressure imposed on the system.
- xii. Care should be taken to ensure that no carbon particle emitted due to exhaust leaking enters and deposits on alternator windings and on open connections.
- xiii. Support to exhaust piping: Exhaust piping should be supported in such manner that load of exhaust piping is not exerted to turbo charger.

9. Cooling System

System should be designed for ambient temperature of 40 Deg. Celsius Coolant should be used mixed with additive (in suitable proportion) as per recommendation of OEM/ Manufacturer for various engine models. Radiator fan flow should be free from any obstruction. DG Set shall be heat exchanger cooled.

10. Alternator

a. Synchronous Alternator:

Self-excited, screen protected, self-regulated, brush less alternator, Horizontal foot mounted in Single/Double bearing construction (specify one only) suitable for the following:

Rated PF	: 0.8 (lag)
Rated voltage	: 415
Rated frequency	: 50 Hz
No. of phases	: 3
Enclosure	: SPDP
Degree of protection	: IP-23
Ventilation	: Self ventilated air cooled
Ambient Temperature	: 40°C Maximum
Insulation class	: H
Temperature Rise	: Within class F limits at rated load
Voltage Regulation	: $\pm 1\%$
Voltage variation	: $\pm 5\%$
Winding Pitch	: 2/3 Pitch
Stator Winding	: Double layer concentric

Wave form distortion	: No load < 1.8%, non-distorting balanced linear load < 5%
Overload duration/ capacity:	: 10% for one hour in every 12 hours of continuous use.
Frequency variation	: As defined by the Engine Governor (+/- 1%)
Total Harmonic Factor	: Better than 2%.
Excitation	: Separately excited.
Type of AVR	: Electronic
Type of Bearing and Lubrication	: Anti-friction bearing with Grease Lubrication arrangements.
Standard	: BS:5000/IS-4722 & IEC: 34 as amended Up to date.

b. Alternator should be able to deliver output rating at actual site conditions.

c. Excitation: The alternator shall be brushless type and shall be SELF/separately excited self-regulated having static excitation facility. The rectifier shall be suitable operation at high ambient temperature at site. The excited unit shall be mounted on the control panel or only alternator assembly.

d. Fault Tripping: In the event of any fault e.g. over voltage/ high bearing temperature high winding temperature or an external fault, the AVR/ genset controller shall remove the excitation voltage to the alternator. An emergency trip shall also be provided.

e. Standards: the alternator shall be in accordance with the following standards as are applicable. IS: 4722/BS: 2613 / 1970. the performance of rotating electrical machine. IS: 4889/ BS: 269 rules for method of declaring efficiency of electrical machine.

f. Performance: Voltage dip shall not exceed 20% of the rated voltage for any step load or transient load as per ISO:8528 (part I). The winding shall not develop hot spots exceeding safe limits due to imbalance of 20% between any two phases from no load to full load.

The generator shall preferable by capable of withstanding a current equal 1.5 times the rated current for a period of not more than 15 second as required vide clause 14.1.1 of IS 4722: 1992. The performance characteristics of the alternator shall be as below:

Efficiency at full load 0.8 PF	Not less than 93.5%
Total distortion factor	Less than 3%

(i) 10% overload	One hour in every 12hrs of continuous use
(ii) 50% overload	15 seconds

- g. Terminal Boxes:** terminal boxes shall be suitable for terminating minimum copper/ aluminium cables. The terminal box shall be suitable to withstand the mechanical and terminal stresses developed due to any short circuit at the terminals.
- h. Earth terminals:** 2 nos. earth terminals on opposite site with vibration proof connections, nonferrous hardware etc. with galvanized plate and passivated washer of minimum size 12mm dia hole shall be provided.
- i. Battery/ Electrical System:** - Batteries should be placed on insulated stands and relatively at cool place. The battery should be (four Nos) 180 AH capacity 12 Volt, 27 plate and the size of copper conductor battery cable should be of 50 sq.mm or suitable size as per engine manufacturer.

11. Cabling

Copper armoured cable between alternator to AMF panel should be done. Multi core armoured copper cable should be used for inter connecting the engine controls with the switch gear and other equipments. It is recommended to support output cables on separate structure on ground so that weights of cables should not fall on alternator/ base rail.

12. Foundation scope:

This section covers details of foundation for DG set with or without acoustic enclosures.

A. Genset Location and foundation:

- DG Sets shall be installed in open area at site near substation building.
- For DG Sets installed outside in open area- A PCC (1:2:4, grade) foundation of weight 5 times the operating weight of the Genset with enclosure or as recommendation by the Genset manufacturer OEM/OEA, whichever is higher, is required to provided and is included in scope of work for SITC of Genset. 300mm of this foundation height should be above the ground level. The length and breadth of foundation should be at least 250 mm more on all sides than the size of enclosure. Genset should be mounted on AVM's inside the enclosure.
- Design of the foundation as recommended by the OEM shall be submitted by the contractor before execution of work along with the drawings as mentioned in section above. Foundation

should be designed considering safe bearing capacity of soil. Vibration isolators (AVMs) should be provided to reduce vibration transmission to the surrounding structure.

13. ACOUSTIC ENCLOSURE

Scope: This section covers technical requirement of the acoustic enclosure.

i. As per CPCB norms, restriction has been imposed for new DG sets up to 1000 KVA for noise level (see Appendix 'II'). Therefore, in terms of these norms, acoustic enclosure should be type tested at the climatic conditions specified in para 2.1.4 through one of the authorized laboratories.

ii. Installation

- a. Acoustic enclosure shall be supplied with built in Anti Vibration Mountings (AVMs). As such Genset can be installed directly on the level surface.
- b. Exhaust piping outlet should not be turned towards window/ventilator of home or occupied building. Provision of rain cap should be ensured.
- c. The acoustic enclosure placement s technical should be such that there is no restriction in front of air inlet and outlet from canopy.

iii. Service Accessibility.

- a. Genset/Engine control panel should be visible from outside the enclosure.
- b. Routine/periodical check on engine/alternator (filter replacement and tappet setting etc) should be possible without dismantling acoustic enclosure.
- c. For major repairs/overhaul, it may be required to dismantle the acoustic enclosure. Sufficient space should be available around the Genset for inspection and service.

iv. General Design Guidelines.

- a. To avoid re-circulation of hot air, durable sealing between radiator and canopy is must.
- b. Ventilation fans are must for the Gensets cooled by heat exchanger/cooling tower system. Exhaust piping inside the enclosures must be lagged (except below).
- c. Temperature rise inside the enclosure should not be more than 5°C for maximum ambient above 40° C and it should be below 10° C for ambient below 40° C.
- d. There should be provision for oil, coolant drain and fill. Fuel tank should have provision for cleaning.
- e. The enclosure should be designed to meet the total air requirement for the D.G Set at full load at site condition as recommendation by the engine manufacture.

v. Specifications for Acoustic Enclosure.

- a) The acoustic enclosure shall be designed and manufactured confirming to meet stringent MoEF / CPCB norms & relevant standards suitable for outdoor installation exposed to weather conditions, and to limit overall noise level to 75 dBA at a distance of 1mtr at 75% load under free field conditions.
- b) The construction should be such that it prevents entry of rain water splashing into the Enclosure and allows free and quick flow of rain water to the ground in the event of heavy air. The detailed construction shall confirm to the details as under.
- c) The enclosure shall be fabricated out of the CRCA sheet of thickness not less than 1.6 mm on the outside cover with inside cover having not less than 0.6mm thick perforated powder coated CRCA sheet.
- d) The hinged doors shall be made from not less than 16 SWG (1.6 mm) thick CRCA sheet and will be made air tight with neoprene rubber gasket and heavy-duty locks.
- e) All sheet metal should be processed through 9-tank process.
- f) The enclosure should be powder coated.
- g) The enclosure should accommodate the daily service fuel tank of the DG Set to make the system compact. There should be provision of gauge, which should show the level of the fuel even when the DG set is not running. The gauge should be calibrated. The fuel tank should be filled from the outside as in automobiles and should be with a lockable cap.
- h) The batteries should be accommodated in the enclosure in battery rack.
- i) The canopy should be provided with high enclosure temperature safety device.
- j) The acoustic lining should be made up of high-quality insulation material i.e. glass / mineral wool/PU foam appreciated thinness & density for sound absorption as per standard design of manufacturer 's to reduce the sound level as per latest CPCB norms. The insulation material shall be covered with fine glass fibre cloth and would be supported by perforated M.S. Sheet duly powder coated/GI sheet/aluminium sheet.
- k) The enclosure shall be provided with suitable size & No. of hinged type doors along the length of the enclosure on each side for easy access inside the acoustic enclosure for inspection, operation, and maintenance purpose. Sufficient space will be provided inside the enclosure on all sides of the DG set for inspection, easy maintenance, and repairs.
- l) The canopy should be as compact as possible with good aesthetic look.
- m) The complete enclosure shall be of modular construction.
- n) The forced ventilation shall be as per manufacturer design using either engine radiator fan or additional blower fan(s). If the acoustic enclosure is to be provided with forced ventilation, then suitable size of axial flow fan (with motor and AutoStart arrangement) and suitable size axial flow exhaust fan to take the hot air from the enclosure complete with necessary motors and auto start arrangement should be provided. The forced ventilation arrangement should be provided with auto stop arrangement to stop after 5 minutes of the stopping of DG Sets.

- o) The acoustic enclosure should be suitable for cable connection. Such arrangements on acoustic enclosure should be waterproof and dust-proof confirming to IP-65 protection.
- p) The inside of enclosure should be provided with at least two nos. 20W-LED Tube light luminaries controlled by a 5 A switch for adequate lighting during servicing etc. of the DG Set. The power supply to this luminaire should be from the load side of the AMF Panel so that it can remain energized under all conditions.

14. AMF Control Panel

Design, supply, installation, testing & commissioning of AMF control panel with suitable rating of ATS shall be fabricated out of 1.6 mm thick sheet steel, totally enclosed, dust, damp, and vermin proof free standing floor mounted type & front operated. It shall be made into sections such that as far as feasible, there is no mixing of control, power, DC & AC functions in the same section and they are sufficiently segregated except where their bunching is necessary. Hinged doors shall be provided preferably double leaf for access for routine inspection from the rear. The panel shall be 3B separation, complete in all aspect as per relevant IS / IEC Specifications. There is no objection to have single leaf hinged door in the front, all indication lamps, instruments meter etc. shall be flushed in the front. The degree of protection required will be IP-54.

2.7 Sandwich insulated Bus Duct Trunking & Rising Mains (IP-66 rating, Class F insulation):

It shall be selected as per CPWD specifications for Electrical Works Part-1, Part – IV, Section -6, ECBC-2017, NBC-2016, and conforming to IEC-61439 Part 1 & Part 6 requirements. However, minimum/indicative rating of bus duct & rising main for substation & buildings are shown in SLD electrical drawing of the building. **(Ref. SLD drawing No. RITES/BLD/EL/TEN/216 to 217).**

2.8 Sub- Station Earthing Scheme: Earthing system comprising of earth electrode, earth conductor, earth bus, protective conductor etc. for the buildings shall be as per provision laid down in NBC 2016 / BIS Standards IS 3043:2018 and IS 732:2019 and CPWD specifications for electrical works. Earthing system should be designed such as to maintain required earth resistance as specified in the above standards.

2.9 All earthing shall be done with **copper plate earthing** unless otherwise specially mentioned for Substation equipment. Copper plate earthing system comprising of earth electrode, earth conductor, earth bus, protective conductor etc. Earthing for building panels shall be as per provision laid down in CPWD specifications part – I, 2013. Earthing system should be designed such as to maintain earth resistance as specified in CPWD specifications. Earth resistance shall be checked / tested in harsh climatic conditions. EPC contractor shall arrange for checking of

the earth resistance to engineer from RITES /OIL.

2.10 Protection of Substation & Buildings against Lightning: EPC contractor shall design the lightning protection scheme as per CPWD Specifications for electrical works (Part-I), NBC-2016 for the substation and all the buildings under the scope of this work. The drawing for the same shall be submitted to Engineer-In charge for approval.

2.11 Internal Lighting & other services for Substation: The EPC contractor shall provide LED lighting fixtures, BLDC ceiling fans, exhaust fans, emergency lighting, fire suppression system in panels, illuminated signages, safety equipment & other necessary services as required for electrical substations as per CPWD specifications.

Note:

- (i) The drawings of the substation layout plan and technical data sheets/specifications of its equipment i.e., Dry type transformers, HT & LT Panels, floor distribution panels, rising mains, APFC Panels if required, HT & LT cables, DG set, lighting system, earthing, safety equipment, 30 minutes. Battery backup system for HT Panel, internal/External lighting, feeder pillars, fire suppression system etc. must be got approved from Engineer-in-charge before fabrication work / procurement.
- (ii) No extra payment will be made to the EPC Contractor for electrical equipment testing at site.

3. AUTOMATIC LINEAR PNEUMATIC HEAT DETECTION TUBE GAS SUPPRESSION SYSTEM (WITH HEAT DETECTION TUBE)

A. GENERAL SITE CONDITIONS:

This section includes provision of Automatic Linear Pneumatic Heat Detection Tube gas suppression system of approved clean gases agent having global warming potential (GWP) ≤ 1 , zero ozone depletion potential (ODP) and atmospheric lifetime less than 10 days complying NFPA-2001 (latest edition). Letter of OEM authorization is compulsory for supply required by the contractor.

B. STANDARD & CODE

Following Indian / international standard specifications, codes of practice updated shall apply to the equipment and the work covered by the scope of this contract.

NFPA 2001	Standard on clean agent fire extinguishing system
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UL 521	UL Standard for Heat Detectors for Fire Protective Signaling Systems
LPS 1666	Direct Low-Pressure application for fixed fire suppression
1) GCR 2016	2) Gas Cylinder Rules 2016

C. SCOPE OF WORK

This scope of broadly covers the following:

- i. Design and provide clean Agent Automatic Linear Pneumatic Heat Detection Tube Fire Gas Suppression system in a uniform concentration within electrical panel/ section(s) in accordance with NFPA2001 and system complying theLPS-1666 Standard tested at NABL approved national test house laboratory.
- ii. Provide sufficient amount of extinguishing agent FK 5-1-1-2/IG-541, considering the following key points when computing volume to verify suitability and to establish design limitations:
 - Volume of hazard area.
 - Specific volume of clean agent.
 - Discharge time and flow rates.
 - Design Concentration and design factors.
 - Detector/ discharge tubing placement in co-ordination with panel / equipment drawings
- iii. The system shall be designed to enable a single container heat detection tube installation run to protect a maximum single volume of 2m³.
- iv. Up to 4 Heat detection tube runs maybe connected to a single container where no single protected volume exceeds a 2m³ of volume.
- v. The maximum heat detection tube length from the container outlet to the end of any single detection tube shall not exceed 10m.
- vi. Provide all engineering design and materials for a complete agent suppression system including storage cylinders with steel bracket, extinguishing agent, detection tube, cylinder valve and associated accessories including but not limited to; adaptors, pressure switch, tube fittings complete as required for successful & safe operation of the system.
- vii. Provision of all necessary safety requirements such as - warning signs, discharge alarm etc. as a

part of the system.

- viii. Master control unit shall consist of hooter cum flasher, inbuilt 12 V battery pack (of minimum 30 minutes back up), two NO & NC contact, charging circuit, extension of power supply from nearest power point, connection / inter-connection complete suitable for system activation.
- ix. The necessary nomenclature such as pressurization level, agent volume, gross /net weight of cylinder and filling date etc. shall be clearly marked on cylinder.
- x. All doors and holes in the enclosed / equipment should be closed or sealed to maintain the tightness of enclosure. Therefore, executing agency has to coordinate with manufacturer of electrical panels for provision of holes to run the tube and brackets for mounting the tube during panel fabrication. The entry of tube inside the panel shall be through suitable size of connectors.
- xi. Contractor has to submit detailed drawings & gas calculations sheet based on NFPA 2001 (Latest Edition) & specifications for approval of Engineer-in-Charge.

D. SYSTEM DESCRIPTION

- i. The detection tube shall be installed throughout the compartments of panel viz bus bar section, switchgear section, cable alleys and back of the panel section (if provided) as per system design requirement. The location and spacing of tube shall be above the hazard, to be protected.
- ii. The detection tube shall be fixed with PESO approved cylinder & valve. The valve shall be at top of cylinder. The tube shall be pressurized with dry nitrogen to 15 Bar. In case of reach of pre-determined temperature as specified in UL certification, the tube shall rupture at a point and gas shall be released from tube over the protected area in electrical panel section (s)/ compartment in uniform pattern.
- iii. The detection tube shall be capable of working even when contaminated with oil, dust, and debris as long as the contamination will allow the heat to pass through the tube.
- iv. The cylinder shall be equipped with brass valve, pressure gauge (to monitor agent pressure) and isolation valve for maintenance purposes. The cylinder valve shall be with a safety burst disc of 28 Bar rupture Pressure.
- v. The Extinguishing Agent shall be stored in cylinder as liquefied compressed gas, super pressurized with dry nitrogen at (195-220) psi.
- vi. Pressure switch shall have necessary indication of discharge of gas with NO & NC contact for interaction with Main Fire Alarm Panel. switch points 8 bars.
- vii. With the system activation, master control unit with hooter cum flasher shall generate audio visual alarm signal for electrical panel/ compartment is covered.
- viii. System shall be so designed & provided that it does not affect the IP ratings of electrical panels.

E. SYSTEM COMPONENTS

- i. Petroleum and explosives safety organization (PESO) approved Cylinder of steel construction with standard red epoxy paint finish. Cylinders shall be accompanied by original manufacturers test certificate confirming the contents of the cylinder and filled in PESO Approved plant.
- ii. Each cylinder shall have pressure gauge and integrated low-pressure switch to provide visual and electrical supervision of the cylinder pressure. The low-pressure switch shall be wired to the Audio-Visual Alarm to provide audible and visual trouble alarm in the event of drop of pressure. The pressure gauge shall be colour coded to provide an easy, visual indication of cylinder pressure.
- iii. The cylinder shall be equipped with brass valve, pressure gauge (to monitor agent pressure) and isolation valve for maintenance purposes. The cylinder valve shall be with a safety burst disc of 28 Bar rupture Pressure.
- iv. Furnish a welded steel bracket with each cylinder assembly for holding the cylinders in a saddle with a front bracket piece that secures the cylinders.
- v. Cylinder shall be provided with a certificate provided by the company who charge with the gas mixture. The certificate shall be secured around the cylinder with chain fastener.
- vi. The Detection Tube shall be UL approved red / black in Colour and pressurized as per system design requirement.
- vii. The integrated Pressure Switch (CE/UL/LPCB marked) should be integral part of valve having NO / NC contact.
- viii. The control box enclosure shall be at least IP 54 rated from OEM of gas suppression system provider complying standard. The unit shall have LED indication for system Healthy, Mains Healthy and Gas Release. The unit shall also have signal silence button. Third Party test reports for IP and Functionality of the system to be submitted for approval complete as required.

F. EXTINGUISHING AGENT

Extinguishing Agent shall be UL approved FK-5-1-12/IG-541 in technology as applicable at the time of execution of work shall be acceptable with –

- a. Extinguishing agent shall be UL listed/ FM approved.
- b. The ozone depletion potential should be zero.
- c. Global warming Potential should be less than equal to 1
- d. The clean agent shall be specified in NFPA 2001 having 99.9% purity and Certificate of conformity from OEM to be provided.

- e. The extinguishing agent should be filled in a PESO approved filling station.

G. INSTALLATION

- i. The system shall be installed as per approved shop drawings.
- ii. The installation/final connections shall carry out in direct supervision of representative of Manufacturer/ specialized agency.
- iii. Cylinder shall be located so that they are not subjected to mechanical, chemical, or other damage.
- iv. All system components shall be capable of withstanding heat of fire and severe weather conditions.
- v. Detection Tube to be properly secured inside the panel by Clips/ Tieck.
- vi. The Detection Tube outside the panel should be protected in flexible conduit.
- vii. In section certificate should be pasted on cylinder clearly marking next due date of inspection.

H. TESTING AND ACCEPTANCE

Testing of the system shall be done at site by holding the Nitrogen Pressure at 14 Bar into the detection tube for 24 hrs. to check for any pressure leakage of the system. If there is no pressure leakage of the system then, the installed system is accepted as healthy and the isolation valve shall be opened.

I. DOCUMENTATION

The contractor to submit the shop drawings & technical submittals of automatic Linear Pneumatic Heat Detection tube fire suppression system for approval of EIC and got approved by the Engineer-in-charge before commencement of work / procurement of any items.

- a) General arrangement (GA) drawing of each system shall be supported with:
 - Shop drawings shall clearly show routing of tube inside the panel as well as in equipment compartments as per system design requirements, location & fixing arrangement of cylinder & its system components, cut out details in panels etc. as required.
 - Control & Schematic Diagram
 - Bill of Materials/quantities indicating makes, technical specs, quantity etc.
 - Data sheets (where applicable)
 - Control logic and write up (where applicable)
 - Technical leaflets / specifications.
- b) Design/ selection sheet of **Automatic Linear Pneumatic Heat Detection Tube** gas suppression system referring relevant standard to be submitted for approval of Engineer-in-charge.

- c) The shop drawings shall also be submitted for manufacture of panel as well as execution of the works at site.

4. PASSENGER/STRETCHER ELEVATORS (LIFTS)

The EPC Contractor shall carry out design, supply, installation, testing & Commissioning of Elevator (Lift) Works including lift license of all elevators (lifts). Passenger cum Bed elevators (lifts)/stretcher elevators (lifts) and Passenger Elevators (Lifts) shall be provided for Residential housing blocks and Guest house building in the complex. The installation shall be carried out as per statutory rules & regulations stipulated by CPWD/NBC-2016 and IS Codes that govern the installations of the elevator (lift). The voltage and frequency of the supply shall subject to variation permissible under Indian Electricity Act and Rules.

Passenger Elevators (Lifts)/Bed Elevators (lifts) (Machine room Less type)			
Sr.No.	Name of Building	Details of Passenger Elevators (Lifts)	Quantity
1.	Residential Building-G+6 (08 Block)	15 passengers elevator (lift), Speed - 1.5m/sec., Stop - 7 stops	08 Nos.
2.	Residential Building-G+6 (08 Block)	15 passengers/Stretcher elevator (lift), Speed -1.5m/sec., Stop – 7stops	08 Nos.
3.	Guest House Building-G+3	15 passenger/stretcher elevator (Lift), Speed- 1 m/sec., Stop- 4 stops	01 Nos.

The work shall be executed as per CPWD General Specifications for Electrical Works (Part III Lifts & Escalators -2003), NBC-2016, ECBC-2017 (Latest amendments), as per relevant IS as per directions of Engineer-in-charge.

IP-based CCTV camera (1 No.) (minimum 2MP) shall be provided in all Elevator (Lift) Cars of all buildings for surveillance/monitoring of elevator (lift) control panel. The CCTV cameras, installed in the elevators (lifts), shall be connected/ integrated with the IP based CCTV system with suitable cabling, hardware items etc. as required. Suitable warning mentioning “You are under CCTV surveillance” shall be posted inside the elevator (lift) car. These cameras may be located at ceiling level and at any location which cannot be easily accessed, noticed, or tampered with. The cable used

for connecting in car camera to monitoring system, shall be of trailing grade. Ordinary co-axial or twisted pair wires shall not be tied along with trailing cables. Specialized trailing cables shall be used in lift applications. Wireless CCTV cameras complete with transmitter/ receiver & all accessories as required, may also be provided alternatively.

- Lift Well, Car Size, Elevator (Lift) Pit Depth, Overhead, and Clear Entrance Width & Height dimensions shall conform to NBC 2016 / IS Code as amended up to date. All elevators (lifts) shall be Machine room less type.
- Anti-skid granite or Stainless Steel 304 chequered plate flooring of suitable thickness shall be provided in all the elevators (lifts).
- Car enclosure & doors shall be made of SS 304 sheet of required thickness.
- All elevator (lift) shall have necessary provisions & door opening as required for physically challenged person.
- Power supply to each elevator shall be connected to emergency source.
- Elevator (Lift) Car operating Panel shall be equipped with Braille buttons. Automatic rescue device and emergency lighting shall be provided in each elevator supported by independent rechargeable batteries. However, the battery capacity should be adequate so as to operate the ARD at least seven times a day provided the duration between usage is at least 30 minutes.
- All Elevator (Lifts) shall have provision of integrated with BMS System for normal operation status & operation in case of fire. Elevator (Lift) shall be compatible with Modbus /BACnet communication protocol or as per OEM requirement.

GENERAL

The equipment and installation covered by these specifications shall conform to codes of practice in force and highest standards of workmanship and materials. This work shall be done in accordance with the provisions of the Local Lifts Authority rules and shall also conform to requirements of local municipal by laws, and subsequent provisions, as also any state or local Act in force and latest Indian Standard 14665 and all latest applicable BIS, NBC code and 'CPWD General Specifications for Electrical Works (Part III, Lifts & Escalators) 2003' as amended up to date.

The Entire electrical installation shall be done in accordance with the Indian Electricity Act 2003, Indian Electricity Rules 1956 as amended to-date. The Electrical wiring shall strictly comply with IS: 732 and latest applicable BIS and NBC code. The electrical works shall also conform to CPWD General Specification for Electrical Work Part-I (Internal) 2023 as amended up to date. The Contractor shall follow all Statutory Requirements as well as best trade practices in the manufacture & installation of elevators (lifts). The Contractor shall arrange to obtain the statutory approval of the Inspectorate of Lifts as may be required for commissioning of the elevators (lifts) and handover for operation after satisfactory tests.

WORKS TO BE ARRANGED

The following items shall be provided to the Elevator (Lift) Contractor under instructions of the Department to suit the requirements of the lift Contractor.

- i. Hoist ways and pits of specified dimensions (within normal building tolerances). The elevator (lift) shall be Machine Room Less type as per architectural drawings, Specific panel for Elevator (Lift) shall be in contractor scope.
- ii. Floor, wall and ceiling finishes in hoist ways, pits including painting (except painting of equipment and materials supplied by lift Contractor) and waterproofing, as well as maintenance windows in elevator (lift) well room.
- iii. Cables from Panel Board through the hoist ways terminating in and including individual Main Switches of required rating for 3 phase and single-phase supply including necessary earthing.
- iv. 3 phase power supply for group testing and commissioning of elevators (lifts) after erection is completed.
- v. Lighting installation within hoist ways and pits as required by the lift Contractor including 1-phase main switch at machine room.
- vi. Ventilation system of machine rooms with heavy duty exhaust fan in each machine room as per the requirement of NBC / BIS codes.
- vii. Providing of hoisting beam in the machine room for hoisting of equipment during erection and to facilitate maintenance in future.

LIFTS CONTRACTOR'S RESPONSIBILITIES: Ancillary Works

- i. All cabling, wiring and earthing from 3 - phase main DB in machine room to Lift Contractor's equipment.
- ii. Sill tracks including sill supports, supporting protection at all landings.
- iii. Screen guards, facia plates and other protection for installation.
- iv. To carry out minor civil work, such as chipping & making openings in slabs, grouting of foundation bolts in shaft, pit and machine room, modification and making rail bracket, hall buttons indicators and laying of sills in positions. Or any other work required for smooth operation/commissioning of elevators (lifts). All chiseling and cutting of pockets and making good.
- v. Ensuring safety against accidents including barricading all openings and caution signs.

- vi. Scaffolding and other Tools & Tackles required for installation in the hoist way required for erection of elevators (lifts).
- vii. All other items necessary for satisfactory execution & completion of works, whether specified or not.
- viii. Power shall be provided at incoming of main DB for elevators (lifts). Main DB in the machine room shall be provided by the lift contractor. From main DB to lifts, cables shall be in the scope of lift contractor.
- ix. Floor gratings, steps / ladders for pits as required by the lifts Contractor. Contractor shall furnish the details of these items in the layout drawing for elevators (lifts) to submit after award of the job.
- x. Temporary power supply connection(s) for erection work shall be arranged by the lift Contractor.

TRACTION MACHINE

The machine shall be worm geared traction type with motor (steel worm, bronze gears, steel sheave shaft & Ferro molybdenum sheave), electro-mechanical type of brake and driving sheave mounted in proper alignment on a single heavy cast iron base or steel bedplate. The worm shaft shall be fitted with roller bearings to take end thrust. The sheave shaft shall also be fitted with roller bearings to ensure proper alignment. All shafts shall be provided with well-designed keys. Rotating parts shall be statically and dynamically balanced. The drive sheave shall be designed with machined V-grooves to ensure adequate traction with minimum wear on rope. All sheaves including deflector sheaves, where used, shall conform to I. S. 14665 (Part 4 sections 3) Adequate and dust – proof lubrication shall be provided for all bearings and worm gears. The brake shall be suitably curved and provided with fire proof friction lining. The operation of brake shall be smooth, gradual and with minimum noise. The brake shall be designed to be of adequate size and strength to stop and hold the car at rest with rated load. The brake shall be capable of operation automatically by various safety devices, current failure and by the normal stopping of the car. The brake shall be released electrically. It shall also be possible to release the brake manually so as to move the lift car in short stops.

For manual operation of elevators (lifts), up & down direction of the movement of the car shall be clearly marked on the motor or traction machine. A warning plate in bold signal red colour to switch off the mains supply before releasing the brake and operating the wheel shall be prominently displayed.

HOIST MOTOR

The motor shall be suitable for 415 Volts +10% to –20%, 50 Hz. + 5%, 3 Phase A.C. Supply. The motor must be designed for arduous lift duty, rapid reversals and constantly repeated starts & stops as defined in the relevant codes of practice. All windings must be heavily insulated, adequately

impregnated for tropical climate and mechanically strengthened and must be specifically designed to have a high starting torque and low starting current characteristics within the limits acceptable to electricity supply co. requirements and I.E. Rules. The motor shall be designed in such a way as to withstand occasional overloading above its rated capacity and shall have overload protection. The motor shall have good speed regulation under different conditions of load and shall be designed to give a noiseless and vibration-free operation. Insulation shall be class F.

MOTOR CONTROL AND DRIVE

The lift motor shall be controlled by a variable voltage variable frequency (V.V.V.F.) micro-processor control system which shall control and monitor every aspect of lift operation at all stages of the car motion cycle on real time basis. The A.C. V.V.V.F. drive system shall control A.C. voltage and frequency concurrently with the hoist motor to regulate the elevator's (lift's) actual performance to match closely the ideal speed pattern, obtain maximum efficiency of operation and provide a very smooth ride. Frequency shall range fully between zero and rated value.

The Controller shall be provided with a self-diagnostic programmed to keep downtime to a minimum possible. The controller shall intelligently adjust door times in response to car calls, hall calls and "Door Open" button operation. An Inspector's changeover test switch and set of test buttons shall be provided in the controller. Operation of the Inspector's changeover switch shall make both the car and landing buttons inoperative and permit the elevators (lift) to be operated in either direction from machine room for test purposes by pressing corresponding test buttons in the controller. It shall not, however, interfere with the emergency stop switches inside the car or on the top of the car.

GUIDES AND FASTENINGS

- Guide-rails for car and counterweight shall consist of machined mild steel Tee sections, erected plumb, and securely fastened to the lift well framing by heavy steel brackets, suitably spaced, to limit deflection of guide rails to 3 mm under normal working conditions.
- The guide-rails shall be of suitable section with ends tongued and grooved, forming matched joint and shall relate to steel fish plates.
- Guide-rails shall cover the full height of the hoist way and pit, such that It shall be not be possible for any of the car or counter weights shoes to run off the guides.
- Guides shall be designed to withstand the action of safety gear when stopping a counter weight or fully loaded car.

- The max. Deviation from true plumb and alignment of guide rails shall be 2 mm.
- All support framing shall be rigid and shall be designed to restrict displacement of the point of support of brackets to 3 mm under normal working conditions.
- The whole guide rail installation, including expansion joints, shall be designed for a smooth ride.
- The guide-rails shall be protected during storage and installation with a rust inhibiting coating which shall be cleaned off on completion of installation.
- Guide-shoes shall be adjustable type & mounted so as to provide continuous contact with guide rails under all conditions.

Guide shoes shall be provided at top and bottom of each side of car and counterweight and shall be designed for quiet operation. Additional guide shoes shall be provided on each side of buffer frame in case of oil buffers. Each elevator (lift) shall be equipped with roller guides for up and down travel. There shall not be any metal-to-metal contact between Car and rail. Roller shall be mounted on ball bearings to provide quiet operation and excellent ride quality. (It is not required in case the design varies however the ride quality shall not be compromised for any other design).

SAFETY

In addition to other specifications, the elevator (lift) shall be provided with safety devices as follows: -

- Against overload.
- Safety gear on car so that in the event of rope breaking or loosening, the car will be brought to rest immediately by means of grips on the guides.
- The over speeding car shall be automatically brought to a gradual stop on guide rails and power supply to the hoist motor shall be switched off.
- Over speed centrifugal governor operating the safety gear in case of over speeding of car in the down direction.
- Car gate lock so that in the event of car gate is opened when passengers are in the car, the elevator (lift) will be brought to rest.
- Over travel limit switches at top and bottom limits of travel to disconnect the power supply and apply brakes to stop the car within a defined safe distance in case of over travel in either direction.

- Ultimate terminal switches to stop the car automatically within top & bottom clearances independently of normal over travel limit switches but with buffers operative.
- Protective guards to counterweights in pit, rope sheaves and wherever required.
- Toe guard apron to the car platform.

CAR

A. Cabin Size

The internal clear dimensions of the cabin shall not be less than those specified in IS 14665-Part I, NBC & CPWD General specifications for electric work (Lifts). The car shall be so mounted on the frame that vibration and noise transmitted to the passengers inside is minimized.

B. Frame and Safety Device

The car frame shall consist of mild steel channel/structural steel top and bottom securely riveted or bolted and substantially reinforced and braced so as to relieve the car enclosure of all strains when the safety device comes into action due to over speed or when the capacity loaded car is run on the buffer springs at normal speed.

The safety device mounted on the bottom members of the frame operated by a centrifugal speed governor shall be arranged to bring the car to a gradual stop on the guide rails in the event of excessive descending speed; and provision shall be made to shut off the power supply to the motor.

C. Buffers

Substantial spring buffers (2 Nos.) shall be furnished and installed in the pit under the car and counterweight. These buffers shall be mounted on RCC Pedestals in the pit. The car buffer spring must be of correct design to sustain the car with capacity load without damage should the car terminal limits become inoperative. The car buffers must be located symmetrically with reference to center of car.

The Contractor may alternatively offer oil type buffers. The plunger shall be mild steel, designed for a very high factor of safety and accurately machined. A toughened rubber bumper shall be fitted to the plunger top to cushion the impact of steel buffer plates attached under the car and the counterweight. An oil gauge shall be provided to check the oil level.

D. Counterweight

The elevator (lift) shall be suitably counter-balanced for smooth and economical operation. Cast iron weights shall be contained in a structural steel frame properly guided with suitable guide shoes (minimum 4 Nos.). It shall be equal to the total weight of elevator (lift) plus approx. 50% of the

contract load. Substantial expanded metal counter-weight screen guard shall be furnished and installed at the bottom of hoist way, as required by Lift Inspector.

E. Hoisting and Governor Ropes

Bright steel wire ropes with fibre cores suitable for Lift duty as per BIS Code shall be used for hoisting ropes. Not less than 3 independent suspension ropes shall be provided and designed to share load equally by means of adjustable shackle rods with equalizer springs at each end of hoisting ropes. Each rope shall have adequate section to provide a minimum factor of safety of 4 based on the max. Force on the rope. Governor ropes shall be similar to hoisting ropes. Their ends shall be securely attached to the car and to the safety gear. The governor ropes shall be tensioned by a weight loaded device in the pit.

F. Enclosure

The car enclosure shall be as specified in technical data sheet. The cabin floor, roof and walls shall be free of distortion and undue deflection as per IS 14665 – Part 4, Section 3.

G. Brakes

D.C. brakes will be spring-applied and electrically released. They shall be designed to provide smooth stops under variable loads.

H. Doors

Provision shall be made for vertical and horizontal fine adjustment of doors as per the specifications given in technical data sheet.

H. Door Operators

The door operators shall be VVVF inverter controlled heavy duty A. C. motor, allowing variable opening and closing speeds, and full synchronization of car and landing doors.

I. Travelling Cables

The traveling cables shall be multi-core with high conductivity stranded conductors specifically designed for lift duty. The cables shall be provided with retaining straps and individual cable clamps.

J. Emergency Lighting

A self-contained, non-maintained emergency light with a trickle boost charger shall be provided.

K. Intercom

An Intercom system shall be provided between the car; machine room and Fire Console room linked to EPABX.

L. Manual Cranking Facility

Manual cranking facility shall be provided in the machine room to facilitate evacuation of passengers in case of power failure. The manual mode shall be in addition to automatic car failure operation specified elsewhere.

M. Emergency Stop Switch

A stop switch in the machine room / top of car shall be provided for use by maintenance crew to cancel all car and landing calls for a particular elevator (lift).

N. Maintenance Switch

On operation of the maintenance switch located on top of the car by the maintenance crew, the car shall travel at slow speed not exceeding 0.85 m / sec by continuous operation of a button.

O. Landing Door Interlocks

Electrical interlocks shall be provided to ensure that the car does not operate unless all doors are closed and unless the car reaches a landing zone.

P. Overload Indicator

An overload indicator with buzzer shall be provided in the cabin to indicate to the passengers that the car will not start as it is overloaded.

Q. Other Features

All features specified in the BIS/NBC/CPWD and in the enclosed technical specifications shall be provided.

R. Elevator (Lift) for Disabled

All the Passengers elevators (lifts) shall be suitable for use by disabled persons. The following additional facilities shall be provided in this elevator (lift):

- Full length handrails shall be provided on the rear and side wall panels.
- The door closing time shall be set for min. 5 seconds and the door closing speed shall not exceed 0.25 m/sec.
- The “door open” and “door closed” announcements shall be audibly made in the car.

- Braille signs / buttons.

S. Operating Panels, Buttons & Switches

Main and secondary car operating panels, buttons and switches shall be located on one of the two front wall panels next to the car door and as specified in the Schedule of elevators (lifts) & as per approved G.A. drawings. All buttons and switches shall be clearly legible with fade-proof text and figures, and shall be easily accessible, (especially for disabled persons in the elevator (lift) designated for them).

T. WORKS TESTS

The following tests shall be carried out at Works:

- High voltage works tests of equipment which is not already tested in accordance with appropriate IS codes.
- Buffer test.

U. TESTS ON COMPLETION

The following tests shall be carried out to the satisfaction of the Engineer-in-Charge.

- Insulation resistance and earth test for all electrical apparatus.
- Continuous operation of the elevator (lift) under full load conditions and simulated starts and stops (150 nos. per hour each) for one hour at the end of which time the service temperature of the motor and the operating coils shall be tested. This shall be as per B.I.S. specification.
- The car shall be loaded until the weight on the rope is twice the combined weight of the car and the specified load. The load must be carried on for about 30 minutes, without any sign of weakness, temporary set or permanent elongation of the suspension rope strands.
- **The following items shall be tested:**
 - a. Levelling accuracy at each landing in conditions of fully loaded and empty car.
 - b. No load current and voltage readings both on 'Up' and 'Down' Circuits.
 - c. Full load current and voltage readings both on 'Up' and 'Down' Circuits.
 - d. One and quarter load current and voltage readings both on 'Up' and 'Down' Circuits.
 - e. Stalling current and voltage and time taken to operate overload.

- f. Overload protection.
- g. Gate sequence relays, if provided and installed.
- h. Car and landing door interlocks.
- i. Collective control and priority sequences, if installed.
- j. Safety gear mechanism for car and counterweight with fully loaded car and also with only 68 kg load.
- k. Speeds on Up and Down travel with full load, half load and empty car.
- l. Door contacts.
- m. Final terminal stopping device.
- n. Normal terminal stopping device.
- o. Car and counterweight buffers with contract load and contract speed.
- p. Operation of controllers.
- q. Manual operation of elevator (lift) at mid-way travel.
- r. Emergency operation.

V. Tests on completion shall also be performed to the satisfaction of Inspector of Lifts and a certificate will be obtained from the 'Lift Inspector' by the EPC contractor.

W. STATUTORY APPROVALS

All statutory approvals from commencement to commissioning of elevators (lifts) shall be obtained by the EPC Contractor from the Inspector of Lifts and / or other authorities. However, the client will provide all necessary assistance for providing documents, drawings and certificates pertaining to other contractors, if required.

The contractor shall pay necessary fees in connection with the approval of installation of lifts.

X. FEATURES REQUIRED FOR VVVF LIFTS

- a. Group / Independent / Attendant Operation.

It shall be possible to group specified cars in a group wherever required with dynamic disposition of cars as required by the traffic pattern. A smart car dispatching system with ring communication

shall be provided for optimum passenger comfort and lift performance under all traffic conditions. Any defective car shall be automatically eliminated from the group. Each car shall be provided with a key switch for independent operation housed in a service cabinet. In this mode, the elevator (lift) shall respond only to car calls. Hall calls will not be registered. It should be possible for an attendant to operate any car.

b. Fireman's Switch

A fireman's toggle switch shall be provided in a break glass for the specified elevator (lift) at ground floor to enable firemen to bring the elevator (lift) non-stop to ground floor from any location and to cancel hall calls until the car is operated on attendant control.

c. Emergency Power Operation

In case of power failure, standby power equipment shall enable elevators (lifts) to reach a pre-determined floor, in a pre-determined sequence, and then permit operation of one or more elevators (lifts) on emergency power. A trickling battery shall be provided to supply power to light fixtures, fan, alarm, and intercom.

d. Profile Generator

A profile generator or similar device shall be provided to use the car at an optimum speed level and to improve leveling accuracy.

e. Predictive Car Selection

Once a hall call is registered, a dynamic car algorithm shall transfer the call to an optimally selected car to provide the maximum traffic efficiency.

f. Home Landing Facility

A car shall return to a pre-determined landing after the last call is answered.

g. Door Safety

Multi-beam infrared / ultrasonic electronics curtains shall be provided to scan the doorway and reverse the door closing in case of any obstruction.

h. Double Door Operation

If both up and down calls are registered at a hall which is the last registering hall in the direction of the car, the elevator (lift) shall travel to that hall and open / close the doors. After this, the car shall reverse its travel and shall open / close the doors again unless no car calls are registered at that floor.

i. Nudging Door Operation

When the doors remain open for more than a predetermined period, a buzzer shall sound and the door shall close automatically. The door sensing device shall be rendered inoperative but the Door Open button and the safety shoe shall remain operative.

j. Selective floor Service

Programming for selective floors services shall be software driven.

k. Manual Cranking & Slow speed Travel

A manual cranking facility shall be provided. Slow speed operation shall be possible from machine room and car top

l. Auto Fan Off

In case no calls are registered for a pre-set time, the cabin fan shall be automatically switched off.

m. Automatic Rescue Device

In case of mains power failure and Lift control system failure, the Elevator's (Lift's) own rechargeable and maintenance free battery power shall move the car to the nearest floor and the door shall open automatically for automatic rescue of passengers. A battery run-down indicator shall be provided.

Y. MAINTENANCE DURING DEFECTS LIABILITY PERIOD

Comprehensive maintenance during Defects Liability Period inclusive of periodic servicing, prompt attention to client complaint, prompt rectification of all malfunctions and equipment failures, replacement of defective equipment / parts, replacement of light fittings, lubrication including lubricants, maintaining correct alignment, and leveling of cars and ensuring smooth running, starts and stops etc. all complete to the client satisfaction shall be done.

5. TECHNICAL SPECIFICATION OF LV SYSTEM

A. STRUCTURED CABLING

OEM Eligibility Criteria for Passive cabling system

Sr. No.	Eligibility Criteria	Compliance
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1	OEM Should be registered in INDIA.	
2	All LAN cable and component (Copper & Fiber) should be from single OEM.	
3	The OEM should have at least one RCDD certified person in India.	
4	Data Sheets of all proposed products should be available on the OEM public website. The data sheets provided on the OEM public website and submitted data sheets should be the same.	
5	The OEM products when installed should carry and provide 10 years of end-to-end channel performance warranty.	
6	The product quoted for cabling should be ROHS complied (ROHS logo to be in the data sheet).	
7	The Cabling product quoted should be in accordance latest global standard to EIA/TIA, IEEE, ISO/IEC and should be mentioned the data sheet.	
8	All Product should when delivered to be submitted with FTC (Factory Test Certificates).	
9	Given Specifications are Minimum Criteria and OEM/Contractor may suggest products which are at least meeting or exceeding the product specifications.	
10	Note: Failing to comply with any of these terms and condition will lead to rejection of the offer	

a. 6 Core Outdoor Armoured Single mode Fiber Cable

Sr. No.	Technical Parameters	Compliance (Yes/No)
1	Armoured 06 Core Single mode (OS2) 9/125 Fiber Cable, ITU G.652.D, G.657A1, Outdoor ECCS Armored Fiber Cable with PBT Loose Tube Filled with Thixotropic Jelly, Uni-Tube, Glass Yarn, Water Swellable Tape Under Armor, UV-HDPE Jacket	
2	Application: The fiber cable should have Bend insensitive features to be installed in Outdoor, Duct, Trenches & Underground application.	

3	Features: The fiber cable should have excellent features of Tensile, Crush, Water Prevention with Jelly & Moisture Barrier Tape	
4	Cable Construction	
5	Uni-tube (Central Loose Tube) with color coded fibers as per EIA/TIA 598	
6	Loose Tube: PBT Loose Tube filled with Thixotropic Jelly	
7	Loose Tube Diameter: 2.5mm Nominal	
8	Armoring: Corrugated ECCS Tape Armouring (0.155 mm Nominal thickness)	
9	Outer Sheath: UV Resistance HDPE	
10	Moisture Barrer: Water Blocking Tape Under Armor	
11	Strength Member over Central Tube: Glass Yarns	
12	Physical / Mechanical Characteristics	
13	Outer Diameter: 8.5 +/- 1.0 mm	
14	Nominal Jacket Thickness: 1.5 mm	
15	Tensile Strength: ≥ 2220 Newton (IEC 60794-1-2-E1)	
16	Bending Radius: $\leq 20 \times OD$ (IEC 60794-1-2-E11) (OD=Cable Outer Diameter)	
17	Crush Resistance: ≥ 2200 Newton/100mm (IEC 60794-1-2-E3)	
18	Water Penetration: Meets IEC 60974-1-2 (24 Hr, 3Meter Sample, 1Meter Height)	
19	Weight: ≤ 85 Kg/km	
20	Environmental Characteristics	
21	Operating Temperature: IEC 60794-1-2-F1 @ -30°C to +70°C	
22	Storage Temperature: IEC 60794-1-2-F1 @ -10°C to +60°C	
23	Installation Temperature: IEC 60794-1-2-F1 @ -30°C to +70°C	

24	Safety: ROHS to be mentioned in data sheet	
25	Optical Characteristics	
26	Fiber Type: SM (9/125) OS2 as per G.657 A1	
27	Max. Attenuation: 0.36 dB/km @ 1310nm , 0.23 dB/km @ 1550nm	
28	Mode Field Diameter @ 1310nm : 8.8 +/- 0.4 μ m	
29	Dispersion: ≤ 3.5 ps/nm.km & ≤ 17.5 ps/nm.km	
30	Fiber cut of Wavelength: ≤ 1320	
31	Cable Cut of Wavelength: ≤ 1260	
32	Zero Dispersion Wavelength: 1300-1324 nm	
33	Zero Dispersion Slope: ≤ 0.090 ps/nm ² .km	
34	Coating Diameter: 250 ± 15 μ m	
35	Cladding Diameter: 125 ± 0.7 μ m	
36	Fiber Curl: ≥ 4 m radius curve	
37	Cladding non-circularity: $\leq 1\%$	
38	Mode Field Concentricity error: ≤ 0.8 μ m	
39	Coating/Cladding Concentricity Error: ≤ 12 μ m	
40	Packaging: Wooden Spool of Min. 2KM roll	
	Make as approved by Engineer-in-charge	

b. 24 Port single mode Fiber Panels

S. No.	Technical Parameters	Compliance (Yes/No)
1	24 Port 1U x 19" LIU Loaded with single mode OS2 LC or SC Adapters & LSZH Pigtails along with Splice Trays	
2	The Fiber Panel shall have Telescopic Sliding Shelf for easy smooth maintenance add/move/changes.	

3	The optical fiber Pigtails shall be factory loaded inside each individual Port of the panel. Pigtails shall be LSZH with Single mode 9/125µm fiber	
4	Panel shall have Min. 4 No. of Cable Entry Slots at back of the Panel supplied with Cable Holders inside Panel.	
5	Completely Enclosed without any open area to avoid any Rodent Entry	
6	Pigtail Parameters Loaded inside Panel shall meet below mentioned requirements: -	
7	Pigtail Buffer Jacket Material: LSZH complying to IEC 61034-1, IEC-60332-1, IEC-60754-1	
8	Connector Insertion loss should be better than 0.35 dB	
9	Return Loss ≥ 50 for UPC and ≥ 65 for APC	
10	Attenuation: 1310/1550: 0.3 /0.2 dB/KM	
11	RoHS Complied, Meets ANSI/TIA 568.3-D	
12	Repeatability: ≤ 0.2 DB 1000 Times Mating Cycles	
	Make as approved by Engineer-in-charge	

c. 6 Port Single mode Fiber Panels

S. No.	Technical Parameters	Compliance (Yes/No)
1	6 Port 1U x 19" LIU Loaded with Single mode OS2 LC or SC Adapters & LSZH Pigtails along with Splice Trays	
2	The Fiber Panel shall have Telescopic Sliding Shelf for easy smooth maintenance add/move/changes.	
3	The optical fiber Pigtails shall be factory loaded inside each individual Port of the panel. Pigtails shall be LSZH with Single mode 9/125µm fiber	
4	Panel shall have Min. 4 No. of Cable Entry Slots at back of the Panel supplied with Cable Holders inside Panel.	

5	Completely Enclosed without any open area to avoid any Rodent Entry	
6	Pigtail Parameters Loaded inside Panel shall meet below mentioned requirements: -	
7	Pigtail Buffer Jacket Material: LSZH complying to IEC 61034-1, IEC-60332-1, IEC-60754-1	
8	Connector Insertion loss should be better than 0.35 dB	
9	Return Loss ≥ 50 for UPC and ≥ 65 for APC	
10	Attenuation: 1310/1550: 0.3 /0.2 dB/KM	
11	RoHS Complied, Meets ANSI/TIA 568.3-D	
12	Repeatability: ≤ 0.2 DB 1000 Times Mating Cycles	
	Make as approved by Engineer-in-charge	

d. LC-LC Single mode Armoured Fiber Patch Cord

S. No.	Technical Parameters	Compliance (Yes/No)
1	Fiber Patch Cord, LC - LC, Spiral Steel Armoured, 9/125 μ m OS2 Single mode Round Cord (4.0 +/- 0.2mm), Duplex	
2	Spiral Steel Armor (≤ 1.8 mm Diameter) , Aramid Yarn, Tight Buffer Fiber	
3	Connectors: The optical fiber patch leads shall comprise of Single mode 9/125 μ m fiber G652.D	
4	Connector Insertion loss ≤ 0.4 dB	
5	Return Loss ≥ 50 for UPC and ≥ 55 for APC	
6	Jacket Material: LSZH complying to IEC 61034-1, IEC-60332-1, IEC-60754-1	
7	Length: 3 Meter	
8	Attenuation: 1310/1550: 0.35/0.22 dB/KM	
9	Operating Temperature: -20°C to +70°C	

10	Repeatability: $\leq 0.2\text{DB}$ 10 Times Mating Cycles	
11	Tensile Strength: 750N	
12	Crush Resistance: 1000N/100mm	
13	RoHS Complied, Meets ANSI/TIA 568.3-D	
	Make as approved by Engineer-in-charge	

e. Cat6A UTP LSZH Cable

S. No.	Technical Parameters	Compliance (Yes/No)
1	Cat 6A, 23 AWG ($0.57 \pm 0.03\text{mm}$) Bare Copper Solid Conductor UTP Unshielded LSZH Jacket, ANSI/TIA 568 C.2 Category 6A, Cable shall not have any kind of Non Metallic Barrier Tape or Metallic Shield inside	
2	Suitable for 10GBASE-T with Min. 500Mhz of Bandwidth	
3	4 Twisted Pair alongside PE / PVC Cross Separator	
4	Conductor: 23AWG Solid Annealed Bare Copper	
5	Conductor Diameter: $0.57 \pm 0.03\text{mm}$	
6	Insulation: High Density Polyethylene, Diameter $1.08 \pm 0.05\text{mm}$	
7	Jacket: LSZH complying to IEC 60332-1 Flame Rating, IEC 60754-1/2 for Halogen Acid Test, IEC 61034-2 for Smoke Density Test	
8	Nominal Jacket Thickness: Min. 0.80mm	
9	Cable Outer Diameter: $7.2 \pm 0.5\text{ mm}$	
10	Operating Temperature: -20°C to $+75^{\circ}\text{C}$	
11	CPR Euro class: Class Eca	
12	IEEE 802.3bt Type 1, Type 2, Type 3, Type 4 PoE Complied	
13	Cable shall not have any kind of Non-Metallic Barrier Tape or Metallic Shield inside	

14	Flame Rating: IEC 60332-1	
15	Temperature Index & Oxygen Index should meet ASTM D 2863	
16	Bend Radius: 8 X Cable Diameter (Min.) or better	
17	Conductor Resistance: $\leq 7.5 \Omega / 100m$	
18	Resistance Unbalance: 5% Max	
19	Mutual Capacitance: $\leq 5.6nF/100m$	
20	Propagation Velocity: 69% (nominal)	
21	Dielectric Strength: DC 1500 V /Min.	
22	Min. Elongation at Break of Insulation: 300%	
23	Min. Jacket Tensile Strength: 10 N/mm ² (Mpa)	
24	RoHS Complied	
	Make as approved by Engineer-in-charge	

f. Cat6A Unshielded Patch Cord, 1 Meter & 3 Meter

S. No.	Technical Parameters	Compliance (Yes/No)
1	Cat6A Patch Cord U/UTP Unshielded LSZH 24AWG Patch Cord, Diameter < 6.2mm, 3 Meter, Operating Temperature -20 to +60 Deg C	
2	Cable Construction of Patch Cord: U/UTP LSZH	
3	There shall be NO individual or overall foil / braid shield in the cable. Cable shall be completely Unshielded with U/UTP Construction because other related components are also Unshielded in the Solution.	
4	Number of conductors: 8 (4 Twisted Pairs)	
5	Conductor Material: Bare Copper (24AWG Stranded)	
6	Cable Overall Nominal diameter: ≤ 6.2 mm	

7	Insulation: HDPE	
8	Cat 6A patch cord plugs to have round cable holder strain relief transparent boot to avoid bending.	
9	Jacket: LSZH (Low Smoke Zero Halogen) with Flame Rating IEC 60332-1 or better	
10	Operating Temperature: -20 to + 60 °C	
11	Plug should have high repeatability cross talk performance with Min. 750 Cycles.	
	Make as approved by Engineer-in-charge	

g. Cat6A Unshielded Modular Jack

S. No.	Technical Parameters	Compliance (Yes/No)
1	Category 6A RJ45 Unshielded Modular Jack, ISO/IEC 11801:2nd edition, EN 50173-1, ANSI/TIA/EIA 568-C.2 Category 6A, IEC 60603-7 (603-7), Universal 110 Impact Tool & Toolless (Both) Termination, Min. 500Mhz Bandwidth, Without Any OEM Proprietary Tool Termination	
2	Suitable for 10GBASE-T applications with Min. 500Mhz Bandwidth.	
3	Compatible with RJ standard plugs (RJ11, RJ12, RJ45), PCB Based Universal 110 Impact Tool & Toolless (both) based connection for installation of cables	
4	Each Jack should be supplied with one Strain relief boot for Pairs and separate additional bend limiting boot for Cable with side split clip with the I/O and a cable tie for proper holding of cable to the JACK.	
5	The Bend limiting boot should have additional clip or locking facility to project the Cable and it should be inbuilt with the JACK.	
6	The Jack should be universal toolless and tool-based connection of installation cables AWG 24 – 23 with Solid Conductor cables.	

7	IDC termination should feature Color Coding according to EIA/TIA 568-A/B, Gold-Plated Contacts, >= 750 mating cycles, >= 200 insertion cycle at IDC	
8	Material: RoHS complied	
9	Housing material: Polycarbonate/Flame Retardant PVC (UL-94-V0)	
10	Should be UL Certified as per UL 1863	
11	DC/AC Voltage Endurance - DC1000V/AC750V (1min)	
12	DC Resistance: Max. 0.3 Ohms	
13	Plug Retention Force: 30 lbs	
14	Operating Temperature: -10 to +60 °C	
15	Current Rating: 1.5 Amp	
16	The Modular Jack shall not require any proprietary Termination Tool for termination of Cable.	
	Make as approved by Engineer-in-charge	

h. 24Port Patch Panel

S. No.	Technical Parameters	Compliance (Yes/No)
1	24-Port 1U Unloaded Universal Modular Straight Patch Panel preloaded with cable support Bar (with slots to tie individual cable properly at the Support Bar (Shuttered IO not recommended, because when shutter gets malfunctioned then entire IO is of no use)	
2	The Patch panel should be universal (stainless steel rear metal frame) and should be able to support both for UTP & STP Solutions	
3	Patch panel should be equipped with cable strain relief retention tray (Cable Support Bar) with slots to tie individual cable properly at the Support Bar	

4	Each port of the panel should have individual transparent labelling point	
5	Panel should be inbuilt transparent spring shutter for dust protection on each port. (Shuttered IO not recommended, because when shutter gets malfunctioned then entire IO is of no use)	
6	Should be RoHS & UL 94V-0 complied	
7	Panel shall be UL 1863 Rated (Certificate to be provided along with Bid)	
	Make as approved by Engineer-in-charge	

i. Face Plates

S. No.	Technical Parameters	Compliance (Yes/No)
1	Style (Square) Keystone-Type Shuttered Faceplates, 1, 2 & 4 port configurations	
2	UK Style (Square) Keystone-Type Faceplates, 1, 2 & 4 port configurations, White Color	
3	Should be featured with spring shutter for each port	
4	Elegant 2 Piece (2 Plate) design for better aesthetics	
5	Cover and Base Plastic Material to meet Min. ABS-UL94-V2	
6	Suitable for both Flush and Wall mount gang box	
7	Dimensions: 86 x 86 x 12.8 (mm)	
	Make as approved by Engineer-in-charge	

B. IP-PBX SYSTEM

IP Telephony System Architecture:

The IP telephony system must be based on a pure IP technology that is a software-only solution.

The IP telephony system must support unified communication (UC) server & gateways architecture for SIP, Digital and Analog trunks connectivity. The system must be capable of supporting Analog, Digital, IP Telephones, and SIP based video desk phones. SIP phones and SIP trunks should directly register on the Telephone server. Card based solution is not acceptable for SIP phone and SIP trunk registration.

The communication servers must work in an Active-Active redundancy mode. It should be possible to define servers load balancing mode. All servers should work together in load balancing mode with defined user capacity i.e., all servers should be active with call processing with predefined SIP phones / gateways register on any of the server for load distribution. If any Server Fails in the Cluster adjacent server should automatically take the load of the failed sever along with load of gateway and end points without breaking on-going calls. Redundant Server/Hot stand-by mode of working is not acceptable. Also, solution with single server with standalone EPABX type gateways at each location will not be accepted.

At each location two servers i.e. main Active call control server along with additional active servers for redundancy to be provided. Both servers must be provided in a cluster mode. If one cluster server fails, the other cluster servers in the network must be able to take the complete load of the calls automatically (without any manual intervention) and without dropping any existing calls (IP, TDM & PRI) or data (CDR, CTI). Management of both servers in cluster should be from same web page. Both servers should have same database.

The LAN connectivity of 100 mbps speed between each location will be provided by customer. Bidder to ensure server cluster connectivity between all the locations to maintain system availability of 99.9% all the time.

The telephony system must be able to register SIP phones/SIP video phones directly to it.

The OEM offered unified communication and call manager application must have valid trademark registration certificate as per Govt. of India trademark act 1999, Section 23, rule No. 62(1). Vendor will submit a copy of the same certified by OEM.

It should be possible to install Telephony system in VMware EXSi 6.5 or higher.

All Data (Numbers, COS, Routing, Applications) should reside in all the Servers. Database replication in all servers should be automatic and real time. Should support N+1 Redundancy Architecture as well as 1+1 redundancy Architecture. Should support Remote Survival Nodes.

In case of failure of one server, the SIP Phones, SIP trunk, SIP Gateways should register with second Server automatically. System Diagnostics should be done in Server. Hot Standby for SIP Phones, trunks and Gateways i.e. SIP Phones, trunks and Gateways should register automatically to next available telephony server.

COTS - commercial off-the-shelf Servers should be used for telephony system. OEM made or proprietary servers will not be accepted.

Card based processor servers or PCM/TDM legacy systems with card-based processor systems are also strictly not allowed. Telephony system should use Linux Operating System. System should support CSTA phase III Protocol. Full continuation for call signaling and media must be supported. Calls must not be disconnected and control must remain throughout the swap to an alternate server including full call control (transfer, conference actions, continuation of CDR data for the existing call).

Load Balancing of end points must be possible by the administrator. There must be no restriction on the number of endpoints being backed up in case of one server failure. UC platform servers must provide full failover and redundancy.

Vendor must submit valid latest Type test **TEC-GR (Vide TEC-GR spec. No.: TEC/GR/SW/PBX-005/01/SEP.2016)** (Generic Requirement) approval certificate issued by Telecommunication Engineering Centre (TEC), Department of Telecommunication, Govt. of India tested with IPv4 & IPv6 for both SIP terminals and SIP Trunks from days 1 for the model of IP-PBX with Server & Media Gateway system quoted. Notarized copy of the same is required to submit along with the technical bid.

Centralized administration should be there. Login to be done from anywhere in the network on single IP address.

SYSTEM SHOULD SUPPORT THE FOLLOWING SIP RFCS:

- RFC 3261 (SIP: Session Initiation Protocol)
- RFC 3262 (Reliability of Provisional Responses in Session Initiation Protocol)
- RFC 3263 (Locating SIP Servers)
- RFC 3264 (An Offer/Answer Model with Session Description Protocol (SDP))
- RFC 3265 (Specific Event Notification)
- RFC 2327 (SDP- Session Description Protocol)
- RFC 1889 and 1890 (RTP/RTCP)
- RFC 3515 (REFER)
- RFC 2833 (DTMF over IP)

SCALABILITY:

It should be possible to add more sites and users without the need to change the software and existing configuration.

The system must be scalable to at least 25,000 endpoints in single cluster architecture. The proof of reference document from OEM is mandatory to be submitted along with technical bid.

The call signaling server must handle traffic at a minimum of 200K BHCC.

Also Call Manager system must support unlimited SIP trunks within the same application with within same server.

The system must be modular, scalable, and distributable.

The IP PBX system must be tested ready with IPv6 as per Govt of India guidelines dated. January 2021. Test certificate issued from TEC to be submitted along with a technical bid. Self-declaration will not be accepted.

System Survivability:

The UC platform must consist of one or many servers where each server in the cluster provides complete 100% application functionality.

In case of a failed server, all endpoints registered with that server need to register instantly with a different server in the cluster with no interruption to on-going calls.

Media Gateways must have survival mechanisms that allow them to maintain 100% of the telephony services for their users in case of failure when the signaling with the call server drops.

The life cycle of the entire system being provided must be at least Ten (10) years.

The system gateway must be able to restart automatically without human intervention when the external DC power supply is resumed after complete power failure (even after the batteries are discharged).

The telephony system must be capable of providing 99.999% availability.

Distributed Architecture:

The UC platform must have distributed architecture and centralized control for all the sites in the network.

The proposed solution must support Hybrid cloud solution in order to guarantee business continuity with overall survivability regardless of a failure at any single location.

The proposed solution must enable part of the cluster to be hosted in a Cloud Service Provider (CSP) to run all applications.

The proposed solution must have built in redundancy using a cloud solution to provide automatic disaster recovery option.

The proposed solution should have provision to be installed using an image of the application an easily implemented on the Cloud Service Provider servers.

Quality of Service (QOS):

The voice and signaling frames must be marked [tagged] in order to be recognized.

Server – Physical Attributes:

COTS – Commercial Off-the-Shelf servers must be used.

The redundant server must have separate hardware, not sharing elements like hard drives and RAM etc., to avoid a single point of failure.

The server should have AC power supply.

The system must be based on server gateway architecture with external appliance servers

No card-based processor systems / soft switch should be quoted.

The call processor must run on Linux OS.

Minimum Server Specifications:

The CPU must be from the Intel® Xeon® processor – Quad Core

The server must have at least 16 GB RAM

The server must have Hard Drives (1TB) of storage

The server must have a Dual 1GB network interface.

Form Factor for physical server should be 1 U

Should be Compatible to work with VMware EXSi 6.5 or higher

VoIP Media Gateways:

The VoIP media gateways shall be capable of being centrally managed via the telephony management application. The system should support multiple gateways.

- The system gateway should be able to restart automatically without human intervention when the external ac power supply is resumed after complete power failure.
 - The system gateways should support the following type of extensions:
 - Analog
 - Digital
- The system gateways should support the following type of trunks:

- Analog: E&M (2W), E&M (4W), DC loop signaling, decadic, DTMF
- Digital: 2Mb stream with the following signaling protocols (Digital CEPT, R2MFC)
- Standard ISDN PRI
- ISDN (30B+D / 23B+D)
- ISDN QSIG (30B+D)

System Security:

Administration of the system should be using HTTPS

- It should support the Interop with leading SBC
- System should use TLS (Transport Layer protocol) to encrypt SIP, HTTP, FTP and SRTP (Secure Real-time Transport Protocol) and SRTCP to encrypt RTP and RTCP
- System Audit Logs for 30 days.
- Certificate management.
- System should have auto Provisioning profiles contain pre-configured sets of features that must automatically polls and updates registered phones with the latest phone firmware and configuration files.

Mobility:

The system should have Call Back feature. If the user dials his own extension from predefined number (mobile/landline) then system should disconnect the call and then system should call the user to provide the dial tone so that user can make intercom or PSTN calls.

The system should have Call Through feature. If the user dials his own extension from predefined number, then system should provide dial tone to make intercom or o/g calls.

The system should have one number (Forking, reach-me-anywhere) feature. Users should be able to receive calls on any of their phones, from almost anywhere. An incoming call rings on all or specific phones until the user answers the call. The user can transfer the call to his own desk phone, whether the answering device is an internal device, an external phone, or a cellular handset. If the answering phone is an external device, the call automatically becomes an authorized mobility call.

The system should support SIP Client on smart phone for Android as well as for iOS.

SIP Endpoints:

All SIP phones must support the standard SIP protocol. No proprietary protocols are allowed to be used.

SIP phones must support the configuration of programmable buttons with functions such as Break-in, Conference call, Deflect, silent monitoring and more. SIP phone should register directly on the telephony server.

SIP phones must work in conjugation with the following applications:

1. Contact Centre (Agents Phones)
2. Attendant Console
3. Managed Audio Conferencing
4. Managed Video Conferencing
5. UC clients

Automatic Call Distribution (ACD):

Busy ACD Group announcement

Hunt Group Release

IVR-ACD

Log In / Log Out

Multiple Announcements:

1. Mandatory announcement - All incoming callers to an ACD/HUNT group must be able to hear an introductory announcement in its entirety usually explaining about the company, product, or campaign.
2. First announcement - If all agents are busy, callers must be able to hear this announcement once usually informing them that their call has been placed in queue. (The system must be able to cut short this announcement if an agent becomes available to attend to the caller.)
3. Music - If no agents are available after the first announcement (or no First and Periodic announcers have been configured), the caller must be able to hear background music while in queue.
4. Periodic announcement - Alternating with background music, these announcements can also be played to callers in queue according to the Periodic Announcement Interval (see above) until the ACD/HUNT call is answered.

Release / Resume

Wait Queue

Wrap-Up Time

Automatic Release of ACD Agent

Automatic Call Distribution (ACD) Extended Overflow

Zone Page:

A phone user must be able to simultaneously broadcast a message over all types of endpoints.

The maximum quantity of endpoints in one zone should not be less than 100.

System Administration:

- System administration should be web based.
- All programming of system should be done through a web-based GUI interface.
- The administrator should have Dynamic Profiles.
- The system should allow for complete multi-level administration. The administrator must be able to define at least five (5) different administration level profiles that can be applied to allow subsets of users to access and manage pages in the systems Web Portal

Certification Requirements:

The OEM must have **ISO 9001, ISO 45001, ISO 270001, and ISO 90003 certification** in all the company's activities. Valid duly certified copies of the same is required to be submitted along with technical bid.

System Features:

- ANI (Caller ID) Restriction
- ARS (Automatic Route Selection)
- Auto Attendant
- Call Forward at Night/Holiday
- Call Forward Destinations
- Call Forward for Undefined Stations
- Call Forward on Busy
- Call Forward on DND (Do Not Disturb)
- Call Forward on Logout
- Call Forward on No Answer
- Caller id-based routing for individual extension
- Deflect (Divert) Call
- Digit Train Conversion
- Direct-In-Dial

- Direct-In-Line (DIL)
- Hot Line
- Interactive Voice Response (IVR)
- Least Cost Routing
- Look Ahead Routing (LAR)
- Numbering Plan
- Personal Routing Rules based on caller id and DNIS
- Predetermined Night Answer
- Toll Restriction – Digit Analysis
- Toll Restriction – Trunk Groups
- Trunk to Trunk Connection
- Trunk Transfer Restriction
- Classes of Service
- Night Answer Central Bell / UNA Pickup
- Page Queue
- Recall
- Recall / Incomplete Destination
- Speed Dial Public (System) and Private
- Virtual Numbers
- Music On Hold
- each User should support up to 6 devices i.e. SIP phone / analog phone / soft phone / mobile client etc.
- Voice Page
- Silent Monitor
- Zone Page
- Barge In
- Connection to MS Teams using SBC
- Wake up

Extension Features:

- Answer Call Waiting by Transfer
- Auto Set Relocate
- Auto-Answer
- Automatic Disconnect
- Automatic Number Identification (ANI) Display
- Browse Personal Directory
- Busy Lamp Field

- Call Forward All
- Call Hold
- Call Log
- Call Parking and Call Pickup
- Call Waiting
- Caller ID Control
- Caller-ID Screening
- Caller id-based routing for individual extension
- Calling Number and Name
- Camp-on Idle
- Configurable DSS Buttons
- Direct Dial without Off Hook (Hands Free)
- Directed Call Pickup
- Display Automatic Number Identification (ANI)
- Display Dialed Number and Name
- Display Dynamic Call Divert Information
- Display Select Hold Display
- Display Time/Date Function
- Do Not Disturb (DND)
- DSS/BLF
- Elapsed Time Display
- Group Call Pickup
- Hands Free
- Hands-Free Announce and Reply (Idle State)
- Last Number Redial
- Login and Logout
- Message Waiting Indication
- Multi Appearance (Call Waiting)
- Multiple Line Appearance
- On-Hook Dialing
- Placing Multiple Calls on Hold
- Privacy – ANI Restriction
- Reminder/wakeup Call
- Restrictions – Station
- System Non-Exclusive Hold
- Transfer with Consultation
- Transfer without Consultation (Blind)
- Voice Page

- Emergency Preemption
- Listen to Paging while in a call (Busy Condition)
- ULA - User Line Appearance (ULA)

Emergency Conference Communication:

The Emergency conference communication should be supplied and installed as per below specification.

Emergency conference resource should be provided from the same telephony system OEM and have the facility to automatic dial out to connect up to 120 participants in a single conference. System should also have 120 party managed meet me conference. It should be possible to further divide 120 party conference bridge into any combination like 10 X 10 party, 5 x 20 party etc. if required. The meet me conference should be secured means to enter to the conference bridge; the user should enter he password.

The UC call manager system must provide total capacity to manage conference of 2 X 120 party (240 Party) with 2 servers in active / active redundancy mode.

The emergency communication management should be using Web Browser/HTML5 based GUI based interface from Windows PC and Touchscreen monitor. Windows 10 PC with touch screen monitor must be supplied along with the required solution.

If SIP camera is connected on same LAN network, it should have the ability to view the video (of the location) from a camera to the Dispatcher screen and create an alarm.

It should have the ability to view video using streaming with HTML5.

It should have the ability to view video using Adobe Flash Player

The Group Operator should have following features as below:

The Group Operator must be able to add / remove members

The Group Operator must be able to add other conference members

The Group Operator must be able to mute / unmute (User, None, All)

The Group Operator must be able to lock / unlock the conference

The Group Operator must be able to close the conference

It must be possible to dial out a pre-defined group (or multi-groups) of participants/numbers by simply pressing the pre-assigned key.

Each pre-set conference must have its own unique dial number such that when this group number is dialed; all the number stations will ring simultaneously.

Any combination of stations and external numbers must be able to be defined as members of the Group Call.

Participants may join a conference in the audible or in the mute mode, if in mute mode, the right to speak must be selectively offered to attendees per their request by a special signal sent to the Group Operator by the attendees.

Attendees must be able to be added or excluded at any time by the Group Operator

The conference must be terminated when the Group Operator leaves (auto terminate if all members left are muted).

The Group Operator must be able to barge into an existing user call based on pre-emption predefined rules.

Group operator must have two SIP phones so that if one phone is busy in conference, the second phone can be used to add participants.

Both group operator SIP phones should be controlled by web-based conference management GUI for telephony feature like answer, hold, transfer etc. if required in future.

The same Group operator should also function as operator console.

Mute Ring Button – When a call comes to the Dispatcher phone, the Dispatcher can mute the ring, the call will continue.

Add Department attribute to user – In the Phone Book, add a department name for the user. The Dispatcher can do a search based on the Department

Configurable Background – Change the background of the Dispatcher screen from the Application.

Join Incoming Calls to an open Conference – When a call arrives to the Dispatcher, he can add that call to an open conference.

Stage Conference from the Dispatcher screen – the Dispatcher can create an ad-hoc conference from the Dispatcher screen.

Change Meet me Access Code – The password of a conference can be changed by the Dispatcher to block or allow callers to join a conference

Specification of Soft client (Mobile/PC):

The soft client should be from the same OEM of IP telephony system. Wi-Fi facility for smart phone to be provided by the customer.

Soft client should be available for latest Windows PC, Android Phone and IOS phone

It should be freely downloadable from Google Play store / Apple store

It should support following features:

- Make a call
- Hold
- Retrieve
- Transfer
- Video call

"Presences (User Select) -

Change status: Available, meeting, DND etc.

Dial users' number - Internal, External, Mobile

- Instant messaging - IM with another soft client User, Search on IM sessions, Save IM sessions
- 3-way audio conference
- call diverts
- view missed call, answered calls and outgoing calls
- Record on Demand (ROD) support
- Speed Dial Public (System) and Private
- Contacts synchronized with the PBX directory

Help desk Specification:

General capabilities:

- The Help desk solution must be embedded within the same servers, not installed on a separate server and should be from the same OEM of the telephony system. The help desk facility must be supplied by vendor within the same UC call manager application from day1.
- The system must be an All-in-one solution that provides a one server solution for UC&C. 5 agent license along with one supervisor should be supplied and installed for help desk solution.
- Single server deployment with intuitive and central management capabilities should support true multimedia.
- Help desk managers must be able to easily prioritize customers and incoming contacts regardless of the media used.
- The same set of business and routing rules can be applied to voice / chat calls, emails, and faxes if required.
- The help desk must support multi-layer routing including Priority, Skill Based, Statistical, Business Rules, and Customer Defined Values.

- Help desk must have embedded IVR, enabling managers to design routing plans and accurately assess help desk activity trends.
- The IVR application must be a GUI application that can be managed by the customer.
- The customer must have the ability to build new self-services applications like new IVR flow for new service.
- Customer must have the ability to define/change routing rules by himself based on customer's profile.
- The help desk must support Outbound, Call-back and Campaigns – including preview, progressive and automated outbound dialing.
- The supervisor must be able to see the status of help desk agents in real-time in his PC like logout, busy, free, release, non ACD etc. in graphical form in pie chart / bar chart.
- **Help desk facilities:**
- Real-time Monitoring – must provide supervisors with statistical information about the current status of the help desk with on line refresh (1sec). The application must include pre-defined list of reports and the customer (end user) should be able to choose reports as needed.
- The Real Time application must provide the ability to build/change the workspace for each user and by user (not vendor or distributor).
- The RT must provide the ability to move agents to/from different groups/queues for current login only.
- **Historical Reports** – Supervisor must be able to collect all information from call entry to call termination. Call profile details for internal investigation purposes should be part of the contact center solution. Supervisor should be able to generate reports for last 3 months by selecting date, time, agent id, group id etc. for agent performance like login / logout / break time, total call answered, average speed answer, wrap up time, IVR usage etc. these reports should be from helpdesk application, not from call billing software.

The help desk solution must have an embedded Management Information System (MIS) suite that monitors all help desk activities, generating reports that summarize the past performance of the system over a given time period, and providing statistical analysis of the help desk within a specified period. Real-time and historical reports provide:

Help desk agent should be able to do following activities from agent application installed on PC:

- Login/Logout from group
- Release/Resume
- Ready
- Wrap-up Code
- Release for Break
- Release for Meeting
- Control Wrap Up

- Supervisor Help
- Agent Board
- Answer
- Hold
- Retrieve
- Hang Up
- Integration Capabilities with Sales Force, Microsoft dynamics, SAP CRM & Oracle & SIEBEL

Integrated Voice Mail Specification:

Integrated 8 port Voice mail system should be supplied along with the system as per mentioned in the schedule of requirement at respective locations. The user should be able to review voice message from his phone. Voice mail should support message wait indication

The Voice mail system solution should be built-in solution from same OEM of UC Call manager application. Third party or hardware-based Voice mail should not be accepted. Voice mail solution should be provided from day1.

User should be able to listen to, delete, save, reply to, and forward through the phone

Should support below mentioned Unified Messaging Features:

- Messages to email
- Multilingual Support
- MWI – Message Waiting Indication
- Voicemail Activation via Soft Keys
- Users should be able to check and handle voice messages from all devices, including the email client/ Webmail client and telephone.
- Voicemail should support the following:
- Microsoft Exchange – 2007 and higher

Analog Media Gateway – (24/96 FXO/FXS) specification:

Media Gateway should have min 96 FXS /96 FXO ports and should be from the same OEM of telephony system. It should be possible to place the gateway at any location in the LAN network without any distance limitation between server and gateway.

- Voice Processing – voice codecs: G.711, G.729A, G.723.1, GSM, iLBC; echo cancellation: G.168 with 64ms echo tail; dynamic jitter buffer; voice activity detector (VAD)
- Call Handling – configurable dialing plan, up to 500 routing rules
- Fax Processing – T.30, T.38 fax relay

- DTMF – RFC 2833, SIP INFO, audio
- Gateway Configuration – Web based user interface
- User Features – caller ID, call forward, call transfer, call forking, hotline, speed Dialing, busy tone detection
- Protocol – SIP (RFC3261, etc.)
- Standards – caller ID detection (FSK/DTMF), configurable call progress tone plan
- Primary and Backup – the gateway can be configured and controlled in server clusters
- LED indicators– power, system status, network status, line status
- FXS/FXO Connector – RJ-45
- Ethernet Connector – Dual RJ-45, 10/100/1000
- Power Input – Dual AC or Dual DC (-48 VDC) Power Supply in redundancy
- Operation Humidity – 10% to 90% (non-condensing)
- Operation Temperature – 0 to 40°C

PRI Gateway specification:

PRI gateway should have Configuration – 1X PRI (30 Channels) / 2X PRI (60 Channel)/ 4 X PRI (120 Channel) and should be from the same OEM of telephony system. It should be possible to place the gateway at any location in the LAN network without any distance limitation between server and gateway.

- Voice Processing – G.711, G.729A, G.723.1, GSM, iLBC; echo cancellation: G.168 with 64ms echo tail; dynamic jitter buffer; VAD and CNG
- Calling Control – called/calling party number translation; second stage Dialing; auto Dialing with DTMF; ring back tone generation and detection; voice announcement
- Voice Proxy – RTP voice proxy function for NAT/firewall traversal
- Fax Relay – T.30 transparent mode, T.38 fax relay
- Call Handling – configurable dialing plan, up to 500 routing rules
- Configuration Interface – Web Utility
- Remote Management – Telnet, HTTP, TR069
- PSTN – ISDN PRI standard: ANSI, NI-2, DMS, 5ESS
- SIP – RFC3261, RFC2976, RFC3515, RFC3581
- DTMF – tone detection; DTMF relay: RFC2833, INFO (SIP)
- Play ring-back tone
- Ethernet – RJ-45, 10/100/1000 Base-T
- Trunking Interface – RJ-45
- Power Input – Dual AC or Dual DC (-48 VDC) power supply in redundancy
- Operation Humidity – 10% to 90% (non-condensing)
- Operation Temperature – 0 to 40°C

- Form Factor for gateway should not be more than 2U

Specification of PC based Operator Console:

SIP phone for operator console should be from the same OEM of IP telephony system

- minimum 3.5" 340x160-pixel graphical LCD with backlight
- up to 6 VoIP accounts
- Minimum 8 one touch programmable keys with LED
- These 8 keys should be programmed up to 21 paperless keys with 3 page view
- option to connect to 3 DSS module, 20 key per DSS module (up to 60 keys using 3 page view)
- key programming from telephony system web not from telephone web
- Local phonebook up to 100 entries
- Auto provision via FTP/TFTP/HTTP/HTTPS for mass deployment
- SRTP/ HTTPS/ TLS, 802.1x
- Volume adjustment, ring tone selection, Headset, Wall-Mountable
- IPv4 / IPv6
- Codec: G.722, G.711, G.723.1, G.729, G.729A, G.726, iLBC
- VAD, CNG, AEC, PLC, AJB, AGC
- Full-duplex hands-free speakerphone
- SIP v1 (RFC2543), v2 (RFC3261)
- NAT Traversal: STUN mode or 3rd party SBC
- DTMF: In-Band, RFC2833, SIP Info
- IP Assignment: Static/DHCP
- Wi-Fi support
- 1xRJ9 handset port
- USB port
- LED for call and message waiting indication
- Dual-port Gigabit Ethernet
- Power over Ethernet (IEEE 802.3af), Class 2

Expansion module:

- 4.3" 272 x 480-pixel color screen
- 20 physical keys on each page with dual-color LEDs
- Three-page keys with illuminated LED for fast switching between pages.
- Supports up to 3 modules daisy-chain
- Wallpaper, screensaver and power saving

Specification of Type-1 SIP Phone:

SIP phone should be from the same OEM of IP telephony system

Minimum 128x48-pixel or more graphical LCD with backlight

2 VoIP accounts

4 soft keys

4 Function keys (Phonebook, MWI, Headset, Redial)

Auto provision via FTP/TFTP/HTTP/HTTPS for mass deployment

SRTP/ HTTPS/ TLS, 802.1x

Volume adjustment, ring tone selection, Headset, Wall-Mountable

IPv4 / IPv6

Codec: G.722, G.711, G.726, G.729, G.729A, iLBC, opus

VAD, CNG, PLC, AGC, AEC, RTCP-XR (RFC3611), VQ-RTCPXR (RFC6035)

Full-duplex hands-free HD speakerphone

SIP v2 (RFC3261)

Support for 3rd party SBC

DTMF: In-Band, RFC2833, SIP Info

IP Assignment: Static/DHCP

1xRJ9 handset port

LED for call and message waiting indication

Dual-port Gigabit Ethernet

Power over Ethernet

Specification of Type-2 IP Phone:

SIP phone should be from the same OEM of IP telephony system

- 3.5-inch (480x320) Colour Screen with backlight
- 4 VoIP accounts
- Local phonebook up to 100 entries
- Auto provision via FTP/TFTP/HTTP/HTTPS for mass deployment

- SRTP/ HTTPS/ TLS, 802.1x
- PoE, Headset, Wall-Mountable
- Volume adjustment, ring tone selection
- Voicemail, MWI
- Call Park, call pickup
- Narrowband codec: G.711, G.723.1, G.726, G.729AB
- Wideband Codec: G.722, AMR-WB, Opus
- VAD, CNG, AEC, PLC, AJB, AGC
- Full-duplex hands-free HD speakerphone
- SIP V2 (RFC3261)
- 3rd party SBC support
- DTMF: In-Band, RFC2833, SIP Info
- IP Assignment: Static/DHCP/PPPoE
- 1xRJ9 handset port
- 1xRJ9 headset port
- LED for call and message waiting indication
- 2xRJ45 10/100M Ethernet ports
- Power over Ethernet

Call Billing Software specification:

The system must have call billing software that is able to log all calls made and received by an extension (either internal or external).

All CDR (Call Detail Records) must be sent on-line from both servers to external Windows based Billing system for complete recording of Internal, External and Network calls to generate various types of traffic reports.

- The following details must be provided:
- Reports must indicate the call start date.
- Reports must indicate the call start time
- Reports must indicate the elapsed time
- Reports must indicate the trunk group ID
- Reports must indicate the trunk ID
- Reports must indicate the charged user ID
- Reports must indicate the caller ID

Make as approved by Engineer-in-charge
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Technical specification of core switch

S. No.	Detailed Minimum Technical Specifications	Compliance (Yes / No)
A	Solution Requirement	
	The Switch should support non-blocking Layer 2 switching and Layer 3 routing	
	These switches should support stacking/VC /Equivalent technology for min 8 Members	
B	Hardware and Interface Requirement	
	Min 48 10G SFP+ Ports, 4x100G/40G QSFP28 Ports ,2x40G QSFP+ Ports. Fiber module should be from the same make. should Come with stacking/VSS or equivalent accessories for min 400 Gbps between 2x switches.	
	The Switch should have non-blocking architecture	
	switching capacity of 1.92Tbps, forwarding performance 1400 Mpps,	
	Should support non-blocking hardware architecture	
	All interfaces should provide wire speed forwarding for both Fiber and copper modules	
	Support for at least 4000 VLANs	
C	Layer2 & Layer 3 Features	
	It should support IGMP snooping v1, v2 & v3	
	It should have Static Routing, RIP, VRRP, OSPF, BGP, EIGRP, VxLAN, PIM-SM, PIM-DM, OSPFv3, BGP 4+ ,MPLS from day 1	
	Switch should support Jumbo Frame	
	Dynamic Host Configuration Protocol (DHCP) snooping	
	Switch should support LLDP capabilities	
	Should support Port Isolation, Port binding against IP and MAC, should have technology like Sticky MAC feature for traffic utilization, DHCP Snooping, Option 82,	
	Should have mechanism for DDoS attack prevention, SYN flood protection, MD5, SHA-256	
	Should support Secure Shell (SSH) Protocol and Simple Network Management Protocol Version 3 (SNMPv3).	
	Should support less than 50 Millisecond convergence for ring based on G.8032/Equivalent, Support for ISSU, MPLS	
	Switch needs to have console port for administration & management	
	Management using CLI, GUI using Web interface should be supported	
	FTP/TFTP for upgrading the operating System	
	IEEE 802.1x, Radius support	

	IEEE 802.1D Spanning-Tree Protocol, UDLD	
	IEEE 802.1p class-of-service (Cos) prioritization, DSCP, TOS	
	IEEE 802.1Q VLAN, QinQ	
	IEEE 802.3u 100BASE-TX specification	
D	Manageability	
	Switch should be able to support management via CLI, Web interface, All switches & wireless component and switch should have same operating system. Same access points and switches should have option to be managed by cloud infrastructure without replacing the hardware in future.	
	ZTP (Zero touch Provisioning), NTP, RSPAN, DHCP Server, DHCP Relay, 802.3ah, 802.1ag, sflow/Netflow	
	Switch should be manageable through both IPv4 & IPv6. IPv4/IPv6 Dual Stack, ISATAP Tunnel	
	Operating temperature 0-40 Degree C	
E	Safety & Compliance	
	RoHS, CE, FCC	
	Make as approved by Engineer-in-charge	

Technical Specification of Distribution Switch

S. No.	Detailed Technical Specifications	Compliance (Yes / No)
A	Solution Requirement	
	The Switch should support non-blocking Layer 2 switching and Layer 3 routing	
	These switches should support stacking/VC /Equivalent technology for min 12 Members	
B	Hardware and Interface Requirement	
	Min 24x10GBPS SFP+, 4x 40G/100G(QSFP28) Port	
	The Switch should have non-blocking architecture. RPS.	
	800 Gbps or higher Backplane capacity and minimum 600 Mpps of forwarding rate	
	Should support non-blocking hardware architecture	
	All interfaces should provide wire speed forwarding for both Fiber and copper modules	
	Support for at least 4000 VLANs & 32k MAC address	
C	Layer2 & Layer 3 Features	
	It should support IGMP snooping v1, v2 & v3	
	It should have Static Routing, RIP, VRRP, OSPF & BGP from day 1	
	Switch should support Jumbo Frame	
	Dynamic Host Configuration Protocol (DHCP) snooping	
	Switch should support LLDP capabilities	

	Should support Port Isolation, Port binding against IP and MAC, should have technology like Sticky MAC feature for traffic utilization, DHCP Snooping, Option 82,	
	Should have mechanism for DDoS attack prevention, SYN flood protection, MD5, SHA-256	
	Should support Secure Shell (SSH) Protocol and Simple Network Management Protocol Version 3 (SNMPv3).	
	Should support less than 50 Millisecond convergence for ring based on G.8032/Equivalent , Support for ISSU ,	
	Switch needs to have console port for administration & management	
	Management using CLI, GUI using Web interface should be supported	
	FTP/TFTP for upgrading the operating System	
	IEEE 802.1x, Radius support	
	IEEE 802.1D Spanning-Tree Protocol, UDLD	
	IEEE 802.1p class-of-service (CoS) prioritization, DSCP, TOS	
	IEEE 802.1Q VLAN,QinQ	
	IEEE 802.3u 100BASE-TX specification	
D	Manageability	
	Switch should able to support management via CLI, Web interface	
	ZTP(Zero touch Provisioning),NTP, RSPAN, DHCP Server , DHCP Relay.	
	Switch should be manageable through both IPv4 & IPv6.IPV4/ipV6 Dual Stack , ISATP Tunnel	
	Operating temperature 0-50 Degree C	
E	Safety & Compliances	
	RoHS, CE, FCC	
	Make as approved by Engineer-in-charge	

Technical Specification of 24 POE Port Access PoE Switch

1	Product details - Please specify
1.1	Please mention Make, Model No. and Part Code.
2	Architecture & Port Density
2.1	The Switch should have 24 x 100M/1G RJ45 PoE/PoE+ ports with 4 x 1G SFP/10G SFP+/25G SFP28 Slots from Day 1. Required stacking cable with min. length of 1 or 3mtr based on requirement to be considered per switch
2.2	The Switch should support Virtual Switching System (VSS) or Virtual Chassis (VC) or equivalent Switch Clustering/Stacking feature, where the Switch Clustering feature should combine multiple switches into a single network element.

2.3	The Switch should be configured with Inter Switch Links (ISLs) between the Access Switches, to achieve a minimum of 100 Gbps of Inter Switch Bandwidth, from Day 1.
3	Performance
3.1	Switching Bandwidth: The Switch should provide Switch Fabric Bandwidth Capacity of 240 Gbps or more.
3.2	Forwarding Capacity: The Switch should provide Packet Forwarding Capacity of 180 Mpps or more.
4	Layer 2 features
4.1	Should support up to 32K MAC addresses or more.
4.2	Should support Jumbo Frames (9K bytes or more).
4.3	Should support 4K Active VLANs, with the following features.
•	Dual-Mode VLANs
•	MAC-based VLANs
•	Dynamic MAC-based VLAN Activation
•	Dynamic Voice VLAN Assignment
•	Dynamic VLAN Assignment
•	VLAN mapping or VLAN Translation to translate CVLANs to SVLANs
4.4	Should support Spanning Tree Protocols, with the following features.
•	802.1D Spanning Tree
•	802.1W Rapid Spanning Tree Protocol (RSTP)
•	802.1s Multiple Spanning Tree
•	802.1s Multiple Spanning Tree enhancement (MSTP+)
•	Fast Port Span, Fast Uplink Span, and Single-instance Span
•	Compatibility with PVST/PVST+, PVRST+ and PVST+
•	BPDU Guard

•	Root Guard for STP & MSTP
•	Port Loop Detection
•	Spanning Tree path cost method changes
•	MSTP path-cost configuration
•	Flexlink or equivalent
4.5	Should support Link Aggregation Groups (LAG), with the following features.
•	Static LAG
•	802.3ad Link Aggregation Control Protocol (Dynamic LAG)
•	Dynamic insertion and removal of ports
•	Support for LAG between different default port speeds
4.6	Should support 802.1q Tunneling, with the following features.
•	802.1ad (Q-in-Q) tagging
•	Q-in-Q BPDU tunneling
•	Selective Q-in-Q
4.7	Should support Private VLANs, with the following features.
•	PVLANs with dual mode support
•	PVLAN with LAG
4.8	Should support VLAN Registration Protocol, with the following features.
•	Multiple VLAN Registration Protocol (MVRP)
•	MVRP with Per-VLAN STP and Per-VLAN RSTP
4.9	Should support the following Layer 2 Switching Features.
•	Remote Fault Notification (RFN)
•	Link Fault Signaling (LFS)
•	Uni-Directional Link Detection (UDLD)

5	Layer 3 features
5.1	Should support up to 16K IPv4 routes or more.
5.2	Should support the following Basic IPv4 and IPv6 Layer 3 Routing features.
•	IPv4 and IPv6 Static Routes
•	RIP v1/v2, RIPng
•	ECMP
•	Port-based Access Control Lists
•	Layer 3/Layer 4 ACLs
•	Host routes
•	Virtual Interfaces
•	Routed Interfaces
•	Route-only Support
•	Routing Between Directly Connected Subnets
5.3	Should support the following Advanced IPv4 and IPv6 Layer 3 Routing features.
•	IPv4 and IPv6 Dynamic Routes
•	OSPF v2, OSPF v3 (IPv6)
•	PIM-SM, PIM-SSM, PIM-DM, PIM passive (IPv4/IPv6 multicast routing functionality)
•	Policy Based Routing (PBR)
•	Virtual Route Redundancy Protocol VRRP v2 (IPv4)
•	Virtual Route Redundancy Protocol VRRP v3 (IPv6)
•	Non-Stop Routing (NSR)
•	GRE
•	IPv6 over IPv4 tunnels
•	Multi VRF (IPv4 and IPv6) with Inter-VRF route leaking using static routes

•	DHCP Server (IPv4 & IPv6)
•	MSDP
6	Quality of Service (QoS) & Traffic Management
6.1	Should support the following Quality of Service (QoS) features.
•	ACL Mapping and Marking of ToS/DSCP (CoS)
•	ACL Mapping and Marking of 802.1p
•	ACL Mapping to Priority Queue
•	Classifying and Limiting Flows Based on TCP Flags
•	DiffServ Support
•	Honoring DSCP and 802.1p (CoS)
•	MAC Address Mapping to Priority Queue
•	Dynamic Buffer Allocation for QoS Priorities
•	Separate QoS Queuing for Unicast and Multicast
•	Priority Queue Management using Weighted Round Robin (WRR), Strict Priority (SP), and a combination of WRR and SP
6.2	Should support the following Traffic Management features.
•	ACL-based Inbound Rate Limiting and Traffic Policies
•	Broadcast, Multicast, and Unknown Unicast Rate Limiting
•	Inbound Rate Limiting per Port
•	Outbound Rate Limiting per Port and per Queue
7	Software Defined Networking (SDN)
7.1	Should support the following SDN features and functionality.
•	OpenFlow v1.0
•	OpenFlow v1.3
•	Hybrid Switch Mode (OpenFlow enabled on per-port basis)

•	Hybrid Port Mode with Layer 2 Mode, Layer 3 Mode, and Simultaneous Layer 2/Layer 3 Mode
•	Support for Multiple Controllers
8	Security
8.1	Should support the following Security features
•	Layer 2 ACLs (MAC)
•	Layer 3 ACLs (IPv4 & IPv6)
•	Layer 4 ACLs
•	Binding IPv4, IPv6, and MAC ACLs to VLAN
•	802.1X Authentication and Accounting
•	MAC Authentication and Accounting
•	Web Authentication
•	DHCP Snooping
•	Dynamic ARP Inspection
•	Neighbor Discovery (ND) Inspection
•	Protection against Denial of Service (DoS) Attacks
•	Authentication, Authorization, and Accounting (AAA)
•	Port Security with Secure MAC Address Limiting
•	Unknown Unicast Flood Blocking (UUFb)
•	Advanced Encryption Standard (AES) with SSHv2
•	RADIUS/TACACS/TACACS+
•	Secure Copy (SCP)
•	Secure Shell (SSHv2)
•	Change of Authorization (CoA) RFC 5176
•	Trusted Platform Module

•	Protected Ports
•	IPv6 RA Guard
•	RADSEC (RFC 6614)
•	Encrypted Syslog (RFC 5425)
8.2	The Switch should support the following Flexible Authentication features.
•	802.1x with Dynamic ACL Assignment
•	802.1x with Dynamic VLAN Assignment
•	802.1x and MAC Authentication on the same port
•	Flexible Authentication together with Dynamic ARP Inspection (IPv4 and IPv6) with Dynamic ACLs
•	Flexible authentication together with DHCPv4 and DHCPv6 Snooping with Dynamic ACLs
•	802.1x Authentication together with IP Source Guard Protection
•	MAC Authentication together with IP Source Guard Protection
•	MAC Authentication together with Dynamic VLAN Assignment
•	MAC Authentication together with Dynamic ACLs
•	MAC Authentication together with 802.1x
•	802.1x together with Denial of Service (DoS) Attack Protection
•	Periodic Reauthentication for MAC Authentication
•	Periodic Reauthentication for 802.1x
8.3	The Switch should support Cisco ISE, Aruba ClearPass, and Ruckus Cloud path for 802.1X Authentication, MAC Authentication, Dynamic VLAN Assignment, Dynamic ACL Assignment, External Web Authentication and Change of Authorization (CoA).
9	Monitoring & Manageability
9.1	Should support manageability using Network Management Software with Web based Graphical User Interface (GUI) and Cloud based Management Solutions.

9.2	The Switch should support the following Monitoring & Management features.
•	ERSPAN
•	RSPAN
•	Virtual Cable Tester (VCT)
•	Digital optical monitoring
•	LEDs On/Off Command
•	Cisco Discovery Protocol (CDP) for IPv4 and IPv6 Traffic
•	Automation with Ansible
•	REST API
•	Switch Cloud Management
•	SNMP v1, v2, and v3
•	Mirroring based on ACL, MAC ACL and VLAN
•	Configuration Archive, Replace & Roll back
•	IP DHCP binding scalability of minimum 2000 Devices
•	Software Defined Video-over-Ethernet (SDVoE) compliance or equivalent
9.3	The Switch should support these additional features.
•	Fan less Mode for quiet zero noise, switch operation
•	Energy Efficient Ethernet (802.3az)
•	Configuration Archive and Replace
•	External USB storage
•	System backup to USB
•	In-Service Software Upgrade (ISSU) for stacking
•	Ethernet in the First Mile for OAM (IEEE 802.3ah standard)
9.4	Should support Integrated Standard based Command Line Interface (CLI), Telnet, TFTP, HTTP access to switch management/monitoring.

9.5	Should support NetFlow or sFlow or equivalent.
10	Physical Attributes, Memory, PoE, Power Supply and Fans
10.1	The Switch should have minimum 4MB Packet Buffer, 4GB Main Memory and 8GB Flash Memory.
10.2	The Switch should have a PoE Power Budget of 370W from day 1.
10.3	The switches should support physical stacking of more than 8nos per stack
10.4	Operating Temperature: 0 - 50 C, Humidity 10% to 95 % Non-Condensing
11	Mandatory Compliance:
11.1	All categories of Switches, Transceivers, Switch OS, WAP's should be from same OEM
11.2	The Switch OS should be EAL/NDPP Certified. The Latest Updated Maintenance Common Criteria Report (Evaluation and Validation) should be submitted.
11.3	3Yrs of end user support warranty should be considered.
11.4	Proposed switches, access points, accessories like stacking cable, transceivers etc should from same OEM for interoperability and management.
11.5	Bidder must submit Part Coded Bill of Materials of the offered Product
11.6	Manufacturers Authorization Letter Specific to this tender must be submitted. Tender submitted without MAF will be rejected.
12	Warranty
12.1	The Switch should be quoted with 3 Years of TAC Support and Lifetime (Till End of Support) for Hardware Warranty with NBD Hardware Replacement.
12.2	The Switches should be compatible with the existing smart zone setup in terms of for management & troubleshooting. The required software, management licenses etc. should be offered
12.3	The Switches should be compatible with the existing smart zone setup for performing scheduled firmware upgradations.
12.4	Switches backup and restore configuration should be done through existing smart zone network management solution

	Make as approved by Engineer-in-charge
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Technical Specification of Firewall /UTM

1	2x10/100/1000 Base T ports with 2x 100/1000 SFP Ports in Combo, Min 1 x10 Gbps ports either in Copper or Fiber, Min 4 Gbps L2/L3 Throughput, Min concurrent session 2 million, Min 1 x set of Bypass Ports, Min 1x USB ports, should have dual AC power supply from day 1. Firewall, Switches, Wireless Components, Fiber module should be from the same OEM for seamless integration.
2	The firewall should support transparent (Layer 2) firewall or routed (Layer 3) firewall Operation
3	Supports WLAN applications and 3G wireless access
4	OSPF, RIP, and OSPFv3, IPv6 static routing, route advertisements, tunnel translation
5	Anti-Virus & Integrated IPS Features Set
6	IPv6-enabled inspection services for applications based on HTTP, FTP, SMTP, ICMP, TCP, and UDP
7	The Appliance should have USB Ports and dedicated bypass ports in addition to requested data ports
8	Network security appliance should support "Stateful" policy inspection technology. It should also have application intelligence for commonly used TCP/IP protocols like telnet, ftp etc.
9	The proposed vendor must have a track record of continuous improvement in threat detection (IPS)
10	Firewall should provide application inspection for LDAP, SIP, SNMP, FTP, SMTP, HTTP, DNS, ICMP, DHCP, RPC, SNMP
11	Allows secure deployment of next-generation IPv6 networks, as well as hybrid environments that require simultaneous, dual stack support of IPv4 and IPv6, should have feature to scan even up to 15 layers of compressed files.
12	IPSEC and SSL VPN, Application control settings
13	Antivirus settings, User and Group settings

14	SSL VPN tunnel, GRE Tunnel, malicious URL filtering, URL category filtering
15	The proposed solution should be able to replace the web page when the web page matches the Web Filtering blocking criteria. Min 2500 predefined attack signatures.
16	The solution shall allow administrators to create multiple new local URL filtering categories besides dynamic categories
17	real-time statistics of CPU, memory, and interface traffic, NTP synchronization
18	RADIUS server entries, TACACS+ server entries
19	The IPSEC VPN and SSL VPN capability. Firewall, Router, Wireless Components, Fiber module should be from the same OEM for seam less integration.
	Make as approved by Engineer-in-charge

C. IP CCTV Surveillance System

Cat6 UTP LSZH Cable

S. No.	Technical Parameters	Compliance (Yes/No)
1	Cat 6, 23 AWG Bare Copper Solid Conductor UTP Unshielded LSZH, in accordance to ANSI/TIA 568 C.2 Category 6, Flame Rating IEC 60332-1, 250 Mhz Bandwidth, No Shield/Non-Metallic Barrier Tape,	
2	Suitable for 1GBASE-T with Min. 250Mhz of Bandwidth	
3	4 Twisted Pair alongside PE / PVC Cross Separator	
4	Conductor: 23AWG Solid Annealed Bare Copper	
5	Conductor Diameter: $0.55 \pm 0.025\text{mm}$	
6	Insulation: High Density Polyethylene, Diameter $0.95 \pm 0.05\text{mm}$	
7	Jacket: LSZH complying to IEC 60332-1 Flame Rating, IEC 60754-1/2 for Halogen Acid Test, IEC 61034-2 for Smoke Density Test	
8	Euroclass : Eca	
9	Cable Outer Diameter: $5.8 \pm 0.3 \text{ mm}$	

10	Operating Temperature: -20°C to +60°C	
11	Cable shall not have any kind of Non-Metallic or Metallic Shield Barrier Tape inside	
12	Breaking Strength: $\geq 400\text{N}$	
13	Temperature Index & Oxygen Index should meet ASTM D 2863	
14	Bend Radius: 4 X Cable Diameter (Min.) or better	
15	Conductor Resistance: $\leq 8.0 \Omega / 100\text{m}$	
16	Resistance Unbalance: 5% Max	
17	Mutual Capacitance: $\leq 5.6\text{nF}/100\text{m}$	
18	Propagation Velocity: 69% (nominal)	
19	Dielectric Strength: DC 1500 V /Min.	
20	Min. Elongation at Break of Insulation: 400%	
21	Min. Jacket Tensile Strength: $\geq 10 \text{ N/mm}^2$ (Mpa)	
22	RoHS Complied	
	Make as approved by Engineer-in-charge	

Cat6 Unshielded Patch Cord, 1 Meter & 3 Meter

S. No.	Technical Parameters	Compliance (Yes/No)
1	Cat6 Patch Cord U/UTP Unshielded LSZH 24AWG Patch Cord, Diameter $< 6.0\text{mm}$, 3 Meter, Operating Temperature -20 to +60 Deg C	
2	Cable Construction of Patch Cord: U/UTP LSZH	
3	Number of conductors: 8 (4 Twisted Pairs)	
4	Conductor Material: Bare Copper (24AWG Stranded)	
5	Cable Overall Nominal diameter: $\leq 6.0 \text{ mm}$	

6	Insulation: HDPE	
7	Cat 6 patch cord plug to have round cable holder strain relief transparent boot to avoid bending.	
8	Jacket: LSZH (Low Smoke Zero Halogen) with Flame Rating IEC 60332-1 or better	
9	Operating Temperature: -20 to + 60 °C	
10	Plug should have high repeatability cross talk performance with Min. 750 Cycles.	
	Make as approved by Engineer-in-charge	

24Port Patch Panel

Sr. No.	Technical Parameters	Compliance (Yes/No)
1	24-Port 1U Unloaded Universal Modular Straight Patch Panel preloaded with cable support Bar (with slots to tie individual cable properly at the Support Bar (Shuttered IO not recommended, because when shutter gets malfunctioned then entire IO is of no use)	
2	The Patch panel should be universal (stainless steel rear metal frame) and should be able to support both for UTP & STP Solutions	
3	Patch panel should be equipped with cable strain relief retention tray (Cable Support Bar) with slots to tie individual cable properly at the Support Bar	
4	Each port of the panel should have individual transparent labelling point	
5	Panel should be inbuilt transparent spring shutter for dust protection on each port. (Shuttered IO not recommended, because when shutter gets malfunctioned then entire IO is of no use)	
6	Should be RoHS & UL 94V-0 complied	
7	Panel shall be UL 1863 Rated (Certificate to be provided along with Bid)	

	Make as approved by Engineer-in-charge	
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Cat6 Unshielded Modular Jack

Sr. No.	Technical Parameters	Compliance (Yes/No)
1	Category 6 RJ45 Unshielded Modular Jack, ISO/IEC 11801:2nd edition, EN 50173-1, ANSI/TIA/EIA 568-C.2 Category 6, IEC 60603-7, Universal 110 Impact Tool & Toolless (Both) Termination, Min. 250Mhz Bandwidth, Without any OEM Proprietary Tool Termination	
2	Suitable for 1000BASE-T applications in acc. with IEEE 802.3an up to 250 MHz .	
3	Compatible with RJ standard plugs (RJ11, RJ12, RJ45), PCB Based Universal 110 Impact Tool & Toolless (both) based connection of installation cables	
4	Each Jack should be supplied with one Strain relief boot for Pairs and separate additional bend limiting boot for Cable with side split clip with the I/O and a cable tie for proper holding of cable to the JACK.	
5	The Bend limiting boot should have additional clip or locking facility to project the Cable and it should be inbuilt with the JACK.	
6	The Jack should be universal toolless and tool-based connection of installation cables AWG 24 – 23 with Solid Conductor cables.	
7	IDC termination should feature Color Coding according to EIA/TIA 568-A/B, Gold-Plated Contacts, >= 750 mating cycles, >= 200 insertion cycle at IDC	
8	Material: RoHS complied	
9	Housing material: Polycarbonate/Flame Retardant PVC (UL-94-V0)	
10	Should be UL Certified as per UL 1863	
11	DC/AC Voltage Endurance - DC1000V/AC750V (1min)	

12	DC Resistance: Max. 0.3 Ohms	
13	Plug Retention Force: 30 lbs	
14	Operating Temperature: -10 to +60 °C	
15	Current Rating: 1.5 Amp	
16	The Modular Jack shall not require and Proprietary Termination Tool for termination of Cable.	
	Make as approved by Engineer-in-charge	

Face Plates

S. No.	Technical Parameters	Compliance (Yes/No)
1	Style (Square) Keystone-Type Shuttered Faceplates, 1, 2 & 4 port configurations	
2	UK Style (Square) Keystone-Type Faceplates, 1, 2 & 4 port configurations, White Color	
3	Should be featured with spring shutter for each port	
4	Elegant 2 Piece (2 Plate) design for better aesthetics	
5	Cover and Base Plastic Material to meet Min. ABS-UL94-V2	
6	Suitable for both Flush and Wall mount gang box	
7	Dimensions : 86 x 86 x 12.8 (mm)	
	Make as approved by Engineer-in-charge	

Technical Specification of Dome Camera

S.N o.	Features	Specification
	General	
1	Imaging Device	1/2.7" Progressive scan CMOS

2	Maximum resolution	5MP - (2592X1944)
3	Video Compression	H.265/H.265+/H.264/H.264+/MJPEG
4	Video Quality Adjustment	8Kbps ~ 8Mbps
5	ONVIF	ONVIF profile S,G & T
6	Minimum Illumination	Color 0.05lux @ F1.2 ; B/W 0 lux (IR ON)
7	IR Range	45 Mtr.
8	Day/Night Method	AUTO ICR
9	Signal-to-Noise ratio	50 db minimum
10	Electronic shutter Range	Auto/Manual, 1/5 - 1/100000s
11	Dynamic Range	120 db minimum
12	Back light compensation	HLC, WDR(120db)
13	Digital Noise reduction	3D Noise Reduction
14	Analytics	Face Detection, Customer Flow statistics, Heat Map, Human & Vehicle Detection, Pedestrian Intrusion Detection & Line cross Detection. Queue length detection, regional population statistics, Stationary Object Detection, Tempering detection, Rare Sound Detection, License plate detection.
15	Privacy Zone	4 Configurable windows
16	Region of Interest	8
17	Motion Detection	On/OFF (1 to 8 Level)
18	Image setting	Compression, 3D digital noise reduction (ON/OFF) color, brightness, sharpness, contrast, white balance, exposure control, backlight compensation
19	Alarm	I/O, 1-IN and 1-OUT
20	Audio	1 In & 1 Out
21	Audio Code type	G711A/G711U
22	Audio Type	Two Way
23	Electrical	
24	Network	RJ-45 (10/100Base-T)
25	Power Input	PoE 802.3af/12VDC
26	Power Consumption	Max 4.55W
27	Local Storage	Yes 256GB
28	Cloud Storage	Yes
29	Cloud Overwrite	OFF/Automatic/1/3/7/14/30/90Days
30	Video Type	RF/AVI/MP4

	Mechanical	
31	Lens	2.7 - 13.5 mm auto focus motorized optical zoom
32	Angular Field of View	Horizontal: 107-33.6°, Vertical: 54.7-18.9°, Diagonal: 133.7-37.6°
	Video	
33	Video Encoding	H.265, H.264 Smart Codec bit rate control (CBR & VBR)
34	Video Quality Adjustment	8Kbps ~ 8Mbps
35	Video Stream	Quad-Stream (Main/Sub/Mobile/Fourth)
36	Maximum frame Rate	50Hz: 25fps@5MP/4MP/3MP/2MP/960P/720P/VGA/QVGA
		60Hz: 30fps@5MP/4MP/3MP/2MP/960P/720P/VGA/QVGA
37	Multiple streaming	Quad Stream Main Stream: 2592x1944, 2592x1520, 2048x1536, 1920x1080, 1280x960, 1280x720 Sub Stream: 1920x1080, 640x480, 320x240 Mobile Stream: 640x480, 320x240 Fourth Stream: 640x480, 320x240
38	Supported Protocols	IPv4, IPv6, IPv4 DNS, IPv6 DNS, TCP/IP, HTTP, HTTPS, UPnP, RTSP/RTP/RTCP, IGMP, SMTP, DHCP, NTP, FTP, DNS, DDNS, SNMP, NFS/SMB/CIFS, SSL/TLS, P2P.
39	Security	IP address filter; complicated password; Authenticated username and password, AES 256
40	Mobile Surveillance	Support mobile surveillance through app on android and i-phone
41	Supported browser	IE8,9,10,11, Chrome 44/lower, Firefox 51/lower, Safari 11/lower
42	Video backup file type	RF, AVI, MP4
43	Snapshot	Yes
44	Hardware Reset	YES
45	User	
46	Max. User	7 users
47	Multicast	Unlimited
48	Security Access	Multiple user access level with password protection IP filtering, HTTPS.
49	Environmental	
50	Operating Temperature	-30~+55°C
51	Operating Humidity	less than ≤ 95% RH
52	Housing	IP66 & IK10
53	Camera & Housing type	Dome, Metal
54	Certification	UL, CE, FCC, BIS
		Make as approved by Engineer-in-charge

Technical Specification of Bullet Camera

Sr.No.	Features	Specification
	General	
1	Imaging Device	1/2.7" Progressive scan CMOS
2	Maximum resolution	5MP - (2592X1944)
3	Video Compression	H.265/H.265+/H.264/H.264+/MJPEG
4	Video Quality Adjustment	8Kbps ~ 8Mbps
5	ONVIF	ONVIF profile S, G & T
6	Minimum Illumination	Color 0.01lux @ F1.2 ; B/W 0 lux (IR ON)
7	IR Range	45 Mtr.
8	Day/Night Method	AUTO ICR
9	Signal-to-Noise ratio	50 db minimum
10	Electronic shutter Range	Auto/Manual, 1/5 - 1/100000s
11	Dynamic Range	120 db minimum
12	Back light compensation	HLC, WDR (120db)
13	Digital Noise reduction	3D Noise Reduction
14	Analytics	Face Detection, Customer Flow statistics, Heat Map, Human & Vehicle Detection, Pedestrian Intrusion Detection & Line cross Detection. Queue length detection, regional population statistics, Stationary Object Detection, Tempering detection, Rare Sound Detection, License plate detection.
15	Privacy Zone	4 Configurable windows
16	Region of Interest	8
17	Motion Detection	On/OFF (1 to 8 Level)
18	Image setting	Compression, 3D digital noise reduction (ON/OFF) color, brightness, sharpness, contrast, white balance, exposure control, backlight compensation
19	Alarm	I/O, 1-IN and 1-OUT
20	Audio	1 In & 1 Out
21	Audio Code type	G711A/G711U
22	Audio	Two Way
23	Electrical	
24	Network	RJ-45 (10/100Base-T)
25	Power Input	PoE 802.3 af/12VDC
26	Power Consumption	Max 6.72W
27	Local Storage	Yes 256GB
28	Net HDD	Yes
29	Cloud Storage	YES
30	Cloud Overwrite	OFF/Automatic/1/3/7/14/30/90Days
	Video Type	RF/AVI/MP4

31	Mechanical	
32	Lens	2.7 - 13.5 mm auto focus motorized optical zoom
	Angular Field of View	Horizontal: 107 - 33.6°, Vertical: 54.7 - 18.9°, Diagonal: 133.7 - 37.6°
33	Video	
34	Video Compression	H.265/H.265+/H.264/H.264+/MJPEG
35	Video Encoding	H.265, H.264 Smart Codec bit rate control (CBR & VBR)
36	Video Quality Adjustment	8Kbps ~ 8Mbps
	Video Stream	Quad-Stream (Main/Sub/Mobile/Fourth)
37	Multiple streaming	Quad Stream Main Stream: 2592x1944, 2592x1520, 2048x1536, 1920x1080, 1280x960, 1280x720 Sub Stream: 1920x1080, 640x480, 320x240 Mobile Stream: 640x480, 320x240 Fourth Stream: 640x480, 320x240
38	Maximum frame Rate	50Hz: 25fps@5MP/4MP/3MP/2MP/960P/720P/VGA/QVGA
39		60Hz: 30fps@5MP/4MP/3MP/2MP/960P/720P/VGA/QVGA
40	Supported Protocols	IPv4, IPv6, IPv4 DNS, IPv6 DNS, TCP/IP, HTTP, HTTPS, UPnP, RTSP/RTP/RTCP, IGMP, SMTP, DHCP, NTP, FTP, DNS, DDNS, SNMP, NFS/SMB/CIFS, SSL/TLS,
41	Security	IP address filter; complicated password; Authenticated username and password, AES 256
42	Mobile Surveillance	Support mobile surveillance through app on android and i-phone
43	Supported browser	IE8,9,10,11, Chrome 44/lower, Firefox 51/lower, Safari 11/lower
44	Video backup file type	RF, AVI, MP4
45	Hardware Reset	YES
46	User	
47	Max. User	7 users
48	Multicast	Unlimited
49	Security Access	Multiple user access level with password protection Ip filtering, HTTPS.
50	Environmental	
51	Operating Temperature	-30~+55°C
52	Operating Humidity	less than ≤ 95% RH
53	Housing	IP67 & IK10
54	Camera & Housing type	Bullet, Metal
55	Certification	UL, CE, FCC, BIS
		Make as approved by Engineer-in-charge

Technical Specification of PTZ Camera

Sr.No.	Features	Specification
	General	
1	Flash	128MB NAND
2	RAM	256MB*2
3	Imaging Device	1/2.7" Progressive scan CMOS
4	Maximum resolution	5MP - (2592X1944)
5	Video Compression	H.265/H.265+/H.264/H.264+/MJPEG
6	Video Quality Adjustment	8Kbps ~ 8Mbps
7	ONVIF	ONVIF profile S, G & T
8	Minimum Illumination	AGC On: Color: 0.5 lux, B/W: 0.05 lux, 0 lux with IR
9	IR Range	200 Mtr.
10	Zoom	33X optical Zoom ,12 X Digital Zoom
11	Day/Night Method	AUTO ICR
12	Signal-to-Noise ratio	50 db minimum
13	Electronic shutter Range	Auto/Manual, 1/5 - 1/100000s
14	Dynamic Range	120 db minimum
15	Back light compensation	HLC, WDR (120db)
16	Digital Noise reduction	3D Noise Reduction
17	Analytics	Heat Map, Human & Vehicle Detection, Pedestrian Intrusion Detection & Line cross Detection, Stationary Object Detection, Tempering detection, Auto Tracking, Face Recognition & Detection
18	Privacy Zone	4 Configurable windows
19	Region of Interest	8
20	Motion Detection	ON/OFF (8 Level)
21	PTZ	Pelco D, Pelco P, Protocol Support
22	PAN	360 ° endless, preset speed 200°
23	Tilt	-10°~95° (Auto Flip)
24	Pan speed	0.1 to 300°/Sec
25	Tilt Speed	0.1 to 200°/Sec
26	Pattern	4 Pattern
27	Tour	4 Tour/ 32 preset each tour
28	Presets	255
29	PTZ Operation	1 sequence, 1 cruise, Auto pan
30	BNC	1 X BNC
	Image setting	Compression, 3D digital noise reduction (ON/OFF) colour, brightness, sharpness, contrast, white balance, exposure control, backlight compensation

31	Defog	Yes
32	Alarm	I/O, 1-IN and 1-OUT
	Audio Compression	G711A, G711U, G726_16K
33	Audio	Two Way
34	Electrical	
35	Network	RJ-45 (10/100Base-T)
36	Power Input	PoE 802.3af & at/12VDC
	Local Storage	micro SDHC Max 256GB
37	Cloud Storage	Yes
38	Mechanical	
39	Focal Length	4.3-141.9mm
40	Angular Field of View	59° - 2.4° (Wide - Tele)
41	Video	
42	Video Encoding	H.265, H.264 Smart Codec bit rate control (CBR & VBR)
43	Video Quality Adjustment	8Kbps ~ 8Mbps
44	Video Stream	Quad-Stream (Main/Sub/Mobile/Fourth)
45	Maximum frame Rate	50Hz: 25fps@5MP/4MP/3MP/2MP/960P/720P/VGA/QVGA
		60Hz: 30fps@5MP/4MP/3MP/2MP/960P/720P/VGA/QVGA
46	Multiple streaming	Quad Stream Main Stream: 2592x1944,2592x1520,2304x1296 2048x1536,1920x1080,1280x960, 1280x720 Sub Stream: 1920x1080,1280x720, 640x480,320x240 Mobile Stream: 640x480, 320x240 Fourth Stream: 640x480, 320x240
47	Supported Protocols	IPv4, IPv6, IPv4 DNS, IPv6 DNS, TCP/IP, HTTP, HTTPS, UPnP, RTSP/RTP/RTCP, IGMP, SMTP, DHCP, NTP, FTP, DNS, DDNS, SNMP, NFS/SMB/CIFS, SSL/TLS, P2P
48	Security	IP address filter; complicated password; Authenticated username and password, AES 256
49	Supported browser	IE8,9,10,11, Chrome 44/lower, Firefox 51/lower, Safari 11/lower
50	Video backup file type	RF, AVI, MP4
51	Snapshot	Yes
52	User	
53	User	7 users
54	Multicast	Unlimited

55	Security Access	Multiple user access level with password protection Ip filtering, HTTPS.
56	Environmental	
57	Operating Temperature	-30~+65°C
58	Operating Humidity	less than $\leq 95\%$ RH
59	Housing	IP67 & IK10
60	Camera type	Dome
61	Certification	UL,CE,FCC,BIS
		Make as approved by Engineer-in-charge

Technical Specification of 256 Channel NVR

Video	Resolution	12MP,8MP,5MP, 3MP ,2MP(1080P),1.3MP(960P),1.0MP(720P)
Recording Bandwidth	Inputs	1024M
	out	800M
RAM	32 GB*2	
Flash	NAND 256 GB	
Performance		
Live/Playback	Live View	Up to 256 Channel
	Compression	Audio: G.711A /G.711U Video: H.264/H.265/H.264+/H.265+
	Frame rate	4K:240fps 4MP:480fps 2MP:960fps
	Event	MD (Sensitivity Level 1~8)
		Video loss
		Beep, Email, Push notification to Mobile APP
	Playback function	Fast forward / backward (2x, 4x, 8x,16x)

		Slow forward (1/2x, 1/4x, 1/8x,1/16x)
	Playback	Max 16CH playback
RAID	RAID Type	RAID0, RAID1, RAID5, RAID6, RAID10
	Fisheye	Support
UI/Monitoring	Local	UI5.0
	WEB	IE10/11 Firefox V.52 above Google chrome V.57 above Edge V.79 above
	Mobile viewer	YES
Analytics		Face Detection
		Pedestrian Detection
		Vehicle Detection
		Cross Counting
		Heat Map
		Crowd Density Detection
		Perimeter Intrusion Detection
		Line Crossing Detection
		Queue Legth Detection
		License Plate Detection
		Rare Sound Detection
Interface		
P/W Reset		YES
Monitors	HDMI	3（1024*768, 1280*720, 1280*1024, 1440*900, 1920*1080, 2560*1440, 3840*2160

	Ethernet	2 RJ45 1000 Base-T port
	USB	USB2.0*1(front) USB3.0*2(rear)
	Audio out	1
	E-sata	1
	HDD	16 SATA HDD, each SATA support up to 18TB (Enterprise-Class)
	Line in	1
	RS232	1*RJ45
	RS485	1
	Alarm Input/out	16in /4out
General		
OSD Language		English/French/Spanish/Portuguese/German/Italian/Greek/Danish/Finnish/Russia/Hebrew/Turkish/Bulgarian/Arabic/Korean/Japanese/Thai/Hindi/Romanian/Dutch/Polish
Mechanical design	Dimension (WxDxH)	440*460*149mm
Electrical	Input Voltage/Current	ATX 250W
	Power consumption (excluding hard disk)	8W
Environmental	Operating Temperature/Humidity	Temperature: -10°C~+50°C (14°F ~ 122°F) Humidity: Less than 90%RH
User	512 User with Admin	
Face capture Quantity	capture quantity up to 1,000,000pcs	

Face registered Quantity	up to 10000pcs (support each group up to 2000pcs)
	Make as approved by Engineer-in-charge

Technical Specification of 24 Port PoE Switch

Sr.No.	Specifications
1	Port Configuration: - RJ45 * 24 x 10/100/1000M PoE ports, Uplinks 4 x 1/10G SFP fibre slot ports, Console 1 x RS232 Port.
2	Hardware Performance: - Forwarding Capacity 95.23 Mpps, Switch Capacity 128 Gbps, Mac Table 32K, Jumbo Frames 9.6K.
3	Software Features
A	Ring management: - Supports ITU-T G.8031, Supports ITU-T G.8032. Enable self-recover time in less than 20ms
B	Spanning Tree: - Standard Spanning Tree 802.1d, Rapid Spanning Tree (RSTP) 802.1w, Multiple Spanning Tree (MSTP) 802.1s
C	Port Aggregation: - LACP, Static aggregation, Max 14 aggregation group and 8 ports per group.
D	VLAN: - 4K VLAN based on port, IEEE802.1q, VLAN based on the protocol, VLAN based on MAC, Voice VLAN, QinQ configuration, Port configuration of Access, Trunk, Hybrid
E	DHCP: - DHCP Client, DHCP Snooping, DHCP Server, DHCP Relay
F	Multicast: - MLD Snooping v1/v2, Multicast VLAN, IGMP Snooping v1/v2, Max 1024 multicast groups, Fast log out
G	Layer 3 Features: - L2+ network management, IPV4/IPV6 management, L3 soft routing forwarding, Static route, Default route @ 128 pcs, APR @ 1024 pcs
H	Security: - IP-MAC-VLAN-Port binding, ARP inspection, Anti-DoS attack, AAA & RADIUS, MAC learning limit Mac black holes, IP source protection, IEEE802.1X & MAC address authentication, Broadcast storm control, Backup for host datum, SSH 2.0, SSL, Port isolation, ARP message speed limit, User hierarchical management and password protection
I	ACL: - Port-based Issuing ACL, ACL based on port and VLAN, L2 to L4 packet filtering, matching first 80 bytes, message. Provide ACL based on MAC, Destination, MAC address, IP Source, Destination IP, IP Protocol Type, TCP/UDP Port, TCP/UDP, Port Range, and VLAN, etc.

J	Quality of Service: - Flow-based Rate Limiting, Flow-based Packet Filtering, 8 x Output queues of each port, 802.1p/DSCP priority mapping, Diff-Serv QoS, Priority Mark/Remark, Queue Scheduling Algorithm (SP, WRR, SP+WRR)
K	Interface: - IEEE802.3X (Full-duplex), Port temperature protection setting, Port green Ethernet Energy-saving setting, broadcast storm control based on port speed, the speed limit of the message flow in the access port, The minimum particle size is 64Kbps.
L	Management: - One-key recovery, Cable Diagnose, LLDP Web Management (HTTPS) NTP, System work log, Ping Test, CPU instant utilization status view, Console/AUX Modem/Telnet/SSH2.0 CLI, Download & Management on FTP, TFTP, X modem, SFTP, SNMP V1/V2C/V3, NMS-smart network management system, platform (LLDP+SNMP
4	Power over Ethernet (PoE)
A	PoE Port- Port 1 to 24, PoE Management: PoE working status, Delay start of power supply, PoE output priority configuration, Scheduling of PoE operation and time, Total power limit of PoE power supply, PoE output power allocation, off & af &at
B	PoE network management and fast Ring function: - IEEE802.1Q VLAN, flexible VLAN division and QinQ configuration. QoS, Priority mode based on 802.1P, Port & DSCP, queue scheduling algorithm including EQU, SP, WRR & SP+WRR. ALC, filter data packet through configuring matching rules, processing operation & time permission, and provide flexible and safe access control. IGMP V1/V2 and IGMP Snooping. ERPS/STP/RSTP/MSTP. Static and dynamic aggregation.
5	Environmental Range:- Operating Temperature: -40°C to +80°C, Storage Temperature: -40°C to +85°C, Operating Humidity: 5%~90% RH non-condensing, Altitude: meters < 3000, Feet <10000
6	Heat Dissipation: Low noise fan active cooling
7	Power Supply: - Built-in power supply, AC 100~240V 50-60Hz 5A, Built-in power supply, AC 100~240V 50-60Hz 6.6A
8	Mechanical Stability: - Vibration: IEC 60068-2-6, Shock: IEC 60068-2-27, Freefall: IEC 60068-2-32, Ingress Protection: IP30
9	Certification: - CCC, CE mark, commercial, CE/LVD EN60950, FCC Part 15 Class B, RoHS, Safety: UL, CE
	Make as approved by Engineer-in-charge

D. FIRE ALARM SYSTEM

PART 1 GENERAL

1.1 DESCRIPTION

- A. The fire alarm system shall comply with requirements of NFPA Standard 72 for Protected Premises Signaling Systems except as modified and supplemented by this specification. The system shall be electrically supervised and monitor the integrity of all conductors.
- B. The facility shall have an emergency voice alarm communication system. Digitally stored message sequences shall notify the building occupants that a fire or life safety condition has been reported. Message generator(s) shall be capable of automatically distributing up to eight (8) simultaneous, unique messages to appropriate audio zones within the facility based on the type and location of the initiating event. The Fire Command Center (FCC) shall also support Emergency manual voice announcement capability for both system wide or selected audio zones, and shall include provisions for the system operator to override automatic messages system wide or in selected zones.
- C. The system shall support additional, alternate Fire Command Centers, which shall be capable of simultaneous monitoring of all system events. Alternate Fire Command Centers shall also support an approved method of transferring the control functions to an alternate Fire Command Center when necessary. All Fire Command Centers shall be individually capable of assuming Audio Command functions such as Emergency Paging, audio zone control functions, and Firefighter's Telephone communication functions.
- D. Each designated zone shall transmit separate and different alarm, supervisory and trouble signals to the Fire Command Center (FCC) and designated personnel in other buildings at the site via a multiplex communication network.

1.2 SCOPE

- A. A new intelligent reporting, microprocessor-controlled fire detection system shall be installed in accordance to the project specifications and drawings.
- B. Basic Performance:
 - 1. Alarm, trouble and supervisory signals from all intelligent reporting devices shall be encoded on NFPA Class A Signaling Line Circuits (SLC).
 - 2. Device Circuits (IDC) shall be wired NFPA Class A as part of an addressable device connected by

the SLC Circuit.

3. Notification Appliance Circuits (NAC) shall be wired NFPA Class A as part of an addressable device connected by the SLC Circuit.
4. On Class A configurations a single ground fault or open circuit on the system Signaling Line Circuit shall not cause system malfunction, loss of operating power or the ability to report an alarm.
5. Alarm signals arriving at the FACP shall not be lost following a primary power failure (or outage) until the alarm signal is processed and recorded.
6. Two-way emergency telephone communication circuits shall be supervised for open and short circuit conditions.
7. Two-way emergency telephone (Fire Fighter Telephone) communication shall be supported between the Audio Command Center and up to **thirty (30)** remote Fire Fighter's Telephone locations simultaneously on a conference in multiple FFT Risers.
8. Means shall be provided to connect FFT voice communications to the speaker circuits in order to allow voice paging over the speaker circuit from a telephone handset.
9. The digital audio message generator shall be of reliable, non-moving parts, and support the digital storage of up to 32 minutes of tones and emergency messages, shall support programming options to string audio segments together to create up to 1000 messages, or to loop messages and parts of messages to repeat for pre-determined cycles or indefinitely.
10. The proposed product shall not restrict the buyer to one single organization, nor shall it require any proprietary dongle or other programming tools for after sales & maintenance activity.

1.3 DESIGN INTENT

- a. Main fire alarm panel with digital voice command system, Fire fighters' telephone, zone selector keypad and announcement console – Ground floor – Near passenger lift lobby.
- b. Secondary fire alarm panels- At each level – near lift lobby
- c. Active repeater panels at security cabin
- d. All fire alarm panels connected as pier to pier.
- e. Fire survival cables (750 deg. 2 hours)
- f. Class - A cabling to loop all detectors, devices & MCP's to control panel.
- g. Coverage per detector as per NFPA -2015, considering > 60 ACH
- h. System integration (Soft integration) with all standalone panels such as agent release panels for deluge valves, Pre-action panels, lift switchboard, DG fresh air switchboard, etc.
- i. VESDA (Very Early Smoke Detection Apparatus) to cover the false flooring and room void areas of Data Hall, UPS, Battery rooms and MMR.
- j. Emergency communication system, integral with the Main FACP, including voice alarm system

components, microphones, zone selector keypads and tone generators

- k. Audible Alarm Notifications
- l. Fire fighters telephone system as part of main fire alarm system, which is two-way, supervised voice communication proposed to link between the MFACP and remote fire fighters' telephone stations throughout the building (at all staircases at all levels)

1.4 GENERAL INSTRUCTIONS

- a. Protect from moisture by using appropriate coverings. Store at dry interior locations.
- b. Sequence work to avoid interferences with building finishes and installation of other products.
- c. Supply as maintenance stock, consumable devices, components as recommended by Supplier, but shall not be less than two units of each type/ rating of installed consumable material/ component/ device.
- d. For ease of service all panel I/O wiring terminal blocks shall be removable, plug-in types and have sufficient capacity for #18 to #12 AWG wire. Terminal blocks that are permanently fixed are not acceptable.

1.5 WARRANTY

- A. The fire alarm control panel, voice panels and any head-end equipment shall have a manufacturer's warranty of a minimum of 12months.

1.6 APPLICABLE STANDARDS AND PRODUCT APPROVALS

- A. The specifications and standards listed below form a part of this specification. The system shall fully comply with the latest issue of these standards, if applicable.

- B. National Fire Protection Association (NFPA) - USA:

NFPA 12	Extinguishing Systems (low and high)
NFPA 12A	Halon 1301 Extinguishing Systems
NFPA 13	Sprinkler Systems
NFPA 15	Water Spray Systems
NFPA 16	Foam / Water Deluge and Spray Systems
NFPA 17	Dry Chemical Extinguishing Systems
NFPA 17A	Wet Chemical Extinguishing Systems
NFPA 2001	Clean Agent Extinguishing Systems
NFPA 70	National Electric Code

NFPA 90A	Air Conditioning Systems
NFPA 92A	Smoke Control Systems
NFPA 92B	Smoke Management Systems in Malls, Atria, Large Areas
NFPA 72	National Fire Alarm Code
NFPA 101	Life Safety Code

C. Underwriters Laboratories Inc. (UL) - USA:

UL 268, 7th Edition	Smoke Detectors for Fire Protective Signaling Systems
UL 864, 10th Edition	Control Units for Fire Protective Signaling Systems
UL 2572	Mass Notification Systems
UL 217	Smoke Detectors, Single and Multiple Station
UL 228	Door Closers - Holders for Fire Protective Signaling Systems
UL 268A	Smoke Detectors for Duct Applications
UL 521	Heat Detectors for Fire Protective Signaling Systems
UL 464	Audible Signaling Appliances
UL 38	Manually Actuated Signaling Boxes
UL 1481	Power Supplies for Fire Protective Signaling Systems
UL 346	Waterflow Indicators for Fire Protective Signaling Systems
UL 1076	Control Units for Burglar Alarm Proprietary Protective Signaling Systems
UL 1971	Visual Notification Appliances
UL 2017	Standard for General-Purpose Signaling Devices and Systems
UL60950	Safety of Information Technology Equipment

D. Factory Mutual – USA

E. Local and State Building Codes.

F. All requirements of the Authority Having Jurisdiction (AHJ).

G. The system shall be certified for seismic applications in accordance with the International Building Code (IBC). The basis for qualification of seismic approval shall be via shake table testing.

H. The System shall be FM 6320 (Factory Mutual) approved as a Gas Detection system when employed with the 4-20 monitor module and industry standard 4-20 mA gas detectors.

PART 2.0 PRODUCTS

2.1 MAIN FIRE ALARM CONTROL PANEL OR NETWORK NODE:

- A. Main FACP or network node shall contain a microprocessor based Central Processing Unit (CPU) and power supply. The CPU shall communicate with and control the following types of equipment used to make up the system: intelligent addressable smoke and thermal (heat) detectors, addressable modules, printer, annunciators, and other system-controlled devices.
- B. In conjunction with intelligent Loop Control Modules and Loop Expander Modules, the main FACP shall perform the following functions:
 - 1. Supervise and monitor all intelligent addressable detectors and monitor modules connected to the system for normal, trouble and alarm conditions.
 - 2. Supervise all initiating signaling and notification circuits throughout the facility by way of connection to addressable monitor and control modules.
 - 3. Detect the activation of any initiating device and the location of the alarm condition. Operate all notification appliances and auxiliary devices as programmed.

System Capacity and General Operation

- A. The FACP shall can communicate on a peer-to-peer, inherently regenerative communication format and protocol. The network shall support communication speed up to 100 Mbps and support up to 200 panels / nodes per network.
- B. The control panel shall be capable of expansion via up to 10 SLC loops. Each loop shall support minimum 300 analog/addressable devices for a system capacity of 3000 points. The Fire Alarm Control Panel shall include a full featured operator interface control and annunciation panel that shall include a backlit 600-character liquid crystal display, individual, color coded system status LEDs, and a QWERTY keypad for the control of the fire alarm system. Said LCD shall also support graphic bit maps capable of displaying the company name and logo of either company.
- C. All programming or editing of the existing program in the system shall be achieved without interrupting the alarm monitoring functions of the fire alarm control panel.
- D. The FACP shall be able to provide the following software and hardware features:
 - 1. Pre-signal and Positive Alarm Sequence: The system shall provide means to cause alarm signals to only sound in specific areas with a delay of the alarm from 60 to up to 180 seconds after start

of alarm processing. In addition, a Positive Alarm Sequence selection shall be available that allows a 15-second time period for acknowledging an alarm signal from a fire detection/initiating device. If the alarm is not acknowledged within 15 seconds, all local and remote outputs shall automatically activate immediately.

2. Smoke Detector Pre-alarm Indication at Control Panel: To obtain early warning of incipient or potential fire conditions, the system shall support a programmable option to determine system response to real-time detector sensing values above the programmed setting. Two levels of Pre-alarm indication shall be available at the control panel: alert and action.
3. Alert: It shall be possible to set individual smoke detectors for pre-programmed pre-alarm thresholds. If the individual threshold is reached, the pre-alarm condition shall be activated.
4. Action: If programmed for Action and the detector reaches a level exceeding the pre-programmed level, the control panel shall indicate an action condition. Sounder bases installed with either heat or smoke detectors shall automatically activate on action Pre-Alarm level, with general evacuation on Alarm level.
- b) The system shall support a detector response time to meet world annunciation requirements of less than 3 seconds.
- c) Device Blink Control: Means shall be provided to turn off detector/module LED strobes for special areas.
- d) NFPA 72 Smoke Detector Sensitivity Test: The system shall provide an automatic smoke detector test function that meets the sensitivity testing requirements of NFPA 72.
- e) Programmable Trouble Reminder: The system shall provide means to automatically initiate a reminder that troubles exist in the system. The reminder will appear on the system display and (if enabled) will sound a piezo alarm.
- f) On-line or Off-line programming: The system shall provide means to allow panel programming either through an off-line software utility program away from the panel or while connected and on-line. The system shall also support upload and download of programmed database and panel executive system program to a Personal Computer/laptop. A single change to one CPU database shall not require a database download to other CPUs.
- g) History Events: The panel shall maintain a history file of at least last 5000 events, each with a time and date stamp. History events shall include all alarms, troubles, operator actions, and programming entries. **The control panels shall also maintain a 1000 event Alarm History buffer**, which consists of the 1000 most recent alarm events from the 5000-event history file.
- h) Smoke Control Modes: The system shall provide means to perform Fire Smoke Control Station mode. This mode controls all dampers, smoke extraction fan, fresh air supply fans, etc. during Fire condition. Smoke Control to meet NFPA-92A and 90B and HVAC mode to meet NFPA 90A.

- i) **The system shall provide means for all SLC devices on any SLC loop to be auto programmed into the system by specific address.** The system shall recognize specific device type ID's and associate that ID with the corresponding address of the device.
- j) **Passwords and Users:** The system shall support two password levels, master, and user. Up to 9 user passwords shall be available, each of which may be assigned access to the programming change menus, the alter status menus, or both. Only the master password shall allow access to password change screens.
- k) **Block Acknowledge:** The system shall support a block Acknowledge for Trouble Conditions
- l) **Sensitivity Adjust:** The system shall provide Automatic Detector Sensitivity Adjust based on Occupancy schedules including a Holiday list of up to 15 days.
- m) **Environmental Drift Control:** The system shall provide means for setting Environmental Drift Compensation by device. When a detector accumulates dust in the chamber and reaches an unacceptable level but still below the allowed limit, the control panel shall indicate a maintenance alert warning. When the detector accumulates dust in the chamber above the allowed limit, the control panel shall indicate a maintenance urgent warning.
- n) **Custom Action Messages:** The system shall provide means to enter up to 100 custom action messages of up to 160 characters each. It shall be possible to assign any of the 100 messages to any point.
- o) **Local Mode:** If communication is lost to the central processor the system shall provide added survivability through the intelligent loop control modules. Inputs from devices connected to the SLC and loop control modules shall activate outputs on the same loop when the inputs and outputs have been set with point programming to participate in local mode or when the type codes are of the same type: that is, an input with a fire alarm type code shall activate an output with a fire alarm type code.
- p) **Read status preview - enabled and disabled points:** Prior to re-enabling points, the system shall inform the user that a disabled device is in the alarm state. This shall provide notice that the device must be reset before the device is enabled thereby avoiding activation of the notification circuits.
- q) **Custom Graphics:** When fitted with an LCD display, the panel shall permit uploading of a custom bit-mapped graphic to the display screen.
- r) **Group Decision Making by Smoke Detectors:** The system shall provide means to link one detector with minimum two more detectors in group decision making. The group of minimum three detectors shall work in tandem to take fast and genuine alarm decision mitigating the risk of false alarm. There shall be no requirement of cross zoning or mandatory sequential address setting in the detectors to achieve this function. This shall be a built-in intelligence in the system to take fast & reliable decision on genuine alarm triggering. The alarm event shall be a result of group of detector chamber readings considered as a consolidated alarm signal.

- s) **ACTIVE EVENT:** The system shall provide a Type ID called FIRE CONTROL for purposes of air-handling shutdown, which shall be intended to override normal operating automatic functions. Activation of a FIRE CONTROL point shall cause the control panel to (1) initiate the monitor module Control-by-Event, (2) send a message to the panel display, history buffer, installed printer and annunciators, (3) shall not light an indicator at the control panel, (4) Shall display ACTIVE on the LCD as well a display a FIRE CONTROL Type Code and other information specific to the device.
- t) **NON-FIRE Alarm Module Reporting:** A point with a type ID of NON-FIRE shall be available for use for energy management or other non-fire situations. NON-FIRE point operation shall not affect control panel operation nor shall it display a message at the panel LDC. Activation of a NON-FIRE point shall activate control by event logic but shall not cause any indication on the control panel.
- u) **Mass Notification Override:** The system shall be UL 2572 listed for Mass Notification and shall be capable, based on the Risk Analysis, of being programmed so that Mass Notification/Emergency Communications events take precedence over fire alarm events.
- v) **Security Monitor Points:** The system shall provide means to monitor any point as a type security.
- w) **One-Man Walk Test:** The system shall provide both a basic and advanced walk test for testing the entire fire alarm system. The basic walk test shall allow a single operator to run audible tests on the panel. All logic equation automation shall be suspended during the test and while annunciators can be enabled for the test, all shall default to the disabled state. During an advanced walk test, field-supplied output point programming will react to input stimuli such as Control by Event and logic equations. When points are activated in advanced test mode, each initiating event shall latch the input. The advanced test shall be audible and shall be used for pull station verification, magnet activated tests on input devices, input and output device and wiring operation/verification.
- x) **Control By Event Functions:** CBE software functions shall provide means to program a variety of output responses based on various initiating events. The control panel shall operate CBE through lists of zones. A zone shall become listed when it is added to a point's zone map through point programming. Each input point such as detector, monitor module or panel circuit module shall support listing of up to 10 zones into its programmed zone map.
- y) **Permitted zone types** shall be general zone, releasing zone and special zone. Each output point (control module, panel circuit module) can support a list of up to 10 zones including general zone, logic zone, releasing zone and trouble zone. It shall be possible for output points to be assigned to list general alarm. Non-Alarm or Supervisory points shall not activate the general alarm zone.
- z) **1000 General Zones:** The system shall support up to 1000 general purpose software zones for linking inputs to outputs. When an input device activates, any general zone programmed into that device's zone map will be active and any output device that has an active general zone in

its map will be active. It shall also be possible to use general zone as arguments in logic equations.

- aa) **1000 Logic Equations:** The system shall support up to 1000 logic equations for AND, OR, NOT, ONLY1, ANYX, XZONE or RANGE operators that allow conditional I/O linking. When any logic equation becomes true, all output points mapped to the logic zone shall activate.
- bb) **100 trouble equations per device:** The system shall provide support for up to 100 trouble equations for each device, which shall permit programming parameters to be altered, based on specific fault conditions. If the trouble equation becomes true, all output points mapped to the trouble zone shall activate.
- cc) **Control-By-Time:** A time-based logic function shall be available to delay an action for a specific period of time based upon a logic input with tracking feature. A latched version shall also be available. Another version of this shall permit activation on specific days of the week or year with ability to set and restore based on a 24-hour time schedule on any day of the week or year.
- dd) **Multiple agents releasing zones:** The system shall support up to 10 releasing zones to protect against 10 independent hazards. Releasing zones shall provide up to three cross-zone and four abort options to satisfy any local jurisdiction requirements.
- ee) **Alarm Verification, by device, with timer and tally:** The system shall provide a user-defined global software timer function that can be set for a specific detector. The timer function shall delay an alarm signal for a user-specified time period and the control panel shall ignore the alarm verification timer if another alarm is detected during the verification period.

E. Central Processing Unit

1. The Central Processing Unit shall contain and execute all control-by-event (including Boolean functions including but not limited to AND, OR, NOT, ANYx, and CROSSZONE) programs for specific action to be taken if an alarm condition is detected by the system. Such control-by-event programs shall be held in non-volatile programmable memory, and shall not be lost with system primary and secondary power failure.
2. The Central Processing Unit shall also provide a real-time clock for time annotation, to the second, of all system events. The time-of-day and date shall not be lost if system primary and secondary power supplies fail.
3. The CPU shall be capable of being programmed on site without requiring the use of any external programming equipment. Systems that require the use of external programmers or change of EPROMs are not acceptable.
4. The CPU shall provide an RS-232 interface between the fire alarm control panel and the UL Listed Electronic Data Processing (EDP) peripherals.
5. The CPU shall provide two RS-485 ports for the serial connection to annunciation and control subsystem components.
6. The RS-232 serial output circuit shall be optically isolated to assure protection from earth

ground.

7. In the event of CPU failure, all SLC loop modules shall fallback to **degrade mode**. **Systems not offering degrade mode shall offer Redundant CPU**. Such degrade mode shall treat the corresponding SLC loop control modules and associated detection devices as conventional two-wire operation. Any activation of a detector in this mode shall automatically activate associated Notification Appliance Circuits.

F. Display

- a. The system display shall provide a **600-character** backlit alphanumeric Liquid Crystal Display (LCD). It shall also provide eleven Light-Emitting-Diodes (LEDs) that indicate the status of the following system parameters: AC POWER, FIRE ALARM, PREALARM, SECURITY, SUPERVISORY, SYSTEM TROUBLE, OTHER EVENT, SIGNALS SILENCED, POINT DISABLED, CONTROLS ACTIVE, and CPU FAILURE.
- b. **These characters shall be only for fire alarm / trouble information and not for Logo or other purpose. It shall be UL Listed. Repeater panel displays in FACP are not allowed unless until approved by UL**
- c. The system display shall provide a QWERTY keypad for ease of operation.
- d. The keypad shall have control capability to command all system functions, entry of any alphabetic or numeric information, and **field programming without the use of any external equipment or laptop**. Two different password levels with up to ten (one Master and nine User) passwords shall be accessible through the display interface assembly to prevent unauthorized system control or programming.

G. Loop (Signaling Line Circuit) Control Module:

1. The control panel shall be capable of expansion via up to **10 SLC loops**. Each loop shall support minimum **300 analog/addressable devices** for a system capacity of **3000 points**.
2. The Loop Control Module shall contain its own microprocessor and shall be capable of operating in a local/degrade mode (any addressable device input shall be capable of activating any or all addressable device outputs) in the unlikely event of a failure in the main CPU.
3. Each loop shall maintain 20% spare capacity for future expansion.
4. Each Loop shall be capable of operating as a NFPA Class B circuit in case of single open circuit fault in existing SLC Circuit.
5. The SLC interface board shall receive analog or digital information from all intelligent detectors and shall process this information to determine whether normal, alarm, or trouble conditions

exist for that particular device. Each SLC Loop shall be isolated and equipped to annunciate an Earth Fault condition. The SLC interface board software shall include software to automatically maintain the detector's desired sensitivity level by adjusting for the effects of environmental factors, including the accumulation of dust in each detector. The analog information may also be used for automatic detector testing and the automatic determination of detector maintenance requirements.

H. Network Communication

The FACP shall communicate over a peer-to-peer communication network, inherently over a regenerative communication format and protocol. The network shall support communication speed up to 100 Mbps and support up to **200 Control Panels/ Network Nodes**, over a **single medium** (copper conductor / fiber optic), redundant ring, communication channel for fire alarm, voice evacuation and telephone talk-back system. The system shall support up to **200 such networks** in a single system.

The network card shall have inbuilt Fiber port for terminating Fiber Optic Cable without any third-party converters.

I. Digital Voice Command Center

1. The Digital Voice Command Center located with the FACP, shall contain all equipment required for all audio control, emergency telephone system control, signaling and supervisory functions. This shall include speaker zone indication and control, telephone circuit indication and control, digital voice units, microphone, and main telephone handset.
2. Function: The Voice Command Center equipment shall perform the following functions:
 - a. Operate as a supervised multi-channel emergency voice communication system.
 - b. Operate as a two-way emergency telephone system control center.
 - c. Audibly and visually annunciate any trouble condition for digital tone and voice units required for normal operation of the system.
 - d. Provide all-call Emergency Paging activities through activation of a single control switch.
 - e. As required, provide vectored paging control to specific audio zones via dedicated control switches.
 - f. Provide a factory recorded "library" of voice messages and tones in standard WAV. File format, which may be edited and saved on a PC running a current Windows® operating system.
 - g. Provide a software utility capable of off-line programming for the DVC operation and the audio message files. This utility shall support the creation of new programs as well as editing and

saving existing program files. Uploading or downloading the DVC shall not inhibit the emergency operation of other nodes on the fire alarm network.

- h. The Digital Voice Command shall be modular in construction, and shall be capable of being field programmable without requiring the return of any components to the manufacturer and without requiring use of any external computers or other programming equipment.
- i. The Digital Voice Command and associated equipment shall be protected against unusually high voltage surges or line transients.
- j. Fire, Voice & Telephone data shall flow through single network cable.
- k. The Voice Evacuation System shall be capable of establishing communication between the master voice controller and amplifier over fiber optic cable network without using any third-party media converter.
- l. Failure of Fire Panel CPU shall not affect the operation of DVC. In case DVC are controlled by Fire Panel CPU, a separate panel with dedicated CPU shall be considered for each DVC.

J. Power Supply

- 1. The Main Power Supply shall operate on 120/240 VAC, 50/60 Hz, and shall provide all necessary power for the FACP.
- 2. The Main Power Supply shall provide the required power to the CPU using a switching 24 VDC regulator and shall incorporate a battery charger for 24 hours of standby power using dual-rate charging techniques for fast battery recharge.
- 3. The Main Power Supply shall provide a battery charger for 24 hours of standby using dual-rate charging techniques for fast battery recharge. The supply shall be capable of charging batteries ranging in capacity from 7-**200 amp-hours** within a 48-hour period.
- 4. The Main Power Supply shall provide a very low frequency sweep earth detect circuit, capable of detecting earth faults.
- 5. The Main Power Supply shall be power-limited per UL864 requirements.
- 6. The Main Power Supply shall communicate power supply, line voltage, battery status and charger status to the local LCD display. Any abnormal condition shall be annunciated and logged to the system alarm history log.
- 7. The interface to the power supply from the Fire Alarm Control Panel (FACP) shall be via the Signaling Line Circuit (SLC) or other multiplexed means. Power supplies that do not use an intelligent interface are not suitable substitutes. The required wiring from the FACP to the addressable power supply shall be a single unshielded twisted pair wire.
- 8. The addressable power supply shall supervise for battery charging failure, AC power loss, power brownout, battery failure, NAC loss, and optional ground fault detection. In the event of a

trouble condition, the addressable power supply shall report the incident and the applicable address to the FACP via the SLC.

9. The addressable power supply shall have an AC Power Loss Delay option. If this option is utilized and the addressable power supply experiences an AC power loss, reporting of the incident to the FACP will be delayed. A delay time of zero, two, eight or sixteen hours shall be programmable.
10. The addressable power supply shall have an option for Canadian Trouble Reporting and this option shall be programmable.
11. The addressable power supply mounts in either the FACP backbox or its own dedicated surface mounted backbox with cover.
12. Each of the power supply's four output circuits shall be programmed- for Notification Appliance Circuit or General Purpose 24 VDC power. Any output circuit shall be able to provide up to 2.5 amps of 24 VDC power.
13. The addressable power supply's output circuits shall be individually supervised when they are selected to be either a Notification Appliance Circuit when wired Class "A" or by the use of an end-of-line resistor. When the power supply's output circuit is selected as General 24 VDC power, the circuit shall be individually supervised when an end-of-line relay is used.
14. When selected for Notification Appliance Circuits, the output circuits shall be individually programmable for Steady, March Time, Dual Stage or Temporal.
15. When selected as a Notification Appliance Circuit, the output circuits of the addressable power supply shall have the option to be coded by the use of a universal zone coder.
16. The addressable power supply shall interface and synchronize with other power supplies of the same type. The required wiring to interface multiple addressable power supplies shall be a single unshielded, twisted pair wire.
17. An individual or multiple interfaced addressable power supplies shall have the option to use an external charger for battery charging. Interfaced power supplies shall have the option to share backup battery power.

L. Controls with associated LED Indicators

1. Emergency Two-Way Telephone Control Switches/Indicators.
 - a. The emergency telephone circuit control panel shall include visual indication of active and trouble status for each telephone circuit in the system.
 - b. The telephone circuit control panel shall include switches to manually activate or deactivate each telephone circuit in the system.

M. Field Programming

1. The system shall be programmable, configurable, and expandable in the field **without the need for special tools**. There shall be no firmware changes required to field modify the system time, point information, equations, or annunciator programming/information.
2. All field defined programs shall be stored in non-volatile memory.

N. Specific System Operations

1. Smoke Detector Sensitivity Adjust: A means shall be provided for adjusting the sensitivity of any or all addressable intelligent detectors in the system from the system keypad. **Sensitivity range shall be within the allowed UL window and have a minimum of 9 levels.**
2. Alarm Verification: Each of the intelligent addressable smoke detectors in the system may be independently selected and enabled to be an alarm verified detector. The alarm verification delay shall be programmable from 0 to 60 seconds and each detector shall be able to be selected for verification. The FACP shall keep a count of the number of times that each detector has entered the verification cycle. These counters may be displayed and reset by the proper operator commands.

O. System Point Operations

1. Any addressable device in the system shall have the capability to be enabled or disabled through the system keypad or Graphics User Interface.
2. System output points shall be capable of being turned on or off from the system keypad or the video terminal.
3. Point Read: The system shall be able to display the following point status diagnostic functions without the need for peripheral equipment. Each point shall be annunciated for the parameters listed:
 - a. Device Status.
 - b. Device Type.
 - c. Custom Device Label.
 - d. Software Zone Label.

- e. Device Zone Assignments.
- f. Analog Detector Sensitivity.
- g. All Program Parameters.
- 3. System History Recording and Reporting: The fire alarm control panel shall contain a history buffer that will be capable of storing up to 5000 system events. Each of these events will be stored, with time and date stamp, until an operator requests that the contents be either displayed or printed. The contents of the history buffer may be manually reviewed; one event at a time, and the actual number of activations may also be displayed and or printed. History events shall include all alarms, troubles, operator actions, and programming entries.
- 4. The history buffer shall use non-volatile memory. Systems which use volatile memory for history storage are not acceptable.
- 5. Automatic Detector Maintenance Alert: The fire alarm control panel shall automatically interrogate each intelligent system detector and shall analyze the detector responses over a period of time.
- 6. If any intelligent detector in the system responds with a reading that is below or above normal limits, then the system will enter the trouble mode, and the Intelligent Detector will be annunciated on the system display, and printed on the optional system printer. This feature shall in no way inhibit the receipt of alarm conditions in the system, nor shall it require any special hardware, special tools or computer expertise to perform.
- 7. The system shall include the ability (programmable) to indicate a "pre-alarm" condition. This will be used to alert maintenance personal when a detector is at 80% of its alarm threshold in a 60 second period.

2.2 SYSTEM COMPONENTS

A. Network Control Annunciator

A network control annunciator shall be provided to display all system intelligent points. The NCA shall be capable of displaying all information for **200,000 points on the network**. Network display devices, which are only capable of displaying a subset of network points, shall not be suitable substitutes.

The NCA shall include a 10" (1024 x 600) Color touchscreen display with QWERTY Keypad. Additionally, the network display shall include environmental adjustment controls to maximize LCD legibility and the ability to scroll events by type. i.e. Fire Alarm, Supervisory Alarm, Trouble, etc.

The network control annunciator shall have the ability to display multiple events in order of priority and time of occurrence. Counters shall be provided to indicate the total number of events by type.

The NCA shall mount in any of the network node fire alarm control panels. Optionally, the network display may mount in a backbox designed for this use and shall connect to the network over either a wire or fiber interface.

The network control annunciator shall have an event history buffer capable of storing a minimum of 1000 events in non-volatile memory. Additionally, the NCA shall have a fire alarm history buffer capable of storing a minimum of 200 events in non-volatile memory. Systems that do not protect fire alarm events from being overwritten by other events are not suitable substitutes.

The NCA shall include Three USB connection, USB C, USB B Micro, and USB A, industry standard RS-232 ports for UL864 listed printers and CRT's. These peripheral devices shall print or display network activity.

The network control annunciator shall include control switches for system wide control of Acknowledge, Signal Silence, System Reset, Drill, and local Lamp Test. A mechanical means by which the controls switches are "locked out", such as a key, shall be available.

The NCA shall include long life LEDs to display Power, Fire Alarm, Pre-Alarm, Security Alarm, System Trouble, Supervisory, Signals Silenced, Disabled Points, Other (non-fire) Events, and CPU Failure.

The network control annunciator shall include a Master password and up to nine User passwords. Each password shall be up to eight alpha-numeric characters in length. The Master password shall be authorized to access the programming and alter status menus. Each User password may have different levels of authorization assigned by the Master password.

The NCA shall allow editing of labels for all points within the network; control on/off of outputs; enable/disable of all network points; alter detector sensitivity; clear detector verification counters for any analog addressable detector within the network; clear any history log within the network; change the Time/Date settings; initiate a Walk Test.

The network control annunciator shall support an optional Windows TM based program utility. This utility shall allow the user create an NCA database, upload/download an NCA database, and download an upgrade to the NCA executive. To ensure program validity, this utility shall check stored databases for errors. A compare function shall be included to identify differences between databases.

For time keeping purposes the NCA shall include a time-of-day clock.

2.4 SYSTEM COMPONENTS - ADDRESSABLE DEVICES

A. Addressable Devices – General

1. Addressable devices shall provide an address-setting means using **rotary decimal switches**. Addressable devices that require the address be programmed using a programming utility are not an allowable substitute.
2. Addressable devices shall use simple to install and maintain decade, decimal address switches.
3. Addressable devices, which use a binary-coded address setting method, such as a DIP-switch, are not an allowable substitute. Addressable devices that require the address be programmed using a special tool or programming utility are not an allowable substitute.
4. Detectors shall be intelligent (analog) and addressable, and shall connect with two wires to the fire alarm control panel Signaling Line Circuits.
5. **Addressable smoke and thermal detectors shall provide dual alarm and power/polling bi-colour LEDs.** Both LEDs shall flash green under normal conditions, indicating that the detector is operational and in regular communication with the control panel, and both LEDs shall be placed into steady red illumination by the control panel, indicating that an alarm condition has been detected. If required, the LED flash shall have the ability to be removed from the system program. An output connection shall also be provided in the base to connect an external remote alarm LED.
6. The fire alarm control panel shall permit detector sensitivity adjustment through field programming of the system. The panel on a time-of-day basis shall automatically adjust sensitivity.
7. Using software in the FACP, detectors shall automatically compensate for dust accumulation and other slow environmental changes that may affect their performance. The detectors shall be listed by UL as meeting the calibrated sensitivity test requirements of NFPA Standard 72.
8. The detectors shall be ceiling-mount and shall include a separate twist-lock base with tamper proof feature. Base options shall include a sounder base with a built-in (local) sounder rated at 85 DBA minimum, a relay base and an isolator base designed for Style 7 applications. The system shall also support an intelligent programmable sounder base; the programmable sounder base shall be capable of providing multiple tones based on programming and at a minimum be capable of providing a Temp-4 tone for CO (Carbon Monoxide) activation and a Temp-3 tone for fire activations and be capable of being synchronized with other programmable sounder bases and common area notification appliances; 85 DBA minimum.
9. Detectors shall also store an internal identifying type code that the control panel shall use to identify the type of device (PHOTO, THERMAL).
10. Detectors will operate in an analog fashion, where the detector simply measures its designed environment variable and transmits an analog value to the FACP based on real-

time measured values. The FACP software, not the detector, shall make the alarm/normal decision, thereby allowing the sensitivity of each detector to be set in the FACP program and allowing the system operator to view the current analog value of each detector.

11. Addressable devices shall store an internal identifying code that the control panel shall use to identify the type of device.
12. A magnetic test switch shall be provided to test detectors and modules. Detectors shall report an indication of an analog value reaching 100% of the alarm threshold.
13. Addressable modules shall mount in a 4-inch square (101.6 mm square), 2-1/8 inch (54 mm) deep electrical box. An optional surface mount Lexan enclosure shall be available.
14. Detectors / Bases with connection terminals exposed to Ceiling / False Ceiling shall be provided with Protective Insulation of the same make as of Detectors.

B. Addressable Manual Call Point (Break Glass / Pull Down Type)

1. Addressable manual call point shall send data to the panel representing the state of the manual switch and the addressable communication module status. They shall use a key operated test-reset lock, and shall be designed so that after actual emergency operation, they cannot be restored to normal use except by the use of a key.
2. All operated stations shall have a positive, visual indication of operation and utilize a key type reset.
3. Manual fire alarm boxes shall be constructed of Lexan / ABS Plastic with clearly visible operating instructions provided on the cover. The word FIRE / Fire Sign shall appear on the front of the stations.

C. Addressable Control Module

1. Addressable control modules shall be provided to supervise and control the operation of one conventional circuit of compatible Notification Appliances, 24 VDC powered, polarized audio/visual notification appliances
2. The control module NAC may be wired for Class A/B with a current rating of 2 Amps
3. Audio/visual power shall be provided by a separate supervised circuit from the main fire alarm control panel or from a supervised UL listed remote supply.

D. Isolator Module

Isolator modules shall be provided to automatically isolate wire-to-wire short circuits on an SLC Class A or Class B branch. The isolator module shall limit the number of modules or detectors that may be rendered inoperative by a short circuit fault on the SLC loop segment or branch. At least one isolator module shall be provided for each floor or protected zone of the building.

1. If a wire-to-wire short occurs, the isolator module shall automatically open-circuit (disconnect) the SLC. When the short circuit condition is corrected, the isolator module shall automatically reconnect the isolated section.
2. The isolator module shall not require address-setting, and its operations shall be totally automatic. It shall not be necessary to replace or reset an isolator module after its normal operation.
3. The isolator module shall provide a single LED that shall flash to indicate that the isolator is operational and shall illuminate steadily to indicate that a short circuit condition has been detected and isolated.
4. If Isolator Bases are proposed, Vendor needs to consider Isolator base for all detectors.

E. Serially Connected Annunciator

1. The annunciator shall communicate to the fire alarm control panel via an EIA 485 (multi-drop) two-wire communications loop. The system shall support two 6,000 ft. RS-485 wire runs. Up to 32 annunciators, each configured up to 96 points, may be connected to the connection, for a system capacity of 3,000 points of annunciation.
2. An RS-485 repeater shall be available to extend the RS-485 wire distance in 3,000 ft. increments. The repeater shall be UL864 approved.
3. Each annunciator shall provide up to 96 alarm and 97 trouble indications using a long-life programmable color LED's. Up to 96 control switches shall also be available for the control of Fire Alarm Control Panel functions. The annunciator will also have an "ON-LINE" LED, local piezo sounder, local acknowledge and lamp test switch, and custom zone/function identification labels.
4. The annunciator may be field configured to operate as a "Fan Control Annunciator". When configured as "Fan Control," the annunciator may be used to manually control fan or damper operation and can be set to override automatic commands to all fans/dampers programmed to the annunciator.
5. Annunciator switches may be programmed for System control such as, Global Acknowledge, Global Signal Silence, Global System Reset, and on/off control of any control point in the system.
6. An optional module shall be available to utilize annunciator points to drive RS-485 driven relays. This shall extend the system point capacity by 3,000 remote contacts.
7. The LED annunciator shall offer an interface to a graphic style annunciator and provide each of the features listed above.

F. Programmable Directional Sounders

1. **Shall follow NFPA 72 2019 edition Clause 18.4.8.**
2. Electronic sounders shall operate on 24 VDC nominal.

3. Electronic sounders shall be field programmable without the use of special tools, at a sound level of at least 90 dBA measured at 10 feet from the device.
4. Shall be capable to broadcast preprogrammed Voice Message.
5. Shall be flush or surface mounted as shown on plans.
6. Shall produce broad band directional sound with **20 Hz to 20 Khz frequency band** to guide occupants to safe exists even in complete darkness.

G. Addressable Portable Emergency Telephone Handset Jack

1. Portable emergency telephone handset jacks shall be flush mounted on stainless steel plates as indicated on plans. Handset jacks shall be approved for emergency telephone system application.
2. Insertion of a remote handset plug into a jack shall send a signal to the fire command center which shall audibly and visually indicate the on-line condition, and shall sound a ring indication in the handset.
3. The two-way emergency telephone system shall support thirty-five (35) handsets on line without degradation of the signal.
4. **Remote Telephone Handset shall be capable of making paging announcement across all the zones in the system.**

H. Addressable Fixed Emergency Telephone Handset

1. The telephone cabinet shall be painted red and clearly labeled emergency telephone. The cabinets shall be located where shown on drawings.
2. The handset cradle shall have a switch connection such that lifting the handset off the cradle shall send a signal to the fire command center which shall audibly and visually indicate its on-line (off-hook) condition.
3. The two-way emergency telephone system shall support thirty-five (35) handsets on line (off hook) without degradation of the signal.
4. Remote Telephone Handset shall be capable of making paging announcement across all the zones in the system.

I. Batteries

The battery shall have sufficient capacity to power the fire alarm system for not less than 48 hours in standby plus 2hours of alarm upon a normal AC power failure.

The batteries are to be completely maintenance free. No liquids are required. Fluid level checks for refilling, spills, and leakage shall not be required.

If necessary to meet standby requirements, external battery and charger systems may be used.

PART 3.0 – EXECUTION

3.1 INSTALLATION

- A. Installation shall be in accordance with the NEC, NFPA 72, local and state codes, as shown on the drawings, and as recommended by the major equipment manufacturer.
- B. All conduit, junction boxes, conduit supports and hangers shall be concealed in finished areas and may be exposed in unfinished areas. Smoke detectors shall not be installed prior to the system programming and test period. If construction is ongoing during this period, measures shall be taken to protect smoke detectors from contamination and physical damage.
- E. All fire detection and alarm system devices, control panels and remote annunciators shall be flush mounted when located in finished areas and may be surface mounted when located in unfinished areas.
- F. Manual fire alarm boxes shall be suitable for surface mounting or semi-flush mounting as shown on the plans, and shall be installed not less than 42 inches (1067 mm), nor more than 48 inches (122 mm) above the finished floor.

3.2 CAUSE & EFFECT LOGIC

System shall be programmed as per the attached cause & effect logic.

3.3 TESTING

- The service of a competent, factory-trained engineer or technician authorized by the manufacturer of the fire alarm equipment shall be provided to technically supervise and participate during all of the adjustments and tests for the system. All testing shall be in accordance with NFPA 72.
- A. Before energizing the cables and wires, check for correct connections and test for short circuits, ground faults, continuity, and insulation.
 - B. Close each sprinkler system flow valve and verify proper supervisory alarm at the FACP.
 - C. Verify activation of all waterflow switches.
 - D. Open initiating device circuits and verify that the trouble signal actuates.
 - E. Open and short signaling line circuits and verify that the trouble signal actuates.
 - F. Open and short notification appliance circuits and verify that trouble signal actuates.
 - G. Ground all circuits and verify response of trouble signals.
 - H. Check presence and audibility of tone at all alarm notification devices.
 - I. Check installation, supervision, and operation of all intelligent smoke detectors using the walk test.
 - J. Each of the alarm conditions that the system is required to detect shall be introduced on the system. Verify the proper receipt and the proper processing of the signal at the FACP and the correct activation of the control points.
 - K. When the system is equipped with optional features, the manufacturer's manual shall be consulted to determine the proper testing procedures. This is intended to address such items as verifying

controls performed by individually addressed or grouped devices, sensitivity monitoring, verification functionality and similar.

EFFECT		ANNUNCIATION			VOICE ALARM SYSTEM			NAC ACTIVATION		REQUIRED FIRE SAFETY CONTROL					SUPPLEMEN TARY					
CAUSE	SIGNAL TYPE	ALARM INDICATION & BUZZER IN MFACP	ALARM INDICATION & BUZZER IN NETWORK REPEATER	ALARM INDICATION & BUZZER IN GRAPHICS WORKSTATION	APPROVED ALERT MESSAGE SHALL BE BROADCASTED TO AFFECTED FLOOR FOR 180 SEC IMMEDIATELY & AUTOMATICALLY.	AFTER 180 SEC ALERT MESSAGE SHALL BE OVERRIDDEN BY EVAC MESSAGE. APPROVED EVAC MESSAGE SHALL BE BROADCASTED ON AFFECTED FLOOR FOR 90 SEC & ALERT MESSAGE SHALL BE BROADCASTED ON ONE FLOOR ABOVE & ONE FLOOR BELOW OF AFFECTED FLOOR.	APPROVED EVAC MESSAGE SHALL BE BROADCASTED ON AFFECTED FLOOR IMMEDIATELY & AUTOMATICALLY FOR 90 SEC & ALERT MESSAGE SHALL BE BROADCASTED ON ONE FLOOR ABOVE & ONE FLOOR BELOW OF AFFECTED FLOOR.	AFTER COMPLETION OF EVAC MESSAGE SOUNDER SHALL BE ACTIVATED IMMEDIATELY & AUTOMATICALLY OF AFFECTED UNTIL SYSTEM RESET	AFTER COMPLETION OF EVAC MESSAGE DIRECTIONAL SOUNDER & STROBE SHALL BE ACTIVATED IMMEDIATELY & AUTOMATICALLY AFFECTED FLOOR.	RELEASE MAGNETIC HELD DOORS	UNLOCK EGRESS	SHUTDOWN ELEVATORS	ACTUATE SMOKE EXHAUST	ACTUATE SUPPRESSION SYSTEM PREDISCHARGE ALARMS	INITIATE SUPPRESSING SYSTEM RELEASING SEQUENCE	ACTUATE GRAPHIC SYSTEM DISPLAY FLOOR MAP	PRESSURIZE STAIRWELL	SHUTDOWN AIR HANDLING UNIT	OTHER SYSTEM SHUTDOWN PROCESS	ACTUATE EXTERIOR STROBE
ACTIVATION OF SINGLE DETECTOR	FIRE	☒	☒	☒	☒	☒		☒	☒	☒	☒					☒				
ACTIVATION OF MORE THAN ONE DETECTOR ON SAME AFFECTED FLOOR	FIRE	☒	☒	☒			☒	☒	☒	☒	☒	☒	☒			☒				☒
ACTIVATION MANUAL PULL STATION / CALL POINT	FIRE	☒	☒	☒			☒	☒	☒	☒	☒	☒	☒			☒				☒
ACTIVATION OF FLOW SWITCH	FIRE	☒	☒	☒			☒	☒	☒	☒	☒	☒	☒			☒				☒
ACTIVATION OF DUCT DETECTOR	FIRE	☒	☒	☒						☒	☒							☒		
ACTIVATION OF CO SENSOR AT BASEMENT	SUPERVISORY	☒	☒	☒									☒							
ACTIVATION OF TAMPER SWITCH	SUPERVISORY	☒	☒	☒																
OTHER STATUS																				
FIRE ALARM CONTROL PANEL LOW BATTERY	TROUBLE	☒	☒	☒																
GROUND FAULT	TROUBLE	☒	☒	☒																
LOOP OPEN CIRCUIT	TROUBLE	☒	☒	☒																
LOOP SHORT CIRCUIT	TROUBLE	☒	☒	☒																
DEVICE COMMUNICATION FAIL	TROUBLE	☒	☒	☒																
SMOKE DETECTOR DIRTY	TROUBLE	☒	☒	☒																
VOICE EVACUATION SYSTEM LOW BATTERY	TROUBLE	☒	☒	☒																
SPEAKER OPEN CIRCUIT	TROUBLE	☒	☒	☒																
SPEAKER SHORT CIRCUIT	TROUBLE	☒	☒	☒																
AMPLIFIER FAIL	TROUBLE	☒	☒	☒																
AMPLIFIER OVERLOAD	TROUBLE	☒	☒	☒																
FFT RISER OPEN CIRCUIT	TROUBLE	☒	☒	☒																
FFT RISER SHORT CIRCUIT	TROUBLE	☒	☒	☒																

3.4 FINAL INSPECTION

- A. At the final inspection, a factory-trained representative of the manufacturer of the major equipment shall demonstrate that the system functions properly in every respect.

3.5 INSTRUCTION & TRAINING

- A. Instruction shall be provided as required for operating the system. Hands-on demonstrations of the operation of all system components and the entire system including program changes and functions shall be provided.

- B. The contractor and/or the systems manufacturer's representatives shall provide a typewritten "Sequence of Operation."

3.6 SUBMITTALS

a) Power calculations.

- Battery capacity calculations.
- Supervisory power requirements for all equipment.
- Alarm power requirements for all equipment.
- Justification showing power requirements of the system power supplies.
- Voltage drops calculations for wiring runs in worst case condition.

b) Complete manufacturer's catalogue data including supervisory power usage, alarm power usage, physical dimensions, finish and mounting requirements.

c) Submit panel configuration and interconnection of modules and all other data as required to make an informed judgment regarding product suitability. As a minimum, data shall be submitted on the following:

- Main system including all fire detection with main and secondary control panels.
- Circuit interface panels including all modules.
- Power supplies, batteries, and battery chargers.
- Equipment enclosures.
- Intelligent addressable manual pulls stations, multi-criterion detectors, heat detectors, analogue smoke detectors, alarm monitoring modules, and supervised control modules.
- Audible and visual evacuation signals and devices.
- Software and firmware as required providing a complete functioning system.
- Wiring.
- System driven remote annunciators.
- Interface module and wiring configuration from local system to Fire Command System.

d) Submit copies of UL listing or FM approval data showing compatibility of the proposed devices or appliances and the panels being provided.

e) Submit the following shop drawings.

- Floor plans showing all initiating, end of line, supervisory, indicating appliances, and output control devices; including circuit interface panels, enunciators, printers, Control panel location.
- Raceways, marked for size, conductor count with type and size

- Calculations and mathematical justification for audible devices shall meet the code requirement of 15 dBA above ambient at 10 feet distance for audible warning signals.
 - Wiring diagrams showing points of connection and terminals used for all electrical connections to the system devices and panels.
 - Complete single-line riser diagram showing all equipment and the size type and number of all conductors.
- f) Submit Method Statement for systems component wiring, installation, testing, commissioning, and operating.
- g) Typical installation drawings
- h) Complete operation and maintenance manual with two sets of proposed installation drawings shall be submitted.
- i) Warranty all system components, devices, peripherals, wiring, for Three years from date of practical completion Certificate.
- j) Guarantee all wiring and raceways to be free from inherent mechanical or electrical defects for One years from date of practical completion Certificate.

Make as approved by Engineer-in-charge

Public Address/Voice Alarm System Design Scheme

Part I Design Requirements

Project Overview

[This chapter mainly describes the basic information and requirements for this bid.]

System Design

The design for the Digital Public Address/Voice Alarm System should meet the user and system requirements below. Its key features and design basis are as described in the next chapters.

User Requirements

The broadcasting system uses the same device to play the background music, business announcement and emergency broadcast. It has an emergency call microphone in the fire control center for evacuating the crowds in specific zones when accident happens and a paging microphone in the broadcasting center to broadcast announcements and search notices.

The background music, business announcement and emergency broadcast share the same loudspeakers. In ordinary situation, the loudspeakers are for playing background music and business announcement, but for emergency broadcast instead during the fires.

The broadcasting devices located in the broadcasting center or the specified location according to the user's requirement. The placement of the devices should meet the operating environment requirement and save up floor space. The amplifier capacity should comply with national standards and be with

redundancy. Meanwhile, the emergency broadcast can achieve the linkage of the adjacent layers. Users can select the layers to be linked.

System Design Requirements

Maturity

As the development of modern science and technology, all kinds of the advanced technologies have been applied in the intelligent buildings to save the labor costs, to improve the efficiency and to ensure the intelligence for the modern buildings.

The modern technologies support the public address applications in the intelligent buildings. The global mature and advanced technologies are critical for designing the public address system, and only the broadcasting system incorporating mature and industry-leading technology and high-technology products can ensure the intelligence for buildings.

In the system design, advanced concepts, technologies, and methods need to be adopted; system structure, product designs and wiring easiness also need to be considered.

Functionality and Reliability

Besides the technical advancement mentioned above, functional design, system structure, system performance, manufacturing process and after-sales support are also important as to ensure the reliability and stability of the system operation, maximizing the mean error-free time.

A mature technical platform and the rigorous manufacturing process are the bases of functions' realization. During the runtime, system should be able to discover and eliminate all the functional faults in time. System administrators can easily access the failures and work logs.

Key Design Bases and Indices

Key Design Bases

XXX Bid Document

Architectural Plan

Civil Architectural Electrical Design Code (JGJ/T16-92)

Code for Engineering Design of Generic Cabling System for Building and Campus (GB/T50311-2000)

Standard for Design of Intelligent Building (GB/T50314-2000)

Code for Design of Automatic Fire Alarm System (GBJ116-92)

Automatic control system for fire protection (GB16806-2006)

Technical Code for Public Address System Engineering (GB50526-2010)

Key Design Indices

Sound Field Intensity

The intensity of the sound field is close related to the background noise. In terms of the noise criterion for different environments, the noise criterion in general buildings is 25 ~45db except the parking lot with noise criterion around 55 ~56db. The uniformity of the sound field is the uniformity on the sound pressure level, and its range of variation should better be around 8db. Sound intelligibility is the subjective assessment standard of the sound articulation. The performance indices for the loudspeaker: the frequency response is the key factor that affects the intelligibility. The frequency response of the ceiling loudspeaker within 100Hz ~ 14 KHz is appropriate. The specific indices for the various loudspeakers should accord to the frequency response of the amplifier, which should be better than the loudspeakers. The selection and location for the loudspeakers are mainly determined by factors like maximum sound pressure level, sound field uniformity, transmission frequency characteristic and space size. In terms of the technical standards, Indices of Acoustic Characteristics for Sound Reinforcement System & Civil Architectural Electrical Design Code, the system is designed by third level of the sound reinforcement for the voice and music. The indices are as follows:

Vacant auditoria stable overall sound pressure level $\geq 85\text{dB}$ (Within 250~4000Hz average sound pressure level)

transmission frequency characteristic: 250 ~ 4000Hz, tolerance +4~10dB

Sound Field Non-uniformity $\leq 8\text{dB}$

The sound pressure of the listening point on the vertical axis along the single loudspeaker's projecting direction can be calculated by:

$$LP = L0 + 10 \lg PS - 20 \lg r$$

LP: auditory point sound pressure level (dB), L0: loud speaker sound pressure level (dB SPL), PS: sound pressure power of sound source, also the loudspeaker rated power (W), r: vertical distance of the speaker and the listening point

Background Music: The sound source location of the background system should not be aware by the listeners. The goal for the design is that the tone quality of the music is tender and clear. The design indices: the sound pressure inside is uniform. The average sound pressure = noise level + (3~5) db. The frequency band is 100~12000Hz, and the playback is straight.

Emergency broadcast: The emergency system is designed to make the listener hear clear and correct sound. The design indices: The sound pressure inside is uniform. The average sound pressure = 88 ~94 db. The frequency band is 100 ~ 6000Hz and the playback is straight.

As described above, the noise level can determine the average sound pressure for this design: Background music sound pressure = 60 ~ 70 dB, and Emergency broadcast sound pressure = 88 ~94dB.

Power Amplifier

$$P = K1 \cdot K2 \cdot \Sigma P0$$

P —Total electric power amplifier device output (W); P0— max. electric power of $K_i \cdot P_i$ when every branch broadcasts at the same time

P_i — the speakers rated capacity of the i-th branch; K_i — sync coefficient of the i-th branch

Service broadcasting programs, $K_i=0.2\sim0.4$; Background music system, $K_i=0.5\sim0.6$

The business of broadcasting, $K_i=0.7\sim0.8$; Fire accident broadcasting, $K_i=1.0$ (the maximum electric power broadcast should be in line with the national standards)

K_1 —Line attenuation compensation coefficient: When line attenuation is 1db, K_1 is 1.26; line attenuation is 2db, K_1 is 1.58

K_2 — the aging factor, 1.2~1.

System Function

System Design Composition and Principle

The system has placed broadcasting devices in every plant room. The distributed control device (DCD) is integrated with many functions and supports the connection via Ethernet. It has 8 loudspeaker output circuits to connect the loudspeaker in zones and 4 network connection interfaces to connect to 4 network paging control panel or Ethernet. Each DCD can connect to 4 amplifier channels. The device has integrated amplifier switching matrix to support the redundancy switching and the general dry contact input/output interface to connect the external audio sources and dry contact interface of the linkage fire system. It has a built-in 1G memory to store the audios such as digital voice messages and the alarm tone of the emergency broadcast and to store log files. This device can operate independently without the PC. It encompasses the functions like the audio playing, zone control, fault monitoring, log recording, volume control and amplifier switchover.

The control device for the system customized Paging Station. It has the LCD touch screen for operations, like zone selecting, calling, audio sourcing selecting, and emergency broadcast, monitoring, and internal communication. The functions of the Programmable network paging console can be configured using the configuration software.

Key Features

Distributed Control

This distributed design for the system is based on the building structure to facilitate the connection of the loudspeakers in several nearby zones, which will make the control of the loudspeakers easier. The selected system devices should use the TCP/IP technologies and build on the standard network platform to ensure the extensibility, compatibility for multiple technology platforms and advancement of the system.

Faults detection and Isolation

The Faults detection function for the broadcasting system can automatically examine the host system, power amplifier, power source, communication, and detect the open circuit, short circuit, and grounding fault to generate the fault alarm and log.

When the grounding fault or short circuit occurs, the amplifier or the main controller of the system should isolate the circuit to ensure the operation of the main devices and normal circuits.

Background Music Audio Source

The broadcasting system can use audio sources from devices such as the CD, radio, and MP3 to provide the zones with different audio sources, which can meet the various requirements for different zones. In the tolerant power range, different zones using the same audio source can share an amplifier, reducing the system cost. The background music inputs methods are multiple: both network and local inputs are allowed.

Service Broadcasting

The paging microphone allows users to make paging and broadcast search notices by zone. The paging microphone has a color LCD touch screen. Users can configure the functions directly on the screen. Users can select the zone and audio source, adjust the volume, and enable the emergency broadcast by pressing the button on it. Paging microphones can intercommunicate. 255 broadcast priorities levels available. When the connected distributed control unit is out of order, a paging microphone can continue the intercommunication with other microphones in the network and proceed the remote broadcasting configuration on other control units.

Time Synchronization

Within the system, the controller can specify any main device as the main time source, or choose a third-party system to be the time source. Other devices synchronize their time with the main time source to make sure the time is consistent in the system.

Timed Broadcast

The timed broadcast function of the broadcasting system allows users to set the periods for playing different music or service broadcasts in different zones.

Emergency broadcast

The broadcasting system can be linkage of the fire system to achieve the alarm function in the adjacent layers. The layers for linkage can be configured with the software as the requirements. Users can record the voice message for the emergency broadcast and save it in the host system. The emergency broadcast can start automatically (when linkage of the fire system) and manually. The broadcasting system has its own 1 PPT emergency microphone, which could be used to play emergency broadcasts and evacuate the crowds in specific zones.

When the fire alarm rings, the system can display the fires in planar graph and show the fire zones. According to the fire status, users can call the emergency microphone to make the fire zones enter alarm and evacuation two modes.

Priority

The broadcasting system allows users to set the priorities. Generally, the priority for the broadcast is in the order: Emergency Broadcast > Service Broadcast > Background Music, and for the microphone is Emergency Microphone > Service Broadcast Microphone.

Automatic Amplifier Shift

The system has standby amplifiers. When a main amplifier fails to function, the standby amplifier replaces it automatically. After the main amplifier recovers, the system will use it instead of the standby amplifier in an automatic way.

Broadcast Record

The controller can record sufficient voice message (up to 500Mb), and its contents can be customized. The voice message contains the emergency broadcasts for fires, such as evacuation and all-clear information, which are played by the digital speech synthesis system automatically (linkage of the fire alarm system) or manually.

Internal Communication

The call stations can communicate with each other using the simple communication functions.

The paging microphone in the broadcasting center has a color LCD touch screen. Users can configure the functions according to the actual requirements.

The call station's calling zone can be authorized. Each call station can call the local authorized zones. Several call stations can call the same zone, and one call station can call the zones of other call stations.

Background Music Broadcast

The background music broadcasting functions and local audio sources can be specified in different zones. The administrator can configure the background music according to the time and occasion. The system will automatically switch to the emergency broadcast under emergency.

Program Source

The broadcasting control center has various audio sources, including the professional digital players for background music, CD, FM/AM tuner, remote control microphone for zone calling

Log

The system can record the device fault log and operation log. Users can search information in the logs using time, devices, operators, and log types as search criteria.

Distributed Call Station

The distributed call station can call the specified zones. The call station has remote control function, which can control emergency broadcast, zone audio sources and external devices.

Local & Remote Monitoring

Users can monitor audio sources status via local distributed control units and remote call stations.

Emergency broadcast

Emergency broadcast Requirement

Forced Switchover (Auto/Manual): Once the emergency broadcast is triggered by a remote-control microphone or other external devices (fire alarm system), the other functions (like background music and general broadcast) will be paused, but the system will play the recorded voice message (alarm and evacuation information) for fire emergency or the information from the emergency microphone until the all-clear.

The system will switch to emergency broadcast in 3 seconds.

The signal tones and voice messages (alarm and evacuation information) of the Emergency broadcast system are stored on the voice storage flash or SD card, which will not be ageing or lost. Users can play them in all zones.

Emergency broadcast system should have system testing function as a standard configuration. The testing for the system can enable the operator to ensure the normal operation for the system all the time. The testing of emergency broadcast should be enabled via the software instead of playing the actual emergency broadcasts and causing public's panic.

Emergency broadcast system should have two pre-recorded emergency broadcast messages. The system should be able to edit and record the alarm and evacuation message. The priorities of the emergency broadcast can be programmed. DSM should be with a PTT emergency microphone, with which operators can announce real-time evacuation notices when emergency happens.

Users can listen to stored or audios about to play to avoid playback error message which may lead to crowd panic.

Language kinds should meet the user's requirements. There should be spaces left for the tags to facilitate users to stick related prompts for management.

System can automatically record voice messages sent by emergency microphone. Users can listen to and broadcast the voice message.

The linkage signals of the emergency broadcast use the volt-free contact. All signals are from the fire control center. The number of the cables is the same as the fire control centers number.

Can achieve the linkage of the adjacent layers. Users can select the layers to be linked via software configuration.

The loudspeakers allow the 3-wire forced volume control switchover.

Circuit Detection

It can detect every zone's status to ensure the working order of the circuits and report faults in time. When a short-circuit error takes place, the device isolates the zone where error happens to protect the amplifiers and to cut off its impact on other zones.

Part II Solution

Device Selection

The system should be based on the TCP/IP network technologies with high level of integration and flexible configuration to meet various requirements. It is designed for the emergency broadcast and background music application enabling fire emergency broadcast functions like automatic fire warning, voice message presetting, emergency calling, emergency call recording and circuit monitoring, and service broadcasting functions like zone background music broadcasting, zone call broadcasting, timed music broadcasting and monitoring. The device is accordance with the Automatic Control System for Fire Protection (GB16806) and Standard EN 54-16.

With public address and voice alarm system integrated into one, the system should solve the duplication of investment and security risks and other problems caused by two individual systems. The distributed intelligent systems of the should be capable of accessing the 10/100M Ethernet

network and multiple audio sources from the network. It supports connections in the Ethernet and the networking for several other control systems to perform centralized and distributed management via software. This product should be suitable for medium-scale and large-scale buildings.

Technical Characteristics

High Integration

The controller should be able to integrate audio source storage, network audio reception, voice broadcasting, audio matrix, volume control, monitoring, troubleshooting and main/standby amplifier switch functions together.

Safety

The system should be GB16806/EN54-16 compliant and can work in severe environment. It can be used to broadcast in case of emergency to disperse and evacuate people. It is a great guarantee for people's life security

Ethernet Based Extension

The central controller should be able to connect over the Ethernet without any external device. The configurable Network Paging Microphone could be placed anywhere in the Ethernet, which will enable the following:

The method of distributed installation can save the cost of wiring.

If the band width is sufficient, it can use current internal network.

Provide plentiful network audio sources.

Control the local or global broadcasting by the manner of permissions.

Easy to upgrade.

Multitask Processing Mechanism

To simplify the operations, the system adopts multitask processing mechanism, thus it can process multiple tasks and make it possible to broadcast multiple audio source and broadcast in multiple zones.

Multiple Audio Sources Support

The system supports the audio sources from emergency microphone, network audio, ancillary line input, build-in audio/music files and web microphones.

Prioritized Control

The system supports 255 priorities. Users can configure it in any way they want and realize all kinds of prioritized broadcasting controlling.

Local and Network Monitoring Function

With the built-in loudspeaker of controller or Configurable Network Paging Console, users can monitor each zone's broadcasting. The monitoring volume can be adjusted.

Touch screen

Programmable network paging console (NPC) is equipped with a touch screen. Users can operate on the touch screen to configure multi-zones settings without the need to extend hardware key modules, saving floor space on the desktop.

Troubleshooting Feature

The ASD troubleshooting technology can support the system to automatically detect the failure of the main power, standby power, amplifier circuit, amplifier protection/power, software, communications, and loudspeakers etc. It can also locate the failure for maintainers to repair or replace the fault units.

Loudspeaker Circuit Detection

The SIM loudspeaker circuit detecting technology helps monitor loudspeaker circuits' statuses. When short-circuit or open-circuit faults are found, the device will send a warning automatically. The software can display all the circuit faults for maintenance's convenience. Usually, a loudspeaker's short-circuit fault will activate the amplifier protection, affecting the working order in other zones. SIM solves this problem by automatically isolate the short-circuited zone where fault is found.

Matrix Allocation Function

This system includes the audio source matrix and power matrix to provide two kinds of work mode: low-cost power zones, and amplifiers & zones, which can meet the requirement for low-cost applications or high-power zones.

Input/output Triggering.

This system has input/output interfaces which can be triggered by dry contacts. Engineer staffs can configure the contact input to achieve the broadcasting control by the external device (linkage of the fire system). The contact output can be used to control the external devices for the volume, forced switchover and so on. The contact output interface can be linked of the broadcasting operations.

Voice Alarm

This system is designed in accordance with the EN54-16 Standard and national standard GB16806. It is full-featured for the fire alarm. When the fire occurs, the system will play the preset voice information. Then it can play the evacuation information for the zones with fires and the fire alarms for the nearby zones. The system can set several kinds of language for voice information, which can be played using the operation panel.

Broadcast Delay

Users can configure the time delay function in the broadcasting system. After making broadcasts, users need to wait for the time delay set previously, then the pre-recorded program will be played. This function can be utilized as in the following situation: When fire occurs, the system will pause for the administrator to confirm the fire alarm signal, and then play the emergency broadcasts. It can avoid crowd panic results from mis-operations.

OneClick Technology

To increase operation's convenience, OneClick technology is developed with which broadcasting can be started by clicking only one key. The operation shortcut can be configured with system management software and network paging console.

Internal Communication

The Programmable network paging console should have built-in loudspeakers and microphones for internal communication. Users need to configure this function before using it.

Flexible Configuration

This system allows users to customize the configurations flexibly using the specific software. The engineering technicians and maintainers can configure the built-in audio sources, keys' functions, troubleshooting, audio source play modes, broadcasting priorities and contact input/output and other customized operations the OneClick functions. With careful configurations, the system can meet the needs of both background music broadcasting and emergency alarms. The configuration software for this system is used to configure the hardware. The default configuration for the device contains only the basic functions. Users need to work on each setting, including basic parameters, troubleshooting, broadcast preset and key events and so forth.

Configuration Software features:

Project File Management

It includes:

Create project, save project, open project, and check the latest project file;

Manage the subsystem and configuration files

Property Setting

The property setting is to set the basic property and broadcasting functions. The basic property settings include the zone settings, DSM settings and paging station settings.

Broadcasting Setting

The broadcasting function allows users to set the event- triggered, time-triggered and fire alarm broadcasting.

Output Configuration: After the error checking and building, system can generate the configuration files according to the configured project files.

Audio Source Conversion: It allows users to convert the audio files according to the sampling rate, digitalizing bit and channel number into the audio files in specified formats for the system.

Device Description

Digital Integrated System Manager / controller

The Digital Integrated System Manager is hereinafter referred to as the DSM.

The DSM is controlling equipment in the system designed for extending the number of loudspeaker zones, and can support multiple sound source files for broadcasting. The system integrates with functions such as the sound source file storage system, the network audio broadcasting system, the loudspeaker zone control system, and the system for monitoring and diagnosing faults.

The DSM offers various features, which are listed below:

Supports the manual selection of sound source files, zone buttons, and can be directly operated.

Supports emergency microphone input for emergency broadcast in the event of, for example, a fire evacuation.

Includes different indicator lights that identify system running states.

Performs remote paging and broadcasting operation through the Configurable Network Paging Console.

Includes 8 zone outputs. The number of zones can be configured through the software.

4 auxiliary inputs that can connect to external sound source equipment such as a CD player or tuner 4 auto volume control input ports that can set the phantom power supply and gain of each input.

Auto loudspeaker circuits 'short-circuit and open-circuit detection.

A fuse protects the main power supply. In the case of a power supply short circuit within the DSM, the system automatically disconnects the main power supply.

Built-in loudspeaker that can monitor the zones and sound sources, as well as monitor the network audio through the paging station.

Contains 1GB of built-in flash memory that can store recorded voice files so as to fulfill functions such as voice information broadcasting and voice synthesis and to store LOG files

Can simultaneously broadcast four types of sound sources, such as voice audio sources and external input or network audio sources.

Can automatically record operation and fault logs, and can store up to 10,000 logs of each log type (operation and fault log types).

Supports switching between the main and backup power amplifiers, and is capable of configuring the standby mode.

Contains a self-test function.

Supports automatic fault diagnosis.

Supports broadcasting volume adjustment.

Can define the zone and sound source functions. Button function description labels are also easy to install.

An audio matrix enables broadcasting any audio source in any zone.

Supports fire emergency broadcast mode to improve personnel evacuation efficiency in case of an emergency.

Supports broadcasting designated recorded voice audio at a designated time to allow for unattended broadcasting.

Capable of system extension through an Ethernet network.

Parameter	Value
Power Supply	
Main power supply	~100-240V, 50/60Hz
Backup power supply	DC 21.5V-28.5V
Main power fuse	T2AL 250V
Max. input power	120 W
Rated power	50 W
Audio Input	
Auxiliary input	0dB
Input impedance	20 kΩ
Frequency response	60Hz-16KHz
PTT microphone input	-51dB
SNR	>85dB
Audio Output	
Audio output channels	4个
Output signal	0dB

Parameter	Value
Record output	0dB
Loudspeaker Circuit	
Output channels	8, with circuit fault detection function
Max. output load power	250W
Trigger Input / Output	
Trigger input ports	8
Trigger output ports	8 (NO, NC and COM)
Max. working voltage	AC 250V/DC 30V
Max. working current	2.5A
Others	
Monitoring loudspeaker	10W/8W
Ethernet speed	10M/100M
Ethernet interface number	4
Storage space	1GB
Work Condition	
Humidity	< 95%, without condensing
Working temperature	-10°C~+55°C
Storage temperature	-40°C~+70°C
Specification	
Dimension (W×H×D)	482 mm×88 mm ×420mm
Mount dimension (W×H×D)	580mm×235mm ×552mm
Net weight	9.3Kg
Gross weight	12.5Kg

Pure Final Stage Class: D Digital Power Amplifier

Pure final stage class D power amplifier 500W can be used in this system, which has the following characteristics:

Real-time display of the temperature;

100V or 70V constant-pressure output;

Output volumes can be digitally adjusted

5.8 Unit LED display facilitates you to examine the device status.

Perfect output short circuit protection and over-temperature protection; Automatic press limit control for long-time over loading output;

Cool fan can be warm booted. The completely new design of the build-in short air duct cooling system structure for amplifiers can ensure the high stability performance.

Remote fault reset function such as overloading or short circuit protection.

Parameter	Value
Rated Power	1 × 500 W
Main power supply voltage	~ 100 - 240 V, 50/60 Hz
DC Backup power supply operating voltage range	21.5 V - 28.5 V DC
Power consumption	< 700 W
Main power supply protection	T10 AL 250 V
Speaker output voltage	100 V / 70 V
Frequency response	70 Hz -18 kHz (+1 dB ~ -3 dB)
Input sensitivity	1.414 V _{RMS}
Impedance	20 kΩ
SNR	> 100 dB(with A-Weight)
Non-linear distortion	< 0.05 % (at 1/3 rated power, 1 kHz)

Configurable Network Paging Console

The Configurable Network Paging Console is hereinafter referred to as the “NPC”

The network paging microphone (NPC) connects to the system and related devices to transmit audio and control information through the Ethernet network. The NPC is used for paging, controlling broadcasts, monitoring zones, and for using the bidirectional intercom function.

The NPC has the following characteristics:

Lightweight and innovative patent outward design. Support embedded desktop installation

4.3-inch color LCD touch screen. Can display system status and be operated for zones and groups' division or global paging and broadcast control

Simple and intuitive user interface

Can connect with up to 20 units. Button numbers can be added via software configuration

Built-in monitor loudspeakers. Can utilize functions like zone monitoring and two-way intercom between stations.

Digital audio processing to avoid acoustic fidelity distortion

Can intercommunicate with other paging microphones and DSM in the network even when the connected DSM is not running

3 shortcut buttons: microphone talk mode switch, select all function and emergency broadcast

Parameters	Values
Power supply voltage	DC 12V
AUX input	0 dB
AUX input impedance	10 K Ω
Frequency response	60 Hz – 16 KHz (local input signal)
Microphone input	-51 dB
Audio output	0 dB
Monitoring loudspeaker	2 W/8 Ω
Monitoring sound pressure level	Higher than 65 dB and lower than 115 dB (1 m in front of where the audio is being output)
Operating temperature	-10 °C to +55 °C
Storage temperature	-40 °C to +70 °C
Humidity	<95%, without condensation

Loudspeakers

Wall mount Loudspeaker

6W wall mount Loudspeaker

Parameters	Values
Max power	9 W
Rated power	6 W
Power taps @ 100V	6 W / 3 W
Sound pressure level at 6W/1W (1kHz,1m)	96 dB / 88 dB
Frequency range (-10dB)	110 Hz - 13 kHz
Dispersion angle(1kHz/-6dB)	140°
Rated input voltage	100 V / 70 V
Connection	Cable
Dimensions (diameter x height)	260 x 180 x 120 mm
Weight	1.08 Kg
Case material	ABS

Ceiling mount Loudspeaker

6W Ceiling mount Loudspeaker

Parameters	Values
Max power	9 W
Rated power	6 W
Power taps @ 100V	6 W / 3 W/ 1.5 W
Sound pressure level at 1W (1kHz,1m)	90 +- 5 db
Frequency range (-10dB)	125Hz – 16kHz (-10 dB with reference to 1kHz)
SPL at rated power/1W	95dB / 88dB (1m & 1kHz)
Rated input voltage	100 V / 70 V

Parameters	Values
Connection	Cable
Dimensions (diameter x height)	166mm X 60mm
Weight	500 gm
Case material	ABS

Training

Train the clients to know the system structure and principles.

Train the clients to know and master the installation methods of digital broadcasting systems.

Train the clients to master the configuration methods of digital broadcasting systems.

Train the clients to use digital broadcasting system software.

Train the clients to know basic troubleshooting and maintenance methods of digital broadcasting systems.

Make as approved by Engineer-in-charge

7. Technical Specification of Boom Barrier System

A vehicle access barrier shall mean a rising boom barrier that shall open in case of an impulse with the use of a valid card or Push Button. The barrier shall close either after the vehicle has passed or after an adjustable hold-open time in case the vehicle chooses not to pass through. Appropriate sensors shall be used for safety purpose. Standard boom length covering road should be 6 mtr. It shall meet the following specifications.

Sr.No.	Item	Description
1	Application	Outdoor
2	Drive Unit	Heavy duty Gear Motor
3	IP Rating	67
4	Housing	Barrier Housing Unit: Aluminium Housing Boom: Powder Coated white with Red reflective strips. RAL 9010m or approved colour with red reflective strips.

5	Housing Dimension	Modular
6	Housing Material of Construction	All Aluminium Housing with suitable Base frame in SS 304 for high protection against corrosion.
7	Protection	All Housing and internal parts will be rust & corrosion free metals or alloys of high strength with suitable Epoxy coating as applicable.
8	Housing Dimension (W X D X H)	350mm x 240mm x 1030 mm approximately
9	Boom Specifications	Rectangular/Round
10	Intelligence	The barrier shall use a suitable AC Drive for smooth functioning of the equipment. Possibility for integration via standard user interfaces.
11	Anti-crushing safety	Inbuilt Reverse on obstacle feature
12	Certifications Needed	• ISO -9001:2015 CE, • IP - 67 Certificate from a government accredited lab • Environmental test certified
13	Power Supply	230+/- 10% VAC, 50 Hz.
14	Maximum Power Consumption	60 Watt or Minimum
15	Opening & Closing Time	Not more than 6 sec
16	Operating Temperature	-30 Degree Celsius to + 70 Degree Celsius
17	Safety	Detection of Presence of Vehicle in Loop or in the path of Infrared Safety Sensors available. Loops or Sensors to be used to prevent barriers from closing on the vehicle. Barrier should get closed once the barrier crosses the loop.
18	Duty Cycle	100%
19	Integration	Shall function in integration with, Smart cards, proximity reader-based access control systems etc.
20	Performance Requirement	MCBF- 5 million Cycle

8. CABLE TRAYS

a) Scope

The scope of work shall cover powder coated GI Cable trays inside the building and hot dip galvanized for outdoor, tray support systems, fittings and accessories for laying the power cables, control cables and tele technical cables. Manufacture and finish shall be completed at the factory works.

b) Construction

The cable tray shall be manufactured to comply with the specifications of National Electrical code (NEC). The cable trays shall be of steel as per IS 226 and galvanized and the thickness of galvanization shall be not less than 100 microns. All bolts and nuts also be galvanized and shall confirm to IS 1363 – 60 for quality, threading, and dimensions before galvanizing. Hot dip galvanizing shall confirm to IS 2633, Galvanizing of each member shall be carried out in one complete immersion. The galvanizing shall be uniform, clear, smooth, continuous, and free from acid spots. The quality of Zinc used for galvanizing shall be 98.8% purity.

The cable trays and accessories shall be fabricated out of hot rolled steel sheets, which shall be hot dipped galvanized. The complete assembled cable tray sections shall be corrosion resistant, high strength and with extreme smooth surface.

CABLE TRAYS – TYPES

i) Ladder Type Cable Trays

Ladder type cable trays shall be used for laying the power cables of medium voltage and low voltage. Ladder type cable trays shall be made out Channels for side rails and rungs.

Ladder type cable tray shall consist of a Welded rung assembly of field proven design. It shall consist of side rails of 3.0 mtr. Standard lengths and slotted rung spaced 250mm center to center.

Coupler, Coupler fasteners, tray assembly fasteners must form the part of the equipment. The Ladders should be heavy duty sheet metal constructions yet robust enough to carry a cable load of 50 kg/m on a span of 1.25 mtr.

The design shall be flexible enough to accommodate change of widths at site. Sections of 2mm shall be standard for different widths of trays as detailed in BOQ. Horizontal bends, Vertical internal / external, tees, crosses shall be standard products. Hinged or flexible coupler assemblies for Vertical / Horizontal route changes shall be available as standard as also reducers (offset couplers). Covers shall be of pre-galvanized sheet 18 SWG (1.25mm up to 300w) & 16 SWG 1.6mm on higher widths) construction possible for easy mounting on to trays wherever applicable. Provision to add on earth

flat holding clamps, trays fixing clamps should be part of cable tray design.

ii) **Perforated Cable Tray**

Perforated cable trays shall be used for laying the cables of low voltage and control cable. Perforated cable trays shall be generally of channel type and the perforation in the trays shall be either 8 x 15 mm or 10 X 20 mm oval holes.

Perforated cable trays of 1.6mm thickness and standard lengths 2.5 mtr as detailed in BOQ shall be offered by EPC Contractor if required. Finish shall be as specified earlier.

The cable tray of size 150mm Width and Less than that shall be of Perforated type.

c) **SIZES**

The cable trays shall be of the following standard sizes.

Length: 3000 mm

Width: 300 / 450 / 600 / 800 / 1000 mm as required.

Height: 75 / 100 mm

Thickness of sheet steel: 2.0 mm / 3.0 mm as per width of tray.

The flange and width of the cable trays shall be decided based on the diameter and the number of cables running through each section of the cable tray.

d) **ACCESSORIES**

Cable tray manufacturer shall supply suitable accessories for clamping the cable trays on cable tray supports from ceiling and beams of the building structure. Cable trays shall be supplied with GI coupler plates, hardware, nuts bolts and washers for joining the standard lengths of cable tray section.

e) **LOAD DEFLECTION CRITERIA**

Section and thickness of cable tray as per the manufacturer's standards.

Support span 1500 mm

Cable load for 300 mm	50	Kg / m
Cable load for 400 mm	75	Kg / m
Cable load for 500 mm	85	Kg / m
Cable load for 600 mm	100	Kg / m
Cable load for 750 mm	110	Kg / m
Cable load for 1000 mm	125	Kg / m

In addition to this 70 Kg/m concentrated load at the center span shall be considered. Allowable mid span deflection < 7 mm

f) GALVANIZING

Zinc confirming to grade Zn 98 of IS 209 shall be normally used for galvanizing. As far as practicable, the fabricated tray and accessories shall not be sunk to the bottom of the bath. Tray and accessories shall be galvanized at the lowest possible temperature, which will allow free drainage of zinc from the work piece during withdrawal. The thickness of the coating shall be minimum 100 microns. The zinc deposited shall not be less than approximately 720 gm / sq. m. The thickness shall be checked by a magnetic method as per IS 3203.

Small components handed in baskets shall be centrifuged to remove excess of Zinc immediately after galvanizing while the coating is still in molten condition. Hardware such as nuts and bolts shall be Electro galvanized / zinc passivated.

g) TESTING AND INSPECTION

All the cable trays and accessories shall be offered for inspection at the fabricators works by client / consultant representatives. Physical inspection and tests will be conducted on the trays to check its compliance with the specification and approved drawings.

Elko meter shall be used to measure the thickness of galvanizing. Deflection load test at the manufacturer's works Test span shall be simple beam span with free unrestrained ends.

The cable tray shall be uniformly loaded along the span with cable tray simply supported at the two ends as per the load criteria. Vertical deflection of the tray shall be measured at the two points along the midway between the supports. The average of these two readings shall be considered the vertical deflection of the tray. Vertical deflection shall be within the design criteria.

h) EARTHING

The cable tray shall be provided with earth clamps at every 3000mm for connecting suitable copper strip / wire earth continuity.

9. CABLES

a) Scope

The scope of this section covers the supply, laying, testing, commissioning, and handing over of Overhead/laying on cable tray and underground LT cables as specified in drawings/SLDs approved by Engineer-in-charge. CPWD specifications shall be followed for supply and laying of Medium Voltage XLPE cables.

Medium voltage cables shall be aluminum/copper conductor XLPE insulated, PVC sheathed armoured conforming to IS7098 part-1(1988) for LT cables and IS 7098 part-II (2011) for HT cable as amended up to date. LT Cables shall be rated for a 1100 Volts & HT cables shall be rated for a 11000 Volts.

All Conductor cables shall be as per Following.

Conductors shall be insulated with high quality XLPE base compound. A common covering (bedding) shall be applied over the laid-up cores by extruded sheath of unvulgarized compound. Armoring shall be applied below the outer sheath of PVC sheathing. The outer sheath shall bear the manufacturer's name and trade mark at every meter length. Cores shall be provided with the following colour scheme of PVC insulation.

3) 1 Core	4) Red/ Black/Yellow / Blue
5) 2 Core	6) Red and Black
7) 3 Core	8) Red/ Yellow & Blue
9) 3.5 /4 Core	10) Red/ Black/Yellow / Blue

LAYING

Cables shall be laid as per the specifications given below:

b) Duct system

Wherever are specified such as road crossing, entry to building or in paved area etc. cables shall be laid in underground ducts. The duct system shall consist of a required number of stone ware pipes, GI, CI or spun reinforced concrete pipe with simplex joints and all the jointing work shall be done according to the CPWD building specifications or as per the instructions of the Engineer-In-Charge as the case may be. The size of the pipe shall not be less than 100 mm in diameter for a single cable and shall not be less than 150 mm for more than one cable as on. The pipe shall be laid directly in ground without making any special bed but wherever asbestos cement pipes are used, the pipes shall be encased in concrete of 75 mm thick. The ducts shall be properly anchored to prevent any movement. The top surface of the cable ducts shall not be less than 60 cm. below the ground level. The ducts shall be laid a gradient of at least 1:300. The duct shall be provided manholes of adequate size at regular intervals for drawing the cables. The manhole cover and frame shall be of cast iron and machine finished to ensure a perfect joint. The manhole covers shall be installed flush with the ground or paved surfaces. The duct entry to the manholes shall be made leak proof with lead -wool joints. The ducts shall be properly plugged at the ends to prevent entry of water, rodents, etc.

Suitable duct markers shall be placed along the run of the cable ducts. The duct markers shall at least be 15cm. square embedded in concrete, indicating duct. Suitable cable supports made of angle iron shall be provided in the manholes for supporting the cables. Proper identification tags shall be provided for each cable in the manholes.

c) Cables in outdoor trenches

Cables shall be laid in outdoor trenches wherever called for. The depth & width of the trenches shall be as per cpwd specifications.

The cable shall be pulled over rollers in the trench steadily and uniformly without jerks and strains. The entire cable length shall as far as possible be paved off in one stretch. However, where this is not possible the remainder of the cable may be removed by "Flaking" i.e. by making one long loop in the reverse direction. After the cable has been uncoiled and laid into the trench over the rollers, the cable shall be lifted slightly over the rollers beginning from one end by helpers standing about 10 mtrs. Apart and drawn straight. The cable should then be taken off the rollers by additional helpers lifting the cable and then laid in a reasonably straight line and cable are to be fixed with suitable clamp.

For short cut runs and sizes up to 50 sq.mm of cables up to 1.1 KV grade any other suitable method of direct handling and laying can be adopted with the prior approval of the engineer-in-charge.

When the cable has been properly straightened, the cores are tested for continuity and insulation resistance. The ends of all cables shall be sealed immediately. In the case of PVC cables suitable moisture seal tape shall be used for this purpose.

Cable laid in ground in a single tier formation shall have a 8 cms cushion & covering of 17 cms above the base cushion of clean, dry sand before the protective cover is laid . In the case of vertical multi-tier formation after the first cable has been laid, a sand cushion of 30 cms shall be provided over the initial bed before the second tier is laid. If additional tiers are formed, each of the subsequent tiers also shall have a sand cushion of 30 cms as stated above. The topmost cable shall have final sand covering not less than 17 cms before the protective cover is laid.

Unless otherwise specified, the cables shall be protected by the second-class bricks of not less 20 cms x 10 cms x 10 cms (nominal size) protection covers placed on top of the sand (bricks to be laid breadth wise) for the full length of the cable. Where more than one cable is to be laid in the same trench, this protective covering shall cover all the cables and project at 5 cm over the sides of the end cables. The trenches shall be taken back filled with excavated earth free from stones or other sharp edge debris and shall be rammed and watered, if necessary, in successive layers not exceeding 30 cm, unless otherwise specified.

d) Route marker

LT Cable route marker marked "Cable" shall be provided along with the route of the cable and location of loops. The route markers shall be of tapered concrete slab of 60 x 60 cm at bottom and 50 x 50 cm at top having a thickness of 10cm. Cable marker shall be mounted parallel to and 50cm away from the edge of the trench. Distance of two route marker not more than 20 meters.

e) Cables in indoor trenches

Cables shall be laid in indoor trenches wherever specified. The trench shall be made as per drawing. The dimensions of the trenches shall be determined depending upon the maximum number of cables that is expected to be accommodated and can be conveniently laid. Cables shall be arranged in tier formation in trenches and cables are to be fixed with clamps. Suitable clamps, hooks, and saddles shall be used for securing the cables in position. Wherever specified, trenches shall be filled with fine sand and covered with RCC or steel Chequered trench covers.

f) Cable on Trays/ Racks

Cable shall be laid on cable trays / racks wherever specified. Cable racks/ trays shall be of ladder, trough, or channel design suitable for the purpose. The nominal depth of the trays/ racks shall be minimum 50 mm. The width of the trays shall be as per requirement. The trays' racks shall be completed with end plates, tees, elbows, risers, and all necessary hardware, entire steel trays' racks shall be hot dip galvanized including widths & accessories for external purposes and powder painted for internal purpose. Cable trays shall be erected properly to present a neat and clean appearance. Suitable cleats or saddles made of aluminum strips with PVC covering shall be used for securing the cables to the cable trays. The cable trays shall comply with the following requirements:

- i) The tray shall have suitable strength and rigidity to provide adequate support for all contained cables.
- ii) It shall not present sharp edges or projections injurious to the insulation of wiring/ cables.
- iii) If made of metal, it shall be adequately protected against corrosion or shall be made of corrosion-resistant material.
- iv) It shall have side rails or equivalent structural members.
- v) It shall include fittings or other suitable means for changes in direction and elevation of runs.

g) INSTALLATION

- h) Cable trays should be installed as a complete system. Trays shall be supported properly from the building structure. The entire cable tray system shall be rigid.
- i) Each cable tray run should be completed before the installation of cables.
- ii) In portions where additional protection is required, non-combustible covers/ enclosures shall be used.
- iii) Cable trays shall be exposed and accessible.
- iv) Where cables of different systems are installed on the same cable tray, non-combustible, solid barriers shall be used for segregating the cables.
- v) Cable trays shall be grounded by two nos, earth continuity wires. Cable trays shall not be used as equipment grounding conductors.
- vi) At no place should the cable tray/ rack/ ladder running horizontally rest on any building partition like Brick wall, RC C beams etc. but instead proper MS supports' hangers to be provided at minimum of 1500 mm intervals and at every Turning angle.

Jointing and terminations

Cable jointing shall be done as per the recommendations of the cable manufacturer. All jointing work shall be done only by qualified/ licensed cable jointer.

All jointing pits shall be of sufficient dimensions as to allow easy and comfortable working.

Jointing materials and accessories like conductors, ferrules, solders, flexes, insulation of gland protective tapes, filling compound, jointing box etc. of right quality and correct sizes, confirming to relevant Indian Standards.

Each termination shall be carried out using brass compression glands and cable sockets. Hydraulic crimping tool shall be used for making the end terminations. Cable glands shall be bonded to the earth by using suitable size copper wire/ tape.

The HT Cables shall be conforming to IS-7098 (Part-2) up to the latest amendments.

i) GENERAL

The cables shall be supplied, inspected, laid, tested, and commissioned in accordance with Drawings/SLDs approved by Engineer-in-charge. Specifications, Indian Standard Specifications as per latest amendments and cable manufacturer's instructions. The cables shall be of reputed make.

The recommendations of the cable manufacturer regarding jointing and sealing shall be strictly followed. The installation of cables shall be done by an approved, qualified, and experienced person in this trade

j) MATERIAL

The H.V. cables shall be 11KV, aluminum conductor cross linked polyethylene armored cable laid underground and or in masonry trenches as shown on Drawings. The conductor shall be made of Electrical purity aluminum wires and stranded together and compacted. The cable shall be of 3 Corex300 sqmm. The insulation shall be of high-quality cross-linked polyethylene applied by extrusion process. Both conductor and the insulator are provided with shielding made of semi Conducting compound. Armoring is applied over inner sheath and shall be of flat steel strips. The outer sheath shall be of heat resisting trapdoor (PVC) compound. This shall be of black color.

k) INSPECTION

All cables shall be inspected upon receipt at site and checked for any damage during transit.

l) JOINTS IN CABLES

The EPC Contractor shall take care to see that all the cables received at site are apportioned to various

locations in such a manner as to ensure maximum utilization and avoidance of jointing cable. This apportioning shall be got approved by the Engineer in charge before the cables are cut to lengths. Where joints are unavoidable, the location of such joints shall be got approved by the Engineer in charge.

m) JOINTING BOXES FOR CABLES

Cable joint boxes shall be of appropriate size, suitable for aluminum conductor XLPE insulated cables of 11000 volts ratings and shall be manufactured by make approved by Engineer-in-charge.

n) JOINTING CABLES

All cable joints shall be made in suitable, approved cable joint boxes. Jointing of cables in the joint boxes and the filling in of compound shall be done in accordance with the best practice in trade, in accordance with manufacturer's instructions and in an approved manner. All straight T-joints shall be done in epoxy mould boxes with TROPOLIN/M-SEAL epoxy resin or approved equal. All jointing accessories shall also be manufactured by approved make. All terminal ends of conductors shall be heavily soldered up to at least 50 mm length.

All cables shall be jointed color to color and tested for continuity and insulation resistance before jointing commences. The seals of cables must not be removed until preparations for jointing are completed. Joints shall be finished on the same day as commenced and sufficient protection for the weather shall be arranged. Joints shall be made by means of suitable solder for conductors, the conductors being firmly butted into the connections or thimbles or ferrules and the whole soldered with proper solder and soldering flux or resin. The conductors shall be efficiently insulated with high voltage insulating tape and use of spreaders of approved size and pattern. The joints shall be filled with epoxy compound being topped as necessary to ensure that the box is properly filled.

o) CABLE Termination

Cable termination shall be done in terminal cable box using cable glands and the cable ends sealed with sealing compound. The cable boxes of transformers shall be filled with bituminous compound or as per the direction of Engineer-in-charge.

p) BONDING OF CABLES

Where a cable enters any piece of apparatus, it shall be connected to the casing by means of an approved type of armored clamp and gland. The clamps must grip the armoring firmly to the gland or casing, so that in the event of ground movement no undue stress is passed into the cable conductors.

q) LAYING OF CABLES

H.V. cables shall be laid either buried directly underground or in Masonry/Concrete trenches. The

cable buried under ground shall be at minimum depth of 1.2 mtr. From the ground level. Sand cushion of not less than 80 mm shall be provided both above and below the cable with a protective concrete slab on the top of the sand layer. The cable trench shall be back filled and compacted.

r) PROTECTION OF CABLES:

The cable shall be protected by placing precast in forced 50mm, thick (1:2:4) concrete slabs 200mm wide on the top layer of sand for the length of the cable. Where more than one cable is running in the same trench, the concrete block shall cover all the cables and shall project minimum 80mm on either side of the cables.

Cables under road crossings and any surfaces subjected to heavy traffic shall be protected by running them through GI pipes of suitable size.

s) EXCAVATIONS AND BACK FILL

All excavations and back fill including timbering, shoring and pumping required for the installation of the cables shall be carried out by the EPC Contractor in accordance with the drawings and requirements laid down elsewhere.

Trenches shall be dug true to line and dreads. Back fill for trenches shall be filled in layers not exceeding 150 mm. Each layer shall be properly rammed and consolidated before laying the next layer. The EPC Contractor shall restore all surfaces, roadways, sidewalks curbs, walls or other works cut by excavation to their original condition, satisfactory to the Engineer in charge.

t) ROUTE MARKERS AND WARNING PLATES:

Approved C. I. cable markers shall be provided along the route of the cable at every 20 M Distance and at both ends of road crossing, indicating H.V. cables. Special C.I. markers shall be provided at all buried cable joints indicating Electrical cable joint.

u) TESTING CABLES:

Prior to burying cables, the following tests shall be conducted:

- i) Insulation between phases and between phase and earth for each length of cables, before and after jointing.
- ii) For H.V. cables, high voltage test by applying 17.5 KV DC voltage for 15 minutes for each core and earth.

On completion of cable laying work, the following tests shall be conducted in the presence of the Engineer in charge.

- i) Insulation Resistance test (sectional and overall)

- ii) Continuity resistance test.
- iii) Sheathing continuity test.
- iv) Earth test.

All tests shall be carried out in accordance with relevant Indian standard code of practice and Electricity Rules. The EPC Contractor shall provide necessary instruments, equipment, and staff for conducting the above tests and shall bear all expenses in connection with such tests.

10. BUS TRUNKING/RISING MAINS SYSTEM

SCOPE OF WORK

The scope of work involves Supply, Installation, Testing and Commissioning of 3 phase, 4 wire (100% neutral) with integral earth bus LV Compact Sandwich bus bar trunking of suitable rating as per design. All bus bar trunking products and fittings shall be in accordance with IEC 61439-6 or UL 857.

GENERAL

This specification is intended for design, manufacture, transporting, installation and testing at site of compact sandwich type Aluminum conductor LT Bus Bar Trunking system (Bus duct) suitable for indoor/outdoor installations as specified in the attached data sheet. All bus trunking products and fittings shall be in accordance with IEC 61439 – 6. The degree of protection of the Bus Bar Trunking system shall be IP55 for indoor & IP65 for outdoor in accordance to IEC 60529, and Shall have CE marking.

STANDARDS

Updated and current Indian Standard Specifications and Codes of Practice will apply to the equipment and the work covered by the scope of this contract. In addition, the relevant clauses of the following standards as amended up to date shall also apply. Wherever appropriate Indian Standards are not available, relevant British and/or IEC 61439-6 Standards shall be applicable.

STANDARDS	DESCRIPTION
IS8623(Part-1)1993	Low voltage switch gear and controller assemblies: Part-1 Requirement for type tested and partially type tested assemblies
IS8623(Part-2)1993	Low voltage switch gear and controller assemblies: Part- 2 Particular requirement for bus bar trunking system (busways)
IS8623(Part-3)1993	Low voltage switch gear and controller assemblies: Part-3 Particular requirement for equipment where unskilled persons have access for their use

IEC 61439-6	Low-voltage switch gear and control gear assemblies- Part 6: Bus bar trunking systems (busways)
IEC 61439-1	Low-voltage switch gear and control gear assemblies- Part 1: General rules
IEC 60529	Degrees of protection provided by enclosures (IP Code)
IEC 62262	Protection against Mechanical damage provided by enclosures (IK Code)
Indian Electricity Act 2003	
Indian Electricity Rules 1956	
National Building Code 1994	
National Electric Code 1985	
Code of Practice for Fire Safety of Building (general)	
IS 1641	General Principal and Fire Grading

BUSBARS

Bus bars shall be:

For up to 4000 Amps: Hard drawn silver/tinned (joints) Platted Aluminum Alloy Conductor with Halogen Free insulation.

Rated operating voltage of the bus bar trunking is 1000V, 3 Phase, 4 wire with integral earth. The neutral conductor should have the same cross-sectional area as the phase conductor. The whole bus bar trunking system shall be capable of withstanding the short circuit of the electrical installation without damaging the electrical, mechanical, and thermal stress under fault condition at a service voltage of 1000V, 50Hz. The minimum rated insulation voltage shall be 1000V. The impulse withstand voltage shall be minimum 12 kV. The bus bar, of full range and each rating should pass full type tests specified in IEC 61439-6:2012. The certificate shall be issued by an Indian or international independent testing authority (e.g. LOVAG or UL, ASTA Diamond).

HOUSING/ ENCLOSURE

- The bus bar trunking housing shall be constructed of electro galvanized steel and shall be provided with a suitable protective finish of epoxy paint.
- The enclosures of Bus Ducts shall be totally enclosed for protection against mechanical damage and tested for IK10 as per IEC 62262.
- The bus bar trunking housing shall be totally enclosed, non-ventilated for protection against water and dust accumulation with suitable IP rating as mentioned below.
- It should have passed minimum 1500 hours salt spray test to ensure the anticorrosion ability.
- The absolute temperature at any point of the bus bar trunking enclosure and at joints shall be strictly in conformation to IEC 61439-6.

- f) Enclosures shall be provided with riveted joints applied at intervals specified by the manufacturer along the entire length to eliminate possibility of theft. Ingress protection for the enclosures shall be minimum IP55 for indoor Bus duct as per IEC 60529. Ingress protection for the enclosures shall be minimum IP 65 (without considering any Canopy) for outdoor Bus ducts as per IEC 60529. However, suitable canopy to be provided to cover the outdoor busducts to avoid any mechanical damage by any accidental falling of foreign bodies.
- g) The enclosure of Bus Trunking System shall be fabricated from 1.5 mm thick CRCA sheet steel/GI/Extruded Al and shall be powder coated to suite the aesthetic ambience or as approved after regressive metal treatment process. Enclosure shall be rendered dust proof and vermin proof by adequate gasketing etc. to provide ingress protection.

JOINTS

The bus bar trunking joint shall be with Factory fitted Monobloc. The bolt shall be torque indicating and at earth potential. The bolt shall be two-headed design to indicate when proper torque has been applied and require only a standard long handle wrench to be properly activated. It should be possible to remove any joint connection assembly to allow electrical isolation or physical removal of a bus bar trunking length without disturbing adjacent bus bar trunking lengths. There shall be no bolts passing through the bus bars of the bus way as a part of safety.

INSULATION

Each bus bar shall be insulated with minimum Class F (155 Deg Centigrade) Polyethylene Terephthalate film (Multilayer PET) film with Saturated Triplex material of dry laid mat made of PET on both sides of PET film. Epoxy insulation shall not be permitted. Further, the insulation shall be certified as “halogen-free” by the manufacturer. All plastic components have a V1 Self-extinguishing degree as per UL 94.

FLANGE AND BOXES

- a) Flanged end box shall be provided to accommodate flange end for connecting the bus trunking with the flanges of panels, transformers etc., through flexible connections or as recommended by manufacturer.
- b) The flange end termination on panel should be guaranteed for Ingress protection as per requirement for indoor/outdoor application and should be backed by a type test report from the manufacturer.
- c) Phase matching of bus trunking with equipment shall be done prior to installation.

TAP-OFF BOXES/ PLUG-IN BOXES

- This shall be suitable for supplying three phase loads from 63A to 630A. The Tap Off Boxes

shall be available with Plug in type. Plug in type Tap off Boxes can be installed and removed when busbar is energized.

- The tap-off boxes are made 1.6mm sheet steel with powder coated and provision for fixing MCCB.
- **Tap-off boxes have an interlock with a cover and can be inserted and removed from the outlet only when the cover is in the ‘OPEN’ position that is in the isolated position.**
- All tap-off boxes shall have protective conductor which is the first to make an electrical connection when plugged into the outlet, and the last to disconnect when the tap-off is unplugged.
- All insulating plastic components comply with the incandescent wire test and have self-extinguishing properties.
- Operating voltage and Insulation voltage level must be 1000 Volts and suitable for 50/60 Hz frequency.

SUPPORTS

Both horizontal and vertical spacing between support structures shall be as per the recommendations of the manufacturer. Installation shall be done with all the required hot dipped galvanized iron supports as per the manufacturer’s recommendations.

EARTHING

The entire bus trunking system shall be provided with Cu / GI integral earthing, and with two independent earthing conductors of size, at least 25x5 (mm) throughout the length of the system. The earth flats shall be effectively connected to the enclosure by riveting /bolting. End covers shall be provided as required.

CERTIFICATES/DOCUMENTS TO BE ENCLOSED

- a) The type test certificates from international independent testing authority like LOVAG or UL, ASTA Diamond, KEMA KEUR, for the following are to be enclosed along with the technical bid failing which the offer shall be rejected:
1. Resistance / properties of insulating materials to abnormal heat.
 2. Resistance to flame propagation.
 3. Temperature rise limits
 4. Short Circuit Strength
 5. Effectiveness of the Protective Circuit
 6. Dielectric Properties.
 7. Clearances and creepage distances.
 8. Mechanical operation.

9. Structural strength
 10. Resistance to corrosion
 11. Thermal Stability of Enclosures.
 12. Mechanical impact test -IK 10.
 13. Verification of marking
 14. Ingress Protection.
- b) Test certificate / reports of the Class “F” insulating material from its manufacturer.
- c) Test certificate / reports of salt spray test carried out on the enclosure for a minimum period of 1500hours.

TESTING AT FACTORY

Routine tests like dimensional checks, electric continuity checks, IR test, dielectric withstand test shall be carried out at works / factory as per IEC 61439-6 in the presence of department engineers in order to qualify the product. All routine test reports are to be submitted to the engineers visiting the factory for inspection. The product shall be dispatched from the factory only after obtaining necessary clearances from the department engineers.

INSTALLATION, TESTING AND COMMISSIONING AT SITE

- a) The panel board shall be supplied in properly packed conditions. After ascertaining that there is no damage to the packing, all the items shall be inspected after unpacking. It is the responsibility of the vendor to inspect all the materials for damage, if any, after unloading at site. Installation shall be done with required supports, joints etc. strictly as per the recommendations of the manufacture and IEC standards. Care should be taken by the vendor to ensure that no damage is caused either to the product or the technicians who are deployed for the installation work.
- b) Department shall not be responsible for any such losses that may occur during installation, testing and commissioning works.
- c) All the required commissioning tests shall be carried out in the presence of department engineers.
- d) Installation, Testing & Commissioning shall be done strictly by the manufacturer personnel only. The system shall be commissioned only after the manufacturer's engineer qualifies that the product has been installed and tested as per the manufacturer's recommendations and in conformation with IEC 61439-6-2012.
- e) All the required operating and maintenance manuals, as-built layout of the bus way along with complete details of all the components of the system, necessary protocols to be followed during maintenance etc. shall be handed over to the Department.

GENERAL TERMS

- a) The agency shall submit the fabrication drawings / bus trunk layout drawings to the department for approval. On approval agency shall proceed for fabrication.
- b) All test facilities required shall be provided for the tests prescribed in the schedule by the party without any additional cost at their factory. The tests shall be carried out in the presence of CLIP inspection team before dispatch from their works.
- c) The bus trunk system shall be guaranteed for a period of 12 months from the date of commissioning. The agency shall repair / replace the damaged components without any extra cost during the guarantee period.
- d) The agency shall provide three complete sets of operation and maintenance manuals and test certificates for the operation and maintenance purpose of the department.
- e) The department shall reserve the right to accept / reject any part of work, which is not acceptable as healthy engineering practice of carrying out such work. In case of any failure to carry out the scope of work in time as per the instruction of department, the work shall be carried out / got done by the department at the cost and risk of the agency and the entire amount with supervision charges shall be recovered from the agency.
- f) Payments shall be made based on the measurements taken at site. Payment shall be done only on pro-rata basis upon certification of measurement by the Department Engineer.

TECHNICAL PARAMETERS OF AL SANDWICH BUS TRUNKING

S. No	Description/Parameter	Specifications
1	Make & Model of the bus way	As approved by Engineer In-charge
2	No of Phases	3phase, 4 wire with 100% Neutral and integral earthing
3	Conductor Material	Aluminum
4	Insulation Material	Polyethylene Terephthalate film (PET)
5	Class of Insulation	Class F (155 Deg C) Polyethylene Terephthalate film (PET)
6	Rated Insulation Voltage	1000V
7	Rated Impulse Withstand	12 kV
8	Rated Operation Voltage	415V
9	Rated Operation Current	630A-4000A
10	Rated Operation current of Neutral Conductor	100% of Phase Current

11	Rated Symmetrical Short Circuit Withstand Current for 1 Sec	630- 50kA 800-42KA 1000-50kA 1250-75kA 1600- 80kA 2000- 80kA 2500- 150kA 3200- 160kA 4000A-160kA
12	Integral/External Earth	Yes
13	Material of External Earth Conductor	GI 25 X 5MM
14	Integral Earth Conductor	Integral Earthing required
15	Maximum Absolute Temperature at any point on the Enclosure	To be mentioned by the vendor
16	Maximum Absolute Temperature rise at joints	To be mentioned by the vendor
17	Ingress Protection (without considering canopy)	IP 55 for Indoor, IP65 for Outdoor
18	1500 hr Salt spray test as per IEC 61439-6	Mandatory, necessary documents to be submitted with the offer
19	Seismic Zone certificate	Suitable up to Zone 4
20	Type Test certificate for the product from international laboratories like /LOVCGE/UL, ASTA Diamond. Class F insulation test certificate / report from the insulation	It is mandatory to enclose the documents
	Manufacturer.	
21	Certification Standard	IEC 61439-6-2012
23	Acceptance to inspection, installation, and testing commissioning clauses and other general terms and conditions.	As per Points 10,11,12&13
24	Mechanical Strength,	IK10

11. EARTHING

The work shall be done in general as per CPWD Specifications 2019, Volume-I & II with latest correction slip (if any).

11.1 GENEAL

This section covers the general arrangement of the earthing, i.e. all non-current carrying the electrical installation shall be earthed as per latest IS code and general specifications for electrical works (part -1, internal) of CPWD specifications.

All Metal conduits, trunking, cables trays, switchgear, distribution boards, meters, light fixtures, fans, and all other metal parts forming part of the work shall be bonded together and connected by two separate and distinct conductors to earth electrodes. Earthing shall also be in conformity with the provisions of Rule 32, 61, 62, 67 and 88 of IER 1956. The earth electrode shall not be situated less than 1.5 mtr. From any building and minimum 2 mtr. Between earth electrodes.

11.2 EARTHING SYSTEM

It shall comprise of earth electrodes, earth strips, earth continuity conductor and all Earthing conductors shall be of high conductivity copper, GI or aluminum and shall be protected against mechanical damage and corrosion. The size of earth conductors shall not be less than half that of the largest current carrying conductor. The connection of earth continuity conductors of earth bus and earth electrodes shall be strong and sound and shall be rigidly fixed to the walls, cable trenches, cable trays or conduits and cable by using suitable clamps made of nonferrous metals.

11.3 EARTHING ELECTRODES

Earthing electrodes shall be designed as per the requirement of latest IS codes. The number and size of earth electrodes shall be calculated so that under fault conditions no electrode is loaded above its maximum permissible current density. The resistance of earth electrode shall be as low as possible, the maximum allowable value being one ohm.

Earthing electrodes of plate type may be adopted. The choice of plate or pipe electrode shall be decided according to the anticipated fault level of the network and local soil conditions. Generally, plate electrodes shall be used for substations and large medium voltage network and pipe electrodes for small medium voltage network and installations.

11.4 LOCATION OF EARTH ELECTRODES

Normally an earth electrode shall not be situated less than 1.5 m from any building. Care shall be taken that the excavation for earth electrode may not affect the column footings or foundation of the buildings. In such cases electrodes may be further away from the building.

The location of the earth electrode will be such where the soil has reasonable chance of remaining moist. As far as possible, entrances, pavements, and road ways, are to be avoided for locating earth electrode.

11.5 PLATE ELECTRODE

Plate electrodes shall be made of GI plate of 6 mm thick and 60x60 cm. size OR 3 mm THICK AND 60*60 cm size. The plate shall be buried vertically in ground at depth of not less than 3.5 meters to the top of the plate, the plate being encased in charcoal to a thickness of 15 cm. all around. It is preferable to bury the electrode to a depth where sub -soil water is present. Earth leads to the electrode shall be laid in a GI pipe and connected to the plate electrode with GI bolts, nuts, and washers. A GI pipe of not less than 19 mm dia shall be placed vertically over the plate and terminated in a funnel at 5 cm. below ground. The funnel shall be provided with a wire mesh. The funnel shall be enclosed in masonry chamber of 30 x 30 x 30 cm dimensions. The chamber shall be provided with lockable CI frame cover of 30 x 30 cm size. The earth station shall also be provided with a suitable permanent identification label/ tag.

Note: If copper plate is used it shall be of 3mm thickness.

Pipe electrode shall comprise of a 4.5 Mtr. long 40 mm dia GI pipe buried vertically in a pit of 35 x 35 cm size and filled with alternate layers of charcoal, salt and river sand and connected at the top to a GI pipe of 19 mm with a funnel at the other end, 5 c m above the ground. The earth lead shall be properly fixed to the pipe electrode with brass bolts, nuts and washers. The funnel and earth lead connections shall be enclosed in a masonry chamber of 30 x 30 x 30 cm. dimensions. The chamber shall be provided with a lockable CI frame and CI cover. Proper permanent identification tag/ label shall be provided for each electrode.

11.6 INSTALLATION

All joints shall be reverted and sweated. Joints in the earth bar shall be bolted and the joints faces tinned. Where the diameter of the bolt for connecting earth bar to apparatus exceeds one quarter of the width of the earth bar, the connection to the bolt shall be made with a wider piece of flange of copper jointed to earth bar. These shall be tinned at the point of connection and special care taken to ensure a permanent low resistance contact to iron or steel. All steel bolts, nuts, washers, etc. shall be cadmium plated. Main earth bars shall be spaced sufficiently away from the surface to which they are fixed, such as walls or the side of trenches to allow for easy connections. Copper earth bars shall not be fixed by ferrous fittings. The earthing shall be suitably protected from mechanical injury by galvanized iron within ground shall be buried at least 60 c m deep. The earthing lead shall be securely bolted and soldered to the plate or pipe as the case may be. In the case of the plate, the lead shall be connected by means of cable socket with two bolts and nuts. All washers shall be of the same materials as the plate or pipe. All iron bolts, nuts and washers shall be galvanized.

11.7 Method of Installation of watering arrangement

In the case of plate earth electrode, a watering pipe of 20 mm dia of medium class GI pipe shall be provided and attached to the electrode. A funnel with mesh shall be provided on the top for watering

the pit. In case of pipe earth electrode, a 40 mm x 20 mm reducer shall be used for accessing the funnel. The watering funnel attachment shall be housed in masonry enclosure of not less than 30 cm x 30 cm x 30 cm. A cast iron cover having locking arrangement shall be suitably embedded in the masonry enclosure.

11.8 PRECAUTIONS

Earthing system shall be mechanically robust and the joints shall be capable of retaining low resistance even after passages of fault currents.

Joints shall be soldered, tinned and double riveted. All the joints shall be mechanically and electrically continuous and effective. Joints shall be provided protection against corrosion.

The earthing lead from electrode onwards shall be suitably protected from mechanical injury by a 15 mm dia GI pipe in case of wire and by 40 mm dia medium class GI pipe in case of strips. Portion of this protection pipe within the ground shall be buried at least 30 cm deep (to be increased to 60 cm in case of road crossing and pavements). The portion within the building shall be recessed in walls and floor to adequate depth.

11.9 TESTING

On the completion of the entire installation, the following tests shall be conducted and no earth electrode shall have ohmic resistance of more than 2 ohm and in rocky soil not more than 3 ohms.

- a. Earth resistance of electrodes
- b. Impedance of earth continuity conductors as per IEE regulations.
- c. Effectiveness of earthing as per IEE regulations.

All meters, instruments and staff required for the tests shall be provided by the EPC Contractor.

Earthing system comprising of Copper/ GI (as reqd. as per CPWD specification) earthelectrode, earth conductor, earth bus, protective conductor etc. for each building, Equipment, structures and system shall be as per provision laid down in CPWD specifications Part-I-2023. Earthing system should be designed such as to maintain earth resistance as specified in CPWD specifications. Earth resistance shall be checked/tested in harsh climatic conditions.

Metallic body of all the equipment shall be earthed confirming CPWD specifications for Electrical works (Part-I), Internal 2023 corrected up to date.

Armouring of cables shall be connected to equipment and panel at both the ends.

All the light and fan points should be properly earthed with 1.5 sq.mm FRLSH PVC insulated copper wire.

After completing the work, necessary test results as envisaged in CPWD General Specifications Part-I (Internal)-2023, CPWD General Specifications Part-II (External)-2023 & Indian Electricity Rules 2005, shall be recorded and submitted to the department. The results shall be within the permissible limits.

12. LIGHTNING PROTECTION SYSTEM

a) STANDARDS

The following Indian Standard specifications and Codes of Practice will apply to the equipment and the work covered by the scope of the contract. In addition, the relevant clauses of the Indian Electricity Act 1910 and Indian Electricity Rules 1956 as amended up to date shall apply. Wherever appropriate Indian Standards are not available relevant British and/ or IEC Standards shall be applicable.

BIS certified equipment shall be used as a part of the contract in line with Government regulations. Necessary test certificates in support of the certification shall be submitted prior to supply of the equipment.

It is to be noted that updated and current Standards shall be applicable irrespective of those listed below.

Code of Practice for the protection of buildings and allied structures against Lightning IS 2309: 1989, IS/ IEC 62305 Code of Practice for Earthing IS 3043: 1987.

b) GENERAL

The Lightning Protective System shall comprise of Air Terminations, Down Conductors, Earth Terminations etc. as required. The System shall preferably use the same conducting material throughout and will comply to the detailed specifications detailed here in after.

The entire lightning system should be mechanically strong to withstand the mechanical force produced in case of a lightning stroke.

c) MATERIALS

The materials of which the protective system is composed shall be resistant to corrosion or be adequately protected against corrosion. The material shall be as specified as below and shall comply to the following requirements:

- i) **Copper** - When solid or stranded copper wire is used it shall be of the grade ordinarily required for commercial electrical work generally designated as being of 98% conductivity when annealed, conforming to Indian Standard specifications.
- ii) **Galvanized Steel** - Galvanized steel used shall be thoroughly protected against corrosion by hot dipped Zinc coating. The material coating shall with stand the test specified in 182309:1968.
- iii) The strips to be used shall be in maximum lengths available as manufactured normally avoiding unnecessary joints.

d) AIR Termination

- i) **Vertical Air Termination:** Vertical air terminations shall comprise of finials made of 25 mm dia. GI tube with multiple prongs at the top. Vertical terminations where provided shall project 30cms above the project salient point or network on which it is fixed.
- ii) **Horizontal Air Terminations:** Horizontal air terminations should be so interconnected that no part of the roof is more than 9.0 meter away from the nearest horizontal conductor. For a flat roof horizontal air termination along the outer perimeter of the roof is to be used. For a roof of larger area, a network of parallel horizontal conductors shall be installed. Horizontal air terminations should be coursed along contours such as ridges, parapets and edges of the flat roofs and where necessary over flat surfaces in such away as to join each air termination to the rest and should themselves form a closed network.

All metallic finials, chimneys, duct, vent pipes, railings, gutters, and the like on or above the main surface of the roof of the structure should be bonded to and form part of the air termination network.

- iii) **DOWN CONDUCTORS:** Conductors shall be of material as specified in the Specifications. These shall be distributed around the outside walls of the structure and shall preferable be run along the corners and other projections. Elevator (Lift) shafts shall not be used for fixing the down Conductors. The routing of the down Conductors shall be such that it is accessible for inspection, testing and maintenance.

- e) **TESTING JOINTS AND BENDS:** The lightning protective system should have as few joints in it as possible. Wherever joints in the down conductor above ground level are necessary they shall be mechanically and electrically effective.

In the down conductor below ground level there shall be no joints.

The joints may be clamped, screwed, bolted, riveted, sweated brazed or welded. Bolted joints should be used on test points or on bonds to existing metal.

Each down conductor should be provided with a testing joint in a position convenient for testing but in accessible.

- f) **FASTENERS:** Conductors shall be securely attached to the building by fasteners which shall be substantial in construction, not subject to breakage. These shall be of galvanized steel or

other suitable materials with suitable precautions to avoid corrosion. The method and nature of the fixing should be simple, solid and permanent. The lightning conductors shall be secured at not more than 1.20 m apart for horizontal run and 1.00 m for vertical run.

- g) **EARTH TERMINATION:** Each down conductor shall have an independent earth termination and all earth terminations should be interconnected.
- h) **EARTH ELECTRODES:** Earth electrodes shall be constructed and installed as laid down in the IS 3043.
- i) **Plate earth Electrode:** The plate electrodes shall be of Copper or G.I. as per requirement. The minimum dimensions of the electrode shall be G.I. 600 mm x 600mm x 6 mm thick and for copper 600 mm x 600 mm x 3 mm thick. The electrode shall be buried in ground with its face vertical and top not less than 3 m belowground level.
- ii) **Earth Electrode Pit:** In the case of plate earth electrode, a watering pipe of 20 mm dia of medium class G.I. Pipe shall be provided and attached to the electrode. A funnel with mesh shall be provided at the top of this pipe for watering the earth. The watering funnel attachment shall be covered in masonry enclosure of not less than 300 x 300 x 300 mm. A cast iron/ M.S. frame with cover having locking arrangement shall be suitably embedded in the masonry enclosure.

i) Location of Earth Electrode

The following guidelines shall be followed for locating the earth electrodes

- a) An earth electrode shall not be situated less than 2 meters from any building.
- b) The excavations for electrode shall not affect the column footings or foundations of the buildings. In such cases electrode may be further away from the building.
- c) The location of the earth electrode shall be such where the soil has reasonable chance of remaining moist, as far as possible.
- d) Entrances, pavements, and road ways shall not be used for locating the earth electrode.
- e) Earth Pits shall be painted with colour code as per IS standard mentioning individual resistance value, combined resistance value, testing date and next testing date etc.

j) EARTH RESISTANCE

The whole of the lightning protective system should have a combined resistance to earth not exceeding 5 ohms before any bonding has been affected to metal or on a surface or to surface below ground.

13. ELECTRICAL VEHICLE CHARGING

Supply, installation, testing & commissioning of EV charging power socket as per drawings approved by Engineer-In-Charge shall be provided by EPC Contractor for charging of EV vehicle in BQ Housing Complex, Duliajan, Assam.

Technical specifications of EV charger are placed as under:

(A) Technical Specification 7.4kw AC Type 2 Charger		
1	AC Supply System	230V (+-15%), 1 Phase 3 Wire
2	Input Frequency	50Hz ($\pm 3\%$)
4	Output Voltage AC	230V (+-15%)
5	Output Current	32 A max Output current limit is configurable from 6A to 32A through OCPP remotely
6	Output Power	7.4 kW max
7	Output connector compatibility	IEC 61851-1 / IEC 62196 – Type 2 AC EV connector
8	Power Lines	L1, N, PE
9	Control Lines	CP, PP
10	Length of charging cable	5 Meters
11	Vehicle Connector protective cap	Yes
12	EMI/EMC for EVSE	Compliance as per IEC 61851-21-2
13	Error codes	Defined in user manual
14	Overvoltage protection	Above 275V AC, Protection with auto recovery
15	Overcurrent protection	Software controlled protection enabled in the charger. External MCB to be installed externally during installation.
16	Leakage current protection	Type A & Type B RCD protection 30mA AC and 6mA DC current detection
17	Continuous Earth wire monitoring	Yes
18	CP diode detection	Yes
20	Insulation Resistance	Greater than 1 M Ohm

21	Dielectric Withstand Voltage	Test voltage of 1615V AC compliance as per IEC 60664-1
22	EFT	+/- 4kV
23	Surge protection	Differential mode: +/- 2kV Common mode: +/- 4kV
24	ESD	Air discharge: +/- 8kV Contact discharge: +/- 4kV
25	Operating temperature	-35C to +55C
26	Storage temperature	-40C to +80C
27	Certified energy meter	Class 1 certified
28	RS485 Port	Yes
29	Dynamic Power Management capability	Yes
30	Energy Management Services	Yes
31	Power Backup	Optional
33	Cooling	Air cooled
34	Mechanical Strength	IK08 according to IEC62262
35	IP Rating	IP 65
36	Dimension (mm)	390*277*110
37	Weight (Kg)	8
38	Mounting	Wall mount / Pole mount
40	Visual Indicators	LED panel
41	Messages displayed LED	Vehicle plugged in / plugged out Charger available/charging Fault conditions
42	Emergency Switch	Optional
43	User authentication	RFID/ Mobile app
44	Connectivity options	Wi-Fi, BLE and SIM (Optional)
45	SD card storage	Yes
46	OCPP	OCPP 1.6J
47	Backoffice Connectivity	Yes
49	Standards / Certification	IEC62262, IEC 60664-1, IEC61851-1, IEC62196, IEC 61851-21-2, ARAI certified as per IS17017 and AIS038 Third Party CE Certified, EMC Class B

(A) Technical Specification 22kw AC Type 2 Charger		
1	AC Supply System	415V (+-15%), 3 Phase 5 Wire
2	Input Frequency	50Hz (±3%)
3	Output Parameters	
4	Output Voltage AC	415V (+-15%)

5	Output Current	32 A max Output current limit is configurable from 6A to 32A through OCPP remotely
6	Output Power	22 kW max
7	Output connector compatibility	IEC 61851-1 / IEC 62196 – Type 2 AC EV connector
8	Power Lines	L1, L2, L3, N, PE
9	Control Lines	CP, PP
10	Length of charging cable	5 Meters
11	Vehicle Connector protective cap	Yes
12	EMI/EMC for EVSE	Compliance as per IEC 61851-21-2
13	Error codes	Defined in user manual
14	Overvoltage protection	Above 275V AC, Protection with auto recovery
15	Overcurrent protection	Software controlled protection enabled in the charger. External MCB to be installed externally during installation.
16	Leakage current protection	Type A & Type B RCD protection 30mA AC and 6mA DC current detection
17	Continuous Earth wire monitoring	Yes
18	CP diode detection	Yes
19	General	
20	Insulation Resistance	Greater than 1 M Ohm
21	Dielectric Withstand Voltage	Test voltage of 1615V AC compliance as per IEC 60664-1
22	EFT	+/- 4kV
23	Surge protection	Differential mode: +/- 3kV Common mode: +/- 6kV
24	ESD	Air discharge: +/- 8kV Contact discharge: +/- 4kV
25	Operating temperature	-35C to +55C
26	Storage temperature	-40C to +80C
27	Certified energy meter	Class 1 certified
28	RS485 Port	Yes
29	Dynamic Power Management capability	Yes
30	Energy Management Services	Yes
31	Power Backup	Optional
32	Mechanical	
33	Cooling	Air cooled
34	Mechanical Strength	IK08 according to IEC62262
35	IP Rating	IP 65
36	Dimension (mm)	390*277*110
37	Weight (Kg)	10

38	Mounting	Wall mount / Pole mount
39	User Interface and Software	
40	Visual Indicators	LED panel
41	Messages displayed LED	Vehicle plugged in / plugged out Charger available/charging Fault conditions
42	Emergency Switch	Optional
43	User authentication	RFID/ Mobile app
44	Connectivity options	Wi-Fi, BLE, SIM, Ethernet
45	SD card storage	Yes
46	OCPP	OCPP 1.6J, Future Upgradable to OCPP 2.0.1
47	Backoffice Connectivity	Yes
48	Safety Standards & Certifications	
49	Standards / Certification	IEC62262, IEC 60664-1, IEC61851-1, IEC62196, IEC 61851-21-2, ARAI certified as per IS17017 and AIS038 Third Party CE Certified, EMC Class B

14 Three phase four wire smart energy meter of accuracy class 1.0 with communication facility and remote connect/ disconnect switch.

Scope of Work

These specifications cover the design, manufacturing, supply, installation, testing and commissioning of three phase, 4 wires Smart Energy Meter with bidirectional communication facility, Meter should work on prepaid / Postpaid mode & remote connect/disconnect switch including business Desktop PC Latest Version & Printer. The meter shall communicate with Head End System (HES) on any one of the communication technologies mentioned in IS16444 Part 1 or latest, as per the requirement. Three Phase Smart energy meters installed with Polycarbonate Box.

The technical parameters of three phase energy meter are given below:

Sr.No.	Particulars	Meter Specification
1.	Connection type	3-phase 4-wire, direct connected
2.	Applicable standards	Meter will comply to IS:16444, IS-15884, IS-13779 & I S-15959 as amended up to date.
3.	Reference Voltage	240 V L-N, +/- 30% of Vref
4.	Current rating	10-100 A
5.	Accuracy	Class 1.0
6.	Frequency	50 Hz, $\pm 5\%$
7.	Metrology LED	Two LED for active and reactive energy
8.	Switch category	UC1 or better
9	Burden	
a	Voltage circuit	As per IS 16444
b	Current circuit	As per IS 16444

10.	Parameters to be Measured per phase	Instantaneous -V, I, kW, Power Factor and Cumulative Active Energy.
11.	Enclosure material	
a	Meter case, terminal block	Engineering Plastic
b	Sealing	Break to open enclosure, Additional sealing provision on meter body, ETBC and communication module.
12.	Environmental	
a	Ingress protection	IP 54
b	Insulation class	Protective class II
c	Operating temperature	-10 °C to +60 °C
d	Storage temperature	-25 °C to +70 °C
e	Humidity	95%, non-condensing
13.	Warranty	10 years.

BASIC FEATURES: The Smart Meter would have the following minimum basic features placed as under:

- Bidirectional Communication.
- Integrated Load limiting /connect/disconnect switch with performance equivalent to UC1 or better.
- Tamper event detection, recording and reporting.
- Power event alarms as per IS 16444 Part 1.
- Remote firmware upgrade.
- The meter should have dust proof capacitive touch keypad on meter body.
- Meter should have BLE compatibility for offline reading in case of failure of WAN.
- Pre-paid features as well as postpaid features (Configurable).
- TOD (Time of Day/Time of Use) feature.
- On demand reading through head end system (HES).
- Quality of supply (QoS) monitoring.
- Intuitive, legible, and user-friendly display.
- 24x7 customer support in toll free number for customer support for each supplied meter & reconfiguration of the replacement of meters in case of any defect.
- Energy meter with IP54 enclosure and mobile app with all accessories.

GENERAL STANDARDS APPLICABLE FOR METERS

Sr. No.	Standard	Title
1	IS 13779 with latest amendments	AC Static Watt-hour Meter class 1& 2

2	IS 15884 with latest amendments	Alternating Current Direct Connected Static Prepayment Meters for Active
3	IS 16444 Part 1 with latest amendments	A.C. Static Direct Connected Watt Hour Smart Meter Class 1 and 2- Specification
4	IS 15959 Part 1 & Part 2 with latest amendments	Data Exchange for Electricity Meter Reading, Tariff and Load Control- Companion Standards

COMMUNICATION - Meter shall have the ability to communicate with Head End System (HES) on any one of the communications technologies mentioned in IS16444 Part 1 in a secure manner. In case of Plug in type communication module, the meter shall log communication module removal /non responsive event with snapshot.

REMOTE CONNECT/DISCONNECT/LOAD LIMITING: Remote Connect/disconnect/Load control facilities would be as per IS 16444 part 1.

OTHER SPECIFICATIONS:

Online Monitoring and control: Meter should have feature to send and receive data from software anywhere.

Alarm: Mobile should have provision to provide to show notification of load and balance.

Anti-Temper features: Will be provided Phase miss, 35KV spark discharge, Magnetic tamper, Top cover open, Neutral disturbance as per CEA Regulation.

Overload/Overcurrent condition: The meter shall disconnect the load in case of overload/overcurrent condition. The maximum load/power at which the load is disconnected shall be set by the user. Overload and Maximum current recorded shall be logged with date time. The meter shall be reset once the overload/overcurrent condition is nullified.

Prepaid/Postpaid Feature: Meter will be configured for postpaid operation along with overload Disconnection feature. Meter shall have provision for configuring as prepaid meter also if required in future.

Data retention: Non-volatile memory (non-battery backed up) with data retention in absence of power.

Meter Housing: High grade engineering plastic, IP54, 2 nos. provision for sealing meter cover to meter base, 1 nos. provision for sealing terminal cover.

Communication: Meter should have communication module to perform low-cost long-range communication for period of 10 years.

End to End Solution: Meter, software, mobile app should be single make for hassle free operation.

Test Report - type test report from Govt Lab/NABL/ ILAC accredited lab. Make as approved by Engineer-in-charge.

The following item is also included in the scope of EPC contractor

Sr. No.	Item	Description	Quantity	Remarks / specification
1	Desktop PC	Compatible desktop for running vending software system.	1	P-4 dual core, 8-GB RAM, hard disk 500 GB, 1-CD/DVD combo drive R/W type, 4-USB Port including 15" monitor of same make, video card, keyboard, cordless mouse.
3	Printer	For printing reports	1	Printer with 12 pages per minute capacity, 600 x 600 dpi with HP, REt, 1200 dpi effective output quality (600 x 600 x 2 dpi with REt, with duty cycle 3,000 single-sided pages per month for report printing complete as required
4	UPS	For power backup in case of an outage	1	1KVA capacity UPS
5	Internet connectivity	For accessing web-based vending system	1	Stable internet connectivity with a broad bandwidth recommended.

Special Technical Criteria

Warranty: The contractor should provide warranty of 10 years from month of supply for Energy meters supplied. Contractor must submit the necessary supporting documents from OEM regarding the warranty period of 10 years from month of supply.

Data exchange protocol from smart meter to HES shall be based on open protocol like TCP/IP or DLMS/COSEM.

OEM should provide their own software for recharging of prepaid meters. The OEM's application software should be user friendly.

OEM should have Valid MTCTE certificate for quoted smart meter issued by Dept of Telecommunication & ISO 27001 certifications (Cyber Audited CERT security

The Smart Meter OEM should have in-house NABL accredited lab.

All critical hardware and software components of the smart energy meter (IO Module/MFM/Gateways) should be manufactured/developed by the same Indian OEM. Further, in case of use of imported or components of any other manufacturers, the contractor shall take full responsibility for repair/replacement/service for such components.

Scope includes providing Business Desktop PC Latest Version & Printer also provided by EPC contractor

OEM Should be provided Make in India Certificate

This item includes SIM Charges for three-years and support for the hardware, software, software hosting, network monitoring and maintenance for three phase smart energy meters. This would start from the First year and would include App facility, Tariff design, 24*7 customer support in toll free number for Customer support for each supplied meter & reconfiguration of the replacement of meters

in case of any defect. After completion of the two years, the charges for SIM/portal/maintained will be borne by end user on yearly basis.

The EPC contractor provide the smart Energy meter to each unit of residential Quarters. The detail of Energy meter placed as under:

Sr. No.	Description	Qty	Unit
A			
1.	Supply of Three Phase dual mode, Prepayment meter with Polycarbonate box and following specifications: • AC Three Phase, Four Wire, Tamper Proof, Static Watt-Hour Meter of Accuracy Class 1.0, 240 V, 50 Hz, current 10 A-100A, conforming to relevant Indian Standards like IS-13779, IS-15884, IS-16444, IS-15959 or relevant IEC standards with dual mode (Credit/Prepay) support, Advanced Revenue Protection Features, In-build billing data calculation, Microprocessor and LED based circuitry With 10 years' product manufacturer warranty. Meter should – • Have load limiting switch with the load setting configurable remotely or through app wirelessly over BLE (Bluetooth low energy) • Have Bidirectional communication module to accommodate NBIoT cellular communication technology for remote connection disconnection • Support switching (Credit to Prepay and Prepay to Credit) through authenticated command remotely and also wirelessly over BLE technology Model: Liberty 500/Equivalent	192	Nos.
2.	Installation & Commissioning of Three Phase Smart meters with Polycarbonate Box	192	Nos
3.	Hosting and Configuration of the Web based Software with following modules: • Consumer and Meter Management Module- for creating and managing the consumer information • Tariff management module- for define and manage the tariffs.	192	Nos

	- Consumer App- to view energy data and billing data on mobile application android and IOS apps. - Training to vending System Operator, training of consumer Tariff Designing, Customer Database definition, Web based Vending system etc.		
4.	Annual subscription charges for vending transactions and SIM card (for 2 years). Scope includes providing SIM and Cloud Server for data management along with Smart meters. Note: after 2 years the subscription needs to be renewed.	384	Nos

15. EMERGENCY LIGHT & ILLUMINATED SIGNAGE

The works shall include Design, supplying, installation, testing & commissioning of emergency lighting system and **Illuminated Signage for residential buildings block**, community center & guest house etc. as required.

1.0. Signages

- The size and design of signages required for the buildings shall be as per NBC 2016.
- The size, colour and formatting shall be as per IS: 9457 and IS: 12349
- The fixing of Signages should be with SS screws drilled wherever required with clean finish.
- Earth Pits shall be painted with colour code as per IS standard mentioning individual resistance value, combined resistance value, testing date and next testing date etc.
- The minimum and indicative quantity requirement of building for signage is as follows:**

S. No	Description	Location
1.	SITC and design of Fire Evacuation Route Plan with location of Fire Extinguishers, MCP, Hooter, First Aid Box, Safe Assembly Area & other information providing as per NBC requirements, mounting display board 5mm thick PVC board Auto Glow sheets and transparency sheets of size and layout as per IS: 9457 with details of Fire Escape Plan in floor layout complete of each building respectively as required and clearly indicate "YOU are HERE"	Shall be placed at every floor of the building at common areas, Elevator (Lift) lobby, corridors and fire towers as per the requirements of NBC-2016.
2.	Supply and fixing of 5mm thick PVC board Auto Glow sheets and transparency sheets of size and layout as per IS: 9457 with display of "Exit staircase" Sign Boards indication as per NBC complete etc. as required.	At every landing of stair cases and fire tower.

3.	Supply and fixing of 5mm thick PVC board Auto Glow sheets and transparency sheets of size and layout as per IS: 9457 with "Fire Alarm Call point" indication as per NBC complete etc. as required.	At each Fire Alarm call point placed.
4.	Supply and fixing of 5mm thick PVC board Auto Glow sheets and transparency sheets of readable size and layout as per IS: 9457 with "Refuge Area" indication as per NBC complete etc. as required.	At each refuge Area
5.	Supply and fixing of 5mm thick PVC board Auto Glow sheets and transparency sheets of readable size and layout as per IS: 9457 with "In case of Fire, Do not use lift" indication as per NBC complete etc. as required.	At every Lift lobby at every landing.
6.	Supply and fixing of 5mm thick PVC board Auto Glow sheets and transparency sheets of size and layout as per IS: 9457 with "Fire Hose Reel" indication as per NBC complete etc. as required.	At every landing of Fire shafts, above the fire shaft door.
7.	Supply and fixing of CCTV surveillance board of size 12"x9", with indication of "You are under CCTV surveillance" with auto glow and HP latex vinyl print and acrylic lamination on both sides, to be at required locations as per NBC complete etc. as required.	At all points where indoor and outdoor CCTV cameras are installed. The quantity shall be proportional to number of cameras provided.
8.	Supply and fixing of 3mm thick PVC board with auto glow and HP latex vinyl print of size 12"x 4" signages with for displaying "Danger 440 Volt" with symbol indication as per NBC complete etc. as required.	At each electrical Panel.
9.	Supply and fixing of 5mm thick PVC board Auto Glow sheets and transparency sheets of size and layout as per IS: 9457 "Fire Extinguisher" indication as per NBC complete etc. as required.	At each fire extinguisher placed

- Above signage list are indicative. EPC contractor has to provide signage as per site requirement and instruction of Engineer In-charge.

EPC Contractor shall develop shop drawings for all types of signage as per statutory requirements, for the project and shall submit to the Engineer in charge for approvals before executing the works. The contractor will arrange samples and required mock-ups as instructed by the Engineer in charge.

16. UNINTERRUPTED POWER SUPPLY (UPS)

The scope of works includes Supplying, Installation, Testing and commissioning of Three phase UPS system with batteries.

16.1 GENERAL

This specification describes a Three phase continuous duty, on-line, uninterruptible power system, hereafter referred to as the UPS. The UPS shall operate in conjunction with the existing building electrical system to provide power conditioning, back-up and distribution for critical electrical loads. The UPS shall consist of, as required by the project, the UPS module, battery cabinet(s), and accessories as described in this specification.

60kVA Modular Online UPS- Input supply: Three Phase, Output supply: Three Phase N+1 (Make- Approved by Engineer-in-charge)

Supplying, installation, Testing & Commissioning of 60kVA capacity at full load (Unity Power Factor) at operating temperature 0 to 40°C, Online double conversion true sine wave Uninterrupted hot swappable modular (UPS) system with N+1 modules (Module size should not be more than 25% of UPS capacity, N+1 where N indicates Total number of module to match UPS capacity & +1 indicated one additional module for redundancy) For battery redundancy UPS is capable for the replacement or addition of battery modules without shutting down the entire system, battery modules can be replaced without transferring the load to BYPASS. The UPS shall include battery bank suitable for 30 minutes back up (Minimum 48,000VAH) on full load (Battery shall be VRLA SMF in ABS Container) and Static Bypass switch along with manual bypass must be available with each module. Suitable isolation transformer of same make of UPS OEM for additional protection against neutral faults etc. UPS shall have inbuilt phase sequence correction. The UPS systems offered is to be of the latest technology with Digital Control Microprocessor based for reliable operation, using IGBT's both for the rectifier & inverter (3 Level) with PWM. The quality of design, manufacturing and inspection process should confirm to the relevant international standards such as IEC/EN/UE. The operating efficiency of the UPS systems shall be >96% while operating on battery mode and delivering quality power to the 100% non-linear loads. Current total harmonic effect (ITHD) on the input grid shall be < 3%. Output power factor shall be unity (i.e. kW rating should be as same as kVA rating). UPS shall be suitable for incoming supply AC: 3Phase 400V +/- 20%, 50 Hz +/- 5 Hz, AC Output voltage: 3Phase 415 Volt, 50 Hz +/- 1%, output THDv-< 1% on nonlinear load. Overload capacity of 120% for 10 mins, noise level less than 50db at 1 meter distance, protections: Input Under voltage over voltage, abnormal out voltage, battery over charging, output over current, short circuit protection, battery deep discharge protection. UPS must comply with low voltage electromagnetic compatibility (EMC) achieved as per EN6204 Part I and Part 2, it shall be a Voltage and Frequency Independent (VFI)-type UPS. Communication RS232/RS485/SNMP port open protocol for BMS integration. Required battery racks and interconnecting copper conductor cables of suitable size and connectors and all required accessories are inclusive of the cost. Forced air-cooling with integral inbuilt fans for each module with

redundancy (if one fan fail UPS should be able to handle at least 80% of the load. The system shall be complying IEC 62040- 1,2 & 3, ISO14001 & ISO 450001. Display Panel to display UPS status like a) Input: Voltage, current, Frequency. b) Bypass: Voltage, Frequency. c) Output: Voltage, frequency, Current. d) Battery: Voltage, Capacity etc. UPS should have capacity of Event Logging & Statistical Data: UPS should capture and display up to 1000 events like: Over temperature / DC Bus Fail / Fan Fail / Fuse Fail / Overload / Short-circuit / Device Fail / Inverter Fail / Rectifier Fail / Bypass Fail, etc. e) Statistical Data: No. of power failures / Transfers to Bypass / Total Running time, etc. f) Mains Mode of Operation /Battery Mode of Operation / Bypass feeding the load / UPS Fault /Battery charging and discharging, overload, battery voltage and battery capacity. g) Audible Alarms: Mains Failure, Battery Low Alarm, UPS Overload, Fault, Shutdown, Input Over, Under Voltage, Output Over, Under Voltage, Battery Over, Under Voltage, Overload and short circuit, Over Temperature. The UPS should carry test certificate of same model offered from any NABL test lab. UPS OEM should have service centre in the same state to provide service & maintenance to UPS time to time (GST of OEM of the same state need to be provide). Warranty of 1 year on UPS and 2 years on battery need to provide with UPS.

17. FIRE FIGHTING SYSTEM

17.1 Introduction

Fire safety in buildings has become very important consideration in construction and maintenance. The aim of fire safety measure is to provide protection to life of occupants and property in the event of fire in the building. National Building Code of India-2016, Vol-1, Part-4 deals in detailed measures to be adopted for fire safety in buildings. The measures depend upon the occupancy, use, height, and area of the building.

The classification of building type of occupancies into different Groups in accordance with NBC-2016, Vol-1, Part-4 is as under:

Residential housing Block

- (i) Type of the Building -- Group A4 (Apartment Houses)
- (ii) Max. Heights of buildings -- Height Between 15 to 35 Meters (Categories as per NBC)
- (iii) Pressure Zoning -- Towers will be designed only for one zone.

Community center building

- (i) Type of the Building -- Group D (Assembly Building)
- (ii) Max. Heights of buildings -- Less Than 10m in Height and up to 300 Person (Categories as per NBC)
- (iii) Pressure Zoning -- Building will be designed only for one zone.

Guest House building

- (i) Type of the Building -- Group A5 (Hotel Building)

(ii) Max. Heights of buildings -- Less Than 15m in Height and Area Exceeding 300Sqm and up to 1000 Sqm (Categories as per NBC)

(iii) Pressure Zoning --Building will be designed only for one zone.

- The scope of work includes design, planning, approval of drawings from competent authority, approval from fire department, supply, installation, testing and commissioning, handing over and taking relevant NOC from the local bodies as applicable.
- The entire fire safety installation shall be compliant with the NBC-2016, relevant BIS codes / standard and as per the direction of local fire authorities for the entire Complex to ensure the highest safety standard and uniformity of system. Further, before property is opened to public, the fire protection shall be fully operated and tested under simulated conditions to demonstrate compliance with the standards, codes, and guidelines.
- Fire Fighting system with Down comer cum Wet riser, Fire sealing of floors in all shafts, electric & diesel fire pump, internal & external fire hydrants, hose reel system and all the relevant components as per the requirement of NBC 2016.
- Portable fire extinguishers, Sand buckets and first aid boxes etc. shall be provided as per the approval of Engineer-In-Charge.

17.2 NBC REQUIREMENT AND PROPOSALS

A. The building falls under Residential building under Apartment houses A-4 having height between 15 to 35 meters for apartment buildings. As per the NBC-2016, Table 7, following are the basic minimum requirement for Fire Fighting Installation of residential towers having heights between 15 to 35 meters:

Residential housing Blocks.

Sl. No.	Type of installation	Requirements (as per NBC-2016)	Proposal for the Project
1.	Fire Extinguishers	Required	Provided – 1 set of 2 nos. fire extinguishers within the Fire Hydrant Cabinet of following type - 1 no. 4.5 Kg CO2 (having 90% base material) fire extinguisher. 1 no. 9 kg DCP (Having 90% Base material) fire extinguisher. In addition to above 4.5 Kg CO2 gas & 9 Kg ABC Powder

			type (having 90% base material) extinguishers will also be provided inside Lift machine room, parking areas, electrical panel rooms, Pump rooms, security rooms and ESS / DG areas.
2.	Hose Reel	Required	Provided – 20 mm dia and 30-meter-long hose reel drum with 5 mm shut off nozzle to be provided with each hydrant valve on each floor inside fire hydrant cabinet. The hose reel shall be connected directly to the riser.
4.	Wet Riser	Not Required	Not Provided.
5.	Down Comer	Required	Provided. 100 mm dia down comer has been proposed along with followings inside Fire Hose Cabinet (FHC): (a) single headed 63 mm dia hydrant valve (b) 20 mm dia 30-meter-long hose reel with nozzle (c) 2 nos. 63 mm dia. 15-meter-long hose pipes. (d) one branch pipe with nozzle (e) Fire man's axe.
6.	Yard Hydrants	Not Required	Not Provided.
7.	Automatic Sprinkler System	Not Required	Not Provided.
8.	Manual Fire Alarm System	Required.	Provided. Manual call points and electronic hooters are provided on all floors near all exit locations and other suitable locations. The public address and talk

			back system are also provided at suitable locations, if height of building is above 24 meters.
9.	Automatic Fire Detection & Alarm System	Not Required.	Not Provided. However, Smoke and heat detectors are provided in all electrical shafts, electrical panel / meter rooms, lift machine rooms, pump rooms etc. to provide early warning from a probable risk area, if building height is above 30 meters.
10.	Under Ground Water Tank	Not Required	Not Provided.
11.	Terrace Tank	25 KL	Provided – 25 KL over each block terrace
12.	Ground Level Fire Pumps	Not Required.	Not Provided.
13.	Terrace Level Fire Pump	Required – of capacity 900 lpm with minimum pressure of 3.5 kg/sq cm.	Provided – of capacity 900 lpm

- B. The building falls under Residential building under Hotels A-5 having height Less Than 15 meters and Floor Area Exceeding 300sqm but not more 1000sqm on any of the floor for hotel buildings. As per the NBC-2016, Table 7, following are the basic minimum requirement for Fire Fighting Installation of Hotel buildings height Less Than 15 meters and Floor Area Exceeding 300sqm but not more 1000sqm on any of the floor:

Guest House building

Sl. No.	Type of installation	Requirements (as per NBC-2016)	Proposal for the Project
1.	Fire Extinguishers	Required	Provided – 1 set of 2 nos. fire extinguishers within the Fire Hydrant Cabinet of following type - 1 no. 4.5 Kg CO ₂ (having 90% base material) fire extinguisher. 1 no. 9 kg DCP (Having 90%

			Base material) fire extinguisher. In addition to above 4.5 Kg CO2 gas & 9 Kg ABC Powder type (having 90% base material) extinguishers will also be provided inside Lift machine room, parking areas, electrical panel rooms, Pump rooms, security rooms and ESS / DG areas.
2.	First aid Hose Reel	Required	Provided – 20 mm dia and 30-meter-long hose reel drum with 5 mm shut off nozzle to be provided with each hydrant valve on each floor inside fire hydrant cabinet. The hose reel shall be connected directly to the riser.
4.	Wet Riser	Required	Provided. 100 mm dia wet riser has been proposed along with followings inside Fire Hose Cabinet (FHC): (a) single headed 63 mm dia hydrant valve (b) 20 mm dia 30-meter-long hose reel with nozzle (c) 2 nos. 63 mm dia. 15-meter-long hose pipes. (d) one branch pipe with nozzle (e) Fire man's axe.
5.	Down Comer	Not Required	Not Provided.
6.	Yard Hydrants	Not Required	Not Provided.
7.	Automatic Sprinkler System	Not Required	Not Provided.
8.	Manual Fire Alarm System	Required.	Provided. Manual call points and electronic hooters are provided on all floors near all exit locations and other suitable

			locations. The public address and talk back system are also provided at suitable locations, if height of building is above 24 meters.
9.	Automatic Fire Detection & Alarm System	Required.	Provided Smoke and heat detectors are provided in all guest rooms, dining hall, kitchen, electrical shafts, electrical panel / meter rooms, lift machine rooms, pump rooms etc. to provide early warning from a probable risk area.
10.	Under Ground Water Tank	Required- 50KL	Provided 50 KL in two compartments of 25KL Each Underground.
11.	Terrace Tank	10 KL	Provided – 10 KL
12.	Ground Level Fire Pumps	Required	Provided. 1 No. 1620LPM Main hydrant Fire Pump. 1 No. 180 LPM Jockey Pump 1 No. 1620 LPM Diesel Engine Driven Fire Pump.
13.	Terrace Level Fire Pump	Required	Provided – of capacity 450 lpm

- C. The building falls under Assembly Building D1-D5 having height less than 10m and up to 300 persons for assembly buildings. As per the NBC-2016, Table 7, page 45, following are the basic minimum requirement for Fire Fighting Installation of assembly building having height less than 10m and up to 300 persons:

Community centre building.

Sl. No.	Type of installation	Requirements (as per NBC-2016)	Proposal for the Project
1.	Fire Extinguishers	Required	Provided – 1 set of 2 nos. fire extinguishers within the Fire Hydrant Cabinet of following type -

			1 no. 4.5 Kg CO2 (having 90% base material) fire extinguisher. 1 no. 9 kg DCP (Having 90% Base material) fire extinguisher. In addition to above 4.5 Kg CO2 gas & 9 Kg ABC Powder type (having 90% base material) extinguishers will also be provided inside Lift machine room, parking areas, electrical panel rooms, Pump rooms, security rooms and ESS / DG areas.
2.	Hose Reel	Required	Provided – 20 mm dia and 30-meter-long hose reel drum with 5 mm shut off nozzle to be provided with each hydrant valve on each floor inside fire hydrant cabinet. The hose reel shall be connected directly to the riser.
4.	Wet Riser	Not Required	Not Provided.
5.	Down Comer	Required	Provided. 100 mm dia down comer has been proposed along with followings inside Fire Hose Cabinet (FHC): (a) single headed 63 mm dia hydrant valve (b) 20 mm dia 30-meter-long hose reel with nozzle (c) 2 nos. 63 mm dia. 15-meter-long hose pipes (d) one branch pipe with nozzle (e) Fire man's axe.
6.	Yard Hydrants	Not Required	Not Provided.
7.	Automatic Sprinkler System	Not Required	Not Provided.
8.	Manual Fire Alarm	Required.	Provided.

	System		Manual call points and electronic hooters are provided on all floors near all exit locations and other suitable locations. The public address and talk back system are also provided at suitable locations, if height of building is above 24 meters.
9.	Automatic Fire Detection & Alarm System	Not Required.	Not Provided.
10.	Under Ground Water Tank	Not Required	Not Provided.
11.	Terrace Tank	20 KL	Provided – 20 KL at terrace.
12.	Ground Level Fire Pumps	Not Required.	Not Provided.
13.	Terrace Level Fire Pump	.	Provided – of capacity 450 lpm.

Note:

- (i) Pressure gauge to be provided at hydraulically remotest point to ensure the required 3.5 kg/cm² pressure in the fire water line.
- (ii) Synthetic Fire hose type-III (63mm dia & 15m) to be provided as per IS:636.
- (iii) All the firefighting equipments shall be ISI marked and certificate for the same shall be provided.
- (iv) A First Aid Box shall be provided on the ground floor of each building, at a clearly visible and easily accessible location for all occupants.
- (v) Floor evacuation plans to be provided for each floor of Residential Housing Blocks, Community center & Guest House.
- (vi) IG-541 gas-based fire suppression system and fire detection system to be installed in the substation shall be provided.

17.3 ADDITIONAL CONDITION AND SPECIFICATION FOR FIRE FIGHTING SYSTEM

- a. The scope of firefighting system installation in all the buildings to be constructed shall be as per NBC 2016. Planning, Design & preparation of Drawings, obtaining approvals from the department, NOC from Fire department, Supply, installation, testing and commissioning of Fire

Fighting system (Wet riser and down comer), and fire extinguishers etc. as per the requirement of NBC 2016 with amendments, updated BIS codes, Fire bye-laws of Govt. Assam, CPWD specifications for the buildings as described in the details of specifications. After completion of work getting fire clearance (if required) from the local bodies, Chief Fire Officer etc.

b. Condition for designing

- EPC contractor shall be Design and preparation of the working drawings for internal and external firefighting system. The system drawings should clearly show Down Comer or Wet riser system (as applicable), fire extinguishers, UG sump, Fire pump house, Terrace tank, terrace pump, fire pumps, yard hydrants, first aid hose reel, ring main etc. in line with the statutory requirements. Sizes of all equipment required and prepare detailed specifications, obtain necessary NOC/permissions from the statutory / local fire authority / bodies etc. as required. Check and approve detailed drawings and data sheet of suppliers/ manufacturers and incorporation of modification of any working / architectural/structural drawings.
- Fire Fighting System requirement given in drawings are minimum and indicative. Final design shall be submitted by the EPC contractor as per NBC 2016 which shall be not on lesser provision as mentioned in Appendix-A: Minimum Requirements for Firefighting Installations (Based on Table 7 of Vol-1, Part-4 of NBC 2016) in CPWD General Specifications for Electrical Works Part V (Wet Riser & Sprinkler System) 2020 as amended up to date.
- Municipal Regulation of the city shall also be taken in to consideration while selecting and designing firefighting system for a building.

17.4 Wet Riser cum Down Comer: -

17.5 Pressure at hydraulically remote hydrant and at the highest hydrant shall not be less than 3.5 kgf/cm². The pressure at the hydrants shall however not exceed 7 kgf /cm².

17.6 The pipe line will be designed in such a way that it should be possible to get discharge at any location. Design parameters shall be as under: -

- (i) Maximum flow velocity : 2.5 mps
- (ii) Maximum Friction : 5 m per 100 m run

17.7 Main Fire Pumps (Both Electrical as well as diesel) shall be selected for:

- (i) Discharge : 1620 lpm/2280 lpm/2850 lpm

- (ii) Head : 35 m + Height of terrace level hydrant above pump level + 6% of the maximum length of pipe from pump discharge to any hydrant at terrace level.

17.8 Terrace pump shall be selected for:

- (i) Discharge : 450 lpm/900 lpm
- (ii) Head : 20 m+ 6% of the maximum length of pipe from terrace pump to any hydrant at terrace level.

17.9 Pressurization pump (Jockey Pump) shall be selected for:

- (i) Discharge : 180 lpm
- (ii) Head : 35 m + height of terrace level hydrant above pump level.

17.10 No. of Risers: - Number of risers will be decided to fulfill the following conditions: -

- (i) No corner of the building is farther than 30 m from nearest riser.
- (ii) Horizontal distance between two risers shall not be more than 50 m.
- (iii) Normally one riser is provided for every 1000 sq.mtr of plinth area or part thereof.

However, the number of risers can be suitably increased to meet the given situation.

17.11 **Down Comer:** - A minimum of two terrace pumps (electrical) shall be provided. One pump shall be act as standby.

- All down comer pipes shall be inter connected at the terrace level. In case terraces are not interconnected, all building will be treated as individual buildings.
- Fire service inlet shall be provided with each Riser/down comer for facilitating pumping of water from fire service tenders.

17.12 **Control system in case of Downcomer:** - The starting of terrace pump shall be automatic i.e. with the opening of any hydrant valve or hose reel on any floor, the pump will start automatically with fall in line pressure. In addition, start/stop push button shall be provided at ground floor near internal hydrant for starting the pump manually. Where fire control room has been

provided, remote operation of terrace pump may be done from fire control room in place of near internal hydrant. The control panel for terrace pumps shall be provided near the pumps in a suitable enclosure to avoid unauthorized operation.

17.13 Fire Pump

Pumps conforming to IS 12469 shall be exclusively used for Firefighting purposes. The pumps shall be centrifugal type direct driven with a 3 phase, 415 V + 10%, 50 Hz, A.C. motor. The standby fire pump shall be driven by diesel engine. The pumps may be either of horizontal split casing (HSC) type with operating speed not exceeding 1500 rpm, or solid casing with operating speed not exceeding 3000 rpm as specified in the tender documents.

The main fire pump and terrace pump shall be suitable for continuous operation in the system. The jockey pump shall be suitable for intermittent operation to buildup pressure in the system on account of leakage. The head and discharge requirements shall be as specified in the tender documents. The head shall be suitable for the system and shall take into consideration the pressure drops across the various components in the water circuit as well as the frictional losses.

Pump shall be capable of discharging not less than 150 percent of the rated discharge at a head of not less than 65 percent of the rated head. The shut off head shall not exceed 120 percent of the rated head in the case of horizontal Pumps

17.14 MATERIAL AND CONSTRUCTION

- (vii) The centrifugal pumps shall conform to IS 1520.
- (viii) The pump casing shall be of heavy section close grained cast iron and designed to withstand 1.5 times the working pressure. The casing shall be provided with shaft seal arrangement as well as flanges for suction and delivery pipe connections as required.
- (ix) The impeller shall be of bronze, brass, or stainless steel. This shall be shrouded type with machined collars. Wear rings, where fitted to the impeller, shall be of the same material as the impeller. The impeller surface shall be smooth finished for minimum frictional loss. The impeller shall be secured to the shaft by a key.
- (x) The shaft shall be of stainless steel and shall be accurately machined. The shaft shall be balanced to avoid vibrations at any speed within the operating range of the pump.
- (xi) The shaft sleeve and wearing ring etc. shall be of bronze, brass, or stainless steel.
- (xii) The bearings shall be ball or roller type suitable for the duty involved. These shall be grease lubricated and shall be provided with grease nipples/cups.
- (xiii) The bearings shall be effectively sealed against leakage of lubricant or entry of dust or water.
- (xiv) The shaft seal shall be mechanical type, so as to allow minimum leakage. A drip well shall be provided beneath the seal.
- (xv) The pumps shall be directly coupled to the motor/diesel engine shaft through a flexible coupling

protected by a coupling guard.

- (xvi) The pump and motor/diesel engine shall be mounted on a common robust bed plate fabricated from mild steel section. The bed plate shall have rigid, flat and true surfaces to receive the pump and motor/diesel engine mounting feet. The pump will be perfectly aligned with the motor/engine so as to avoid any vibration during operation at all variations of load.

17.15 ACCESSORIES

Each pump shall be provided with the following accessories: -

- Sluice valves on suction and delivery.
- Reducers, as may be required to match the sizes of the connected pipe work.
- Non-return valve at the delivery.
- Pressure gauge at delivery side between pump and the non-return valve.
- Flexible coupling/connections shall be provided between Pump sets and Valves on suction and delivery sides of all the pump sets.

Note:

- i) No butterfly valves shall be installed inside the pump room.
- ii) The size of the non-return valve and cut off (Sluice valve) shall not be less than the size of the initial delivery pipe.

17.16 INSTALLATION

- The pump and motor/engine assembly shall be mounted and arranged for ease of maintenance and to prevent transmission of vibration and noise to the building structure or to the pipe work.
- The pump and motor/engine assembly shall be installed on suitable RCC foundation. The length and width of the foundation shall be such that 100 mm space is left all around the base frame. The height of foundation shall be so decided that the total weight of foundation block is 1.5 times the operating weight of the pump assembly. The foundation shall be isolated from the floor by vibration isolating pads. Angle iron frame of size 35 mm x 35mm x 5 mm shall be provided on the top edges of the foundation.
- More than one pump and motor assembly shall not be installed on a single base or cement concrete block.
- The suction/discharge pipe shall be independently supported and their weight shall not be transferred to the pump. It should be possible to disconnect any pump for repairs without disturbing the connecting pipe line.
- A minimum clearance of 1 m around the main pumps shall be provided. For jockey pump-clearance of 75 cm shall be adequate.
- Sufficient space is to be left in front for the radiator of diesel engine for free discharge of hot air. Arrangement for discharging hot air to outside the pump house shall be provided so that hot air does not stagnate in the pump house.

17.17 Fire Service inlet: -

In order to facilitate feeding of water in the system by fire service, a 2/3-way 63 mm diameter collecting head shall be provided and connected with each riser/down comer and the ring main with non-return valve and butterfly/slucie valve. This should be located at a place where fire brigade tender can reach.

17.18 DIESEL ENGINE FOR FIRE PUMP

A. SCOPE

This scope of works shall be covers the details of requirements of a diesel engine for main fire pump to act as standby.

B. GENERAL

The diesel engine shall be suitable for automatic operation complete with necessary automatic starting gear, battery system and shall be complete with all accessories. Both engine and pump shall be assembled on a common bed plate, fabricated from mild steel channel.

C. DRIVE

The pump shall be only direct driven by means of a flexible coupling. The coupling between the engine and the pump shall allow each unit to be removed without disturbing the other. Coupling guard shall be provided. The speed shall be 1500 RPM.

D. DIESEL ENGINE

- a) **Environment conditions-** The engine shall be suitable to operate under the conditions of environment at site.
- b) **Engine Rating-** The engine shall be multi cylinder/vertical 4 stroke cycle, water cooled, developing suitable HP at the operating speed specified to drive the fire pump. Continuous capacity available for the load shall be exclusive of the power requirement of auxiliaries of the diesel engine, and after correction for altitude, ambient temperature and humidity for specified environment conditions. The engine rating shall be suitable to drive the pump at 150 percent of its rated discharge with at least 65 percent of rated head or 20% in excess of the maximum brake

horsepower required to drive the pump at its duty point, whichever is higher. The engine shall have 10% overload capacity for one hour in any period of 12 hours continuous run.

The engine shall be:

- naturally aspirated, supercharged or turbo-charged and rather air or water-cooled.
- provided with an in-built tachometer to indicate rpm of the engine.
- suitable for cold starting for which suitable heaters shall be provided in lubricating oil.
- able to develop full load within 15 seconds from the receipt of the signal to start.
- The diesel engine shall conform to BS 649/ IS 1601/ IS 10002, amended up to date.

c) **Engine Accessories-** The engine shall be complete with following accessories:

- Fly wheel dynamically balanced.
- Direct coupling for pump and coupling guard.
- Radiator with hoses, fan, water pump, drive arrangement and guard.
- Air cleaner dry type.
- Fuel service tank with necessary pipe work.
- Fuel filter.
- Pump for lubricating oil and lubricant oil filter.
- Electric starting battery 12 V/24 V with 2 Nos. batteries.
- Exhaust silencer with necessary pipe work.
- Governor.
- Instrument panel housing all the gauges, including Tachometer, hour meter and starting switch with key (for manual starting).
- Necessary safety controls.
- Winterization arrangement.
- Hand operated semi rotary pump for filling the service tank.
- A standard kit of tools (this shall be always kept on hand).

d) **Cooling System-** The engine shall be radiator water cooled. The radiator assembly shall be mounted on the engine. The radiator fan shall be driven by the engine as its auxiliary with multiple fan belts. When half the belts are broken, the remaining belts shall be capable of driving the fan. Cooling water shall be circulated by means of an auxiliary pump of suitable capacity driven by the engine in a closed circuit.

- e) **Fuel System-** The fuel system shall be gravity fed from the fuel tank to the engine driven fuel pump. The engine fuel tank shall be mounted either adjacent to the engine or suitably wall mounted on brackets. The fuel filter shall be suitably located to permit easy servicing.

The fuel tank shall be of welded steel construction (3mm thick) and of capacity sufficient to allow the engine to run on full load for at least 8 hours. The tank shall be complete with necessary floor mounted supports, level indicator (protected against mechanical injury) inlet, outlet, overflow connections and drain plug and piping to the engine fuel tank. The outlet should be so located as to avoid entry of any sediments into the fuel line to the engine.

Any valve in the fuel feed pipe between the fuel tank and the engine shall be placed adjacent to the tank and it shall be locked in the open position.

All fuel tubing to the engine shall be with M.S.'C' class pipe with flexible hose connections where required. Pipe joints shall not be soldered and plastic tubing shall not be used.

The following shall be provided:

- A sludge and sediment trap shall be provided.
- An inspection and cleaning hole
- Means to enable the entire fuel system to be bled of air (Air relief cocks are not allowed; screwed plugs are permitted).

- f) **Lubricating Oil System-** Forced feed lubricating Oil system shall be employed for positive lubrication. Necessary lubricating oil filters shall be provided, located suitably for convenient servicing.

- g) **Starting System-** The starting system shall comprise of necessary batteries 12 Volts/ 24 Volts, starter motor of adequate capacity and axle type gear to match with the toothed ring on the fly wheel. Suitable protection to protect starting motor from excessively long cranking runs shall be suitably integrated with engine protection system.

- The capacity of the battery shall be suitable for meeting the needs of the starting system.
- The battery capacity shall be adequate for 10 consecutive starts without recharging with cold engine under full compression.
- Three attempt starting facility shall be provided. If the engine fails to start after third attempt, the engine shall be locked out and suitable audio-visual alarm shall be given to indicate engine failure. The starter motor used for automatic starting may also be used for manual starting provided there are separate batteries for manual starting.
- The scope shall cover all cabling, terminals, initial charging etc.

- h) **Exhaust System-** The exhaust system shall be complete with residential grade silencer suitable for outdoor installation and silencer piping shall be extended up to 1 m, outside pump house duly insulated with 50 mm thick glass wool and 1.0 mm thick aluminium sheet cladding.

Retrofitted emission-control equipment shall be used having a minimum specified PM-capturing efficiency of at least 70%, type approved by one of the five CPCB recognized labs. (Recommendations of National Clean Air Program 2019 launched by Ministry of Environment, Forest, and Climate Change)

- i) **Engine shut down mechanism-** This shall be manually operated and shall return automatically to the starting position after use.
- j) **Governing System-** The engine shall be provided with an adjustable governor to control the engine speed within 5% of its rated speed under all conditions of load up to full load. The governor shall be set to maintain rated pump speed at maximum pump load.
- k) **Engine Instrumentation** - Engine instrumentation shall include the following: -
- Lub.oil pressure gauge
 - Lub.oil temperature gauge
 - Water temperature gauge
 - Tachometer
 - Hour meter

The instrumentation panel shall be suitably mounted on the engine.

- l) **Engine protection devices** - Following engine protection and automatic shutdown facilities shall be provided:(i) Low lub. oil pressure.
- High cooling water temperature.
 - High lub. oil temperature.
 - Over speed shut down
- m) **Pipe work-** All pipe lines with fittings and accessories required shall be provided for fuel oil, lub.oil and exhaust systems.
- n) **Anti vibration mounting-** Suitable vibration mounting duly approved by engineer-in-charge shall be employed for mounting the unit so as to minimize transmission of vibration to the structure.
- o) **Battery Charger-**

Battery of diesel engine operated fire pump shall have separate charger from emergency power supply circuit. Necessary float and boost charger shall be incorporated in the control section of power and control panel with manual selection of boost charge, to keep the battery under trim condition. Voltmeter to indicate the state of charge of the batteries shall be provided. Where separate batteries are provided for automatic and manual starting, the charging equipment shall be capable of trickle

charging both the batteries simultaneously. Equipment shall be provided to enable the state of charge of the batteries to be determined.

- p) The engine installation shall be approved by the representative of engine manufacturer (who shall carry out after sales service under AMC).
- q) The following spare parts shall be supplied with the engine and kept on hand:
 - Two sets of fuel filters, elements, and seals;
 - Two sets of lubricating oil filters, elements, and seals;
 - Two sets of belts (where used);
 - One complete set of engine-joints, gaskets, and hoses;
 - Two injector nozzles;
 - One complete set of piston rings for each cylinder; and
 - One inlet valve and one exhaust valve).

17.19 PIPE MATERIALS

Pipes shall be of the following materials:

- Mild steel heavy class (C-class) conforming to IS:1239 for sizes up to 150 mm.
- Welded black steel pipe, Class 2, conforming to IS: 3589, for sizes greater than 150 mm. These pipes shall be factory rolled and fabricated from minimum 6mm thick M.S. Sheet for pipes up to 350 mm diameter and from minimum 7 mm thick M.S. sheet for pipes of 400 mm diameter and above.

MS pipes may be allowed for extension of existing systems which are laid with CI pipes.

- Cast iron double flanged pipe, Class-A conforming to IS 1536 or IS: 1537 (to be provided only in underground application).

Note: For pipe work of Automatic Sprinkler System inside the building, Stainless Steel Pipes, and fittings of grade AISI 304 as per JIS standard 3448 are also permitted particularly where replacement of pipes is not easy like areas above false ceiling etc., subject to the condition that these pipes with associated fittings are suitable to safely withstand the system test pressures.

- GI Pipe medium Class (B-class) conforming to IS:1239 (For Drain)
- Cadmium plated steel nuts/bolts/washers shall be used.
- Flex drop of stainless-steel metallic pipe with mounting accessories, frame for installation on false ceiling.

17.20 ANTI-CORROSIVE PROTECTION ON UNDER GROUND MS PIPE

Corrosion protection tape shall be wrapped on MS pipes to be buried in ground. This corrosion protection tape shall comprise of coal tar/asphalt component supported on fabric of organic or inorganic fibre and minimum 4 mm thick and conform to requirement of IS: 10221-Code of practice for coating and wrapping of underground mild steel pipe line. Before application of corrosion protection tape all foreign matter on pipe shall be removed with the help of wire brush and suitable primer shall be applied over the pipe thereafter. The primer shall be allowed to dry until the solvent evaporates and the surface becomes tacky. Both primer and tape shall be furnished by the same manufacturer. Corrosion protection tape shall then be wound around the pipe in spiral fashion and bounded completely to the pipe. There shall be no air pocket or bubble beneath the tape. The overlaps shall be 15 mm and 250 mm shall be left uncoated on either end of pipe to permit installation and welding. This area shall be coated in situ after the pipe line is installed. The tapes shall be wrapped in accordance with the manufacturer's recommendations. If application is done in cold weather, the surface of the pipe shall be pre- heated until it is warm to touch and traces of moisture are removed and then primer shall be applied and allowed to dry.

Holiday Testing for wrapping and coating is essential. Holiday testing may preferably be carried by flexible and detachable ring probe, which will enable the entire 360° of the surface of the pipe to be scanned.

At least 10 percent of all the welded joints shall be radio graphically tested and half of the joints radio graphed shall be the field joints.

17.21 VALVE

Each pump shall be provided with a non-return valve and a sluice valve on the delivery side; the sluice valve being installed on the upstream side of the nonreturn valve. A pressure gauge shall also be provided between the pump and the non-return valve. The size of the non-return valve and cut off (sluice) valve shall not be less than the size of the initial delivery pipe and, in no case, less than the delivery outlet of the pump. No butterfly valves shall be installed inside the pump room.

- Sluice valve shall conform to IS: 780.
- Butterfly valve, wherever used, shall conform to IS:13095.

All valves shall be suitable to with-stand the pressure in the system and rating shall be PN 1.6. All valves shall be right-handed (i.e. handle or key shall be rotated clock wise to close the valve), the direction of opening and closing shall be marked and an open/shunt indicator fitted.

- The material of valves shall be as under:

Body	:	Cast iron
Disc	:	Cast Bronze or Stainless Steel
Seat	:	Either integral or Nitrile rubber

O-ring : Nitrite/ Silicon

- Non return valves shall be swing check type in horizontal run and lift check type in vertical run of pipes.
- Air release valves shall be of gunmetal body.

Valves in fixed firefighting installations shall have supervisory switch with it signaling to fire alarm panel or to have chain(s), pad lock(s), label, and temper-proof security tag(s) with serial number to prevent tempering/unauthorized operation. These valves shall be kept in their intended 'open' position.

17.22 STRAINERS

Stainless steel strainers shall have minimum 1 mm thick screen with 3 mm perforations. Strainers shall be provided with flanges.

17.23 ORIFICE PLATE

Orifice plate shall be made of 6 mm thick stainless steel and shall have an identification tag projecting beyond any flange between which it is clamped. The orifice shall be plain central hole without burs and diameter not less than one-half of the internal diameter of the pipe to which it is fitted.

17.24 INSTRUMENTS

- Pressure gauge of appropriate range and 150 mm diameter size shall be provided.
- The pressure gauge shall be duly calibrated before installation and shall be complete with shut off valve.

17.25 AIR VESSEL

Air vessel shall be provided on top of each riser and shall be fabricated out of 8 mm thick M.S. Sheet. The ends shall be dished. This shall be of 250 mm diameter, 1.2 m high and installed vertically on suitable legs. The legs shall be provided with M.S. Plate of size 75 mm x 75 mm x 5 mm at the bottom so that the legs do not puncture the roof. The legs shall be grouted in CC foundation. Flange connection shall be provided for connection with wet riser pipe. Air release valve and pressure gauge with shut off valve shall be provided. The air vessel shall be tested at 25 kgf/cm² pressure before installation.

17.26 FLUSHING ARRANGEMENT

Flushing connections with isolation valves should be provided at suitable locations in the firewater

ring main.

17.27 PRESSURE TESTING

- All piping shall be tested to hydrostatic test pressure of at least one and a half times the maximum operating pressure, but not less than 10 kgf/sq.cm for a period not less than 24 hours. While Hydro Testing, inclusion of cut-off valves in the mains to be tested can be avoided. All leaks and defects in joints revealed during the testing shall be rectified to the satisfaction of the Engineer-in-Charge.
- Piping repaired subsequent to the above pressure test shall be re-tested in the same manner.
- System may be tested in sections and such sections shall be securely capped.
- Pressure gauges may be capped off during pressure testing of the installation.

17.28 PIPE SUPPORTS

For installing pipes vertically or horizontally inside the building standard pipe supports of reputed make shall be used. Following supports shall be used.

- Split pipe support clamps with rubber lining for vertical, horizontal and roof hanging.
- Clevis Hangers for horizontal supports to adjust varying heights.
- Sprinkler Hangers for horizontal supports for pipes from 15 mm dia to 150 mm dia.

Fasteners and fully threaded rods shall be used for installing the pipe supports. The sizes of pipe supports and installation shall be in accordance with manufacturer's recommendations.

For pipes of size 100 mm and above, with the prior approval of Engineer-in-Charge, 'U' clamp with dash fastener may be used for supporting horizontal pipe from ceiling.

17.29 ELECTRICAL WORK

a) SCOPE

This scope of work shall be covers the requirements for the electrical works associated with firefighting installations, namely, motors, switch boards, power cabling, control wiring, earthing, and remote control-cum-indicating panels.

b) GENERAL

- Unless otherwise specified in the tender specifications, all equipment and materials for electrical works shall be suitable for operations on 415 V / 240 V + 10% (3 phase/single phase), 50 Hz AC system.
- All electrical works shall be carried out complying Central Electricity Authority (Measures Relating to Safety and Electric Supply) Regulations, 2010 and NEC 2011, as amended up to date.

- All parts of electrical works shall be carried out as per appropriate CPWD General Specifications for Electrical works, namely, Part I (Internal) 2023, Part II (External) 2023 work, and Part IV (Sub-station)- 2013 all as amended up to date.
- All materials and components used shall conform to the relevant IS specifications amended to date.

c) POWER SUPPLY

Power supply to following systems and equipment, where provided, shall be from normal and emergency (standby generator) power sources with changeover facility:

- Fire pumps.
- Pressurization and smoke venting; including its ancillary systems such as dampers and actuators.
- Terrace pump
- Fireman's lifts (including all elevators (lifts)).
- Exit signage lighting.
- Emergency lighting.
- Fire alarm system.
- Public address (PA) system (relating to emergency voice evacuation and annunciation).
- Magnetic door holds open devices.
- Lighting in fire command center and security room.

The generator shall be capable of taking starting current of all the fire and life safety systems and equipment as above. Where parallel HV/LV supply from a separate substation fed from different grid is provided with appropriate transformer for emergency, the provision of generator may be waived in consultation with the Authority.

The electric supply to the pumping set(s) shall be entirely independent of all other equipment in the premises that is even when the power throughout the entire premises is switched off, the supply to the pump shall continue to be available un-interrupted. This can be achieved by taking the connection for the pump(s) from the incoming side of the main L.T. breaker. In case, where parallel HV/LV supply from a separate substation fed from different grid is provided with appropriate transformer for emergency connected to a common bus bar, the connection may be taken through the bus bars.

The power supply to the panel/distribution board of these fire and life safety systems shall be through fire proof enclosures or circuit integrity cables or through alternate route in the adjoining fire compartment to ensure supply of power is reliable to these systems and equipment. It shall be ensured that the cabling from the adjoining fire compartment is protected within the compartment of vulnerability. The location of the panel/distribution board feeding the fire and life safety system shall

be in fire safe zone ensuring supply of power to these systems.

Circuits of such emergency system shall be protected at origin by an automatic circuit breaker so set as to permit the motor to be overloaded during an emergency to the maximum limit permissible by the manufacturer. Further, the no volt coil/the under-voltage release of that circuit breaker shall be removed. Master switches controlling essential service circuits shall be clearly labelled.

- d) Independent supply shall be provided for water supply pumps if installed in the same pump house.
- e) If the fire pump house is away from the sub-station building, the route of the cable shall not pass under the building or permanent structure. Cable shall be laid along the route which is safe from fire.

Sufficient spare power shall always be available to always drive pumping sets throughout the year. Suitable capacity ACBs/SDFUs shall be provided in the electrical panel for extending supplies to fire pumps. Such switches shall be suitably marked "FIRE SWITCH" and shall not be switched-off without permission/intimation to appropriate authority. In case any maintenance/repair work is to be carried out on the electrical panel where from supplies to fire pumps have been extended, alternative arrangement shall be made to ensure that power supply to fire pumps continue to be available for operation any time.

f) MOTORS

The motors shall be squirrel cage AC induction type. The motors shall be suitable for continuous duty and rating necessary to drive the pump at 150 percent of its rated discharge with at least 65 percent rated head. The motor shall be totally enclosed fan cooled type conforming to protection clause IP 21 of IS: 4691. The class of insulation shall be 'F.' The synchronous speed shall be 1500/3000 rpm as per requirement of the pump. The motor shall conform to IS:325.

g) MOTOR STARTER

- The motor starter shall conform to IS: 1822 "Motor starters of voltage not exceeding 1000 volts" and shall be air insulated and suitable for 415 V, $\pm 10\%$, 50 Hz, 3 phase AC supply and shall be integrated in the panel.
- Starter for the motor shall be direct on line (D.O.L) for motors up to and including 7.5 H.P. rating and automatic star-delta type for motors of higher ratings unless otherwise specified in the tender specifications. However, for main Fire Pump & Sprinkler Pump Soft Starters may be used.
- Each starter shall be provided with the following protections: (a) Thermal overload on all the three phases with adjustable settings, (b) Independent single-phase preventer. (Current sensing type).

- Adequate number of extra NO/NC contacts for interlocks, indicating lamps, remote operation etc. shall be provided on the starter/contactors. (v) Under voltage/No volt trip shall not be provided.

h) SWITCH BOARDS

- The main switch board shall be floor mounted, free standing or wall mounted cubical type and shall be factory built fabricated by one of the approved switch board manufacturers. The board shall be fabricated from 2.0 mm thick CRCA sheet and powder coated after 7 tank treatment process. The board shall be fabricated with IP 42 degree of protection. It shall be suitable for termination of the incoming cable(s) from bottom.
- The capacity of switch gear shall be suitable for the requirements of motor fed/ controlled. Starting currents shall be duly considered.
- Switch fuse units shall be used up to and including 32 A and SDFU shall be used for 63 A and above. ACB shall be used for 630A and above ratings.
- All Switch fuse units/SDFUs shall be of AC 23 duty as per IS: 4064-1978 as amended up to date. They shall be complete with suitable HRC cartridge type fuses.
- Switch boards shall house starters for motors with independent current sensing type single phase preventor for each starter.
- Volt meter with selector switch, a set of indicating lamps and fuses for voltmeter and lamps shall be provided. Ammeter with CTs, and selector switch shall be provided with each motor starter. Instruments shall be flush mounted with the panel and have a class index not higher than 1.0. The instruments and accessories shall be provided whether or not specifically indicated in the tender specifications.
- The fabrication of switchboard shall be taken up only after the drawings for the fabrication of the same are approved by the Engineer-in-charge.
- Switchboards shall be fabricated as per specifications indicated in sub-para above.
- The layout shall be designed for convenient connections and interconnections with the various switchgear. Connections from individual compartments to cable alleys shall be such as not to shutdown healthy circuits in the event of maintenance work becoming necessary on a defective circuit.
- Care shall be taken to provide adequate clearances between phase bus bars as well as between phase bus bars, neutral and earth.
- Where terminations are done on the bus bars by drilling holes therein, extra cross section shall be provided for the bus bars. Alternatively, terminations may be made by clamping.
- Provision shall be made for proper termination of cables at the switchboards such that there is no strain either on the cables, or on the terminators. Cables connected to the upper tiers shall be duly clamped within the switchboard.
- Identification labels shall be provided against each switchgear and starter compartment, using aluminium engraved labels.

- Metallic danger board conforming to relevant IS shall be fixed on each electrical switchboard.

i) SYSTEM CONTROLLER

For controlling operation of pumps and indicating fault, system controller shall be provided. The system controller shall consist of relays, timer, contactors etc. and shall be designed to operate the fire pumps with interlocking and fault indication as described. Annunciation window shall be provided to indicate following faults:

- Low water level in UG tank.
- Low water level in terrace tank.
- Main pump failed to start.
- Main pump failed during operation.
- Diesel pump failed to start.
- Diesel pump failed during operation.
- Supply to Main Pump failed.
- Supply to Pressurization Pump failed.
- Supply to Terrace Pump failed.

Suitable sensors, differential pressure switches, monitors shall be provided at respective locations. The control system shall be operational on 12 Volt/24 Volt DC starting batteries of engine. Battery chargers shall be provided to ensure that the batteries remain charged. Batteries shall be sealed maintenance free type.

j) REMOTE INDICATING PANEL

- The remote indicating panel shall be provided in the fire control room. This panel shall have necessary status indication of all electric motors.
- Back indication to show the status of operation of all the motors, pressure in the system, water level in underground and overhead tank etc. shall be provided.
- Panel shall be fabricated from not less than 1.6 mm thick CRCA sheet and powder coated after 7 tank treatment process. The panel shall be dust, damp and vermin proof. This shall be of wall mounting type. This shall be complete with necessary termination arrangements, multicore cables, tag blocks, control transformer, designation plastic labels, double earth studs etc. as required.

k) POWER CABLING

- Unless otherwise specified, the power cables shall be XLPE insulated, PVC outer sheathed aluminium conductor, armoured cables 1100 V grade. The power cables shall be of 2 cores for

single phase, 4 core for sizes up to and including 25 sq.mm for 3 phase and 3-1/2 core for sizes higher than 25 sq.mm for 3phase.

- Alternatively, XLPE/PVC insulated copper cable (single core/multicore armoured/un-armoured) of grade 1100 V shall be used.
- **For main power cable(s)** from LT Room to Fire Pumps Panel, if the cable(s) is taken in cable trench duly filled with sand & with proper distancing from other cables within the cable trench, or in fire rated shaft, then the cable(s) of above type be used. However, if the cable(s) is likely to be exposed to fire, then fire survival cable(s) shall be used.
- Power cables shall be of sizes to meet the starting and running current of motors fed and shall be as approved by the Engineer-in-Charge, after taking into consideration the load, the length of cabling.
- Cables shall be laid in suitable metallic trays suspended from ceiling, or mounted on walls. Cable ducts shall not be provided in pump rooms. Cable trays shall be of perforated steel sheet with adequate structural strength and rigidity. Necessary supports and suspenders for cable trays shall be provided by the contractor as required.

l) CONTROL WIRING

- Control wiring shall be done using ISI marked PVC insulated and PVC sheathed, 2.5 sq.mm, 250 V grade, armoured multi-core copper conductor cable. The control cable shall also be laid in the same manner as power cable.
- The number and size of the control cables shall be such as to suit the control system design adopted by the contractor.
- Runs of control wires within the-switchboard shall be neatly bunched and suitably supported/clamped. Means shall be provided for easy identification of the control wires.
- Control wiring shall correspond to the circuitry/sequence of operations and interlocks approved by Engineer-in-Charge.

m) EARTHING

- Provision of earth electrodes and the type of earthing shall be as specified in the tender specifications.
- The earth work shall be carried out in conformity with CPWD Specifications for Electrical works (Part-I), Internal2023 as amended up to date.
- Metallic body of all motors, medium voltage equipment and switch boards shall be connected by two separate and distinct earth conductors to the earth stations of the installations. Looping of such body earth conductors is acceptable from one equipment, or switch board to another.
- The size of earth conductors for body earthing of equipment shall be 2 Nos. 6 mm dia copper wire/2 Nos. 25 x 3 mm G.I. strip

- Armouring of cables shall be connected to the body of the equipment/switch board at both the ends. Compression type glands shall be used for all such terminations in the case of PVC/XLPE cables.

n) PAINTING

All panels shall be supplied with the manufacturer's standard finish painting or as approved by Engineer-in-charge.

17.30 IMPORTANT INDIAN STANDARDS FOR FIRE FIGHTING WORK	
IS 1239-1968 (Part-I)	Specifications for mild steel tube, tubular, and other steel pipe fittings.
IS 1239-1968 (Part-II)	Specifications for mild steel tube, tubular, and other steel pipe fittings.
IS 1536-1976	Specification for centrifugally Cast (Spun) Iron pressure pipes with flanges for water, gas, and sewage.
IS 1538 (Part 1 to 23)	Specification for Cast Iron fittings for pressure pipes for water, gas and sewage.
IS 1726-1960	Code for cast iron manhole frame and cover.
IS 3589-1981	Specification for electrically welded steel pipes for water, gas and sewage.
IS 4736-1986	Galvanizing G.I. Pipes
IS 636-1988	Non percolating flexible firefighting delivery hose (third revision)
IS 694-1990	PVC insulated cables for working voltages up to and including 1.100 volts (third revision)
IS 778-1984	Copper alloy gate, globe and check valves for water works purposes (fourth revision) (Amendment 2)
IS 780-1984	Sluice valves for water works purposes (50 to 300 mm) size (sixth revision) (amendment 3)
IS 884-1985	Specification for first-aid hose-reel for fire Fighting (for fixed installations) (first revision) (with amendment No.1)

IS 900-1992	Code of practice for installation and maintenance of induction motors (second revision)
IS 901-1988	Specification for couplings, double male and double female, instantaneous pattern for Fire Fighting (third revision)
IS 902-1992	Suction hose coupling for Fire Fighting of purposes (third revision)
IS 903-1984	Specification of fire hose delivery couplings branch pipe, nozzles, and nozzle spanner (third revision) (Amendment 5)
IS 937-1981	Specification for washers for water fittings for Fire Fighting purposes (revised) (with amendment No, 1)
IS 1520-1980	Horizontal centrifugal pumps for clear cold, fresh water (second revision)
IS 1536-1976	Horizontally cast-iron pressure pipes for water, gas & sewage (first revision) (with Amendments No. 1 to 4)
IS 1554-1988 Part I	PVC insulated (heavy duty) electric cables (working voltage up to and including 1100 volts (third revision)
IS 1554-1988 Part II	PVC insulated (heavy duty) electric cables (working voltage from 3.3 KV up to and including 11 KV (second revision)
IS 1648-1961	Code of practice for fire safety of buildings (General) Fire Fighting equipment and its maintenance (with amendment No.1)
IS 3624-1987	Pressure and vacuum gauges (Second revision)
IS 4736-1968	Hot-dip zinc coatings on steel tubes (with Amendment No.1)
IS 5290-1983	Specification for landing valves (second revision) (with Amendments No.6)
IS 5312- 1984 Part I	Swing check type reflux valves Part I-single door pattern (with amendments nos. 1 & 2)
IS 5312- 1986 Part II	Swing check type reflux valves Part II-Multi door pattern (with amendments nos. 1 & 2)

IS 7285	Seamless cylinders for storage of gas at high pressure.
IS 2189-1962	Code of practice for Automatic Fire alarm system
IS 2195-1962	Specification for heat sensitive fire detectors
IS 732-1973	Code of practice for electrical wiring installation

17.31 Water Storage Tank

The effective capacity of Under-ground static & Terrace Water Storage Tanks and capacity of Fire Pumps etc. shall also be as specified in Table-7 of NBC- 2016. The same has been reproduced as Appendix-A in CPWD GENERAL SPECIFICATIONS FOR ELECTRICAL WORKS PART-V (WET RISER & SPRINKLER SYSTEMS) 2020.

17.32 Other Requirements:

- (a) The agency must study specifications, provision for fire-fighting system for various buildings, NBC code – 2016, local byelaws, and additional conditions carefully. The work shall be executed in close co-ordination with the progress of building work.
- (b) Contractor shall provide all tools, equipment, metering and testing devices required for the purpose. On award of work, Contractor shall submit a detailed proposal giving methods of testing and gauging the performance of the equipment to be supplied and installed under this Contract.
- (c) All tests shall be made in the presence of the Engineer-in-charge or his representative or any inspecting authority and results recorded. At least five working days' notice in writing shall be given to the inspecting parties before performing any test.
- (d) Water flow rate of all equipment and in pipe lines through valves shall be adjusted to design conditions. Complete results of adjustments shall be recorded and submitted.
- (e) Contractor shall ensure proper balancing of the hydraulic system and for the pipes / valves installed in his scope of work by regulating the flow rates in the pipe line by valve operation. The contractor shall also provide permanent Tee connection (plug) in water supply lines for ease of installing pressure gauge & rotameters. Contractor shall also supply all required pressure gauge & rota meter for system commissioning and balancing. The balancing shall be to the satisfaction of Engineer in charge.
- (f) Orifice plates made of minimum 6 mm thick SS sheet with suitable hole to regulate the pressure shall be provided with all hydrant valves in the system.
- (g) Three copies of all test results shall be submitted to the Engineer in A4 size sheet paper in two weeks after completion of the tests
- (h) All firefighting panels for (firefighting system, wet riser system, sprinkler system etc.) shall be TTA (Totally type tested Assembly) conforming to IEC 61439 Part 1 and Part 2.
- (i) MS pipe with red colour coating shall be used for the building. Sprinkler system shall be designed

with 2 pipe system (One for Sprinkler Riser and one for Sprinkler drain pipe).

- (j) Pumps shall be installed in positive suction only as per NBC 2016.
- (k) All the firefighting pipes laid underground shall be wrapped with appropriate thickness of not less than 5 mm bituminous compound/coal tar/asphalt tape coating.
- (l) Entire refuge area shall be provided with sprinklers as per NBC 2016.
- (m) Automatic sprinkler system and wet riser system in basement and other areas in the building as per NBC 2016 shall be in the scope of contractor.
- (n) All the electrical rooms, AHU rooms, kitchen area, substation etc. as mentioned in NIT drawings (shall be provided with Providing and fixing of Portable (FCO) Automatic Fire Prevention Apparatus very high quality bio degradable, ozone friendly UL (Underwriters Laboratories) listed, built in HDPE (High-density polyethylene) container to automatically extinguish fire when comes in contact with fire light in weight, compact in size extremely easy to use ABC powder type with automatic and manual operated CE certified with latest EN 1568-3:2008 & EN 1568- 4:2008. The product shall be covering 5 years warranty as per guideline.

17.33 Specification

The works shall be executed as per CPWD's General specification for Electrical Works, Part-I (Internal-2013); Part-II (External)-1994; Part-V (Wet Riser and Sprinkler System for Fire Fighting Installation)-2020, IE Rules, provision laid down in NBC – 2016, ECBC 2017 as amended up to date and latest building byelaws of local body / Local Municipal Corporation and various Indian Standards amended up to date, and as per direction of Engineer-in-Charge. The additional specifications are to be read above and in case of any variations; specifications given along the tender shall apply.

17.34 Indian standards

- a) IS-3844:1989 Code of practice for installation and Maintenance of Internal fire Hydrant and Hose reels
- b) IS-13039:2014 Code of practice for External hydrant system IS-8757 Glossary of terms associates Fire safety. IS: 554 Dimensions for pipe threads where pressure tight joints are required on the thread
- c) IS:638 Sheet rubber jointing and rubber
- d) IS-884 Specification for first-aid hose reel for the fighting.
- e) IS-901 Couplings, double male and double female, instantaneous pattern for fire fighting
- f) IS-902 Specification for suction hose couplings for firefighting purpose. IS-903 Specification for fire hose delivery couplings, branch pipe, nozzles, and nozzle spanner
- g) IS-904 Specification for two-way and three – way suction collection heads for firefighting purposes.
- h) IS-908 Specification for fire hydrant, stand post type IS-636 Non percolating flexible firefighting equipment.

- i) IS-1646 Code of practice for fire safety of buildings (general): Electrical installations.
- j) IS-5290 Specification for landing valves.
- k) IS-10221 Anti corrosion treatment for underground MS pipes.
- l) IS-8090 Specification for couplings, branch pipe, nozzle, used in hose reel tubing for firefighting.
- m) IS 2190:2010 Selection, installation, and maintenance of First-Aid Fire extinguishers.

17.35 Pre-commissioning

On completion of the installation of all pumps, piping, valves, pipe connections, insulation etc. the Contractor shall proceed as follows: Prior to start-up and hydraulic testing, the Contractor shall clean the entire installation including all fitments, pipe work and the installation and keep them in a new condition. All pumping systems shall be flushed and drained at least once through to get rid of contaminating materials. All pipes shall be rodded to ensure clearance of debris, cleaning and flushing shall be carried out in sections as the installation becomes completed. All strainers shall be inspected and cleaned out or replaced.

When the entire systems are reasonably clean, a pre-treatment chemical shall be introduced and circulated for at least 8 hours. Warning signs shall be provided at all outlets during pre-treatment. The pre-treatment chemical shall: remove oil, grease and foreign residue from the pipe work and fittings; Pre-condition the metal surfaces to resist reaction with water or air.

Establish an initial protective film; After pre-treatment, the system shall be drained and refilled with fresh water and left until the system is put into operation. Details and procedures of the pre-treatment shall be submitted for approval. Check all clamps, supports and hangers provided for the pipes. Fill up pipes with water and apply hydrostatic pressure to the system as given in the relevant section of the specification. If any leakage is found, rectify the same and retest the pipes.

17.36 Fire Protection System

- Check all hydrant valves by opening and closing: any valve found to be open shall be closed. Check all the piping under hydro test.
- Check that all suction and delivery connections are properly made for all pump sets. Check rotation of each motor after decoupling and correct the same if required.
- Test run each pump set.
- All pump sets shall run continuously for 8 hours (if required, temporary piping back to the tank to be made).

17.37 Testing and Commissioning of wet riser system

Pressurize the fire hydrant system by running the jockey pump and after it attains the shut-off pressure of the pump. Then, open bypass valve and allow the pressure to drop in the system. Check that the

jockey pump cuts-in and cuts-out at the preset pressure. If necessary, adjust the pressure switch for the jockey pump. Close by-pass valve. Open hydrant valve and allow the water to flow into the fire water tank in order to avoid wastage of water. The main fire pump shall cut-in at the preset pressure and shall not cutout automatically on reaching the normal line pressure. The main fire pump shall stop only by manual push button. However, the jockey pump shall cut-out as soon as the main pump starts.

Switch off the main fire pump and test check the Diesel engine driven pump in the same manner as the electrically driven pump. When the fire pumps have been checked for satisfactory working on automatic controls, open fire hydrant valves simultaneously and allow the hose pipe to discharge water into the fire tank to avoid wastage. Check each landing valve, male and female couplings, and branch pipes, for compatibility. Any fitting which is found to be incompatible and do not fit into the other properly shall be replaced by the Contractor. Each landing valve shall also be checked by opening and closing under pressure. Check all annunciations by simulating the alarm conditions at site.

17.38 Final acceptance tests

Following commissioning and inspection of the entire installation, and prior to issue of the Completion Certificate, the Contractor shall carry out final acceptance tests in accordance to a program to be agreed with the Engineer-in-charge.

If the results of the acceptance tests show that plant, systems, and/or equipment fail to perform to the efficiencies or other performance figures as given in this Specification, the Contractor shall adjust, modify and if necessary, replace the equipment without further payment in order that the required performance is obtained.

Where acceptance tests are required by the relevant Authorities having jurisdiction, these tests shall be carried out by the Contractor prior to the issue of Completion Certificate to the acceptance of the Authorities.

17.39 Rejection of installation / plant

Any item of plant or system or component which fails to comply the requirements of this specification in any respect whatsoever at any stage of manufacture, test, erection or on completion at site may be rejected either in whole or in part as considered necessary/appropriate. Adjustment and/or modification work as required by the Engineer-in-charge so as to comply the Authority's requirements and the intent of the Specification shall be carried out by the EPC Contractor at his own expense and to the satisfaction of the Authority/Engineer-in-charge.

After works have been accepted, the Contractor may be required to carry out assist in carrying out

additional performance tests as reasonably required by the Engineer-in-charge /Employer.

17.40 Handing over of documents

All testing and commissioning shall be done by the Contractor to the entire satisfaction of the Engineer-in-Charge and all testing and commissioning documents shall be handed over to him.

The Contractor shall also hand over all maintenance and operation manuals, all certificates, and all other documentation as per the terms of the contract to the Engineer-in-Charge.

17.41 Portable Fire Extinguishers

The EPC contractor shall be provided the fire extinguishers as per the requirement of NBC-2016, ECBC-2017, CEA etc. as required and drawings approved by Engineer-In-Charge for fully functional of the BQ housing Complex.

Providing and fixing of Carbon-dioxide type fire extinguishers consisting of welded M.S. cylindrical body, squeeze lever discharge valve fitted with pressure indicating gauge internal discharge tube 30 cm long high-pressure discharge hose, discharge nozzle, suspension bracket conforming to IS: 15683 finished externally with red enamel paint and fixed to wall with brackets complete with internal charge.
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Capacity 4.5 kg

Providing and fixing of ABC Power type fire extinguishers consisting of welded M.S. cylindrical body, squeeze lever discharge valve fitted with pressure indicating gauge internal discharge tube 30 cm long high-pressure discharge hose, discharge nozzle, suspension bracket. Conforming to IS: 15683 finished externally with red enamel paint fixed to wall with brackets complete with internal charge.
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Capacity 4.5 kg

17.42 ADDITIONAL CONSIDERATIONS:

- (a) All tests shall be made in the presence of the Engineer-in-Charge or his representative or any inspecting authority.
- (b) Water flow rates of all equipment and in pipe lines through valves shall be adjusted to design conditions. Complete results of adjustments shall be recorded and submitted.
- (c) Contractor shall ensure proper balancing of the hydraulic system and for the pipes / valves installed in his scope of work by regulating the flow rates in the pipe line by valve operation. The contractor shall also provide permanent Tee connection (plug) in water supply lines for ease of installing pressure gauge, temperature gauge & rotameters. Contractor shall also supply all required pressure gauge, temperature gauge & rota meter for system commissioning and balancing. The balancing shall be to the satisfaction of Consultant / Engineer-in- Charge.
- (d) Three copies of all test results shall be submitted to the Engineer in A4 size sheet paper in two

weeks after completion of the tests.

- (e) Hydrants in building shall be located /installed near staircase in such a fashion so that fire man takes the hydrant and then enters in fire affected area.
- (f) Fire Extinguishers shall be provided as per the direction of Engineer-in-charge.

17.43 All FHC (Fire Hose Cabinet) shaft doors, will be provided by EPC contractor as per the National Building Code (NBC) of India, must adhere to specific fire resistance and safety requirements to ensure they remain functional during a fire. These doors are crucial for providing access to firefighting equipment and maintaining a safe passage for emergency responders Key Requirements for FHC Shaft Doors as per NBC:

Fire Resistance Rating:

FHC shaft doors are typically required to have a minimum fire resistance rating of 120 minutes (2 hours), indicating they can withstand fire exposure for that duration while maintaining their integrity.

Construction:

The doors are usually constructed with fire-resistant materials like steel and incorporate features such as fire-resistant infills (like honeycomb core or mineral wool) and specialized hardware.

Compliance:

FHC shaft doors should comply with relevant Indian Standards (IS) such as IS 3614 and also align with NBC guidelines for fire safety.

18. SOLAR POWER GENERATION SYSTEM

The scope of work shall include designing, Planning, supplying, Installation, testing & commissioning of Solar PV Power Plant for the BQ housing complex based on required type and capacity of Solar PV Power generation.

Reference: Solar Power and Efficiency in Government Buildings, MOHUA

11) S r.No.	12) Description
13) 1.	14) Scope of grid connected Solar PV Power Plant describing the inclusion of Solar modules, DC and AC cabling, inverter, mounting arrangement required, earthing, lightning protection, metering arrangement etc. as required.
15) 2.	16) Details of available terrace area, location, Geo graphical coordinates, required efficiency for Solar modules and type, guarantee provisions etc.

18.1 SPECIFICATIONS FOR SOLAR ROOF TOP SYSTEM

The EPC Contractor shall carry out supply, installation, testing & commissioning of Direct Online Grid connected Solar Photo Voltaic Power system **375 kWp** as per GRIHA/ ECBC/NBC 2016 norms. The generated power will be directly connected to the Power grid/Distribution Panel/rising mains of respective Building/ Block for load sharing during day time. The average area requirement per kWp will be 10 sq.mtr on roof top. The Solar Photo Voltaic Power System shall be provided at terrace residential building blocks, guest house and community center building to obtain minimum GRIHA 3-star rating certification for the project. Solar cables shall be used for interconnection of equipment like Solar PV Panels and Inverter etc. Solar cables shall be suitable for outdoor installations. Given Solar quantity is minimum, EPC contractor recalculates solar requirement and approved by Engineer-in-charge as per GRIHA- 3-star rating.

A Grid Tied Solar Rooftop Photo Voltaic (SPV) system consists of SPV array, Module Mounting Structure, Power Conditioning Unit (PCUs) consisting of Maximum Power Point Tracker (MPPT), Inverter, and Controls & Protections, interconnect cables, Junction boxes, AC & DC Distribution boxes and switches etc. should conform to the BIS or IEC or international specifications.

Grid tied SPV system is without battery and should be designed with necessary features to supplement the grid power during day time. The EPC Contractor shall use only the PV modules that are empaneled to the OEM empanelment. The total Solar PV array capacity shall be assembled with minimum 545 Wp (with minimum of 24V) Mono Crystalline MNRE approved solar modules with 144 cells with minimum 20% Module Efficiency. The modules should be tested and certified as per relevant IEC standard as per MNRE guidelines and offered module shall have a power output warranty of 90% of the rated power for 10 years.

18.2 Brief scope of work:

The general scope under this contract includes design with site survey, manufacture, testing, inspection, transportation, erection, carrying out preliminary tests at site, commissioning, performance testing and handing all the equipment installed for grid connected SPV Power plant.

A Grid Tied Solar Rooftop Photo Voltaic (SPV) power plant consists of SPV array, Module Mounting Structure, Power Conditioning Unit (PCU) consisting of Maximum Power Point Tracker (MPPT), Inverter, Controls & protections, interconnect cables, bi-directional energy meter and switches. PV Array is mounted on a suitable structure. Grid tied SPV system is without battery and should be designed with necessary features to supplement the grid power during daytime. Components and parts used in the SPV power plants including the PV modules, metallic structures, cables, junction box, switches, PCUs etc., should confirm to the BIS, IEC, or international specifications, wherever such specifications are available and applicable.

Solar PV system shall consist of following equipment/components.

- Solar PV modules consisting of required number of Crystalline PV modules.
- Grid interactive Power Conditioning Unit with Remote Monitoring System.
- Mounting structures.
- Junction Boxes.
- Earthing and lightening protections.
- IR/UV protected Cables, pipes, and accessories.
- Solar Meter and Bi-directional Energy Meter

The PV array converts the light energy of the sun to DC power. The module mounting structure shall be used to hold module in position. The DC power shall be converted to AC to supply the AC loads connected like computers, lights, fans etc. within the premises and excess power can be exported to utility power grid and bought back when the PV system is not generating. Solar panels shall be integrated with the premises power supply from electricity authority.

General Requirements

Solar generating system shall support electrical load during day time and also export solar array generated power back to grid whenever the load demand is less than the solar power generation during holidays and weekends.

The PV system shall supply power close to unity power factor, thereby improving the grid quality and increasing its capacity.

Solar panels shall be installed on the shade free roof while the Inverter and distribution boards etc shall be housed inside the room provided by the Employer.

Array structure of PV yard and all electrical equipment such as inverters, Distribution Box etc. shall be grounded properly using adequate number of earthing kits.

Suitable marking shall be provided on the bus for easy identification.

The SPV panel shall be mounted facing south & tilted to an angle equal to the latitude where being used for optimum performance.

18.3 Quality and Workmanship

Solar PV modules are designed to last 25 years or more. It is therefore essential that all system components and parts, including the mounting structures, cables, junction boxes, distribution boxes and other parts also have a life cycle of at least 25 years. Therefore, all works shall be undertaken with the highest levels of quality and workmanship. During inspection special attention will be given to neatness of work execution and conformity with quality and safety norms. Non complete works will have to be redone at the cost of the Installer.

System Details

Sr. No.	Brief Description	Units	Make
1	SPV modules for a total capacity of 375KWp as per specification (Meeting IEC specifications, CEC & UL IEC61215, IEC61730)	In sets	Approved by Engineer-in-charge
2	SPV module mounting structure suitable for accommodating 375 KWp capacity SPV modules including foundation as per specifications on ground	As required	Approved by Engineer-in-charge
3	Grid Tie Inverter / Power conditioning unit (PCU) (Meeting IEC specifications, NEC &/ or CEC &/or UL 1741 & 1998 or IEEE1547 CERTIFICATION/SPECIFICATION)	Nos	Approved by Engineer-in-charge
4	DC Distribution Box	Nos	Approved by Engineer-in-charge
5	AC Distribution Box	Nos	Approved by Engineer-in-charge
6	Cables	Sets as required	Approved by Engineer-in-charge
7	Lightning arrester complete set as per specification	Sets as required	Approved by Engineer-in-charge
8	Earthing complete set as per specification	Sets as required	Approved by Engineer-in-charge

18.4 Solar PV Modules Mounting Structure

Hot dip galvanized MS mounting structures shall be used for mounting the modules/ panels/arrays. Each structure should have angle of inclination as per the site conditions to take maximum insolation.

MMS shall be made of hot dip Galvanized steel per ASTM A123. Necessary testing provision for MMS to be made available by Bidder at site. All bolts, nuts, panel mounting clamps fasteners shall be of stainless steel of grade SS 304 and must sustain the adverse climatic conditions.

The Mounting structure shall be so designed to withstand the speed of 150 km/hr for the wind zone of the location where a PV system is to be installed. It may be ensured that the design has been certified by a recognized Lab/ Institution in this regard. Suitable fastening arrangement such as grouting and calming should be provided to secure the installation against the specific wind speed.

The Structure should be made by on-site installation/removable concrete ballast made of PCC (1:2:3).

Structural material shall be corrosion resistant and electrolytically compatible with the materials used in the module frame, its fasteners, nuts and bolts. Aluminium structures also can be used which can withstand the wind speed of respective wind zone. Necessary protection towards rusting need to be provided either by coating or anodization.

The prospective Installer shall specify installation details of the solar PV modules and the support structures with layout drawings and array connection diagrams. The work shall be carried out as per the designs approved by the Engineer-in-charge.

The fasteners used should be made up of stainless steel. The structures shall be designed to allow easy replacement of any module. The array structure shall be so designed that it will occupy minimum space without sacrificing the output from the SPV panels. Regarding civil structures the EPC contractor need to take care of the load bearing capacity of the roof and need arrange suitable structures based on the quality of roof.

The minimum clearance of the structure from the roof level should be minimum 300 mm or as per the direction of Engineer-in-charge.

18.5 Solar Grid Tie Inverter / Power conditioning unit (PCU)

PCU shall supply the DC energy produced by array to DC bus for inverting to AC voltage using its MPPT (Maximum Power Point Tracking) control to extract maximum energy from solar array and produce 415V AC, 3 phase, 50 Hz to synchronize with the local grid.

MPPT Solar charge controller, inverter and associated control and protection devices etc. all shall be integrated into PCU.

PCU shall provide 3 phase, 415 $\pm$ 10% V, 50 $\pm$ 0.5 Hz supply on AC side with THD<3%.

Efficiency of PCU shall not be less than 97.5%

Degree of protection: IP 65/67

It shall have protection features such as over voltage, short circuit, over temperature etc. The inverter shall be efficient based on PWM with IGBT/MOSFET reliable power-based design.

The PCU shall be capable of complete automatic operation, including wake-up, synchronization & shut down. PCU shall have facility to display its basic parameters of the system.

The Inverter monitoring portal should be on supplier website with their logo.

No. of independent MPPT	Minimum 3
DC Power	$\geq 50\text{KW}$
No. of DC inputs	10-12
Max. input voltage	1000V
MPPT Voltage Range	250-960V
Rated input voltage	600V
Max. input current per string	12A
Rated output Power	$\geq 50\text{KW}$
Max. output current per string	80A
Nominal Grid Voltage	3/N/PE, 3P/E, 230/400Vac
Grid Voltage Range	310-480V
Nominal Frequency	50Hz
Total Harmonic Distortion	$< 3\%$
Power Factor	1 (adjustable ± 0.8)
Max. efficiency	$> 98.5\%$
European weighted efficiency	$> 98\%$
DC Reverse Polarity protection	Yes
DC Switch	Yes
DC SPD	Yes
Safety Protection	Anti-islanding, Ground fault monitoring
Operating Temperature	-25C to 60C
Supported Communication Interfaces	RS485, Wi-Fi, SD Card
Compliance	IEC62116, IEC61727, IEC61683, IEC60068 (1,2,14,30), IEC62109-1/2, IEC61000
Warranty of Inverter	Years

18.6 DC Distribution Box

DC distribution board shall be provided in between solar array and Inverter. DCDB shall have sheet from enclosure of dust & vermin proof. DCDB should be IP65 rated & suitable capacity MCBs be provided for controlling the DC power output to the PCU along with necessary surge arrestors. IP65

18.7 AC Distribution Box

An AC distribution box shall be mounted close to the solar grid inverter. AC Distribution Panel Board should be IP65 rated & control the AC power from PCU/inverter, and should have necessary surge arrestors. Interconnection from ACDB to mains at LT Bus bar while in grid tied mode.

It shall comprise the following components and cable terminations:

Incoming 4 core (single-phase/three-phase) cable from the solar grid inverter

Incomers MCB /MCCB & Outgoing MCCB/ACBs as per required capacity.

AC circuit breaker, 2-pole/4-pole

AC surge protection device (SPD), class1 plus 2

Multi-function Meter

Outgoing for the cable to the building electrical distribution board.

Cables

All cable shall be supplied conforming to IEC60227/IS694&IEC60502/ IS1554. Voltage rating: 1100V- AC, 1,500V-DC

For the DC cabling, XLPO insulated and sheathed, UV stabilized single core flexible copper cables shall be used. Multi-core cables shall not be used.

For the AC cabling, XLPE insulated and PVC sheathed single or multi-core flexible copper cables shall be used. Outdoor AC cables shall have a UV-stabilized outer sheath.

The total voltage drop on the cable segments from the solar PV modules to the solar grid inverter shall not exceed 2.0%.

The total voltage drop on the cable segments from the solar grid inverter to the building distribution board shall not exceed 2.0%

The DC cables from the SPV module array shall run through a UV- stabilized PVC conduit pipe of adequate diameter.

Cables and wires used for the interconnection of solar PV modules shall be provided with solar PV connectors (MC4) and couplers.

The minimum DC cable size shall be 4.0 mm² copper.

Cable from Inverter to ACDB should be copper of suitable size.

Cable from ACDB to LT Panel should be Copper of suitable size.

The following colour coding shall be used for cable wires:

DC positive: red (the outer PVC sheath can be black with a red line marking)

DC negative: black

AC single phase: Phase: red; neutral: black

AC three phase: Phases: red, yellow, blue; neutral: black

Earth wires: green

Cable conductors shall be terminated with tinned copper end-ferrules to prevent fraying and breaking of individual wires transited termination of the DC and AC cables at the Solar Grid Inverter shall be done as per instructions of the manufacturer, which in most cases will include the use of special connectors.

Descriptions	Specification
Solar Cable - From Module to Inverter	1 C X 4/6 Sq. mm Tinned Copper Cables, UV resistant, Solar Cable, TUV certified; 1800 Vdc
AC Cables (Inverter to AC Distribution Box)	4 C X 10/16/25/50/70/95/ suitable Sq. mm, Power cables Cu conductor, XLPE insulation, un-armoured, PVC outer sheath, 1100V (Flexible Type)
AC Cables (AC Distribution Box to Main LT Panels)	3.5 C X 25/35/50/70/95/120/300/400/500 / Suitable Cable Sq. mm, Power cable Al / Cu conductor, XLPE insulation, armoured, PVC outer sheath, 1100V

18.8 Earthing

The PV module structure components shall be electrically interconnected and shall be grounded.

Earthing shall be done in accordance with IS 3043-1986, provided that earthing conductors shall have a minimum size of 10.00 mm² copper cable. Sleeves to be provided for the earth conductors in underground connections to earth electrodes.

Separate dedicated and inter connected earth electrodes must be used for the earthing of the solar PV system support structure with a total earth resistance not exceeding 5 Ohm.

The earth electrodes shall have a precast concrete enclosure with a removable lid for inspection and maintenance. The entire earthing system shall comprise non-corrosive components.

The earthing shall be maintenance free type earthing and shall be done through at least 1.5-meter Electrode.

1CX10 /16/25/35 suitable Sqmm, Copper conductor Un-armoured Cable for Earthing AC & DC System.
Chemical gel Kit for Earthing
Lighting Arrestor suitable as per site requirement and as per approval of Engineer-in-charge.
25 X 5 GI Strip for Earthing Lighting Arrestor.

Installation & Commissioning: The installation shall be done by the supplier/manufacture who is responsible for its performance and direction of installation & ensures structural stability. The contractor shall conduct a detailed site assessment before quoting. The PV installer shall obtain data specific to the site, rather than relying on general data. While making foundation design, due consideration shall be given to weight of the module assembly, maximum wind speed at the site etc.

The supplier of panel & inverter should have a warehouse in the state of Assam to support the department in case of replacements. Inverter & PV panel supplier should have service centres in the state of Assam.

Documentation: The supplier shall provide easy-to-use illustrated installation and operation manual for easy installation and trouble-free usage. Manual shall contain complete system details such as array layout, schematic of the system, working principle etc, clear instruction on regular maintenance, trouble shooting of the solar generating system & emergency shutdown procedure.

The PV array converts the light energy of the sun to DC power. The module mounting structure shall be used to hold module in position. The DC power shall be converted to AC to supply the AC loads connected like computers, lights, fans etc. within the premises and excess power can be exported to utility power grid and bought back when the PV system is not generating. Solar panels shall be integrated with the premises power supply from electricity authority.

19. MASTER ANTENNA TELEVISION SYSTEM (MATV)

The system shall include supply, installation, testing & commissioning of MATV System (Wiring and conduiting) in residential building & guest house building as directed by Engineer-In-Charge. A central Antenna (or satellite dish) receives the signal, and then a network of cables and amplifiers it to each TV point of Residential buildings & guest house buildings.

The scope shall be limited to supply and installation of MATV conduiting & wiring with TV Outlet sockets as per CPWD specifications and approved GFC Drawings.

The Master antenna /Dish shall be provided individually & separately for all residential building

blocks. It shall be provided at the top most convenient point in respective building with a suitable room/space on the top most floor or terrace for housing the amplifier unit. From the amplifier rooms, conduits should be laid in recess/surface or in the shaft to facilitate drawing co-axial cable to individual rooms/areas. Suitable Tap Off boxes shall be provided in every room. The specifications of the same are listed below:

- a) **Conduiting-** Conduiting for MATV system shall be carried out in MS / Heavy Duty PVC Conduit.
- b) **Outlets -** All MATV outlets shall be provided with modular range of cover plate, box, and coaxial outlet. Cover plate shall match in shape & finish with other light and power accessories.
- c) **Junction Box:** Suitable size of GI box shall be provided for termination of conduit for MATV system.
- d) **MATV Line Amplifiers:** The MATV Amplifier shall be housed in a high frequency resistant Aluminium housing at the highest level (terrace level). The MATV Amplifier shall have an in-built variable equalizer and Alternator for site signal condition adjustments.
- e) **Coaxial Cables:** The coaxial cable shall be of wideband type.
- f) **Tap Off / Splitter:** These shall be of ultra-wide bandwidth and of hybrid type. These shall have a flat frequency response over the entire operating range. These shall have aluminium cast housing for high frequency radiation resistance. The Tap off shall be in one way, two way and four-way configurations.

20. TECHNICAL SPECIFICATION OF HVAC WORK

20.1. GENERAL

BQ Housing Complex in Duliagan, Assam, comprises a 8 Block of G+6 Residential Building with bedrooms, master bedrooms, study rooms, and living areas cooled by energy-efficient split AC units; a G+3 Guest House featuring guest rooms, a multipurpose hall, entrance lobby, and dining areas also using split ACs for individualized climate control; and a G+1 Community Centre with a gym, indoor sports room, kids' playroom, and hall, equipped with a Variable Refrigerant Volume (VRV) system for efficient cooling of high-occupancy spaces, ensuring year-round comfort at 22–24°C and 50–60% humidity, with sustainable design elements like low-E windows and proper electrical infrastructure to support both systems.

Design, Planning, Supply, Installation, Testing and Commissioning of HVAC work shall be carried out in accordance with General Specifications for Heating, Ventilation & Air-Conditioning (HVAC)-2024, Govt. Building Bylaws, NBC 2016, ECBC 2017, relevant IS Codes, International standards as applicable, in conjunction with DBR Indian Electricity Rules and Acts as amended up to date.

The requirement of the energy efficiency measures shall be complied as per GRIHA 3-star rating or better and ECBC-2017 as amended up to date. Clearances from local authorities as per norms shall be with the scope of work of EPC contractor.

20.2 SCOPE OF WORK

The Heating, Ventilation and Air-Conditioning (HVAC) system shall comprise of following:

- a. Variable Refrigerant Volume (VRV)
- b. HVAC Panel
- c. Split Type Unit for guest house.
- d. Ceiling Suspended Ductable / DX AHU/TFA
- e. High wall units
- f. Cassette units
- g. Variable Frequency Drives
- h. Motor control center.
- i. Thermostats for AHUs
- j. Refrigerant and condensate drain piping with insulation inclusive of all valves and fittings, copper pipe, channel, threaded rod supports, vibration isolators, strainers etc.
- k. Sheet metal ducts, external insulation, acoustic lining, canvas connections, volume control dampers, smoke& fire dampers threaded rod supports as required.
- l. Supply and return air registers, grills, and diffusers.
- m. Insulation of pipes with protective coating.
- n. Vibration isolators.
- o. Automatic controls and instruments viz. breaker, CVT, temperature and pressure gauges with provision of BMS integration.
- p. Fire sealant around ducts / pipes and cables when passing through wall / floor and other openings.
- q. Wiring and earthing from MCC panels to various refrigeration, air conditioning and mechanical ventilation equipment, control wiring and interlocking.
- r. Cutting holes, chases and the like through all types of non-structural walls, and finishing for all services crossings, including sealing, frame work, fire proofing, providing sleeves, cover plates, making good structure and finishes to an approved standard.
- s. Balancing, testing, and commissioning of the entire HVAC and mechanical ventilation installation.
- t. Test reports, list of recommended spares, as-installed drawings, operation, and maintenance manual for the entire HVAC installation.
- u. Complete co-operation with independent commissioning agent and corrective actions on snag list points prepared by such independent commissioning agent.
- v. Multiple trainings of technical staff.
- w. Variable Refrigerant Volume (VRV) type Air-Conditioning System included in these specifications is by means of scroll type compressors for ODU's, high wall type units, cassette units, duct-able units, DX-TFA units and DX AHUs. Therefore, only vapor compression type refrigeration equipment's are covered herein.

- x. These General Specifications shall be supplemented with tender specifications as may be required for work. The tender specifications, wherever they differ from these General Specifications, shall have not over-riding value and shall be followed for that work.

20.3 WORKS TO BE DONE BY THE CONTRACTOR

Unless otherwise mentioned in the tender documents, the following works shall be done by the contractor and therefore, their cost shall be deemed to be included in their tendered cost-whether specifically indicated in the schedule of work or not: -

- i) Foundations for equipment including foundation bolts and vibration isolation spring/pads,
- ii) Suspenders, brackets, and floor/ wall supports for suspending/supporting ducts and pipes,
- iii) Suspenders and/or cable trays for laying the cables,
- iv) Sealing of all floor slab/ wall openings provided by the Client or contractor for pipes and cables, from fire safety point of view, after laying of the same.
- v) Painting of all exposed metal surfaces of equipment and components with appropriate colour as specified.
- vi) Making openings in the walls/ floors/ slabs or modification in the existing openings wherever provided for carrying pipe line, ducts, cables etc.
- vii) Providing wooden/ metallic frames for fixing grills/ diffusers.
- viii) Under deck insulation of top floor slab, if top floor is air-conditioned and if specified in schedule of work.
- ix) Making good all damages caused to the structure during installation and restoring the same to their original finish.
- x) Any other work which is not mentioned above but required for completeness of work / system shall be in the scope of bidder.

20.4 MACHINERY FOR ERECTION

All tools and tackles required for unloading / handling of equipment and materials at site, their assembly, erection, testing and commissioning shall be the responsibility of the contractor.

20.5 QUALITY OF MATERIALS AND WORKMANSHIP

- The components of the installation shall be of such design so as to satisfactorily function under all conditions of operation.
- The entire work of manufacture/fabrication, assembly and installation shall conform to sound engineering practice. The entire installation shall be such as to cause minimum transmission of noise and vibration to the building structure.
- All equipment and materials to be used in work shall be manufactured in factories of good

repute having excellent track record of quality manufacturing, performance and proper after sales service.

20.6 TERMINOLOGY

20.6.1 Air Conditioning

The process of treating air so as to control simultaneously its temperature, humidity, purity, distribution and air movement and pressure to meet the requirements of the conditioned space.

20.6.2 Coil by Pass

The coil by pass factor is the percentage of air passing over the coil that is not affected by the coil.

20.6.3 SHGC

The SHGC is the fraction of incident solar radiation admits through a window both directly and absorbed and subsequently released. SHGC is express as a number between 0 to 1. The lower a window solar heat coefficient the less solar heat is transmitted.

20.6.4 Relative Humidity

Ratio of the actual water vapours in the air as compared to the maximum amount of water that may be contained at its dry bulb temperature. When the air is saturated, dry bulb, wet bulb and dew point temperatures are all equal.

20.6.5 Dry-Bulb Temperature

The temperature of air as registered by an ordinary thermometer.

20.6.6 Wet-Bulb Temperature

The temperature registered by a thermometer whose bulb is covered by a wetted wick and exposed to a current of rapidly moving air.

20.6.7 Dew Point Temperature

The temperature at which condensation of moisture begins when the air is cooled at same pressure.

20.6.8 Humidity

It is the amount of water vapour present in a certain volume of air.

20.6.9 Enthalpy

A thermal property indicating the quantity of heat in the air above an arbitrary datum in kilo

joules per kg of dry air (or in Btu per pound of dry air).

20.6.10 Psychrometry

Psychrometry is the science involving thermo dynamic properties of moist air and the effect of atmospheric moisture on materials and human comfort. It also includes methods of controlling thermal properties of moist air.

20.6.11 Psychrometric Chart

A Psychrometric chart graphically represents the thermodynamic properties of moist air. If two properties are known, all the other properties can be determined with the help of psychometric chart.

20.6.12 Positive Ventilation

The supply of outside air by means of a mechanical device, such as a fan.

20.6.13 Atmospheric Pressure

The pressure of air exerted on the surface of earth by the atmospheric column is called atmospheric pressure. At sea level, the atmospheric or barometric pressure is 760mm column of mercury (29.92 in Hg/ 406.8-inch water column/ 101.325 Kpa).

Generally atmospheric pressure is used as a datum for indicating the system pressures in air-conditioning and accordingly, pressures are mentioned above the atmospheric pressure or below the atmospheric pressure considering the atmospheric pressure to be zero. A 'U' tube manometer will indicate zero pressure when atmospheric pressure is measured.

20.6.14 Indoor Air Quality (IAQ)

Indoor air quality refers to the nature of conditioned air that circulates throughout the space/ area where one works or lives, i.e. the air we breathe when we are indoors. IAQ refers not only to comfort which is affected by temperature, humidity, and odours but also to harmful biological contaminants and chemicals present in the conditioned space.

Bad Indoor Air Quality can be a serious health hazard. Carbon dioxide (CO₂) has been recognized by ASHRAE as the surrogate ventilation index or the only measurable variable for the indoor air contaminants.

20.6.15 Thermal Transmittance

Thermal transmission through unit area of the given building unit divided by the temperature difference between the air or some other fluid on either side of the building unit in 'steady state' conditions.

20.6.16 Shade Factor

The ratio of instantaneous heat gain through the shading device to that through a plain glass sheet of 3 mm thickness.

20.6.17 Sensible Heat Factor (SHF)

Sensible heat factor is the ratio of sensible heat to total heat, where total heat is the sum of sensible and latent heat.

20.6.18 Supply Air

The air that has been passed through the conditioning apparatus and taken through the duct system and distributed in the conditioned space is termed as supply air.

20.6.19 Return Air

The air that is collected from the conditioned space and returned to the conditioning equipment is termed as return air.

20.6.20 Re-Circulated Air

The return air that has been passed through the conditioning apparatus before being re-supplied to the space is called re-circulated air.

20.6.21 Duct System

A continuous passageway for the transmission of air which in addition to the ducts, may include duct fittings, dampers, plenums, and grilles & diffusers.

20.6.22 Plenum

An air compartment or chamber to which one or more ducts are connected and which forms part of a distribution system.

20.6.23 Supply and Return Air Grilles & Diffusers

Grilles and diffusers are the devices fixed in the air-conditioned space for distribution of conditioned supply air and return of air collected from the conditioned space for re-circulation.

20.6.24 Fire Separation Wall

The wall provides complete separation of one building from another or part of a building from another part of the same building to prevent any communication of fire of any access or heat transmission to wall itself which may cause or assist in the combustion of materials of the side opposite to that portion which may be on fire.

20.6.25 Refrigerant

The fluid used for heat transfer in a refrigerating system, which absorbs heat at a low temperature and low pressure of the fluid and rejects heat at a higher temperature and higher pressure of the fluid, usually involving changes of state of the fluid.

20.6.26 Global Warming Potential (GWP)

Global Warming can make our planet and its climate less hospitable and more hostile to human life. It is, therefore, necessary to reduce emission of greenhouse gases such as Co₂, Sox, Nox and refrigerants. The potential of a refrigerant to contribute to Global Warming is called its GWP. Long atmospheric life time of refrigerants results in Global Warming unless the emissions are controlled.

20.7 REFERENCE STANDARDS

- National Building Code of India -2016
- ECBC -2019
- Relevant Bureau of Indian Standards (BIS) Codes
- ASHRAE (American Society of Heating Refrigerating & Air Conditioning Engineers) Handbooks
 - Application -2019
 - Refrigeration -2018
 - Fundamentals -2017
 - Systems & Equipment -2016
- ISHRAE (Indian Society of Heating Refrigerating & Air Conditioning Engineers) HVAC Handbook-2017
- Duct construction standards as per relevant BIS codes & SMACNA standards.
- Air filters as per ASHRAE Standard 52.1 -1992
- Indoor Air Quality as per ASHRAE Standard 62.1 – 2019
- ASHRAE Standard 90.1-2007
- Motors, cabling, wiring and accessories as per BIS codes / IE Rules / IS codes.
- Recommended practices of Equipment suppliers.

20.8 ABBREVIATIONS

DBT Dry Bulb Temperature

WBT Wet Bulb Temperature

BF By Pass Factor

DP Dew Point

SHF Sensible Heat Factor

RH Relative Humidity

T Temperature

NBC National Building Code

SHGC Solar Heat Gain Coefficient

U Value Over All Heat Transfer Coefficient

CFM Cubic Feet Per Minute

DX Direct Expansion

F Degree Fahrenheit

20.9 BASIS OF DESIGN

20.9.1 OUTDOOR DESIGN CONDITION

Site Location: Duliajan, Assam.

Geographic location: Latitude 17°27' N & Longitude 78°28' E

Altitude: 545 M above mean sea level.

Summer

Dry Bulb Temperature: 102.9 Deg. F

Wet Bulb Temperature: 71.9 Deg. F

Relative Humidity: 21.7%

Monsoon

Dry Bulb Temperature: 92.9 Deg. F

Wet Bulb Temperature: 78.1 Deg. F

Relative Humidity: 52.2%

Winter

Dry Bulb Temperature: 56.80Deg. F

Mean coincident Wet Bulb Temperature: 52.8Deg. F

Relative Humidity 77.3%

20.9.2 INDOOR DESIGN CONDITIONS

Following indoor design conditions for centrally air-conditioned space shall be adopted.

SPACE	INDOOR CONDITIONS		NOISE CRITERIA	FRESH AIR
	Temp. (°F)	Relative Humidity	NC	AS PER ASHRAE 62.1-2019
DORMITORY	73.4±2°F	55%	40	5 CFM / person + 0.06 CFM / Sq. Ft.
DINNING ROOM	73.4±2°F	55%	40	7.5 CFM / person + 0.18 CFM / Sq. Ft.
ROOM-1	73.4±2°F	55%	40	5 CFM / person + 0.06 CFM / Sq. Ft.
MULTIPURPOSE RM	73.4±2°F	55%	40	7.5 CFM / person + 0.18 CFM / Sq. Ft.
KIDS SPORTS ROOM	73.4±2°F	55%	40	0.3 CFM / Sq. Ft.
INDOOR SPORTS ROOM	73.4±2°F	55%	40	0.3 CFM / Sq. Ft.
INDOOR GYM ROOM	73.4±2°F	55%	40	0.3 CFM / Sq. Ft.
MEETING HALL	73.4±2°F	55%	40	5 CFM / person + 0.06 CFM / Sq. Ft.
COMMON HALL	73.4±2°F	55%	40	5 CFM / person + 0.06 CFM / Sq. Ft.
BEDROOM	73.4±2°F	55%	40	5 CFM / person + 0.06 CFM / Sq. Ft.
LIVING RM	73.4±2°F	55%	40	5 CFM / person + 0.06 CFM / Sq. Ft.
Study Room	73.4±2°F	55%	40	5 CFM / person + 0.06 CFM / Sq. Ft.

20.9.3 MECHANICAL VENTILATION

Toilet	10 ACPH
Storage Rooms	6 ACPH
Pump Room	25 ACPH
Smoke exhaust in all public area, Ground Floor & above	12 ACPH (as per NBC -2016)

20.9.4 (Assembly U value)

CLIMATE ZONE	Description	Unit		Remarks
		Maximum U-factor (BTU/ Hr. Sq. Ft. Deg. F)	Maximum SHGC	
Composite	Façade	1.13	0.56	Single glass
	Wall Exposed to Sun	0.35	-	230 mm thick
	Wall Partition	0.40	-	230 mm thick
	Floor Partition	0.40	-	150 mm thick
	Roof Exposed to Sun	0.12	-	Insulated Roof
	Roof Shaded	0.07	-	Insulated Roof

Note: The above values are to be fine-tuned with energy simulation outputs and based on the actual selection of Glass by Architect or Client at the detailed design stage.

20.9.5 Occupant Density/Lighting Load/Equipment Load:

SPACE	Occupancy Sq. Ft/Person	Equipment Load W/Sq. Ft	Lighting Load W/Sq. Ft
DORMITORY	63	1.0	1.2
DINNING ROOM	23	1.0	1.2
ROOM-1	132	1.0	1.2
MULTIPURPOSE RM	31	1.0	1.2
KIDS SPORTS ROOM	32	1.0	1.4
INDOOR SPORTS ROOM	44	1.0	1.3

INDOOR GYM ROOM	34	1.5	1.4
MEETING HALL	15	1.5	1.4
COMMON HALL	33	1.0	1.5
BEDROOM	70	1.5	1.1
LIVING RM	64	1.0	1.1
STUDY RM	36	1.0	1.1

20.10 DESIGN PARAMETERS OF EQUIPMENT

20.10.1 Variable Refrigerant Volume (VRV)

Performance rating of the VRV units shall be based on following design parameters:

Temperature of Ambient air: 95.0 deg. F

Refrigerant for VRV Units: R-410a

Maximum Allowable IKW / TR @ 100% Load at AHRI: 0.75

COP at AHRI@100% Load: 3.4

20.10.2	Air Handling Units and Its Components		
	Maximum velocity across pre-filters	:	110 M / Min.
	Maximum face velocity across cooling coils	:	150 M / Min.
	Maximum fan outlet velocity for fans above 300 mm dia.	:	550 M / Min.
	Maximum fan speed for fans up-to 300 mm dia.	:	1450 RPM
	Maximum fan speed for fans above 300 mm dia.	:	1000 RPM
	Maximum motor speed	:	1450 RPM
20.10.3	Centrifugal Fans		
	Maximum fan outlet velocity for fans up-to 450 mm dia.	:	550 M / Min.
	Maximum fan outlet velocity for fans above 450 mm dia.	:	700 M / Min.
	Maximum fan speed for fans up-to 450 mm dia.	:	1450 RPM
	Maximum fan speed for fans above 450 mm dia.	:	1000 RP

20.10.4 Design parameter for selection of Axial Flow Fans shall be:

Maximum fan outlet velocity (Smoke)	:	1050 M/Min
Maximum fan speed (Smoke)	:	1450 RPM
Maximum fan outlet velocity (Normal)	:	550 M/Min
Maximum fan speed (Normal)	:	1000 RPM

20.10.5 Design parameter for selection of Air Washer shall be:

Maximum fan outlet velocity for fans up to 450 mm dia.	:	550 M/Min
Maximum fan outlet velocity for fans above 450 mm dia.	:	700 M/Min
Maximum fan speed		
a. Fan above 450 mm dia.	:	1000 RPM
b. Fans up to and including 450 mm dia.	:	1450 RPM
Maximum fan motor speed	:	1450 RPM
Maximum Velocity Across Cooling Pad	:	500 RPM
Efficiency of Cooling Pad	:	90%
Minimum Thickness of Cooling Pad	:	200MM

20.10.6	Duct Design		
	Maximum flow velocity		M/Sec.
	Maximum Friction		inch per 100 Ft
	Maximum Velocity at supply air outlet		

20.11 PROPOSED HVAC SYSTEM & HEAT LOAD SUMMARY

(A) HEAT LOAD SUMMARY OF COMMUNITY CENTRE

HEAT LOAD SUMMARY OF COMMUNITY CENTRE																							
S. No.	Description	Area (Sqft)	Height(Ft)	Occ. (Sqft/Per)	Equip. Load (W/Sqft)	Light Load (W/Sqft)	No. of Persons	Fresh Air (CFM)	Coolin g Load (TR)		Heatin g Load (KW)	Deh. CFM (Summer)	IDU Selection						ODU Selection				
									Summer	Monsoon	Winter		CFM	TR	Type	Motor (KW)	Nos	Total (KW)	Cap. HP	Nos.	TOTAL HP	INPUT POWER KW	TOTAL (KW)
A	GROUND FLOOR																						
1	COMMON HALL	3368	11.2	37	1.0	1.5	90	652	13.83	11.21	13.50	6098	812	1.6	4-WAY CASSETTE	0.2	10	2	26	1	26	19.5	20
2	TFA	0	11.2	0	0.0	0	0	652	1.98	3.46	-6.00	101	635	4	TFA UNIT	0.4	1	0.40					
3	SOCIETY OFFICE	180	11.2	45	1.0	1.1	4	31	1.24	0.97	-1.20	544	794	2.00	SPLIT HIWALL UNIT	2.60	1	2.6					
4	ATM	114	11.2	57	1500.0	1	2	17	2.38	2.11	0.50	1154	TENANT SCOPE										
	Total	3662					96	1352	19.4	17.8	6.8	7896					12	5	26	1	26	19.5	20
B	FIRST FLOOR																						
1	KIDS SPORTS ROOM	568	11.6	33	1.0	1.4	17	170	4.27	3.93	1.80	1486	865	2.60	4-WAY CASSETTE	0.20	2	0.4	28	1	28	21	21
2	INDOOR SPORTS ROOM	833	11.6	49	1.0	1.3	17	250	5.50	5.05	0.30	1991	1183	3.20	4-WAY CASSETTE	0.30	2	0.6					
3	INDOOR GYM ROOM	908	11.6	36	1.5	1.4	25	272	7.51	7.01	2.80	2774	830	2.00	4-WAY CASSETTE	0.20	4	0.8					
4	MEETING HALL	326	11.6	18	1.5	1.4	18	110	3.01	2.73	0.80	1194	812	1.60	4-WAY CASSETTE	0.20	2	0.4					
	Total	2635					77	802	20.3	18.7	5.7	7445					10	2		1	28		21
	G.Total	6297					173	2154	40	36	13	15341					22	7		2	54		41

(B) HEAT LOAD SUMMARY OF GUEST HOUSE BUILDING

HEAT LOAD SUMMARY-GUEST HOUSE BLOCK																		
S. No.	Description	Area (Sqft)	Height(Ft)	Occ. (Sqft/Per)	Equip. Load (W/Sqft)	Light Load (W/Sqft)	No. of Persons	Fresh Air (CFM)	Cooling Load (TR)		Heating Load (KW)	Deh. CFM (Summer)	HVAC Unit Selection					
									Summer	Monsoon			Winter	CFM	TR	Type	Motor (KW)	Nos
A	GROUND FLOOR																	
1	RECEPTION AREA	339	10.3	56	1.0	0.5	6	50	2.77	2	-1.80	1266	540	1.50	SPLIT HIWALL	2.0	2	3.9
2	DORMITORY-1	440	10.3	63	1.0	1.2	7	61	2.53	2	-0.90	1121	540	1.50	SPLIT HIWALL	2.0	2	3.9
3	DORMITORY-2	440	10.3	63	1.0	1.2	7	61	2.48	1.9	-1.00	1095	540	1.50	SPLIT HIWALL	2.0	2	3.9
4	DINNING ROOM	902	10.3	23	1.0	1.2	40	462	7.34	7	-1.80	2677	540	1.50	SPLIT HIWALL	2.0	4	7.8
	Total	2121					60	635	15.1	12.8	-5.5	6159					10	19.5
B	FIRST FLOOR																	
1	DORMITORY-3	440	10.3	63	1.0	1.2	7	61	2.53	2	-0.90	1121	540	1.50	SPLIT HIWALL	2.0	2	3.9
2	DORMITORY-4	440	10.3	63	1.0	1.2	7	61	2.48	1.9	-1.00	1095	540	1.50	SPLIT HIWALL	2.0	2	3.9
3	ROOM-1	264	10.3	132	1.0	1.2	2	26	1.59	1.1	-1.20	746	794	2.00	SPLIT HIWALL	2.6	1	2.6
4	ROOM-2	264	10.3	132	1.0	1.2	2	26	1.49	1.1	-0.90	695	794	2.00	SPLIT HIWALL	2.6	1	2.6
5	ROOM-3	209	10.3	104	1.0	1.2	2	23	1.46	1	-1.50	682	794	2.00	SPLIT HIWALL	2.6	1	2.6
6	MULTIPURPOSE RM	629	10.3	31	1.0	1.2	20	263	4.52	4.2	-1.30	1720	540	1.50	SPLIT HIWALL	2.0	4	7.8
7	STAFF ROOM	207	10.3	103	1.0	1.2	2	22	1.45	1	-1.40	681	794	2.00	SPLIT HIWALL	2.6	1	2.6
	Total	2452					42	483	15.5	12.2	-8.2	6741					12	26.0
C	SECOND FLOOR																	
1	ROOM-4	264	10.3	132	1.0	1.2	2	26	1.69	1.2	-1.70	795	794	2.00	SPLIT HIWALL	2.6	1	2.6
2	ROOM-5	264	10.3	132	1.0	1.2	2	26	1.59	1.1	-1.20	746	794	2.00	SPLIT HIWALL	2.6	1	2.6
3	ROOM-6	264	10.3	132	1.0	1.2	2	26	1.59	1.1	-1.20	746	794	2.00	SPLIT HIWALL	2.6	1	2.6
4	ROOM-7	264	10.3	132	1.0	1.2	2	26	1.49	1.1	-0.90	695	794	2.00	SPLIT HIWALL	2.6	1	2.6
5	ROOM-8	209	10.3	104	1.0	1.2	2	23	1.46	1	-1.50	682	794	2.00	SPLIT HIWALL	2.6	1	2.6
6	ROOM-9	264	10.3	132	1.0	1.2	2	26	1.70	1.2	-1.70	802	794	2.00	SPLIT HIWALL	2.6	1	2.6
7	ROOM-10	264	10.3	132	1.0	1.2	2	26	1.56	1.1	-1.00	732	794	2.00	SPLIT HIWALL	2.6	1	2.6
8	ROOM-11	264	10.3	132	1.0	1.2	2	26	1.56	1.1	-1.00	732	794	2.00	SPLIT HIWALL	2.6	1	2.6
9	ROOM-12	264	10.3	132	1.0	1.2	2	26	1.56	1.1	-1.00	732	794	2.00	SPLIT HIWALL	2.6	1	2.6
10	ROOM-13	207	10.3	103	1.0	1.2	2	22	1.45	1	-1.40	681	794	2.00	SPLIT HIWALL	2.6	1	2.6
	Total	2524					20	251	15.7	11.1	-12.6	7343					10	26.0
D	THIRD FLOOR																	
1	ROOM-14	264	10.3	132	1.0	1.2	2	26	1.69	1.2	-1.70	795	794	2.00	SPLIT HIWALL	2.6	1	2.6
2	ROOM-15	264	10.3	132	1.0	1.2	2	26	1.59	1.1	-1.20	746	794	2.00	SPLIT HIWALL	2.6	1	2.6
3	ROOM-16	264	10.3	132	1.0	1.2	2	26	1.59	1.1	-1.20	746	794	2.00	SPLIT HIWALL	2.6	1	2.6
4	ROOM-17	264	10.3	132	1.0	1.2	2	26	1.49	1.1	-0.90	695	794	2.00	SPLIT HIWALL	2.6	1	2.6
5	ROOM-18	209	10.3	104	1.0	1.2	2	23	1.46	1	-1.50	682	794	2.00	SPLIT HIWALL	2.6	1	2.6

6	ROOM-19	264	10.3	132	1.0	1.2	2	26	1.70	1.2	-1.70	802	794	2.00	SPLIT HIWALL	2.6	1	2.6
7	ROOM-20	264	10.3	132	1.0	1.2	2	26	1.56	1.1	-1.00	732	794	2.00	SPLIT HIWALL	2.6	1	2.6
8	ROOM-21	264	10.3	132	1.0	1.2	2	26	1.56	1.1	-1.00	732	794	2.00	SPLIT HIWALL	2.6	1	2.6
9	ROOM-22	264	10.3	132	1.0	1.2	2	26	1.56	1.1	-1.00	732	794	2.00	SPLIT HIWALL	2.6	1	2.6
10	ROOM-23	207	10.3	103	1.0	1.2	2	22	1.45	1	-1.40	681	794	2.00	SPLIT HIWALL	2.6	1	2.6
	Total	2524					20	251	15.7	11.1	-12.6	7343					10	26.0
	Grand Total	9622					142	1621	61.9	47.3	-38.9	27586					42	97.5

20.12 INSPECTION AND TESTING

20.12.1 Initial Inspection & testing

- i) Initial inspection of materials & equipment at manufacturer's works as per requirements shall be done by the engineer-in-charge or his representative. For item/equipment requiring initial inspection at manufacturer's works, the contractor will intimate the date of testing of equipment at the manufacturer's works before dispatch. The contractor shall give sufficient advance notice regarding the dates proposed for such tests to the department's representative(s) to facilitate his presence during testing. RITES Consultant / Engineer-in-charge or his representative at his discretion may witness such testing. Equipment will be inspected at the manufacturer/authorized dealer's premises, before dispatch to the site by the contractor.

20.12.2 Final Inspection & Testing

Final Inspection & testing will be done by the RITES Engineer-in-Charge/Client or his representative as per technical data of the system installed.

20.12.3 Safety measures

All equipment shall incorporate suitable safety provisions to always ensure safety of the operating personnel. The initial and final inspection reports shall bring out explicitly the safety provisions incorporated in each equipment.

- 20.12.4 Electrical works including electrical motors, panels etc. shall be for inspected necessary clearance obtained from relevant statutory body by HVAC contractor at their own cost.

20.13 RUNNING IN PERIOD & DATE OF ACCEPTANCE

- i) After the installation work has been completed by the contractor, he will conduct tests and adjust as may be necessary to satisfy himself that the plant including low side equipment is capable of continuous running. There after he will offer to the department a running-in period of 7 days subject to a minimum aggregate of 120 hrs at his own cost. The duty cycle of the plant during this running in period shall be same as that specified in the tender documents. In case of multiple compressor installations, all the compressors should be run by rotation. The plant will be operated and a log of all parameters will be maintained during this period. The contract or will be free to carry out necessary adjustments etc. during this period without stopping the plant. Record of inside conditions will be made during this period to check the same are as per NIT requirements. The plant will be said to have successfully completed the running-in-period, if no break down or abnormal/ unsatisfactory operation of any machinery occurs during this period. After this the plant will be made available for beneficial use. After the plant has operated without any major break down/ trouble and inside conditions are maintained as per requirements for the above specified running in period, it shall be taken over by the Client subject to

guarantee clause mentioned below. This date of taking over of plant after trouble free operation during the running in period shall be the date of acceptance.

- ii) Any loss of refrigerant or oil during the running in period shall be made good by the contractor free of charge.
- iii) Capacity test of the chilling unit & other major equipment shall be carried out as and when conditions become stabilized as per details given in the prescribed Performa.
- iv) Seasonal testing may be carried out as & when outside conditions become suitable.

20.14 GUARANTEE

- i) The contractor shall guarantee the complete system to maintain the specified conditions under all conditions of ambient and internal loads subject to the condition that designed outside conditions & designed internal loads are not exceeded. Also, the inlet / outlet temperatures at the specified flow of refrigerant in the Split/VRV unit shall be guaranteed.
- ii) All equipment shall be guaranteed for a period of 12 months from the date of acceptance and taking over of the installation by the Client against unsatisfactory performance and/or breakdown due to defective design, material, manufacture, workmanship, or installation. The equipment or component or any part thereof so found defective during the guarantee period shall be repaired or replaced free of cost to the satisfaction of the Engineer-in-charge. In case it is felt by the Client that undue delay is being caused by the contractor in doing this, the same will be got done by the Client at the risk & cost of the contractor. The decision of Engineer-in-charge in this regard shall be final
- iii) Any leakage of refrigerant and/or oil due to defective design, manufacture, workmanship, or installation during the guarantee period shall be made good by the contractor free of charge.

20.15 TENDER DRAWINGS, DRAWINGS FOR APPROVAL & COMPLETION DRAWINGS

20.15.1 Tender Drawings

The drawings appended with the tender documents are intended to show the areas to be conditioned, space allotted for various equipment, tentative duct, cable, and pipe routes. The equipment offered shall be suitable for installation in the spaces shown in these drawings.

20.15.2 Drawings for approval on award of the work

The contractor shall prepare & submit three sets of following drawings and get them approved from the Engineer-in-charge before the start of the work. The approval of

drawings however does not absolve the contractor not to supply the equipment/materials as per agreement, if there is any contradiction between the approved drawings and agreement.

- a) Lay out drawings of the equipment to be installed in various rooms such as VRV ODU layout and other equipment.
- b) Drawings including section, showing the details of erection of entire equipment including their foundations, water basin for the air washers, etc.
- c) Plumbing drawings showing the layout of entire piping, dia. & length of pipes, valves and isometric drawings showing connections to various equipment.
- d) Ducting drawings showing sizes, locations of dampers, grilles & diffusers.
- e) Electrical wiring diagrams for all electrical equipment and controls including the sizes and capacities of the various cables and equipment,
- f) Dimensioned drawings of all electrical and control panels,

Drawings showing the details of all insulations and vapour barrier works,

Drawings showing detail of supports for pipes, cable trays, ducts etc.

any other drawings relevant to the work.

20.15.3 Completion Drawings

Three sets of the following laminated drawings shall be submitted by the contractor while handing over the installation to the Client. Out of this one of the sets shall be laminated on a hard base for display. In addition, one set will be given on compact disc.

- a) installation drawings giving complete details of all the equipment including their foundations,
- b) Indoor Units installation drawings.
- c) Plumbing layout drawings including insulation giving sizes and lengths of all the pipes and the sizes and locations of all pipes, and including isometric drawings for the entire piping layout and Sizes of all the pipe connections to the various equipment.
- d) Duct layout drawings with their sizes and locations, and sizes and locations of all dampers, grills & diffusers.
- e) Line diagram and layout of all electrical control panels giving switchgear ratings and their disposition, cable feeder sizes and their layout,
- f) Control wiring drawings with all control components and sequence of operations to explain the operation of control circuits,

20.15.4 AFTER SALES SERVICES

The contractor shall ensure adequate and prompt after sales service in the form of maintenance, spares, and personnel as and when required and shall minimize the breakdown period. In case of equipment supplied by other manufacturers the firm shall furnish a guarantee from the manufacturer for the same before the plant is taken over.

20.15.5 DOCUMENTS TO BE FURNISHED ON COMPLETION OF INSTALLATION

Three sets (both hard and soft copies) of the following documents shall be furnished to the department by the contractor on completion of work: -

- a) Completion drawings
- b) 3 sets of manufacturer's technical catalogues of all equipment and accessories.
- c) Operation and maintenance manual of all major equipment detailing all adjustments, operation, and maintenance procedure.

20.15.6 TRAINING

Upon commissioning and final handover of the installation, the Contractor shall submit (within 4 weeks) 3 copies of operating instructions, maintenance and service manuals, part lists and all final drawings and diagrams, indexed and bound together in hard cover ring binder.

The Contractor shall conduct a training program for designated Employer's personnel. These courses shall be carried out during normal office hours and not be less than 1 month's duration. The date of commencement of training shall be mutually agreed upon and, in any case, shall be within two weeks of handover of installation.

The training program shall cover all operating and maintenance aspects of the system, inclusive of detailed explanation and demonstration of each and every piece of equipment and an overview of the system network.

All instruction manuals, tools, transportation, etc. association with the training program shall be provided by the Contractor. Such cost shall be deemed to have been included in the EPC CONTRACTOR program.

20.16 VARIABLE REFRIGENAT VOLUME SYSTEM (VRV/VRF)

General

The scope of modular type Variable Refrigerant Flow/Variable Refrigerant Volume (VRF/VRV) air-cooled outdoor units, suitable for cooling and heating (heat pump), conforming to these specifications and in accordance with the Schedule of Quantities and approved drawings. The system shall operate on 400V $\pm$ 10%, 3-phase, 50 Hz AC power supply and use R-410A refrigerant. The prices quoted shall include all equipment, ancillary materials, and items required to fulfil the intent and purpose as laid down in the specifications and drawings approved by Engineer-In-Charge. The installation location for outdoor units shall be verified to ensure proper air circulation for optimal performance.

The air-conditioning system shall be based on 100% full inverter technology, capable of cooling and heating, with air-cooled outdoor units equipped with hermetically sealed inverter-type scroll compressors. The system shall consist of one or multiple outdoor units with single/multiple refrigerant piping circuits and multiple indoor units of various types, controlled by individual or centralized microprocessor-based controllers.

20.16.1 OUTDOOR UNIT

1. The outdoor unit shall be factory-assembled with a weatherproof casing constructed from heavy-gauge galvanized iron (GI) steel panels, baked with synthetic resin paint for corrosion resistance. The unit shall be completely factory-wired, tested with all necessary controls, and pre-filled with the first charge of R-410A refrigerant before delivery to the site. The tentative capacity of outdoor units shall be as per the Bill of Quantities (BOQ), with final quantities determined based on site conditions and approved design parameters. The service panel shall be easily removable for access to electrical components and the compressor section.
2. The outdoor unit shall feature hermetically sealed, inverter-based scroll compressors (OEM-manufactured, i.e., the compressor shall be of the same make as the VRF unit). Units with a capacity of 10 HP and above shall have a minimum of two compressors, all of which shall be inverter-type in each module.
3. The VRF system shall support the following refrigerant piping limits:
 - Equivalent piping length between indoor and outdoor units shall be extendable up to 190 meters.
 - Maximum height difference between outdoor and indoor units shall be 50 meters (outdoor unit above indoor unit).
 - Maximum level difference between indoor units shall be up to 30 meters.
4. The outdoor unit shall operate within the following ambient condition ranges:
 - Cooling Mode: -5°C to 50°C (unit shall function without tripping at 50°C ambient temperature).
 - Heating Mode: -20°C to 15°C. The Coefficient of Performance (COP) at AHRI conditions (35°C DB ambient, 27°C DB/19°C WB indoor) shall not be less than 3.1 at 100% load, and the Integrated Energy Efficiency Ratio (IEER) shall not be less than 6.5. These values shall be certified by the original equipment manufacturer (OEM) with sealed and signed documentation. No deviations from the specified COP and IEER shall be accepted.
5. The outdoor unit shall be factory-tested and pre-charged with R-410A refrigerant. The unit shall include an automatic refrigerant quantity judgment function to verify sufficient refrigerant in the cycle.
6. The compressor section shall be housed in a separate acoustic sheet metal enclosure to minimize sound levels during operation.

7. The scroll compressors shall be 100% inverter-based, highly efficient, and capable of responding to variations in cooling or heating load requirements. Compressor capacity regulation shall range from 10% to 100%, with a frequency range of 11 Hz to 110 Hz, using Insulated Gate Bipolar Transistor (IGBT) with Intelligent Power Module (IPM) technology for precise control in 0.1 Hz increments.
8. The unit shall incorporate free-phase technology to automatically correct phase sequence in case of phase reversal, ensuring continuous operation.
9. The unit shall feature Enhanced Vapor Injection (EVI) technology to improve COP, enable operation in high ambient conditions during cooling, and ensure smooth performance in low ambient conditions during heating.
10. Outdoor unit fans shall have a minimum of 25 speed steps to optimize energy consumption across various ambient conditions.

20.16.2 COMPRESSOR

The compressor shall be a 100% inverter-based scroll compressor (OEM-manufactured), highly efficient, with capacity regulation from 10% to 100% and a frequency range of 11 Hz to 110 Hz. The inverter technology shall utilize IGBT with IPM for precise capacity control in 0.1 Hz increments. The compressor shall be hermetically sealed and equipped with all necessary safety devices as outlined in clause 1.4 to ensure safe operation.

20.16.3 REFRIGERANT CIRCUIT

Each refrigeration circuit (using R-410A refrigerant only) shall include hermetically sealed scroll compressors, condenser coils, electronic expansion valves, solenoid valves, 4-way valves, shutdown valves, service ports, accumulators, and all other components essential for safe and satisfactory operation. The accumulator shall prevent liquid refrigerant from damaging the compressor.

20.16.4 HEAT EXCHANGER

The heat exchanger shall be a multi-pass, cross-finned tube type with highly efficient aluminium fins mechanically bonded to oxygen-free copper tubes. The aluminium fins shall be coated with an anti-corrosion resin film or blue film for enhanced durability. The unit shall include direct-driven, low-noise, multi-step propeller fans arranged for top discharge.

20.16.5 SAFETY DEVICES

The outdoor unit shall be equipped with the following minimum safety devices to ensure safe operation:

- High-pressure switch
- Low-pressure switch
- Fan motor safety thermostat
- Inverter overload protector
- Overcurrent relay

- Under-voltage monitor
- Fusible plugs
- Single-phasing preventer
- Thermal overload protection
- Crankcase heater

Additional safety devices shall be provided as required to ensure reliable and safe system operation.

20.16.6 INSTALLATION

The outdoor units shall be installed on suitable foundations with polymer-based vibration isolators to minimize transmission of vibration to the building structure. The installation shall comply with approved shop drawings and ensure proper air circulation around the units. The units shall be positioned to avoid recirculation of discharged air and ensure accessibility for maintenance.

20.16.7 PAINTING

All outdoor units shall be supplied with the manufacturer's standard weather-resistant finish, primed and coated with synthetic resin paint suitable for outdoor applications. The paint shall provide protection against UV radiation, moisture, and corrosion.

20.17 INDOOR UNITS

20.17.1 Indoor Unit

- I. Each unit shall have electronic control valve to control refrigerant flow rate respond to load variation of the room.
- II. The fan shall be multi blade type, statically & dynamically balanced to ensure low noise and vibration free operation of the system. The fan shall be direct driven type, mounted directly on motor shaft having support from housing.
- III. The cooling coil shall be made out of seamless copper tubes and having continuous aluminium fins. The tubes shall be hydraulically/ mechanically expanded for minimum thermal contact resistance with fins.
- IV. Indoor unit shall have cleanable type filter fixed to an integrally moulded/ moulded plastic frame. The filter shall be slide in and neatly insert able type.

20.17.2 Indoor 4-way cassette type Units: -

General

The scope of this section comprises the supply, installation, testing, and commissioning of 4-way flow cassette type indoor units for Variable Refrigerant Flow/Variable Refrigerant Volume (VRF/VRV) systems, conforming to these specifications and in accordance with the Schedule of Quantities and approved drawings. The units shall be ceiling-mounted, suitable for operation on single-phase 230V $\pm$ 10%, 50 Hz AC power supply, and designed for cooling and heating applications as part of a VRF/VRV system. The prices quoted shall include all equipment, ancillary materials, and items required to fulfil the intent and purpose as laid down in the specifications and approved drawings. The cooling capacity of the indoor units shall be rated at air inlet conditions of 27°C DB and 19°C WB.

20.17.3 CASSETTE TYPE INDOOR UNIT

1. The cassette type indoor unit shall be a 4-way air throw type ceiling-mounted, and equipped with a synthetic washable media pre-filter, fan section with low-noise fan/dynamically balanced blower, multi-speed motor, coil section with direct expansion (DX) copper coil, electronic expansion valve, outer cabinet, in-built drain pump, grille, necessary supports, vibration isolators, and cordless remote control, complete as required.
2. The unit shall have a maximum height of 330 mm to ensure compatibility with ceiling spaces. The design shall allow for connecting fresh air intake without requiring a special chamber and without increasing the total height beyond 330 mm.

3. The unit shall include an in-built drain pump suitable for a vertical lift of up to 850 mm, ensuring effective condensate removal. The drain pump shall be factory-installed and tested for reliable operation.
4. The unit shall feature a minimum of 4-speed fan control to allow flexible airflow adjustment based on space requirements. The fan shall be dynamically balanced to minimize vibration and noise.
5. The noise level of the unit at the highest operating speed shall not exceed 48 dB(A), measured at a vertical distance of 1.5 meters from the grille of the unit, ensuring quiet operation suitable for indoor environments.
6. The unit shall be equipped with a motion sensor as an optional feature, to be supplied wherever required as per the Bill of Quantities (BOQ) or drawings. The motion sensor shall enable energy-saving operation by detecting occupancy and adjusting unit performance accordingly.
7. The pre-filter shall be synthetic, washable, and designed for easy removal, cleaning, and reinstallation without requiring specialized tools. The filter shall effectively capture dust and particulate matter to maintain indoor air quality.
8. The unit shall include an automatic forced shutdown provision to stop operation in case of a fire, triggered by a signal from the Building Management System (BMS), ensuring compliance with fire safety requirements.
9. The outer cabinet shall be constructed from high-quality, corrosion-resistant material (e.g., galvanized steel or ABS plastic), finished with a durable, aesthetically pleasing coating suitable for indoor environments. The grille shall be designed for uniform air distribution and easy access to internal components for maintenance.
10. The unit shall be factory-tested for cooling/heating performance, air leakage, noise levels, and electrical safety, with test certificates provided upon request. The unit shall be pre-charged with R-410A refrigerant as part of the VRF/VRV system.

20.17.4 COIL AND REFRIGERANT CIRCUIT

The coil section shall consist of a direct expansion (DX) copper coil with aluminium fins, mechanically bonded for efficient heat transfer. The refrigerant circuit shall include an electronic expansion valve for precise control of refrigerant flow, compatible with R-410A refrigerant. The circuit shall be designed to integrate seamlessly with the VRF/VRV outdoor unit's refrigeration system.

20.17.5 FAN AND MOTOR

The fan section shall include a low-noise, dynamically balanced blower driven by a multi-speed motor with a minimum of 4 speed settings. The motor shall be energy-efficient, suitable for continuous duty, and provided with class 'F' insulation and IP-44 protection. The motor shall

be factory-wired with thermal overload protection to ensure safe operation.

20.17.6 SAFETY AND CONTROL FEATURES

The indoor unit shall be equipped with the following minimum safety and control features:

- Thermal overload protection for the fan motor.
- Electronic expansion valve for precise refrigerant control.
- BMS interface for fire signal integration and forced shutdown.
- Optional motion sensor for energy-saving operation (where specified).
- Microprocessor-based controller for integration with the VRF/VRV system's centralized or individual control system.

Additional safety features shall be provided as required to ensure reliable and safe operation.

20.17.7 INSTALLATION

The cassette type indoor units shall be installed in accordance with approved shop drawings, ensuring proper alignment and secure mounting to the ceiling structure. The units shall be fitted with polymer-based vibration isolators to minimize transmission of vibration and noise to the building structure. The installation shall allow for easy access to the filter, drain pump, and internal components for maintenance. Proper clearance shall be maintained for fresh air intake and air distribution.

20.17.8 PAINTING

The indoor unit's outer cabinet and grille shall be supplied with the manufacturer's standard finish, coated with a durable, corrosion-resistant paint or material suitable for indoor environments. The finish shall be aesthetically pleasing and resistant to fading or discoloration.

20.18 Ducted / Concealed Split unit type indoor units: -

General

The scope of this section comprises the supply, installation, testing, and commissioning of ceiling-mounted Ductable type indoor units for Variable Refrigerant Flow/Variable Refrigerant Volume (VRF/VRV) systems, conforming to these specifications and in accordance with the Schedule of Quantities and approved drawings. The units shall be suitable for operation on single-phase 230V $\pm$ 10%, 50 Hz AC power supply and designed for cooling and heating applications as part of a VRF/VRV system. The prices quoted shall include all equipment, ancillary materials, and items required to fulfil the intent and purpose as laid down in the specifications and approved drawings. The cooling capacity of the indoor units shall be rated at air inlet conditions of 27°C DB and 19°C WB.

20.18.1 CEILING MOUNTED DUCTABLE TYPE INDOOR UNIT

1. The ceiling-mounted Ductable type indoor unit shall be equipped with a washable synthetic media pre-filter, fan section with low-noise fan/dynamically balanced blower, multi-speed motor, coil section with direct expansion (DX) copper coil, electronic expansion valve, outer cabinet, drain pan, corded remote control, vibration isolators, necessary supports, and other accessories, complete as required.
2. The unit shall be designed to deliver the minimum capacity and external static pressure as specified in the Schedule of Quantities or drawings, ensuring adequate airflow for ducted applications.
3. The pre-filter shall be synthetic, washable, and designed for easy removal, cleaning, and reinstallation without requiring specialized tools. The filter shall effectively capture dust and particulate matter to maintain indoor air quality.
4. The fan section shall include a low-noise, dynamically balanced blower with a multi-speed motor offering a minimum of 4 speed settings to allow flexible airflow adjustment based on duct design and space requirements. The fan shall be optimized for the specified external static pressure.
5. The noise level of the unit at the highest operating speed shall not exceed 50 dB(A), measured at a distance of 1.5 meters from the unit's air outlet, ensuring quiet operation suitable for indoor environments.
6. The unit shall include an automatic forced shutdown provision to stop operation in case of a fire, triggered by a signal from the Building Management System (BMS), ensuring compliance with fire safety requirements.
7. The outer cabinet shall be constructed from high-quality, corrosion-resistant material (e.g., galvanized steel), finished with a durable coating suitable for indoor environments. The cabinet shall be insulated to minimize heat loss and condensation.
8. The drain pan shall be constructed from corrosion-resistant material (e.g., stainless steel or coated steel) and designed to collect and drain condensate effectively. The pan shall be sloped to prevent water stagnation and include a drain connection compatible with standard piping.
9. The unit shall be factory-tested for cooling/heating performance, air leakage, noise levels, and electrical safety, with test certificates provided upon request. The unit shall be pre-charged with R-410A refrigerant as part of the VRF/VRV system.
10. The unit shall be designed for concealed ceiling installation, with provisions for duct connections and access panels for maintenance of filters, coils, and other components.

20.18.2 COIL AND REFRIGERANT CIRCUIT

The coil section shall consist of a direct expansion (DX) copper coil with aluminium fins, mechanically bonded for efficient heat transfer. The refrigerant circuit shall include an electronic expansion valve for precise control of refrigerant flow, compatible with R-410A refrigerant. The circuit shall be designed to integrate seamlessly with the VRF/VRV outdoor

unit's refrigeration system.

20.18.3 FAN AND MOTOR

The fan section shall include a low-noise, dynamically balanced blower driven by a multi-speed motor with a minimum of 4 speed settings. The motor shall be energy-efficient, suitable for continuous duty, and provided with class 'F' insulation and IP-44 protection. The motor shall be factory-wired with thermal overload protection to ensure safe operation.

20.18.4 SAFETY AND CONTROL FEATURES

The indoor unit shall be equipped with the following minimum safety and control features:

- Thermal overload protection for the fan motor.
- Electronic expansion valve for precise refrigerant control.
- BMS interface for fire signal integration and forced shutdown.
- Microprocessor-based controller for integration with the VRF/VRV system's centralized or individual control system, operated via a corded remote control.

Additional safety features shall be provided as required to ensure reliable and safe operation.

20.18.5 INSTALLATION

The ceiling-mounted Ductable type indoor units shall be installed in accordance with approved shop drawings, ensuring proper alignment and secure mounting within the ceiling space. The units shall be fitted with polymer-based vibration isolators to minimize transmission of vibration and noise to the building structure. Duct connections shall be properly sealed to prevent air leakage, and the installation shall allow for easy access to the filter, drain pan, and internal components for maintenance. Adequate clearance shall be maintained for air intake and discharge.

20.18.6 PAINTING

The indoor unit's outer cabinet shall be supplied with the manufacturer's standard finish, coated with a durable, corrosion-resistant paint or material suitable for indoor environments. The finish shall be resistant to fading, discoloration, and moisture, ensuring long-term durability.

20.18.7 High Wall Units:

This section covers the supply, installation, testing, and commissioning of a minimum capacity VRV/VRF high wall type indoor unit.

EQUIPMENT SPECIFICATIONS

a) Design and Features

- The unit shall be a high wall type indoor unit, designed for efficient wall-mounted installation.
- It shall include:
 - Washable synthetic media pre-filter.
 - Fan section with a low noise fan/dynamically balanced blower.
 - Multi-speed motor for adjustable airflow.
 - Coil section with DX copper coil.
 - Electronic expansion valve for precise refrigerant flow.
 - Durable outer cabinet.
 - Integrated drain pan.
 - Necessary accessories.
 - Cordless remote control for user convenience.

b) Electrical Specifications

- The unit shall operate on a single-phase 230 V $\pm$ 10%, 50 Hz AC power supply.

20.18.8 Centralized system remote controller: -

A multifunctional compact centralized system remote controller shall be provided with the system. It should be user friendly.

It should be able to control *at least 128* indoor units connected at a station. The centralized system remote controller shall perform the following minimum functions:

- I. Starting/stopping of Air conditioners as a zone or group or individual units.
- II. Temperature setting for each indoor unit or zone.
- III. Switching between temperature controls modes, switching of fan speed.
- IV. Monitoring of operation status such as operation mode & temperature setting of individual indoor units, maintenance information, trouble-shooting information.
- V. Controller should be capable for scheduling of both Indoor and Outdoor units as per the requirement of user i.e. for cyclic time management, weekly scheduling of operation etc. to save on power with minimum men interface.

20.19 SPLIT HIGH WALL INVERTER AIR CONDITIONER UNIT

20.19.1 EQUIPMENT SPECIFICATIONS

a) Design and Features

- The unit shall be a high wall type inverter split air conditioner, comprising one indoor unit and one outdoor unit, designed for efficient wall-mounted installation.
- The unit shall incorporate an inverter-driven rotary compressor, utilizing environmentally friendly, ozone non-depleting R32 refrigerant or equivalent.
- The indoor unit shall be equipped with:
 - i. Synthetic washable media pre-filter for effective air filtration.
 - ii. Fan section with a low-noise, dynamically balanced blower.
 - iii. Multi-speed motor for adjustable airflow to suit varying cooling needs.

- iv. DX copper coil with high-quality inner-grooved copper tubes for enhanced thermal conductivity (20%-30% improvement).
- v. Electronic expansion valve for precise refrigerant flow control.
- vi. Durable outer cabinet with anti-corrosive coating (e.g., blue fin or gold fin technology) to withstand extreme weather conditions.
- vii. Integrated drain pans for efficient condensate management.
- viii. Grill and necessary supports for secure installation.
- ix. Vibration isolation components to minimize operational noise.
- x. Full-function, battery-operated cordless remote control for user convenience.
- xi. The outdoor unit shall feature a 100% copper condenser coil with anti-corrosive coating for extended lifespan and reliable performance in all weather conditions.

20.19.2 Electrical Specifications

- The unit shall operate on a single-phase 230 V $\pm$ 10%, 50 Hz AC power supply.
- Power cable shall be 3x4 Sqmm, copper, PVC sheathed, with one core dedicated for loop earthing to ensure electrical safety.

20.19.3 SAFETY FEATURES

- The unit shall include an automatic forced shutdown provision triggered by signals from the Building Management System (BMS) in the event of a fire.
- Refrigerant leakage detection system to automatically cease operation, protecting the compressor from high-temperature damage due to refrigerant leaks.
- Built-in protection against voltage fluctuations, compressor overload, and thermal overload to ensure safe operation.

20.19.4 PERFORMANCE REQUIREMENTS

- The cooling capacity of the indoor unit shall be rated at air inlet conditions of 27°C dry bulb and 19°C wet bulb temperatures.
- The unit shall achieve a minimum BEE-5-star rating for energy efficiency, with an ISEER value compliant with 5-star standards (e.g., ≥ 5.02 as per industry benchmarks).
- The unit shall be capable of operating efficiently at ambient temperatures up to 52°C.

20.19.5 INSTALLATION REQUIREMENTS

- The indoor unit shall be mounted at a suitable height as per site conditions or as directed by the Engineer-In-Charge.
- The outdoor unit shall be mounted on a wall bracket or a raised platform of suitable length and width, positioned at a height of 45 cm from the ground, as per site requirements or as directed by the Engineer-In-Charge.
- The unit shall be supplied with standard-length refrigerant piping, drain piping, and control cables. Refrigerant piping shall be well-insulated to prevent condensation.
- Drain pipes shall be routed to the nearest shaft with adequate slope to ensure proper drainage, as per the specified schedule.
- All finishing works, including packet filling, wall colony work, or any other civil work inside or outside the building, shall be completed to a high standard.
- Only air conditioner units and associated accessories from the approved makes list (as attached) shall be accepted.

20.19.6 STABILIZER SPECIFICATIONS

- The stabilizer for split high wall inverter air conditioner shall meet the following specifications:
 - **Input Voltage:** 230 V, 50 Hz, AC supply.
 - **Working Voltage Range:** 150 V to 280 V, or as per OEM specifications.
 - **Capacity:** ≥ 15 Amps to support the air conditioners power requirements.
 - **Protection Features:**
 - Low and high voltage cut-off protection to safeguard the unit from voltage fluctuations.
 - Built-in thermal overload protection to prevent overheating.
 - Intelligent time delay system to ensure safe startup after power interruptions.
 - **Installation:** Wall-mounting cabinet design for secure and convenient installation.

20.19.6 WARRANTY AND SUPPORT

- The unit shall include a minimum 1-year comprehensive warranty on the product, with an extended 5-year warranty on the compressor and 5 years on the PCB/controller.
- Warranty coverage shall include the compressor, evaporator coil, condenser coil, PCB, swing motor, fan motor, and sensors, with exclusions for air filters, plastic/metal parts, remote handset, labor, gas charging, and damages due to site conditions, pests, or power supply issues.

20.19.7 Y- Joints / Header Joints

Y-Joint / Ref-net separation refrigeration pipe joints and headers shall be in the appropriate orientation to enable correct distribution of refrigerant. The Distribution Joints should be factory insulated with pre-formed sections of Expanded Polystyrene/ Equivalent.

20.19.8 Refrigerant Piping

- a. All refrigerant pipes and fittings shall be soft drawn/ half hard/hard drawn copper tubes and wrought copper fitting suitable for connection with silver solder. Soft Drawn copper pipes shall be used only up to pipe size of 15.86 mm. The thickness of wall of copper tubes shall be 1.2 mm for tube size from 6.4 mm to 28.6 mm and 1.62 mm thickness for tube size above 31.8 mm up to 54 mm.
- b. The Copper Piping & Piping Circuit should be with Minimum Number of joints to avoid the leakage probability, which shall be attained by:
- c. Using preferably less joint piping system e.g. One End Expanded Tubes.
- d. (b) Bending the tubes instead of using elbow joints wherever 90-degree bending is required.
- e. Piping should be routed at site in such a manner, that brazed joints in the refrigeration piping are kept to a minimum.
- f. The makes of tube fittings shall be same as that of tubes
- g. The thickness of fittings used shall be same as that of the pipe.
- h. All joints in copper piping shall be swaged joints using low temperature brazing and/ or silver solder. Before jointing any copper pipe or fittings, its interior shall be thoroughly cleaned by passing a clean cloth via wire or cable through its entire length. The piping shall be continuously kept clean of dirt etc. The nitrogen gas

must be purged for protecting copper pipe during brazing operation. After brazing allow it for natural cooling.

- i. Refrigerant lines shall be sized to limit pressure drop between evaporator and condensing unit to less than 0.2 kg per Sq.cm.
 - j. After the refrigerant piping installation has been completed the refrigerant piping system shall be pressure tested using, Freon mixed with nitrogen at a pressure of 20 Kg per Sq. cm. (High side) and 10 Kg per Sq. cm (Low side). Pressure shall be maintained on the system for 24 hours.
 - k. The system shall then be evacuated to a minimum vacuum of 70 cm. of mercury and held for 24 hours, during which time; change in vacuum shall not exceed 12 cm of mercury.
 - l. All refrigerant piping shall be installed strictly as per the instructions and recommendations of air conditioning equipment manufacturers.
 - m. The refrigerant copper tubes shall be complied with ASTM B280 (Standard specification for seamless copper tube for air conditioning and refrigeration).
 - n. The refrigerant Copper tube fittings shall conform to ASME B16.18/B16.22/B16.26/ B16.50. However, copper tube fitting used for brazed connections in air conditioning and refrigeration system shall comply with ASME standard B16.22. The refrigerant copper tube fitting shall be of same manufacturers as of copper tubes. The thickness of all copper tubes fitting shall be equal to or greater than thickness of copper tubes.
 - o. All pipes must be tested for defects using Eddy Current Testing and other test as per ASTM B280-16.
 - p. Joints in refrigerant piping should be kept as minimum as possible. The use of elbows, sockets, bends etc. shall be avoided and as per requirement the copper tubes shall be suitably bend with automatic bending machine.
 - q. Each copper tube shall be marked (embossed) with manufacturer name, OD in mm, wall thickness in mm, lot number, month & year of manufacturer and conforming standard (as applicable).
 - r. The copper tube fitting shall be marked (embossed) with manufacturer name, OD in mm and wall thickness in mm.
- All refrigerant pipes shall be properly supported and anchored to the building structure using steel hangers, anchors, brackets, and supports which shall be fixed to the building structure by means of inserts or expansion shields of adequate size and number to support the load imposed thereon.
 - To protect XLPE insulation of exposed copper piping from degrading due ultra violet rays & atmospheric condition, it shall be covered poly shield coating with at least two coats of resin and hardener above XLPE insulation. Fibre glass tape shall be helically wound & coated with painted two coats of resin with hardener to give smooth & plain finish.

20.20 **Drain Piping**

- i. The Indoor units shall be connected to condensate drain pipe made of PVC of suitable sizes.

- ii. The pipe shall be laid in proper slope for efficient drainage of condensate water. U trap shall be provided at each indoor unit.
- iii. All pipe supports shall be pre-fabricated & pre-painted.
- iv. Slotted angles for supports should be properly installed with clamps etc.

20.21 Control Cable

The control cable delivered at site should be shielded and should run in PVC conduit of suitable size.

Above data sheet to be filled based on module selected as per BOQ to meet the technical specification. Above technical data sheet shall be duly sign and stamp on OEM letter head.

20.22 TUBE AXIAL FLOW FANS

20.22.1 SCOPE

The scope of Tube Axial Flow Fans conforming to these Specifications and in accordance with the requirement of Drawings and Schedule of Quantities.

20.22.2 TYPE

Fans shall be Tube Axial Flow type as indicated on Drawings and identified in Schedule of Quantities.

20.22.3 CAPACITY

The air-moving capacity of fans shall be as shown on Drawings and in Schedule of Quantities.

20.22.4 AXIAL FLOW FAN

Fan shall be complete with motor, motor mount, belt driven (or direct driven) and vibration isolation type, suspension arrangement as per approved for construction shop drawings.

i. Casing:

Casing shall be constructed of heavy gauge MS sheet. Fan casing, motor mount and straightening vane shall be of welded MS construction. Motor mounting plate shall be minimum 15 mm thick and machined to receive motor flange.

Casing shall have flanged connection on both ends for ducted applications. Support brackets for ceiling suspension shall be welded to the casing for connection to hanger bolts. Straightening vanes shall be aerodynamically designed for maximum efficiency by converting velocity pressure to static pressure potential and minimizing turbulence. Casing shall be bonderized, primed and finish coated with enamel paint.

ii. Rotor:

Rotor hub and blades shall be cast aluminum alloy construction. Blades shall be die-formed aero foil shaped for maximum efficiency and shall vary in twist and width from hub to tip to effect equal air distribution along the blades length. Fan blades mounting on the hub shall be statically and dynamically balanced. Extended grease leads for external lubrication shall be provided. The fan pitch control may be manually readjusted at site upon installation, for obtaining actual air flow values, as specified and quoted.

iii. Motor:

Motor shall be energy efficient squirrel-cage, totally-enclosed fan cooled, standard frame, constant speed, continuous duty, single winding, suitable for $415 \pm 10\%$ volts, $50 \pm 5\%$ cycles, 3 phase AC power supply, provided with class 'F/H' insulation with IP-55 protection. Motor shall be specially designed for quiet operation. The speed of the fan motor shall as identify in Schedule of Quantities. Motor conduit box shall be mounted on exterior of fan casing, and lead wires from the motor to the conduit box shall be protected from the air stream by enclosing in a flexible metal conduit.

20.22.5 Vibration Isolation:

The assembly of fan and motor shall be suspended from the slab by vibration isolation suspension of rubber-in-shear type.

20.22.6 Speed:

The speed of axial flow fans shall not exceed 960 RPM for impeller dia. exceeding 450 mm and shall not be greater than 1440 with impeller dia. less than 450 mm. However, for fans having static pressure of 30 mm WC or above the speed of the fan shall not exceed 1440 rpm for fan with impeller diameter of above 450 mm and 2800 rpm for fan with impeller diameter of 450 mm or less. The first critical speed of rotating assembly shall be at least 25% above the operating speed."

20.22.7 Noise Level:

Noise level should not exceed 75dbA at a distance of 3 meters.

20.22.8 INSTALLATION

The Tube Axial Flow Fans shall be so installed as to transmit minimum amount of vibration to the building structure. Adequate vibration isolation shall be provided by use of rubber/ neoprene pads and/or vibration isolation spring mountings.

20.22.9 PAINTING

All equipment shall be supplied as per manufacturer's standard finish painting.

20.23 IN-LINE FANS

1. SCOPE

Inline fans shall be complete with casing, external rotor motor & impeller; these fans should have, built in thermal contact for motor protection. Direction of discharge and rotation position shall be as per the job requirement and shall be marked on the fan assembly.

2. CASING

Housing shall be constructed of hot rolled GSS sheet metal construction. Housing metal parts shall be either spot-welded or screwed or mounted together with rivets. Indication showing rotation arrow and make; model number and duty conditions of the fan shall be available on the housing.

3. FAN

Fan wheel shall be forward/backward curved type. Fan wheel shall be statically and dynamically balanced.

4. BEARING

Ball bearings are completely maintenance free and can be used in any mounting position, at maximum indicated temperature. The bearing lubricant is suitable for a minimum ambient temperature of minus 15 (admissible for a short time without reaching dew point at -30°C). For application at maximum indicated ambient temperature life expectancy L 10 is 40,000 hours minimum.

5. FAN MOTOR

Fans are supplied with built-in-thermal contact (TK). At the critical high temperature point ("B" = 130°C or "F" = 155°C) the thermal contact will open and break the power supply to the Fan. Fan motor has insulation class "B" or "F" and protection class IP44 or IP54. Drive of fan shall be direct driven. Motor speed shall not exceed 960 RPM.

6. INSULATION

The fans shall be thermally and acoustically insulated with 50 mm layer of mineral wool, with perforated sheet steel on the inner surface.

20.24 PROPELLER FANS

Direct-driven Propeller Fans for exhaust/supply air, conforming to these specifications and in accordance with the requirements of Drawings and Schedule of Quantities.

1.0 TYPE

Fans shall be direct-driven Propeller Fans, as indicated on Drawings and identified in the Schedule of Quantities, suitable for exhaust or supply air applications.

2.0 CAPACITY

The air-moving capacity of the fans shall be as shown on Drawings and specified in the Schedule of Quantities.

PROPELLER FAN Fans shall be complete with a permanent split capacitor or shaded pole motor, mounting plate, and accessories including wire guard, bird screen, and fixed louvers for weather protection, as required.

3.0 Casing and Mounting:

The fan casing shall be constructed of heavy-gauge galvanized steel or corrosion-resistant material, with a robust mounting plate to ensure secure installation. The casing shall include flanged edges for easy connection to ducts or wall openings. Support brackets or mounting frames shall be provided for wall or ceiling installation, as specified. The casing shall be primed and finished with a weather-resistant enamel coating to withstand environmental exposure.

4.0 Blades:

Fan blades shall be constructed from die-formed aluminium or corrosion-resistant steel, designed with an aerodynamic profile for maximum efficiency and uniform air distribution. Blades shall be statically and dynamically balanced to minimize vibration and noise. Blade pitch shall be factory-set but manually adjustable on-site to achieve specified airflow values.

5.0 Motor:

Motors shall be permanent split capacitor or shaded pole type, designed for quiet operation, continuous duty, and single-phase AC power supply ($220 \pm 6\%$ volts, 50 cycles). Motors shall be provided with class 'F' insulation and IP-55 protection, suitable for outdoor or humid environments. The motor conduit box shall be externally mounted on the fan casing, with lead wires protected by a flexible metal conduit to prevent exposure to the airstream.

6.0 Accessories:

Each fan shall include a wire guard and bird screen made of galvanized steel or equivalent, with a mesh size of 12 mm or smaller, to prevent debris ingress. Fixed louvers, constructed from corrosion-resistant material, shall be provided for weather protection in outdoor applications. All accessories shall be factory-fitted and compatible with the fan assembly.

7.0 Vibration Isolation:

Fans shall be installed with rubber or neoprene pads to minimize vibration transmission to the building structure. Vibration isolation mounts may be used where specified in the Drawings.

8.0 Noise Level:

The noise level of the fans shall not exceed 70 dBA at a distance of 3 meters under normal operating conditions.

20.25 DUCTING

1. SCOPE

The general requirements for sheet metal ductwork for air distribution with associated items such as air outlets and inlets, fresh air intake and fire dampers.

2. MATERIAL

2.1 Ducts

- i) All ducts shall be fabricated either from Galvanized Sheet Steel (GSS) conforming to IS: 277. The steel sheets shall be hot dip galvanized with coating of minimum 120 grams per square meter (GSM) of Zinc.
- ii) The thickness of sheets for fabrication of rectangular ductwork shall be as under. The thickness required corresponding to the longest side of the rectangular section shall be applicable for all the four sides of the ductwork.

Longest side (mm)	Minimum sheet thickness
	For GSS
750 mm and below	0.63
751 mm to 1500 mm	0.80
1501 mm to 2250 mm	1.00
2251 mm	1.25

- iii) All sheet metal connections, partitions and plenums required for flow of air through the filters, fans etc. shall be at least 1.25 mm thick galvanized steel sheets, in case of G.I. sheet ducting shall be stiffened with 25 mm x 25 mm x 3 mm angle iron braces.
- iv) Circular ducts, where provided shall be of thickness as specified in IS: 655 as amended up to date.

2.2 Associated Items

- i) Supply/ return air outlets, F.A. grilles and accessories shall be constructed from extruded aluminium sections.
- ii) Flanges for matching duct sections, stiffening angles (braces) and supporting angles shall be of rolled steel sections, and shall be of the following sizes.

Application	Duct Width	Angle size
Flanges	Up to 1000 mm	35 mm x 35 mm x 3 mm
-do-	1001 mm to 2250 mm	40 mm x 40 mm x 3 mm
-do-	More than 2250 mm	50 mm x 50 mm x 3 mm
Bracings	up to 1000 mm	25 mm x 25 mm x 3 mm
-do-	More than 1000 mm	40 mm x 40 mm x 3 mm
Support angles	up to 1000 mm	40 mm x 40 mm x 3 mm
-do-	1001 mm to 2250 mm	40 mm x 40 mm x 3 mm
-do-	More than 2250 mm	Size and type of RS section shall be decided in individual cases

- iii) Hanger rods shall be of mild steel and of at least 10 mm dia. for ducts up to

2250 mm size, and 12 mm dia. for larger sizes.

- iv) All nuts, bolts and washers shall be zinc plated steel. All rivets shall be galvanized or shall be made of magnesium - aluminium alloy. Self-tapping screws shall not be used.

3. CONSTRUCTION

3.1 Ducts

- i) Ducts shall be fabricated at site or factory fabricated and shall be generally as per IS: 655 "Specifications for metal air ducts", unless otherwise deviated in these General Specifications.
- ii) The interior surfaces of the ducting shall be smooth.
- iii) All the ducts up to 600 mm longest side shall be cross broken between flanges by a single continuous breaking. Ducts of size 600 mm and above shall be cross broken by single continuous breaking between flanges and bracings. Alternatively, beading at 300 mm centres for ducts up to 600 mm longest side, and 100 mm centres for ducts above 600 mm size shall be provided for stiffening.
- iv) As far as possible, long radius elbows and gradual changes in shape shall be used to maintain uniform velocity accompanied by decreased turbulence, lower resistance and minimum noise. The ratio of the size of the duct to the radius of the elbow shall be normally not less than 1:1.5.
- v) Flanged joints shall be used at intervals not exceeding 2500 mm. Flanges shall be welded at corners first and then riveted to the duct.
- vi) Stiffening angles shall be fixed to the sides of the ducts by riveting at 1.25 meters from joints for ducts of size 600 mm to 1500 mm, and 0.6 mm from joints for ducts of size larger than 1500 mm. Bracings for ducts larger than 1500 mm can alternatively be by diagonal angles.
- vii) Plenums for filters shall be complete with suitable access door of size 450 mm x 450 mm.

3.1 Air Outlet and Inlets (Supply and Return)

- i) All air outlets and intakes shall be made of extruded aluminium sections & shall present a neat appearance and shall be rigid with mechanical joints.
- ii) Square and rectangular wall outlets shall have a flanged frame with the outside edges returned or curved 5 to 7 mm and fitted with a suitable flexible gasket between the concealed face of the flanges and the finished wall face. The core of supply air register shall have adjustable front louvers parallel to the longer side to give up to 22.5 degrees vertical deflection and adjustable back louvers parallel to the shorter side to achieve a horizontal spread air pattern to at least 45 degrees. Return air grilles shall have only front louvers. The outer framework of the grilles shall be made of not less than 1.6 mm thick aluminium sheet. The louvers shall be of aerofoil design of extruded aluminium section with minimum

thickness of 0.8mm at front and shall be made of 0.8mm thick aluminium sheet. Louvers may be spaced 18 mm apart.

- iii) square and rectangular ceiling outlets/intakes shall have a flange flush with the ceiling into which it is fitted or shall be of anti-smudge type. The outlets shall comprise an outer shell with duct collar and removable diffusing assembly. These shall be suitable for discharge in one or more directions as required. The outer shell shall not be less than 1.6 mm thick extruded section aluminium sheet. The diffuser assembly shall not be less than 0.80 mm thick extruded aluminium section.
- iv) Circular ceiling outlets/intakes shall have either flush or anti smudge outer cone as specified in the tender specifications. Flush outer cones shall have the lower edge of the cone not more than 5 mm below the underside of the finished ceiling into which it is fitted. Anti-smudge cones shall have the outer cone profile designed to reduce dirt deposit on the ceiling adjacent to the air outlet. The metal sheet used for construction of these shall be minimum 1.6 mm thick extruded aluminium sheet.
- v) Linear diffusers shall have a flanged frame with the outside edges returned 3.5 mm and shall have one to four slots as required.
The air quantity through each slot shall be adjustable. The metal sheet used for the construction of these shall be minimum 1.6 mm thick extruded aluminium sheet.
- v) Grilles and diffusers constructed of extruded aluminium sections shall have grille bars set straight, or deflected as required. These shall be assembled by mechanical interlocking of components to prevent distortion. These grilles and diffusers shall have a rear set of adjustable blades, perpendicular to the face blades for deflection purposes.
- vi) All supply air outlets shall be fitted with a volume control device, made of extruded aluminium gate section. The blades of the device shall be mill finish/block shade pivoted on nylon brushes to avoid rusting & rattling noise, which shall be located immediately behind the outlet and shall be fully adjustable from within the occupied space without removing any access panel. The volume control device for circular outlets shall be opposed blade radial /shutter type dampers, or two or more butterfly dampers in conjunction with equalizing grid. Opposed blade dampers shall be used for square and rectangular ceiling/wall outlets and intakes.
- viii) All the products supplied by contractor should supplement in performance by selection curves of product ratings from the manufacturer.
- ix) Laminar supply air diffusers shall be made of 2mm thick powder coated aluminium sheet duly insulated with 5mm thick closed cell polyethylene foam insulation having factory laminated aluminium foil and joints covered with self-adhesive aluminium tape and having holes 2/3 mm dia. including frame work.

3.3 Fresh Air Intakes

- i) Fresh air intake grills shall be made of extruded aluminium sections.

- ii) A flanged frame using RS sections shall be provided on front face to conceal the gap between the louvers and the adjoining wall face. Corners of frame shall be welded. The frame shall be made structurally rigid.
- iii) Louvers made from extruded aluminium section shall be in modular panel form for ease of handling. These shall be free from waves and buckles. Vertical blades shall be truly vertical and horizontal blades shall be truly horizontal. Butt joints in blades shall not be accepted.
- iv) Additional intermediate equally spaced supports and stiffeners shall be provided to prevent sagging/ vibrating of the louvers, at no more than 750mm centers where the louver's length is longer than 750mm.
- v) A bird wire screen made of 12 mm mesh in 1.6 mm steel wire held in angle or channel frame shall be fixed to the rear face of the louver frame by screens.

3.3 Flexible Duct

Pre-Insulated Factory Fabricated Semi-Rigid Flexible Ducts

- i) Pre-insulated ducts shall be factory fabricated, semi-rigid, and constructed from aluminium to ensure durability and corrosion resistance.
- ii) Ducts shall be supported at regular intervals as per site requirements to maintain structural integrity and prevent sagging.
- iii) Ducts shall be provided in modular sections for ease of installation and handling, ensuring a smooth and seamless fit at the site.
- iv) The ducts shall be free from deformations, dents, or irregularities, with a consistent and uniform surface finish.

3.4 Butterfly Control Dampers

- i) Butterfly control dampers shall be made from 1.6 mm galvanized or stainless steel, supplied, installed, tested, and commissioned.
- ii) Opposed blade dampers shall have mill finish/black shade, pivoted on nylon bushings to prevent rust and noise.
- iii) Dampers shall be rigid, free from warping, and painted with corrosion-resistant coating as specified.
- iv) Dampers shall be tested for leakage and performance per AMCA standards, with supports for sizes over 750 mm.

4.0 INSTALLATION OF METALLIC DUCT

4.1 Ducting

- i) The fabrication and installation shall be in a workmanlike manner. Duct work

- shall be rigid and straight without kinks.
- ii) All exposed ducts within the conditioned space shall have slip joints. Flanged joints shall not be used.
- iii) All joints shall be airtight.
- iv) Ducts shall be supported independently from the building structure and adequately, to keep the ducts true to shape. The support spacing shall be not more than 2 m. where ducts cannot be suspended from ceiling, wall brackets or other suitable arrangements, as approved by the Engineer-in-charge shall be adopted. Neoprene or other vibration isolation packing of minimum 6 mm thickness shall be provided between the ducts and the angle iron supports/brackets. Vertical duct work shall be suitably supported at each floor by steel structural members.
- v) Where metal ducts or sleeves terminate in woodwork, tight joints shall be made by means of closely fitting heavy flanged collars. Where ducts pass through brick or masonry openings, wooden frame work shall be provided within the openings and the crossing ducts shall be provided with heavy flanged collars on either side of the wooden frame work, so that duct crossing is made leak-proof.
- vi) Duct connections to the air-handling unit shall be made by inserting a double canvas sleeve 100 mm long. The sleeve shall be securely bonded and bolted to the duct and unit casing.
- ii) Dampers shall be provided in branch duct connections for proper volume control and balancing the air quantities in the system, whether indicated in the drawings or not. Suitable links, levers and quadrants shall be provided for proper operation, control and setting of the dampers. Every damper shall have an indicating device clearly showing the position of the dampers at all times.
- iii) Where electrical heaters are mounted in the duct, these shall be of low temperature totally enclosed type fitted with radiation fins. A removable panel for access to the heaters shall be provided in the duct. Any hole in the duct for electrical wiring must be provided with suitable bushes to avoid leakage. 6 mm thick asbestos board lining shall be provided all around the inside of the duct for a distance of 30 centimeter. on either side of the electrical heaters. A manually reset thermostatic safety switch shall be provided near the duct section having heaters. In addition, the heaters must be interlocked with the connected fan motor of the AHU.

4.2 Air Outlets and Inlets

- i) The locations of the air outlets and intakes shall be shown in the tender drawings and necessary openings and the wooden framework for fixing the grilles shall be provided by the air conditioning contractor. The location of these outlets/inlets is subject to change and the approval of the Engineer-in-charge shall be obtained before finally fixing the grilles/ diffusers in position.
- ii) In installing fresh air intakes, no fixing device shall be visible from the face of the frame. Where louvers are to be fixed in masonry or concrete, fixing shall be with either expanding plugs or raw plugs. Where the louvers are to be fixed in steel or wood, non-ferrous screws or bolts shall be used.
- iii) Supply air outlets and return air intakes shall be anodized/ powder coated aluminium to the desired colour to match the surroundings wall/ceiling. The

fresh air intakes shall be anodized/ powder coated aluminium as approved by the Engineer-in-Charge. The paint colour shall be approved by the Engineer-in-Charge.

- iv) All damages to the finish of the structure during the installation work shall be made good by the air-conditioning contractor before handing over the installation to the Department.
- iv) All Classified areas shall be planned with HEPA filters (H14) at ceiling level for supply air & Return air riser with bottom grille to pick air from lower level.

5.0 BALANCING

The entire air distribution system shall be balanced with the help of an anemometer. The measured air quantities at fan discharge and at the various outlets shall be within ± 5 percent of those specified/quoted. Branch duct adjustments shall be permanently marked after the air balancing is completed so that these can be restored to their correct position if disturbed at any time.

20.26 INSULATION WORK

1. General

The Insulation of Refrigerant pipes, drain water pipes, ducting, hi wall units, duct able units, cassette units and AHUs etc., shall be carried out as per specifications given below:

2. Materials

2.1 The materials to be used for insulation shall be as follows, unless some other material is specifically mentioned elsewhere. The detailed specifications of the materials are listed under respective sub heads.

2.2 Refrigerant Pipe Insulation: Cross linked polyethylene insulation.

2.3 Drain Pipe Insulation: Cross linked polyethylene insulation.

2.4 Duct Insulation: Cross linked polyethylene insulation.

2.5 Acoustic Insulation: Cross linked polyethylene insulation

3. Refrigerant pipe insulation

3.3.1 The material for insulation of refrigerant pipes shall be pipe sections of flexible cross linked polyethylene insulation having a 'K' value of **0.033 W/mk** at a mean temperature of **23°C** and a minimum density of **30 Kg. /cub m.**

3.3.2 The thickness of insulating material for pipe work shall be as under:

Pipe size (mm) Cross linked polyethylene insulation

6.4 mm to 19.1 mm Refrigerant pipes 13 mm
22.2 mm to 41.3 mm Refrigerant pipes 19 mm

-

3.3 Drain Pipe Insulation

3.3.2 The material for insulation of drain pipes shall be pipe sections of flexible cross linked polyethylene insulation having a 'K' value of **0.033 W/mk** at a mean temperature of **23°C** and a minimum density of **30 Kg./cubm**.

3.4 The thickness of insulation shall be 6 mm.

Duct Insulation

3.4.1 Insulation Material

The material for duct insulation shall be Closed Cell Cross Linked Polyethylene (XLPE) Foam, Class O. The insulation shall have:

- A density of $27 \pm 3 \text{ kg/m}^3$ (24–30 kg/m³).
- A thermal conductivity not exceeding 0.033 W/mK at an average temperature of 23°C.
- A temperature range of –40°C to 105°C.
- Fire ratings:
 - Class 1 as per BS 476 Part 6:1989 for fire propagation test.
 - Class 0 as per BS 476 Part 7:1987 for surface spread of flame test.
- Fire approval from CBRI - Roorkee.
- Water vapour permeability not less than 0.024 per inch ($2.48 \times 10^{-14} \text{ kg/m.s. Pa}$, i.e., $\mu > 7000$ for water vapour diffusion resistance).

The material shall be in the form of foam sheets or rolls of uniform thickness, factory-laminated with aluminum foil of mat finish on one side.

3.4.2 The thickness of duct insulation shall be as follows:

- a. Supply air duct in conditioned space (external) : 32 mm thick
- b. Supply air duct in conditioned space (internal) : 19 mm thick
- c. Return air duct in conditioned space (external) : 25 mm thick
- d. Return air duct in conditioned space (internal) : 13 mm thick

3.5 Acoustic Treatment

3.5.1 Insulation Material

The material for acoustic treatment of ducts shall be non-fibrous fire-retardant open cell cross-linked polyethylene foam (Eco sound FR-Open Cell Cross Linked Polyethylene). The insulation shall have:

- A density of $60 \pm 3 \text{ kg/m}^3$ (57–63 kg/m^3).
- A porous surface on one side and a flat surface on the other.

The material shall be in the form of foam sheets or rolls of uniform density, applied using synthetic rubber-based adhesive with low VOC and high strength characteristics.

3.5.2 Insulation Thickness

The thickness of insulation for acoustic lining shall be as follows:

- Supply and return air ducts (initial portion of ducting and main plenum): 15 mm or 25 mm thick non-fibrous fire-retardant open cell cross-linked polyethylene foam.

4. Installation

4.1 Refrigerant Piping

- 4.1.1 The pipe shall be thoroughly cleaned and rendered free from all rust and grease.
- 4.1.2 Two coats of synthetic glue shall be applied on the cleaned pipe surface.
- 4.1.3 The preformed sections of suitable thickness (as per BOQ) insulation shall be fixed tightly to the surface to take care to seal all joints.
- 4.1.4 All longitudinal joints shall be sealed with synthetic glue and covered with adhesive tape of 38 mm along the length. All joints along the circumference would be sealed with 50mm x 4mm to 5mm self-adhesive insulation sheet.

4.2 Drain Piping

- 4.2.1 The pipe shall be thoroughly cleaned with and rendered free from all rust and grease.
- 4.2.2 Two coats of synthetic glue shall be applied on the cleaned pipe surface.
- 4.2.3 Then preformed sections of 6 mm thick insulation shall be installed on the pipe.
- 4.2.3 All longitudinal joints shall be sealed with synthetic glue and covered with adhesive tape of 38 mm along the length. All joints along the circumference would be sealed with 50mm x 4mm to 5mm self-adhesive insulation sheet.
- 4.2.4 All joints shall be covered with 3mm x 50mm PVC self-adhesive.

4.3 Ducting

- 4.3.1 Clean the surface of the ducts with a wire brush and make it free from rust and oil.
- 4.3.2 Apply one coat of cold adhesive to the surface of the duct and to the surface of the insulation sheet.
- 4.3.3 Wait for the adhesive to become tacky dry to touch and then affix the insulation sheet firmly to the duct surface. There should be no air entrapped below the insulation sheet for proper adhesion to the duct.
- 4.3.4 The joints shall be sealed with 50mm wide aluminium tape.
- 4.3.5 The ducts in areas exposed to the weather shall be additionally covered with 7 mil fibre glass cloth applied with Star bond epoxy coating to protect from mechanical damage and UV rays.

4.4 Duct Acoustic Lining

- 4.4.1 The duct surface shall be first cleaned from inside.
- 4.4.2 The acoustic insulation sheets should be fixed between these sections of ducts using adhesive compound and self-adhesive stick pins.
- 4.4.3 The sheet of insulation shall be secured to the duct by means of stick pins as mentioned above.

Appendix-I - Technical Submittal

1. VRV Outdoor Units			
A)	General	Unit	Data
1	Make		
3	Model No. of VRV		
4	Refrigerant		
5	Certification		
6	Dimensions (L x W x H)	mm	
7	Operating Weight	Kg.	
8	Service Clearance Required (L x W x H)	mm	
9	Refrigerant Weight	Kg.	
B)	Compressor		
1	Make		
2	Type of Compressor		
3	Model No. of Compressor		
4	Drive Arrangement		
5	Speed (Operating)	RPM	
6	Speed (Maximum)	RPM	
7	Suction Temperature	Deg. F	
8	Condensing Temperature	Deg. F	
9	Capacity at Design Temperature	TR	
10	Actual Capacity	TR	
11	Nos. Of Compressors/ODU	Nos.	
12	Performance at Constant Condenser Temperature of 95 deg. F		
a)	100% load	IKW / TR	
b)	90% load	IKW / TR	
c)	80% load	IKW / TR	
d)	70% load	IKW / TR	
e)	60% load	IKW / TR	
f)	50% load	IKW / TR	
g)	40% load	IKW / TR	
h)	30% load	IKW / TR	
I)	20% load	IKW / TR	
j)	10% load	IKW / TR	
C)	Motor		
1	Make		
2	Type of Motor		
3	Electrical Characteristics		
4	Type of Starter		
5	Running Current	Amps.	
6	Starting Current	Amps.	
7	Motor RPM	rpm	
8	Insulation Material & Thickness	mm	
D)	Condenser		
1	Make		

2	Type of Condenser		
3	Model No.		
4	Entering Refrigerant Temperature	Deg. F	
5	Leaving Refrigerant Temperature	Deg. F	
6	Fouling Factor	FPS	
7	Refrigerant Flow Rate	USGPM	
9	Cabinet Material		
10	Tube Material & Thickness	mm	
11	Fin Material & Thickness		

2.	<u>Hi Wall /Cassette/Duct-able Units</u>	
2.1	Make	--
2.1.1	Casing	--
2.1.2	Coil	--
2.2	Type : Vertical/Horizontal	--
2.3	Design : Hideaway/With Outer casing.	--
2.4	Dimension (L x B x H)	mxmxm
2.5	Cooling coil	
2.5.1	Coil Area	Sq.m.
2.5.2	No. of rows	Nos.
2.5.3	No. of fins/cm	Nos.
2.5.4	Tube dia. OD	mm
2.5.5	Thickness of tube	mm
2.6	Material of casing CRCA/GI	--
2.7	Air quantity at max speed and 1 M long duct collar.	CMH
2.8	Air quantity at min. Speed and 1.0 m. Long duct collar	CMH
2.9	Whether Auxiliary Drain Pan Provided.	Yes/No.
2.10	No. of fan speed	Nos.
2.11	Make & model of room thermostat.	--

3. ELECTRICAL TECHNICAL DATA SHEETS

For HVAC Panel parameters (To be filled by the EPC Contractor)

Sr. No.	Description	Recommended Specification	Confirmation by the EPC contractor
1	Type of Panel	a. MCC non-draw out type compartmentalized. b. AHU Panels non draw out type, non compartmentalized.	
2	Type of Mounting	Free standing Floor Mounted	
3	Fault kA	50kA -1 Sec	
4	Busbar	Aluminium	
5	Thickness of CRCA sheets:		
a.	Structural members	2mm	
b.	Covers and doors	2mm	
c.	Base channel	MCC - ISMC 100	
d.	Gland plate	3mm	
6	Painting/ Process	Synthetic Enamel Paint As per Nine/seven tank process Oven baked.	
7	Paint shade;		
a.	Inside	RAL - 7032	
b.	Outside	RAL - 7032	
8	Details of bus bars	Electrolytic grade Copper of specified rating for details see constructional features mentioned in specifications	
9	Cable Entry	For MCC & AHU Panels Top or Bottom depending upon location of Panel.	
10	Enclosure Protection/Ventilation	IP -65/67 with louvers for Ventilation.	
11	Control Wiring/Power Wiring	Insulated 660 Volts FRLS Cu wire.	
a.	Voltage Circuit	1.5 sq mm	
b.	Current Circuit	2.5 sq mm	
c.	Minimum size of Power wiring Ckt.	16 sq mm	
12	Maximum Operating Height	2100	
13	Mounting height of relays/ Meters/ Control Switches	Range 350mm to 1900mm	
14	Incoming Feeder	2 Nos. suitable rating MCCBs with multifunction meter (V, I, PF, KWH, frequency)	
15	Outgoing feeder	Suitable rating RCCB/MCB/MCCB as recommended by OEM of VRV system.	

APPENDIX – II

PROFORMA FOR TEST RESULTS NOTES ON TEST INSTRUMENTS AND CAPACITY COMPUTATIONS

Sr. No.	Item	Test Results	
1.	Ambient conditions	D.B. Temp W.B. Temp %RH	- deg. C - deg. C
2.	Compressors	R.P.M Suction pressure Discharge pressure Oil pressure	- Kg/sq.cm -Kg/sq.cm -Kg/sq.cm
3.	Compressor Motors	R.P.M. Voltage Current (i)at 100% load (ii) at partial load (a) (b) (c)	- Volts - amps - amps - amps
4.	VRV Units	Ref. flow rate Ref. temperature Entering Leaving Ref. pressure Entering Leaving	- LPM - deg. C - deg. C -Kg/sq.cm -Kg/sq.cm
5.	Fresh air intakes	Face area Air quantity	- Sq.m - Cu.m/ min
6.	Room conditions at the working plane (No. of readings shall be taken and averaged out)	Temperature D.B. W.B.	- deg. C - deg. C
7..	Controls	Function of each control shall be tested and report furnished	

21 SMART METERING

The EPC Contractor shall be provided smart metering for Energy, water meter (for appropriate bore size) as per NBC2016/ECBC-2017 as amended up to date and as per GRIHA-3-star rating. smart energy meter to be provided for each building unit of residential complex. In addition to any other energy consuming equipment and systems as per GRIHA 3-star rating etc.

21.1 SMART METERING AND MONITORING SYSTEM

The EPC contractor shall be provided smart metering and monitoring for energy and water consumption as per the requirements of GRIHA.

(A) The source metering requirements as per the Griha manual are as follow:

Source	Description
Energy	Ensure regular monitoring of the project's energy consumption by installing digital meters at the following point sources at the project level for: • Utility grid • On-site renewable energy system • Diesel generator set, gas genset, etc. • Each building level (at each apartment level for residential and at each tenant level for retail and office buildings).
Water	Ensure regular monitoring of the project's water consumption by installing digital meters at the following point sources: • Municipal supply • Borewell • Treated water outlet from sewage treatment plant (grey/black) • Captured rainwater for reuse in project

The EPC Contractor must provide the extended metering requirements mentioned below:

Source	Description
Energy	Digital energy meters to be installed at the following points to monitor energy consumption: A. Office/institutional/retail/transit terminals/healthcare/hospitalities: • Heating, ventilation, and air conditioning central plant: Air handling unit, cooling tower, chillers [British thermal unit meters/multi-function metering, and/or distributed units (split/ window ACs)] • Lighting (indoor and outdoor) • Basement parking lighting B. Residential • For basement parking lighting, community/recreation centre, water pumping, outdoor lighting. • Lifts and common areas.
Water	Digital water meters to be installed at the following points to monitor water consumption: • Flushing (at each building for bulk metering) • Domestic (at each building for bulk metering) • Irrigation • Heating, ventilation, and air conditioning

21.2 All smart metering and monitoring systems should be capable of the following:

- Hourly data reporting in near real-time (no more than 15-minute delay).
- Energy mix breakdown and consumption patterns.
- Water consumption patterns from various sources.
- Ability to set energy and water consumption targets, alarms, and pricing.

- Ability to compare historical trends and benchmark data.
- Real-time monitoring with user interface that operates even on mobile devices.

Ensure installation of smart metering systems which are capable of tracking energy and water consumption through a web-hosted portal for all meters.

Communicable Systems	Points
Installation of one-way communicable system	1
Installation of two-way communicable system	3

22. Technical Specification for Gas Pipelines system

Scope of work

The EPC contractor will design, supply, installation, testing and commissioning gas pipeline system (internal & External) for 192 residential units and guest house kitchen. The gas pipeline work carried out as per T4S published by Petroleum and Natural Gas Regulatory Board (PNGRB) as amended up to date. Gas meter will be provided for each residential unit and guest house kitchen. In addition, other consumables required for successful testing and commissioning will also be provided.

1. Definitions

- (1) In these guidelines, unless the context otherwise requires, -
 - “Act” means The Petroleum and Natural Gas Regulatory Board Act, 2006
 - “MRS” means Metering and regulating station used for pressure controlling and metering of natural gas at customer’s premises with or without pressure regulation.
 - “Internal Piping (IP)” means the section of pipeline installed fabricated from the outlet flange of MRS, up to appliances point.
 - “Commissioning” means safe introduction of natural gas into the internal pipeline.
 - “Authorized Personnel” means a competent person of the agency authorized by CGD or PNGRB who is deployed or assigned to carry out a specific job for conducting inspection and testing of the installation in line with these guidelines.
- (2) Words and expressions used and not defined in these guidelines, but defined in the Act or in the rules or regulations made thereunder, shall have the meanings respectively assigned to them in the Act or in the rules or regulations, as the case may be.

2. Applicability

- (1) These guidelines shall apply to the industrial and commercial customers connected by CGD entity with the of city or local natural gas distribution networks as well as customers consuming natural gas other than for domestic use and categorized under non-domestic segment except in transport for automotive use.
- (2) These guidelines cover the general recommendation for material selection, safe installation, inspection / testing are to be carried out prior to commissioning of the

- facilities at the downstream of MRS up to appliance point.
- (3) This guideline should not be applied retrospectively to such aspect of existing installations. However, the reference to this guideline should be taken for all future alteration and modification in existing PNG installation as appropriate deemed fit by CGD entity.
 - (4) The customer shall ensure compliance to this guideline for the health, safety and integrity of the IP downstream of meter or MRS as well as appliances or equipment throughout the operating period.
 - (5) A certificate from PNGRB approved third party agency certifying that material selection, planning & designing, installation, inspection, testing and commissioning as well as health assessment of the facilities downstream of Meter or MRS shall be obtained by EPC contractor.
 - (6) The customer to ensure that any alteration in IP line or any addition or reduction of equipment post gas-in shall be done after approval from OIL or CGD entity or PNGRB authorized third party inspection agencies with intimation to OIL or CGD entity.
 - (7) Wherever underground PE pipeline is laid, PNGRB T4S for CGD network standard shall be referred.

3. Intent

The guideline is intended for ensuring the health, safety, and integrity of internal installation within the customer premises including and not limited to its material selection, planning & designing, installation, inspection, testing and commissioning as well as operation and maintenance of the facilities downstream of Meter or MRS conforming to specifications laid down in the PNGRB Technical Standards for CGD Networks (T4S).

4. Route Selection

In case gas burners are to be connected directly to the regulator supplied by the authorized entity, the line size to be designed in such a way that adequate pressure is available at burners inlet when all burners (existing as well as proposed) are running on full load capacity.

The meter and regulator or MRS shall be installed at the entry of the premise.

In case a separate regulator(s) is/are to be installed before burners then pressure at the farthest point (before the regulator) should not be less than minimum inlet pressure required at the regulator inlet.

The route is selected by the contractor in such manner that it should not create unsafe condition for new installation as well as alteration cases.

The route shall be chosen in such a way that oil stains, hot applications, electrical appliances, electrical cables, AC Outdoor units etc. are at safe distance from gas line path.

End user has to ensure that IP is not used for any purposes which can compromise the safety and integrity of installation.

Due care should be taken during planning stage to accommodate additional equipment or burners adjoining to existing facility without compromising safety and integrity of installations.

The IP line should be installed aboveground due to reasons attributed to the safety, integrity, ease in inspection, O&M and to avoid corrosion.

Safe route should be selected from where firm structure is available. The structural stability to be ensured by end-user prior to installation and ensure fit for intended use.

Safe distance should be maintained from utilities like steam line, oil line & electric HT line, hot zones, etc.

Passing of IP line should be avoided through less or non-ventilated areas, temporary structure, drain ducts.

Installation of pipeline should be avoided at heights where the construction and subsequent inspection of PNG piping becomes unsafe.

Wherever, adequate ventilation is not available at critical public places, customer should carry out risk assessment and implement mitigation measures (one or more of the followings) reduce the risk to ALARP and Onus of operation and maintenance of the same shall remain with customer.

- Ventilator
- Increase inspection frequency
- PNG Gas detection system
- Solenoid and shut off valve
- Any other means to reduce the risk

5. Materials Selection

All materials and equipment forming a permanent part of the IP system shall be constructed according to this guideline and should be qualified for the conditions in which it is to be used. The materials should be conforming, but not limited to, the standards and specification mentioned in Annexure 1.

6. Installations

The installation of IP line shall be ensured as per PNGRB T4S regulations for CGD networks.

- All welding should be carried out by qualified welder as per API 1104.
- Teflon tape should be used for threaded joints. Any sealant agents or chemicals like M-seal, fevikwik, etc. are not permitted.
- Wherever gas pipeline crosses over other pipes and electric cables/electric points necessary measures to be taken to safe guard the gas line. For pipes passing through walls, sealing should be sleeved and filled with cementing.
- Flanged joints should not be provided within 300 mm distance of electrical points or junction points. However, the service line installed in confined spaces like basement, etc. shall be of welded type only.

- All pipe work should be horizontally and vertically aligned.
- Pipe should be firmly tightened on support with U bolts or clamps, having adequate strength to hold the piping system.
- Adequate clamps / pipe supports on the piping system shall be provided such that the safety and integrity of PNG installation is maintained.
- Pipe support or clamps should be firmly fixed on wall or structure. A safe gap of 1" should be maintained between pipe and wall.
- Provide rubber sheet between pipe and support to protect pipe from corrosion. Provide PVC coated "U" clamp to support the pipe of size 2" and higher.
- Pipe support should not be mounted on temporary wall/temporary shed/trees / glass or wooden facades, etc.
- In case provision is made by customer to have alternate fuel source during natural gas supply disruption, then the customer should ensure that both the fuels shall not be used at a time using same IP line (i.e. both the fuel should not be mixed). The customer should get the "Changeover operation methodology" and customized site-specific P&ID approved by CGD entity or PNGRB authorized TPA.
- In case of IP modification or alteration after gas commissioning, the entire pipeline is to be isolated, purged using nitrogen prior to carry out any welding operation.
- Flanges joint should have nuts/stud bolts. The extra length stud bolts to be used such that at least 3 pitches are visible on both sides after placement of washers. All stud bolts and nuts should be galvanized as per ASTM A153.

7. Corrosion protection and Markings

- Pipes / fittings / valves / supports & other equipment shall be painted.
- Special care should be taken to ensure uniform painting thickness throughout entire pipework.
- Direction of gas flow and PNG should be marked along the natural gas pipeline.
- Self-adhesive anti-corrosion tapes or PE sleeve while crossing of walls or slabs or corrosion prone area to protect the pipe from corrosion.

8. Inspection and Testing

- Entire pipe work should be flushed with nitrogen / air to remove spoils, dust, dirt, welding spatter etc. Meter and regulator should be removed before flushing to avoid any potential damage. Flushing should be done from each end point.
- After flushing, the entire pipework should be pneumatically tested, after removing meter and regulators, as per T4S for CGD networks.
- Testing should be done with all appliances valves open and plugged. Each joint, specifically threaded joints should be checked with leak detection method. All necessary precautionary measures should be taken during testing.
- For IP operating at (minimum operating pressure) MOP up to 4 bar, the testing shall be done with 1.5 times design pressure and pressure holding period shall be minimum 60 min.
- After successfully completion of testing, the pressure should be vented out safely, meter and regulators may be kept in place. The testing certificate should be produced and signed by Contractor and Authorized personnel of approved third party inspection agency as per the proforma. (Annexure2 to be filled by TPIA).
- If pressure in Meter or MRS downstream pipeline is operating at the pressure 4 bar or above, Nondestructive examination and soundness testing should be carried out as per T4S for CGD network regulation. All defects identified during NDE shall be repaired.

- For IP operating at MOP up to 4 bar, all welded/ threaded joints are inspected and defects are repaired prior to commissioning.
- The leak survey should be carried out annually to ensure safe control of operation by customer.
- All the pressure tests conducted at every three years shall be witnessed by authorized personnel of approved TPA issuing a certificate for being fit for use. The fit for use certificate shall be submitted to Engineer In-charge of CGD entity for their records and prior to commissioning.

9. Commissioning

Before start of commissioning works, all the relevant details of pre commissioning checks, safety audits including inspection and testing are to be recorded, so as to ensure that operations are carried out in a safe manner.

After completion of the pre-commissioning checks, the pipeline network is considered to be ready for the commissioning.

The relevant records and documents should be made available prior to commissioning, i.e., material test reports, testing reports, equipment test & calibration reports, drawings, etc.

Internal pipeline checks

- Remove blind spade from downstream of isolation valve
- Keep the vent valve in the farthest end in open condition.
- Ensure no open flame and other flammable material is available near the venting area.
- Commission IP line by gradually opening of isolation valve and ensure more than 90% CH₄ by safe venting of natural gas at farthest vent point.

10. Safety Measures

Adequate No. of safety gadgets, personal protective equipment, emergency handling facilities, sign boards, Do's and Don'ts, etc. as required during the installation, testing and commissioning should be in place and available at designated place. The following should be ensured at sites.

- The work area has been cordoned off and required displays are provided.
- A minimum of one (1) dry powder type extinguishers (9 kg) should be available at the time of purging /venting / commissioning activities.
- Smoking, naked light or other sources of ignition should be prohibited. Warning signage/ pictograms like "NO SMOKING" and "NO NAKED LIGHTS" should be prominently displayed around the work site, including vent points during purging and commissioning.
- Signage covering Emergency contact number of entity to be affixed on Meter or MRS.
- Electrical continuity bonds (copper jumpers) are fitted across separated metallic pipes prior to purging operation.
- It should be emphasized that purged gas other than air is potentially dangerous for asphyxiation and information to prevent asphyxiation should be provided in procedure.
- Proper housekeeping shall be maintained by customer around MRS or metering areas.
- In any case, no hazardous or inflammable materials shall be placed in vicinity of PNG installation.

- All installations must have displayed the Dos and DON'Ts based on those provided by CGD entity.

The entity reserves right to discontinue gas supply to customer in the event on any non-compliances or unsafe installation or unsafe practices observed by CGD entity at any point of time during and after commissioning.

Annexure 1: Material Standards and Specifications

Sr. No.	Item	Material Type	Specification
1	Steel pipe / Galvanized pipe	Mild steel (M.S.) or Carbon Steel (C.S)/Galvanized pipe	ERW GI pipes (Class C) as per IS 1239 Part 1 standard or API 5L Gr. B or ASTM A106 or ASTM A333
2	PE pipes and Fitting sand valves (underground)	Polyethylene	PE100/ PE80 grade, as per IS ISO 4437-2, ISO 4437-3 and ISO 4437-4 standards
2	Isolation Valve	Brass/Carbon Steel (C.S)	Up to and including 2": EN 331 More than 2": API 6D
3	Galvanized Fittings	GI Fittings, Mild steel (M.S.) or Carbon Steel (C.S)	IS 1239 (Part-2); IS 1879, ASME B16.9, ASME B16.11, ASTM A234
4	Galvanized Flanges (forged)	Carbon Steel	SORF, 150#, ASME B 16.5, Material as per ASTM A 105 hot dip galvanized
5	Socket weld end fittings	CS Forged, Socket Welding fitting,	ASTM A 234 WPB, Dimensions as per ANSI B16.11, hot Dip galvanized
6	PE fittings	Electro fusion Fitting	ISO 4437-3
7	Flexible Hose	Stainless Steel (S.S.) 316L	SS braided bellow pipe
8	Gasket	CNAF- Spiral wound or non-metallic flat	ASME B16.20/ASME B16.21
9	Galvanized Fasteners	steel	ASTM A193 /A 194 Grade B7 /2 H
10	Galvanizing	Hot dip Zinc coating	IS 4736
11	"U" clamp	GI Coated and PVC sleeve	1"to 4"
12	Welding Electrodes		AWS E 6010 &E6013
13	Stud bolt, Nut	Galvanized	Stud - ASTM A 193 GrB7 Nut – ASTM A 194
14	Filter	Stainless steel	50-micron mesh and forged body. Internal SS 316 & External ASTM A 106.
15	Self-adhesive Anti corrosive Tape for wall crossing	300-micron, 2.5 cm or 5 cm (1" or 2")	As approved by RITES/OIL/CGD entity

Part D

**Technical Specification
for
Signages**

Summary

CPWD SPECIFICATIONS 2019, CODE OF PRACTISE FOR ROAD SIGNS, INDIAN ROAD CONGRESS 2010, IRC-67-2010, NBC-2016 & DIRECTIONS OF THE ENGINEER IN CHARGE		
Sl. no.	ITEMS OF WORKS	SPECIFICATION
	External/Internal Signages	
1	Wall/ Pillar mounted external signage (External)	Providing, Supplying & fixing Single sided Sign Board of max size 600 mm x 900 mm with suitable inner frame and proper support, with sign substrate of suitable size of 4mm ACP sheet with face fully covered by RETRO REFLECTIVE MICRO PRISMATIC Type IV Sheeting and messages and symbols accordingly, fixed to inner frame with the help of Very High Bond tape of 24mm wide and 2.3mm thick. The board is to be fixed on RCC wall/pillar with dash fasteners of SS pipe of S-304 grade. All stainless steel items shall be of S-304 Grade (EXTERNAL SIGNAGES). (Material should be in compliance with Code of practise for Road signs, Indian Road congress 2010, IRC-67-2010). Quantity as under (All in Capital Letters): 02 nos. for each Residential Tower – Eg. “TOWER-A” 01 nos. for Guest House, Community Centre, Electrical Sub-station, Pump Room & UGWT, STP 02 nos. for Guard Room
2	Building/Location identifier (Map with “YOU ARE HERE” sign (External)	Providing, Supplying, fabrication & fixing Single sided Sign board (Information panel/map) of size 1220mm x 1800mm (50mm dia SS pipe of S-304 grade 1.6mm thick) with inner frame of 1120mm x 1700mm (of 19mm x 24mm rectangle, 1.2mm Thick SS pipe of S-304 Grade) along with diagonal support also of 19mm x 24mm rectangle, 1.2mm Thick SS pipe of S-304 Grade, with sign substrate of size 1120mm x 1700mm, of 4mm ACP with face fully covered by RETRO REFLECTIVE MICRO PRISMATIC Type IV Sheeting, and messages and symbols, maps, “YOU ARE HERE” marker, directional arrows (for blocks, other buildings, visitor parkings, exits) accordingly, fixed to inner frame of 1120mm x 1700mm of (19mm x 24mm rectangle, 1.2mm Thick SS pipe of S-304 Grade) with the help of Very High Bond tape of 24mm wide and 2.3mm thick. The board shall have a one leg support of 168mm dia 3 mm thick SS pipe of S-304 grade of height 2520 mm with base plate of 10mm thick SS plate of size 600mm x 600mm with 4 Nos. stiffeners of 10mm SS plate of S-304 grade. The boards shall be fixed on a RCC foundation of appropriate size as per design laid with the help of shuttering and 12 Nos. of foundation bolts of 20mm dia. SS, S-304 grade of 600mm length including excavation for foundation, PCC 1:3:6 all complete. (Material should be in compliance with Code of practise for Road signs, Indian Road congress 2010, IRC-67-2010). Quantity – 02 nos. at suitable location decided by Engineer-in-charge.
3	Room number for Guest House/ House Number for Flats	Manufacturing, Supplying & fixing of S.S.Plate (1.5 mm thick) of size (200 mm x 100 mm) made of 2mm SS sheet with message and letters Plotter / Electronically cut text of vinyl lamination including all nuts, bolts and other accessories.

CPWD SPECIFICATIONS 2019, CODE OF PRACTISE FOR ROAD SIGNS, INDIAN ROAD CONGRESS 2010, IRC-67-2010, NBC-2016 & DIRECTIONS OF THE ENGINEER IN CHARGE		
Sl. no.	ITEMS OF WORKS	SPECIFICATION
		(Material should be in compliance with Code of practise for Road signs, Indian Road congress 2010, IRC-67-2010). For every Flat number of Residential Towers and Room numbers for Guest House. Message & Letters decided by Engineer-in-charge.
4	Shaft/Utility space	Stainless Steel Plate (Grade SS 304), Size: 300 x 200 mm, Thickness: 1.5 mm with both side Laser Engraving / Etching with Name as per Engineer in-charge approval (Black-filled / RETRO REFLECTIVE VINYL with color approved by the Engineer Incharge) Complete with wall / door Mounting Accessories (Studs, Screws, Wall Plugs) Installation on or above the shaft / room door. For Elevators (Lifts), Electrical shaft, LV shaft, Fire shaft, Floor number for each building as applicable. Engraving/etching name to be decided by Engineer-in-charge.

NOTE: For EMERGENCY LIGHT & ILLUMINATED SIGNAGE refer E&M specifications.

Part E

KITCHEN ITEMS SPECIFICAITONS FOR GUEST HOUSE

COMMERICAL KITCHEN ITEMS SPECIFICAITONS FOR GUEST HOUSE

- The equipment to be procured, supplied, installed & commissioned after approval of RITES/OIL.

S. No.	ITEM DESCRIPTION	SIZE	QTY
01	DOUBLE DEEP FAT FRYER		1
	Power: 2.5Kw × 2 Cap.: 5.5 Ltrs. × 2		
02	TABLE FOR FRYER	600X750X850+150	1
	- Top in 18 swg thick SS 304 sheet bent 50mm and 12mm on all sides excepting sides with splash back. - Rear splash raised as per boq above the working height of 850mm and bent 25mm. - Frame work of MS 32x32x3mm angle - Middle & Bottom Shelves in 20 swg thick SS sheet. - Nylon bullet feet of 20mm 150 mm ground clearance.		
03	FOUR BURNER RANGE	750X750X850	1
	- Top in 16swg s.s. 304sheet bent 50mm and 12mm on all sides excepting on sides with splash back; - Frame work of MS 32x32x3mm angle - Front in 20swg s.s. sheet, sides / back in 20swg s.s. sheet - Heavy duty removable grates of cast iron to facilitate easy removal of flue and to minimize heating up of cooking range s.s. top - Burner brackets of 0.9mm thick s.s. sheet - Slide out drip trays of 0.9mm thick GI sheet with 16swg thick s.s. handle. 4 nos. T 35 burners with pilot flame. - Under Shelf in 18-gauge 25mm square pipe - Individual needle control valves for burner and pilot flame; recessed valves; - Burner manifold of Ç' class pipe with 3/8" Copper inlet. - Nylon bullet feet of 20mm		

04	GRIDDLE PLATE	600X750X850	1
	• Plate of 12 mm thick polished MS plate for griddle plate • Front in 20swg s.s. sheet; sides / back in 20swg s.s. sheet • Burner brackets of 25x25x3mm SS angles with facility for easy removal of burners • 2 nos. V shaped Unitech burners • Individual needle control valves for each burner and pilot flame; recessed valves • Burner manifold of Ç' class pipe with 3/8" Copper inlet. • Adjustable bullet feet • Frame work of MS 32x32x3mm angle • Under Shelf in 18-gauge 25mm square pipe • 150mm ground clearance.		
05	WORK TABLE	750X750X850+150	1
	• Top in 18 swg thick SS 304 sheet bent 50mm and 12mm on all sides excepting sides with splash back. • Rear splash raised as per boq above the working height of 850mm and bent 25mm. Middle & Bottom Shelves in 20 swg thick SS sheet. • Nylon bullet feet of 20mm • 150 mm ground clearance.		
06	WORK TABLE WITH SINK	1125X600X850+150	1
	• Top in 16 swg thick SS 304 sheet bent 50mm and 12mm on all sides excepting sides with splash back. • Rear splash raised as per boq above the working height of 850mm and bent 25mm. • Bottom Shelf in 20 swg thick SS 304 sheet. • Adjustable Nylon bullet feet. • 150 mm ground clearance. SINK • One Sink in 16swg SS sheet • 38mm dia CP drain waste.		

07	TWO BURNER CHINESE RANGE	1200X700X850+300	1
	- Recessed top 25 mm in 16swg thick ss sheet bent 50mm and 12mm on all sides excepting sides with splash back. - Rear splash raised 300mm above the working height and bent 25mm - Front in 20 swg s.s.sheet with recessed design. - Sides / back in 20swg s.s. sheet ; both sides with holes for air inlet - Heavy duty round removable grates of cast iron or MS bars ; 400 mm dia. - One T 50 Speed jet Burners with individual pilot flames with needle control valves. - Burner manifold of Ç' class ERW pipe with 3/8" Copper inlet ; Nylon bullet feet of 20mm - Drainage with filter for hygiene, front side to have plumbing pipe with water slope towards drain - Under Shelves in 20 swg thick SS sheet. 150 mm ground clearance.		
08	MIZZA TABLE	600X600X850+150	1
	- Top in 18 swg thick SS sheet bent 50mm and 12mm on all sides excepting sides with splash back. - Rear splash raised as per boq above the working height of 850mm and bent 25mm. Middle & Bottom Shelves in 20 swg thick SS sheet. - Nylon bullet feet of 20mm 150 mm ground clearance.		
09	MAKELINE UNDER COUNTER REF.	1365X700X850	1

	Unit top constructed of # 16ga. S.S.304 top with Front turned down 2" & in 1/2", Rear & both sides turned down 2" & flush with body cladding. Top shall be secured to 32x32x3mm thick M.S. welded angle frame work. Exterior/interior cladding no. 20 ga. S.S.304 sheet with 2" thk. Puff insulation. Front of unit shall be provided two nos. doors with integral handles & cylindrical locking arrangement & magnetic gasket. Internal temperature 0 deg. To + 5 deg. to be maintained. Unit provided four no. removable & adjustable S.S. welded wire shelves on the s.s. steel clips, side channel supported adjustable height. RIGHT side of unit shall be provided with 1.0 h.p. normal temperature type Emerson/any reputed compressor w/ louvered access panels, air cooled single phase, independent controls, digital thermometer & electrical cord. Conventional cooling with thermostat. Bottom of internal tank shall be provided 1" dia drain outlet and removable perforated no. 22 ga. S.S. false bottom & below no. 18 ga. S.s. drip tray w/ integral handle. Unit shall be mounted on 1-1/2" square pipe, 6" high s.s. legs.W/ nylon feet. Top of the unit shall have space to accomodate GN pans		
10	OVERHEAD HEATING ELEMENT	1365X300	1
	Overhead Shelf With Food Warmer. Include as standard equipment four stainless steel shelf mounting tabs. an on-off switch to be installed to either the front or rear of the unit. The infrared heating element shall be tubular metal sheathed.		
11	WORK TABLE	900X700X850+150	1

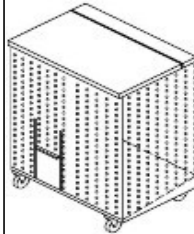
	- Top in 18 swg thick SS sheet bent 50mm and 12mm on all sides excepting sides with splash back. - Rear splash raised as per boq above the working height of 850mm and bent 25mm. Middle & Bottom Shelves in 20 swg thick SS sheet. - Nylon bullet feet of 20mm 150 mm ground clearance.		
12	OVERHEAD SHELF	900X300	1
	One no.s full length # 18 ga. S.S. over head shelves of 12" width, which are mounted on 1" S.S. pipe uprights		
13	THREE BURNER RANGE	1500X700X850	1
	- Top in 16swg thick SS 304 sheet bent 50mm and 12mm on all sides. - Front in 20swg SS 304 sheet with recessed design. - Sides / back in 20swg s.s. sheet - Heavy duty removable grates of cast iron or MS bars ; 400X400mm. - Grates to facilitate easy removal of flue and to minimize heating up of cooking range s.s. top. - Cutouts for grates to be bent as channels; cutouts for grates to be bent as channels. - Burner brackets of 0.9mm thick s.s. sheet. frame work, of 32x32x3mm MS Angles - T 35 canteen Burners with individual pilot flames. - Individual needle control valves for each burner and pilot flame. - Bottom Shelves in 20 swg thick SS sheet. - Burner manifold of C' class ERW pipe with 3/8" Copper inlet; - Adjustable Nylon bullet feet 150 mm ground clearance		

14	WORK TABLE	600X700X850+150	1
	- Top in 18 swg thick SS 304 sheet bent 50mm and 12mm on all sides excepting sides with splash back. - Rear splash raised as per boq above the working height of 850mm and bent 25mm. - Middle & Bottom Shelves in 20 swg thick SS sheet. - Nylon bullet feet of 20mm 150 mm ground clearance.		
15	SS TANDOOR	750X750X850	2
	- Exterior in 18swg SS. sheet - Clay shell. - Steel ring at mouth - Rear splash raised 100mm above the working height of 850 mm and bent 25mm - Rear side of splash to be closed. - Two flexible handles to facilitate the movement 4 nos. 100x50mm casters with ss brackets, two with brakes - Front open-able mouth for cleaning ash		
16	WORK TABLE	450X750X850+150	2
	- Top in 18 swg thick SS sheet bent 50mm and 12mm on all sides excepting sides with splash back. - Rear splash raised as per boq above the working height of 850mm and bent 25mm. - Middle & Bottom Shelves in 20 swg thick SS sheet. - Nylon bullet feet of 20mm 150 mm ground clearance.		
17	TABLE WITH SINK	900X700X850+150	1

	- Top in 16 swg thick SS 304 sheet bent 50mm and 12mm on all sides excepting sides with splash back. - Rear splash raised as per boq above the working height of 850mm and bent 25mm. - Bottom Shelf in 20 swg thick SS 304 sheet. - Adjustable Nylon bullet feet. 150 mm ground clearance. SINK - One Sink in 16swg SS sheet 38mm dia CP drain waste.		
18	UNDER COUNTER REF.	1500X700X850	1
	Unit top constructed of # 16ga. S.S.304 top with Front turned down 2" & in 1/2", Rear & both sides turned down 2" & flush with body cladding. Top shall be secured to 32x32x3mm thick M.S. welded angle frame work. Exterior/interior cladding no. 20 ga. S.S.304 sheet with 2" thk. Puff insulation. Front of unit shall be provided two nos. doors with integral handles & cylindrical locking arrangement & magnetic gasket. Internal temperature 0 deg. To + 5 deg. to be maintained. Unit provided four no. removable & adjustable S.S. welded wire shelves on the s.s. steel clips , side channel supported adjustable height . RIGHT side of unit shall be provided with 1.0 h.p. normal temperature type Emerson/any reputed compressor w/ louvered access panels, air cooled single phase, independent controls, digital thermometer & electrical cord. Conventional cooling with thermostat . Bottom of internal tank shall be provided 1" dia drain outlet and removable perforated no. 22 ga. S.S. false bottom & below no. 18 ga. S.s. drip tray w/ integral handle. Unit shall be mounted on 1-1/2" square pipe , 6" high s.s. legs.W/ nylon feet.		
19	WALL MOUNTED DECK OVEN	1 DECK 2 TRAY	1
	Dimension: 1235×800×500mm Chamber: 860×660×200mm Power: 6.6Kw		

20	TWO DOOR VERTICAL REF.	620X692X1887	1
	Exterior cladding # 20 ga. S.S.304 sheet and Interior cladding # 20 ga. S.S.304 sheet w/ 2" thk. Puff insulation. Front of unit shall be provided with #18 ga. SS insulated spring operated doors with gasket and with two nos. plastic handles, each door hinged towards the sides, supported by 8 mm SS flat. Internal temperature +0 deg. to +5 deg. to be maintained. Refrigerant to be CFC free-R-134A. To provide TWO (2) nos. wire shelving of SS 8mm thick wire, dully spot welded and polished with 18ga SS rack brackets and adjustable bracket assembly. Front of the unit shall have panel for parition into two equal sections. Top of unit shall be provided with 1/3 H.P Emerson/any reputed make hermetically sealed compressor w/ louvered access panels, independent controls, digital control, digital display on / off swith with indicator lights with electrical cord. . Bottom of internal tank shall be provided with 1.5" dia drain outlet and 22 ga. S.S. drip tray with integral handles. Unit shall be mounted on 1-1/2" dia, 6" high s.s. legs. with nylon feet's.		
21	DEEP FREEZER MAKE - WESTERN	525 LTR	1
22	PREPARATION TABLE	900X600X850+150	2
	- Top in 18 swg thick SS 304 sheet bent 50mm and 12mm on all sides excepting sides with splash back. - Rear splash raised as per boq above the working height of 850mm and bent 25mm. - Middle & Bottom Shelves in 20 swg thick SS sheet. - Nylon bullet feet of 20mm 150 mm ground clearance.		
23	OVERHEAD SHELF	900X300	2
24	PLANTERY MIXER	20 LITRES	1
25	JTC BLENDER	T800	1
26	SINGLE SINK	600X600X850+150	1
	Top in 16 swg thick SS 304 sheet bent 50mm and 12mm on all sides excepting sides with splash		

	back. ·Rear splash raised 150mm above the working height of 850mm & bent 25mm. ·Side bracing in 25mm x 18 swg SS square pipe ·Adjustable Nylon bullet feet ·150 mm ground clearance ·Sink in 16 swg SS sheet deep drawn sinks will be preferred. ·38mm dia CP drain waste		
27	PICKUP TABLE WITH OHS	1125X600X850+450+3100	1
	·Top in 18 swg thick SS 304 sheet ·Under Shelf in 20 swg thick SS sheet. Top of unit provided two no.s full length # 18 ga. S.S. overhead shelves of 12" width, which are mounted on 1" S.S. pipe uprights ·Adjustable Nylon bullet feet ·150 mm ground clearance		
28	DISH LANDING TABLE	1350X600X850+150	1
	·Top in 16 swg thick SS 304 sheet bent 50mm and 12mm on all sides excepting sides with splash back. ·Top recessed by 25 mm, 1 Nos of 150mm Ø x100mm height garbage chute. ·Rear splash raised 150mm above the working height of 850mm & bent 25mm. ·3 side bracing in 25mm x 18 swg SS square pipe ·Adjustable Nylon bullet feet ·150 mm ground clearance		
29	GARBAGE BIN	UNDER TABLE	1
30	THREE SINK UNIT	1500X600X850+150	1
	·Top in 16 swg thick SS 304 sheet bent 50mm and 12mm on all sides excepting sides with splash back. ·Rear splash raised 150mm above the working height of 850mm & bent 25mm. ·Side bracing in 25mm x 18 swg SS square pipe ·Adjustable Nylon bullet feet ·150 mm ground clearance ·Sink in 16 swg SS sheet deep drawn sinks will be preferred. ·38mm dia CP drain waste		

31	CLEAN DISH TABLE	600X600X850+150	1
	- Top in 18 swg thick SS 304 sheet bent 50mm and 12mm on all sides excepting sides with splash back. - Rear splash raised as per boq above the working height of 850mm and bent 25mm. - Bottom Shelves in 20 swg thick SS sheet. - Nylon bullet feet of 20mm 150 mm ground clearance.		
32	RACK	750X450X1800	2
	- All welded construction. 5 Nos. of shelves - Shelves in 20swg thick ss. 304sheet - SS hardware - Adjustable Nylon bullet feet 150 mm ground clearance		
33	POTATO ONION BIN MS	600X600X850	2
	- Construction in 20swg thick MS perforated sheet 18/20swg thick M.s. hinged lid at top - Sliding shutters in front for FIFO dispensing - Sloping base - Lid to overhand by 25mm in front so as to be used as handle. - Floor of chute will be at 325mm height. - Chute to be flushed with base. 150 mm ground clearance		
34	MS STORAGE RACK	750X450X1800	4
	- All welded construction. 5 Nos. of shelves - Shelves in 20swg thick Ms. sheet - MS hardware 150 mm ground clearance		
35	AATA /MAIDA DRUM	25 KGS	4
36	PLASTIC PALLET	900X600X150	2
37	EXHAUST HOOD	2100X900X450	1
	- Fabrication in 20/22 swg s.s. 202 sheet - Equipped with grease extraction baffle filters SS		

	· Grease drain pan · Channels for fixing of filters · 150 mm wide		
38	EXHAUST HOOD · Fabrication in 20/22 swg s.s. 202 sheet · Equipped with grease extraction baffle filters SS · Grease drain pan · Channels for fixing of filters	1350X900X450	1
39	EXHAUST HOOD · Fabrication in 20/22 swg s.s. 202 sheet · Equipped with grease extraction baffle filters SS · Grease drain pan · Channels for fixing of filters	1650X900X450	1
40	EXHAUST HOOD · Fabrication in 20/22 swg s.s. 202 sheet · Equipped with grease extraction baffle filters SS · Grease drain pan · Channels for fixing of filters	900X900X450	2
41	EXHAUST HOOD · Fabrication in 20/22 swg s.s. 202 sheet · Equipped with grease extraction baffle filters SS · Grease drain pan · Channels for fixing of filters	900X900X450	1

SECTION - 6

DRAWINGS

N.B: All tender drawings along with their respective list are attached separately

SECTION – 6 A
(Topographic Survey Report/Drawings (Indicative only))

Sl. No.	Name
1.	Topographic Survey Report/Drawings (Indicative only) - attached separately

N.B: Attached separately

SECTION – 6 B
(Geotechnical Investigation Report (Indicative only))

Sl. No.	Name
1.	Geotechnical Investigation Report (Indicative only) - attached separately

N.B: Attached separately

SECTION – 6 C
(Design Basis Report)

Sl. No.	Name
1.	Design Basis Report

N.B: Attached separately

Section 6D:
(Schedule of Finishes)

Sl. No.	Name
1.	Schedule of Finishes for Civil, Sanitary & Plumbing Works

N.B: Attached separately

PART - 3

**GENERAL CONDITIONS OF CONTRACT FOR EPC
PROJECTS**

GENERAL CONDITIONS OF CONTRACT FOR EPC PROJECTS

CONTENTS

Sl. No.	Details
SECTION - 7	Conditions of Contract
SECTION - 8	Clauses of Contract
SECTION - 9	RITES Safety Code
SECTION - 10	RITES Model Rules for Protection of Health and Sanitary Arrangements for Workers
SECTION - 11	RITES Contractor's Labour Regulation

Section – 7

CONDITIONS OF CONTRACT

SECTION - 7

CONDITIONS OF CONTRACT

Definitions:

1. The **EPC Contract** means the documents forming the tender and acceptance thereof and the formal agreement executed between the competent authority on behalf of the Employer and the Contractor, together with the documents referred to therein including these conditions, the specifications, designs, drawings and instructions issued from time to time by the Engineer-in-Charge and all these documents taken together, shall be deemed to form one Contract and shall be complementary to one another.

Meaning of Expressions

2. In the Contract, the following expressions shall, unless the context otherwise requires, have the meanings, hereby respectively assigned to them:
 - (i) The expression **works** or **work** shall, unless there be something either in the subject or context repugnant to such construction, be construed and taken to mean the works by or by virtue of the contract contracted to be executed whether temporary or permanent, and whether original, altered, substituted or additional.
 - (ii) The **Site** shall mean the land, places on, into or where work is to be executed under the Contract or any adjacent land, path or street or where work is to be executed under the Contract or any adjacent land, path or street which may be temporally allotted or used for the purpose of carrying out the Contract.
 - (iii) The **EPC Contractor/ Contractor** shall mean the individual, firm, company or joint venture (if allowed), whether incorporated or not, undertaking the works and shall include the legal personal representative of such individual or the persons composing such firm or company or joint venture (if allowed) or the successors of such firm or company or joint venture (if allowed) and the permitted assignees of such individual, firm, company or joint venture (if allowed).
 - (iv) The **Principal / Employer** means the organization as mentioned in **Schedule 'F'** represented by RITES Ltd. for and on behalf of it, as an Agent/Power of Attorney Holder.
 - (v) The **Engineer-in-charge** means the Engineer Officer of RITES Ltd. acting for and on behalf of the Employer as Agent/Power of Attorney

Holder and who shall supervise and be in-charge of the work and who shall sign the Contract on behalf of the Employer, as mentioned in **Schedule 'F'**.

- (vi) The Engineer means the person to whom the Engineer-in-charge entrusts as his authorized representative his responsibility to act on his behalf and perform any or all the functions of the Engineer-in-Charge under the Contract. Engineer-in-charge can in his own discretion entrust his responsibilities to act on his behalf and perform any or all the functions of the Engineer-in-Charge under the Contract to more than one person (by whatever name designated in RITES like Site Engineer, Controlling Site Engineer or Project Coordinator etc.). The word “Engineer” in this contract document is applicable to any or all such persons to whom Engineer-in- Charge entrusts his responsibilities.
- (vii) Accepting Authority shall mean the authority mentioned in **Schedule 'F'**.
- (viii) **Excepted Risks** are risks due to riots (other than those on account of contractor's employees), war (whether declared or not) invasion, act of foreign enemies, hostilities, civil war, rebellion revolution, insurrection, military or usurped power, any acts of Government, damages from aircraft, acts of God, such as earthquake, lightening and unprecedented floods, and other causes over which the contractor has no control and accepted as such by the Accepting Authority or causes solely due to use or occupation by the Employer of the part of the works in respect of which a certificate of completion has been issued or a cause solely due to the Employer’s faulty design of works.
- (ix) Market Rate shall be the rate as decided by the Engineer-in-charge on the basis of the cost of materials and labour at the site where the work is to be executed plus the percentage mentioned in **Schedule 'F'** to cover all overheads and profits.

Provided that no extra overheads and profits shall be payable on the part(s) of work assigned to other agency(s) by the contractor as per terms of contract.
- (x) Schedule(s) referred to in these conditions shall mean the relevant schedule(s) annexed to the tender documents or the standard Schedule of Rates mentioned in **Schedule 'F'** hereunder, with the amendments thereto issued up to the date of receipt of the tender.
- (xi) Specifications means the technical specifications to be followed by the Contractor on the work to be executed.
- (xii) **Tendered value** means the value of the entire work as stipulated in the letter of award.

- (xiii) **Date of commencement of the work:** The date of commencement of work shall be the date of start as specified in schedule 'F' or the first date of handing over of the site, whichever is later, in accordance with the phasing if any, as indicated in the tender document.
- (xiv) **“Class - I Local supplier”** means a supplier or service provider, whose goods, services or works offered for procurement, meet the minimum local content as prescribed for ‘Class-I supplier’ under the Public Procurement (Preference to Make in India), Order 2017; (Notified vide Order No. P-45021/2/2017-PP (BE-II) issued by Department of Promotion of Industry and Internal Trade dated 16.09.2020), including any amendment or revisions thereof.
- (xv) **“Class - II Local supplier”** means a supplier or service provider, whose goods, services or works offered for procurement, meet the minimum local content as prescribed for ‘Class-II supplier’ but less than that prescribed for ‘Class -I local supplier’ under the Public Procurement (Preference to Make in India), Order 2017;
- (xvi) **Stipulated date of completion:** It is the date arrived at by adding the time allowed for completion as specified in Schedule ‘F’ to the date of commencement of the work.
- (xvii) **Replacement Cost** means (a) tendered value plus (b) cost of materials supplied by the Employer (assessed cost of such materials if the materials are supplied free of cost) plus 10% of both (a) and (b).
- (xviii) **GST** shall mean Goods and Service Tax - Central, State and Inter State.
- (xix) **Force Majeure (FM)** – means conditions beyond control of contracting parties like war, hostility, acts of public enemy, civil commotion, sabotage, serious loss or damages by fire, explosions, epidemics, strikes, lockout or act of God.
- (xx) **RITES means** RITES Ltd., a Public Sector Enterprise under the Ministry of Railways, acting for and on behalf of the Employer as an Agent/Power of Attorney Holder.
- (xxi) **Concurrent delay:** Concurrent delays are those delays occurring in the work concurrently in any combination or combination of all delay fall under different sub clauses 5.2, 5.3 and 5.5.
- (xxii) **Adolescent Person:** A person who has completed his/her fourteenth year of age but has not completed his eighteenth year.
- (xxiii) **Hazardous works:** Hazardous process/works are the works as defined in the clause (cb) of the Factory Act, 1948.

Scope and Performance

3. Where the context so requires, words imparting the singular only also include the plural and vice versa. Any reference to masculine gender shall whenever required include feminine gender and vice versa.

Headings and Marginal Notes

4. Headings and Marginal notes to these General Conditions of Contract shall not be deemed to form part thereof or be taken into consideration in the interpretation or construction thereof or of the Contract.

Documentation to be supplied to the Contractor

5. The Contractor shall be furnished, free of cost one certified copy of the Contract documents except CPWD standard specifications, CPWD Schedule of Rates and such other printed and published documents, together with all drawings as may be forming part of the tender documents. None of these documents shall be used for any purpose other than that of this Contract.

Works to be carried out

6. The work to be carried out under the Contract shall, except as otherwise provided in these conditions, include all labourers, materials, tools, plants, equipment and transport which may be required for in preparation of and for and in the full and entire execution and completion of the works. The descriptions given in the Schedule of Quantities (Schedule A) shall, unless otherwise stated, be held to include wastage on materials, carriage and cartage, carrying and return of empties, hoisting, setting, fitting and fixing in position and all other labours necessary in and for the full and entire execution and completion of the work as aforesaid in accordance with good practice and recognized principles.

Sufficiency of Tender

7. The Contractor shall be deemed to have satisfied himself before tendering as to the correctness and sufficiency of his tender for the works and of the rates and prices quoted in the Schedule of Quantities, which rates and prices shall, except as otherwise provided, cover all his obligations under the Contract and all matters and things necessary for the proper completion and maintenance of the works.

Discrepancies and Adjustment of Errors

8. The several documents forming the Contract are to be taken as mutually explanatory of one another, detailed drawings being followed in preference to small scale drawings and figured dimensions in preference to scale and Special Conditions in preference to General Conditions.

- 8.1 In the case of discrepancy between the Schedule of Quantities, the Specifications and/or the Drawings, the following order of preference shall be observed:
- (i) Description of Schedule of Quantities.
 - (ii) Technical/Particular Specifications and Special Conditions, if any.
 - (iii) Drawings
 - (iv) CPWD Specifications
 - (v) Indian Standard Specifications of BIS.
- 8.2 If there are varying or conflicting provisions made in any one document forming part of the contract, the Accepting Authority shall be the deciding authority with regard to the intention of the document and his decision shall be final and binding on the Contractor.
- 8.3 Any error in description, quantity or rate in Schedule of Quantities or any omission therefrom shall not vitiate the Contract or release the Contractor from the execution of the whole or any part of the works comprised therein according to drawings and specifications or from any of his obligations under the Contract.

9. Signing of Contract

- a) The successful tenderer/contractor, on acceptance of his tender by the Accepting Authority, shall, within 28 days from the stipulated date of start of the work, sign the contract consisting of: -
 - (i) the Notice Inviting Tender and Instructions to Tenderers, all the documents including drawings, if any, forming the tender as issued at the time of invitation of tender and acceptance thereof together with any correspondence leading thereto.
 - (ii) General Conditions of Contract for EPC Project (Part-3) consisting of:
 - (a) Conditions of Contract.
 - (b) Clauses of Contract.
 - (c) RITES Safety Code.
 - (d) RITES Model Rules for the protection of health, sanitary arrangements for workers.
 - (e) RITES Contractor's Labour Regulations.
 - (f) List of Acts and omissions for which fines can be imposed.
- b) Each page of the Contract Agreement should be signed by the Engineer-in-Charge and the Contractor's authorized signatory. If there are any corrections, cuttings, omissions, over writings, insertions etc. (after issue of Tender Document) their number should be clearly mentioned on each

page of the Contract Document before signing.

- c) No payment for the work done will be made unless contract is signed by the contractor.
- d) In the event of successful tenderer being a firm/company, then the agreement shall be signed by all the partners or directors thereof individually. In the event of the absence of any partner/director, it shall be signed on his behalf by a person holding a power of attorney (duly notarized by notary public or board resolution in case of company) authorizing him to do so.

10. Miscellaneous Conditions of Contract

The Contractor shall comply with the provisions of the Apprentices Act 1961, and the rules and orders issued there under from time to time. If he fails to do so his failure will be a breach of the Contract and the Engineer-in-Charge / the Employer may in his discretion without prejudice to any other right or remedy available in law cancel the Contract. The Contractor shall also be liable for any pecuniary liability arising on account of any violation by him of the provisions of the said Act.

Section – 8

CLAUSES OF CONTRACT

SECTION – 8

CLAUSES OF CONTRACT

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CLAUSE 1

Performance Guarantee

- (i) The contractor shall submit an irrevocable Performance Guarantee at specified percentage of the tendered amount as mentioned in **Schedule 'F'**, in addition to other deposits mentioned elsewhere in the contract for his proper performance of the contract agreement, (not withstanding and/or without prejudice to any other provisions in the contract) within period specified in Schedule 'F' from the date of issue of letter of acceptance. This period can be further extended by the Engineer-in-Charge up to a maximum period as specified in schedule 'F' on written request of the contractor stating the reason for delays in submitting the Performance Guarantee, to the satisfaction of the Engineer-in-Charge. This guarantee shall be in the form of Demand Draft, Fixed Deposit Receipt or Bank Guarantee from any of the Scheduled Commercial Banks in accordance with the form **given at Annexure 'A'**. In case a fixed deposit receipt of any Bank is furnished by the contractor to the Employer as part of the performance guarantee and the Bank is unable to make payment against the said fixed deposit receipt, the loss caused thereby shall fall on the contractor and the contractor shall forthwith on demand furnish additional security to the Employer to make good the deficit.
- (ii) The Performance Guarantee shall be submitted by the contractor on format as per the form given at Annexure 'A' and shall be initially valid up to the stipulated date of completion plus minimum 6 months beyond that. In case the time for completion of work gets enlarged, the contractor shall get the validity of Performance Guarantee extended to cover such enlarged time for completion of work. After recording of the completion certificate for the work by the competent authority, the performance guarantee shall be returned to the contractor, without any interest. **However, in case of contracts involving maintenance of building and services/any other work after construction of same building and services/other work, then 50% of Performance Guarantee shall be retained as Security Deposit. The same shall be returned year wise proportionately.**
- (iii) The Engineer-in-Charge shall not make a claim under the performance guarantee except for amounts to which the Employer is entitled under the contract (not withstanding and/or without prejudice to any other provisions in the contract agreement) in the event of:
 - (a) Failure by the contractor to extend the validity of the Performance Guarantee as described herein above, in which event the Engineer-in-Charge may claim the full amount of the Performance Guarantee.
 - (b) Failure by the contractor to pay the Employer any amount due, either as agreed by the contractor or determined under any of the Clauses/Conditions of the agreement, within 30 days of the service of notice to this effect by Engineer-in-Charge.

- (iv) In the event of the contract being determined or rescinded under provision of any of the Clause/Condition of the agreement, the performance guarantee shall stand forfeited in full and shall be absolutely at the disposal of the Employer.
- (v) As per requirement of the client or otherwise specified in the contract, part completion certificate may be issued for the building(s)/ infrastructure project for the part(s) which have been completed in all respect and are ready for use. However, statutory approvals, Completion drawing of various services, wherever required, shall be obtained before handing over of building(s)/ part(s) of the project. Scope of the completed part(s) shall be mentioned in such part completion certificate.

The part completion certificate shall include outstanding balance work that need to be completed in accordance with the provisions of the contract. This part completion certificate shall be recorded by the authority as per contract value of work. After recording of the part Completion Certificate for the work by the competent authority, the proportionate amount of 80% of performance guarantee shall be returned to the contractor, without any interest.

However, in case of contracts involving Maintenance of building and services/any other work after construction of same building and services/ other work, then proportionate amount of 40% of performance guarantee shall be returned to the contractor without any interest after recording the part Completion certificate.

CLAUSE 1A

Recovery of Security Deposit

The person(s) whose tender may be accepted (hereinafter called the contractor) shall permit the Employer at the time of making any payment to him for work done under the contract to deduct a sum at the rate of **5%** of the gross amount of each running and final bill till the sum deducted will amount to security deposit of **5%** of the tendered amount of the work. Such deductions will be made and held by the Employer by way of Security Deposit unless he/they has/have deposited the amount of Security at the rate mentioned above in the form of fixed deposit receipts. In case a fixed deposit receipt of any Bank is furnished by the contractor to the Employer as part of the security deposit and the Bank is unable to make payment against the said fixed deposit receipt, the loss caused thereby shall fall on the contractor and the contractor shall forthwith on demand furnish additional security to the Employer to make good the deficit.

All compensations or the other sums of money payable by the contractor under the terms of this contract may be deducted from or paid by the sale of a sufficient part of his security deposit or from the interest arising there from, or from any sums which may be due to or may become due to the contractor by the Employer on any account whatsoever and in the event of his Security Deposit being reduced by reason of any such deductions or sale as aforesaid, the contractor shall within 10 days make good by submitting, Fixed Deposit Receipt tendered by the State Bank of India or by Scheduled Commercial Banks, endorsed

in favour of the Engineer-in-Charge, any sum or sums which may have been deducted from or raised by sale of his security deposit or any part thereof. The security deposit shall be collected from the running bills and the final bill of the contractor at the rates mentioned above.

The security deposit as deducted above can be released against bank guarantee **(in the format given at Annexure A-1)** issued by a scheduled commercial bank or state bank of India on its accumulations to a minimum of Rs. 5 lac subject to the condition that amount of such bank guarantee, except last one, shall not be less than Rs. 5 lac. The amount shall be released only after receipt of confirmation of the Bank Guarantee from issuing bank. Provided further that the validity of bank guarantee shall be in conformity with provisions contained in clause 17 which shall be extended from time to time depending upon extension of contract granted under provisions of clause 2 and clause 5.

In case of contracts involving maintenance of building and services/any other work after construction of same building and services/other work, then 50% of Performance Guarantee shall be retained as Security Deposit. The same shall be returned year wise proportionately.

CLAUSE 2

Compensation for Delay

If the contractor fails to maintain the required progress in terms of clause 5 or to complete the work and clear the site on or before the stipulated completion date or justified extended date of completion determined as per clause 5 (excluding any extension under Clause 5.5) also considering any extension granted under clauses 12 and 15, he shall, without prejudice to any other right or remedy available under the law to the Employer on account of such breach, pay as compensation, the amount calculated as below.

Compensation for delay of work @ **0.75%** of accepted tendered value per month of delay (to be computed on per day basis)

Provided further that the total amount of compensation for delay to be paid under this condition shall **not exceed 5% (five percent)** of the accepted Tendered Value of work or of the accepted Tendered Value of the Sectional part of work as mentioned in Schedule 'F' for which a separate period of completion is originally given.

The period of delay solely attributable to contractor shall be computed as the time taken by contractor going beyond the 'justified date of completion' as determined by the authority specified in schedule F under clause 5. Further in case where the contractor is entitled to additional time under clause 12 and/or clause 15, that shall also be accounted for while deciding the net period of delay. In case, the authority specified in Schedule F decides to levy compensation during the progress of work, the period of delay attributable to contractor

shall be computed (by such authority) as the period by which the progress is behind the schedule on date of such decision, after due consideration of justified extension at that stage of work.

In case no compensation has been decided by the authority in Schedule 'F' during the progress of work, this shall be no waiver of right to levy compensation by the said authority if the work remains incomplete on final justified extended date of completion.

If the Engineer-in-Charge decides to give further extension of time allowing performance of work beyond the justified extended date the contractor shall be liable to pay compensation for such extended period. The levy of compensation under this clause shall be without prejudice to the right of action by the Engineer-in-Charge under clause 3 or any other clause in contract.

In case action under clause 2 has not been finalized and the work has been determined under clause 3, the right of action under clause 2 shall remain post determination of contract and in such case the levy of compensation shall be for days the progress is behind the schedule on date of determination, as assessed by the authority in Schedule F, after due consideration of justified extension. The compensation for delay, if not decided before the determination of contract, shall be decided after of determination of contract. Further, in such case where the contract has been determined, the total amount of recovery against compensation under clause 2 plus that under clause 3 (i.e., forfeiture of security deposit, performance guarantee) shall not exceed 8% of the accepted tendered value of work.

The amount of compensation may be adjusted or set-off against any sum payable to the Contractor under this or any other contract with the Employer.

In case, the contractor does not achieve a particular milestone mentioned in schedule F, or the re-scheduled milestone(s) in terms of Clause 5, the amount shown against that milestone shall be withheld, to be adjusted against the compensation levied as above. With-holding of this amount on failure to achieve a milestone, shall be automatic without any notice to the contractor. However, if the contractor catches up with the progress of work on the subsequent milestone(s), the withheld amount shall be released. In case the contractor fails to make up for the delay in subsequent milestone(s), amount mentioned against each milestone missed subsequently also shall be withheld. The amount so withheld can be released against BG/FDR **issued by a scheduled commercial bank of equivalent amount in the format given at Annexure A-2 after confirmation of the Bank Guarantee by the issuing bank.** Further, no interest, whatsoever, shall be payable on such withheld amount.

CLAUSE 3

When Contract can be Determined

Subject to other provisions contained in this clause, the Engineer-in-Charge may, without

prejudice to his any other rights or remedy against the contractor in respect of any delay, not following safety norms, inferior workmanship, any claims for damages and/or any other provisions of this contract or otherwise, and whether the date of completion has or has not elapsed, by notice in writing absolutely determine the contract in any of the following cases:

- (i) If the contractor having been given by the Engineer-in-Charge a notice in writing to rectify, reconstruct or replace any defective work or that the work is being performed in an inefficient or otherwise improper or un-workman like manner shall omit to comply with the requirement of such notice for a period of seven days thereafter.
- (ii) If the contractor has, without reasonable cause, suspended the progress of the work or has failed to proceed with the work with due diligence and continues to do so after a notice in writing of seven days from the Engineer-in-Charge.
- (iii) If the contractor fails to complete the work or section of work with individual date of completion on or before the stipulated or justified extended date, on or before such date of completion; and the Engineer in Charge without any prejudice to any other right or remedy under any other provision in the contract has given further reasonable time in a notice given in writing in that behalf as either mutually agreed or in absence of such mutual agreement by his own assessment making such time essence of contract and in the opinion of Engineer-in-Charge the contractor will be unable to complete the same or does not complete the same within the period specified.
- (iv) If the contractor persistently neglects to carry out his obligations under the contract and/ or commits default in complying with any of the terms and conditions of the contract and does not remedy it or take effective steps to remedy it within 7 days after a notice in writing is given to him in that behalf by the Engineer-in-Charge.
- (v) If the contractor shall offer or give or agree to give to any person in the Employer service or to any other person on his behalf any gift or consideration of any kind as an inducement or reward for doing or forbearing to do or for having done or forborne to do any act in relation to the obtaining or execution of this or any other contract for the Employer.
- (vi) If the contractor shall enter into a contract with the Employer in connection with which commission has been paid or agreed to be paid by him or to his knowledge, unless the particulars of any such commission and the terms of payment thereof have been previously disclosed in writing to the Engineer-in-Charge.
- (vii) If the contractor had secured the contract with the Employer as a result of wrong tendering or other non-bonafide methods of competitive tendering or commits breach of Integrity Agreement.
- (viii) If the contractor being an individual, or if a firm, any partner thereof shall at any time be adjudged insolvent or have a receiving order or order for administration of his estate made against him or shall take any proceedings for liquidation or composition (other than a voluntary liquidation for the purpose of amalgamation or reconstruction) under any Insolvency Act for the time being in force or make any conveyance or assignment of his effects or composition or arrangement for the benefit of his creditors or purport so to do, or if any application be made under any Insolvency Act for the time being in force for the sequestration of his estate or if a trust deed be executed by him for benefit of his creditors.

- (ix) If the contractor being a company shall pass a resolution or the court shall make an order that the company shall be wound up or if a receiver or a manager on behalf of a creditor shall be appointed or if circumstances shall arise which entitle the court or the creditor to appoint a receiver or a manager or which entitle the court to make a winding up order.
- (x) If the contractor shall suffer an execution being levied on his goods and allow it to be continued for a period of 21 days.
- (xi) If the contractor assigns (excluding part(s) of work assigned to other agency(s) by the contractor as per terms of contract), transfers, sublets (engagement of labour on a piece-work basis or of labour with materials not to be incorporated in the work, shall not be deemed to be subletting) or otherwise parts with or attempts to assign, transfer, sublet or otherwise parts with the entire works or any portion thereof without the prior written approval of the Engineer -in-Charge.
- (xii) When the contractor has made himself liable for action under any of the aforesaid cases (i) to (xi), the Engineer-in- Charge on behalf of the Employer shall have powers:
 - (a) To determine the contract as aforesaid so far as performance of work by the Contractor is concerned (of which determination notice in writing to the contractor under the hand of the Engineer-in-Charge shall be conclusive evidence). Upon such determination, the Earnest Money Deposit, Security Deposit already recovered, Security deposit payable and Performance Guarantee under the contract shall be liable to be forfeited and shall be absolutely at the disposal of the Employer.
 - (a) (i) **Further, the contractor shall be deemed to be declared a ‘Poor Performer’ as per Clause 48.7 hereinafter (without issuance of any notice in this regard) and shall not be eligible for a period of up to two years, to participate in tenders of other works which may be invited by RITES Ltd.**
 - (b) After giving notice to the contractor to measure up the work of the contractor and to take such whole, or the balance or part thereof, as shall be un-executed out of his hands and to give it to another contractor to complete the work. The contractor, whose contract is determined as above, shall not be allowed to participate in the tendering process for the balance work which may include any new item to complete the work. In the event of above courses being adopted by the Engineer-in-Charge, the contractor shall have no claim to compensation for any loss sustained by him by reasons of his having purchased or procured any materials or entered into any engagements or made any advances on account or with a view to the execution of the work or the performance of the contract. And in case action is taken under any of the provision aforesaid, the contractor shall not be entitled to recover or be paid any sum for any work thereof or actually performed under this contract unless and until the Engineer-in-Charge has certified in writing the performance of such work and the value payable in respect thereof and he shall only be entitled to be paid the value so certified.

CLAUSE 3A

Action in case of delay in start of work

In case, the work including planning, designing and execution as per scope of contract cannot be started due to reasons not within the control of the contractor within 1/8th of the stipulated time for completion of work or 180 days whichever is higher, either party may close the contract by giving notice to the other party stating the reasons. In such eventuality, the Performance Guarantee of the contractor shall be refunded within 30 days.

Neither party shall claim any compensation for such eventuality. This clause is not applicable for any breach of the contract by either party.

CLAUSE 4

Contractor liable to pay compensation even if action not taken under Clause 3

In any case in which any of the powers conferred upon the Engineer-in-Charge by Clause-3 thereof, shall have become exercisable and the same are not exercised, the non-exercise thereof shall not constitute a waiver of any of the conditions hereof and such powers shall notwithstanding be exercisable in the event of any future case of default by the contractor and the liability of the contractor for compensation shall remain unaffected. In the event of the Engineer-in-Charge putting in force all or any of the powers vested in him under the preceding clause he may, if he so desires after giving a notice in writing to the contractor, take possession of (or at the sole discretion of the Engineer-in-Charge which shall be final and binding on the contractor) use as on hire (the amount of the hire money being also in the final determination of the Engineer-in-Charge) all or any tools, plant, materials and stores, in or upon the works, or the site thereof belonging to the contractor, or procured by the contractor and intended to be used for the execution of the work/or any part thereof, paying or allowing for the same in account at the contract rates, or, in the case of these not being applicable, at current market rates to be certified by the Engineer-in-Charge, whose certificate thereof shall be final, and binding on the contractor, clerk of the works, foreman or other authorized agent to remove such tools, plant, materials, or stores from the premises (within a time to be specified in such notice) in the event of the contractor failing to comply with any such requisition, the Engineer-in-Charge may remove them at the contractor's expense or sell them by auction or private sale on account of the contractor and his risk in all respects and the certificate of the Engineer-in-Charge as to the expenses of any such removal and the amount of the proceeds and expenses of any such sale shall be final and conclusive against the contractor.

CLAUSE 5

Time and Extension for Delay

The time allowed for execution of the Works as specified in the Schedule 'F' or the extended time in accordance with these conditions shall be the essence of the Contract. The execution

of the work shall commence from such date as mentioned in Schedule 'F' or from the date of handing over of the site, as notified by the Engineer-in-Charge, whichever is later. If the Contractor commits default in commencing the execution of the work as aforesaid and such default continues even after time period specified in the notice in writing by the Engineer-in-Charge then the performance guarantee shall be forfeited by the Engineer-in-Charge and shall be absolutely at the disposal of the Employer without prejudice to any other right or remedy available in law.

The contract shall stand determined when such decision regarding the forfeiture of the performance guarantee is issued to the contractor.

5.1 The contractor as soon as possible but within 7 (Seven) days issue of letter of award of work shall submit a time and progress chart to the Engineer-in-Charge. Such program shall be made in due consideration of:

- a) Schedule of handing over of site as specified in the Schedule 'F'
 - b) Schedule of issue of design(s) and drawing(s) as specified in the Schedule 'F',
- (i) The contractor shall submit a Time and Progress Chart for each milestone. The Engineer-in-Charge may within 7 (seven) days of receipt of such chart, make modifications thereafter, if any, and communicate the approved chart to the contractor, failing which the chart submitted by the contractor shall be deemed to be approved by the Engineer-in-Charge.

Further, to ensure good progress during the execution of the work, the Contractor shall in all cases in which the time allowed for any work, exceeds one month (save for special jobs for which a separate programme has been agreed upon) complete the work as per milestones given in Schedule 'F' failing which the amounts specified in Schedule 'F' shall be withheld for non-achievement of milestones. The withheld amount may be released when the subsequent milestone(s) is/are achieved. The withheld amount may also be released against submission of Bank Guarantee of equivalent amount issued by a scheduled commercial bank in the format given at **Annexure A-2** after confirmation of the Bank Guarantee by the issuing bank.

The Chart shall be prepared in direct relation to the time stated in the Contract documents for completion of items of the work(s). It shall indicate the forecast of the dates of commencement and completion of various trades of sections of the work and may be amended as necessary by agreement between the Engineer-in-Charge and the Contractor within the limitations of time imposed in the Contract documents.

- (ii) In case of non-submission of time and progress chart by the contractor, the chart prepared by the Engineer-in-Charge shall be deemed to be final.

- (iii) Such program by the Engineer-in-Charge shall not relieve the contractor of any of the obligations under the contract.
- (iv) The contractor shall submit the Time and Progress Chart containing upto date progress of work using the mutually agreed software or in the format decided by Engineer-in-Charge. Such chart shall be submitted by the contractor on or before **5th day** of each month failing which a recovery as mentioned in Schedule 'F' shall be made at the earliest from the next running account bill without any notice in this regard.
- (v) While recording the hindrances in the progress of the work, due consideration should be given to the cause of hindrance. The hindrances shall be segregated in following categories:
 - a) delays due reasons beyond the control of both parties (sub-clause 5.2)
 - b) delays attributable to the Employer and concurrent delays (sub- clause 5.3)
 - c) delays solely attributable to the contractor (sub- clause 5.5)

5.2 Delays due to reasons beyond the control of both parties:

If the work(s) be delayed by: -

- (i) force majeure, or
- (ii) abnormally bad weather, or
- (iii) serious loss or damage by fire, or
- (iv) civil commotion, local commotion of workmen, strike or lockout, affecting any of the trades employed on the work, or
- (v) delay on the part of other contractors or tradesmen engaged by Engineer-in-Charge in executing work not forming part of the Contract, or
- (vi) any other cause like above which, in the reasoned opinion of the Engineer-in-Charge is beyond the Contractor's control.

Then upon the happening of any such event causing delay, the Contractor shall within **03 (three) days** give written notice thereof in writing to the Engineer -in- Charge but shall nevertheless use constantly his best endeavours to prevent or make good the delay and shall do all that may be reasonably required to the satisfaction of the Engineer-in-Charge to proceed with the work (s).

The contractor shall not have no claim on account of any hindrance in case notice(s) are not given by the contractor in writing.

The Engineer-in- Charge, on receipt of such notice(s), after considering the factual ground situation shall either acknowledge or reject the notice(s).

In case of rejection, the reasons(s) for rejection shall be communicated by the Engineer-in-Charge to the agency.

The decision of Engineer-in-Charge with regard to nature of event causing delay, its start date and end date, as has been finalized during acknowledgement of notice, shall be final and binding.

The end date of such events shall be recorded by Engineer-in-Charge either during acknowledgment of notice or subsequent to acknowledgement if end date of hindrance is after the date of acknowledgement of notice.

In absence of notice by the contractor, Engineer-in-Charge or his representative(s) may record events causing delay within **05 (five) days** of occurrence of hindrance in writing further that not recording of events causing delays by the Engineer-in-charge does not ipso facto entitle the contractor for any hindrance.

The contractor shall have no claim of damages against the delays listed in this sub clause i.e., sub clause 5.2. The contractor shall have no claim of damages for extension of time granted or rescheduling of milestone/s for events listed in sub clause 5.2.

5.3 Delays attributable to the Employer and the Concurrent delays:

In case the work is hindered, in the opinion of the contractor, by the Employer or for any reason / event, for which the Employer is responsible, then upon the happening of such event causing delay, the Contractor shall within 03 (three) days give notice in writing thereof to the Engineer-in-Charge but shall nevertheless use constantly his best endeavours to prevent or make good the delay and shall do all that may be reasonably required to the satisfaction of the Engineer-in-Charge to proceed with the work.

The contractor shall not be entitled for any hindrance in case notice(s) are not given by the contractor in writing.

The Engineer-in- Charge, on receipt of such notice(s), after considering the factual ground situation, shall either acknowledge or reject the notice(s).

In case of rejection, the reasons(s) for rejection shall be communicated by the Engineer-in-Charge to the agency.

The decision of Engineer-in-Charge with regard to nature of event causing delay, its start date and end date, as has been finalized during acknowledgement of notice, shall be final and binding.

The end date of such events shall be recorded by Engineer-in-Charge either during acknowledgment of notice or subsequent to acknowledgement if end date of hindrance is after the date of acknowledgement of notice.

In absence of notice by the contractor, Engineer-in-Charge or his representative(s) may record events causing delay within **05 (five) days** of occurrence of hindrance in writing further that not recording of events causing delays by the Engineer-in-charge does not ipso facto entitle the contractor for any hindrance.

Such extension of time or rescheduling of milestone/s shall be without prejudice to any other right or remedy of the parties in contract or in law; provided further that for the concurrent delays(s) under this sub clause and sub clause 5.2 to the extent the delay is covered under sub clause 5.2, the contractor shall be entitled for only extension of time and shall have no claim of damages.

5.4 Rescheduling of milestone(s) and ‘Justified extended date of completion’

The request for rescheduling of the Milestone(s) and extension of time, shall be made by the Contractor in writing once in a month on the basis of hindrances accepted by Engineer-In-Charge under sub-clause 5.2 and sub-clause 5.3 on the prescribed forms i.e., Form of application by the contractor for seeking rescheduling of milestones or Form of application by the contractor **for seeking extension of time (Annexure J) to the authority as indicated in Schedule 'F'**. The Contractor shall indicate in such a request number of days by which rescheduling of milestone/s and / or extension of time is desired.

The authority as indicated in Schedule ‘F’, after examining the request, shall give a fair and reasonable ‘justified extension of time’ for completion of work and simultaneously reschedule the milestones(s), If required so. The authority shall consider all the hindrances accepted by Engineer-in- Charge as per sub-clauses 5.2, 5.3 and 5.5.

The authority shall decide rescheduling of milestones(s) and extension of time within 21 (Twenty-One) days of the request submitted by the contractor in writing.

In event no request by the contractor for rescheduling of milestone(s) and extension of time, the authority as indicated in schedule F, after affording opportunity to the contractor, may give, fair and reasonable extension of time based on hindrances accepted by Engineer-in-Charge and reschedule the milestone(s) once in a month. Such justified extension of time shall determine the ‘justified extended date’ of completion of work.

5.4.1 Provided that the end date of any event causing delay shall not fall beyond the date of request for extension of time or rescheduling of milestone(s) by the contractor. In case end date of event falls beyond the date of submission of said request, then period for extension up to date of application shall be considered in the said request for events eligible for consideration and remaining period shall be applied in subsequent request of extension of time or rescheduling of milestone(s).

Engineer-in-Charge shall finalize/ reschedule a particular mile stone before taking action

against subsequent mile stone. Such extension or rescheduling of the milestones shall be communicated to the Contractor by the authority as indicated in Schedule 'F' in writing, within 21 (twenty-one) days of the date of receipt of such request from the Contractor in writing.

5.5 Delays attributable solely to the contractor:

In case the work is delayed by reasons solely attributable to contractor, then Engineer -in-Charge or his representative(s) may record the event causing delay within 05 (five) days of occurrence of delay. Contractor shall take the notice of the same for necessary action. He may submit his version, if any within 05 (days). Engineer -n-Charge, considering the version of the contractor, will take decision on such recording of the event and the decision of the Engineer-in-Charge shall be final and binding.

The contractor shall be liable for levy of compensation for such delays i.e., for the period beyond the 'justified extended date of completion as determined in Clause 5.4 and this default of contractor shall be dealt in conjunction with Clause 2 of contract document.

In case the work is delayed, due to hindrances attributable to solely to the contractor, beyond the justified extended date (as stated in clause 5.4), the authority indicated in Schedule F, without prejudice to the provisions under Clause 3, may grant extension of time required for completion of work without rescheduling the milestone(s) and extend the date completion.

5.6 Record of hindrances in writing:

The Engineer-in-Charge may require the contractor to give notice of hinderances in writing devised by him. All the notices of hinderances have to be submitted through such written by the contractor timely to claim extension of time/rescheduling of milestone(s). The contractor shall not be entitled for any damages in case such written notices are not given timely by the contractor.

CLAUSE 6

Computerized Measurement Book

Engineer-in-Charge shall, except as otherwise provided, ascertain and determine by measurement the value of work done in accordance with the contract.

All measurements as per the stage payments mentioned in Schedule F having financial value shall be entered by the contractor and compiled in the shape of the Computerized Measurement Book having pages of A-4 size as per the format of the Employer so that a complete record is obtained of all the stages of works performed under the contract.

All such measurements recorded by the contractor or his authorized representative from time to time, during the progress of the work, shall be got checked by the contractor from the Engineer-in-Charge or his authorized representative as per interval or program fixed in consultation with Engineer-in-Charge or his authorized representative. After the necessary corrections made by the Engineer-in-Charge, the measurement sheets shall be returned to the contractor for incorporating the corrections and for resubmission to the Engineer-in-Charge for the dated signatures by the Engineer-in-Charge and the contractor or their representatives in token of their acceptance.

Whenever bill is due for payment, the contractor would initially submit draft computerized measurement sheets and these measurements would be got checked/test checked from the Engineer-in-Charge and/or his authorized representative. The contractor will, thereafter, incorporate such changes as may be done during these checks/test checks in his draft computerized measurements, and submit to the Engineer-in-Charge a computerized measurement book, duly bound, and with its pages machine numbered. The Engineer-in-Charge and/or his authorized representative would thereafter check this MB, and record the necessary certificates for their checks/test checks.

The final, fair, computerized measurement book given by the contractor, duly bound, with its pages machine numbered, should be 100% correct. No cutting or over-writing in the measurements would thereafter be allowed. If at all any error is noticed, the contractor shall have to submit a fresh computerized MB with its pages duly machine numbered and bound, after getting the earlier MB cancelled by the Engineer. Thereafter, the MB shall be taken in the Engineer's Office records, and allotted a number as per the Register of Computerized MBs. This should be done before the corresponding bill is submitted to the Office of Engineer-in-Charge for payment. The contractor shall submit two spare copies of such computerized MB's for the purpose of reference and record by the various officers of the Employer.

The contractor shall also submit to the Engineer-in-Charge separately his computerized Abstract of Cost and the bill based on these measurements, duly bound, and its pages machine numbered along with two spare copies of the bill. Thereafter, this bill will be processed by the Engineer/ Engineer-in-charge and allotted a number as per the computerized record in the same way as done for the measurement book meant for measurements.

The contractor shall, without extra charge, provide all assistance with every appliance, labour and other things necessary for checking of measurements by the Engineer-in-Charge or his representative.

Except where any general or detailed description of the work expressly shows to the contrary, measurements shall be taken in accordance with the procedure set forth in the contract notwithstanding any provision in the relevant Standard Method of measurement or any general or local custom.

The contractor shall give not less than seven days' notice to the Engineer-in-Charge or his authorized representative in charge of the work before covering up or otherwise placing beyond the reach of checking and/or test checking the measurement of any work in order that the same may be checked and/or test checked and correct dimensions thereof be taken before the same is covered up or placed beyond the reach of checking and/or test checking measurement and shall not cover up and place beyond reach of measurement any work without consent in writing of the Engineer-in-Charge or his authorized representative in charge of the work who shall within the aforesaid period of seven days inspect the work, and if any work shall be covered up or placed beyond the reach of checking and/or test checking measurements without such notice having been given or the Engineer-in-Charge's consent being obtained in writing the same shall be uncovered at the Contractor's expense, or in default thereof no payment or allowance shall be made for such work or the materials with which the same was executed.

Engineer-in-Charge or his authorized representative may cause either themselves or through another agency to check the measurements recorded by contractor and all provisions stipulated herein above shall be applicable to such checking of measurements or levels.

It is also a term of this contract that checking and/or test checking the measurements of any item of work in the measurement book and/or its payment in the interim, on account of final bill shall not be considered as conclusive evidence as to the sufficiency of any work or material to which it relates nor shall it relieve the contractor from liabilities from any over measurement or defects noticed till completion of the defects liability period.

CLAUSE 7

Payment on intermediate Certificates to be regarded as Advances

The running account bills shall be submitted by the contractor for the work executed on the basis of such recorded measurements as per Clause 6 on the format of the Employer in triplicate on or before the date of every month fixed for the same by the Engineer-in-Charge. The contractor shall not be entitled to be paid any such interim payment if the gross work done together with net payment/ adjustment of advances for material collected, if any, since the last such payment is less than the amount specified in Schedule 'F', in which case the interim bill shall be prepared on the appointed date of the month after the requisite progress is achieved.

However, to expedite the progress of work, Engineer-in-Charge, on the request of contractor, may make interim payment(s) even before the net payment limit specified in schedule 'F' is achieved. In such case(s) no interest / compensation shall be recoverable from contractor. Such payment by Engineer-in-Charge shall not be construed as waiver of limit specified in schedule 'F' for subsequent interim payment(s).

Engineer-in-Charge shall arrange to have the bill verified in the event of the failure of the contractor to submit the bills, no claims whatsoever due to delays on payment including that of interest shall be payable to the contractor. Payment on account of amount admissible shall be made by the Engineer-in-Charge certifying the sum to which the contractor is considered entitled by way of interim payment at such rates as decided by the Engineer-in-Charge.

An amount of ad-hoc payment not less than 75% of the net amount of the bill under check, shall be made within 10 working days of submission of the bill. The remaining payment is also to be made after final checking of the bill within 28 working days of submission of bill by the contractor.

All such interim payments shall be regarded as payment by way of advances against final payment only and shall not preclude the requiring of bad, unsound and imperfect or unskilled work to be rejected, removed, taken away and reconstructed or re-erected. Any certificate given by the Engineer-in-Charge relating to the work done or materials delivered forming id the revised tendered amount (worked out on the basis of quoted rate of individual items) part of such payment, may be modified or corrected by any subsequent such certificate(s) or by the final certificate and shall not by itself be conclusive evidence that any work or materials to which it relates is/are in accordance with the contract and specifications. Any such interim payment, or any part thereof shall not in any respect conclude, determine or affect in any way powers of the Engineer-in-Charge under the contract or any of such payments be treated as final settlement and adjustment of accounts or in any way vary or affect the contract.

Pending consideration of extension of date of completion, interim payments shall continue to be made as herein provided without prejudice to the right of the Employer to take action under the terms of this contract for delay in the completion of work, if the extension of date of completion is not granted by the competent authority.

Since the Contract is being entered into with the Contractor by RITES acting for and on behalf of the Employer as Agent/Power of Attorney Holder, no Bill will be paid unless sufficient funds have been made available by the Employer.

CLAUSE 7A

No Running Account Bill shall be paid for the work till the applicable labour licenses, registration with EPFO, ESIC and BOCW Welfare Board, whatever applicable are submitted by the contractor to the Engineer-in-Charge.

CLAUSE 7B

PAYMENT TO THIRD PARTY

If the exigencies of the work so demand, the Engineer-in-Charge may allow payment to a third party, who is creditor to the contractor, after fulfilling the following conditions:

- a) The Contractor gives an authority letter addressed to the Engineer-in-Charge on a non-judicial stamp paper of Rs. 100/- in the format given below.

I/We authorize the Engineer-in-Charge to pay directly on my/our behalf to (name of the third party) an amount of Rs..... (Rupees in words) for the work done or supplies made by (name of the third party). I/We shall be responsible for the quality and quantity of the same under the provisions of the Agreement No..... dated

Signature of the Contractor

- b) The total payment to the third party (or parties) shall not exceed 10% of the Agreement cost of the work.

CLAUSE 8

Completion Certificate and Completion Plans

Within ten days of the completion of the work or on part completion of one or more buildings(s) out of independent building in a project or infrastructure project, as per requirement of client or otherwise specified in schedule F, the contractor shall give notice of such completion to the Engineer-in-Charge and within thirty days of the receipt of such notice, the Engineer-in-Charge shall inspect the work and shall furnish the contractor with a part or final completion certificate as the case may be indicating defects (a) to be rectified by the contractor and/or (b) for which payment will be made at reduced rates. But no final certificate of completion shall be issued, nor shall the work be considered to be complete until the contractor shall have removed from the premises on which the work shall be executed all scaffolding, surplus materials, rubbish and all huts and sanitary arrangements required for his/their work people on the site in connection with the execution of the works as shall have been erected or constructed by the contractor(s) and cleaned off the dirt from all wood work, doors, windows, walls, floor or other parts of the building, in, upon, or about which the work is to be executed or of which he may have had possession for the purpose of the execution; thereof, and not until the work shall have been measured by the Engineer-in-Charge. If the contractor shall fails to comply with the requirements of this Clause as to removal of scaffolding, surplus materials and rubbish and all huts and sanitary arrangements as aforesaid and cleaning off dirt on or before the date fixed for the completion of work, the Engineer-in-Charge may at the expense of the contractor remove such scaffolding, surplus materials and rubbish etc., and dispose off the same as he thinks fit and clean off such dirt as aforesaid, and the contractor shall have no claim in respect of scaffolding or surplus materials or final cleaning work as aforesaid except for any sum actually realized by the sale thereof.

CLAUSE 8A

Completion Plans to be submitted by the Contractor

The contractor shall submit completion plans for Internal and External Civil, Electrical and Mechanical Services within thirty days of the completion of the work.

In case, the contractor fails to submit the completion plan as aforesaid, he shall be liable to pay a sum of 0.1 % (zero-point one percent) of Tendered Value or limit prescribed in Schedule F whichever is more as may be fixed by the authority as mentioned in Schedule F and in this respect the decision of the that authority shall be final and binding on the contractor.

CLAUSE 9

Payment of Final Bill

The final bill shall be submitted by the contractor in the same manner as specified in interim bills within three months of physical completion of the work or within one month of the date of the final certificate of completion furnished by the Engineer-in-Charge whichever is earlier. No further claims shall be made by the contractor after submission of the final bill, and these shall be deemed to have been waived and extinguished. Payments of those items of the bill in respect of which there is no dispute and of items in dispute, for quantities and rates as approved by Engineer-in-Charge, will, as far as possible be made within the period of **03 (Three) months** the period being reckoned from the date of receipt of the bill by the Engineer-in- Charge or his authorized Engineer.

The Contractor will not however be entitled to any compensation or claims or damages by way of interest etc. in case of delay in payment.

In case the Contractor fails to submit his final bill along with supporting documents within the time stipulated above for such submission, the Engineer-in-Charge may prepare or cause to be prepared the final bill within six months of the date of final certificate of completion; in such event no claim whatsoever on account of difference in quantities or rates and due to delay in payment, including interest, shall be payable to the Contractor.

CLAUSE 9 A

Payment of Contractor's Bill to Banks

Payments due to the contractor may, if so desired by him, be made to his bank, registered financial, co-operative or thrift societies or recognized financial institutions instead of direct to him provided that the contractor furnishes to the Engineer-in-Charge (1) an authorization in the form of a legally valid document such as a power of attorney conferring authority on the bank; registered financial, co-operative or thrift societies or recognized financial institutions to receive payments and (2) his own acceptance of the correctness of the amount made out as being due to him by the Employer or his signature on the bill or other claim

preferred against the Employer before settlement by the Engineer-in-Charge of the account or claim by payment to the bank, registered financial, co-operative or thrift societies or recognized financial institutions. While the receipt given by such banks; registered financial, co-operative or thrift societies or recognized financial institutions shall constitute a full and sufficient discharge for the payment, the contractor shall whenever possible present his bills duly receipted and discharged through his bank, registered financial, co-operative or thrift societies or recognized financial institutions.

Nothing herein contained shall operate to create in favour of the bank; registered financial, co-operative or thrift societies or recognized financial institutions any rights or equities vis-a-vis the Employer

CLAUSE 10A

Materials to be provided by the Contractor

The contractor shall, at his own expense, provide all materials, required for the works other than those which are stipulated to be supplied by the Employer.

The contractor shall, at his own expense and without delay, supply to the Engineer-in-Charge samples of materials to be used on the work and shall get these approved in advance. All such materials to be provided by the Contractor shall be in conformity with the specifications laid down or referred to in the contract. The contractor shall, if requested by the Engineer-in-Charge furnish proof, to the satisfaction of the Engineer-in-Charge that the materials so comply. The Engineer-in-Charge shall within thirty **(30)** days of supply of samples or within such further period as he may require intimate to the Contractor in writing whether samples are approved by him or not. If samples are not approved, the Contractor shall forthwith arrange to supply to the Engineer-in-Charge for his approval, fresh samples complying with the specifications laid down in the contract. When materials are required to be tested in accordance with specifications, approval of the Engineer-in-Charge shall be issued after the test results are received.

The Contractor shall at his cost submit the samples of materials to be tested or analyzed and shall not make use of or incorporate in the work any materials represented by the samples until the required tests or analysis have been made and materials finally accepted by the Engineer-in-Charge. The Contractor shall not be eligible for any claim or compensation either arising out of any delay in the work or due to any corrective measures required to be taken on account of and as a result of testing of materials.

The contractor shall, at his cost, make all arrangements and shall provide all facilities as the Engineer-in-Charge may require for collecting, and preparing the required number of samples for such tests at such time and to such place or places as may be directed by the Engineer-in-Charge and bear all charges including testing charges. The Engineer-in-Charge or his authorized representative shall always have access to the works and to all workshops

and places where work component is being prepared or from where materials, manufactured articles or machinery are being obtained for the works and the contractor shall afford every facility and every assistance in obtaining the right to such access.

The Engineer-in-Charge shall have full powers to require the removal from the premises of all materials which in his opinion are not in accordance with the specifications and in case of default, the Engineer-in-Charge shall be at liberty to employ at the expense of the contractor, other persons to remove the same without being answerable or accountable for any loss or damage that may happen or arise to such materials. The Engineer-in-Charge shall also have full powers to require other proper materials to be substituted thereof and in case of default, the Engineer-in-Charge may cause the same to be supplied and all costs which may attend such removal and substitution shall be borne by the Contractor.

If stipulated in Schedule 'F', the contractor shall at his own expense, provide a material testing lab at the site for conducting routine field tests. The lab shall be equipped at least with the testing equipment as specified in **Annexure 'B-1'** except where otherwise stipulated in Schedule F.

The Contractor shall comply with special conditions relating to procurement, testing and storage of cement and steel as given in **Annexure 'B-2'**.

CLAUSE 10 B

Secured Advance on Materials

- (i) The contractor, on signing an indenture in the form given at **Annexure 'I'**, shall be entitled to be paid during the progress of the execution of the work up to 75% of the assessed value of any materials, which are in the opinion of the Engineer-in-Charge non-perishable, non-fragile and non-combustible and are in accordance with the contract and which have been brought on the site in connection therewith and are adequately stored and/or protected against damage by weather or other causes but which have not at the time of advance been incorporated in the works. When materials on account of which an advance has been made under this sub-clause are incorporated in the work, the amount of such advance shall be recovered/deducted from the next payment made under any of the clause or clauses of this contract.

Such secured advance shall also be payable on other items of perishable nature, fragile and combustible with the approval of the Engineer-in-Charge provided the contractor provides a comprehensive insurance cover for the full cost of such materials. The decision of the Engineer-in-Charge shall be final and binding on the contractor in this matter. No secured advance, shall however, be paid on high-risk materials such as ordinary glass, sand, petrol, diesel.

Mobilization Advance

- (ii) Mobilization advances not exceeding 10% of the tendered value may be given, if requested by the contractor in writing within **six months** of the order to commence the work. Such advance shall be released in two or more instalments to be determined by the Engineer-in-Charge. The amount of any instalment shall not exceed 5% of the tendered amount of the work. The first instalment of such advance shall be released by the Engineer-in-charge to the contractor on his request. The second and subsequent instalment(s) shall be released by the Engineer-in-Charge only after the contractor furnishes a proof of the satisfactory utilization of full amount of the earlier instalment(s) to the satisfaction of the Engineer-in-Charge. The mobilization advance will be utilized for the following:

Establishment of site office for contractor and Employer staff, testing lab, testing lab equipment, labour campus & basic amenities, services for labour/staff, cement godown, reinforcement yard and stores etc.

Any other item as mentioned in NIT by the NIT approving authority.

90% of the price of new items and 50% of the depreciated price of old items will be considered for assessing the utilized amount of mobilization advance.

Expenditure incurred on items/ materials which are measurable and payable under agreement/extra items as well as materials eligible for secured advance will be excluded from utilized amount of mobilization advance, if any.

The assessment of Engineer-in- Charge in this respect shall be final and binding.

However, clause 10B (ii) and clause 10B (iii) are mutually exclusive i.e. both cannot be operated for the same material simultaneously.

Before any installment of advance is released, the contractor shall execute Bank Guarantee Bonds not more than 6 (six) in number from Scheduled Commercial Bank as per the form given in **Annexure 'C'** for the amount equal to 110% of the amount of advance and valid for the period till recovery of advance. This (Bank Guarantee from Scheduled Commercial Bank on prescribed format for the amount equal to 110% of the balance amount of advance) shall be kept renewed from time to time to cover the balance amount and likely period of complete recovery.

Plant, Machinery & Shuttering Material Advance: - Not Applicable

- (iii) An advance for plant, machinery & shuttering material required for the work and brought to site by the Contractor may be given if requested by the contractor in writing within one month of bringing such plant and machinery to site. Such advance shall be given on such plant and machinery which in the opinion of the Engineer-in-charge will

add to the expeditious execution of work and improve the quality of work. The amount of advance shall be restricted to 5% percent of the tender value. In the case of new plant and equipment to be purchased for the work, the advance shall be restricted to 90% of the price of such new plant and equipment paid by the contractor for which the contractor shall produce evidence satisfactory to the Engineer-in-Charge. In the case of second hand and used plants and equipment, the amount of such advance shall be limited to 50% of the depreciated value of plant and equipment as may be decided by the Engineer-in-Charge. The contractor shall, if so required by the Engineer-in-Charge, submit the statement of value of such old plant and equipment duly approved by a Registered Valuer recognized by the Central Board of Direct Taxes under the Income-Tax Act, 1961. No such advance shall be paid on any plant and equipment of perishable nature and on any plant and equipment of a value less than Rs. 50,000/-. Seventy five percent of such amount of advance shall be paid after the plant & equipment is brought to site and balance twenty five percent on successfully commissioning the same.

Leasing of equipment shall be considered at par with purchase of equipment and shall be covered by tripartite agreement with the following:

1. Leasing company which gives certificate of agreeing to lease equipment to the Contractor.
2. Engineer-in-Charge, and
3. The Contractor

This advance shall further be subject to the condition that such plant and equipment (a) are considered by the Engineer-in-Charge to be necessary for the works; (b) and are in working order and are maintained in working order; (c) hypothecated to the Employer as per proforma of Hypothecation Deed given in **Annexure 'D'** before the payment of advance is released. The contractor shall not be permitted to remove from the site such hypothecated plant and equipment without the prior written permission of the Engineer-in-Charge. The contractor shall be responsible for maintaining such plant and equipment in good working order during the entire period of hypothecation failing which such advance shall be entirely recovered in lump sum. For this purpose, steel scaffolding and form work shall be treated as plant and equipment. The contractor shall insure the Plant and Machinery for which mobilization advance is sought and given, for a sum sufficient to provide for their replacement at site. Any amounts not recovered from the insurer will be borne by the contractor.

The Plant, Machinery & Shuttering Material Advance will be utilized for the following:

For purchase of any type of tool, plant and machinery required for execution of work such as concrete batch mix plant, mixtures, transit mixtures, loader, excavators, dumpers, DG sets, vibrators, hot mix bitumen plant, paver, rollers and any other equipment etc.

Barricading of site and procurement of centring / shuttering / staging material etc.

Any other item as mentioned in NIT by the NIT approving authority.

90% of the price of new items and 50% of the depreciated price of old items will be considered for assessing the utilized amount of Plant, Machinery & Shuttering Material Advance.

The assessment of Engineer-in-Charge in this respect shall be final and binding.

However, clause 10B (ii) and clause 10B (iii) are mutually exclusive i.e. both cannot be operated for same material simultaneously.

Interest & Recovery

- (iv) The mobilization advances and plant and machinery advance in (ii) & (iii) above bear **simple interest at the rate of 8 percent per annum** and shall be calculated from the date of payment to the date of recovery, both days inclusive, on the outstanding amount of advance. Recovery of such sums advanced shall be made by the deduction from the contractors bills commencing after first ten percent of the gross value of the work is executed and paid, on pro-rata percentage basis to the gross value of the work billed beyond 10% in such a way that the entire advance is recovered by the time eighty percent of the gross value of the contract is executed and paid, together with interest due on the entire outstanding amount up to the date of recovery of the installment.
- (v) If the circumstances are considered reasonable by the Engineer-in-Charge, the period mentioned in (ii) and (iii) for request by the contractor in writing for grant of mobilization advance and plant and equipment advance may be extended at the discretion of the Engineer-in-Charge.

CLAUSE 10 CC:

Price Adjustment for Works: **DELETED**

CLAUSE 10D

Dismantled Material Employers' Property:

The contractor shall treat all materials obtained during dismantling of a structure, excavation of the site for a work, etc. as Employer's property and such materials shall be disposed off to the best advantage of Employer according to the instructions in writing issued by the Engineer-in-Charge.

CLAUSE 11

Work to be Executed in Accordance with Specifications, Drawings, Orders as per Mode mentioned in Schedule 'F'

The contractor shall execute the whole and every part of the work in the most substantial and workmanlike manner both as regards materials and otherwise in every respect in strict accordance with the specifications. The contractor shall also conform exactly, fully and faithfully to the design, drawings and instructions in writing in respect of the work issued by the Engineer-in-Charge. Contractor shall be furnished free of charge one copy of the contract documents.

The contractor shall comply with the provisions of the contract and with the care and diligence execute and maintain the works and provide all labour and materials, tools and plants including for measurements and supervision of all works, structural plans and other things of temporary or permanent nature required for such execution and maintenance in so far as the necessity for providing these, is specified or is reasonably inferred from the contract. The Contractor shall take full responsibility for adequacy, suitability and safety of all the works and methods of construction.

The 3D Architectural Models of the project will be prepared on any BIM applications with minimum LOD 350 and IFC format of the same shall be made available for interoperability of building information to all stake holders of the project.

The 3D Architectural BIM Model Information(s) shall be used for structural, MEP services and horticulture design. The comprehensive BIM model with minimum LOD 350 of project shall have all the architectural, structural, horticulture and MEP services element.

All architectural and MEP drawings (2 D) shall be generated from BIM model. All the structural drawings shall be in conformity with BIM model.

Project or work shall be executed using 2D drawings generated from BIM model.

The provisions of BIM shall be applicable in the project only if, the same is mentioned in **schedule 'F'**.

CLAUSE 12

DEVIATIONS/ VARIATIONS EXTENT AND PRICING

The Engineer-in-Charge shall have power (i) to make alterations in, additions to or substitutions for the original scope of work as defined in the contract, that may appear to him to be necessary or advisable during the progress of the work, and (ii) to omit a part of the works in case of non-

availability of a portion of the site or due to any other reason (save except Clause-13), the contractor shall be bound to carry out the works in accordance with any instructions given to him in writing signed by the Engineer-in-Charge and such alterations, omissions, additions or substitutions shall form part of the contract as if originally provided therein and any change in the scope of work as defined in the contract, which the contractor may be directed to do in the manner specified above as part of the works, shall be carried out by the contractor on the same terms and conditions in all respects, except for increase/decrease in the cost and additional time due to change of scope, to be determined based on the sub-clauses hereunder:

12.1 The time for completion of the works shall, in the event of any deviations resulting in additional cost over the tendered value sum being ordered, be extended, if requested by the contractor, as follows:

- i) In the proportion which the additional cost of the altered, additional or substituted work, bears to the original tendered value plus
- ii) 25% of the time calculated in (i) above or such further additional time as may be considered reasonable by the Engineer-in-Charge.

12.2 PAYMENT OF DEVIATIONS/ VARIATIONS BEYOND 0.25% OF THE ACCEPTED TENDERED AMOUNT.

In case there is any change in scope as defined in the contract, the contractor shall carry out the changes as per direction of Engineer-in-Charge and nothing extra shall be payable to the contractor on account of same if the additional cost of such work is up to 0.25% (zero-point two five percent) of the accepted tendered amount and worked out as per sub-clause 12.3 below. Variations/deviations up to 0.25% (zero-point two five percent) of the accepted tendered amount shall be deducted from overall variations/deviations for making payment.

12.3 DETERMINATION OF RATES

In the event, there is any deviations/variations in work as defined in the contract, the contractor shall submit the complete proposal to Engineer-in-charge within 15 days duly supported with: -

- (a) Analysis of rates for items involved, along with relevant documents, rates of materials, tools/plants and labour, etc.
- (b) The impact, if any, which the deviations/variations is likely to have on the project completion schedule,

On receipt of such proposal, either individually or covering group of items, the Engineer-in-charge shall examine the proposal regarding its admissibility and finalize the proposal/rates within 45 days after receipt of proposal with all requisite details and documents from the contractors, after giving due consideration to the proposal, analysis and rates of materials and labours, etc.

12.3.1 The increase/decrease in the rates due to deviations/variations shall be decided based on

the following criteria: -

(i) Pricing of Deviations

- (a) If the item of work as stipulated in the schedule of quantity/ scope of work deviates on plus side, then the rate for the deviated quantity shall be paid at the agreement rate up to the deviation limit as specified in schedule "F" with the same terms & conditions of the contract. Beyond deviation limit as specified in the schedule "F", rate shall be payable on market rates to be determined based on the relevant documents and prevailing market rates, as per Para (ii) below
- (b) If the item of work as stipulated in the schedule of quantity/scope of work deviates on minus side, then the amount for such deviated quantity shall be deducted proportionately at the agreement rate.

(ii) Pricing of variations

If there are changes in the quantity/specifications/ alterations/ substitutions/ additions, etc. in the items, other than mentioned in para-(i) above, the rates shall be determined based on detailed analysis of rates with original stipulated scope of items & newly proposed/ provided items. The difference of rates so determined shall be payable to/recoverable from the contractor. The rates for both the components i.e., materials & labour shall be based on prevailing market rates. The rate finalized by the Engineer-in-Charge shall be final and binding.

12.3.2 In case of either non-submission of timely proposal or incomplete proposal by the contractor for deviations/variations, the Engineer-in-Charge shall give final opportunity to the Contractor to submit the complete proposal for change of cost within next 15 days. In case of non-submission or further incomplete submission by the contractor within the stated period, the Engineer-in-Charge shall initiate the proposal and decide the change of cost. In such case the proposal finalized by the Engineer-in-Charge shall be final and binding on the contractor.

12.4 Restrictions on Deviations/Variations

- (i) Work(s) due to deviations/variations shall be executed only after getting the instructions of Engineer-in-charge, save except to meet any work of emergent nature.
- (ii) Notwithstanding anything to the contrary in this clause 12, any change arising from default of the contractor in the performance of his obligations under this agreement shall not be deemed to be deviations/ variations and shall not result in any adjustment of the contract price or the project completion schedule.

CLAUSE 13

Foreclosure of Contract due to Abandonment or Reduction in Scope of Work

If at any time after acceptance of the tender or during the progress of work, the purpose or object for which the work is being done changes due to any supervening cause or in the opinion of the Engineer-in-Charge the project cannot be carry forward either in part or in

full due to the reasons attributable to the contractor or otherwise, and as a result of which the work has to be abandoned or reduced in scope, the Engineer-in-Charge shall give notice in writing to that effect to the contractor stating the decision as well as the cause for such decision. The decision of the Engineer-in-charge shall be final and the contractor shall act accordingly in the matter. The contractor shall have no claim to any payment of compensation or otherwise whatsoever, on account of any profit or advantage which he might have derived from the execution of the works in full but which he did not derive in consequence of the foreclosure of the whole or part of the works.

The contractor shall be paid at contract rates, full amount for works executed at site and, in addition, a reasonable amount as certified by the Engineer-in-Charge for the items hereunder mentioned which could not be utilized on the work to the full extent in view of the foreclosure;

- (i) Any expenditure incurred on preliminary site work, e.g. temporary access roads, temporary labour huts, staff quarters and site office; storage accommodation and water storage tanks.
- (ii) The Employer shall have the option to take over contractor's materials or any part thereof either brought to site or of which the contractor is legally bound to accept delivery from suppliers (for incorporation in or incidental to the work) provided, however the Employer shall be bound to take over the materials or such portions thereof as the contractor does not desire to retain. For materials taken over or to be taken over by the Employer, cost of such materials as detailed by Engineer-in- Charge shall be paid. The cost shall, however, take into account purchase price, cost of transportation and deterioration or damage which may have been caused to materials whilst in the custody of the contractor.
- (iii) Reasonable compensation for transfer of T & P from site to contractor's permanent stores or to his other works, whichever is less. If T & P are not transported to either of the said places, no cost of transportation shall be payable.
- (iv) Reasonable compensation for repatriation of contractor's site staff and imported labour to the extent necessary.

The contractor shall, if required by the Engineer- in-Charge, furnish to him, books of account, wage books, time sheets and other relevant documents and evidence as may be necessary to enable him to certify the reasonable amount payable under this condition.

The reasonable amount of items on (i), (iii) and (iv) above shall not be in excess of 2% of the cost of the work remaining incomplete on the date of closure, i.e. total stipulated cost of the work as per accepted tender less the cost of work actually executed under the contract and less the cost of contractor's materials at site taken over by the Employer as per item (ii) above. Provided always that against any payments due to the contractor on this account or otherwise, the Engineer-in-Charge shall be entitled to recover or be credited with any outstanding balances due from the contractor for advance paid in respect of any tool, plants and materials and any other sums which at the date of termination were recoverable by the Employer from the contractor under the terms of the contract.

In the event of action being taken under Clause 13 to reduce the scope of work, the contractor may furnish fresh Performance Guarantee on the same conditions, in the same manner and at the same rate for the balance tendered amount and initially valid up to the extended date of completion or stipulated date of completion if no extension has been granted plus minimum 180 days beyond that. Wherever such a fresh Performance Guarantee is furnished by the contractor the Engineer-in-Charge may return the previous Performance Guarantee.

CLAUSE 14

Pre-Construction activities

The data supplied by the Employer, if any, are for General Guidance only. The contractor shall be responsible for carrying out pre- construction activities for construction of work as defined in the tender documents. The contractor shall also carry out site investigations to verify site details / Data at his own cost.

Since start of pre-construction activities contractor has to deploy Building Information Model (BIM) professional having knowledge of using Building Information Model (BIM) of minimum LOD 350.

CLAUSE 15

Suspension of Work

- i) The contractor shall, on receipt of the order in writing of the Engineer-in-Charge, (whose decision shall be final and binding on the contractor) suspend the progress of the works or any part thereof for such time and in such manner as the Engineer-in-Charge may consider necessary so as not to cause any damage or injury to the work already done or endanger the safety thereof for any of the following reasons:
 - a) on account of any default on the part of the contractor or;
 - b) for proper execution of the works or part thereof for reasons other than the default of the Contractor; or
 - c) for safety of the works or part thereof. The Contractor shall, during such suspension, properly protect and secure the works to the extent necessary and carry out the instruction given in that behalf by the Engineer-in- Charge.
- ii) If the suspension is ordered for reasons (b) and (c) in sub-para (i) above:
 - a) the contractor shall be entitled to an extension of time equal to the period of every such suspension plus 25%, for completion of the item or group of items of work for which a separate period of completion is specified in the contract and of which the suspended work forms a part, and;
 - b) If the total period of all such suspensions in respect of an item or group of items or

work for which a separate period of completion is specified in the contract exceeds thirty days, the contractor shall, in addition, be entitled to such compensation as the Engineer-in-Charge may consider reasonable in respect of salaries and/or wages paid by the contractor to his employees and labour at site, remaining idle during the period of suspension, adding thereto 2% to cover indirect expenses of the contractor provided the contractor submits his claim supported by details to the Engineer-in-Charge within fifteen days of the expiry of the period of 30 days.

- iii) If the works or part thereof is suspended on the orders of the Engineer-in-Charge for more than three months at a time, except when suspension is ordered for reason (a) in sub para (i) above, the contractor may after receipt of such order serve a written notice on the Engineer-in-Charge requiring permission within fifteen days from receipt by the Engineer in-Charge of the said notice, to proceed with the work or part thereof in regard to which progress has been suspended and if such permission is not granted within that time, the contractor, if he intends to treat the suspension, where it affects only a part of the works as an omission of such part by the Employer or where it affects whole of the works, as an abandonment of the works by the Employer, shall within ten days of expiry of such period of 15 days give notice in writing of his intention to the Engineer-in-Charge. In the event of the contractor treating the suspension as an abandonment of the contract by the Employer, he shall have no claim to payment of any compensation on account of any profit or advantage which he might have derived from the execution of the work in full but which he could not derive in consequence of the abandonment. He shall, however, be entitled to such compensation, as the Engineer-in-Charge may consider reasonable, in respect of salaries and/or wages paid by him to his employees and labour at site, remaining idle in consequence adding to the total thereof 2% to cover indirect expenses of the contractor provided the contractor submits his claim supported by details to the Engineer-in-Charge within 30 days of the expiry of the period of 3 months.

CLAUSE 16

Action in case Work not done as per specifications

All works under or in course of execution or executed in pursuance of the contract, shall at all times be open and accessible to the inspection and supervision of the Engineer-in-Charge, his authorized subordinates in charge of the work and all the superior officers, officer of the Quality Assurance Unit of the Employer or any organization engaged by the Employer for Quality Assurance and of the Chief Technical Examiner's Office, and the contractor shall, at all times, during the usual working hours and at all other times at which reasonable notice of the visit of such officers has been given to the contractor, either himself be present to receive orders and instructions or have a responsible agent duly accredited in writing, present for that purpose. Orders given to the Contractor's agent shall be considered to have the same force as if they had been given to the contractor himself.

If it shall appear to the Engineer-in-charge or his authorized subordinates in charge of the work or to the Chief Engineer-in-charge of Quality Assurance or his subordinate officers or the officers of the organization engaged by the Employer for Quality Assurance or to the Chief Technical Examiner or his subordinate officers, that any work has been executed with unsound, imperfect, or unskillful workmanship, or with materials or articles provided by him for the execution of the work which are unsound or of a quality inferior to that contracted or otherwise not in accordance with the contract, the contractor shall, on demand in writing which shall be made within twelve months (six months in the case of work costing Rs. 10 Lac and below except road work) of the completion of the work from the Engineer-in-Charge specifying the work, materials or articles complained of notwithstanding that the same may have been passed, certified and paid for forthwith rectify, or remove and reconstruct the work so specified in whole or in part, as the case may require or as the case may be, remove the materials or articles so specified and provide other proper and suitable materials or articles at his own charge and cost. In the event of the failing to do so within a period specified by the Engineer-in- Charge in his demand aforesaid, then the contractor shall be liable to pay compensation at the same rate as under clause 2 of the contract (for non-completion of the work in time) for this default.

In such case the Engineer-in-Charge may not accept the item of work at the rates applicable under the contract but may accept such items at reduced rates as the authority specified in **schedule 'F'** may consider reasonable during the preparation of on account bills or final bill if the item is so acceptable without detriment to the safety and utility of the item and the structure or he may reject the work outright without any payment and/or get it and other connected and incidental items rectified, or removed and re-executed at the risk and cost of the contractor. Decision of the Engineer-in-Charge to be conveyed in writing in respect of the same will be final and binding on the contractor.

CLAUSE 17

Damages and Defects Liability

17.1 During Progress of Work

If the contractor or his working staff or workers damage any part of the work in the scope of contract, or any building, road, road kerb, fence, enclosure, water pipe, cables, drains, electric or telephone post or wires, trees, grass, grassland, cultivated ground, etc. contiguous to the premises on which the work or any part of it is being executed, the contractor shall make good the same at his own cost.

Contractor shall repair/replace and restore the damaged structures/services in a time bound manner as required and as directed by Engineer-in-Charge. Contractor shall not be given any benefit of hinderance caused in the execution of the work owing to such damaged structure/service and time taken in its restoration by the contractor.

17.2 During Defect Liability Period

The contractor shall be responsible for all the defects and deficiencies in the work within the scope of this contract, **during the defect liability period which shall be for 24 Months** after the date of actual completion of work as recorded by Engineer-in-Charge. The liability of contractor for defects and deficiencies may arise due to:

- (a) Improper planning and design of the project, if in the scope of contract.
- (b) Works, Tools, Plant & Machinery, Materials or Workmanship no being in accordance with this contract.
- (c) Improper upkeep & maintenance during construction of the work.
- (d) Improper upkeep, operation and/or maintenance during defect liability period, if these are in the scope of this contract.
- (e) Failure by the contractor to comply with any other obligation under this contract. Such defects and deficiencies shall be made good by the contractor at his own cost after getting instructions/ notice from the Engineer-in-Charge within the time period specified in such instructions/notice.

However, contractor need not wait for instructions/ notice from Engineer-in-Charge for rectification of defects in work which come to his notice, and he should initiate action for needful rectification of defect on priority under intimation to Engineer-in-Charge, to avoid any untoward incident.

17.3 Structural Soundness

The contractor shall follow the good engineering practice for safety, serviceability and structural soundness of the building/ structure/ road work etc. as covered in the scope of contract.

17.3.1 Structure Design in the Scope of Contract

The contractor shall have obligation to rectify all defects in the structural elements or any other part of building/ structure/ road etc. due to design deficiency at his own cost for 10 (Ten) years from the date of completion as recorded in the completion certificate by the Engineer-in-Charge. Such defects shall be made good by the contractor at his own cost after getting instructions/ notice from the Engineer-in-Charge within the time period specified in such instructions/ notice and as per methodology duly approved by the Engineer-in-Charge.

17.3.2 Structure Design not in the Scope of Contract – Not Applicable

~~The Contractor shall not be liable for design deficiency.~~

17.3.3 Liability for Execution

The contractor shall be fully liable for any deficiency in structural soundness of work owing to execution of the work under the scope of this contract. The contractor shall

have obligation to rectify all defects in the structural elements owing to any deficiency in execution of work at his own cost for 10 (Ten) years from the date of completion as recorded in the completion certificate by Engineer-in-Charge. Such defects shall be made good by the contractor at his own cost after getting instructions/ notice from Engineer-in-Charge within the time period specified in such instructions/ notice and as per methodology duly approved by Engineer-in-Charge.

17.4 Methodology for Rectification of Defects

The design, methodology and quality of rectification of defects carried out by the contractor shall be as per sound engineering practice.

17.5 Contractor's failure to rectify defects as defined in the sub-clauses 17.1, 17.2, 17.3 & 17.4

In the event that the contractor fails to repair or rectify the defect or deficiency within the period specified by the Engineer-in-Charge, the Engineer-in-Charge shall be entitled to get the same repaired, rectified or remedied at the contractor's cost and recover such amount from any dues like performance guarantee, security deposits etc. available with Engineer-in-Charge. Engineer-in-Charge may take action for debarment of contractor from tendering in the RITES/Employer by following due process. For inaction or failure to rectify the defects covered under sub clause 17.3 within specified time limit, the Engineer-in-Charge may also initiate legal and/or other actions under other applicable laws in addition to other remedies available under the contract.

17.6 Release of Security Deposit

Fifty percent (50%) of the security deposit of the contractor shall be retained for a period of 18 (Eighteen) months from the date of completion of work as per completion certificate issued by the Engineer-in-Charge or till the final bill has been passed whichever is later. This balance fifty percent (50%) security deposit shall be released after completion of defect liability period.

Provided further, that the security deposit shall be released within a month of its due date as stated above only if satisfactory performance is observed during the said period and after deduction of Employer/Government dues from the contractor, if any.

CLAUSE 18 A

Recovery of Compensation paid to Workmen

In every case in which by virtue of the provisions under sub- section (1) of section 12 of the Workmen's Compensation Act. 1923, the Employer is obliged to pay compensation to a workman employed by the contractor, in execution of the works, the Employer will recover from the contractor, the amount of the compensation so paid and, without prejudice to the rights of the Employer under sub- section(2) of section 12 of the said Act, the Employer shall be at liberty to recover such amount or any part thereof by deducting it from the security deposit or

from any sum due by the Employer to the contractor whether under this contract or otherwise. The Employer shall not be bound to contest any claim made against it under sub- section (1) of section 12, of the said Act, except on the written request of the contractor and upon his giving to the Employer full security for all costs for which the Employer might become liable in consequence of contesting such claim.

CLAUSE 18 B

Ensuring Payment and Amenities to Workers if Contractor fails

In every case in which by virtue of the provisions of the Contract Labour (Regulation and Abolition) Act, 1970, and the Contract Labour (Regulation and Abolition) Central Rules, 1971, the Employer is obliged to pay any amounts of wages to a workman employed by the contractor in execution of the works, or to incur any expenditure in providing welfare and health amenities required to be provided under the above said Act, and the rules under Clause 19H or under the RITES Contractor's Labour Regulations, or under the Rules framed by Government from time to time for the protection of health and sanitary arrangements for workers employed by Employer's Contractors, the Employer will recover from the contractor, the amount of wages so paid or the amount of expenditure so incurred; and without prejudice to the rights of the Employer under sub-section(2) of Section 20, and sub-section (4) of Section 21, of the Contract Labour (Regulation and Abolition) Act, 1970, the Employer shall be at liberty to recover such amount or any part thereof by deducting it from the security deposit or from any sum due by the Employer to the contractor whether under this contract or otherwise. The Employer shall not be bound to contest any claim made against it under sub-section (1) of Section 20, sub-section (4) of Section 21, of the said Act, except on the written request of the contractor and upon his giving to the Employer full security for all costs for which the Employer might become liable in contesting such claim.

CLAUSE 19

Labour Laws to be complied by the Contractor

The contractor shall comply with the provisions of the Contract Labour (Regulation and Abolition) Act, 1970, and the Contract Labour (Regulation and Abolition) Central Rules, 1971. The contractor shall also obtain a valid licence under the said Act before the commencement of the work, and continue to have a valid license until its completion.

The contractor shall also comply with provisions of the Inter-State Migrant Workmen (Regulation of Employment and Conditions of Service) Act, 1979.

The contractor shall also abide by the provisions of the Child and Adolescent Labour (Prohibition and Regulation) Act, 1986, amended by Amendment Act No. 35 of 2016 and thereafter time to time.

The contractor shall also comply with the provisions of the building and other Construction

Workers (Regulation of Employment & Conditions of Service) Act, 1996 and the building and other Construction Workers Welfare Cess Act, 1996.

The contractor shall also comply with the provisions of Sexual Harassment of Women at Workplace (Prevention Prohibition and Redressal) Act, 2013 and amendment thereafter time to time.

Any failure to fulfil these requirements shall attract the penal provisions of this contract arising out of the resultant non-execution of the work.

CLAUSE 19 A

Employment of Under-age Labour

No labour below the age of eighteen years shall be employed on the work. However Adolescent Persons can be employed on non-hazardous works/process.

CLAUSE 19 B

Payment of wages:

- (i) The contractor shall pay to labour employed by him either directly or through subcontractors, wages not less than fair wages as defined in the RITES Contractor's Labour Regulations or as per the provisions of the Contract Labour (Regulation and Abolition) Act, 1970 and the contract Labour (Regulation and Abolition) Central Rules, 1971, wherever applicable. Gazette Notification 19.01.2017, S.O 188 (E) extra ordinary part 2 – sec. 3 (ii) amended time to time.

Thus, higher of the wages either notified by Govt. of India, Ministry of Labour and/or that notified by the local administration of the State Govt. both relevant to the place of work and the period of reckoning shall be paid by the contractor to the labourer.

- (ii) The contractor shall, notwithstanding the provisions of any contract to the contrary, cause to be paid fair wage to labour indirectly engaged on the work, including any labour engaged by his sub-contractors in connection with the said work, as if the labour had been immediately employed by him.
- (iii) In respect of all labour directly or indirectly employed in the works for performance of the contractor's part of this contract, the contractor shall comply with or cause to be complied with the RITES contractor's Labour Regulations from time to time in regard to payment of wages, wage period, deductions from wages recovery of wages not paid and deductions unauthorizedly made, maintenance of wage books or wage slips publication of scale of wage and other terms of employment, inspection and submission of periodical returns and all other matters of the like nature or as per the provisions of the Contract Labour (Regulation and Abolition) Act, 1970, and the Contract Labour (Regulation and Abolition) Central Rules, 1971, wherever applicable.

- (iv) (a) The Engineer-in-Charge concerned shall have the right to deduct from the moneys due to the contractor any sum required or estimated to be required for making good the loss suffered by a worker or workers by reason of non-fulfilment of the conditions of the contract for the benefit of the workers, non-payment of wages or of deductions made from his or their wages which are not justified by their terms of the contract or non-observance of the Regulations.
- (b) Under the provision of Minimum Wages (Central) Rules, 1950, the contractor is bound to allow to the labours directly or indirectly employed in the works one day rest for 6 days continuous work and pay wages at the same rate as for duty. In the event of default, the Engineer-in-Charge shall have the right to deduct the sum or sums not paid on account of wages for weekly holidays to any labours and pay the same to the persons entitled thereto from any money due to the contractor by the Engineer-in-Charge concerned.

In the case of Union Territory of Delhi, however, as the all-inclusive minimum daily wages fixed under Notification of the Delhi Administration No.F.12(162) MWO/DAB/ 43884-91, dated 31-12-1979 as amended from time to time are inclusive of wages for the weekly day of rest, the question of extra payment for weekly holiday would not arise.

- (v) The contractor shall comply with the provisions of the Payment of Wages Act, 1936, Minimum Wages Act, 1948, Employees Liability Act, 1938, Workmen's Compensation Act, 1923, Industrial Disputes Act, 1947, Maternity Benefits Act, 1961, and the Contractor's Labour (Regulation and Abolition) Act 1970, Gazette Notification 19.01.2017, S.O 188 (E) extra ordinary part 2 -sec.3 (ii) and the modifications thereof or any other laws relating thereto and the rules made there under from time to time.
- (vi) The contractor shall indemnify and keep indemnified the Employer against payments to be made under and for the observance of the laws aforesaid and the RITES Contractor's Labour Regulations without prejudice to his right to claim indemnity from his sub-contractors.
- (vii) The laws aforesaid shall be deemed to be a part of this contract and any breach thereof shall be deemed to be a breach of this contract.
- (viii) Whatever is the minimum wage for the time being, or if the wage payable is higher than such wage, such wage shall be paid by the contractor to the workmen directly without the intervention of Jamadar and that Jamadar shall not be entitled to deduct or recover any amount from the minimum wage payable to the workmen as and by way of commission or otherwise.
- (ix) The contractor shall ensure that no amount by way of commission or otherwise is deducted or recovered by the Jamadar from the wage of workmen.

CLAUSE 19 C

Compliance with provisions in Safety Code

In respect of all labour directly or indirectly employed in the work for the performance of the contractor's part of this contract, the contractor shall at his own expense arrange for the safety provisions as per RITES Safety Code framed from time to time and shall at his own expense provide for all facilities in connection therewith. In case the contractor fails to make arrangement and provide necessary facilities as aforesaid, he shall be liable to pay a penalty as decided by the authority mentioned in Schedule 'F' for each default and in addition, the Engineer-in-Charge shall be at liberty to make arrangement and provide facilities as aforesaid and recover the costs incurred in that behalf from the contractor.

CLAUSE 19 D

Submission of Fortnightly Labour Report

The contractor shall submit by the 4th and 19th of every month, to the Engineer-in-Charge, a true statement showing in respect of the second half of the preceding month and the first half of the current month respectively: -

- (1) the number of labourers employed by him on the work,
- (2) their working hours,
- (3) the wages paid to them,
- (4) the accidents that occurred during the said for night showing the circumstances under which they happened and the extent of damage and injury caused by them, and
- (5) the number of female workers who have been allowed maternity benefit according to Clause 19F and the amount paid to them.

Failing which the contractor shall be liable to pay to the Employer, a sum as decided by the authority mentioned in Schedule F for each default or materially incorrect statement. The decision of the Engineer-in-Charge shall be final in deducting from any bill due to the contractor, the amount levied as fine and be binding on the contractor.

CLAUSE 19 E

Compliance with the Rules on Health and Sanitary arrangements

In respect of all labour directly or indirectly employed in the works for the performance of the contractor's part of this contract, the contractor shall comply with or cause to be complied with all the rules framed by RITES/Employer from time to time for the protection of health and sanitary arrangements for workers employed by the Employer and its contractors.

CLAUSE 19 F

Maternity Benefits

Leave and pay during leave shall be regulated as follows:

1. Leave:

- (i) in the case of delivery - maternity leave not exceeding 8 weeks, 4 weeks up to and

including the day of delivery and 4 weeks following that day,

- (ii) in the case of miscarriage - upto 3 weeks from the date of miscarriage.

2. Pay:

- (i) in the case of delivery - leave pay during maternity leave will be at the rate of the women's average daily earnings, calculated on total wages earned on the days when full time work was done during a period of three months immediately preceding the date on which she gives notice that she expects to be confined.
- (ii) In the case of miscarriage - leave pay at the rate of average daily earning calculated on the total wages earned on the days when full time work was done during a period of three months immediately preceding the date of such miscarriage.

3. Conditions for the grant of Maternity Leave:

No maternity leave benefit shall be admissible to a woman unless she has been employed for a total period of not less than six months immediately preceding the date on which she proceeds on leave.

- 4. The Contractor shall maintain a register of Maternity (Benefit) in the Prescribed form as shown in Appendix I and II, and the same shall be kept at the place of work.

CLAUSE 19 G

Action in case of Non-observance of Rules and Regulations

In the event of the contractor(s) committing a default or breach of any of the provisions of the RITES's, Contractor's Labour Regulations and Model Rules for the protection of health and sanitary arrangements for the workers as amended from time to time or furnishing any information or submitting or filing any statement under the provisions of the above Regulations and' Rules which is materially incorrect, he/they shall, without prejudice to any other liability, pay to the Employer a sum as decided by the authority mentioned in Schedule 'F' for every default, breach or furnishing, making, submitting, filing such materially incorrect statements and in the event of the contractor(s) defaulting continuously in this respect, the penalty may be enhanced to as decided by the authority mentioned in Schedule 'F' per day for each day of default subject to a maximum of 5 percent of the estimated cost of the work put to tender. The decision of the Engineer-in-Charge shall be final and binding on the parties.

Should it appear to the Engineer-in-Charge that the contractor(s) is/are not properly observing and complying with the provisions of the RITES's Contractor's Labour Regulations and Model Rules and the provisions of the Contract Labour (Regulation and Abolition) Act 1970, and the Contract Labour (R& A) Central Rules 1971, for the protection of health and sanitary arrangements for work-people employed by the contractor(s) (hereinafter referred as "the said

Rules") the Engineer-in-Charge shall have power to give notice in writing to the contractor(s) requiring that the said Rules be complied with and the amenities prescribed therein be provided to the work-people within a reasonable time to be specified in the notice. If the contractor(s) shall fail within the period specified in the notice to comply with and/observe the said Rules and to provide the amenities to the work-people as aforesaid, the Engineer-in-Charge shall have the power to provide the amenities hereinbefore mentioned at the cost of the contractor(s). The contractor(s) shall erect, make and maintain at his/ their own expense and as per approved standards all necessary huts and sanitary arrangements required for his/their work-people on the site in connection with the execution of the works, and if the same shall not have been erected or constructed, according to approved standards, the Engineer-in-Charge shall have power to give notice in writing to the contractor(s) requiring that the said huts and sanitary arrangements be remodelled and/or reconstructed according to approved standards, and if the contractor(s) shall fail to remodel or reconstruct such huts and sanitary arrangements according to approved standards within the period specified in the notice, the Engineer-in-Charge shall have the power to remodel or reconstruct such huts and sanitary arrangements according to approved standards at the cost of the contractor(s).

CLAUSE 19 H

Provision of Labour Camp with Amenities

The contractor(s) shall at his/their own cost provide his/their labour with a sufficient number of huts (hereinafter referred to as the camp) of the following specifications on a suitable plot of land to be approved by the Engineer-in-Charge.

- i) a) The minimum height of each hut at the eaves level shall be 2.10m (7ft.) and the floor area to be provided will be at the rate of 2.7 sq.m. (30 sq.ft.) for each member of the worker's family staying with the labourer.
- b) the Contractor shall in addition construct suitable cooking places having at minimum area of 1.80m x 1.50m (6'x5') adjacent to the hut for each family.
- c) The Contractor(s) shall also construct temporary latrines and urinals for the use of the labourers each on the scale of not less than four per each one hundred of the total strength, separate latrines and urinals being provided for women.
- d) The Contractor shall construct sufficient number of bathing and washing places, one unit for every 25 persons residing in the camp. These bathing and washing places shall be suitably screened.
- ii) a) All the huts shall have walls of sun-dried or burnt bricks laid in mud mortar or other suitable local materials as may be approved by Engineer-in-Charge. In case of sun-dried bricks, the walls should be plastered with mud gobi on both sides. The floor may be kutcha but plastered with mud gobi and shall be at least 15 cm (6") above the surrounding

- ground. The roofs shall be laid with thatch or any other materials as may be approved by the Engineer-in-Charge and the Contractor shall ensure that throughout the period of their occupation the roofs remain water-tight.
- b) The Contractor shall provide each hut with proper ventilation.
 - c) All doors, windows, and ventilators shall be provided with suitable leaves for security purposes.
 - d) There shall be kept an open space of at least 7.2m (8 yards) between the rows of huts which may be reduced to 6m (20 ft.) according to the availability of site with the approval of the Engineer-in-Charge. Back to back construction will be allowed.
- iii) **Water Supply** - The Contractor shall provide adequate supply of water for the use of labourers. The provisions shall not be less than two gallons of pure and wholesome water per head per day for drinking purposes and three gallons of clean water per head per day for bathing and washing purposes. Where piped water supply is available, supply shall be at stand posts and where the supply is from well or river, tanks which may be of metal or masonry, shall be provided. The Contractor shall also at his own cost make arrangements for laying pipe lines for water supply to his labour camp from the existing mains wherever available, and shall pay all fees and charges therefor.
- iv) The site selected for the camp shall be high ground, removed from jungle clearances, if required.
- v) **Disposal of Excreta** - The Contractor shall make necessary arrangements for the disposal of excreta from the latrines by trenching or incineration which shall be according to the requirements laid down by the Local Health Authorities. If trenching or incineration is not allowed the Contractor shall make arrangements for the removal of the excreta through the Municipal Committee/authority and inform it about the number of labourers employed so that arrangements may be made by such committee/ authority for the removal of the excreta. All charges on this account shall be borne by the Contractor and paid direct by him to the Municipality/authority. The Contractor shall provide one sweeper for every eight seats in case of dry system.
- vi) **Drainage** - The Contractor shall provide efficient arrangements for draining away sullage water so as to keep the camp neat and tidy.
- vii) The Contractor shall make necessary arrangements for keeping the camp area sufficiently lighted to avoid accidents to the workers.
- viii) **Sanitation** - The Contractor shall make arrangements for conservancy and sanitation in the labour camps according to the rules of the Local Public Health and Medical Authorities.

CLAUSE 19 I

Compliance with instructions on removal from site of undesirable person

The Engineer-in-Charge may require the contractor to dismiss or remove from the site of the work any person or persons in the contractors' employ upon the work who may be incompetent or misconduct himself and the contractor shall forthwith comply with such requirements. In respect of maintenance/repair or renovation works etc. where the labour have an easy access to the individual houses, the contractor shall issue identity cards to the labourers, whether temporary or permanent and he shall be responsible for any untoward action on the part of such labour.

CLAUSE 19 J

Unauthorized occupation of building during construction

It shall be the responsibility of the Contractor to see that the building under construction is not occupied by anybody unauthorisedly during construction and is handed over to the Engineer-in-Charge with vacant possession of complete building. If such building though completed is occupied illegally, then the Engineer-in-Charge shall have the option to refuse to accept the said building/ buildings in that position. Any delay in acceptance on this account will be treated as the delay in completion and for such delay, a compensation shall be levied as per Clause 2 of the agreement, by the Engineer-in-Charge whose decision shall be final both with regard to the justification and quantum and be binding on the contractor.

However, the Engineer-in-Charge, through a notice, may require the contractor to remove the illegal occupation any time on or before construction and delivery.

CLAUSE 19 K

Employment of skilled/semi-skilled workers

The contractor shall at all stages of work, deploy formally certified skilled workforce or shall give a commitment to the effect that they would ensure that all his workers would be skilled through Recognition of Prior Learning (RPL) within two months from the date of commencement of the work under the project, at the cost of the contractor.

Contractor may approach Ministry of Skill Development and Entrepreneurship at cpsu-skillindia@gov.in through Engineer in Charge for support in getting the workers skilled through Recognition Prior Learning (RPL)

Clause 19L

Contribution of EPF and ESI

The ESI and EPF contributions on the part of employer in respect of this contract shall be paid by the contractor. These contributions on the part of the employer paid by the contractor shall be reimbursed by the Engineer-in-charge to the contractor on actual basis. The verification of deployment of labour will be done through biometric attendance system or any other suitable method by the Engineer-in-Charge. The applicable and eligible amount of EPF & ESI shall be reimbursed preferably within 7 days but not later than 30 days of submission of documentary

proof of payment provided same are in order.

Clause 19 M

Sexual Harassment of Women at Workplace

The contractor shall comply with all provision(s) and guideline(s) of Sexual Harassment of Women at Workplace (Prevention Prohibition and Redressal) Act, 2013 and amendment thereafter time to time or any other rules framed under any labour law affecting women worker(s).

CLAUSE 20

Minimum Wages Act to be complied with

The contractor shall comply with all the provisions of the Minimum Wages Act, 1948, and Contract Labour (Regulation and Abolition) Act, 1970, Gazette Notification 19.01.2017, S.O 188 (E) extra ordinary part 2-sec. 3(ii) amended from time to time and rules framed there under and other labour laws affecting contract labour that may be brought into force from time to time.

CLAUSE 21

Work not to be sublet, Action in case of insolvency

The contract shall not be assigned or sublet without the written approval of the Engineer-in-Charge. And if the contractor shall assign or sublet his contract, or attempt to do so, or become insolvent or commence any insolvency proceedings or make any composition with his creditors or attempt to do so, or if any bribe, gratuity, gift, loan, perquisite, reward or advantage pecuniary or otherwise, shall either directly or indirectly, be given, promised or offered by the contractor, or any of his servants or agent to any public officer or person in the employ of the Employer in any way relating to his office or employment, or if any such officer or person shall become in any way directly or indirectly interested in the contract, the Engineer-in-Charge on behalf of the Employer shall have power to adopt the course specified in Clause 3 hereof in the interest of the Employer and in the event of such course being adopted, the consequences specified in the said Clause 3 shall ensue.

CLAUSE 22

QUALITY ASSURANCE AND SUPERVISION FOR EXECUTION PART OF WORK

22.1 Quality of Materials and Workmanship

- (i) The Contractor shall ensure that the materials and workmanship are in accordance with the requirements specified in this Agreement, Specifications and Standards and Sound Engineering practice. The work shall be of the specified quality and standard, both in

respect of ingredients as well as the intended functions it is supposed to perform for service life.

- (ii) The Contract warrants that all materials shall be new, unused, not reconditioned, unless otherwise allowed as per contract or by Engineer-in-Charge, and in conformity with Specification and Standards, Applicable Laws and Sound Engineering Practice, and that the Contractor shall not use any materials which are generally recognized as being deleterious under Sound Engineering Practice.

22.2 Quality Assurance System

The Contractor shall devise a quality assurance mechanism to ensure compliance with the provisions of this Agreement (the "Quality Assurance Plan" or "QAP").

- (i) The Contractor shall, submit to the Engineer-in-Charge, its Quality Assurance Plan 15 (fifteen) days in advance of start of the execution stage specified in the Tender Document. The Engineer-in-Charge shall convey its comments to the Contractor within a period of 7 (seven) days of receipt of the QAP stating the modifications, if any, required and the Contractor shall incorporate those in the QAP conforming with the provisions of this clause. The QAP shall include the following:
 - (a) Contractor's Organization & structure, duties and responsibilities of individual key personnel, quality policy of contractor, procedure for control of non-conformities and corrective action, inspections and documentation.
 - (b) Internal quality audit system.
 - (c) Machinery, Shuttering, other Tool & plants, etc. required to be deployed at site.
 - (d) Method statement of important activities. These can be submitted as per the sequencing of the activities of the work.
 - (e) Quality control mechanism including sampling and testing of Materials, test frequencies, standards, acceptance criteria, testing facilities, reporting, recording and interpretation of test results, approvals, proforma for testing and calibration in accordance with the Specifications and Standards and Sound Engineering Practice, and Material Lot size, number of required tests and frequency of testing for different construction materials.

All the relevant and applicable codes, specifications and standards, as well as the acceptance criteria for various items of work, workmanship, materials and process employed needs to be mentioned.
 - (f) Checklist for various items and materials.
 - (g) Formats for site documentation, monthly reports on implementation of QAP.

(ii) Sampling of materials

All samples of materials including Cement Concrete Cubes shall be taken by the QA engineers deployed by the Contractor and shall be witnessed by the Engineer-in-Charge or his authorized representatives as specified in the Tender Document. All the necessary assistance, facilities and safety shall be provided by the contractor. Cost of sample of materials and testing charges shall

be borne by the contractor, and he/she is responsible for safe custody of samples to be tested at site.

(iii) Testing of Materials

The contractor shall establish temporary field laboratory of adequate size with all necessary facilities. Field laboratory shall be equipped with the testing equipment for conducting routine field tests as per this contract. It will also have copies of standards, BIS codes, IRC codes, relevant publications.

All the tests in field lab setup at construction site shall be carried out by the QA Staff deployed by the contractor and shall be witnessed by the Engineer-in-Charge or his authorized subordinates as specified in the Tender Document. The contractor shall provide all necessary facility to them for witnessing the tests in the field laboratory. In general, contractor shall carry out 90% of field tests in site laboratory and 10% tests shall be got carried out from outside laboratory as indicated below. Contractor shall endeavour to obtain test reports for tests conducted from outside laboratory in a reasonable time.

(iv) Maintenance of Register of Test –

- (a) All the entries in the register of test are to be made by the designated QA Engineers of the contractor and same is to be regularly reviewed by the field officers as well as the Engineer-in-Charge. The contractor shall allow inspection of such records any time as desired by Engineer-in-Charge or his authorized representative.
- (b) All the tests carried out at construction site or outside laboratories are to be maintained by the contractor in the prescribed format in the test registers provided by the contractor and duly authenticated by Engineer-in-Charge. The test reports shall also be maintained in hard file.
- (c) Contractor is responsible for maintenance and safe custody of all the test registers and test records.
- (d) Mandatory test conducted as per approved proforma shall be attached with each Running bill. Submission of copy of all test registers and material at site register along with each alternate Running Account Bill and with Final Bill is mandatory.

(v) Maintenance of Material at Site (MAS) Register

MAS register of the key materials including Cement and Steel Registers shall be maintained in the proforma approved by Engineer-in-Charge. All the entries in the MAS registers are made by the designated staff of the contractor and same is regularly reviewed by the field officers as well as the Engineer-in-Charge. Contractor is responsible for maintenance and safe custody of MAS registers.

- (vi) The Contractor shall procure all relevant codes, publications, apparatus and instruments, fuel consumables, water, electricity, labour, materials, samples and qualified personnel as are necessary for examining and testing the Works, Materials and workmanship in

- accordance with the Quality Assurance Plan.
- (vii) All the cost of testing including cost of samples, packaging, transportation, testing charges of Construction, materials and workmanship under this clause shall be borne by the contractor.
 - (viii) The contractor shall submit monthly quality progress report on implementation of the provisions of Quality Assurance Plan on the format approved by the Engineer-in-Charge.

22.3 Samples

The Contractor shall at its own expense and without delay, provide the samples of materials and relevant information like Manufacturer's test reports, standard samples of manufactured materials and samples of such other Materials as the Engineer-in-charge may require for review and approvals in accordance with clause 10A of GCC before actual use.

22.4 Test

- (i) For determining that the Works conform to the Specifications and Standards, the Engineer-in-Charge shall require the Contractor to carry out or cause to be carried out tests at such time and frequency and in such manner as specified in this Agreement in accordance with sound engineering practice for quality assurance. Frequency and the manner in which tests shall be conducted shall be in the following order of preference:
 - (a) Contract provisions.
 - (b) CPWD specifications.
 - (c) BIS codes
 - (d) IRC codes
 - (e) MoRTH Specifications.
 - (f) International Codes.
 - (g) Manufacturer's specifications.

Outside tests shall be conducted at Government labs /IITs/ NITs and other approved laboratories by the Engineering-in-Charge for testing of materials.

- (ii) The Contractor shall, with due diligence, carry out all the tests in accordance with the Agreement and furnish the results thereof to the Engineer-in-Charge. The Engineer-in-Charge or his authorized representative shall witness or participate during the testing as specified in Tender Document. The contractor shall provide necessary assistance for witnessing/ participating in the field tests.
- (i) In the event that results of any tests conducted under this clause establish any defects or deficiencies in the Works, the Contractor shall carry out remedial measures at its own cost and furnish a report to the Engineer-in-Charge in this regard. The Engineer-in-Charge shall require the Contractor to carry out or cause to be carried out tests to determine that such remedial measures have brought the works into compliance with the Specifications and standards and the procedure shall be repeated until such Works conform to the

Specifications and Standards.

22.5 Method Statement

The 'Method statement' is a statement by which the construction procedures for important activities are stated, checked, and approved. The method statement shall be prepared for important activities as identified by the contractor as mentioned in QAP or any other activity as instructed by Engineer-in-Charge. The 'Method statement' should have a description of the item with elaborate procedure in steps to implement the same, the specifications of the materials involved, equipment to be deployed, measures for ensuring safety, their testing and acceptance criteria, precautions to be taken, mode of measurement, etc. The Contractor shall, at least 15 (fifteen) days prior to the commencement of activities, submit to the Engineer-in-Charge for review, the method statements proposed to be adopted for executing the various items of the work. The Engineer-in-Charge shall complete the review and convey its comments, if any, to the Contractor within a period of 07 (seven) days from the date of receipt of the proposed methodology from the Contractor.

22.6 Inspection & review by the Engineer-in-Charge and External Audit.

The Engineer-in-Charge, his authorized subordinates, senior officers of Employer/RITES, QA unit or any other third party may inspect and review the progress and quality of the work and issue appropriate directions to the Contractor for taking remedial action in the event the work is not in accordance with the provisions of this Agreement. The work may be inspected at any time/stage by external inspection teams like CTE or TE, Third Party Quality assurance agency, Employer/RITES team etc. may conduct inspection of the quality of the works. The findings of the inspections shall be notified to the Contractor for taking remedial action in accordance with the agreement. The Contractor shall provide all assistance as may be required by the inspection teams in the conduct of its inspection hereunder.

Suitable actions shall be taken as per the provisions contained in the relevant clauses of the agreement, if the work is not found to be as per specifications or quality as specified in the agreement.

22.7 Inspection of records

The Engineer-in-Charge or his authorized representative shall have right to inspect the records of the Contractor relating to the works.

22.8 Inspection of Works

- (i) The Engineer-in-Charge and his authorized subordinates shall at all times;
 - (a) have full access to all parts of the site and to all places from which natural materials are being obtained for use in the works: and
 - (b) during production, manufacture and construction at the site and at the place of

production, be entitled to examine, inspect, measure and test the materials and workmanship and to check the progress of the manufacturer of Materials.

- (ii) The Contractor shall give the Engineer-in-Charge and its authorized representative access, facilities, and safety equipment for carrying out their obligations under this Agreement.

22.9 Examination of work before covering up/ Test Check of item of Work

In respect of the work which the Engineer-in-Charge or his authorized representatives are required to examine, inspect, measure or test before it is covered up or put out of view or any part of the work is placed thereon, the Contractor shall give notice to the Engineer-in-Charge whenever any such work is ready and before it is covered up. The Engineer-in-Charge shall then either carry out the examination, inspection or testing without unreasonable delay within 7 days, or promptly give notice to the Contractor that the Engineer-in-Charge does not require him to do so. Provided, however, that if any work is of a continuous nature where it is not possible or prudent to keep it uncovered or incomplete, the Contractor shall notify the schedule of carrying out such work to give sufficient opportunity, not being less than 3 (three) business days' notice, to the Engineer-in-Charge to conduct its inspection, measurement or test while the work is continuing. Provided further that in the event the Contractor receives no response from the Engineer-in-Charge within a period of 3 (three) business days from the date on which the Contractor's notice hereunder is delivered to the Engineer-in-Charge, the Contractor shall be entitled to assume that the Engineer-in-Charge would not undertake the said inspections.

22.10 Rejection

- (i) If, as a result of an examination, inspection, measurement or testing, any Plant, Materials, design or workmanship is found to be defective or otherwise not in accordance with the provisions of this Agreement, the Engineer-in-Charge may reject such piece of work, Plant, Materials, design or workmanship by giving notice to the Contractor, with reasons. The Contractor shall then promptly make good the defect and ensure that the rejected item complies with the requirements of this Agreement.
- (ii) If the Engineer-in-Charge requires a piece of work, plant, Material, design or workmanship to be retested, the tests shall be repeated on the same terms and conditions, as applicable in each case. If the rejection and retesting cause the RITES/Employer to incur any additional costs, such costs shall be recoverable by the Engineer-in-Charge from the Contractor and may be deducted by the Engineer-in-Charge from any amount due to be paid to the Contractor.
- (iii) The Contractor shall not be entitled to any extension of time on account of rectifying any defect or retesting as specified in this clause.
- (iv) Examination, inspection, measurement or testing of any plant, Material, design or workmanship by the Engineer-in-Charge or its failure to convey its observations or to examine, inspect, measure or test shall neither relieve the Contractor of its obligations and liabilities under this Agreement in any manner nor shall the Engineer-in-Charge be liable for the same in any manner.

22.11 Remedial work

- (i) Notwithstanding any previous test or certification, the Engineer-in-Charge may instruct the Contractor to:
 - (a) remove from the site and replace any piece of work, plant or materials which are not in accordance with the provisions of this Agreement.
 - (b) remove and re-execute any work which is not in accordance with the provisions of this Agreement and the Specification and Standards; and
 - (c) execute any work which is urgently required for the safety of the Project, whether because of an accident, event or otherwise.
- (ii) If the Contractor fails to comply with the instructions issued by the Engineer-in-Charge under aforesaid para, within the time specified in the notice or as mutually agreed, the Engineer-in-Charge may get the work executed by another agency. The cost so incurred by the Engineer-in-Charge for undertaking such work shall, without prejudice to the rights of the Engineer-in-Charge to recover damages in accordance with the provisions of this Agreement, be recoverable from the Contractor and may be deducted by the Engineer-in-Charge from any amount due to be paid to the Contractor.

22.12 Quality Control Records

The Contractor shall hand over authenticated copy of all its quality control records and documents to the Engineer-in-Charge before the Completion Certificate is issued.

22.13 Video recording

During the Construction Period, the Contractor shall provide to the Engineer-in-Charge for every calendar quarter, a video recording which will be compiled into a 15 (fifteen) minutes digital video covering the status and progress of work in that quarter. Video recording should show different activities, stage of work, quality assurance activities etc. including animation, graphs, digital maps, commentary, subtitles, etc. spread over the quarter. The video recording shall be provided to the Engineer-in-Charge no later than 15 (fifteen) days after the close of each quarter to be reckoned from next full month of date of start of work.

22.14 Suspension of unsafe Construction Works

- (i) Upon recommendation of the Engineer-in-Charge to this effect, or on his own volition in cases of emergency or urgency, the Engineer-in-Charge may by notice require the Contractor to suspend forthwith the whole or any part of the Works if, in the reasonable opinion of Engineer-in-Charge, as the case may be, such work threatens the safety of the Users and or other persons on or about the Project. Provided, however, that in case of an emergency, the Engineer-in-Charge may *suomoto* issue the notice referred to hereinabove.
- (ii) The Contractor shall, pursuant to the notice under above para, suspend the Works or any part thereof for such time and in such manner as may be specified by the Engineer-in-

Charge and thereupon carry out remedial measures to secure the safety of suspended works, the Users, other persons and vehicles on or about the Project. The Contractor by notice require the Engineer-in-Charge to inspect such remedial measures forthwith and request for revocation of suspension. Upon reviewing the remedial measures, the Engineer-in-Charge shall either revoke such suspension or instruct the Contractor to carry out such other and further remedial measures as may be necessary and reasonable, and the procedure set forth in this Clause shall be repeated until the suspension hereunder is revoked.

- (iii) Subject to other provisions of the agreement, all reasonable cost incurred for maintaining and protecting the Works or part thereof during the period of suspension (the "Preservation Costs") shall be borne by the contractor, if in the opinion of Engineer-in-Charge suspension is on account of reasons attributable to the contractor.
- (iv) If suspension of Work is for reasons not attributable to the Contractor, the Engineer-in-Charge shall determine any Time Extension, if required, in accordance with the provisions of clause-5.

22.15 Online maintenance of Site records including testing records.

The Engineer-in-Charge may require the contractor to upload all the site records in any online system devised by him. The contractor shall have to ensure that all the required site records, as desired by the Engineer-in-Charge shall be uploaded in this online system. Nothing extra on this account shall be payable to the contractor. In case these records are to be maintained in any online module then contractor shall comply with this.

CLAUSE 23

Changes in firm's Constitution to be intimated

Where the contractor is a partnership firm/company/LLP, the previous approval in writing of the Engineer-in-Charge shall be obtained before any change is made in the constitution of the firm/company/LLP. Where the contractor is an individual or a Hindu undivided family business concern, such approval as aforesaid shall likewise be obtained before the contractor enters into any partnership agreement where under the partnership firm would have the right to carry out the works hereby undertaken by the contractor. If previous approval as aforesaid is not obtained, the contract shall be deemed to have been assigned in contravention of Clause 21 hereof and the same action may be taken, and the same consequences shall ensue as provided in the said Clause 21.

CLAUSE 24: Deleted

CLAUSE 25

Settlement of Disputes & Arbitration

- 1) Except where otherwise provided in the Contract all questions and disputes relating to the meaning of the specifications, design, drawings and instructions herein before mentioned and as to the quality of workmanship or materials used on the work or as to any other question, claim, right, matter or thing whatsoever in any way arising out of or relating to the Contract, designs, drawings, specifications, estimates, instructions, orders or these conditions or otherwise concerning the works or the execution or failure to execute the same whether arising during the progress of the work or after the cancellation, termination, completion or abandonment thereof shall be dealt with as mentioned hereinafter:
- 2) If the Contractor considers any work demanded of him to be outside the requirements of the Contract, or disputes any drawings, record or decision given in writing by the Engineer on any matter in connection with or arising out of the Contract or carrying out of the work, to be unacceptable, he shall promptly within 15 days, request the Engineer-in-Charge in writing for written instruction or decision. Thereupon, the Engineer-in-Charge shall give his written instructions or decision within a period of one month from the receipt of the Contractor's letter.
- 3) If the Engineer-in-Charge fails to give his instructions or decision in writing within the aforesaid period or if the Contractor is dissatisfied with the instructions or decision of the Engineer-in-Charge, the Contractor may, within 15 days of the receipt of the Engineer-in-Charge's decision or within 15 days of expiry of one months from the date of Contractor's letter to the Engineer-in-Charge, whichever is earlier, refer to conciliation through Expert Committee ("EC") to be constituted by the Appointing Authority specified in Schedule 'F' as provided hereunder:

3.1 **Proposal for EC**

The contractor shall give notice to the Appointing Authority for appointing the EC clearly bringing out the points of dispute and the amount claimed with documents in support of the claim.

3.2 **Constitution of EC**

- (i) The Appointing Authority will have the sole discretion to constitute EC which will comprise three Members. One of the three Members will be designated Presiding Member by the Appointing Authority. The Appointing Authority will have authority to reconstitute an EC to fill any vacancy or if any EC Member fails to fulfil his functions or fails to attend two or more consecutive meetings of the EC. The Presiding Member shall convene meetings of the Committee and keep minutes of the same.
- (ii) Upon constitution of the EC, the Appointing Authority will issue the appointment letter to EC Members and endorse copies thereof to the parties concerned.
- (iii) The EC members shall give a declaration of independence and impartiality to both the

parties before commencement of the EC proceedings.

3.3 **Proceedings before EC**

- i) The claimant shall submit its statement of claims to EC members, the other party to the contract and the Employer within 30 days of the issue of the appointment letter to EC members.
- ii) The respondent shall file its reply and counter claim (if any) within 30 days of the receipt of the statement of claims.
- iii) Parties may file their rejoinder/additional documents if any in support of their claim/counter claim within next 15 days. No documents shall be allowed thereafter, except with the permission of EC.
- iv) EC will commence its meetings only after completion of the pleadings. The Presiding Member shall fix the date, time and venue for the first meeting within 15 days of completion of pleadings in consultation with other Members. Thereafter, the next and subsequent meetings of the Committee shall be fixed with mutual consultations during such meetings.
- v) Two Members will constitute a valid quorum and the meeting can take place to proceed in the matter. EC recommendation will be valid if signed by all Members present in the meeting. Further, Members shall make efforts for unanimous recommendations.
- vi) The parties shall be represented by their in-house employees/executives. No party shall bring any advocate or outside consultant/advisor/agent. Ex-officers of RITES who have handled the matter in any capacity are not allowed to attend and present the case before EC on behalf of Contractor. However, ex-employees of parties may represent their respective organizations.
- vii) Solicitation or any attempt to bring influence of any kind on either EC Members or Employer/ Engineer – in – Charge is completely prohibited in conciliation proceedings and Employer/ Engineer – in - Charge reserves the absolute right to close the conciliation proceedings at its sole discretion if any such attempt is made by the Contractor or its representatives.
- viii) Parties agree to rely only upon documentary evidence in support of their claims and not to bring any oral evidence in the EC proceedings.
- ix) EC will give full opportunity of hearing to the parties before giving its recommendations.
- x) EC will conclude its proceedings in maximum 5 meetings and give its recommendations within 90 days of its first meeting. EC will give its recommendations to both the parties recommending possible terms of settlement. The Appointing Authority may extend the time/ number of meetings in exceptional cases, if EC requests for the same with sufficient reasons.

- xi) EC members will be paid fees and provided facilities as detailed in **Annexure - 'F4'**.
- xii) Parties shall not claim any interest on claims/counter claims from the date of notice invoking conciliation till execution of settlement agreement, if so arrived at. In case, parties are unable to reach a settlement, no interest shall be claimed by either party for the period from the date of notice invoking conciliation till the date of EC recommendations and 30 days thereafter in any further proceeding.
- xiii) The venue of the EC meeting shall be the place specified by the Presiding Member of EC.
- xiv) Legally, parties are under no obligation to refer a dispute to conciliation or continue with conciliation proceedings. Parties are free to terminate the conciliation proceedings at any stage as provided under the Arbitration and Conciliation Act, 1996 duly amended.

3.4 Actions after EC Recommendations

- i) The recommendations of EC are non-binding, and the parties may decide to accept or not to accept the same. Parties are at liberty to accept the EC recommendation with any modification they may deem fit.
- ii) The Parties shall give their response, if any, to the Appointing Authority and the other Party within 15 days of receiving EC recommendation.
- iii) If the recommendations are acceptable to the parties partly or fully, Appointing Authority will consider and take a decision on EC recommendations and communicate its decision to the parties. If decision of Appointing Authority is acceptable to the parties, a settlement agreement under Section 73 of the Arbitration and Conciliation Act, 1996 will be signed within 15 days of acceptance by the parties.
- iv) The timelines mentioned in the above sub-clauses are with an objective to achieve expeditious conclusion of EC proceedings. However, it does not mean that any action beyond the timelines will be invalid. However, the parties concerned will make all efforts to complete the actions within the stipulated time.
- v) The parties shall keep confidential all matters relating to the conciliation proceedings including minutes of EC meeting and recommendations of EC. Parties shall not rely upon them as evidence in any Forum / arbitration / court proceeding whether or not such proceedings relate to the dispute that is the subject of the conciliation proceedings,
 - a) Views expressed or suggestions made by other party in respect of a possible settlement of the dispute

- b) Admissions made by the other party in the course of the EC proceedings,
 - c) Proposals made by the EC,
 - d) The fact that the other party had indicated his willingness to accept a proposal for settlement made by the EC.
 - vi) Confidentiality extends also to the settlement agreement, except where its disclosure is necessary for purposes of implementation and enforcement. This stipulation will not apply to disclosure made by Employer to Govt. of India or its authorities, if required.
 - vii) Subject to terms and conditions contained in the above paras, the provisions of the Part III of Arbitration and Conciliation Act, 1996 shall be applicable to the conciliation proceeding and the parties and the EC members shall be bound by the same.
4. If any party is dissatisfied with the decision given by the Appointing Authority after considering recommendations of the EC, it shall, within a period of 30 days from receipt of the decision, give notice in writing to the Appointing Authority for appointment of arbitrator, specifying the issues which the Party is dissatisfied with the decision of the EC and final claims in Form **Annexure 'F2'**, failing which the said decision shall be final, binding and conclusive and not referable to adjudication by the arbitrator.
5. In case conciliation under sub-clause 3 is not resorted to and if the contractor (s) does /do not prefer his/their specific and final claims in writing in Form **Annexure 'F2'**, within a period of 90 days after receiving the intimation from RITES that the final bill is ready for payment, he/they will be deemed to have waived his/their claim (s) and the Employer shall be discharged and released of all liabilities under the contract in respect of these claims.
6. **Appointment of Arbitrator:** Except where the decision has become final, binding and conclusive in terms of Sub Clause (4) or (5) above, disputes or difference shall be referred for adjudication through arbitration by a sole arbitrator appointed by the Appointing Authority. The Arbitration Clause shall not be applicable (a) if the agreement /contract has not been duly executed and/or (b) in case the total value of all the claims of the Contractor submitted in Form **Annexure 'F2'** exceed Rs. 2,00,00,000/- (Rupees Two Crore only)
7. Sub Clause 6 shall not be applicable in case the dispute is (i) between two Public Sector Enterprises or (ii) between a Public Sector Enterprise and a Government Department. In such a case the provisions of Sub Clause 31 hereinafter shall apply.
8. The parties may waive off the applicability of sub-section 12(5) of Arbitration and conciliation Act 1996; if they agree for such waiver, they may do so in writing, after dispute(s) having arisen between them, in the format given under **Annexure 'F3'** of these conditions.
9. **Appointment of Arbitrator where applicability of Section 12 (5) of Arbitration and**

Conciliation Act. has been waived off:

- i) The Arbitral Tribunal shall consist of a Sole Arbitrator who shall be a RITES Officer not below the level of a General Manager in RITES or retired officer of Executive Director or above and empanelled as an arbitrator with RITES. The Sole Arbitrator shall be appointed within 60 days from the day when a written and valid demand for arbitration is received by the Appointing Authority.
- ii) For this propose, RITES will send a panel of at least four (4) names of RITES Officers of GM level or above which may also include the name(s) of retired RITES Officer(s) of ED level or above, empanelled to work as RITES Arbitrator to the contractor within 60 days from the day when a written and valid demand for arbitration is received by the Appointing Authority.
- iii) Contractor will be asked to suggest to the Appointing Authority at least 2 (two) names out of the panel for appointment as Arbitrator within 30 days from the date of dispatch of the request by RITES. The Appointing Authority shall appoint Sole Arbitrator from the selected two names. The Appointing Authority shall complete this exercise of constituting the Arbitral Tribunal within 30 days of the receipt of names from the contractor.

However, if the Contractor fails to suggest any two names out of the list given to him for appointment as Sole Arbitrator, as aforesaid, then the Appointing Authority may appoint any one out of the list referred in (ii) above, as the sole arbitrator which decision shall be binding on the contractor.

10. Appointment of Arbitrator where applicability of section 12 (5) of A&C Act has not been waived off:

- i) The Arbitral Tribunal shall consist of a Sole Arbitrator who shall be an Officer retired from Central/State Government or a Central PSE (other than RITES Ltd.) not below the rank of SAG Officer, empanelled in RITES as arbitrators. For this purpose, the Appointing Authority will send a list of at least four (4) names from the said panel of retired officers duly indicating their retirement date to the contractor within 60 days from the day when a written and valid demand for arbitration is received by the Appointing Authority.
- ii) Contractor will be asked to suggest at least 2 (two) names out of the list given to him for appointment as Arbitrator within 30 days from the date of dispatch of the list by the Appointing Authority. The Appointing Authority shall appoint at least one of them as the Sole Arbitrator. The Appointing Authority shall complete this exercise of constituting the Arbitral Tribunal within 30 days from the receipt of the names of contractor.

However, if the Contractor fails to suggest any two names out of the list given to him for appointment as Sole Arbitrator, as aforesaid, then the Appointing Authority may appoint any one out of the list referred in (i) above, as the sole arbitrator whose decision shall be binding on the Contractor.

11. If the arbitrator appointed as above refuses to act as arbitrator, withdraws from his office as arbitrator, or vacates his/her office or is unable or unwilling to perform his/her functions as arbitrator for any reason whatsoever or dies or in the opinion of the Appointing Authority, fails to act without undue delay, the Appointing Authority shall appoint new arbitrator to act in his/her place in the same manner in which the earlier arbitrator had been appointed. Such re-constituted Tribunal may, at its discretion, proceed with the reference from the stage at which it was left by the previous arbitrator.
12. The Arbitral Tribunal shall have power to call for such evidence by way of affidavits or otherwise as the Arbitral Tribunal shall think proper, and it shall be the duty of the parties hereto to do or cause to be done all such things as may be necessary to enable the Arbitral Tribunal to make the award without any delay. The proceedings shall normally be conducted on the basis of documents and written statements.
13. Before proceeding into the merits of any dispute, the Arbitral Tribunal shall first decide and pass its orders over any plea submitted/objection raised by any party, if any regarding appointment of Arbitral Tribunal, validity of arbitration agreement, jurisdiction and scope of the Tribunal to deal with the dispute (s) submitted to arbitration, applicability of time "limitation" to any dispute, any violation of agreed procedure regarding conduct of the arbitral proceeding or plea for interim measures of protection and record its orders in day to day proceedings. A copy of the proceedings duly signed by the Members of tribunal present during the hearing, shall be provided to the parties.
14. The claimant shall submit his claim stating the facts supporting the claim along with all the relevant documents and the relief remedy sought against each claim within a period of 30 days from the date of appointment of the Arbitral Tribunal.
15. The Other Party/Parties shall submit its/their counter - statement and counter claim(s), if any, within a period of 60 days of receipt copy of claims from Claimant/Tribunal unless extension has been granted by Tribunal.
16. **Place of Arbitration:** The seat of arbitration shall be in Guwahati. The venue of arbitration shall be the place as agreed in writing by both the parties and consented by the Sole Arbitrator.
17. Except dispute and claim preferred under Sub-Clause (4) and (5) of Clause 25, no new claim shall be added during proceedings by either party. Arbitration will adjudicate only upon the matters referred to it by the Appointing Authority and not on Excepted Matters/ Non-Arbitrable Disputes as per the Contract. However, a party may file Counter Claim

- and amend or supplement the original claim or defence thereof during the course of arbitration proceeding subject to acceptance by Tribunal having due regards to the delay in making it.
18. **Obligation during pendency of Arbitration:** Work under the Contract shall unless otherwise directed by the Engineer-in-Charge continue during the Arbitration proceeding and no payment due or payable by the Employer shall be withheld on account of such proceedings, provided however, it shall be open for the Arbitral Tribunal to consider and decide whether or not such work should continue during arbitration proceedings.
 19. While appointing arbitrator under Sub-Clause 9 and 10(i) above, due care shall be taken that he/she is not the one who had opportunity to deal with the matters to which the contract relates or who in the course of his/her duties as RITES Officer expressed views on all or any of the matters under dispute or differences. The proceedings of the Arbitral Tribunal or the award made by such Tribunal will, however, not be invalid merely for the reason that arbitrator, in the course of his/her service, had opportunity to deal with the matters to which the contract relates or who in the course of his/her duties expressed views on all or any of the matters under dispute.
 20. The arbitral award shall state item wise, the sum and reasons upon which it is based. The analysis and reasons shall be detailed enough so that award could be inferred therefrom.
 21. A party may apply for corrections of any computational errors, any typographical or clerical errors or any other errors of similar nature occurring in the award of a Tribunal and interpretation of a specific point of award to Tribunal within 60 days of receipt of the award.
 22. A party may apply to the Tribunal within 60 days of receipt of award to make an additional award as to claims presented in the arbitral proceedings but omitted from the arbitral award.
 23. **Deleted**
 24. Where the arbitral award is for the payment of money, no interest shall be payable on whole or any part of the money for any period till the date on which the award is made. The arbitral tribunal shall indicate i) the Award amount payable on the date of award ii) the period within which it is to be paid and iii) Simple rate of interest applicable beyond the stipulated free period for making payment of Award amount.
 25. The cost of arbitration shall be borne equally by the respective parties. The cost shall inter alia include fee of the arbitrator(s) as per the following schedule and the fee shall be borne equally by both the parties. The cost of the reference and of the award

(including the fees, if any, of the arbitrator) shall be in the discretion of the arbitrator who may direct to any by whom and in what manner, such costs or any part thereof shall be paid and fix or settle the amount of costs to be so paid. Notwithstanding anything contrary provided in any Act, Law or Agreement, the scale of Arbitrators' fee shall not exceed the following:

Scale of Arbitrator's Fee as per The Fourth Schedule of The Arbitration and Conciliation Act, 1996

Sl. No.	Sum in dispute	Model fee
1.1.	Upto Rs.5,00,000/-	Rs.45,000/-
2.2.	Above Rs.5,00,000/- and upto Rs.20,00,000/-	Rs.45,000/- plus 3.5 percent of the claim amount over and above Rs.5,00,000/-
3.3.	Above Rs.20,00,000/- and upto Rs.1,00,00,000/-	Rs.97,500/- plus 3 percent of the claim amount over and above Rs. 20,00,000/-
4.4.	Above Rs.1,00,00,000/- and upto Rs.10,00,00,000/-	Rs.3,37,500/- plus 1 percent of the claim amount over and above Rs.1,00,00,000/-
5.5.	Above Rs.10,00,00,000/- and upto Rs.20,00,00,000/-	Rs.12,37,500/- plus 0.75 percent of the claim amount over and above Rs.10,00,00,000/-
6.6.	Above Rs.20,00,00,000/-	Rs.19,87,500/- plus 0.5 percent of the claim amount over and above Rs.20,00,00,000/- with a ceiling of Rs.30,00,000/-

Note: In the event the arbitral tribunal is a sole arbitrator, he shall be entitled to an additional amount of twenty-five percent on the fee payable as per the above.

26. In case of a serving officer of RITES acting as arbitrator, the fee payable shall be 50% of the fee worked out at the above scale.
27. Further, the fee payable to the arbitrator would be governed by the above schedule irrespective of the fact whether the arbitrator is appointed by the Appointing Authority or by the Court of law.
28. Deleted
29. Deleted
30. Subject to the provisions of Arbitration and Conciliation Act 1996 as amended up to the

date of receipt of tenders, and the rules there under and relevant clauses of General Conditions of contract (GCC) and any statutory modifications there of shall apply to the appointment of arbitrators and arbitration proceedings under this Clause.

31. In case the dispute or difference relating to the interpretation and application of the provisions of commercial contract(s) between Central Public Sector Enterprises (CPSEs) / Port Trusts inter se and also between CPSEs and Government Departments / Organizations (excluding disputes concerning Railways, Income Tax, Customs & Excise Departments), such dispute or difference shall be taken up by either party for resolution through AMRCD as mentioned in Government of India, Ministry of Heavy Industries & Public Enterprises, Department of Public Enterprises OM No. 4(1)/2013-DPE(GM) /FTS-1835 dated 22-05-2018 as amended.

32. It is a term of this Contract that the party invoking arbitration shall give a list of disputes with amount claimed in respect of each such dispute along with the notice for appointment of arbitrator in the form at **Annexure 'F2'**. It is a term of this Contract that **"Excepted matters"** or matters where the decision of the Engineer-in-Charge or any higher authority has been stipulated as **"Final and Binding"** in various Clauses of Contract, stand specifically excluded from the purview of disputed claims of the contractor.

33. It is also a term of this Contract that no person other than a person appointed by such Appointing Authority as aforesaid should act as arbitrator and if for any reason that is not possible, the matter shall not be referred to arbitration at all. It is also a term of this Contract that if the Contractor does not make any specific and final claims in writing within 90 days of receiving the intimation from the Engineer-in-Charge that the final bill is ready for payment, the claim of the Contractor shall be deemed to have been waived and absolutely barred and the Employer shall be discharged and released of all liabilities under the Contract in respect of these claims.

34. **Signing of "No Claim" certificate**

The Contractor shall not be entitled to make any claim or initiate any legal action, suit or proceeding whatsoever against the Employer under or by virtue of or arising out of the Contract, nor shall the Employer entertain or consider any such claim if made by the Contractor after he shall have signed a "No Claim Certificate" in favour of the Employer in such form as stipulated by the Employer, after the works are finally measured up. The Contractor shall be debarred from disputing the correctness of any item covered by the "No Claim Certificate" or demanding a reference to arbitration in respect thereof.

35. **Parties to be impleaded in the conciliation and arbitration proceedings**

In case of any claims or legal proceedings by the Contractor, the Employer as well as RITES Ltd acting as Agent to the Employer will be impleaded as parties to the

Conciliation or Arbitration or legal proceedings.

36. Arbitration shall be conducted in accordance with the provisions of the Arbitration and Conciliation Act, 1996 (26 of 1996) as amended and any statutory modifications or re-enactment thereof and the rules made there under and for the time being in force shall apply to the arbitration proceeding under this clause, except for cases falling under Sub Clause 31.
37. It is also a term of this Contract that the arbitrator shall adjudicate on only such disputes as are referred to him by the appointing authority and give separate award against each dispute and claim referred to him and in all cases the arbitrator shall give reasons for the award.
38. It is also a term of the Contract that the arbitrator shall be deemed to have entered on the reference on the date he/she issues notice to both the parties calling them to submit their statement of claims and counter statement of claims. The fees, if any, of the arbitrator shall, if required to be paid before the award is made and published, be paid half and half by each of the parties.

CLAUSE 26

Contractor to Indemnify the Employer against Patent Rights

The contractor shall fully indemnify and keep indemnified the Employer against any action, claim or proceeding relating to infringement or use of any patent or design or any alleged patent or design rights and shall pay any royalties which may be payable in respect of any article or part thereof included in the contract. In the event of any claims made or action brought against the Employer in respect of any such matters as aforesaid, the contractor shall be immediately notified thereof and the contractor shall be at liberty, at his own expense, to settle any dispute or to conduct any litigation that may arise there from, provided that the contractor shall not be liable to indemnify the Employer if the infringement of the patent or design or any alleged patent or design right is the direct result of an order passed by the Engineer-in-Charge in this behalf.

CLAUSE 27

Withholding and lien in respect of sum due from Contractor

- (i) Whenever any claim or claims for payment of a sum of money arises out of or under the contract or against the contractor, the Engineer-in-Charge or the Employer shall be entitled to withhold and also have a lien to retain such sum or sums in whole or in part from the security deposit, if any deposited by the contractor and for the purpose aforesaid, the Engineer-in-Charge or the Employer shall be entitled to withhold the security deposit, if any, furnished as the case may be and also have a lien over the same pending finalization

or adjudication of any such claim. In the event of the security being insufficient to cover the claimed amount or amounts or if no security has been taken from the contractor, the Engineer-in-Charge or the Employer shall be entitled to withhold and have a lien to retain to the extent of such claimed amount or amounts referred to above, from any sum or sums found payable or which may at any time thereafter become payable to the contractor under the same contract or any other contract with the Engineer-in-Charge of the Employer or any contracting person through the Engineer-in-Charge pending finalization of adjudication of any such claim.

It is an agreed term of the contract that the sum of money or moneys so withheld or retained under the lien referred to above by the Engineer-in-Charge or Employer will be kept withheld or retained as such by the Engineer-in-Charge or Employer till the claim arising out of or under the contract is determined by the arbitrator (if the contract is governed by the arbitration clause) by the competent court, as the case may be and that the contractor will have no claim for interest or damages whatsoever on any account in respect of such withholding or retention under the lien referred to above and duly notified as such to the contractor. For the purpose of this clause, where the contractor is a partnership firm or a limited company, the Engineer-in-Charge or the Employer shall be entitled to withhold and also have a lien to retain towards such claimed amount or amounts in whole or in part from any sum found payable to any partner/limited company as the case may be, whether in his individual capacity or otherwise.

- (ii) The Employer shall have the right to cause an audit and technical examination of the works and the final bills of the contractor including all supporting vouchers, abstract, etc., to be made after payment of the final bill and if as a result of such audit and technical examination any sum is found to have been overpaid in respect of any work done by the contractor under the contract or any work claimed to have been done by him under the contract and found not to have been executed, the contractor shall be liable to refund the amount of over-payment and it shall be lawful for Employer to recover the same from him in the manner prescribed in sub-clause (i) of this clause or in any other manner legally permissible; and if it is found that the contractor was paid less than what was due to him under the contract in respect of any work executed by him under it, the amount of such under payment shall be duly paid by Employer to the contractor, without any interest thereon whatsoever.

Provided that the Employer shall not be entitled to recover any sum overpaid, nor the contractor shall be entitled to payment of any sum paid short where such payment has been agreed upon between the Engineer-in-Charge on the one hand and the contractor on the other under any term of the contract permitting payment for work after assessment by the Engineer-in-Charge.

CLAUSE 28

Lien in respect of claims in other Contracts

Any sum of money due and payable to the contractor (including the security deposit returnable to him) under the contract may be withheld or retained by way of lien by the Engineer-in-Charge or the Employer or any other contracting person or persons through Engineer-in-Charge against any claim of the Engineer-in-Charge or Employer or such other person or persons in respect of payment of a sum of money arising out of or under any other contract made by the contractor with the Engineer-in-Charge or the Employer or with such other person or persons. It is an agreed term of the contract that the sum of money so withheld or retained under this clause by the Engineer-in-Charge or the Employer will be kept withheld or retained as such by the Engineer-in-Charge or the Employer or till his claim arising out of the same contract or any other contract is either mutually settled or determined by the arbitration clause or by the competent court, as the case may be and that the contractor shall have no claim for interest or damages whatsoever on this account or on any other ground in respect of any sum of money withheld or retained under this clause and duly notified as such to the contractor.

CLAUSE 29

Employment of coal mining or controlled area labour not Permissible

The contractor shall not employ coal mining or controlled area labour falling under any category whatsoever on or in connection with the work or recruit labour from area within a radius of 32 km (20 miles) of the controlled area. Subject as above the contractor shall employ imported labour only i.e., deposit imported labour or labour imported by contractors from area, from which import is permitted.

Where ceiling price for imported labour has been fixed by State or Regional Labour Committees not more than that ceiling price shall be paid to the labour by the contractor.

The contractor shall immediately remove any labourer who may be pointed out by the Engineer in-Charge as being a coal mining or controlled area labourer. Failure to do so shall render the contractor liable to pay to the Employer a sum calculated at the rate of Rs.100/- per day per labourer. The certificate of the Engineer-in-Charge about the number of coal mining or controlled area labourer and the number of days for which they worked shall be final and binding upon all parties to this contract.

It is declared and agreed between the parties that the aforesaid stipulation in this clause is one in which the public are interested within the meaning of the exception in Section 74 of Indian Contract Act, 1872.

Explanation: - Controlled Area means the following areas:

Districts of Dhanbad, Hazaribagh, Jamtara - a Sub-Division under Santhal Pargana

Commissionary, Districts of Bankura, Birbhum, Burdwan, District of Bilaspur.

Any other area which may be declared a Controlled Area by or with the approval of the Central Government.

CLAUSE 30

Water for Works

The contractor(s) shall make his/their own arrangements for water required for the work and nothing extra will be paid for the same. This will be subject to the following conditions.

- (i) That the water used by the contractor(s) shall be fit for construction purposes to the satisfaction of the Engineer-in-Charge.
- (ii) The Engineer-in-Charge shall make alternative arrangements for supply of water at the risk and cost of contractor(s) if the arrangements made by the contractor(s) for procurement of water are in the opinion of the Engineer-in- Charge, unsatisfactory.

CLAUSE 31

Alternate water Arrangements:

The contractor shall be allowed to construct temporary wells in Employer land for taking water for construction purposes only after he has got permission of the Engineer-in- charge in writing. No charges shall be recovered from the contractor on this account, but the contractor shall be required to provide necessary safety arrangements to avoid any accidents or damage to adjacent buildings, roads and service lines. He shall be responsible for any accidents or damage caused due to construction and subsequent maintenance of the wells and shall restore the ground to its original condition after the wells are dismantled on completion of the work.

CLAUSE 32

Employment of Technical Staff and employees

Contractors Superintendence, Supervision, Technical Staff & Employees

- (i) The contractor shall provide all necessary superintendence during execution of the work and all along thereafter as may be necessary for proper fulfilling of the obligations under the contract.

The contractor shall immediately after receiving letter of acceptance of the tender and before commencement of the work, intimate in writing to the Engineer-in-Charge, the name(s), qualifications, experience, age, address(s) and other particulars along with certificates, of the principal technical representative to be in charge of the work and other technical representative(s) who will be supervising the work. Minimum requirement of

such technical representative(s) and their qualifications and experience shall not be lower than specified in Schedule 'F'. Even if the contractor (or partner(s) in case of firm/company) is himself / herself an Engineer, it is necessary on the part of the contractor to Employ principal technical representative / technical representative (s) as per stipulation in Schedule 'F'

The Engineer-in-Charge shall within 3 days of receipt of such communication intimate in writing his approval or otherwise of such a representative(s) to the contractor. Any such approval may at any time be withdrawn and in case of such withdrawal, the contractor shall appoint another such representative(s) according to the provisions of this clause. Decision of the tender accepting authority shall be final and binding on the contractor in this respect. Such a principal technical representative and other technical representative(s) shall be appointed by the contractor soon after receipt of the approval from Engineer-in-charge and shall be available at site before start of work.

All the provisions applicable to the principal technical representative under the Clause will also be applicable to other technical representative(s). The principal technical representative and other technical representative(s) shall be present at the site of work for supervision at all times when any construction activity is in progress and also present himself/themselves, as required, to the Engineer-in-Charge and/or his designated representative to take instructions. Instructions given to the principal technical representative or other technical representative(s) shall be deemed to have the same force as if these have been given to the contractor. The principal technical representative and other technical representative(s) shall be actually available at site fully during all stages of execution of work, during recording/checking/test checking of measurements of works and whenever so required by the Engineer-in-Charge and shall also note down instructions conveyed by the Engineer-in-Charge or his designated representative(s) in the site order book and shall affix his/their signature in token of noting down the instructions and in token of acceptance of measurements/checked measurements/ test checked measurements. The representative(s) shall not look after any other work. Substitutes, duly approved by Engineer-in-Charge of the work in similar manner as aforesaid shall be provided in event of absence of any of the representative(s) by more than two days.

If the Engineer-in-Charge, whose decision in this respect is final and binding on the contractor, is convinced that no such technical representative(s) is/are effectively appointed or is/are effectively attending or fulfilling the provision of this clause, a recovery (non-refundable) shall be effected from the contractor as specified in Schedule 'F' and the decision of the Engineer-In-Charge as recorded in the site order book and measurement recorded checked/test checked in Measurement Books shall be final and binding on the contractor. Further if the contractor fails to appoint suitable technical Principal technical representative and/or other technical representative(s) and if such appointed persons are not effectively present or are absent by more than two days without duly approved substitute or do not discharge their responsibilities satisfactorily, the Engineer-in-Charge shall have full powers to suspend the execution of the work until such

date as suitable other technical representative(s) is/are appointed and the contractor shall be held responsible for the delay so caused to the work. The contractor shall submit a certificate of employment of the technical representative(s) (in the form of copy of Form-16 or CPF deduction issued to the Engineers employed by him) along with every on-account bill/ final bill and shall produce evidence if at any time so required by the Engineer-in-Charge.

- (ii) The contractor shall provide and employ on the site only such technical assistants as are skilled and experienced in their respective fields and such foremen and supervisory staff as are competent to give proper supervision to the work.

The contractor shall provide and employ skilled, semiskilled and unskilled labour as is necessary for proper and timely execution of the work.

The Engineer-in-Charge shall be at liberty to object to and require the contractor to remove from the works any person who in his opinion misconducts himself, or is incompetent or negligent in the performance of his duties or whose employment is otherwise considered by the Engineer-in-Charge to be undesirable. Such person shall not be employed again at works site without the written permission of the Engineer-in-Charge and the persons so removed shall be replaced as soon as possible by competent substitutes.

- (iii) For works with estimated cost more than Rs. 10 Crores and stipulated time period more than 6 months:

The contractor shall ensure that at least one deployed technical representative shall be trained in courses related to CPWD specifications, labour laws, safety rules etc. of duration not less than 5 working days either through National CPWD Academy (NCA) or National Institute of Construction Management and Research (NICMAR) or CIDC or any other similar reputed and recognized Institute managed or certified by State/ Central Government. The training cost and other cost related to training shall be borne by the contractor. The contractor shall ensure that at least one technical representative is trained within six months of start of work. The time period of six months can be relaxed by the Engineer-in-Charge depending upon the frequency of training course organized by NCA.

If the contractor fails to ensure that at least one technical representative is trained in the above-mentioned course till completion of work or one year from start of work, whichever is earlier, then a non-refundable recovery of **Rs. 50,000/-** shall be made from the bill of the contractor. Decision of Engineer-in-Charge in this regard shall be final and binding on the contractor.

- (iv) Building Information Model (BIM) professional shall be deployed by the contractor for the work wherever BIM is applicable.

The BIM professional shall be available exclusively for this work from stipulated date of

start and till actual date of completion of project.

The recovery shall be made from bill of contractor in case of non-deployment of BIM Professional as mentioned in **Schedule 'F'** of Tender Document without giving any notice in writing.

The decision of Engineer-in-Charge in this respect shall be final and binding on the Contractor.

CLAUSE 33

Levy/Taxes/Cesses payable by Contractor

A GST: The contractor will examine the various provisions of The Central Goods and Services Tax Act, 2017(CGST)/ Integrated Goods and Services Tax Act, 2017(IGST)/ Union Territory. Goods and Services Tax Act, 2017(UTGST)/ respective state's State Goods and Services Tax Act (SGST) also, as notified by Central/State Govt. and as amended from time to time and applicable taxes before bidding.

- a) The contractor will ensure that full benefit of Input Tax Credit (ITC) likely to be availed by them is duly considered while quoting rates, and the price to be quoted shall be inclusive of all taxes, duties, levies and cesses etc. but exclusive of GST, as applicable, which shall be paid extra in addition to Contract Price.
- b) The successful contractor who is liable to be registered under GST shall submit GST Registration Number with a copy of the GST Registration Certificate to the Employer immediately after the award of contract.
- c) The Contractor, so appointed, shall duly comply with the provisions of GST law, as applicable to him and to the contract, and issue E-Invoice/ Tax Invoice with declaration for non-applicability of E-Invoice, Debit/Credit Notes, Refund voucher as per law failing which no payment shall be released.
- d) The Contractor shall be responsible for deposition of applicable GST to the concerned authority, file all statutory returns well within the time to transfer the ITC to the employer and to comply with the applicable provisions of law.
 - i) Payment of GST on Running Account (RA) bill shall be made on regular basis taking the GST Compliance behavior of the contractor (i.e. regular/timely deposit of taxes to the credit of the Government and filing of regular and proper return under GST, transfer of ITC etc.).
 - ii) Before the payment of GST in next RA bill (unless both the invoices are raised in the same tax period, for which time limit for deposit of taxes have not elapsed and/or time limit for filing statutory return is not over) employer shall get itself assured of reflection of GST, as to all paid/reimbursed against earlier RA bills, in its GSTR-2B or through documentary evidence for deposit of GST.
 - iii) In case of final bill, GST amount shall only be paid/reimbursed by the Employer after thorough review of its GSTR-2B as to all GST paid in the contract and transfer of ITC.

- iv) Regarding Payment/reimbursement of GST against Invoices/RA bills of the contractor, the decision of the Engineer-In-Charge shall be final and binding on the contractor.
- v) Any increase or decrease of GST on outward supply/services, under the contract, shall be on employer's account.

- B. The contract price shall be inclusive of all (save the specific exclusions as enumerated in this contract) taxes, fees, license charges, deposits, duties, tolls, royalty, Octroi, commissions or other charges which may be levied on the input goods/services consumed and output goods/services delivered in the course of Works Contract excluding GST on output supply of Goods/Services.**
- C. The contractor shall deposit royalty and obtain necessary permit for supply of the red bajri, stone, kankar etc. from local authorities. If pursuant to or under any law, notification or order any royalty, cess or the like becomes payable by the Employer and was not paid any time by the contractor to the State Government, Local Authorities in respect of any material used by the contractor in the works, then in such a case, it shall be lawful to the Employer and it will have the right and be entitled to recover the amount so paid in the circumstances, as aforesaid, from dues of the contractor.
- D. All Payment under the contract shall be subject to applicable taxes deductible at source i.e. Income Tax, GST, Labour Welfare Cess and any other taxes.
- E. In case of contract awarded on behalf of employer, the GST Registration Number /TAN etc. of the employer will be used for Invoicing/Tax Deducted at Sources as per respective Act.

CLAUSE 33 A

Royalty Payable on minor minerals – Employer to be indemnified.

Where the State Government has enacted any legislation to the effect that the user of minor minerals such as red bajri, stone, kankar etc., will be responsible to ensure that for the minerals used in the work, the quarry operator extracting the minerals has paid royalty as due to the State Government, before any Bill is paid to the Contractor, the Contractor should furnish a declaration to the Employer confirming that royalty as due has been paid to the State Government for the minerals used in the works. The Contractor should also indemnify the Employer against any legal action that may be instituted against the Employer by the State Government consequent on violation of the provisions in the relevant Act by the Contractor. A copy of the format for the Indemnity Bond is at **Annexure 'G'**.

CLAUSE 33 B

Price adjustment due to change in the rate of Royalty for earth, sand, stone dust and stone etc.

If, after the stipulated date of opening of tenders (including extension, if any, to the said date), any law, regulation, ordinance, order or bye-law having the force of law is enacted, promulgated, abrogated or changed (which shall be deemed to include any change in interpretation or application by the competent authorities) that subsequently affects the rate of royalty payable by the contractor, the contract price shall be correspondingly increased or decreased after the contractor makes payment of royalty at such revised rates.

CLAUSE 34

Conditions for reimbursement of levy/taxes if levied after receipt of Tenders

- (i) All tendered rates shall be inclusive of all taxes, levies or cesses payable under respective statutes excluding of GST on output supply of Goods / Services. However, if any further tax or levy or cess is imposed by Statute or there is any variation on output supply of Goods / Services, after the last stipulated date for the receipt of tender including extension if any and the contractor thereupon necessarily and properly pays such taxes/ levies/ cesses, the Contractor shall be reimbursed the amount so paid, provided such payment, if any, is not, in the opinion of Engineer-in-Charge (whose decision shall be final and binding on the contractor) attributable to delay in execution of work within the control of Contractor.
- (ii) The Contractor shall keep necessary books of accounts and other documents for the purpose of this condition as may be necessary and shall allow inspection of the same by a duly authorized representative of the Employer and/or the Engineer-in-Charge and further shall furnish such other information/document as the Engineer-in-Charge may require from time to time.
- (iii) The Contractor shall, within a period of 30 days of the imposition of any such further tax or levy or cess, give a written notice thereof to the Engineer-in-Charge that the same is given pursuant to this condition, together with all necessary information relating thereto.

CLAUSE 35

Termination of Contract on death of Contractor

Without prejudice to any of the rights or remedies under this contract, if the contractor dies, the Engineer-in-charge on behalf of the Employer shall have the option of terminating the contract without compensation to the contractor.

CLAUSE 36

If a relative working in the Employer's Organization, then the Contractor not allowed to tender

The Contractor shall not be permitted to tender for works in the Employer's RITES concerned SBU Unit (responsible for award and execution of Contracts) in which his near relative is posted

as Associated Finance Officer between the grades of AGM(F) and JM (F) or as an officer in any capacity between the grades of the GGM/GM and Engineer or equivalent (both inclusive). He shall also intimate the names of persons who are working with him in any capacity or are subsequently employed by him and who are near relatives to any Gazetted Officer or equivalent officers in the Employer's organization. Any breach of this condition by the **Contractor would render him liable to be removed from the approved list of Contractors if any of this Employer.**

Any breach of this condition by the contractor shall be treated as concealing of information and appropriate action will be taken for banning of business dealing with such contractors as per clause 11.2 B of Section 1 (NIT and ITT).

NOTE: By the term "near relatives" is meant wife, husband, parents and grandparents, children and grandchildren, brothers and sisters, uncles, aunts and cousins and their corresponding in-laws.

CLAUSE 37

No Gazetted Engineer to work as Contractor within one year of retirement

No engineer of gazetted rank or other gazetted officer employed in engineering or administrative duties in an engineering department of the Employer shall work as a contractor or employee of a contractor for a period of one year after his retirement from the Employer's service without the previous permission of the Employer in writing. This contract is liable to be cancelled if either the contractor or any of his employees is found at any time to be such a person who had not obtained the permission of the Employer as aforesaid, before submission of the tender or engagement in the contractor's service, as the case may be.

CLAUSE 38

Theoretical consumption of Material

- (i) After completion of the work and also at any intermediate stage in the event of non-reconciliation of materials issued theoretical quantity of materials used in the work shall be calculated on the basis and method given hereunder: -
 - (a) Quantity of cement & bitumen shall be calculated on the basis of quantity of cement & bitumen required for different items of work as shown in the Schedule of Rates mentioned in Schedule 'F'. In case any item is executed for which standard constants for the consumption of cement or bitumen are not available in the above-mentioned schedule/statement or cannot be derived from the same shall be calculated on the basis of standard formula to be laid down by the Engineer-in-Charge.
 - (b) Theoretical quantity of steel reinforcement or structural steel sections shall be taken as the quantity required as per design or as authorized by Engineer-in-Charge, including authorized lappings, chairs etc. plus 3% wastage due to cutting into pieces, such theoretical quantity being determined and compared with the actual, each diameter wise, section wise and category wise separately.

- (c) Theoretical quantity of G.I. & C.I. or other pipes, conduits, wires and cables, pig lead and G.I./M.S. sheets shall be taken as quantity actually required and measured plus 5% for wastage due to cutting into pieces (except in the case of G.I./M.S. sheets it shall be 10%), such determination & comparison being made diameter wise & category wise.
- (d) For any other material as per actual requirements.

Over the theoretical quantities of materials so computed a variation shall be allowed as specified in Schedule 'F' For non-scheduled items, the decision of the Engineer-in-Charge regarding theoretical quantities of materials which should have been actually used, shall be final and binding on the contractor.

- (ii) The said action under this clause is without prejudice to the right of the Employer to take action against the contractor under any other conditions of contract for not doing the work according to the prescribed specifications.

CLAUSE 39

Compensation during warlike situations

The work (whether fully constructed or not) and all materials, machines, tools and plants, scaffolding, temporary buildings and other things connected therewith shall be at the risk of the contractor until the work has been delivered to the Engineer-in-Charge and a certificate from him to that effect obtained. In the event of the work or any materials properly brought to the site for incorporation in the work being damaged or destroyed in consequence of hostilities or warlike operation, the contractor shall when ordered (in writing) by the Engineer-in-Charge to remove any debris from the site, collect and properly stack or remove in store all serviceable materials salvaged from the damaged work and shall be paid at the contract rates in accordance with the provision of this agreement for the work of clearing the site of debris, stacking or removal of serviceable material and for reconstruction of all works ordered by the Engineer-in-Charge, such payments being in addition to compensation upto the value of the work originally executed before being damaged or destroyed and not paid for. In case of works damaged or destroyed but not already measured and paid for, the compensation shall be **assessed by the Engineer-in-Charge**. The contractor shall be paid for the damages/destruction suffered and for restoring the material at the rate based on analysis of rates tendered for in accordance with the provision of the contract. The certificate of the Engineer-in-Charge regarding the quality and quantity of materials and the purpose for which they were collected shall be final and binding on all parties to this contract.

Provided always that no compensation shall be payable for any loss in consequence of hostilities or warlike operations (a) unless the contractor had taken all such precautions against air raid as are deemed necessary by the A.R.P. (Air Raid Precaution) Officers or the Engineer-in-Charge (b) for any material etc. not on the site of the work or for any tools, plant, machinery, scaffolding, temporary building and other things not intended for the work. In the event of the

contractor having to carry out reconstruction as aforesaid, he shall be allowed such extension of time for its completion as is considered reasonable **by the Engineer-in-Charge**.

CLAUSE 40

Apprentices Act provisions to be complied with

The contractor shall comply with the provisions of the Apprentices Act, 1961 and the rules and orders issued there under from time to time. If he fails to do so, his failure will be a breach of the contract and the Engineer-in-Charge may, in his discretion, cancel the contract. The contractor shall also be liable for any pecuniary liability arising on account of any violation by him of the provisions of the said Act.

CLAUSE 41

Release of Security deposit after labour clearance

The Security Deposit of the work shall be refunded if no labour complaint has been received from the labour officer till the due date of its payment. If a labour complaint is received during this period, the Engineer-in-Charge shall, after issue of notice in this regard to the contractor, deduct the amount required to settle the complaint from his security deposit and refund the balance amount.

In case, if part completion certificate of work is recorded then security deposit shall be released only after recording final completion certificate of the work and after completion of defect liability period whichever is later or specified otherwise in the contract.

ADDITIONAL CLAUSES OF CONTRACT

CLAUSE 41 A

Release of Security Deposit after clearance by State Government Authority for Mineral Extraction.

Security Deposit of the work shall not be refunded till the Contractor produces a Clearance Certificate from the State Government Department responsible for collection of royalty on extraction of minor minerals like red bajri, stone, kankar etc., in confirmation that royalty as due has been paid for the minor minerals used in the works. As soon as the Provisional Certificate of Physical Completion is issued, unless the Contractor has already obtained the Royalty Clearance Certificate and submitted to the Engineer-in-Charge, he shall apply for the Clearance Certificate to the Department concerned under intimation to the Engineer-in-Charge. The Engineer-in-Charge, on receipt of the said communication, shall write to the Department concerned to intimate if any complaint is pending against the contractor on non-payment of royalty charges for the minerals used on the works. If no complaint is pending on record till after 3 months after completion of the works and/or no communication is received from the

Department concerned to the effect till Four months after the date of reference to the Department, it will be deemed to have received the Clearance Certificate and the Security Deposit will be released if otherwise due.

CLAUSE 42

GENERAL OBLIGATIONS OF THE CONTRACTOR

42.1 Contractor's General Responsibilities

- a) The Contractor shall, subject to the provisions of the Contract and with due care and diligence, execute and maintain the Works and provide all labour, including the supervision thereof, materials, constructional plant and all other things, whether of a temporary or permanent nature, required in and for such execution and maintenance, so far as the necessity for providing the same is specified in or is reasonably to be inferred from the Contract.
- b) The Contractor shall take full responsibility for the adequacy, stability and safety of all site operations and methods of construction, provided that the Contractor shall not be responsible, except as may be expressly provided in the Contract, for the design or specification of the permanent works, or for the design or specification of any temporary works prepared by the Employer.
- c) The Contractor shall carry out all the works strictly in accordance with the drawings, details and instructions of the Engineer-in-Charge. In the opinion of the Engineer-in-Charge if any changes are to be made, the same shall be intimated in writing to the Contractor and the Contractor shall carry out the same. The Engineer-in-Charge's decision in this regard shall be final and not open to arbitration.

42.2 Giving Notices and Payment of Fees

This Contract shall, in all respects, be construed and operated as an Indian Contract and shall be subject to Indian laws in force from time to time. The Contractor should conform to all laws of the land and the regulations and bye-laws of any local authority and of any water or lighting companies with whose systems the structure is proposed to be connected. The Contractor shall give all notices required by the said acts, regulations or bye-laws and pay all fees in connection therewith. The Contractor shall protect and indemnify the Employer against all claims or liabilities arising from his actions in violation of such laws, ordinances, regulations and bye-laws by him. The Engineer will repay or allow to the Contractor all such sums as the Engineer-in-Charge shall certify to have been properly payable by the Employer and paid by the Contractor in respect of such fees.

42.3 Compliance with Statutes, Regulations, etc. and Change of Law

The Contractor shall conform in all respects with the provisions of any such Statute, Ordinance or Law as aforesaid and the regulations or byelaws of any local or other duly constituted authority, which may be applicable to the Works and with such rules and regulations of public bodies and companies as aforesaid and shall keep the Employer indemnified against all penalties and liabilities of every kind for breach of any such Statute, Ordinance or Law, Regulation or Bye-Laws.

The contract price shall be adjusted to take account of any increase or decrease in cost resulting from a change in the applicable laws in India (including the introduction of new laws and the repeal or modification of existing laws) or in the judicial or official governmental interpretation of such laws, made after the last date of submission of tender, which affect the contract price and the performance of obligations under the contract.

42.4 Fossils etc.

All fossils, coins, articles of value or antiquity and structures and other remains or things of geological or archeological interest discovered on the site of the Works shall, as between the Employer and the Contractor, be deemed to be the absolute property of the Employer. The Contractor shall take reasonable precautions to prevent his workmen or any other persons from removing or damaging any such article or thing and shall immediately upon discovery thereof, and, before removal, acquaint the Engineer of such discovery and carry out, at the expense of the Employer, the Engineer-in-Charge's orders as to the disposal of the same.

42.5 Opportunities for other Contractors

The Contractor shall, in accordance with the requirements of the Engineer, afford all reasonable opportunities for carrying out their work to any other Contractors employed by the Employer and their workmen and to the workmen of the Employer and of any other duly constituted authorities, who may be employed in the execution on or near the site of any work not included in the Contract or of any Contract which the Employer may enter into in connection with or ancillary to the Works.

42.6 Interference with Traffic and adjoining properties

All operations necessary for the execution of the Works shall, so far as compliance with the requirements of the Contract permits, be carried on so as not to interfere unnecessarily or improperly with the convenience of the public, or the access to use and occupation of public or private roads and footpaths to or of properties whether in the possession of the Employer or of any other person. The Contractor shall save harmless and indemnify the Employer in respect of all claims, proceedings, damages, costs, charges and expenses whatsoever arising out of, or in relation to, any such matters in so far as the Contractor is responsible, therefore.

42.7 Highways Traffic

42.7.1 Extraordinary Traffic

The Contractor shall use every reasonable means to prevent any of the highways or bridges communicating with or on the routes to the Site from being damaged or injured by any traffic of the Contractor or any of his Sub-Contractors and, in particular, shall select routes, choose and use vehicles and restrict and distribute loads so that any such extraordinary traffic as will inevitably arise from the moving of plant and materials from and to the Site shall be limited, as far as reasonably possible, and so that no unnecessary damage or injury may be occasioned to such highways and bridges.

42.7.2 Special Loads

Should it be found necessary for the Contractor to move one or more loads of constructional plants, machinery or pre-constructed units or parts of units of work over part of a highway or bridge, the moving whereof is likely to damage any highway or bridge unless special protection or strengthening is carried out, then the Contractor shall, before moving the load on to such a highway or bridge, give notice to the Engineer of the weight and other particulars of the load to be moved and his proposals for protecting or strengthening the said highway or bridge. Thereafter, within fourteen days, the Engineer shall direct, in writing the course of action for the Contractor. The Contractor shall be paid for the cost of the necessary modifications only if the specific items requiring such modification are provided in the Bills of quantities.

42.7.3 Settlement of Extraordinary Traffic Claims

If during the execution of the works or at any time thereafter, the Contractor shall receive any claim arising out of the execution of the Works, in respect of damage or injury to highways or bridges, he shall immediately report the same to the Engineer.

The Contractor shall negotiate the settlement of and pay all sums due in respect of such claim and shall indemnify the Employer in respect thereof and in respect of all claims, proceedings, damages, costs, charges and expenses in relation thereto.

42.8 Watching and Lighting

The Contractor shall in connection with the Works, provide and maintain at his own cost all lights, guards, fencing and watching when and where necessary or as required by the Engineer or by any duly constituted authority, for the protection of the Works, or for the safety and convenience of the public or others.

42.9 Way leaves etc.

The Contractor shall bear all costs and charges for special or temporary way leaves required by him in connection with access to the Site. The Contractor shall also provide at his own cost any additional accommodation outside the Site required by him for the purposes of the Works.

42.10 Site Office for the Employer

The Contractor will provide free of cost temporary furnished accommodation as detailed in **Schedule 'F'** for the Engineer and his staff, at the site of work. On expiry of the maintenance period, the Site Office shall be dismantled and site cleared unless the Employer directs otherwise. The furniture will be returned to the Contractor at whatever conditions they are an expiry of maintenance period.

42.11 Electricity Supply required at the works

42.11.1 Arrangement

The Contractor shall make his own arrangement for electricity required for the work and nothing extra will be paid for the same.

OR

42.11.1A A Electric power, if available, may be made available for the work by the Employer at one or more points within the site or near the site on specified terms and conditions. The Contractor shall arrange at his own cost the necessary cabling/wiring, Switch Board and the switch gear etc. and shall be responsible for their safe maintenance. Cost of Electricity supplied shall be fixed by the Employer from time to time.

(**Clause 42.11.1A** will be applicably only if so stipulated in **Schedule 'F'**).

42.11.2 Conditions

- a) Distribution arrangements shall be done by the Contractor at his cost as per approved layout. He shall provide required clearances for overhead lines to facilitate easy movement of machinery. These overhead lines shall be shifted and rerouted at the Contractor's cost during execution of work if the same are found to obstruct any other work of any agency working at site or requires to be shifted due to unforeseen reasons.
- b) On completion of the Work, the Contractor shall, to the satisfaction of the Engineer, remove all wiring installed by him and make good, any disturbance or damage done.
- c) The Contractor shall employ a certified and licenced Electrician for carrying out this work.

42.12 Land for Contractor's Offices, Godown, Workshop

- a) Subject to the availability, the Employer may allot at his own discretion and convenience land for the construction of the Contractor's site office, godowns, workshop and assembly yard near the site. Allotment of such land shall not confer any tenancy rights

to the Contractor. The Contractor shall construct and maintain the same at his cost. All these temporary works shall be well ventilated, lighted and provided with water, electricity and sanitary arrangement to the approval of the Engineer.

- b) The Contractor shall remove immediately on completion of the work such buildings and make good, to the satisfaction of the Engineer, all the damages sustained.
- c) The Contractor shall not use the land for any purpose other than that for or in connection with the Contract.
- d) In case the Contractor requires any land additional to what is made available/allotted to him, the Contractor shall make arrangements for the same at his own cost.

42.13 Land for Labour Accommodation

The Employer shall not provide any land for setting up of the Labour Camp and the Contractor shall make his own arrangements.

OR

42.13.A Subject to availability, on specific request of the Contractor, space for labour accommodation at site may be provided free of rent to the extent possible. However, it will be the responsibility of the Contractor to look after all the requirements of the labour camp such as water, sanitation, health, lighting, general living condition and welfare. In case the Contractor requires any land additional to what is made available/allotted to him, the Contractor shall make arrangements for the same at his own cost.

(Clause 42.13.A will be applicably only if so stipulated in **Schedule 'F'**).

42.14 Excavated Material: DELETED

42.15 Production of vouchers etc. by the Contractor

- i) The Contractor shall, whenever required produce or cause to be produced for examination by the Engineer-in-Charge any quotation, invoice, cost or other account, book of accounts, voucher, receipt, letter, memorandum, paper of writing or any copy of or extract from any such document and also furnish information and returns verified in such manner as may be required in any way relating to the execution of this Contract or relevant for verifying or ascertaining cost of execution of this Contract and the decision of the Engineer-in-Charge on the question of relevancy of any documents, information or return being final and binding on the parties. The Contractor shall similarly produce vouchers etc. if required to prove to the Engineer-in-Charge that the materials supplied by him, are in accordance with the specifications laid down in the Contract.
- ii) If any portion of the work in a Contract is being carried out by a Sub Contractor or any subsidiary or allied firm or Company in terms of permission granted under **Clause 21**

of Contract, the Engineer-in-Charge shall have power to secure the book of such sub-Contractor or any subsidiary or allied firm or company through the Contractor and such book shall be open to the Engineer-in-Charge's inspection.

- iii) The obligations imposed by sub clause (i) and (ii) above are without prejudice to the obligations of the Contractor under any statute, rules or orders binding on the Contractor.

42.16 Law governing the Contract

The Contract shall be governed by the law for the time being in force in the Republic of India.

42.17 Court Jurisdiction

Unless otherwise specified in 'Schedule 'F', the Courts of Guwahati India shall have the exclusive jurisdiction to entertain any suit, application or other proceedings in respect of anything arising under this Contract or any award or awards made by the Arbitrator where Arbitration & Conciliation Act 1996 is applicable shall be filed in the concerned Courts in Guwahati only.

CLAUSE 43

INSURANCES TO BE TAKEN BY THE CONTRACTOR & EMPLOYER TO BE INDEMNIFIED

43.1 Insurance of Works etc.

43.1.1 The Contractor shall effect Contractor's all risk insurance policy (CAR policy) in the joint names of the Employer and the Contractor, the name of the former being placed first in the policy, covering the following:

- a) The works at their Replacement Cost.
- b) All plants and equipments and other things brought to the site by the Contractor shall be insured for a sufficient amount not less than Rs.5 lacs or 1% of the Tendered Value, whichever is more.
- c) Employer's building rented to the Contractor if the building or part thereof is used by the Contractor for the purpose of storing or using materials of combustible nature, on which the decision of the Engineer-in-Charge shall be final and binding.

43.1.2 The insurance shall be against all losses or damages from whatever causes, other than excepted risks, as defined in Clause 2 of Conditions of Contract, for which the

Contractor is responsible under the Contract. The insurance cover shall be for the period of Contract stipulated in **Clause 43.7** below and also for the period of maintenance, for loss or damage arising from a cause prior to commencement of the period of maintenance, and for any loss or damage, occasioned by the Contractor in the course of any operations carried out for the purpose of complying with his obligations during Maintenance Period under Clause 17 of Clauses of Contract.

- 43.1.3** Such insurance shall be effected with an insurer and with terms approved by the Employer. The Contractor shall, whenever required, produce the policy or policies and the receipts for payment of the current premiums.

43.2 Third Party Insurance

- 43.2.1** Before commencing the execution of the Works from the date specified in Clause 5 above the Contractor shall insure against the liability for any material or physical damage, loss injury or death which may occur to any property or life including that of the Employer or to any person, including any employee of the Employer, by or arising out of the execution of the Works or in the carrying out of the Contract. The sum insured will be for an amount specified in **Schedule 'F'**. After each occurrence, Contractor shall pay additional premium necessary to make insurance valid for four occurrences always. The Insurance Policy should cover this amount at all times till issue of Completion Certificate by the Engineer-in-Charge.

- 43.2.2** Such insurance shall be effected with an insurer and its terms approved by the Employer. The Contractor shall, whenever required, produce before Engineer the policy or policies of insurance and the receipts of payment of the current premiums.

43.3 Workmen's Insurance

The Employer shall not be liable for any payment in respect of any damages or compensation payable according to law in respect or in consequence of any accident or injury or loss of life to any workman or other person in the employment of the Contractor or any sub-Contractor, except an accident or injury resulting from any act or default of the Employer, his agents or servants. The Contractor shall insure against such liability with an insurer approved by the Employer for a sum as per established norms during the entire period stipulated in Clause 43.7 below.

43.4 Recovery from the Contractor

Without prejudice to the other rights of the Employer against the Contractor in respect of such default, the Employer shall be entitled to deduct from any sums payable to the Contractor the amount of any damages, compensation costs, charges and other expenses paid by the Employer, and which are payable by the Contractor under this clause.

43.5 Shortfall in payment to Contractor by the Insurance Company

In all cases where the Contractor has taken any insurances, he shall not be entitled to reimbursement by the Employer of any shortfall or deficiency in the amount payable by the Contractor towards settlement of claims and that paid by the insurer in settlement of same claim.

43.6 Insurance by Sub-Contractors

Without prejudice to his liability under this clause the Contractor shall also cause all Sub-Contractors to effect, for their respective portions of the works, similar policies of insurance in accordance with the provisions of this clause and shall produce or cause to produce to the Employer such policies. The Contractor shall not permit a Sub- Contractor to commence work at the site unless the said insurance Policies are submitted. In the event of failure of the Sub-Contractor to take out such a policy of Insurance before commencing the works at the site, the Contractor shall be responsible for any claim or damage attributable to the said Sub-Contractor.

43.7 Period of Policies

All the insurance covers mentioned above shall be kept alive during the period of the Contract from the date specified for start of the work in terms of **Clause 5** above until the Contractor obtains a Maintenance Certificate from the Employer as explained in **Clause 17**.

43.8 Remedy on Contractor's Failure to Insure

If the Contractor shall fail to effect and keep in force the insurances referred to above, or any other insurance which he may be required to effect under the terms of the Contract, then and in any such case the Employer on advice of the Engineer-in-Charge may effect and keep in force any such insurance and pay such premium or premiums as may be necessary for that purpose and from time to time deduct the amount so paid by the Employer as aforesaid from any moneys due or which may become due to the Contractor, or recover the same as debt due from the Contractor.

43.9 Damage to Persons and Property - Employer to be indemnified

The Contractor shall indemnify the Employer against all losses and claims in respect of injuries or damages to any person or material or physical damage to any property whatsoever which may arise out of or in consequence of the execution and maintenance of the works and against all claims, proceedings, damages, costs, charges and expenses whatsoever in respect of or in relation thereto, except any compensation or damages for or with respect to:

- a) The permanent use or occupation of land by the works or any part thereof.
- b) The right of the Employer to execute the works or any part thereof on, over, under, in

or through any land.

- c) Injuries or damage to persons or property caused by Excepted risks or resulting from any act or neglect of the Employer, his agents, servants or other Contractors, not being employed by the Contractor or for or in respect of any claims, proceedings, damages, costs, charges and expenses in respect thereof or in relation thereto. Where the injury or damage was contributed to by the Contractor, his servants or agents, such part of the compensation as may be just and equitable having regard to the extent of the responsibility of the Employer, his servant or agent or other Contractors, for the damage or injury.

A copy of the Format for Indemnity Bond is at Annexure 'H'.

CLAUSE 44

SAFETY AND SECURITY

44.1 Codes etc. to be complied with

The Contractor shall ensure and arrange at his cost fire and the safety provisions, as per safety code of RITES. (*Refer Section No. 9 of General Conditions of Contract*), Indian Standards Institution, safety manuals of the Employer, if any, and such provisions as are locally in force from time to time for all labour, directly or indirectly employed in the works for performance of this Contract. The Contractor will indemnify the Employer from any consequence arising due to Contractor's failure in respect of safety provisions.

Following Codes may be referred to in this connection:

IS 5916	Safety code for construction involving use of hot bituminous materials.	IS 7293
	Safety code for working with construction machinery	
IS 7969	Safety code for handling and storage of building materials.	
IS 8989	Safety code for erection of concrete framed structures.	
IS 13415	Protective barriers in and around buildings - Code of Safety	
IS 13416	Preventive measures against hazards at work places- Recommendations (Parts - 1 to 5)	

44.2 First Aid & Industrial Injuries

- 44.2.1 First aid facilities at easily accessible place shall be provided by the Contractor as per provisions of Labour Act or Rules of the Authority controlling the area where work is carried out.
- 44.2.2 The Contractor shall make arrangements with hospitals for ambulance service and for treatment of industrial injuries to meet eventualities leading to the need for such facilities. The Engineer shall be informed of their telephone numbers and addresses of

the Hospitals.

44.2.3 Details of all critical industrial injuries shall be reported promptly to the Engineer.

44.2.4 Report shall cover type, nature, cause, physician's report and action for prevention of those types again.

44.3 General Safety Rules

44.3.1 Smoking within plant, restricted areas, closed areas, near storage place of lubricant oil and fuel etc. is strictly prohibited.

44.3.2 The Contractor shall erect and maintain barricades required in connection with his operation to guard or protect

- (a) Excavation
- (b) Hoisting/lifting
- (c) Slab openings
- (d) Hazardous areas
- (e) Employer's existing property likely to be subjected to damage by the Contractor's operations
- (f) Unloading spots

44.4 Accidents - Precautions at Worksite

No materials on the sites shall be so stacked or placed as to cause danger or inconveniences to any person or to the public. The Contractor shall provide all necessary fencing and lights to protect the public from accidents and shall be bound to bear expenses of defense of every suit, action or other proceedings at law, that may be brought by any person, for injury sustained, owing to neglect of the above precautions and to pay any damages and costs which may be awarded in any such suit, action or proceeding, to any such person or which may, with the consent of the Contractor be paid to compromise any claim by any such person.

44.5 Electrical Equipments – Precautions

All temporary and permanent electrical installations, power distribution and supply required for execution of Work shall be carried out conforming to existing industrial and domestic safety rules and regulations. Important specific points to be noted are as under,

- (i) Meter room and main switches should be freely accessible at all times and fully protected against all weather conditions.
- (ii) Power distribution system shall be identifiable with display marking on switches.
- (iii) All power distribution shall be carried out with coated, adequately insulated and

of appropriate current/load rating cables. It shall be securely routed for this purpose. No loose, naked, hanging wires shall be permitted.

- (iv) Over load protection devices shall be installed whenever and wherever heavy current/load consuming construction plant or machinery susceptible to hazard is in use and as directed by the Engineer-in-charge.
- (v) Metallic plugs and sockets shall be used in field work. Switch board shall be in close proximity so as to have quick control over the supply.
- (vi) Proper and adequate earthing connection should be provided for all installations, plant and machinery and distribution system.
- (vii) Hand lamps and inspection lamps shall be adequately insulated and guarded with wire mesh and should have proper plugs for use.
- (viii) Security and illuminatory light shall be secured firmly and protected to withstand all weather conditions.

44.6 Maintenance of Safety Devices

All scaffolds, ladders and other safety devices mentioned or described herein shall be maintained in a safe condition and no scaffold, ladder or equipment shall be altered or removed while it is in use. Adequate washing and maintenance facilities shall be provided at or near places at work.

44.7 Personal Safety

All necessary personal safety equipment as considered adequate by the Engineer shall be available for use of persons employed on the Site and maintained in a condition suitable for immediate use and the Contractor shall take adequate steps to ensure proper use of equipment by those concerned.

- a) Workers employed on mixing asphaltic materials, cement, and lime mortars/ concrete shall be provided with protective footwear and protective gloves.
- b) Those engaged in handling any materials which are injurious to eyes shall be provided with protective goggles.
- c) Workers employed on erection works, etc. shall be provided with helmets, safety belts etc.
- d) Workers employed on concrete finishing, welding, painting and other works above 2 metres height shall be provided with a suitable safety belt, as per Factory Rules of the locality.

44.8 Storing Fuel, Oil and Lubricant

The Contractor shall take approval from the Safety Officer of the Employer for storing the lubricants, oil and fuel at site for running the machinery required for the construction.

44.9 Fire Extinguishing

Suitable, sufficient number of fire extinguishers for all types of fire, shall be provided at work site. In addition, sufficient number of fire buckets filled with water and sand shall also be provided. The fire fighting equipment as outlined above shall be dispersed in a suitable and purposeful manner.

44.10 Fire Precautions

The Contractor shall comply with regulations of the controlling authority in force at the site of the works relating to the precautions to be taken against fire hazards.

44.11 Protection arrangements at work site

Adequate protection against any form of damage or deterioration shall be provided for in all sections of the works. This shall include protective tapes, casings, guard rails and the likes, which shall be provided as necessary. Particular care shall be taken to protect finished surfaces during the execution of adjacent in-situ work. The Contractor shall carryout all steps necessary and comply with the directions and instructions of the Engineer to his satisfaction.

44.12 Safety Arrangements for labour

The Contractor shall, at his own expense, arrange for the safety provisions as given above and as required by the Engineer, in respect of all labour directly or indirectly employed for performance of the work and shall provide all facilities in connection therewith. In case the Contractor fails to make arrangements to provide necessary facilities as aforesaid, the Engineer shall be entitled to do so and recover the cost thereof, from the Contractor.

44.13 Safety Manual

The Contractor shall submit a Safety Manual indicating the safety measures proposed to be adopted in light of above provisions, for approval of the Engineer-in- Charge.

44.14 Accidents – Reporting

The Contractor shall, within twenty-four (24) hours of the occurrence of any accident

on, or about the Site, or in connection with the execution of the Works, report such accident to the Engineer and to the appropriate authority wherever such report is required by law. The Contractor will indemnify the Employer from all accident cases.

44.15 Security Measures

The Contractor shall be responsible at his cost for security of Works for the duration of the Contract and shall provide and maintain continuously adequate security personnel to fulfil these obligations. The requirements of security measures shall include, but not limited to, maintenance of Law and order at site, provision of all lighting, guard, flagmen, and other measures necessary for protection of Works within the camps and elsewhere at site, for all materials delivered to the site and all persons employed in connection with the Works continuously throughout working and non-working periods including nights, Sundays, holidays, for the duration of the Contract. At work sites in close proximity of traffic corridors where public are likely to come close to the work area, suitable fencing as directed by the Engineer should be provided.

CLAUSE 45: DELETED

CLAUSE 46

ECOLOGICAL BALANCE

The Contractor shall maintain ecological balance by preventing deforestation, water pollution and defacing of natural landscape. The Contractor shall so conduct his construction operations as to prevent any avoidable destruction, scarring or defacing of natural surrounding in the vicinity of work. In respect of ecological balance, the Contractor shall observe the following instructions for which no extra payments will be made.

- a) Where destruction, scarring, damage or defacing may occur as a result of operations relating to Permanent or Temporary Works, the same shall be repaired, replanted or otherwise corrected at Contractor's expense. All work areas shall be smoothened and graded in a manner to conform to natural appearance of the landscape as directed by the Engineer.
- b) All trees and shrubbery, which are not specifically required to be cleared or removed for construction purposes, shall be preserved and shall be protected from any damage that may be caused by Contractor's construction operations and equipment or by their Employees/Workers. The removal of trees or shrubs will be permitted only after prior approval of the Engineer. Special care shall be exercised where trees or shrubs are exposed to injuries by construction equipment, blasting, excavating, dumping, chemical damage or other operation and the Contractor shall adequately protect such trees by use of protective barriers or other methods approved by the Engineer. Trees shall not be used for anchorage.

The Contractor shall be responsible for injuries to trees and shrubs caused by his operations and Employees/Workers. The terms "injury" shall include, without limitation, bruising, scarring, tearing and breaking of roots, trunks or branches. All injured trees and shrubs shall be restored as nearly practicable, without delay, to their original condition at Contractor's expense.

- c) Where trees have to be necessarily cut for progressing temporary or permanent works, the Contractor shall arrange for compensatory afforestation as may be required by Environmental Rules and Regulations.
- d) In the conduct of construction activities and operation of equipments, the Contractor shall utilize such practicable methods and devices as are reasonably available to control, prevent and otherwise minimize air/ noise pollution.
- e) Excessive emission of dust into the atmosphere will not be permitted during manufacture, handling and storage of concrete aggregates/fly ash/ earth/building materials and the Contractor shall use such methods and equipment as are necessary for collection and disposal or prevention of dust during these operations. The Contractor's method of storing and handling cement shall also include means of eliminating atmospheric discharge of dust. Equipment and vehicles that give objectionable emission of exhaust gases shall not be operated. Burning of materials resulting from cleaning of trees branches, combustible construction materials and rubbish may be permitted only when atmospheric conditions for burning are considered favourable.
- f) Special care must be exercised in ensuring that the labour housed in labour camp within the work site area do not indulge in any activity like drinking alcohol, taking drugs etc and other activities that may affect the ecological balance such as cutting of shrubs for fuel, creating open air nuisance etc.

CLAUSE 47

EXECUTION OF WORKS

47.1 Mobilisation

Period of Mobilisation shall be 15 days counting from the stipulated date of start of work as mentioned in Letter of Acceptance by the Engineer-in-Charge. The Contractor shall carry out following activities within this period stated. He shall submit to the Engineer within 4 days of stipulated date of start, the proposed layout of locating offices, stores, godowns, yards, water, electric network etc. for approval of the Engineer.

Minimum following activities shall be completed within the mobilization period of 15 days or such extended period as approved by the Engineer.

- i) Site office of the Contractor
- ii) Line out including establishing of grid line levels and its approval from the Engineer.
- iii) Submitting list of proposed specialized sub-Contractors as may be required for approval of the Engineer.
- iv) Tapping electric and water connections
- v) One cement godown and steel yard
- vi) Obtaining insurance policies as per the Contract
- vii) Obtaining labour licences, as required
- viii) Obtaining approval of local authorities and complying with any statutory requirements prior to actual start of work.
- ix) Establishing water and electric network within site.
- x) Submitting construction programmed as detailed in Clause 5 of Contract and its approval by the Engineer-in-Charge.

47.2 Setting out of Works

The Contractor shall be responsible for the true and proper setting-out of the Works in relating to original points, lines and levels of reference given by the Engineer in writing and for the correctness, subject as above mentioned, of the position, levels, dimension and alignment of all parts of the Works and for the provision of all necessary instruments, appliances and labour in connection therewith. If, at any time during the progress of the Works, any error shall appear or arise in the position, levels, dimensions or alignment of any part of the Works, the Contractor, on being required so to do by the Engineer shall, at his own cost, rectify such error to the satisfaction of the Engineer. The checking of any setting-out or of any line or level by the Engineer shall not in any way relieve the Contractor of his responsibility for the correctness thereof and the Contractor shall carefully protect and preserve all bench-marks, sight-rails, pegs and other things used in setting-out the works. The Contractor shall use latest equipments like Total Station/Theodolite and Auto level etc for setting out the works.

47.3 Boreholes and Explanatory Excavation: DELETED

47.4 Temporary Works

47.4.1 The Contractor is entirely responsible for the design, construction, maintenance and removal of all temporary works employed in carrying out the Contract. Within a reasonable time (and in any case not less than fifteen days) before he intends to commence construction of any temporary works, the Contractor shall submit full particulars including drawings of the same, for the approval of the Engineer. The Engineer's approval will in no way relieve the Contractor of his responsibility for the safety of the Works, operators, adjoining property, structures or services and compliance with appropriate regulations and codes of practice. Documents for Temporary works supporting adjoining buildings, property and public utilities and roads shall also be submitted to the appropriate authority for their approval if requested/

required.

- 47.4.2 The temporary works shall be designed and constructed in such a manner as to enable the permanent structures to be built around them without detriment to their effectiveness and due allowance will be deemed to have been made for all necessary adjustments thereto to enable the Works to proceed.
- 47.4.3 Timber shoring, boards, struts or similar items shall not be left in position upon completion of the Works without the written consent of the Engineer.
- 47.4.4 All services or utilities on or adjoining the site which are required to be maintained operational shall be protected from movement, subsidence or damage from any cause whatsoever by adequate temporary props, struts, shores and protective screens to the approval of the Engineer and the agent of the service or utility.
- 47.4.5 The Contractor shall make safe and reinstate all areas affected by temporary works.
- 47.4.6 The Contractor shall use properly designed and manufactured steel staging platforms for carrying out work above 3.0m height. All required staging for supporting, centring, shuttering of beams, slab, masonry work, etc. shall be carried out strictly as per the Supplier's instructions or approved arrangement. It is to be noted that designing of such work shall be carried out by the Contractor and shall be submitted for approval of the Engineer. No work above 3.0 m shall be permitted without compliance of this condition.

47.5 Plant, Temporary Works & Materials - Exclusive use

All constructional plants, temporary works and materials provided by the Contractor shall, when brought on to the site, be deemed to be exclusively intended for the execution of the Works and the Contractor shall not remove the same or any part thereof, except for the purpose of moving it from one part of the site to another, without the consent, in writing, of the Engineer, which shall not be unreasonably withheld.

47.6 Use of Site only for Works

The Contractor shall not use any portion of the Site for purpose not connected with the works without the prior written approval of the Engineer. He shall maintain permanent and Site access roads free of spillage and shall not interfere with the flow of traffic. Also same shall apply to terraces and other developed areas.

47.7 Name Board at Site

The Contractor shall prepare and display name board at site as per design approved by the Engineer. It shall have

- Name of Project
- Name of Employer
- Name of Consulting Architect (if any)
- Name of Project Management Consultant (if any)
- Name of Contractor

47.8 Site Drainage/Cleaning/Nuisance

- 47.8.1 All water which may accumulate on the Site during the progress of the works or in trenches and excavation, shall be removed from the site to the satisfaction of the Engineer at the Contractor's cost.
- 47.8.2 The Site shall be maintained free from rubbish. Proper stacking of scaffolding materials, shuttering material, bricks/brick bats, steel pieces, etc. needed for work on day to day basis shall be organized, Heaps in unplanned manner and disorderly fashion shall not be permitted. The Engineer's decision in this matter shall be final.
- 47.8.3 The Contractor shall not, at any time, cause or permit any nuisance on the site or do anything which shall cause unnecessary disturbance or inconvenience to the Employer, tenants or occupants of other properties near the site and to the public in general.

47.9 Disposal of Rubbish

- i) The Contractor shall cart away from site and deposit where directed by the Engineer all refuse, etc. arising from the Works both as it accumulates and at completion of the Works at the direction of the Engineer.
- ii) It is the responsibility of the Contractor to obtain a certificate from the local authorities concerned to the effect that all rubbish arising out of Contractor's activities at the construction site or any other offsite activities borrow pits and/or disposal area (s) has been properly disposed off.

47.10 Shift Working

The Contractor shall be allowed to work in three shifts with prior approval of the Engineer.

47.11 Urgent Repairs

If, by reason of any accident or failure, or other event occurring to, in, or in connection with the Works or any part thereof, either during the execution of the Works or during the period of Maintenance, any remedial or other Work or repair shall, in the opinion of the Engineer, be urgently necessary for the safety of the Works and the Contractor is unable or unwilling at once

to do such work or repair, the Employer may employ and pay other persons to carry out such work or repair as the Engineer may consider necessary. If the Work or repair so done by the Employer is work which, in the opinion of the Engineer, the Contractor was liable to do at his own expense under the Contract, all expenses properly incurred by the Employer in so doing shall be recoverable from the Contractor by the Employer or may be deducted by the Employer from any moneys due or which may become due to the Contractor. Provided always that the Engineer, shall, as soon after the occurrence of any such emergency as may be reasonably practicable, notify the Contractor thereof in writing.

47.12 Contractor to search

The Contractor shall, if required by the Engineer in writing, search under the directions of the Engineer for the cause of any defect, imperfection or fault appearing during the progress of the Works or within the Period of Maintenance (Defect Liability period). If such defect, imperfection or fault shall be one for which the Contractor is liable, the cost of the work carried out in searching as aforesaid shall be borne by the Contractor and he shall in such case repair, rectify and make good such defect, imperfection or fault at his own expense in accordance with the provisions of Clause 17 of Clauses of Contract.

CLAUSE 48

PROGRAMME AND PERFORMANCE

48.1 Review of Programme

If at any time it should appear to the Engineer that the actual progress of work does not conform to the approved programme the Contractor shall produce at the request of the Engineer, a revised programme showing modifications to the approved programme, necessary to ensure completion of the work within the time for completion stipulated in the Contract. The submission to and approval by the Engineer-in-Charge of such programme or the furnishing of such particulars shall not relieve the Contractor of any of his duties or responsibilities or obligations under the Contract. The Engineer-in-Charge shall have full power and authority during progress of work, to issue such instructions as may be necessary for the proper and adequate execution and maintenance of the Work. The Contractor shall carry out and be bound by the same.

48.2 Progress Reports

The Contractor shall submit periodic Progress Reports including coloured progress photographs as per the frequency and in the Proforma laid down by the Engineer indicating the details of actual Progress vis-à-vis planned progress of various components of work, for the period and up to end of the period, slippage if any, action proposed to be taken to pull back the arrears, deployment of machinery and plant, statement showing extra and substituted items submitted by Contractor and of any other item stipulated by the Engineer.

48.3 Maintenance of Records & Registers

The Contractor shall maintain Registers and Records in the format laid down by the Engineer. These Registers and Records shall be open for inspection by the Employer/ Engineer-in-charge and Engineer at all times. An important Register to be maintained is the Hindrance Register which will be an essential document for dealing with applications for extension of time by the Contractor.

48.4 Site Co-ordination Meetings

The Contractor will attend the Site Co-ordination Meetings with Engineer, the Consultant and other Contractors as fixed by the Engineer from time to time to discuss all issues relating to the works in general and progress and quality in particular. All costs incidental to such interaction shall be borne by the Contractor.

48.5 Site Order Book

A site order book shall be kept at the site of the work. As far as possible, all orders regarding the works are to be entered in this book. All entries therein shall be signed by the Engineer or his representative and the Contractor or his representative. In important cases, the Engineer-in-charge will countersign the entries, which have been made. The site order book shall not be removed from the work site except with written permission of the Engineer and the Contractor or his representative shall be bound to take note of all instructions and directions meant for the Contractor as entered in the site order book without having to be called on separately to note them. The Contractor shall ensure compliance of the noting in site order book within three days of the noting. In case of failure to do so, the corresponding work shall be stopped and work already done shall not be accepted.

48.6 Progress Photographs

During the Construction stage the Contractor shall take adequate number of coloured photographs showing the progress of various stages of the Work as directed by the Engineer. Size of photographs will be 125 mm x 250 mm. Photographs shall be supplied with negatives to the Engineer. Each photograph shall be attested with date of photograph, location of work and brief description of what it shows. These photographs shall be from locations as fixed by the Engineer at start of work.

48.7 Poor Performance and Consequences Thereof

The Engineer-in-Charge, without prejudice to his other rights against the contractor, may declare the contractor to be a poor performer after giving him written notice of seven days in any of the following cases:

- i) The contractor abandons the work for 30 days or more without valid reasons.
- ii) Progress on the work is behind schedule by 25% or more of the stipulated contract period for reasons solely attributable to the contractor.
- iii) Completion of work is delayed by 25% or more of the stipulated contract period for reasons solely attributable to the contractor.
- iv) The contractor repeatedly (twice or more) makes claims on frivolous grounds or goes to court or seeks arbitration for such claims.
- v) The contractor repeatedly (twice or more) seeks extra-contractual financial support from RITES for completing the work.
- vi) If there are more than two instances of financial failure of contractor in making timely payments to his labour or sub contractors or to suppliers of materials.

Upon declaration of the contractor as 'poor performer', his name will be placed in RITES' 'Negative List of Contractors' and he will not be eligible for a period of upto two years, to participate in tenders of other works which may be invited by RITES.

CLAUSE 49

INSPECTION OF WORK

49.1 Site Access

- 49.1.1 It is for the Contractor to provide safe and guarded access for the Engineer as detailed in the safety code, For inspection of works, the Contractor shall arrange at his cost required ladders, scaffolding materials, steel measuring tapes, plumb levels, theodolite or any other instrument required by the Engineer for his use at site.
- 49.1.2 The Engineer and any person authorized by him shall at all times have access to the Works and to all workshops and places where work is being prepared or from where materials, manufactured articles or machinery are being obtained for the Works and the Contractor shall afford every facility for and every assistance in or in obtaining the right to such access.

49.2 Examination of Work before covering up

No works shall be covered up or put out of view without the approval of the Engineer and the Contractor shall afford full opportunity for the Engineer to examine and measure any Work which is about to be covered up or put out of view and to examine foundations before Permanent Work is placed thereon. The Contractor shall give due notice to the Engineer whenever any such Works or foundations are ready or about to be ready for examination and the Engineer shall, without unreasonable delay, unless he considers it unnecessary and advises the Contractor accordingly, attend for the purpose of examining and measuring such works or of examining such foundations.

49.3 Uncovering and Making Opening

The Contractor shall uncover any part or parts of the Works or make openings in or through the same as the Engineer may from time to time direct and shall reinstate and make good such part or parts to the satisfaction of the Engineer. If any such part or parts have been covered up or put out of view after compliance with the requirement in Clause 49.2 above and are found to be executed in accordance with the Contract, the expenses of uncovering, making openings in or through, reinstating and making good the same shall be borne by the Employer, but in any other case all costs shall be borne by the Contractor.

CLAUSE 50: DELETED

CLAUSE 51

SAMPLE FLOOR

The Contractor shall construct one sample floor/ unit in each type of flats/ non- residential building and get the same approved from Employer including approval of fittings, fixture, finishing items and colour scheme. Employer shall give approval for sample floor/ unit within fifteen days from the date of its completion in all respects including rectification of defects, if any.

- (a) All fitments and fixtures used, such as electrical fittings, water supply items, sanitary fittings, woodwork and joinery will be as per Contract agreement.
- (b) The sample floor/unit shall act as a guideline for the construction and finishes of all other flats/floors/units.
- (c) The sample floor/unit shall be completed in all respects including all fittings and fixtures within a period as specified in 'Schedule F' from the date of start, failing which a penalty of Rs. **2000/-** (Rupees Two Thousand Only) per day of delay shall be levied.

Provided always that provision of Clause 51 shall be applicable only when so provided in **Schedule 'F'**.

ANNEXURE A
(REFER CLAUSE 1 (i))

Form of Performance Security Bank Guarantee Bond

In consideration of the Employer having agreed under the terms and conditions of contract made vide his Letter of Acceptance (LOA) No dated between (the Employer) represented by RITES Ltd for and on behalf of the Employer as an Agent/Power of Attorney Holder and (hereinafter called "the said Contractor) for the work.....(hereinafter called the said Agreement") the Contractor having agreed to production of a irrevocable Bank Guarantee for Rs. - - - - - (Rupees - - - - - Only) as a Security/Guarantee for compliance of his obligations in accordance with the terms and conditions in the said Agreement,

1. We (indicate the name of the Bank) (hereinafter referred to as " the Bank" hereby undertake to pay to the RITES LTD acting for and on behalf of the Employer as an Agent/Power of Attorney Holder, an amount not exceeding Rs..... (Rupees.....only) on demand by RITES Ltd for and on behalf of the Employer as an Agent/Power of Attorney Holder.
2. We ----- (indicate the name of the Bank) do hereby undertake to pay the amounts due and payable under this Guarantee without any demur, merely on a demand from by RITES Ltd for and on behalf of the Employer as an Agent/Power of Attorney Holder stating that the amount claimed is required to meet the recoveries due or likely to be due from the said Contractor. Any such demand made on the Bank shall be conclusive as regards the amount due and payable by the Bank under this Guarantee. However, our liability under this Guarantee shall be restricted to an amount not exceeding Rs.----- (Rupees -----

3. We, the said Bank further under take to pay to the Employer represented by RITES Ltd. for and on behalf of the Employer as an Agent/Power of Attorney Holder any money so demanded notwithstanding any dispute or disputes raised by the Contractor in any suit or proceeding pending before any court or Tribunal relating there to, our liabilities under this present being absolute and unequivocal. The payment so made by us under this Bond shall be a valid discharge of our liability for payment thereunder and the Contractor shall have no claim against us for making such payment
4. We - - - - - (Indicate the name of the Bank) further agree that the Guarantee herein contained shall remain in full force and effect during the period that would be taken for the

performance of the said Agreement and that it shall continue to be enforceable till all dues of the Employer represented by RITES Ltd for and on behalf of the Employer as an Agent/ Power of Attorney Holder under or by virtue of the said Agreement have been fully paid and its claims satisfied or discharged or till Engineer-in-Charge on behalf of the Employer represented by RITES Ltd for and on behalf of the Employer as an Agent/Power of Attorney Holder certified that the terms and conditions of the said Agreement have been fully and properly carried out by the said Contractor and accordingly discharges this Guarantee.

5. We ----- (indicate the name of the Bank) further agree with the Employer represented by RITES Ltd for and on behalf of the Employer as an Agent/Power of Attorney Holder that the Employer shall have the fullest liberty without our consent and without affecting in any manner our obligations hereunder to vary any of the terms and conditions of the said Agreement or to extend time of performance by the said Contractor from time to time or to postpone for any time or from time to time any of the powers exercisable by the Employer against the said Contractor (s) and to forbear or enforce any of the terms and conditions relating to the said Agreement and we shall not be relieved from our liability by reason of any such variation, or extension being granted to the said Contractor or for any forbearance, act of omission on the part of the Employer or any indulgence by the Employer to the said Contractor or by any such matter or thing whatsoever which under the law relating to sureties would, but for this provision, have effect of so relieving us.
6. This Guarantee will not be discharged due to the change in the constitution of the Bank or the Contractor.
7. This Guarantee will neither be cancelled nor revoked by the bank without the written authorization of the beneficiary (RITES Ltd.). For this purpose the beneficiary would inform the Bank of their authorized signatories together with the specimen signatures.
8. This Guarantee shall be valid up to -----unless extended on demand by the Employer represented by RITES Ltd for and on behalf of the Employer as an Agent/Power of Attorney Holder. Notwithstanding anything mentioned above, our liability against this guarantee is restricted to Rs... ----- - (Rupees_____Only) and unless a claim in writing is lodged with us within six months of the date of expiry or the extended date of expiry of this Guarantee, all our liabilities under this Guarantee shall stand discharged.

Dated the day of for (Indicate the name of bank)

Note: To be put in sealed cover by Bank and addressed to the concerned officer of RITES Ltd.

ANNEXURE A-1

(Refer Clause 1A)

Form of Bank Guarantee for replacing Security Deposit from cash to B.G.

In consideration of the Employer having agreed under the terms and conditions of contract made vide his Letter of Acceptance (LOA) No-----dated-----between ---- -----(the Employer) represented by RITES Ltd for and on behalf of the Employer as an Agent/Power of Attorney Holder and ----- (hereinafter called "the said Contractor) for the work (hereinafter called the said Agreement") the Employer having agreed to release a part of Security Deposit against production of a irrevocable Bank Guarantee by the said contractor for Rs. ----- (Rupees ----- Only) as a Security/Guarantee for compliance of his obligations in accordance with the terms and conditions in the said Agreement,

1. We ----- (indicate the name of the Bank) (hereinafter referred to as " the Bank" hereby undertake to pay to the RITES LTD acting for and on behalf of the Employer as an Agent/Power of Attorney Holder, an amount not exceeding Rs. - (Rupees only) on demand by RITES Ltd for and on behalf of the Employer as an Agent/Power of Attorney Holder.
2. We ----- (indicate the name of the Bank) do hereby undertake to pay the amounts due and payable under this Guarantee without any demur, merely on a demand from by RITES Ltd for and on behalf of the Employer as an Agent/Power of Attorney Holder stating that the amount claimed is required to meet the recoveries due or likely to be due from the said Contractor. Any such demand made on the Bank shall be conclusive as regards the amount due and payable by the Bank under this Guarantee. However, our liability under this Guarantee shall be restricted to an amount not exceeding Rs.----- (Rupees -----only).
3. We, the said Bank further under take to pay to the Employer represented by RITES Ltd. for and on behalf of the Employer as an Agent/Power of Attorney Holder any money so demanded not withstanding any dispute or disputes raised by the Contractor in any suit or proceeding pending before any court or Tribunal relating there to, our liabilities under this present being absolute and unequivocal. The payment so made by us under this Bond shall be a valid discharge of our liability for payment thereunder and the Contractor shall have no claim against us for making such payment.
4. We ----- (Indicate the name of the Bank) further agree that the Guarantee herein contained shall remain in full force and effect during the period that would be taken for the performance of the said Agreement and that it shall continue to be enforceable till all dues of the Employer

represented by RITES Ltd for and on behalf of the Employer as an Agent/Power of Attorney Holder under or by virtue of the said Agreement have been fully paid and its claims satisfied or discharged or till Engineer-in-Charge on behalf of the Employer represented by RITES Ltd for and on behalf of the Employer as an Agent/Power of Attorney Holder certified that the terms and conditions of the said Agreement have been fully and properly carried out by the said Contractor and accordingly discharges this Guarantee.

5. We----- (indicate the name of the Bank) further agree with the Employer presented by RITES Ltd for and on behalf of the Employer as an Agent/Power of Attorney Holder that the Employer shall have the fullest liberty without our consent and without affecting in any manner our obligations hereunder to vary any of the terms and conditions of the said Agreement or to extend time of performance by the said Contractor from time to time or to postpone for any time or from time to time any of the powers exercisable by the Employer against the said Contractor (s) and to forbear or enforce any of the terms and conditions relating to the said Agreement and we shall not be relieved from our liability by reason of any such variation, or extension being granted to the said Contractor or for any forbearance, act of omission on the part of the Employer or any indulgence by the Employer to the said Contractor or by any such matter or thing whatsoever which under the law relating to sureties would, but for this provision, have effect of so relieving us. This Guarantee will not be discharged due to the change in the constitution of the Bank or the Contractor.
6. This Guarantee will not be discharged due to the change in the constitution of the Bank or the Contractor.
7. This Guarantee will neither be cancelled nor revoked by the bank without the written authorization of the beneficiary (RITES Ltd.). For this purpose the beneficiary would inform the Bank of their authorized signatories together with the specimen signatures.
8. This Guarantee shall be valid up to -----unless extended on demand by the Employer represented by RITES Ltd for and on behalf of the Employer as an Agent/Power of Attorney Holder. Notwithstanding anything mentioned above, our liability against this guarantee is restricted to Rs... (Rupees Only) and unless a claim in writing is lodged with us within six months of the date of expiry or the extended date of expiry of this Guarantee, all our liabilities under this Guarantee shall stand discharged.

Dated the day of for (Indicate the name of bank)

Note: To be put in sealed cover by Bank and addressed to the concerned officer of RITES Ltd.

ANNEXURE A-2

(Refer Clause 2)

Form of Bank Guarantee for releasing the withheld amount for non-achievement of milestones

In consideration of the Employer having agreed under the terms and conditions of contract made vide his Letter of Acceptance (LOA) No dated-----between ----- -(the Employer) represented by RITES Ltd for and on behalf of the Employer as an Agent/Power of Attorney Holder and ----- (hereinafter called "the said Contractor) for the work (hereinafter called the said Agreement") the Employer having agreed to release the amount withheld from the contractor for not achieving the milestone(s) specified in the said Agreement against production of a irrevocable Bank Guarantee by the said contractor for Rs. -----
--- (Rupees ----- Only) as a Security/Guarantee for compliance of his obligations in accordance with the terms and conditions in the said Agreement,

1. We ------(indicate the name of the Bank) (hereinafter referred to as "the Bank" hereby undertake to pay to the RITES LTD acting for and on behalf of the Employer as an Agent/Power of Attorney Holder, an amount not exceeding Rs . - ------(Rupees only) on demand by RITES Ltd for and on behalf of the Employer as an Agent/Power of Attorney Holder.
2. We ------(indicate the name of the Bank) do hereby undertake to pay the amounts due and payable under this Guarantee without any demur, merely on a demand from by RITES Ltd for and on behalf of the Employer as an Agent/Power of Attorney Holder stating that the amount claimed is required to meet the recoveries due or likely to be due from the said Contractor. Any such demand made on the Bank shall be conclusive as regards the amount due and payable by the Bank under this Guarantee. However, our liability under this Guarantee shall be restricted to an amount not exceeding Rs.----- (Rupees -----
3. We, the said Bank further under take to pay to the Employer represented by RITES Ltd. for and on behalf of the Employer as an Agent/Power of Attorney Holder any money so demanded notwithstanding any dispute or disputes raised by the Contractor in any suit or proceeding pending before any court or Tribunal relating there to, our liabilities under this present being absolute and unequivocal. The payment so made by us under this Bond shall be a valid discharge of our liability for payment thereunder and the Contractor shall have no claim against us for making such payment.

4. We (Indicate the name of the Bank) further agree that the Guarantee herein contained shall remain in full force and effect during the period that would be taken for the performance of the said Agreement and that it shall continue to be enforceable till all dues of the Employer represented by RITES Ltd for and on behalf of the Employer as an Agent/Power of Attorney Holder under or by virtue of the said Agreement have been fully paid and its claims satisfied or discharged or till Engineer-in-Charge on behalf of the Employer represented by RITES Ltd for and on behalf of the Employer as an Agent/Power of Attorney Holder certified that the terms and conditions of the said Agreement have been fully and properly carried out by the said Contractor and accordingly discharges this Guarantee.
5. We ----- (indicate the name of the Bank) further agree with the Employer presented by RITES Ltd for and on behalf of the Employer as an Agent/Power of Attorney Holder that the Employer shall have the fullest liberty without our consent and without affecting in any manner our obligations hereunder to vary any of the terms and conditions of the said Agreement or to extend time of performance by the said Contractor from time to time or to postpone for any time or from time to time any of the powers exercisable by the Employer against the said Contractor (s) and to forbear or enforce any of the terms and conditions relating to the said Agreement and we shall not be relieved from our liability by reason of any such variation, or extension being granted to the said Contractor or for any forbearance, act of omission on the part of the Employer or any indulgence by the Employer to the said Contractor or by any such matter or thing whatsoever which under the law relating to sureties would, but for this provision, have effect of so relieving us.
6. This Guarantee will not be discharged due to the change in the constitution of the Bank or the Contractor
7. This Guarantee will neither be cancelled nor revoked by the bank without the written authorization of the beneficiary (RITES Ltd.). For this purpose the beneficiary would inform the Bank of their authorized signatories together with the specimen signatures.
8. This Guarantee shall be valid up to unless extended on demand by the Employer represented by RITES Ltd for and on behalf of the Employer as an Agent/Power of Attorney Holder. Notwithstanding anything mentioned above, our liability against this guarantee is restricted to Rs... ----- (Rupees ----- Only) and unless a claim in writing is lodged with us within six months of the date of expiry or the extended date of expiry of this Guarantee, all our liabilities under this Guarantee shall stand discharged.

Dated the day of for (Indicate the name of bank)

Note: To be put in sealed cover by Bank and addressed to the concerned officer of RITES Ltd.

ANNEXURE B-1
(REFER CLAUSE 10 A)

FIELD LABORATORY AND FIELD-TESTING

INSTRUMENTS LIST OF EQUIPMENT FOR FIELD TESTING LABORATORY

For Building Works

1. Balances
 - i) 7 kg. to 10 kg. Capacity, Semi-Self indicating type-Accuracy 10 gm.
 - ii) 500 gm. Capacity, Semi-Self indicating type-Accuracy 1 gm.
 - iii) Pan Balance – 5 kg. Capacity, accuracy 10 gm
 - iv) Platform Balance – 100 kg capacity
2. Ovens-Electricity operated, thermostatically controlled upto 110o C- Sensitivity 1oC. 1Nos
3. Sieves: as per IS 460-1962.
 - i) I.S. Sieves – 450 mm internal dia of sizes 100mm, 80mm, 63mm, 50mm, 40mm, 25mm, 20mm, 12.5mm, 10mm, 6.3mm, 4.75mm complete with lid and pan.
 - ii) I.S. Sieves – 200mm internal dia (brass frame) consisting of 2.36mm, 1.18mm, 600 microns, 425 microns, 300 microns, 212 microns, 150 microns, 90 microns, 75 microns with lid and pan.
 - iii) Sieve shaker capable of 200 mm and 300 mm dia sieves, manually operated with timing switch assembly.
- 5) Equipment for slump test-Slump Cone, Steel Plate, tamping rod, steel scale, scoop all as per IS: 7320.
- 6) Dial gauges 25mm travel – 0.01 mm/ division. Least count – 2 nos.
- 7) 100 tonnes compression testing machine, electrical-cum manually operated.
- 8) Graduated measuring cylinders 200ml capacity – 3 Nos.

- 9) Enamel trays (for efflorescence test for bricks)
 - i) 300 mm x 250 mm x 40 mm - 2 nos.
 - ii) Circular plates of 250 mm dia - 4 nos.
- 10) Cube moulds for concrete 150 mm x 150 mm x 150 mm with Base – 24 Nos.
- 11) Plate as per IS: 516 Cube moulds for cement 7.06 cm x 7.06 cm x 7.06 cm - 6 nos.
- 12) Cube Vibrating Table Electricity operated with Table size 50 cm x 50 cm and 150 kg load carrying capacity - 1 No.
- 13) Unit weight measure IS: 1199 - 1 No.
- 14) Pycnometer 1 litre capacity with 6mm dia at Apex - 2 Nos.
½ litre capacity - 2 Nos.
- 15) Vicat Apparatus with Dashpot complete as per IS: 5313 - 1 No.
- 16) Blaines' Air Permeability Apparatus Complete as per S: 5516 - 1 No.
- 17) Tools for non-destructive testing of concrete.
- 18) Soil testing equipment for Sieve Analysis and determination of C.B.R.Value.

FIELD TESTING INSTRUMENTS

- 1) Steel tapes – 5m & 30 m
- 2) Vernier Calipers
- 3) Micrometer Screw 25mm gauge
- 4) A good quality plumb bob
- 5) Spirit minimum 30 cms long with 3 bubbles for horizontal / vertical measurement
- 6) Wire gauge (circular type) disc
- 7) Foot rule
- 8) Long nylon thread
- 9) Rebound hammer for testing concrete
- 10) Dynamic penetrometer

- 11) Magnifying glass
- 12) Screw driver 30 cms long
- 13) Ball pen hammer, 100 gms.
- 14) Plastic bags for taking samples
- 15) Moisture meter for timber
- 16) Earth resistance tests: for Electrical Divisions
- 17) Meggar
- 18) Glass Beaker 100 cc

ANNEXURE B-2
(REFER CLAUSE 10A)

In all contracts where issue of cement and steel by the Employer is not stipulated, the following special conditions shall be applicable.

1) Special conditions for cement

The contractor shall procure 43 grade (conforming to IS 8112 latest version) ordinary Portland cement or 53 grade OPC (conforming to IS 12269, latest version or PPC (Confirming to IS1489 (Part-I)), for RCC work of mix richer than M30) from reputed manufacturers of cement having a valid Certification Marks Licence from Bureau of Indian Standards (BIS). At the time of bid submission, the tenderers may submit a list of names of cement manufacturers along with a copy of their valid Certification Marks Licence from BIS, which they propose to use in the work. The Tender Accepting Authority reserves the right to accept or reject name(s) of cement manufacturer(s) which the tenderer proposes to use in the work if satisfactory evidence in respect of cement manufacturers' licence issued by BIS, is not furnished. No change in the tendered rates will be accepted if the Tender Accepting Authority does not accept the list of cement manufacturers, given by the tenderer, fully or partially. In case the names of brand / cement manufacturers are not submitted at the time of bid submission, the brand / cement manufacturer meeting the criterion mentioned above, shall be got approved from the Engineer-in-Charge before use of cement of such brand / manufacturer on the work.

The supply of cement shall be taken in 50 kg. Bags/bulker imprinted with manufacture's name, ISI marking, brand/quality and quantity of the product. Samples of cement arranged by the contractor shall be taken by the Engineer-in-Charge and got tested in accordance with provisions of relevant BIS codes.

In case the test results indicate that the cement arranged by the contractor does not conform to the relevant BIS codes, the same shall stand rejected, and it shall be removed from the site by the contractor at his own cost within a week's time of written order from the Engineer-in-Charge to do so.

The cement shall be brought at site in bulk supply of approximately 50 tonnes or as decided by the Engineer-in-Charge.

Unless otherwise directed by the Engineer-in-Charge, the cement godown of the capacity to store a minimum of 2000 bags of cement shall be constructed by the contractor at site of work for which no extra payment shall be made.

Double lock provision shall be made to the cement godown. The keys of one lock shall remain

with the Engineer-in-Charge or his authorized representative and the keys of the other lock shall remain with the contractor. The contractor shall be responsible for the watch and ward and safety of the cement godown. The contractor shall facilitate the inspection of the cement godown by the Engineer-in-Charge.

The cement shall be got tested by the Engineer-in-Charge and shall be used on the work only after satisfactory test results have been received. The contractor shall supply free of charge the cement required for testing including its transportation cost to testing laboratories. The cost of **Mandatory and Non-Mandatory** tests shall be borne by the contractor.

The actual issue and consumption of cement on work shall be regulated and proper accounts maintained as provided in **clause 10 A of the contract**. The theoretical consumption of cement shall be worked out as per procedure prescribed in **clause 38 of the contract** and shall be governed by conditions laid therein. In case the cement consumption is less than theoretical consumption including permissible variation, recovery at the rate so prescribed shall be made. In case of excess consumption, no adjustment needs to be made.

The cement brought to the site and the cement remaining unused after completion of the work shall not be removed from site without the written permission of the Engineer- in-Charge.

The damaged cement shall be removed from the site immediately by the contractor on receipt of a notice in writing from the Engineer-in-Charge. If he does not do so within 3 days of receipt of such notice, the Engineer-in-Charge shall get it removed at the cost of the contractor.

Similar conditions for cement of other types like slag cement etc., will apply.

Special Conditions for Steel

1. The Contractor shall procure TMT bars of Fe415 /Fe500 /Fe550 grade (as specified) from any producer approved/empanelled by BIS / RDSO of Indian Railways. TMT bars shall meet the provisions of IS: 1786: 2008 pertaining to Fe 415D or Fe 500D or Fe 550D grade of steel as specified in the tender.
2. The contractor shall have to obtain and furnish manufacturer's test certificates to the Engineer-in-Charge in respect of all supplies of steel brought by him to the site of work.
3. Samples shall also be taken and got tested by the Engineer-in-Charge as per the provisions in this regard in relevant BIS codes and/or the applicable specifications. In case the test results indicate that the steel arranged by the contractor does not conform to the specifications, the same shall stand rejected, and it shall be removed from the site of work by the contractor at his cost within a week time or written orders from the Engineer-in- Charge to do so.
4. The steel reinforcement bars shall be stored by the contractor at site of work in such a way as to prevent distortion and corrosion, and nothing extra shall be paid on this account. Bars of different sizes and lengths shall be stored separately to facilitate easy counting and checking.

5. For checking nominal mass, tensile strength, bend test, re-bend test etc. specimens of sufficient length shall be cut from each size of the bar at random, and at frequency not less than that specified below :

Size of bar	For consignment below 100 tonnes	For consignment above 100 tonnes
Under 10mm dia bars	One sample for each 25 tonnes or part thereof	One sample for each 40 tonnes or part thereof
10 mm to 16 mm dia bars	One sample for each 35 tonnes or part thereof	One sample for each 45 tonnes or part thereof
Over 16 mm dia bars	One sample for each 45 tonnes or part thereof	One sample for each 50 tonnes or part thereof

6. The contractor shall supply free of charge the steel required for testing including its transportation to testing laboratories. The cost of **Mandatory and Non-Mandatory** tests shall be borne by the Contractor.
7. The actual issue and consumption of steel on work shall be regulated and proper accounts maintained as provided in **clause 10 A of the contract**. The theoretical consumption of steel shall be worked out as per procedure prescribed in **clause 38 of the contract** and shall be governed by conditions laid therein. In case the consumption is less than theoretical consumption including permissible variations necessary recovery shall be made. In case of excess consumption, no adjustment needs to be made.
8. The steel brought to site and the steel remaining unused shall not be removed from site without the written permission of the Engineer-in-Charge.

ANNEXURE 'C'
(REFER CLAUSE 10 B ii)

PROFORMA FOR BANK GUARANTEE FOR MOBILIZATION ADVANCE

(On Non-Judicial Stamp Paper of Appropriate Value)

To,

RITES LTD.

.....

1. In consideration of, RITES Ltd. Acting for and on behalf of as Agent/Power of Attorney Holder of..... (hereinafter called "The Employer") (which expression shall unless repugnant to the subject or context include its successors and assigns) having agreed under the terms and conditions of the Contract Agreement No..... dated..... with* in connection with the work of(hereinafter called "the said Contract") to make at the request of the Contractor' a lumpsum advance of Rs..... /- (Rupees only)..... for utilizing it for the purpose of the Contract on his furnishing a Guarantee acceptable to the Employer, we,..... Bank incorporated under..... and having one of our branches at(hereinafter referred to as "the said Bank") do hereby guarantee the due recovery by the Employer of this said advance with interest thereon as provided according to the terms and conditions of the Contract. If the said Contractor fails to utilize the said advance for the purpose of the Contract and / or the said advance together with Interest thereon as aforesaid is not fully recovered by the Employer, we,..... Bank hereby unconditionally and irrevocably undertake to pay to RITES Ltd. on demand and without demur to the extent of the said sum of Rs..... /- (Rupees only), any claim made by the Employer on us for the loss or damage caused to or suffered by the Employer by reason of the Employer not being able to recover in full the said sum of Rs. /- (Rupees _ _ _ only) with interest as aforesaid.
2. We, Bank further agree that the Employer shall be the sole judge of and as to whether the said Contractor has not utilized the said advance or any part thereof for the purpose of the Contract and the extent of loss or damage caused to or suffered by the Employer on account of the said advance together with interest not being recovered in full and the decision of the Employer that the said Contractor has not utilized the said advance or any part thereof for the purpose of the Contract and as to the amount or amounts of loss or damage caused to or suffered by the Employer shall be final and binding on us.
3. We, , the said Bank, further agree that the Guarantee herein contained shall remain in force and effect during the period that would be taken for the performance of the said Contract and till the said advance with interest has been fully recovered and its claims satisfied or discharged and till the Employer certifies that the said advance with interest has been fully recovered from

the said Contractor, and accordingly **shall have no claim under this Guarantee after 30 (thirty) days from the date of satisfactory completion of the said Contract** (as per the mutually agreed Work Schedule) i.e. upto and inclusive of..... (date) unless a notice of the claim under this Guarantee has been served on the Bank before the expiry of the said period i.e.(date) in which case the same shall be enforceable against the Bank notwithstanding the fact, that the same is enforced after the expiry of the said period.

4. The Employer shall have the fullest liberty without affecting in any way the liability of the Bank under this Guarantee or Indemnity, from time to time, to vary any of the terms and conditions of the said Contract or the advance or to extend time of performance by the said Contractor or to postpone for any time and from time to time any of the powers exercisable by it against the said Contractor and either to enforce or forbear from enforcing any of the terms and conditions governing the said Contract or the advance available to the Employer and the said Bank shall not be released from its liability under these presents by any exercise by the Employer of the liberty with reference to the matters aforesaid or by reasons of time being given to the said Contractor or any other forbearance, act or omission on the part of the Employer or any indulgence by the Employer to the said Contractor on any other matter or thing whatsoever which under the law relating to sureties would, but for this provision, have the effect of so releasing the Bank from its such liability.
5. It shall not be necessary for the Employer to proceed against the Contractor before proceeding against the Bank and the Guarantee herein contained shall be enforceable against the Bank notwithstanding any security, which the Employer may have obtained or obtain from the Contractor shall at the time when proceedings are taken against the Bank hereunder, be outstanding or unrealized.
6. We, the said Bank, lastly undertake not to revoke this Guarantee during its currency except with the previous consent of the Employer in writing and agree that any change in the Constitution of the said Contractor or the said Bank shall not discharge our liability hereunder.

If any further extension of this Guarantee is required the same shall be extended to such required periods on receiving instructions from the Contractor M/s on whose behalf this Guarantee is issued.

Notwithstanding anything contained herein before our liability under this Guarantee is restricted to Rs. (Rupees only) together with interest Our undertaking shall commence from the date of execution and shall remain in force upto Dated this day of In presence of

For and on behalf of (the Bank)

Signature

Name.....

Designation Authorization No.

Seal of the Bank

WITNESS

1.....

2.....

The above Guarantee is accepted by the Employer For RITES Ltd. For and on behalf of
..... (the Employer) as Agent/Power of Attorney Holder

NOTES ON PARA 1

***For Proprietary Concerns**

Shri son of resident of carrying on
business under the name and style of At (hereinafter called
"the said Contractor" which expression shall unless the context requires otherwise include his
heirs, executors, administrators and legal representatives).

***For Partnership Concerns**

*M/s a partnership firm with its office (hereinafter
called "the said Contractor" which expression shall unless the context requires otherwise
include their heirs, executors, administrators and legal representatives); the names of their
partners being.

(i) Shri S/o

(ii) Shri S/o

***For Companies**

* M/s a company under the Companies Act, 1956 and having its
registered office at in the State of (hereinafter called
"the said Contractor" which expression shall unless the context requires otherwise include its
administrators, successors and assigns).

ANNEXURE 'D'
(REFER CLAUSE 10 B iii)

**PROFORMA FOR HYPOTHECATION DEED FOR PLANT & MACHINERY
ADVANCE**

(On Non-Judicial Stamp Paper of Appropriate values in the name of the Contractor)

This indenture made the day of BETWEEN* of the one part and RITES LIMITED acting for and on behalf of as Agent/Power of Attorney Holder of (hereinafter called "the Employer" which expression shall unless the context requires otherwise include his successors and assigns) of the other part.

WHEREAS under Provision in Clause 10B (iii) of the Clauses of Contract of Agreement No. dated (hereinafter called the "Contract") between the Employer and the Contractor, the Contractor has applied to the Employer for an advance of Rs. (Rupees only) for Plant & Machinery described in the schedule hereto specifically acquired by the Contractor for the Works and brought to the site.

AND WHEREAS one of the conditions on which the said Advance of Rs. is to be granted by the Employer to the Contractor is that the Contractor shall hypothecate the plant and machinery described in the Schedule hereto in favour of the Employer as security for the due repayment of the said advance along with interest as due.

AND WHEREAS the Contractor has represented that he is the Owner of the Plant and Machinery described in the Schedule hereto and the same is free from encumbrances.

NOW THIS INDENTURE WITNESSETH THAT in pursuance of the said Contract and in consideration of the promises the Contractor doth hereby hypothecate, assign and transfer to the Employer the Plant and Machinery described in the Schedule hereto with the intent that the same shall remain and form security for repayment to the Employer of the said advance of Rs. together with the interest thereon at 10% per annum.

1. The Contractor hereby agrees, declares and covenants with the Employer as follows:-

- a) The Contractor shall repay to the Employer the said Advance of Rs. (Rupees only) together with Interest thereon as aforesaid by and agrees that the said Advance be recovered by the Employer by making deductions in the manner provided in Clause 10B(iv) of the Clauses of Contract.
- b) The Contractor has paid in full the purchase price of the Plant and Machinery described in the Schedule hereto and each and every one of them is the absolute property of the Contractor and that the same have not been sold, pledged, mortgaged or transferred or in any way dealt

with by the Contractor.

- c) So long as any amount remains payable to the Employer by the Contractor in respect of the said Advance of Rs. and interest thereon the Contractor shall not sell, pledge, hypothecate, transfer part with or in any way deal with the Plant and Machinery described in the Schedule hereto.
- d) If the said advance of Rs. along with the interest as due shall not be repaid by the Contractor or recovered in the manner described above by the Employer by (due date) due to any reason whatsoever or the said Contract has been determined earlier or cancelled or if the Contractor shall sell, pledge, mortgage, transfer, par with or in any way deal with the said plant and machinery or any part thereof or the Contractor or any of the partners is adjudged insolvent or the Contractor firm is to be wound up or the Contractor makes any composition or arrangement with its creditors or the Contractor shall commit breach of any of the terms and conditions or covenants as herein contained or if any of the said plant and machinery or any other property whatsoever belonging to the Contractor has been sold or attached for a period of not less than 21 days in execution of the decree of any court for payment of money, the whole of the said advance of Rs. or such part thereof as may have remained unpaid or unrecovered together with interest thereon shall forthwith become due and payable.
- e) The Employer may on the happening of any of the events mentioned in the preceding clause (d) or in the event of the said advance and interest thereon or any part thereof becoming due and payable and has not been paid or recovered or cannot be recovered as provided in the said conditions, seize and take possession of the said plant and machinery (and either remain in possession thereof without removing the same or else may remove the same) and sell the said plant and machinery or any of them either by public auction or private Contract and may out of the sale proceeds retain the balance of the said advance and interest thereon remaining unpaid and unrecovered and all costs, charges and expenses and payments incurred or made in maintaining, defending or protecting the rights of the Employer hereunder and shall pay over the surplus, if any, to the Contractor.
- f) The Contractor shall at all times during the continuance of this security as stipulated in Para 2 below, the Contractor shall at his cost insure and keep insured the plant and machinery described in the Schedule hereto for the full value thereof in the joint names of the Contractor and the Employer with an insurance company to be approved by the Engineer-in-Charge against the risk of loss or damage from whatever cause arising other than the Excepted Risks. During the Continuance of the security the Contractor shall pay all premia and sums of money necessary for keeping such insurance alive and the insurance policy and receipts in original for premia paid shall be deposited with the Engineer-in-Charge. The Contractor shall assign all his rights, title and interest in the policy to the Employer.
- g) The Contractor shall not permit or suffer the said plant and machinery or any par thereof to be destroyed or damaged or used or to be used or to deteriorate in a greater degree than it would deteriorate by reasonable wear and tear thereof in the performance of the Contract.

- h) In the event of any damage or loss happening to the said plant and machinery or any part thereof from whatever cause other than the Excepted risks the Contractor shall forthwith have the same repaired or replaced as the case may be or arrange for payment of the entire amount recovered or to be recovered from the insurance company to the Employer towards the payment of the said advance of Rs.along with interest as due.
2. Upon repayment or recovery in full of the amount secured on account of this hypothecation deed the said plant and machinery secured hereunder shall stand released from hypothecation but this is without prejudice to the right of the Employer under any other conditions of the Contract.

SCHEDULE ABOVE REFERRED TO

Sl. No.	Particulars Advance of Plant & Machinery	New or Second hand	Nos	Purchase price/considered reasonable by Engineer-in-Charge	% of Col.5 sanctioned	In (Rs.)
1	2	3	4	5	6	7

IN WITNESS WHEREOF the parties hereto have executed these presents on the day and year first above written.

Signed and delivered

(Contractor)

Signed by Shri.....

(Name & Designation) of
RITES LIMITED

For and on behalf of the Employer
as Agent/Power of Attorney Holder

in the presence of

1.

2

in the presence of

1.

2

NOTES ON PARA 1

*For Proprietary Concerns

Shrison of resident of carrying on business under the name and style of at(hereinafter called "the said Contractor" which expression shall unless the context requires otherwise include his heirs, executors, administrators and legal representatives).

*For Partnership Concerns

M/s. a partnership firm with its office at (hereinafter called "the said Contractor" which expression shall unless the context requires otherwise include their heirs, executors, administrators and legal representatives); the names of their partners being

- i) Shri S/o
- ii) Shri S/o etc.

*For Companies

(*M/s. a company under the Companies Act, 1956 and having its registered office at in the State of hereinafter called the said Contractor' which expression shall unless the context requires otherwise include its administrators, successors and assigns).

ANNEXURE – E1

GUARANTEE BOND IN RESPECT OF WATER PROOFING WORKS

(On Non-Judicial Stamp Paper of Rs. 10)

The Agreement made this day of Two thousand and between (hereinafter called the Guarantor of the one part) and (hereinafter called the “Employer”) on the other part.

WHEREAS THIS agreement is supplementary to a Contract (hereinafter called the Contract) dated and made between the Guarantor of the one part and the Employer acting through RITES Ltd. as Agent/Power of Attorney Holder of the other part, whereby the Contractor, inter alia, undertook to render the buildings and structures in the said Contract cited completely water and leak-proof.

And whereas Guarantor agreed to give a guarantee to the effect that the said structures will remain water and leak-proof for **10 (ten) years** from the date of giving of water proofing treatment.

Now the Guarantor hereby guarantees that water proofing treatment given by him will render the structures completely leak proof and the minimum life of such water proofing treatment shall be **10 (ten) years** to be reckoned from the date after the maintenance period prescribed in the Contract.

Provided that the Guarantor will not be responsible for leakage caused by earthquake or or misuse of roof or alteration and for such purpose.

- a) Misuse of roof shall mean any operation which will damage water-proofing treatment, like chopping of firewood and things of the same nature which might cause damage to the roof.
- b) Alteration shall mean construction of an additional storey or a part of the roof or construction adjoining to existing roof whereby water-proofing treatment is removed in parts.
- c) The decision of the Employer with regard to cause of leakage shall be final.

During this period of Guarantee, the Guarantor shall make good all defects and in case of any defect being found, render the building water proof to the satisfaction of the Employer at his cost and shall commence the work for such rectification within seven days from the date of issue of the notice from the Employer calling upon him to rectify the defects failing which the work shall be got done by the Employer by some other Contractor at the Guarantor's cost and risk. The decision of the Employer as to the cost, payable by the Guarantor shall be final and binding.

That if Guarantor fails to execute the water proofing or commits breach there under then the

Guarantor will indemnify the Employer and his successors against all loss, damage, cost expense or otherwise which may be incurred by him by reason of any default on the part of the Guarantor in performance and observance of this supplementary agreement. As to the amount of loss and/or damage and/or cost incurred by the Employer the decision of the Employer will be final and binding on the parties.

In Witness whereof these present have been executed by the Guarantor and by (Name & Designation of Officer of RITES Ltd.) acting for and on behalf of the Employer as Agent/Power of Attorney Holder on the day, month and year first above written.

SIGNED, SEALED and delivered by GUARANTOR in the presence of:-

- 1.
- 2.

SIGNED for and on behalf of THE EMPLOYER as Agent/Power of Attorney Holder by in the presence of: -

- 1.
- 2.

Copy to the Employer

GUARANTEE BOND FOR ANTI TERMITE TREATMENT WORKS

(On Non-Judicial Stamp Paper of Rs. 10)

The Agreement made this day of Two Thousand between M/s (hereinafter called the Guarantor of the one part) and the (hereinafter called the Employer of the other part).

Whereas this agreement is supplementary to the Contract (hereinafter called the Contract) dated made between the Guarantor of the one part and the Employer acting through RITES Ltd. as Agent/Power of Attorney Holder of the other part, whereby the Contractor, interalia, undertook to render the Buildings and structures in the said Contract cited completely Termite proof.

And whereas the Guarantor agreed to give a Guarantee to the effect that the said structures will remain Termite-proof for **(10) ten years** to be reckoned from the date after the maintenance period prescribed in the Contract expires.

During this period of guarantee the Guarantor shall make good all defects and for that matter, shall replace at his risk and cost, such wooden members as may be damaged by termites and in case of any other defect being found he shall render the building termite proof at his cost to the satisfaction of the Engineer-in-Charge and shall commence the works of such rectification within seven days from the date of issuing notice from the Employer calling upon him to rectify the defects failing which the work shall be got done by the Employer by some other Agency at the Guarantor's cost and risk and in the latter case the decision of the Employer as to the cost recoverable from the Guarantor shall be final and binding.

That if the Guarantor fails to execute the Anti Termite Treatment or commits breaches thereunder, then the Guarantor will indemnify the Employer and his successors against all loss, damage, cost expense or otherwise which may be incurred by him by reason of any default on the part of the Guarantor in performance and observance of this supplementary agreement. As to the amount of loss and / or damage and / or cost incurred by the Employer, the decision of the Employer will be final and binding on the parties.

In witness whereof these present have been executed by the Guarantor and by (Name & Designation of Officer of RITES Ltd). for and on behalf of the Employer as Agent/Power of Attorney Holder on the day, month and year first above written.

SIGNED, SEALED and delivered by the GUARANTOR in the presence of: -

- 1.
- 2.

SIGNED for and on behalf of the Employer as Agent/Power of Attorney by
(RITES LTD.) in the presence of: -

- 1.
- 2.

Copy to the Employer

ANNEXURE 'F1'
(REFER CLAUSE 25 (3))

FORM OF CONTRACTOR'S NOTICE CONCILIATION
THROUGH EXPERT COMMITTEE (EC)

To,

(Appointing Authority)
RITES.
Gurgaon

Dear Sir,

In terms of clause 25 of the agreement, particulars of which are given below,
I/ We hereby give notice to you to appoint an Expert Committee (EC) for
settlement of disputes mentioned below:

1. Name of applicant
2. Whether applicant is Individual / Prop. Firm / Partnership Firm / Ltd. co.
3. Full address of the applicant
4. Name of the work and Contract number in which arbitration sought
5. Name of the RITES Office which entered into Contract
6. Contracted amount of the work
7. Date of Contract
8. Date of start of work
9. Stipulated date of completion of work
10. Actual date of completion of work (if completed)
11. Total number of claims made
12. Total amount claimed

13. Date of intimation of Final Bill being ready (if work is completed)
14. Date of payment of Final Bill (if work is completed)
15. Amount of Final Bill (if work is completed)
16. Date of request made to Engineer- in-Charge's for decision
17. Date of receipt of Engineer-in-Charge's decision

Specimen signature of the applicant
(only the person/authority who signed
the Contract should sign)

I/We certify that the information given above is true to the best of my/our knowledge. I/We enclose following documents.

1. Statement of claims with amount claimed against each claim.
- 2.
- 3.
- 4.

Yours faithfully,

(Signature)

Copy in duplicate to

1. Engineer-in-Charge, RITES
2. Employer

ANNEXURE 'F2'
(REFER CLAUSE 25 (4))

FORM OF CONTRACTOR'S NOTICE FOR APPOINTMENT OF ARBITRATOR

To,

(Appointing Authority)
RITES.
Gurgaon

Dear Sir,

In terms of clause 25 of the agreement, particulars of which are given below, I/ We hereby give notice to you to appoint an arbitral tribunal for settlement of disputes mentioned below:

1. Name of applicant
2. Whether applicant is Individual / Prop. Firm / Partnership Firm / Ltd. co.
3. Full address of the applicant
4. Name of the work and Contract number in which arbitration sought
5. Name of the RITES Office which entered into Contract
6. Contracted amount of the work
7. Date of Contract
8. Date of start of work
9. Stipulated date of completion of work
10. Actual date of completion of work (if completed)
11. Total number of claims made
12. Total amount claimed
13. Date of intimation of Final Bill being ready (if work is completed)
14. Date of payment of Final Bill (if work is completed)
15. Amount of Final Bill (if work is completed)
16. Date of request made to Engineer- in-Charge's for decision
17. Date of receipt of Engineer-in-Charge's decision

18. Date of request to Appointing Authority for conciliation through Expert Committee
19. Date of receipt of Appointing Authority's decision on the recommendations given by the Expert Committee
20. Date of serving this notice to the Appointing Authority

Specimen signature of the applicant
(only the person/authority who signed
the Contract should sign)

I/We certify that the information given above is true to the best of my/our knowledge. I/We enclose following documents.

1. Statement of claims with amount claimed against each claim.
- 2.
- 3.
- 4.

Yours faithfully,

(Signature)

Copy in duplicate to

1. Engineer-in-Charge, RITES
2. Employer

ANNEXURE 'F3'
(REFER CLAUSE 25 (8))

Agreement towards Waiver under Section 12(5) and Section 31-A(5) of Arbitration and Conciliation (Amendment) Act

I/we (Name of agency/contractor) with reference to agreement no raise disputes as to the construction and operation of this contract, or the respective rights liabilities, withholding of certificate and demand arbitration in respects of following claims;

Brief of claim:

- (i) Claim 1 - Details at Annexure-
- (ii) Claim 2 -
- (iii) Claim 3 -

I/we (post of Engineer) with reference to agreement no hereby raise disputes as to the construction and operation of this contract, or the respective rights liabilities, withholding of certificate and demand arbitration in respects of following claims;

I/we do/do not agree to waive off applicability of section 12(5) of Arbitration and conciliation (Amendment) Act.

Signature of Claimant Signature of Respondent

Agreement under Section 31 A (5)

I/we (Name of Claimant) with reference to agreement no hereby waive off the applicability of Section 31-A (2) to 31-A (4) of the Arbitration and Conciliation (Amendment) Act. We further agree that the cost of arbitration will be shared by the parties as per Clause 25(38) of GCC.

Signature of Claimant Signature of Respondent

Strike out whichever not applicable.

Certification by Arbitrators appointed under Clause 25 of General Conditions of Contract for Works

1. Name:
2. Contract Details:
3. Prior experience (Including Experience with Arbitrations):
4. I do not have more than five on-going Arbitration cases with me.
5. I hereby certify that I have retired from w.e.f. and empanelled as Arbitrator with RITES.
6. I have no any part or present relationship in relation to the subject matter in dispute, whether financial, business, professional or other kind.

Or

I have part or present relationship in relation to the subject matter in dispute, whether financial, business, professional or other kind. The list of such interests is as under;

7. I have no any part or present relationship in relation with or interest in any the parties whether financial, business, professional or other kind, which is likely to give rise to justifiable doubts as to my independence or impartiality in terms of the Arbitration and Conciliation Act- 1996.

Or

I have past or present relationship in relation with or interest in any the parties whether financial, business, professional or other kind, which is likely to give rise to justifiable doubts as to my independence or impartiality in terms of the Arbitration and Conciliation Act- 1996. The details of such relationship or interest are as under

8. There are no concurrent circumstances which are likely to affect my ability to devote sufficient time to the arbitration and in particular to finish the entire arbitration within twelve months.

Or

There are circumstances which are likely to affect my ability to devote sufficient time to the arbitration and in particular to finish the entire arbitration within twelve months. The list of such circumstances is as under;

Name and Signatures of Arbitrator

ANNEXURE 'F4'
(REFER CLAUSE 25 (3.3 xi))

Fees and Facilities to the EC Members

EC members shall be entitled for the following fees and facilities (All the expenditure incurred in the EC proceedings shall be shared by the Parties in equal proportions. The parties shall maintain account of expenditure and present to the other for the purpose of sharing on conclusion of the EC proceedings):

Sl. No.	Fees/ Facility	Entitlement
1	Fees (for one member)	Rs. 20,000 per meeting subject to maximum of Rs.3,00,000 for the whole project. In addition, the Presiding Member shall be paid an additional amount of Rs. 2000/- per meeting towards secretarial expenses in writing minutes / EC recommendations.
2	Transportation in the city of the meeting	Car as per actual on certification basis subject to a maximum of Rs.2000/- per meeting.
3	Venue for meeting	RITES conference rooms/Hotels
4	Facilities to be provided to the out - stationed member -	
4 (i)	Travel from the city of residence to the city of meeting	Economy class air tickets/ A.C first class train tickets/ Luxury car/ reimbursement of actual fare.
4 (ii)	Transport to and fro airport / railway station in the city of residence	Car as per actual.
4 (iii)	Stay for out stationed members	4 Star Hotel.
4 (iv)	Transport in the city of meeting	Car as per actual.

ANNEXURE 'G'
(REFER CLAUSE 33 A)

INDEMNITY BOND

(ON NON – JUDICIAL STAMP PAPER OF APPROPRIATE VALUE)

NAME OF THE WORK:

Know all men by these presents that I/We (Name of the Contractor with address) do hereby execute Indemnity Bond in favour of RITES having their office at acting for and on behalf of (Employer) as Agent/Power of Attorney Holder.

On this day of

Whereas M/s. RITES have been appointed as the Consultant for the Project Management for and on behalf of (the Employer) for the work of under Contract No. Dated

This Deed witnesseth as follows

I/We (Name of the Contractor) hereby do indemnity and save harmless RITES having its office at from any claims by the State Government Department responsible for collection of royalty for minor minerals like red bajri, stone, kankar etc., on non-payment of royalty as due for the minor minerals used in the execution of the contract, in terms of provisions of the relevant State Act and Rules made thereon.

IN WITNESS WHEREOF THE

HAS SET HIS/HER HANDS ON THIS DAY OF

SIGNED AND DELIVERED BY THE AFORESAID

IN THE PRESENCE OF WITNESSES

- 1)
- 2)

ANNEXURE 'H'
(REFER CLAUSE 43.9)

INDEMNITY BOND

(ON NON – JUDICIAL STAMP PAPER OF APPROPRIATE VALUE)

NAME OF THE WORK:

Know all men by these presents that I/We (Name of the Contractor with address) do hereby execute Indemnity Bond in favour of RITES having their office at acting for and on behalf of (Employer) as Agent/Power of Attorney Holder.

On this day of

WHEREAS M/s. RITES appointed as the Consultants for the Project Management for and on behalf of (the Employer) for the work of under Contract No dated with us.

THIS DEED WITNESSETH AS FOLLOWS:

I/We, (Name of Contractor) hereby do indemnify and save harmless RITES having their office at from

1. Any third party claims, civil or criminal complaints/liabilities site mishaps and other accidents or disputes and/or damages occurring or arising out of any mishaps at the site due to faulty work, negligence, faulty construction and/or for violating any law, rules and regulations in force, for the time being while executing/executed civil works by me/us.
2. Any damages, loss or expenses due to or resulting from any negligence or breach of duty on the part of me/us or any sub-Contractor/s if any, servants or agents.
3. Any claims by an employee of mine/ours or of sub-Contractors if any, under the workman compensation act and employers' Liability act, 1939 or any other law rules and regulations in force for the time being and any acts replacing and/or amending the same or any of the same as may be in force at the time and under any law in respect of injuries to persons or property arising out of and in the course of execution of the Contract work and/or arising out of and in the course of employment of any workman/employee.
4. Any act or omission of mine/ours or sub-Contractor/s if any, our/their servants or agent which may involve any loss, damage, liability, civil or criminal action.
5. The contractor shall indemnify and hold RITES harmless from and against any losses that it

might suffer in pursuance of the indemnity provided by the contractor in the tender conditions or consequent contract or breach of obligations mentioned under the tender or consequent tender.

IN WITNESS WHEREOF THE HAS SET HIS/THEIR HANDS ON
THIS DAY OF SIGNED AND DELIVERED BY THE AFORESAID

IN THE PRESENCE OF WITNESSES:

1.
2.

ANNEXURE 'I'
(REFER CLAUSE 10B)

INDENTURE FOR SECURED ADVANCES

This Indenture made the day of 20..... BETWEEN (hereinafter called the Contractor which expression shall where the context so admits or implies be deemed to include his executors administrators and assigns) of the one part and the Employer represented by RITES Ltd. as is Agent/Power of Attorney Holder (hereinafter called the Employer which expression shall where the context so admits or implies be deemed to include his successors in office and assigns) of the other part.

WHEREAS by an agreement dated (hereinafter called the said agreement) the Contractor has agreed AND WHEREAS the Contractor has applied to the Employer that he may be allowed advances on the security of materials absolutely belonging to him and brought by him to the site of the works the subject of the said agreement for use in the construction of such of the works as he has undertaken to execute at rates fixed for the finished work (inclusive of the cost of materials and labour and other charges) AND WHEREAS the Employer has agreed to advance to the Contractor the sum of Rupees on the security of materials the quantities and other particulars of which are detailed in Accounts of Secured Advances attached to the Running Account Bill for the said works signed by the Contractor on and the Employer has reserved to himself the option of making any further advance or advances on the security of other materials brought by the Contractor to the site of the said works. Now THIS INDENTURE WITNESSETH that in pursuance of the said agreement and in consideration of the sum of Rupees on or before the execution of these presents paid to the Contractor by the Employer receipt whereof the Contractor doth hereby acknowledge) and of such further advances (if any) as may be made to him as aforesaid the Contractor doth hereby covenant and agree with the Employer and declare as follows:-

- (1) That the said sum of Rupees So advanced by the Employer to the Contractor as aforesaid and all or any further sum or sums advanced as aforesaid shall be employed by the Contractor in or towards expediting the execution of the said works and for no other purpose whatsoever.
- (2) That the materials detailed in the said Account of Secured Advances which has been offered to and accepted by the Employer as security are absolutely the Contractor's own property and free from encumbrances of any kind and the contractor will not make any

application for or receive a further advance on the security of materials which are not absolutely his own property and free from encumbrances of any kind and the Contractor indemnifies the Employer against all claims to any materials in respect of which an advance has been made to him as aforesaid.

- (3) That the materials detailed in the said Account of Secured Advances and all other materials on the security of which any further advance or advances may hereafter be made as aforesaid (hereinafter called the said materials) shall be used by the Contractor solely in the execution of the said works in accordance with the directions of the Engineer-in-Charge and in the term of the said agreement.
- (4) That the Contractor shall make at his own cost all necessary and adequate arrangements for the proper watch, safe custody and protection against all risks of the said materials and that until used in construction as aforesaid the said materials shall remain at the site of the said works in the Contractor's custody and on his own responsibility and shall at all times be open to inspection by the Engineer-in-Charge or any office authorized by him. In the event of the said materials or any part thereof being stolen, destroyed or damaged or becoming deteriorated in a greater degree than is due to reasonable use and wear thereof the Contractor will forthwith replace same with other materials of like quality or repair and make good the same as required by the Engineer-in-Charge.
- (5) That the said materials shall not on any account be removed from the site of the said works except with the written permission of the Engineer-in-Charge or any officer authorized by him on that behalf.
- (6) That the advances shall be repayable in full when or before the Contractor receives payment from the Employer of the price payable to him for the said works under the terms and provisions of the said agreement. Provided that if any intermediate payments are made to the Contractor on account of work done than on the occasion of each such payment the Employer will be at liberty to make a recovery from the Contractor's bill for such payment by deducting therefrom the value of the said materials then actually used in the construction and in respect of which recovery has not been made previously, the value for this purpose being determined in respect of each description of materials at the rates at which the amounts of the advances made under these presents were calculated. **Provided further that, the repayment of the advance will be affected after expiry of a period of 120 days since payment of advance, whether the material is consumed in the work or not.**
- (7) That if the Contractor shall at any time make any default in the performance or observance in any respect of any of the terms and provisions of the said agreement or of these presents the total amount of the advance or advances that may still be owing to the Employer shall immediately on the happening of such default be repayable by the Contractor to the Employer together with interest thereon at twelve per cent per annum from the date or respective dates of such advance or advances to the date of repayment and with all costs

charges, damages and expenses incurred by the Employer in or for the recovery thereof or the enforcement of this security or otherwise by reason of the default of the Contractor and the Contractor hereby covenants and agrees with the Employer to repay and pay the same respectively to him accordingly.

- (8) That the Contractor hereby charges all the said materials with the repayment to the Employer of the said sum of Rupees and any further sum or sums advanced as aforesaid and all costs charges, damages and expenses payable under these presents PROVIDED ALWAYS and it is hereby agreed and declared that notwithstanding anything in the said agreement and without prejudice to the powers contained therein if and whenever the covenant for payment and repayment herein before contained shall become enforceable and the money owing shall not be paid in accordance therewith the Employer may at any time thereafter adopt all or any of the following courses as he may deem best:-
- (a) Seize and utilize the said materials or any part thereof in the completion of the said works on behalf of the Contractor in accordance with the provisions in that behalf contained in the said agreement debiting the Contractor with the actual cost of effecting such completion and the amount due in respect of advances under these presents and crediting the Contractor with the value of work done as if he had carried it out in accordance with the said agreement and at the rates thereby provided. If the balance is against the Contractor he is to pay same to the Employer on demand.
 - (b) Remove and sell by public auction the seized materials or any part thereof and out of the moneys arising from the sale retain all the sums aforesaid repayable or payable to the Employer under these presents and pay over the surplus (if any) to the Contractor.
 - (c) Deduct all or any part of the moneys owing out of the security deposit or any sum due to the Contractor under the said agreement.
- (9) That except in the event of such default on the part of the Contractor as aforesaid interest on the said advance shall not be payable.
- (10) That in the event of any conflict between the provisions of these presents and the said agreement the provisions of these presents shall prevail and in the event of any dispute or difference arising over the construction or effect of these presents the settlement of which has not been herein before expressly provided for the same shall be referred to the Accepting Authority whose decision shall be final and the provision of the Indian Arbitration Act for the time being in force shall apply to any such reference.

In witness whereof the said And
by the order and under the direction of the Employer have hereunto set their respective hands
the day and year first above written.

Signed, sealed and delivered by the said contractor in the presence of

{Signature..... }

{Name }

{Address }

Witness

Signed by
by the order and direction of the Employer in the presence of

{Signature..... }

{Name }

{Address }

Witness

ANNEXURE J

**FORM OF APPLICATION BY THE CONTRACTOR FOR SEEKING
RESCHEDULING OF MILESTONE/EXTENSION OF TIME**

1. Name of Contractor
2. Name of work as given in the Agreement
3. Agreement No.
4. Estimated amount put to tender
5. Date of commencement of work as per agreement
6. Period allowed for completion of work as per agreement
7. Date of completion as stipulated in agreement
8. Period for which extension of time has been given previously:
 - a) 1st extension vide No. Dated From _____ To _____ (____ Days)
 - b) 2nd extension vide No. Dated From _____ To _____ (____ Days)Total extension previously given.
9. Reasons for which extensions have been previously given (Copies of the previous sanctions should be attached)
10. Period for which extensions is applied for now From _____ To _____ (____ Days)
11. Hindrances on account of which extension is applied for with dates on which hindrances occurred and the period for which these are likely to last. (Details are to be filled in for each Hindrance in a tabular form)
 - a) Serial No.
 - b) Nature of hindrance
 - c) Date of occurrence
 - d) Items of work which could not be executed on account of this hindrance
 - e) Date of removal of hindrance
 - f) Overlapping period if any, with reference to item _____
 - g) Net delay on account of this hindrance
 - h) Weightage of this hindrance
 - i) Net extension applied for
 - j) Remarks, if any.

Total period on account of hindrances mentioned above _____ Months _____ Days

12. Extension of time required for extra work
 - a) Details of extra work and the amount involved:-
 - b) Proportionate period of extension of time based on tendered value on account of extra work.

13. Total extension of time required for 11 & 12

Submitted to the Site
Engineer

Signature of Contractor Dated

Annexure-K

**No Claim Certificate
(Refer Clause 1)
(On Company letter Head)**

To,

**The Engineer in Charge
RITES Ltd.**

NO CLAIM CERTIFICATE

Sub: Contract Agreement no. ----- dated -----for the work of -----

We have received the sum of Rs. -----(Rupees -----only) in full and final settlement of all the payments due to us for the work of..... under the above-mentioned contract agreement, between us and We hereby unconditionally, and without any reservation whatsoever, certify that with this payment, we shall have no claim whatsoever, of any description, on any account, against, against aforesaid contract agreement executed by us. We further declare unequivocally, that with this payment, we have received all the amounts payable to us, and have no dispute of any description whatsoever, regarding the amounts worked out as payable to us and received by us and that we shall continue to be bound by the terms and conditions of the contract agreement, as regards performance of the contract.

Date:

Place:

Yours faithfully, Signatures of contractor or
Officer authorised to sign the contract documents
on behalf of the contractor/JV
(Company stamp)

Section - 9

RITES SAFETY CODE

SECTION - 9

rites SAFETY CODE

1. Suitable scaffolds should be provided for workmen for all works that cannot safely be done from the ground, or from solid construction except such short period work as can be done safely from ladders. When a ladder is used, an extra mazdoor shall be engaged for holding the ladder and if the ladder is used for carrying materials as well suitable footholds and hand-hold shall be provided on the ladder and the ladder shall be given an inclination not steeper than $\frac{1}{4}$ to 1 ($\frac{1}{4}$ horizontal and 1 vertical.)
2. Scaffolding of staging more than 3.6 m (12ft.) above the ground or floor, swung or suspended from an overhead support or erected with stationary support shall have a guard rail properly attached or bolted, braced and otherwise secured at least 90 cm. (3ft.) high above the floor or platform of such scaffolding or staging and extending along the entire length of the outside and ends there of with only such opening as may be necessary for the delivery of materials. Such scaffolding or staging shall be so fastened as to prevent it from swaying from the building or structure.
3. Working platforms, gangways and stairways should be so constructed that they should not sag unduly or unequally, and if the height of the platform or the gangway or the stairway is more than 3.6 m (12ft.) above ground level or floor level, they should be closely boarded, should have adequate width and should be suitably fastened as described in (2) above.
4. Every opening in the floor of a building or in a working platform shall be provided with suitable means to prevent the fall of person or materials by providing suitable fencing or railing whose minimum height shall be 90 cm. (3ft.)
5. Safe means of access shall be provided to all working platform and other working places. Every ladder shall be securely fixed. No portable single ladder shall be over 9m. (30 ft)in length while the width between side rails in rung ladder shall in no case be less than 29 cm. (11 1/2") for ladder upto and including 3m. (10 ft.)in length. For longer ladders, this width should be increased at least $\frac{1}{4}$ " for each additional 30 cm. (1 foot) of length. Uniform step spacing of not more than 30 cm shall be kept. Adequate precautions shall be taken to prevent danger from electrical equipment. No materials on any of the sites or work shall be so stacked or placed as to cause danger or inconvenience to any person or the public. The contractor shall provide all necessary fencing and lights to protect the public from accident and shall be bound to bear the expenses of defense of every suit, action or other proceedings at law that may be brought by any person for injury sustained owing to neglect of the above precautions and to pay any damages and cost which may be awarded in any such suit, action or

- proceedings to any such person or which may, with the consent of the contractor, be paid to compensate any claim by any such person.
6. Excavation and Trenching – All trenches 1.2m. (4ft.) or more in depth, shall at all times be supplied thereof, with at least one ladder for each 30 m. (100 ft) in length or fraction thereof, Ladder shall extend from bottom of the trench to at least 90 cm. (3 ft) above the surface of the ground. The side of the trenches which are 1.5 m.(5 ft.) or more in depth shall be stepped back to give suitable slope or securely held by timber bracing, so as to avoid the danger of sides collapsing. The excavated materials shall not be placed within 1.5 m. (5 ft.) of the edges of the trench or half of the depth of the trench whichever is more. Cutting shall be done from top to bottom. Under no circumstances undermining or undercutting shall be done.
 7. Demolition - Before any demolition work is commenced and also during the progress of the work,
 - (i) All roads and open areas adjacent to the work site shall either be closed or suitably protected.
 - (ii) No electric cable or apparatus which is liable to be a source of danger, or a cable or apparatus used by the operator shall remain electrically charged.
 - (iii) All practical steps shall be taken to prevent danger to persons employed from risk of fire or explosion or flooding. No floor, roof or other part of the building shall be so overloaded with debris or materials as to render it unsafe.
 8. All necessary personal safety equipment as considered adequate by the Engineer-in-Charge should be kept available for the use of the person employed on the site and maintained in a condition suitable for immediate use, and the contractor should take adequate steps to ensure proper use of equipment by those concerned: -

The following safety equipment shall invariably be provided.

 - (i) Workers employed on mixing asphaltic materials, cement and lime mortars shall be provided with protective footwear and protective goggles.
 - (ii) Those engaged in white washing and mixing or stacking of cement bags or any material which is injurious to the eyes, shall be provided with protective goggles.
 - (iii) Those engaged in welding works shall be provided with welder's protective eye-shields.
 - (iv) Stone breaker shall be provided with protective goggles and protective clothing and seated at sufficiently safe intervals.
 - (v) When workers are employed in sewers and manholes, which are in active use, the contractors shall ensure that the manhole covers are opened and ventilated atleast for an hour before the workers are allowed to get into the manholes, and the

manholes so opened shall be cordoned off with suitable railing and provided with warning signals or boards to prevent accident to the public. In addition, the contractor shall ensure that the following safety measure are adhered to:-

- (a) Entry for workers into the line shall not be allowed except under supervision of the JE or any other higher officer.
- (b) At least 5 to 6 manholes upstream and downstream should be kept open for at least 2 to 3 hours before any man is allowed to enter into the manhole for working inside.
- (c) Before entry, presence of Toxic gases should be tested by inserting wet lead acetate paper which changes colour in the presence of such gases and gives indication of their presence.
- (d) Presence of Oxygen should be verified by lowering a detector lamp into the manhole. In case, no Oxygen is found inside the sewer line, workers should be sent only with Oxygen kit.
- (e) Safety belt with rope should be provided to the workers. While working inside the manholes, such rope should be handled by two men standing outside to enable him to be pulled out during emergency.
- (f) The area should be barricaded or cordoned off by suitable means to avoid mishaps of any kind. Proper warning signs should be displayed for the safety of the public whenever cleaning works are undertaken during night or day.
- (g) No smoking or open flames shall be allowed near the blocked manhole being cleaned.
- (h) The malba obtained on account of cleaning of blocked manholes and sewer lines should be immediately removed to avoid accidents on account of slippery nature of the malba.
- (i) Workers should not be allowed to work inside the manhole continuously. He should be given rest intermittently. The Engineer-in-Charge shall decide the time up to which a worker may be allowed to work continuously inside the manhole.
- (j) Gas masks with Oxygen Cylinder should be kept at site for use in emergency.
- (k) Air-blowers should be used for flow of fresh air through the manholes. Whenever called for, portable air blowers are recommended for ventilating the manholes. The Motors for these shall be vapour proof and of totally enclosed type. Non sparking gas engines also could be used but they should be placed at least 2 metres away from the opening and on the leeward side protected from wind so that they will not be a source of friction on any

inflammable gas that might be present.

- (l) The workers engaged for cleaning the manholes/sewers should be properly trained before allowing to work in the manhole.
 - (m) The workers shall be provided with Gumboots or non-sparking shoes bump helmets and gloves non sparking tools safety lights and gas masks and portable air blowers (when necessary). They must be supplied with barrier cream for anointing the limbs before working inside the sewer lines.
 - (n) Workmen descending a manhole shall try each ladder stop or rung carefully before putting his full weight on it to guard against insecure fastening due to corrosion of the rung fixed to manhole well.
 - (o) If a man has received a physical injury, he should be brought out of the sewer immediately and adequate medical aid should be provided to him.
 - (p) The extent to which these precautions are to be taken depend on individual situation but the decision of the Engineer-in-Charge regarding the steps to be taken in this regard in an individual case will be final.
9. The Contractor shall not employ men and women below the age of 18 years on the work of painting with products containing lead in any form. Wherever men above the age of 18 are employed on the work of lead painting, the following precaution should be taken: -
- i) White lead, sulphate of lead or product containing this pigments, shall not be used in painting operations.
 - ii) Measures shall be taken, wherever required in order to prevent danger arising from the application of paint in the form of spray.
 - iii) Measures shall be taken, wherever practicable, to prevent danger arising out of or from dust caused by dry rubbing down and scraping.
 - iv) Adequate facilities shall be provided to enable working painters to wash during and on cessation of work.
 - v) Overall shall be worn by working painters during the whole of working period.
 - vi) Suitable arrangement shall be made to prevent clothing put off during working hours being spoiled by painting materials.
 - vii) Cases of lead poisonings and suspected lead poisoning shall be notified and shall be subsequently verified by medical man appointed by competent authority of RITES.
 - viii) RITES may require, when necessary medical examination of workers.
 - ix) Instructions with regard to special hygienic precautions to be taken in the painting

trade shall be distributed to working painters.

- x) No paint containing lead or lead products shall be used except in the form of paste or ready made paint.
 - xi) Suitable face masks should be supplied for use by the workers when paint is applied in the form of spray or a surface having lead paint is dry rubbed and scraped.
 - xii) Overall, shall be supplied by the contractors in the workmen and adequate facilities shall be provided to enable the working painters to wash during and on the cessation of work.
10. When the work is done near any place where there is risk of drowning, all necessary equipment should be provided and kept ready for use and all necessary steps taken for prompt rescue of any person in danger and adequate provision should be made for prompt first aid treatment of all injuries likely to be sustained during the course of the work.
11. Use of hoisting machines and tackle including their attachments, anchorage and supports shall conform to the following standards for conditions:
- (i) (a) These shall be of good mechanical construction, sound materials and adequate strength and free from patent defects and shall be kept repaired and in good working order.
 - (b) Every rope used in hoisting or lowering materials or as a means of suspension shall be of durable quality and adequate strength, and free from patent defects.
 - (ii) Every crane driver or hoisting appliance operator shall be properly qualified and no person under the age of 21 years should be in charge of any hoisting machine including any scaffolding winch or give signals to operator.
 - (iii) In case of every hoisting machine and of every chain ring hook, shackle swivel and pulley block used in hoisting or as means of suspension, the safe working load shall be ascertained by adequate means. Every hoisting machine and all gear referred to above shall be plainly marked with the safe working load. In case of a hoisting machine having a variable safe working load each safe working load and the condition under which it is applicable shall be clearly indicated. No part of any machine or any gear referred to above in this paragraph shall be loaded beyond the safe working load except for the purpose of testing.
 - (iv) In case of Employer's machines, the safe working load shall be notified by the Electrical Engineer-in-Charge. As regards contractor's machines the contractors shall notify the safe working load of the machine to the Engineer-in-Charge whenever he brings any machinery to site of work and get it verified by the Electrical Engineer concerned.
12. Motors, gearing, transmission, electric wiring and other dangerous parts of hoisting

- appliances should be provided with efficient safeguards. Hoisting appliances should be provided with such means as will reduce to the minimum the risk of accidental decent of the load. Adequate precautions should be taken to reduce to the minimum the risk of any 12. Motors, gearing, transmission, electric wiring and other dangerous parts of hoisting appliances should be provided with efficient safeguards. Hoisting appliances should be provided with such means as will reduce to the minimum the risk of accidental decent of the load. Adequate precautions should be taken to reduce to the minimum the risk of any.
13. All scaffolds, ladders and other safety devices mentioned or described herein shall be maintained in safe condition and no scaffold, ladder or equipment shall be altered or removed while it is in use. Adequate washing facilities should be provided at or near places of work.
 14. These safely provisions should be brought to the notice of all concerned by display on a notice board at a prominent place at work spot. The person responsible for compliance of the safety code shall be named therein by the contractor.
 15. To ensure effective enforcement of the rules and regulations relating to safety precautions the arrangements made by the contractor shall be open to inspection by the Labour Officer or Engineer-in-Charge of the department or their representatives.
 16. Notwithstanding the above clauses from (1) to (15) there is nothing in these to exempt the contractor from the operations of any other Act or Rule in force in the Republic of India

Section - 10

**UTES MODEL RULES FOR PROTECTION OF HEALTH
AND SANITARY ARRANGE- MENTS FOR WORKERS
EMPLOYED BY RITES OR ITS CONTRACTORS**

SECTION – 10

UTES Model Rules for the Protection of Health and Sanitary Arrangements for Workers Employed by RITES or its Contractors

1. APPLICATION

These rules shall apply to all buildings and construction works in charge of RITES in which twenty or more workers are ordinarily employed or are proposed to be employed in any day during the period during which the contract work is in progress.

2. DEFINITION

Workplace means a place where twenty or more workers are ordinarily employed in connection with construction work on any day during the period during which the contract work is in progress.

3. FIRST-AID FACILITIES

- (i) At every workplace, there shall be provided and maintained, so as to be easily accessible during working hours, first-aid boxes at the rate of not less than one box for 150 contract labour or part thereof ordinarily employed.
- (ii) The first-aid box shall be distinctly marked with a red cross on white background and shall contain the following equipment: -
 - (a) For work places in which the number of contract labour employed does not exceed 50- Each first-aid box shall contain the following equipments :-
 - 1. 6 Small sterilised dressings.
 - 2. 3 medium size sterilised dressings.
 - 3. 3 large size sterilised dressings.
 - 4. 3 large sterilised burn dressings.
 - 5. 1 (30 ml.) bottle containing a two per cent alcoholic solution of iodine.
 - 6. 1 (30 ml.) bottle containing salvolatile having the dose and mode of administration indicated on the label.
 - 7. 1 snakebite lancet.
 - 8. 1 (30 gms.) bottle of potassium permanganate crystals.

9. 1 pair scissors.
 10. 1 copy of the first-aid leaflet issued by the Director General, Factory Advice Service and Labour Institutes, Government of India.
 11. 1 bottle containing 100 tablets (each of 5 gms.) of aspirin.
 12. Ointment for burns.
 13. A bottle of suitable surgical antiseptic solution.
- (b) For workplaces in which the number of contract labour exceed 50. Each first-aid box shall contain the following equipments.
1. 12 small sterilised dressings.
 2. 6 medium size sterilised dressings.
 3. 6 large size sterilised dressings.
 4. 6 large size sterilised burn dressings.
 5. 6 (15 gms.) packets sterilised cotton wool.
 6. 1 (60 ml.) bottle containing a two per cent alcoholic solution iodine.
 7. 1 (60 ml.) bottle containing salvolatile having the dose and mode of administration indicated on the label.
 8. 1 roll of adhesive plaster.
 9. 1 snake bite lancet.
 10. 1 (30 gms.) bottle of potassium permanganate crystals.
 11. 1 pair scissors.
 12. 1 copy of the first-aid leaflet issued by the Director General Factory Advice Service and Labour Institutes /Government of India.
 13. A bottle containing 100 tablets (each of 5 gms.) of aspirin.
 14. Ointment for burns.
 15. A bottle of suitable surgical antiseptic solution.
- (iii) Adequate arrangements shall be made for immediate recoupment of the equipment when necessary.
- (iv) Nothing except the prescribed contents shall be kept in the First-aid box.
- (v) The first-aid box shall be kept in charge of a responsible person who shall always be

readily available during the working hours of the workplace.

- (vi) A person in charge of the First-aid box shall be a person trained in First-aid treatment in the workplaces where the number of contract labour employed is 150 or more.
- (vii) In work places where the number of contract labour employed is 500 or more and hospital facilities are not available within easy distance from the works. First-aid posts shall be established and run by a trained compounder. The compounder shall be on duty and shall be available at all hours when the workers are at work.
- (viii) Where work places are situated in places which are not towns or cities, a suitable motor transport shall be kept readily available to carry injured person or person suddenly taken ill to the nearest hospital.

4. DRINKING WATER

- (i) In every workplace, there shall be provided and maintained at suitable places, easily accessible to labour, a sufficient supply of cold water fit for drinking.
- (ii) Where drinking water is obtained from an Intermittent public water supply, each workplace shall be provided with storage where such drinking water shall be stored.
- (iii) Every water supply or storage shall be at a distance of not less than 50 feet from any latrine drain or other source of pollution. Where water has to be drawn from an existing well which is within such proximity of latrine, drain or any other source of pollution, the well shall be properly chlorinated before water is drawn from it for drinking. All such wells shall be entirely closed in and be provided with a trap door which shall be dust and waterproof.
- (iv) A reliable pump shall be fitted to each covered well, the trap door shall be kept locked and opened only for cleaning or inspection which shall be done at least once a month.

5. WASHING FACILITIES

- (i) In every workplace adequate and suitable facilities for washing shall be provided and maintained for the use of contract labour employed therein.
- (ii) Separate and adequate cleaning facilities shall be provided for the use of male and female workers.
- (iii) Such facilities shall be conveniently accessible and shall be kept in clean and hygienic condition.

6. LATRINES AND URINALS

- (i) Latrines shall be provided in every work place on the following scale namely :-
 - (a) Where females are employed, there shall be at least one latrine for every 25 females.
 - (b) Where males are employed, there shall be at least one latrine for every 25 males.

- Provided that, where the number of males or females exceeds 100, it shall be sufficient if there is one latrine for 25 males or females as the case may be upto the first 100, and one for every 50 thereafter.
- (ii) Every latrine shall be under cover and so partitioned off as to secure privacy and shall have a proper door and fastenings.
 - (iii) Construction of latrines: The inside walls shall be constructed of masonry or some suitable heat-resisting non absorbent materials and shall be cement washed inside and outside at least once a year, Latrines shall not be of a standard lower than borehole system.
 - (iv) **(a)** Where workers of both sexes are employed, there shall be displayed outside each block of latrine and urinal, a notice in the language understood by the majority of the workers “For Men only” or “For Women Only” as the case may be.
(b) The notice shall also bear the figure of a man or of a woman, as the case may be.
 - (v) There shall be at least one urinal for male workers upto 50 and one for female workers upto fifty employed at a time, provided that where the number of male or female workmen, as the case may be exceeds 500, it shall be sufficient if there is one urinal for every 50 males or females upto the first 500 and one for every 100 or part thereafter.
 - (vi) **(a)** The latrines and urinals shall be adequately lighted and shall be maintained in a clean and sanitary condition at all times.
(b) Latrines and urinals other than those connected with a flush sewage system shall comply with the requirements of the Public Health Authorities.
 - (vii) Water shall be provided by means of tap or otherwise so as to be conveniently accessible in or near the latrines and urinals.
 - (viii) Disposal of excreta: - Unless otherwise arranged for by the local sanitary authority, arrangements for proper disposal of excreta by incineration at the work place shall be made by means of a suitable incinerator. Alternately excreta may be disposed of by putting a layer of night soil at the bottom of a pucca tank prepared for the purpose and covering it with a 15 cm. layer of waste or refuse and then covering it with a layer of earth for a fortnight (when it will turn to manure).
 - (ix) The contractor shall at his own expense, carry out all instructions issued to him by the Engineer-in-Charge to effect proper disposal of night soil and other conservancy work in respect of the contractor’s workmen or employees on the site. The contractor shall be responsible for payment of any charges which may be levied by Municipal or Cantonment Authority for execution of such on his behalf.

7. PROVISION OF SHELTER DURING REST

At every place there shall be provided, free of cost, four suitable sheds, two for meals and the

other two for rest separately for the use of men and women labour. The height of each shelter shall not be less than 3 metres (10 ft.) from the floor level to the lowest part of the roof. These shall be kept clean and the space provided shall be on the basis of 0.6 sq.m. (6 sft) per head.

Provided that the Engineer-in-Charge may permit subject to his satisfaction, a portion of the building under construction or other alternative accommodation to be used for the purpose.

8. CRECHES

- (i) At every workplace, at which 20 or more women worker are ordinarily employed, there shall be provided two rooms of reasonable dimensions for the use of their children under the age of six years. One room shall be used as a playroom for the children and the other as their bedroom. The rooms shall be constructed with specifications as per clause 19H (ii) a,b & c.
- (ii) The rooms shall be provided with suitable and sufficient openings for light and ventilation. There shall be adequate provision of sweepers to keep the places clean.
- (iii) The contractor shall supply adequate number of toys and games in the playroom and sufficient number of cots and beddings in the bedroom.
- (iv) The contractor shall provide one ayaa to look after the children in the creche when the number of women workers does not exceed 50 and two when the number of women workers exceed 50.
- (v) The use of the rooms earmarked as creches shall be restricted to children, their attendants and mothers of the children.

9. CANTEENS

- (i) In every workplace where the work regarding the employment of contract labour is likely to continue for six months and where in contract labour numbering one hundred or more are ordinarily employed, an adequate canteen shall be provided by the contractor for the use of such contract labour.
- (ii) The canteen shall be maintained by the contractor in an efficient manner.
- (iii) The canteen shall consist of at least a dining hall, kitchen, storeroom, pantry and washing places separately for workers and utensils.
- (iv) The canteen shall be sufficiently lighted at all times when any person has access to it.
- (v) The floor shall be made of smooth and impervious materials and inside walls shall be lime-washed or colour washed at least once in each year.

Provided that the inside walls of the kitchen shall be lime-washed every four months.

- (vi) The premises of the canteen shall be maintained in a clean and sanitary condition.

- (vii) Wastewater shall be carried away in suitable covered drains and shall not be allowed to accumulate so as to cause a nuisance.
- (viii) Suitable arrangements shall be made for the collection and disposal of garbage.
- (ix) The dining hall shall accommodate at a time 30 per cent of the contract labour working at a time.
- (x) The floor area of the dining hall, excluding the area occupied by the service counter and any furniture except tables and chairs shall not be less than one square metre (10 sqft) per diner to be accommodated as prescribed in sub-Rule 9.
- (xi) **(a)** A portion of the dining hall and service counter shall be partitioned off and reserved for women workers in proportion to their number.
(b) Washing places for women shall be separate and screened to secure privacy.
- (xii) Sufficient tables stools, chair or benches shall be available for the number of diners to be accommodated as prescribed in sub-Rule 9.
- (xiii) **(a)** **1.** There shall be provided and maintained sufficient utensils crockery, furniture and any other equipments necessary for the efficient running of the canteen.
2. The furniture utensils and other equipment shall be maintained in a clean and hygienic condition.
(b) **1.** Suitable clean clothes for the employees serving in the canteen shall be provided and maintained.
2. A service counter, if provided, shall have top of smooth and impervious material.
3. Suitable facilities including an adequate supply of hot water shall be provided for the cleaning of utensils and equipments.
- (xiv) The food stuffs and other items to be served in the canteen shall be in conformity with the normal habits of the contract labour.
- (xv) The charges for food stuffs, beverages and any other items served in the canteen shall be based on 'No profit, No loss' and shall be conspicuously displayed in the canteen.
- (xvi) In arriving at the price of foodstuffs, and other article served in the canteen, the following items shall not be taken into consideration as expenditure namely: -
 - (a) The rent of land and building.
 - (b) The depreciation and maintenance charges for the building and equipments provided for the canteen.
 - (c) The cost of purchase, repairs and replacement of equipments including furniture, crockery, cutlery and utensils.

- (d) The water charges and other charges incurred for lighting and ventilation.
- (e) The interest and amounts spent on the provision and maintenance of equipments provided for the canteen.
- (xvii) The accounts pertaining to the canteen shall be audited once every 12 months by registered accountants and auditors.

10. ANTI-MALARIAL PRECAUTIONS

The contractor shall at his own expense, conform to all anti-malarial instructions given to him by the Engineer-in- Charge including the filling up of any borrow pits which may have been dug by him.

- 11. The above rules shall be incorporated in the contracts and in notices inviting tenders and shall form an integral part of the contracts.

12. AMENDMENTS

RITES may, from time to time, add to or amend these rules and issue directions - it may consider necessary for the purpose of removing any difficulty which may arise in the administration thereof.

Section - 11

**UTES CONTRACTOR'S LABOUR
REGULATIONS**

SECTION – 11

RITIES CONTRACTOR'S LABOUR REGULATIONS

1. SHORT TITLE

These regulations may be called the RITES Contractors Labour Regulations.

2. DEFINITIONS

(i) Workman means any person employed by RITES or its contractor directly or indirectly through a subcontractor with or without the knowledge of RITES to do any skilled, semiskilled or unskilled manual, supervisory, technical or clerical work for hire or reward, whether the terms of employment are expressed or implied but does not include any person: -

(a) Who is employed mainly in a managerial or administrative capacity: or

(b) Who, being employed in a supervisory capacity draws wages exceeding five hundred rupees per mensem or exercises either by the nature of the duties attached to the office or by reason of powers vested in him, functions mainly of managerial nature

: or

(c) Who is an out worker, that is to say, person to whom any article or materials are given out by or on behalf of the principal employers to be made up cleaned, washed, altered, ornamental finished, repaired adopted or otherwise processed for sale for the purpose of the trade or business of the principal employers and the process is to be carried out either in the home of the out worker or in some other premises, not being premises under the control and management of the principal employer.

No person below the age of 18 years shall be employed to act as a workman.

(ii) Fair Wages means wages whether for time or piece work fixed and notified under the provisions of the Minimum Wages Act

from time to time.

- (iii) Contractors shall include every person who undertakes to produce a given result other than a mere supply of goods or articles of manufacture through contract labour or who supplies contract labour for any work and includes a subcontractor.
- (iv) Wages shall have the same meaning as defined in the Payment of Wages Act.

3. Working Hours

- (i) Normally working hours of an adult employee should not exceed 9 hours a day. The working day shall be so arranged that inclusive of interval for rest, if any, it shall not spread over more than 12 hours on any day.
- (ii) When an adult worker is made to work for more than 9 hours on any day or for more than 48 hours in any week, he shall be paid over time for the extra hours put in by him at double the ordinary rate of wages.
- (iii) (a) Every worker shall be given a weekly holiday normally on a Sunday, in accordance with the provisions of the Minimum Wages (Central) Rules 1960 as amended from time to time irrespective of whether such worker is governed by the Minimum Wages Act or not.

(b) Where the minimum wages prescribed by the Government under the Minimum Wages Act are not inclusive of the wages for the weekly day of rest, the worker shall be entitled to rest day wages at the rate applicable to the next preceding day, provided he has worked under the same contractor for a continuous period of not less than 6 days.

(c) Where a contractor is permitted by the Engineer-in-Charge to allow a worker to work on a normal weekly holiday, he shall grant a substituted holiday to him for the whole day on one of the five days immediately before or after the normal weekly holiday and pay wages to such worker for the work performed on the normal weekly holiday at overtime rate.

4. DISPLAY OF NOTICE REGARDING WAGES ETC.

The contractor shall before he commences his work on contract, display and correctly maintain and continue to display and correctly

maintain in a clear and legible condition in conspicuous places on the work, notices in English and in the local Indian languages spoken by the majority of the workers giving the minimum rates of wages fixed under Minimum Wages Act, the actual wages being paid, the hours of work for which such wage are earned, wages periods, dates of payments of wages and other relevant information as per **Appendix 'III'**.

5. PAYMENT OF WAGES

- (i) The contractor shall fix wage periods in respect of which wages shall be payable.
- (ii) No wage period shall exceed one month.
- (iii) The wages of every person employed as contract labour in an establishment or by a contractor where less than one thousand such persons are employed shall be paid before the expiry of seventh day and in other cases before the expiry of tenth day after the last day of the wage period in respect of which the wages are payable.
- (iv) Where the employment of any worker is terminated by or on behalf of the contractor the wages earned by him shall be paid before the expiry of the second working day from the date on which his employment is terminated.
- (v) All payment of wages shall be made on a working day at the work premises and during the working time and on a date notified in advance and in case the work is completed before the expiry of the wage period, final payment shall be made within 48 hours of the last working day.
- (vi) Wages due to every worker shall be paid to him direct or to other person authorised by him in his behalf.
- (vii) All wages shall be paid in current coin or currency or in both.
- (viii) Wages shall be paid without any deductions of any kind except those specified by the Central Government by general or special order in this behalf or permissible under the Payment of Wages Act 1956.
- (ix) A notice showing the wages period and the place and time of disbursement of wages shall be displayed at the place of work and a copy sent by the contractor to the Engineer-in-Charge under acknowledgment.

- (x) It shall be the duty of the contractor to ensure the disbursement of wages through bank account of labour.
- (xi) The contractor shall obtain from the Junior Engineer or any other authorised representative of the Engineer-in-Charge as the case may be, a certificate under his signature at the end of the entries in the “Register of Wages” or the “Wage-cum-Muster Roll” as the case may be in the following form: -

“Certified that the amount shown in column No has been paid to the workman concerned through bank account of labour on at.....”

6. FINES AND DEDUCTIONS WHICH MAY BE MADE FROM WAGES

- (i) The wages of a worker shall be paid to him without any deduction of any kind except the following: -
 - (a) Fines
 - (b) Deductions for absence from duty i.e., from the place or the places whereby the terms of his employment he is required to work. The amount of deduction shall be in proportion to the period for which he was absent.
 - (c) Deduction for damage to or loss of goods expressly entrusted to the employed person for custody or for loss of money or any other deduction which he is required to account, where such damage or loss is directly attributable to his neglect or default.
 - (d) Deduction for recovery of advances or for adjustment of overpayment of wages, advances granted shall be entered in a register.
 - (e) Any other deduction which the Central Government may from time to time allow.
- (ii) No fines should be imposed on any worker save in respect of such acts and omissions on his part as have been approved of by the Chief Labour Commissioner.

Note: - An approved list of Acts and Omissions for which fines can be imposed is enclosed at **Appendix-X**

- (iii) No fine shall be imposed on a worker and no deduction for damage or loss shall be made from his wages until the worker has been given an opportunity of showing cause against such fines or deductions.

- (iv) The total amount of fine which may be imposed in any one wage period on a worker shall not exceed an amount equal to three paise in a rupee of the total wages, payable to him in respect of that wage period.
- (v) No fine imposed on any worker shall be recovered from him by instalment, or after the expiry of sixty days from the date on which it was imposed.
- (vi) Every fine shall be deemed to have been imposed on the day of the act or omission in respect of which it was imposed.

7. LABOUR RECORDS

- (i) The contractor shall maintain a Register of persons employed on work on contract in Form XIII of the CL (R&A) Central Rules 1971 (**Appendix IV**)
- (ii) The contractor shall maintain a Muster Roll register in respect of all workmen employed by him on the work under Contract in Form XVI of the CL (R&A) Rules 1971 (**Appendix V**).
- (iii) The contractor shall maintain a Wage Register in respect of all workmen employed by him on the work under contract in Form XVII of the CL (R&A) Rules 1971 (**Appendix VI**).
- (iv) Register of accident - The contractor shall maintain a register of accidents in such form as may be convenient at the workplace but the same shall include the following particulars:
 - (a) Full particulars of the labourers who met with accident.
 - (b) Rate of Wages.
 - (c) Sex
 - (d) Age
 - (e) Nature of accident and cause of accident.
 - (f) Time and date of accident.
 - (g) Date and time when admitted in Hospital,
 - (h) Date of discharge from the Hospital.
 - (i) Period of treatment and result of treatment.
 - (j) Percentage of loss of earning capacity and disability as assessed by Medical Officer.
 - (k) Claim required to be paid under Workmen's Compensation Act.

- (l) Date of payment of compensation.
- (m) Amount paid with details of the person to whom the same was paid.
- (n) Authority by whom the compensation was assessed.
- (o) Remarks
- (v) The contractor shall maintain a Register of Fines in the Form XII of the CL (R&A) Rules 1971 (**Appendix-XI**). The contractor shall display in a good condition and in a conspicuous place of work the approved list of acts and omissions for which fines can be imposed (**Appendix-X**)
- (vi) The contractor shall maintain a Register of deductions for damage or loss in Form XX of the CL (R&A) Rules 1971 (**Appendix-XII**)
- (vii) The contractor shall maintain a Register of Advances in Form XXIII of the CL (R&A) Rules 1971 (**Appendix-XIII**)
- (viii) The contractor shall maintain a Register of Overtime in Form XXIII of the CL (R&A) Rules 1971 (**Appendix-XIV**)

8. ATTENDANCE CARD-CUM-WAGE SLIP

- (i) The contractor shall issue an Attendance card-cum-wage slip to each workman employed by him in the specimen form at (**Appendix-VII**)
- (ii) The card shall be valid for each wage period.
- (iii) The contractor shall mark the attendance of each workman on the card twice each day, once at the commencement of the day and again after the rest interval, before he actually starts work.
- (iv) The card shall remain in possession of the worker during the wage period under reference.
- (v) The contractor shall complete the wage slip portion on the reverse of the card at least a day prior to the disbursement of wages in respect of the wage period under reference.
- (vi) The contractor shall obtain the signature or thumb impression of the worker on the wage slip at the time of disbursement of wages and retain the card with himself.

9. EMPLOYMENT CARD

The contractor shall issue an Employment Card in Form XIV of the

CL (R&A) Central Rules 1971 to each worker within three days of the employment of the worker (**Appendix-VIII**).

10. SERVICE CERTIFICATE

On termination of employment for any reason whatsoever the contractor shall issue to the workman whose services have been terminated, a Service certificate in Form XV of the CL (R&A) Central Rules 1971 (**Appendix-IX**).

11. PRESERVATION OF LABOUR RECORDS

All records required to be maintained under Regulations Nos. 6 & 7 shall be preserved in original for a period of three years from the date of last entries made in them and shall be made available for inspection by the Engineer-in-Charge or Labour Officer or any other officers authorised by the Central Government or RITES in this behalf.

12. POWER OF LABOUR OFFICER TO MAKE INVESTIGATIONS OR ENQUIRY

The Labour Officer or any person authorised by Central Government on their behalf shall have power to make enquires with a view to ascertaining and enforcing due and proper observance of Fair Wage Clauses and the Provisions of these Regulations. He shall investigate into any complaint regarding the default made by the contractor or subcontractor in regard to such provision.

13. REPORT OF LABOUR OFFICER

The Labour Officer or other persons authorised as aforesaid shall submit a report of result of his investigation or enquiry to the Site Engineer concerned indicating the extent, if any, to which the default has been committed with a note that necessary deductions from the contractor's bill be made and the wages and other dues be paid to the labourers concerned. In case an appeal is made by the contractor under Clause 14 of these regulations, actual payment to labourers will be made by the Executive Engineer after the Superintending Engineer has given his decision on such appeal.

- (i) The Site Engineer shall arrange payments to the labour concerned within 45 days from the receipt of the report from the Labour Officer or the Engineer-in-Charge as the case may be.

14. APPEAL AGAINST THE DECISION OF LABOUR OFFICER

Any person aggrieved by the decision and recommendations of the Labour Officer or other person so authorized may appeal against such

decision to the Engineer-in-Charge concerned within 30 days from the date of decision, forwarding simultaneously a copy of his appeal to the Site Engineer concerned but subject to such appeal, the decision of the officer shall be final and binding upon the contractor.

15. PROHIBITION REGARDING REPRESENTATION THROUGH LAWYER

- (i) A workman shall be entitled to be represented in any investigation or enquiry under these regulations by: -
 - (a) An officer of a registered trade union of which he is a member.
 - (b) An officer of a federation of trade unions to which the trade union referred to in clause (a) is affiliated.
 - (c) Where the employers is not a member of any registered trade union, by an officer of a registered trade union, connected with the industry in which the worker is employed or by any other workman employed in the industry in which the worker is employed.
- (ii) An employer shall be entitled to be represented in any investigation or enquiry under these regulations by: -
 - a. An officer of an association of employers of which he is a member.
 - b. An officer of a federation of associations of employers to which association referred to in clause (a) is affiliated.
 - c. Where the employers are not a member of any association of employers, by an officer of association of employer connected with the industry in which the employer is engaged or by any other employer, engaged in the industry in which the employer is engaged.
- (iii) No party shall be entitled to be represented by a legal practitioner in any investigation or enquiry under these regulations.

16. INSPECTION OF BOOKS AND SLIPS

The contractor shall allow inspection of all the prescribed labour records to any of his workers or to his agent at a convenient time and place after due notice is received or to the Labour Officer or any other

person, authorized by the Central Government on his behalf.

17. SUBMISSIONS OF RETURNS

The contractor shall submit periodical returns as may be specified from time to time.

18. AMENDMENTS

RITES may from time to time add to or amend the regulations and on any question as to the application/Interpretation or effect of those regulations the decision of the Engineer-in-Charge concerned shall be final.

Appendix 'I'

REGISTER OF MATERNITY BENEFITS (Clause 19 F)

Name and address of the contractor

Name and location of the work

Name of the employee	Father's/husband's name	Nature of employment	Period of actual employment	Date on which notice of confinement given
1	2	3	4	5

Date on which maternity leave commenced and ended

Date of delivery/ miscarriage	In case of delivery		In case of miscarriage	
	Commenced	Ended	Commenced	Ended
6	7	8	9	10

Leave pay paid to the employee

In case of delivery		In case of miscarriage		Remarks
Rate of leave pay	Amount paid	Rate of leave pay	Amount paid	
11	12	13	14	15

**SPECIMEN FORM OF THE REGISTER, REGARDING
MATERNITY BENEFIT ADMISSIBLE TO THE CONTRACTOR'S LABOUR**

Name and address of the contractor

Name and location of the work

1. Name of the woman and her husband's name.
2. Designation
3. Date of appointment.
4. Date with months and years in which she is employed.
5. Date of discharged/dismissal, if any.
6. Date of production of certificates in respect of pregnancy.
7. Date on which the women Informs about the expected delivery.
8. Date of delivery/miscarriage/death.
9. Date of production of certificates in respect of delivery/miscarriage.
10. Date with the amount of maternity/death benefit paid in advance of expected delivery.
11. Date with amount of subsequent payment of maternity benefit.
12. Name of the person nominated by the woman to receive the payment of the maternity benefit after her death.
13. If the woman, dies, the date of her death, the name of the person to whom

maternity benefit amount was paid, the month thereof and the date of payment.

14. Signature of the contractor authenticating entries in the register.
15. Remarks column for the use of Inspecting Officer.

Appendix 'III'

Labour Board

Name of work

Name of Contractor

Address of Contractor

Name and address of Employer

Name of RITES' Engineer-in-Charge

Address of RITES' Engineer-in-Charge

Name of Labour Enforcement Officer

Address of Labour Enforcement Officer

Sl. No.	Category	Minimum wage fixed	Actual wage paid	Number present	Remarks

Weekly holiday

Wage period

Date of payment of wages

Working hours

Rest interval

Appendix 'IV'

Form-XIII (See Rule 75)

Register of workmen Employed by Contractor

Name and address of contractor

Name and address of establishment under
which contract is carried on.....

Nature and location of work

Name and address of Principal Employer

Sl. No.	Name and Surname of workman	Age and Sex	Father' s/Husband' s name	Nature of employment/designation	Permanent home address of the workman (Village and Tehsil, Tulak and District)	Local Address	Date of commencement of employment	Signature or thumb impression of the workman	Date of termination of employment	Reasons for terminations	Remarks
1	2	3	4	5	6	7	8	9	10	11	12

Appendix 'V'

Form-XVI (See Rule 78 (2))

(a) Muster Roll

Name and address of contractor

Name and address of establishment
under which contract is
carried on

Nature and location of work

Name and address of Principal Employer For the
Month of fortnight

Sl. No.	Name of workman	Sex	Father's/ husband's name	Dates					Remarks
1	2	3	4	5					6
				1	2	3	4	5	

Appendix 'VI'

Form –XVII (See Rule 78(2))

(a) Register of wages

Name and address of Contractor

Name and address of establishment under which contract is carried on.....

Name and location of work

Name and address Principal Employer

wages period: Monthly/Fortnight

Sl. No.	Name of Workman	Serial No.in the register of workman	designation / nature of work done	No. of days work done	Units of work done	Daily rate of wages/piece rate	mount of wages earned					Deductions if any (indicate nature)	Net amount paid	Signature of thub impression of the workman	Initial of contractor or his representative
							Basic wages	Dearness allowance	Overtime	Other cash	Total				
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16

Appendix 'VII'

Form-XIX [See Rule 78 (2) (b)]

Wages Slip

Name and address of contractor

Name and Father's/Husband's name of workman

Nature and location of work

For the Week/Fortnight/Month ending

1. No. of days worked

2. No. of units worked in case of piece rate workers

3. Rate of daily wages/piece rate
.....

4. Amount of overtime wages

5. Gross wages payable

6. Deduction, if any

7. Net amount of wages paid

Appendix 'VIII'

**Form –XIV [See Rule 76]
Employment Card**

Name and address of Contractor

Name and address of establishment under which contract is carried on

Name and location of work

Name and address Principal Employer

1. Name of the workman
2. Sl. No. in the register of workman employed
3. Nature of employment/ designation
4. Wage rate (with particulars of unit in case of piece work)
5. wage period
6. Tenure of employment
7. Remarks

Signature of contractor

Appendix 'IX'

Form –XV (See Rule 77) Service Certification

Name and address of Contractor

Nature and Location of work

Name address of workman

Age or date of birth

Identification marks

Father's Husband's Name

Name and address of establishment in under which contract is carried on.....

Name and address of Principal Employer.....

Sl. No.	Total Period for which employed		Nature of Work done	Rate of wages (with particulars of unit in case of piece work)	Remarks
	For m	To			
1	2	3	4	5	6

Appendix 'X'

LIST OF ACTS AND OMISSIONS FOR WHICH FINE CAN BE IMPOSED

In accordance with rule 7 (v) of the RITES Contractor's Labour Regulations of Regulations to be displayed prominently at the site of work both in English and local Language.

1. Withful insubordination or disobedience, whether alone or in combination with other.
2. Theft fraud or dishonesty in connection with the contractors besides a business or property of RITES/Employer.
3. Taking or giving bribes or any illegal gratification.
4. Habitual late attendance.
5. Drunkenness lighting, riotous or disorderly or indifferent behaviour.
6. Habitual negligence.
7. Smoking near or around the area where combustible or other materials are locked.
8. Habitual indiscipline.
9. Causing damage to work in the progress or the property of the RITES/Employer or of the contractor.
10. Sleeping on duty.
11. Malingering or slowing down work.
12. Giving of false information regarding name, age father's name, etc.
13. Habitual loss of wage cards supplied by the employers.
14. Unauthorized use of employer's property of manufacturing or making of unauthorized particles at the work place.
15. Bad workmanship in construction and maintenance by skilled workers which is not approved by the RITES/Employer and for which the contractors are compelled to undertake rectifications.
16. Making false complaints and/or misleading statements.
17. Engaging on trade within the premises of the establishments.
18. Any unauthorized divulgence of the collection of any money authorized by the employer.
19. Holding meeting inside the premises without previous sanction of the employers.
20. Threatening or intimidating any workman or employer during the working hours within the premises.
21. Threatening or intimidating any workman or employer during the working hours within the premises.

Appendix 'XI'

Form-XV (See Rule 77)

Form-XII (See Rule 78 (2) (d))

Register of Fines

Name and Address of contractor

Name and Address of establishment in under which contract is carried on

Nature and location of work

Name and address of Principal Employer

Sl. No.	Name of workman	Father' s/Husband' s name	Designation/Nature of employment	Act/Omission for which fine imposed	Date of Offence	Whether workman showed cause against fine	Name of person in whose presence employee' s explanation was heard	Wage period and wages payable	Amount of fine imposed	Date of which fine raised	Remarks
1	2	3	4	5	6	7	8	9	10	11	12

Appendix 'XII'

Form-XX (See Rule 78(2) (d))
Register of Deduction for Damage of Loss

Name and Address of contractor.....

Name and Address of establishment in under which contract is carried on

Nature and location of work

Name and address of Principal Employer

Sl. No.	Name of workman	Father' s/Husband' s name	Designation/Nature of employment	Particulars of damage or loss	Date of Offence	Whether workman showed cause against deduction	Name of person in whose presence employee' s explanation was heard	Amount of fine imposed	No. of Instalments	Date of recovery		Remarks
										First Instalment	Last Instalments	
1	2	3	4	5	6	7	8	9	10	11	12	13

Appendix 'XIII'

Form-XXII (See Rule 78(2) (d))

Register of Advances

Name and Address of contractor.....

Address of establishment in under which contract is carried on.....

Nature and location of work

Name and address of Principal Employer.....

Sl. No.	Name of workman	Father' s/Husband , s name	Designation/Nature of employment	Wage Period and wages payable	Date and amount of advance given	Purpose (s) for which advance made	Number of instalments by which advance to be repaid	Date and amount of each instalment repaid	Date and which last instalment was repaid	Remarks
1	2	3	4	5	6	7	8	9	10	11

Appendix 'XIV'

Form-XXIII (See Rule 78(2) (e))

Register of Overtime

Name and Address of contractor.....

Name and Address of establishment in under which contract is carried on

Nature and location of work

Name and address of Principal Employer

Sl. No.	Name of workman	Father' s/Husband' s name	Sex	Designation/Nature of employment	Date on which Overtime	Total overtime worked or production in case of piece	Normal rate of wages	Overtime rate of wages	Overtime earning	Rate on which overtime paid	Remarks
1	2	3	4	5	6	7	8	9	10	11	12

PART -2
FINANCIAL BID

PART -2

FINANCIAL BID

SCHEDULE (BILL OF QUANTITIES)

Name of Work: Construction of Workman Housing Complex (BQ Area) on EPC mode -II of contract at OIL Duliajan, Assam.

Points to be noted while quoting rates:

1. Where Tender has been invited on Lumpsum Component basis (EPC Mode – II of Contract), bidder shall quote the rates for each component of the schedule
2. Bidders are requested to note that they should necessarily submit their financial bids in the format provided and no other format is acceptable. If the price bid has been given as a standard BOQ format with the tender documents, then the same is to be downloaded and to be filled by all the bidders. Bidders are required to download the BOQ file, open it and complete the Sky Blue coloured (unprotected) cells with their respective financial quotes and other details (such as name of the bidder). No other cells should be changed. Once the details have been completed, the bidder should save it and submit it online, without changing the filename. If the BOQ file is found to be modified by the bidder, the bid will be rejected.
3. Depending on scope and nature of work, additional points may be incorporated by the Authority Competent to approve Tender Documents under item 3.0 of SOP